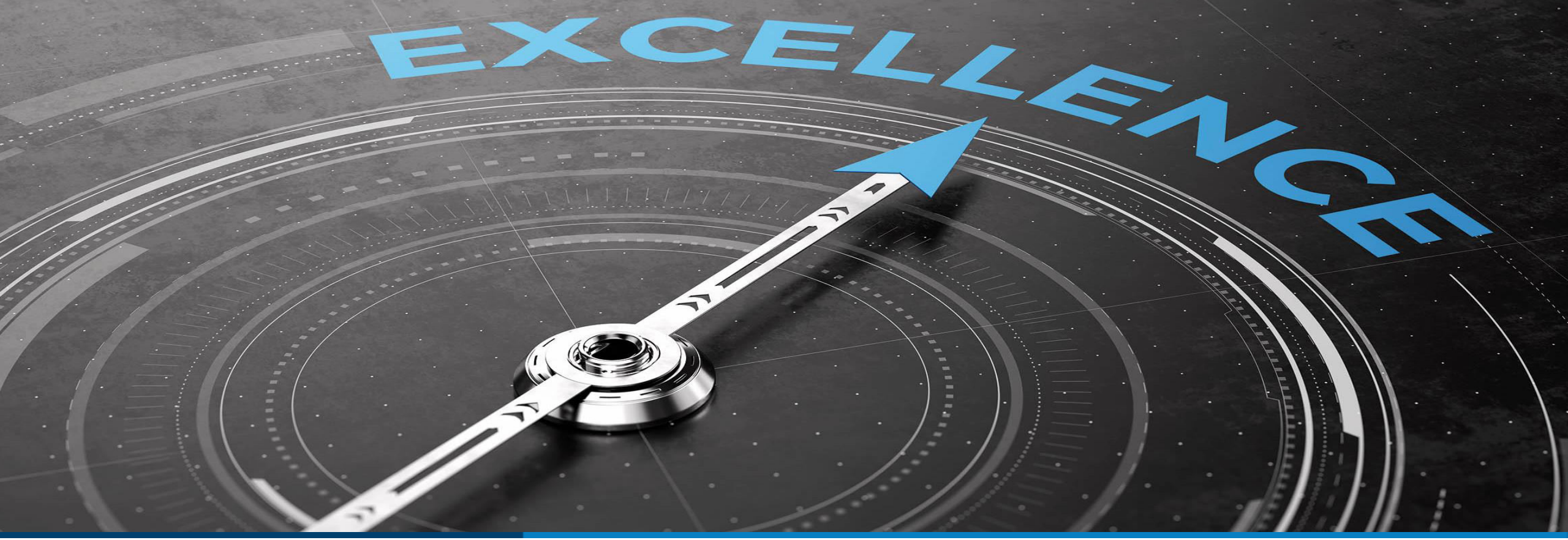


PSQ Green Belt
Manufacturing & Process Quality

Sensors & Actuators I Business Excellence I

Date: May 2021 | M-OPME

Version: Ib

PASSION FOR TECHNOLOGY.

RHEINMETALL

- **Name**
- **Responsibilities**
- **Expectations**



PS Expert



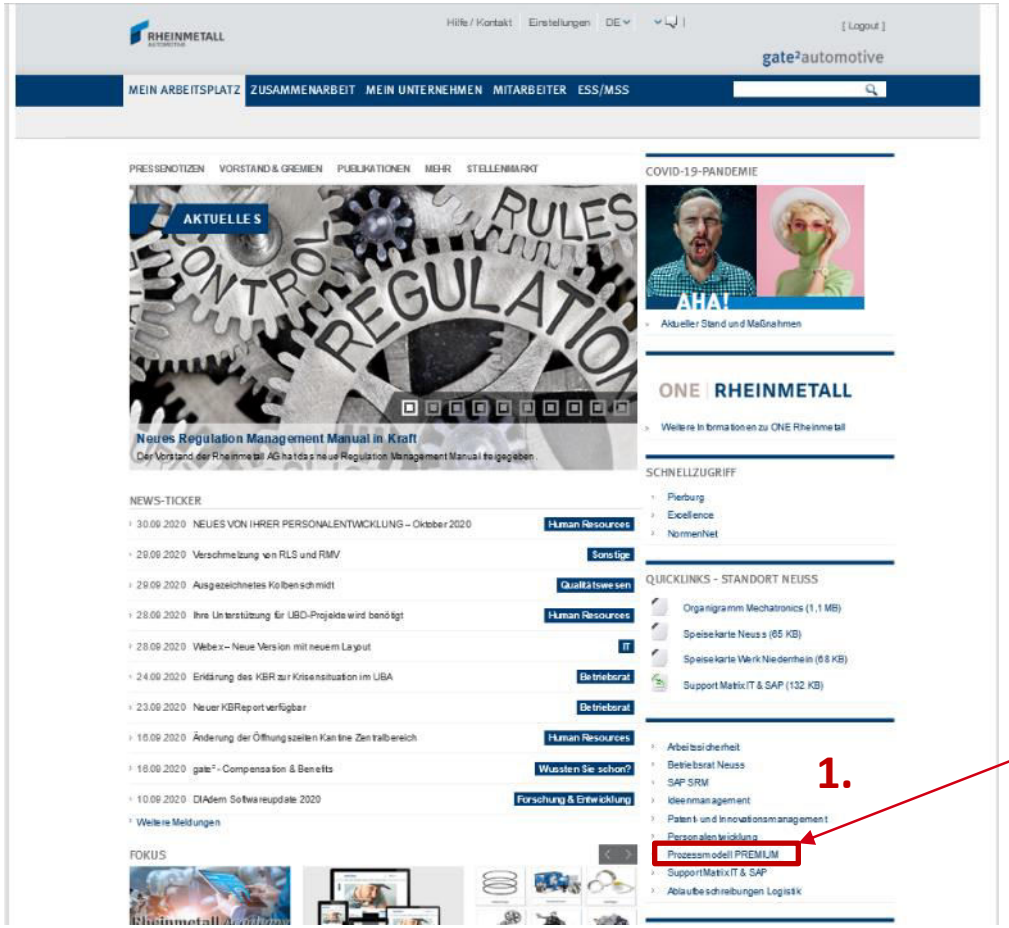
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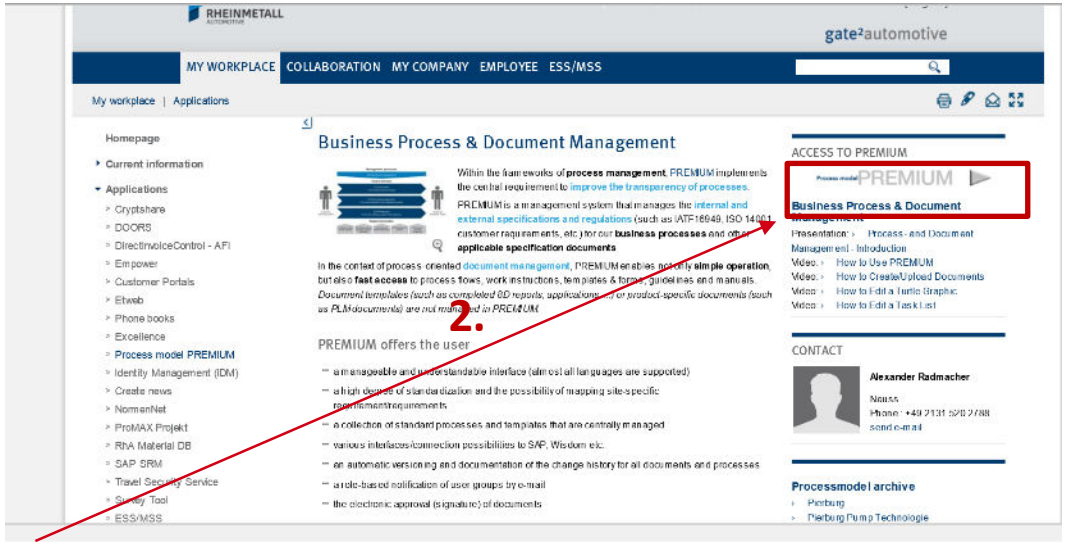
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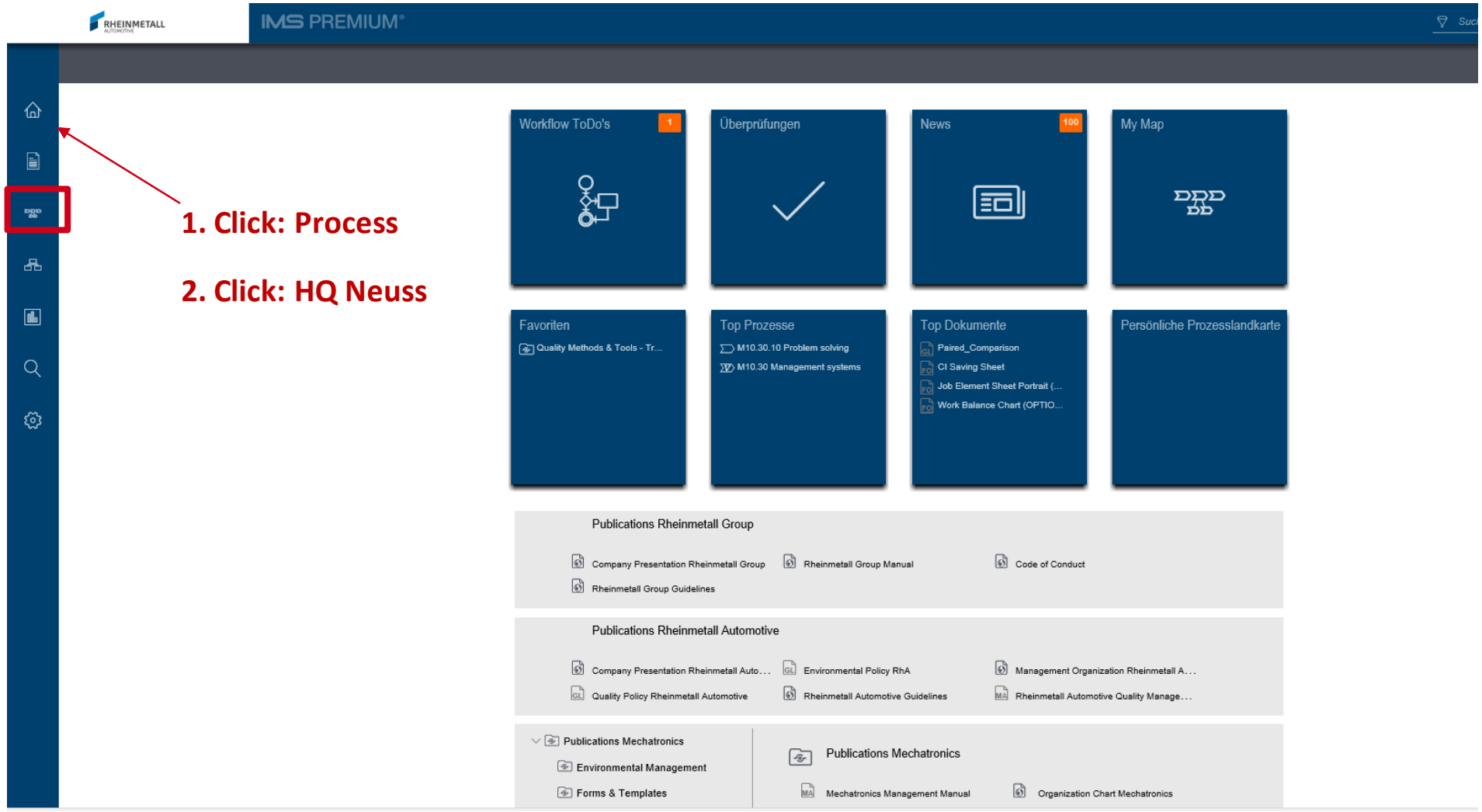
Start Explorer → Process Model Premium



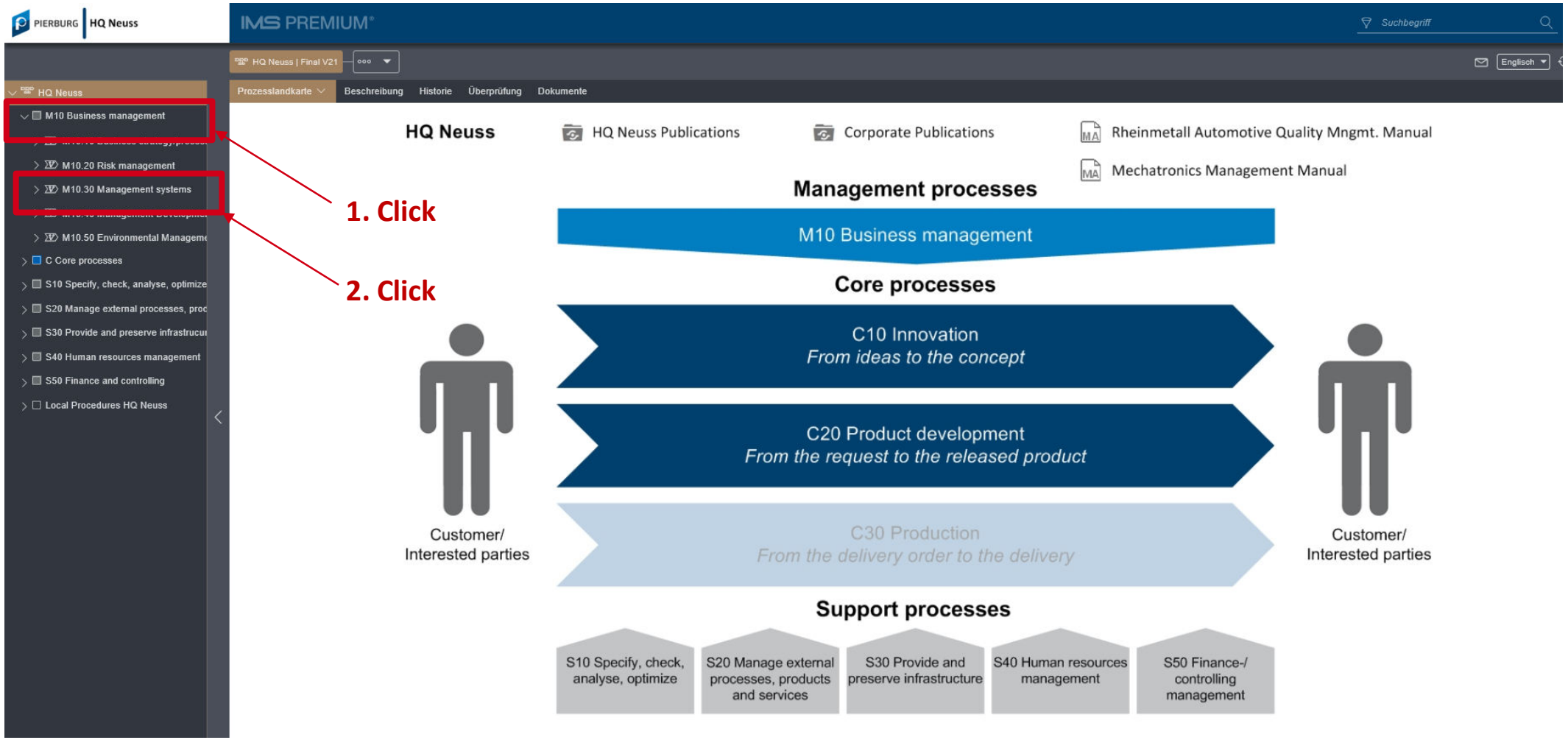


Click: Process Model Premium

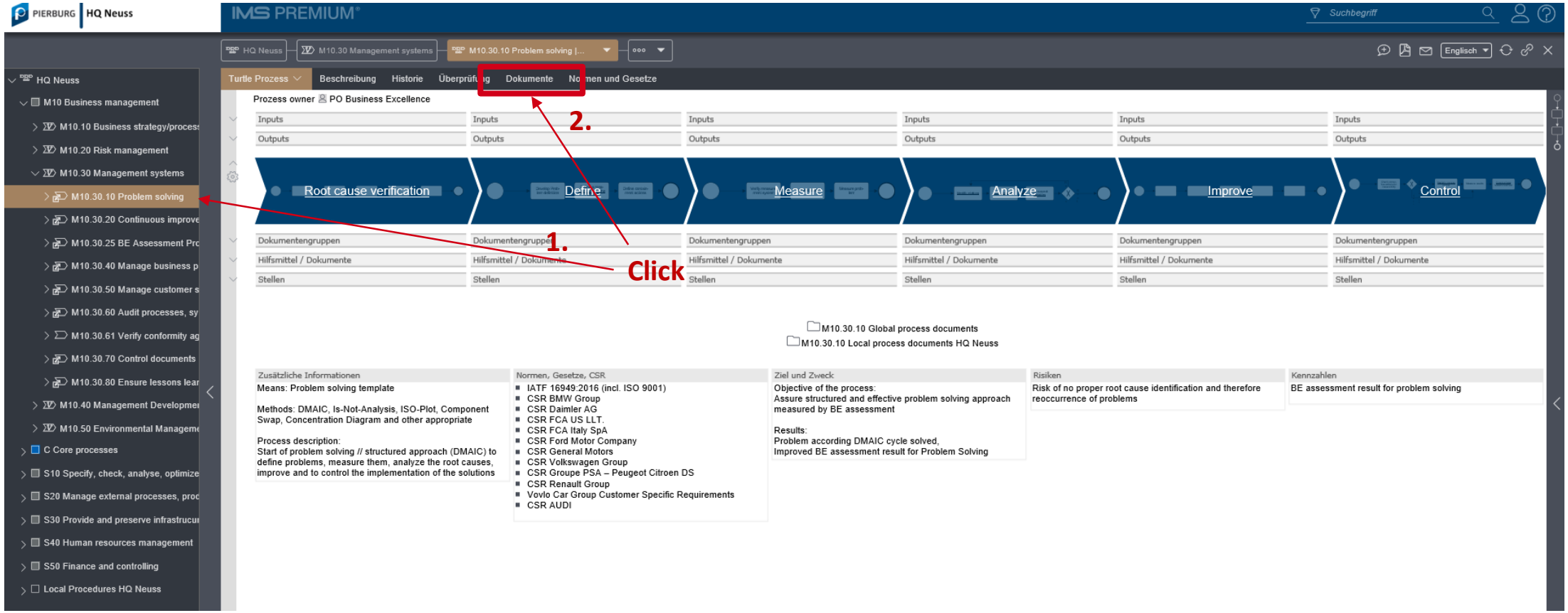
Process Model Premium → IMS



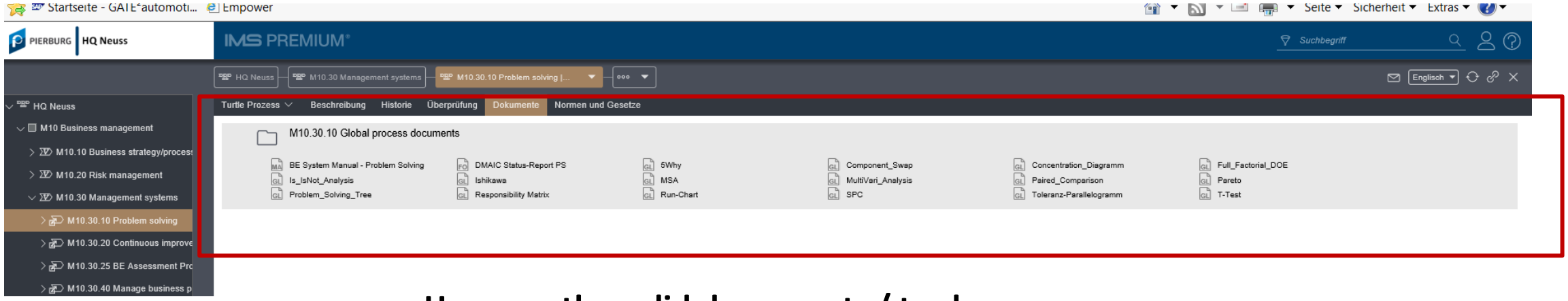
Process Model Premium → IMS



Process Model Premium → IMS



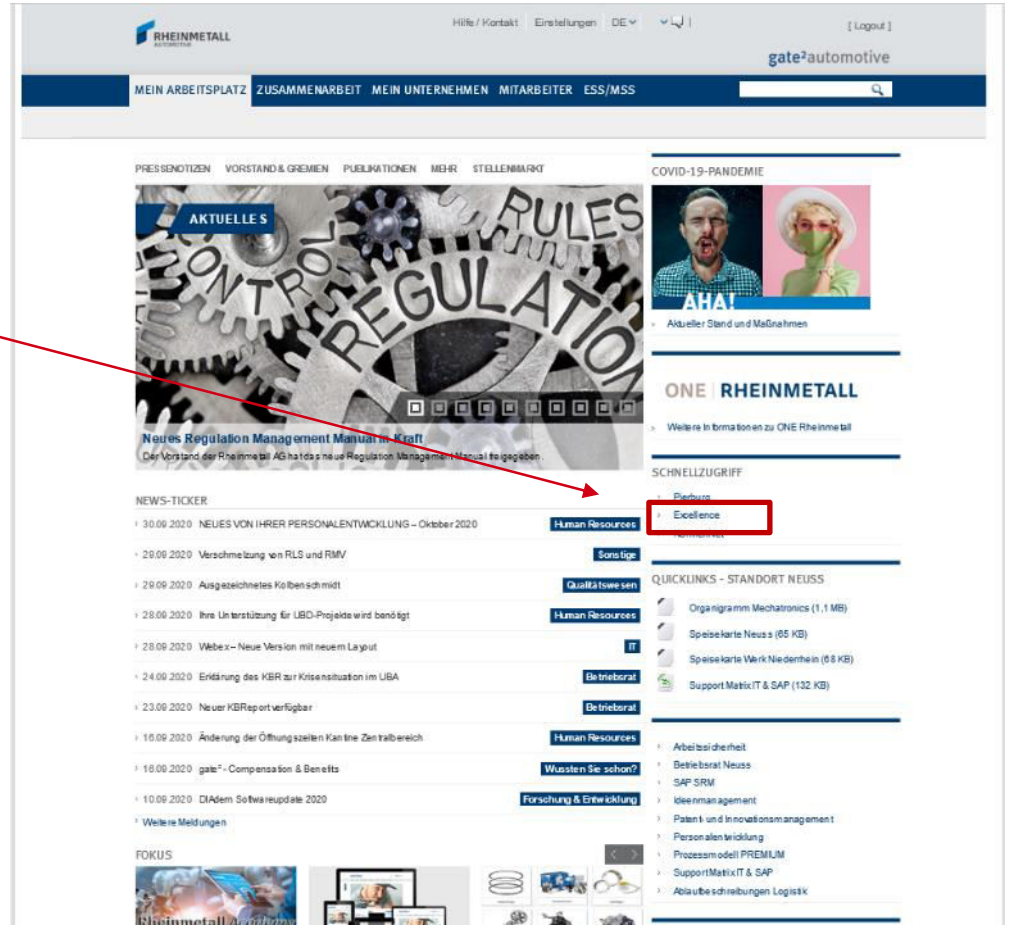
Process Model Premium → IMS → Documents



Here are the valid documents / tools

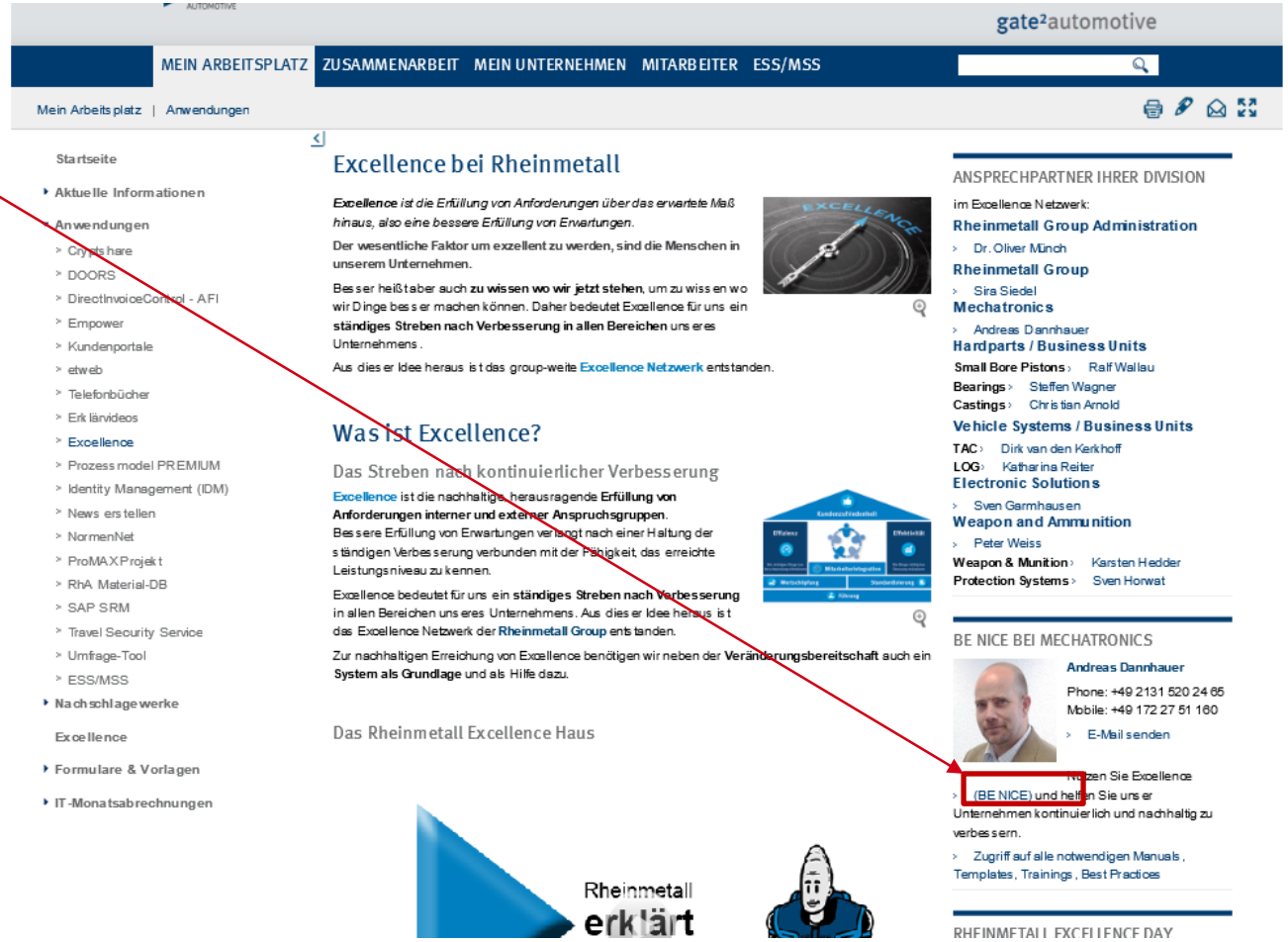
Business Excellence Homepage

Click: Excellence

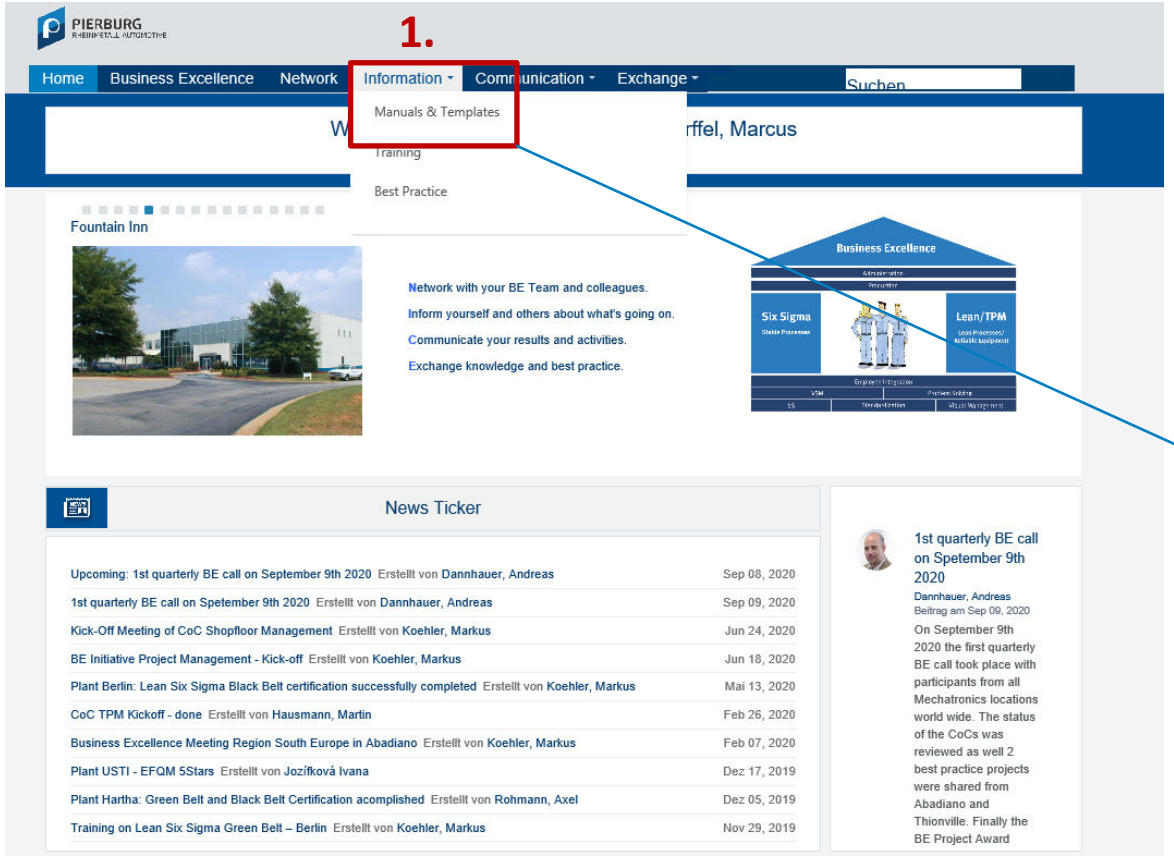


Start Explorer → Excellence

Klick: BE NICE

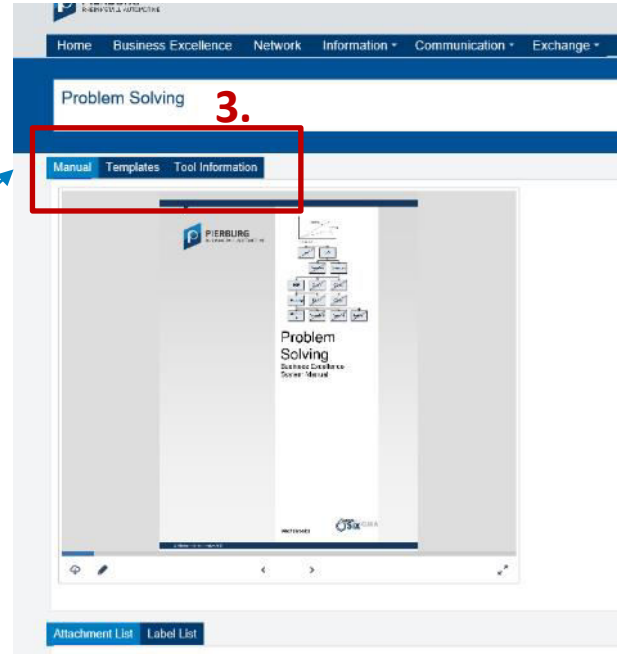


BE Homepage → PS Homepage → Information (Templates, Manual, …)





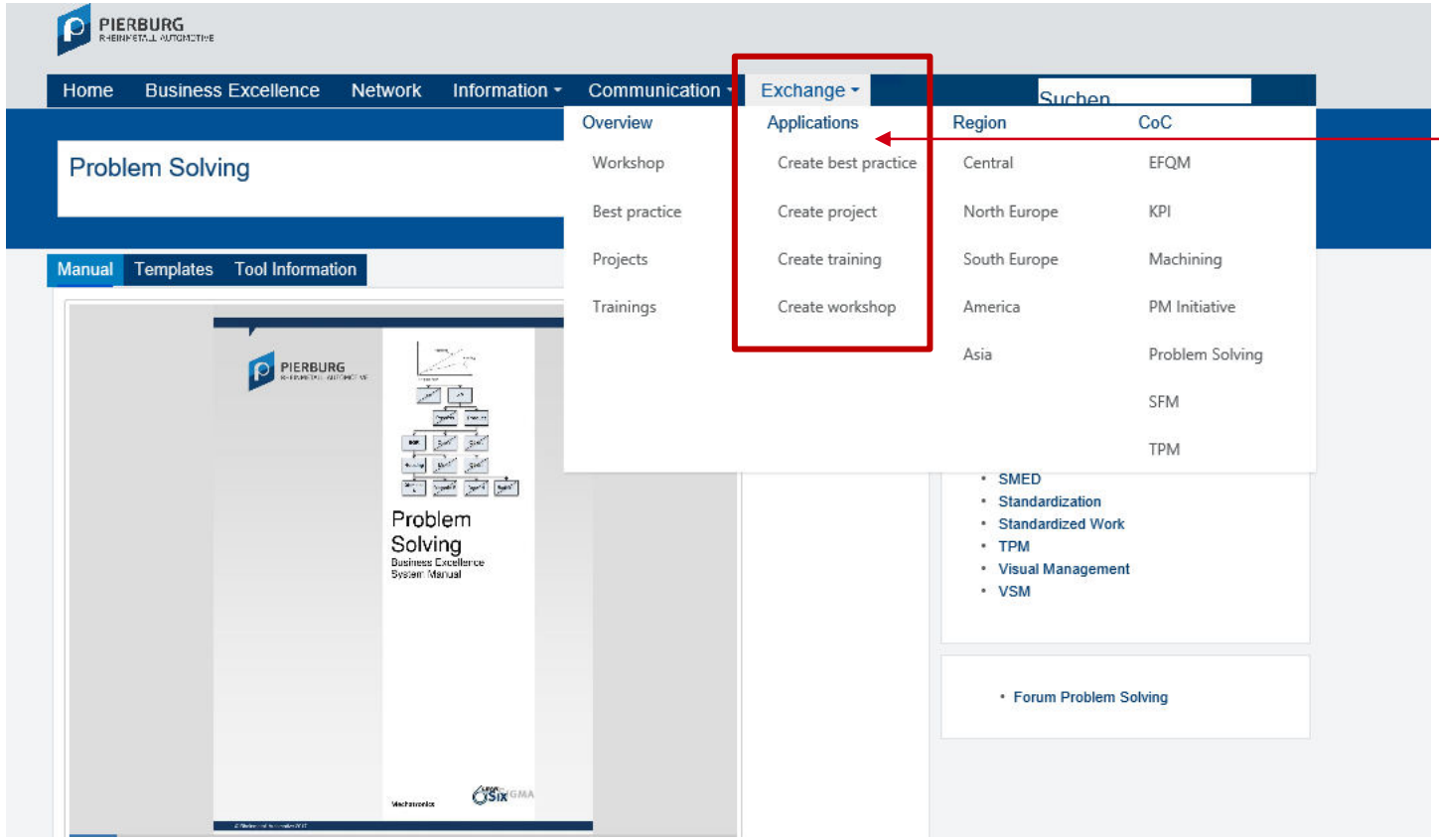
Problem Solving Homepage



Here you will find information about PS:

- **PS manual and others**
- **Links to templates, information about tools**
- **PS Forum**

BE Homepage → PS Homepage → Exchange (Projects, Trainings, …)



Here you can create the following:

- **Best practice**
- **Projects**
- **Training**
- **Workshops**

If you have any questions, please contact BE.

Table of Contents I/II

1 Basics

2 Effective Measurement Systems

3 Feature Strategies

4 Property Strategies

a Example

5 Defect Strategies

6 Malfunction Event Strategies I

Table of Contents II/II

7 Confirmation Testing I

8 Tolerancing / Finishing Projects I

9 Summary

G Glossary

1 Basics

Module 1 Objectives

Basics

Understand:

- Y → X Thinking
- BIG X Paradigm
- DMAIC Phases

Be able to:

- Explain the following:
  - Mone(Y)
  - BIG X
  - BOB and WOW
  - Pareto
  - Dictionary Thinking
  - Process Flow Diagram
  - Problem Description (Cartooning)
  - Problem Definition Tree, Project Definition Tree and Solution Tree
  - Strategy Diagram

Typical problem solving process



Strategy:

- **Trial and Error**
- **A Lot of Action**
- **Frustrating Results**
- **X → Y**

Process:

1. **Gather a group of experts.**
2. **Appoint a facilitator.**
3. **Make a list of suspects.**
4. **Put them in order.**
5. **Assign “Action Items.”**
6. **Monitor the fall out for a shift or two.**
7. **Agree to meet again in a week.**

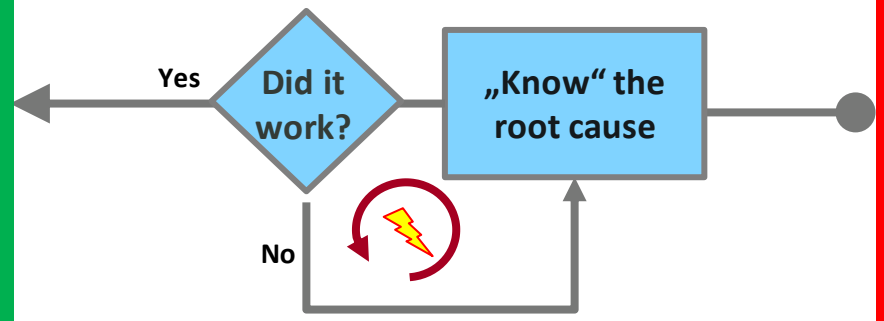
Overview Approaches

Experience Based Approach



Technical Problem

Problem



Root Cause

- Often people tend to "know" what the actual cause is. They jump on the "X" and change something that they think would help
- If the cause is “really known” the duration can be very fast.
- There is a very high risk that this approach become “trial and error”

→ the duration can get very long.

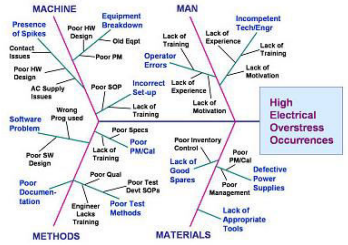
Expert Judgment

Unstructured X → Y

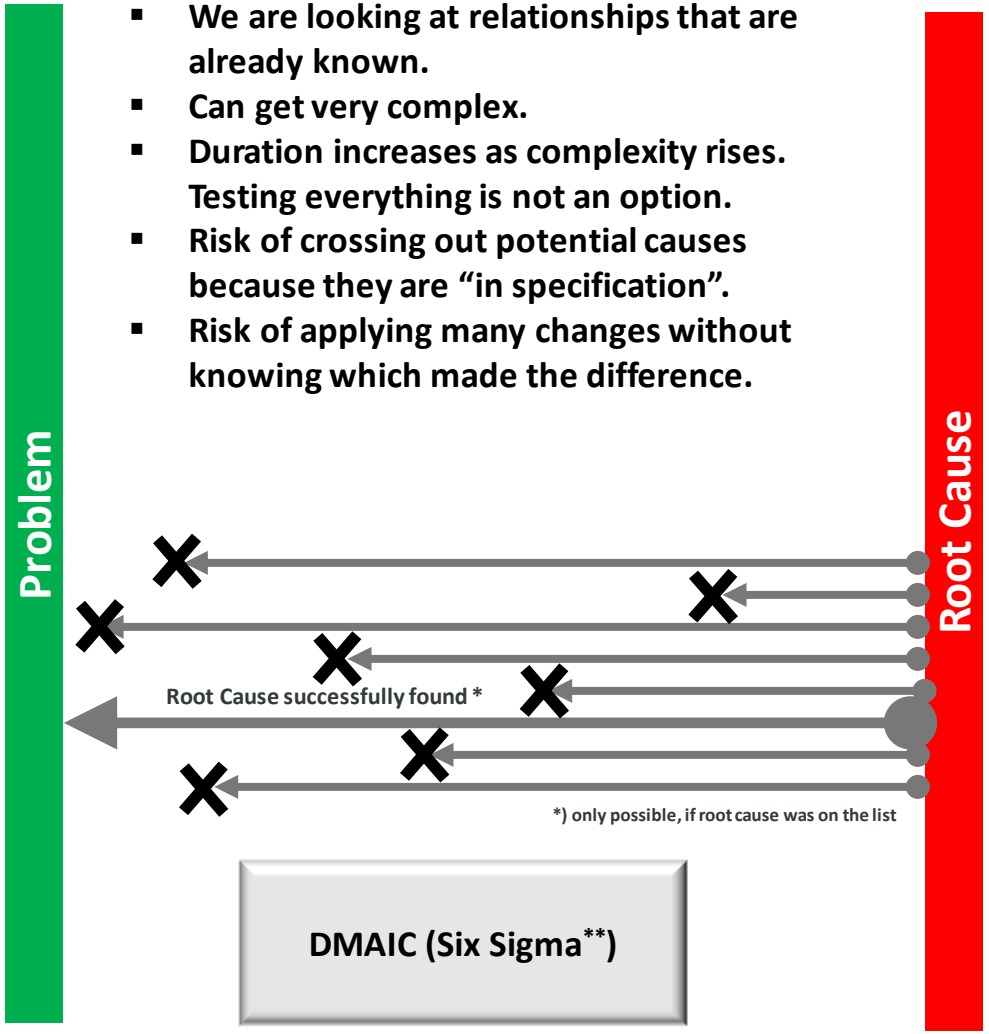
Overview Approaches

Brainstorming Based Approach

Technical Probelm



- We are looking at relationships that are already known.
- Can get very complex.
- Duration increases as complexity rises. Testing everything is not an option.
- Risk of crossing out potential causes because they are “in specification”.
- Risk of applying many changes without knowing which made the difference.



Structured X → Y

** Focus: Data driven approach on factors

X → Y Thinking

Strategy: Change one or more causes (X) to determine whether they create a change a change in effect (Y).

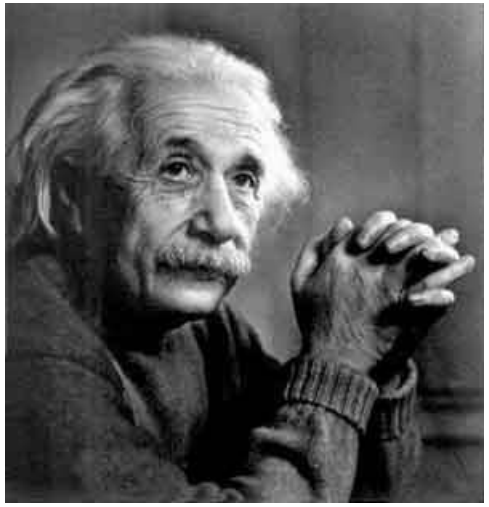
Assumption: I know the relationship between cause and effect.

- Engineering Based
- Math Based

Engineering Based

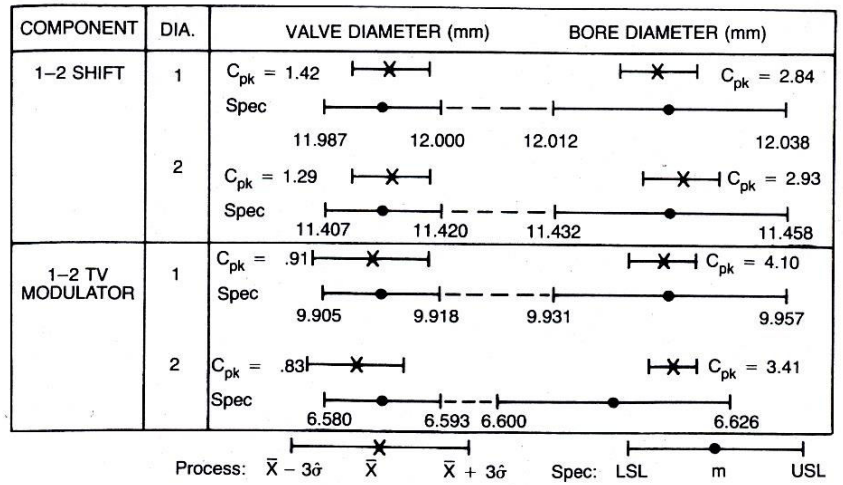
- Talk to experts:
  - Insights into product or process
  - Insights into basic principles:
    - F =ma
    - I =V/R
    - PV = nRT

- Symptoms – perceived root cause – solution.



Math Based

- Talk to data.
- Talk to experts.
- y = f(x)
  $y = a + bx_1, + cx_2 + dx_3 + \ldots$
- Model process:
  - Solve the Problem
  - Optimize the Process
  - Create Best Mathematical Model
- Statistics based techniques:
  - Designed Experiments
  - Analysis of Variance
  - Multiple Regression Analysis

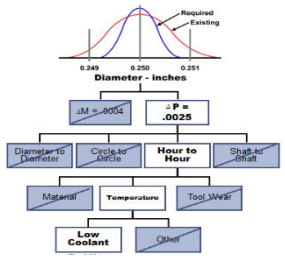


Mathematical relationships are more important then physical relationships.

Overview Approaches

Technical Problem

Contrast Based Approach

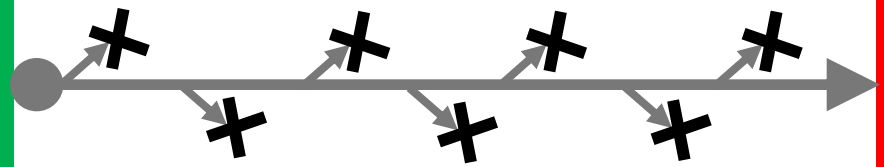


BIG X

Problem

DMAIC (Eliminierung)
5x Warum / KT

- As soon as we have BOBs and WOWs we know that the BIG X is captured.
- The duration is moderate.
- DMAIC has the highest likelihood that the problem will be completely solved.



Root Cause

Structured Y → X

Y → X Thinking

Strategy: Look for differences in various causes (X) at the extremes of the effect (Y) to progressively narrow in on the top few causes (X) driving the variation.

Assumption: I can understand the physics be observing the extremes.

---

1. Identification of **CONTRAST** – talk to the systems/parts.

2. Progressive search – asking questions that **ELIMINATE** primary contributors to that contrast.

3. Factual data to identify the **PARETO** to support the elimination decision.

All 3 must be present for each decision made in a

Project Definition and Solution Tree.

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Evidence Based – Talk to the parts.

- Elimination Strategy is based on evidence gathered by talking to the parts.
- Talk to the parts.
- Use symptoms and engineering knowledge to determine which parts to talk to and how.
- Goal is to find BIG X.
- $\Delta Y = f(\Delta x)$ $\Delta Y = a + b\,\Delta x_1, + c\,\Delta x_2 + d\,\Delta x_3 + \ldots$
- Not interested in modeling the process need to find largest term (BIG X).
- Fastest and surest way to reduce variation and solve problems.



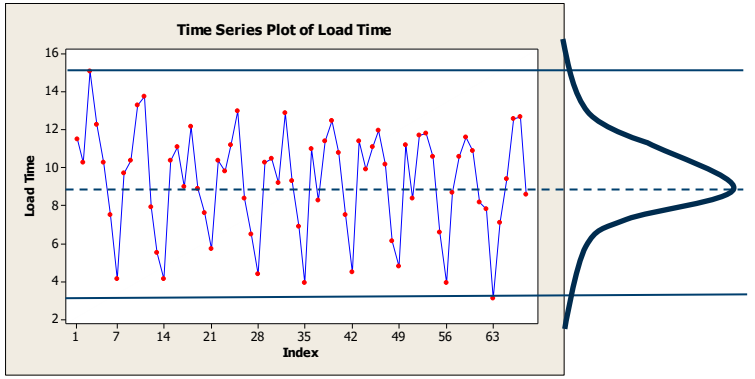
“Keep your eyes open and your mouth shut. Let the parts guide you to the answer.” - Dorian Shainin

Structured Technical Problem Solving
When to Use ES for Technical Problem Solving

General

- We measure the output of a process as the **“Mone(Y)”.**
- The **Mone(Y)** is varying.
- All processes have variation - some more, some less.
- Variation is not generally bad.
- Variation is caused by factors, such as process parameters, material, operators, etc. which we call “**X’s**”.
- To reduce variation we need to understand the impact of the factors **X** to the result **Mone(Y)**.

$$Y = f(x_n)$$

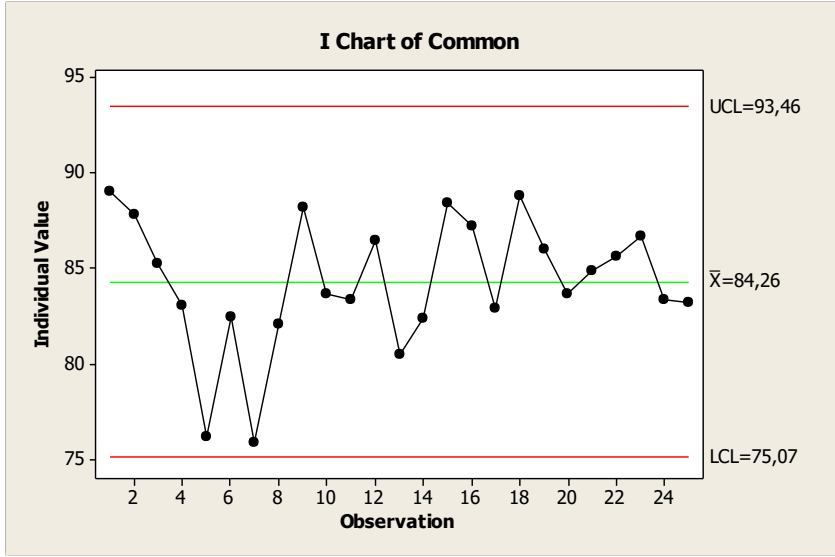


When to Use Elimination Strategy for Technical Problem Solving

Common Cause Variation

- Noise
- Random variation
- In control
- Predictable
- Stable

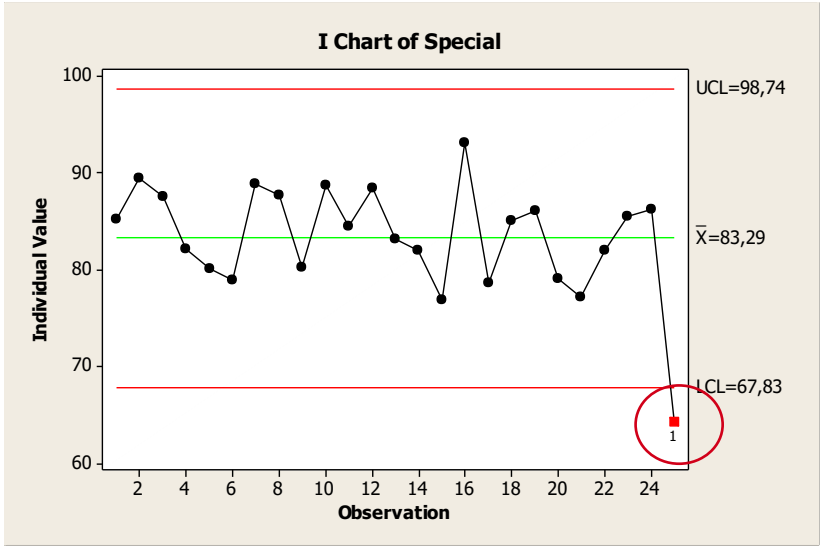
(**X** → **Y** approach)



Special Cause Variation

- Signal
- Not random
- Out of control
- Not predictable
- Unstable

(**Y** → **X** approach)



Overview Approaches

Common Cause(s) Variation

Special Cause(s) Variation

Technical Problem

Experience Based Approach



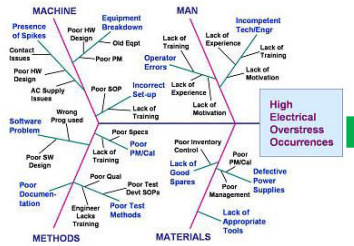
Knowing the Answer

Situational fix of the current Issue.

Trial and Error

Unstructured X → Y

Brainstorming Based Approach



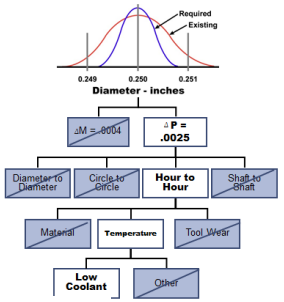
Understanding the system

Iterative elimination of contributing factors.

Six Sigma

Structured X → Y

Contrast Based Approach



BIG X

Understanding the failure

Permanent problem elimination.

Elimination Strategy

Stuctured Y → X

Elimination Strategy vs Six Sigma

Contrast Based Approach

Elimination Strategy (ES)

- The Elimination Strategy investigates the outcome (Mone (Y)) and converges to the root cause (BIG X);
  Strategy: Y → X
- The Elimination Strategy focuses on the main driver (BIG X) for the variation (Mone(Y))
- The Elimination Strategy produces fast results by focusing on the main driver

The ES finds out what factor *IS* causing the problem.

Brainstorming Based Approach

Six Sigma (SS)

- Six Sigma investigates potential causes (Xs) and finds their effect on the outcome (Y);
  Strategy: X → Y
- Six Sigma focuses on all perceived influencing factors
- Six Sigma takes more time to cover many potential factors

Six Sigma finds out what factor *CAN* cause the problem.

Elimination Strategy Overview

The Guiding Principles of Elimination Strategies are:

- Understand the physical world – Learn how the world is organized to develop efficient strategies
- Talk to the system/parts - Look for contrast and leverage
- Progressive search – Start with an open mind and focus on the big contrasts. Small and no contrasts are minor and do not add value.
- Keep it simple

Elimination Strategy Process

1. **Listen to the customer** → Mone(Y)
2. **Definition** of the strategy
3. **Measure** the contrast → E. g. Isograph
4. **Follow** the **large contrasts**
5. **Converge** to the candidate → BIG X candidate
6. **Confirm** to the major influence → BIG X
7. **Implement** control → Control the BIG X

Always make good decisions!



Simple Understanding!

We must understand

- The physical world requires you to understand how it is organized if you are to work with it
- Thinking about the world in terms of physics helps one understand that various combinations of the same inputs can have quite different results:



1. Air
2. Flame
3. Propane



1. Air
2. Propane
3. Flame

Elimination Strategy Concepts
The Pareto Principle

- 20 % of our problems account for 80 % of the cost and customer complaints.
- Seperates the „vital“ few from the „trivial many“.
- Even if we have many problems, we must first focus on the vital few that will provide the best return on investment.

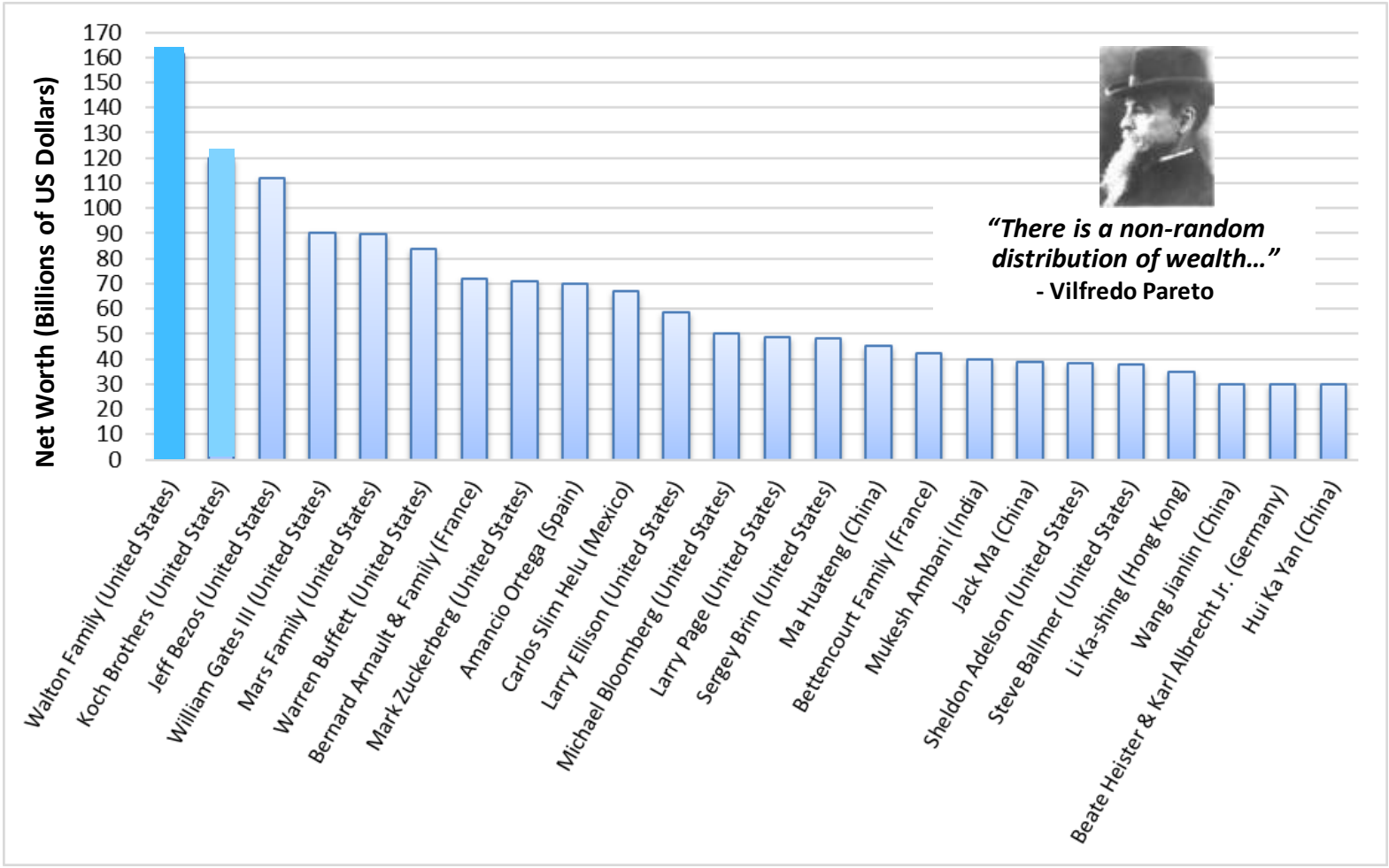


- Vilfredo Pareto
1848 - 1923

Elimination Strategy Concepts

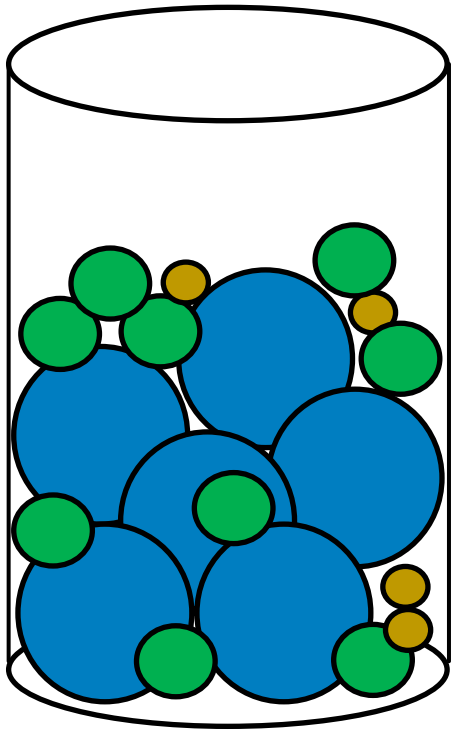
The Pareto Principle

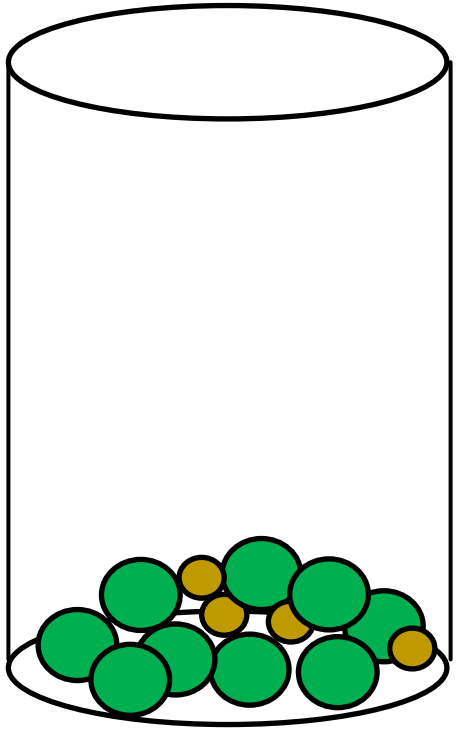
The Richest People in the World (2018)

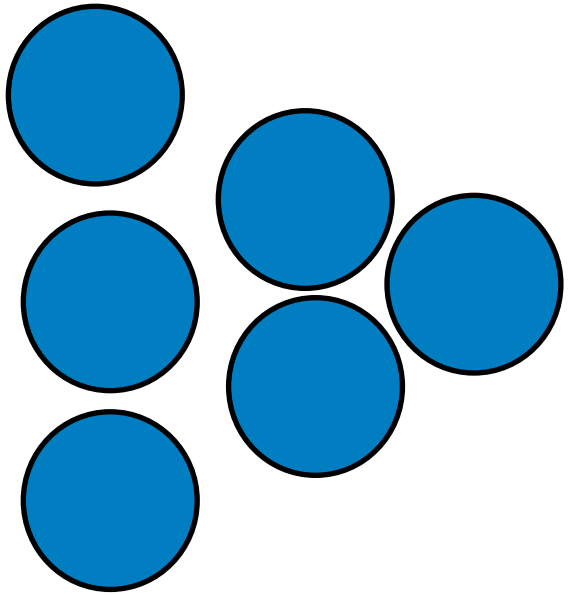


Elimination Strategy Concepts
Pareto Principle – Example – Customers Mone(Y) is Volume

A jar is filled with the following balls: 6 blue, 4 brown, and 9 green balls.
Which color ball removes the most volume if taken out of the jar?







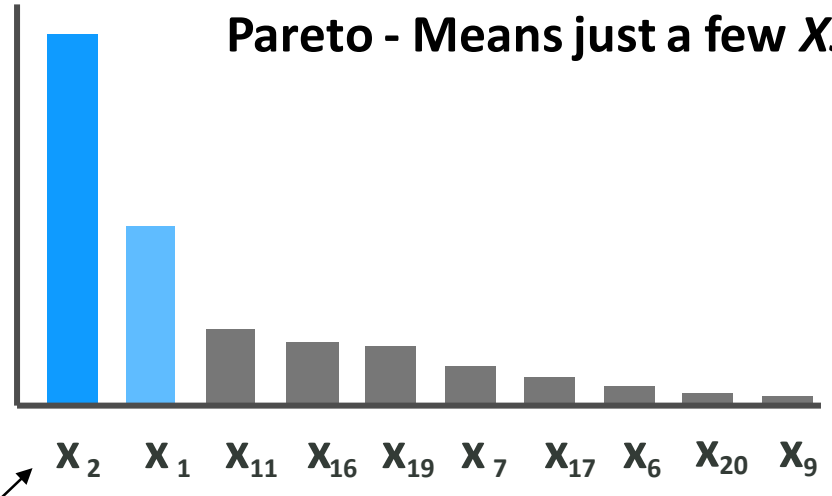
Even though only 6 balls are removed … since the Mone(Y) is volume, it drives a Pareto where the „Blue balls“ are king.

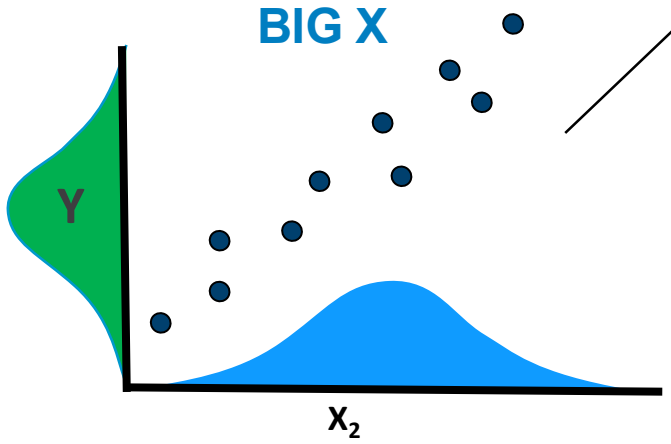
Elimination Strategy Concepts

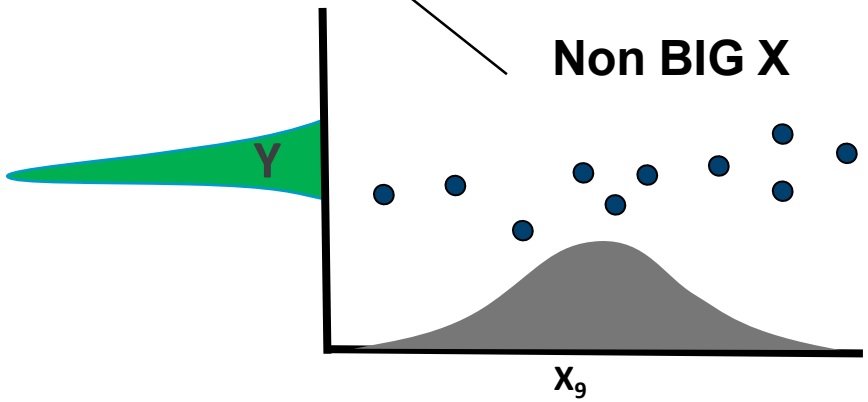
Pareto Principle

ΔY caused by variation in each ***X***

Pareto - Means just a few *Xs* cause most of the *ΔY*







These Xs don't contribute equally to ΔY (the "customer's problem") and their accumulation is not linear.

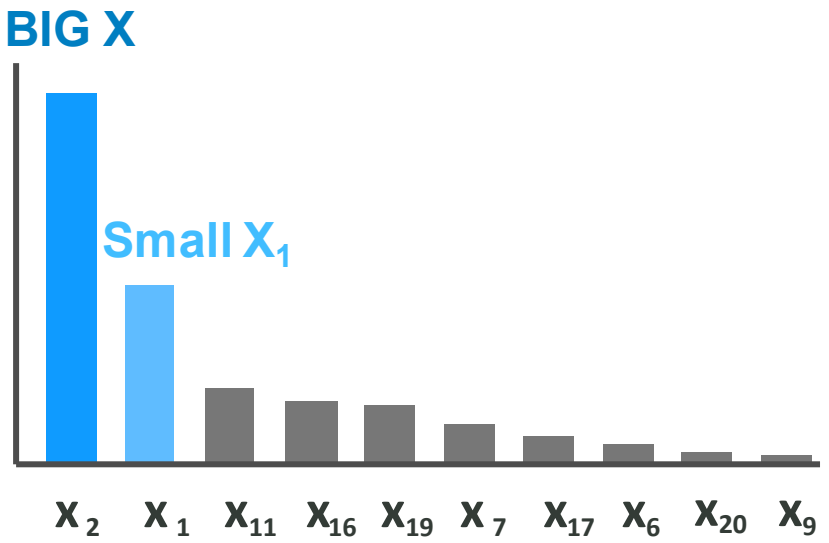
The BIG X is only "BLUE" for a certain amount of time. When discovered and controlled, the BIG X is no longer Blue and the previous light blue takes over.

Elimination Strategy Concepts
Pareto Principle

Pareto - Means just a few *Xs* cause most of the *ΔY*

$$\Delta Y = \sqrt{(\beta_1 \Delta x_1)^2+(\beta_2 \Delta x_2)^2+(\beta_3 \Delta x_3)^2+\ldots+(\beta_{12} \Delta x_1 \Delta x_2)^2+(\beta_{13} \Delta x_1 \Delta x_3)^2+\ldots+\varepsilon^2}$$

ΔY caused by variation in each ***X***



The effects from different causes (Xs) add as the square root of the sum of the squares.

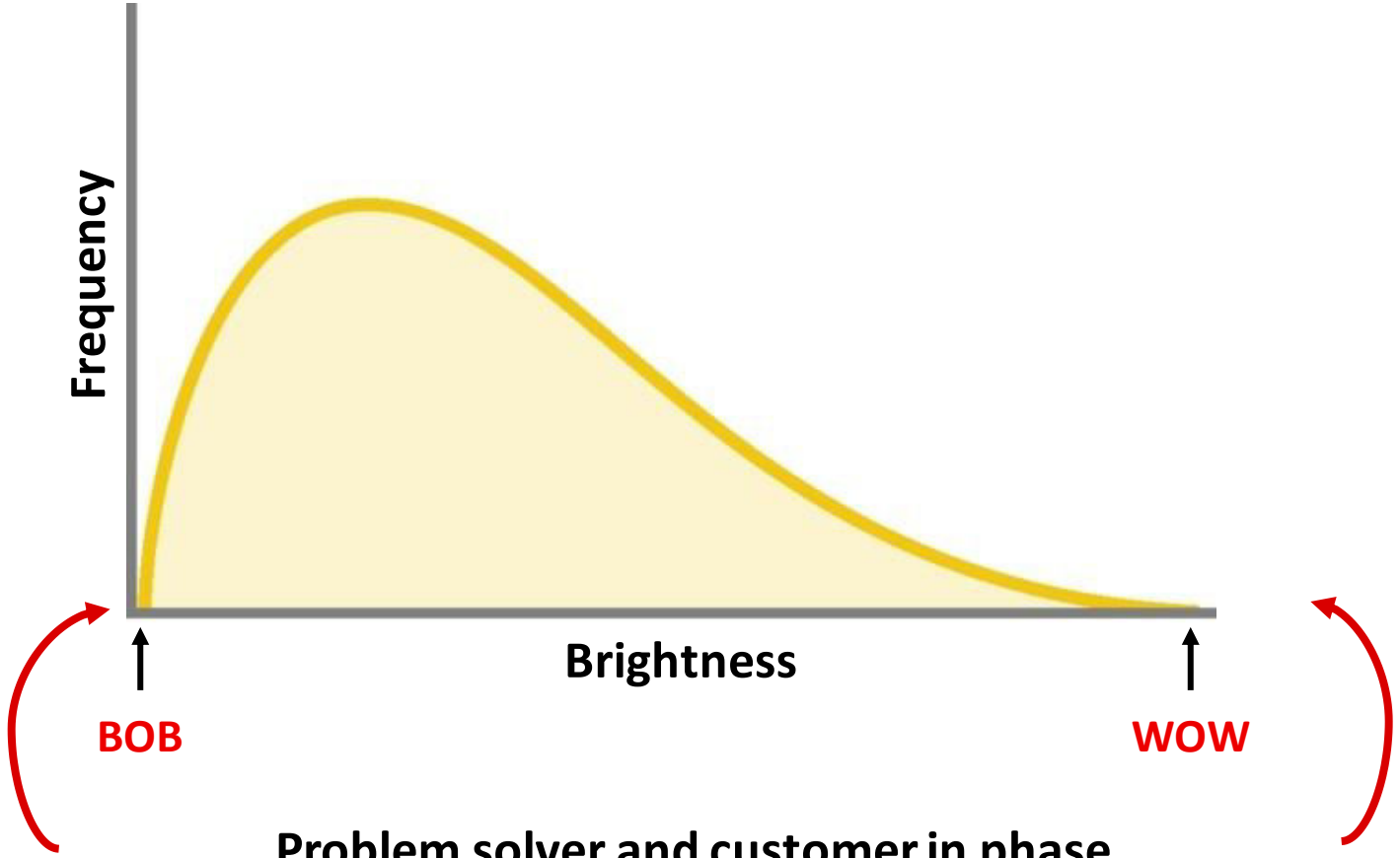
Which X Matters Most? Too many possibilities!

The controlling X (or X's) are:

1	Known	→	start controlling it
2	Suspected	→	confirm it then add controls
3	Unknown	→	find the X, confirm it, control it

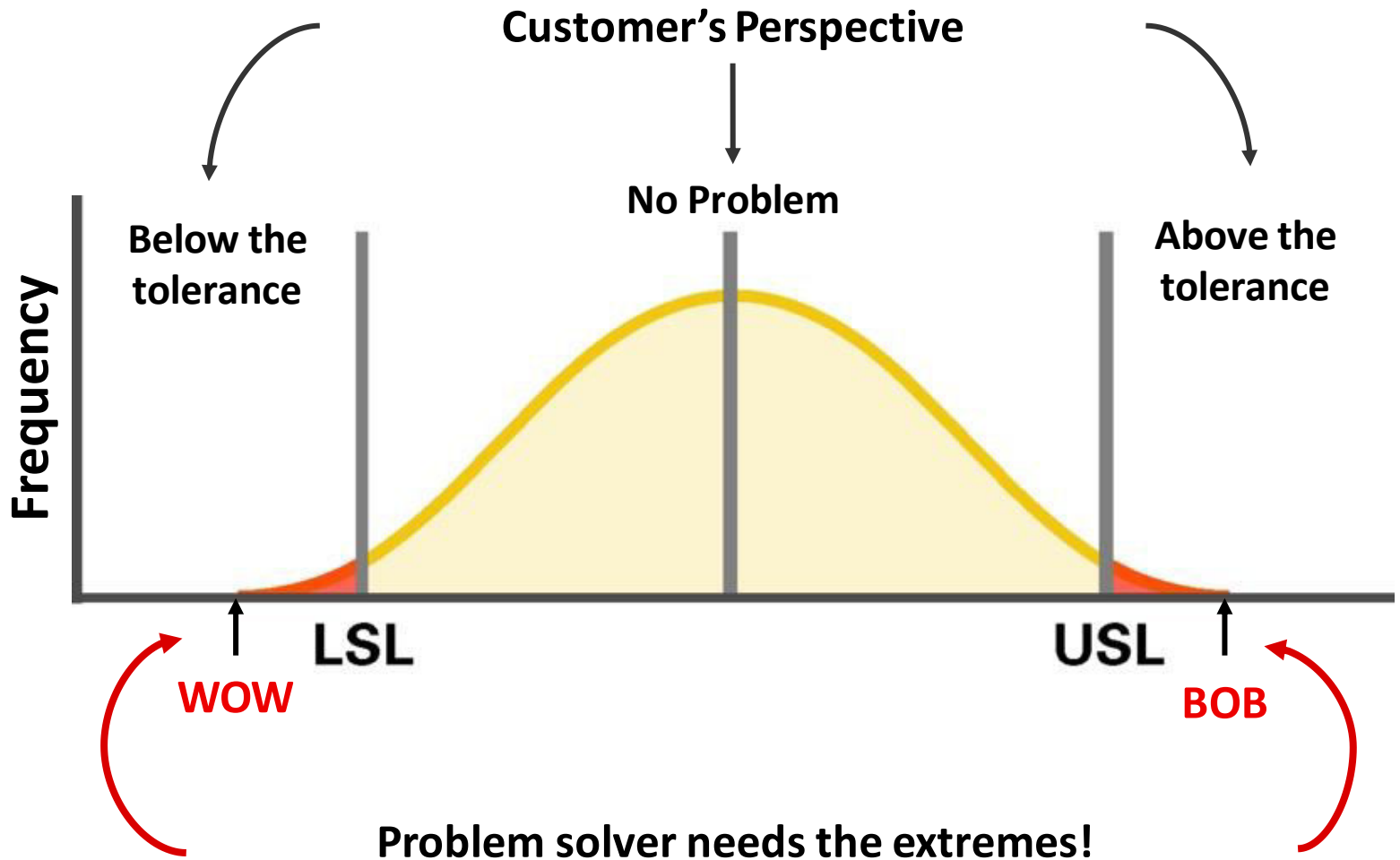
BOB and WOW

- BOB = Best of the Best (extreme opposite of WOW)
- WOW = Worst of the Worst

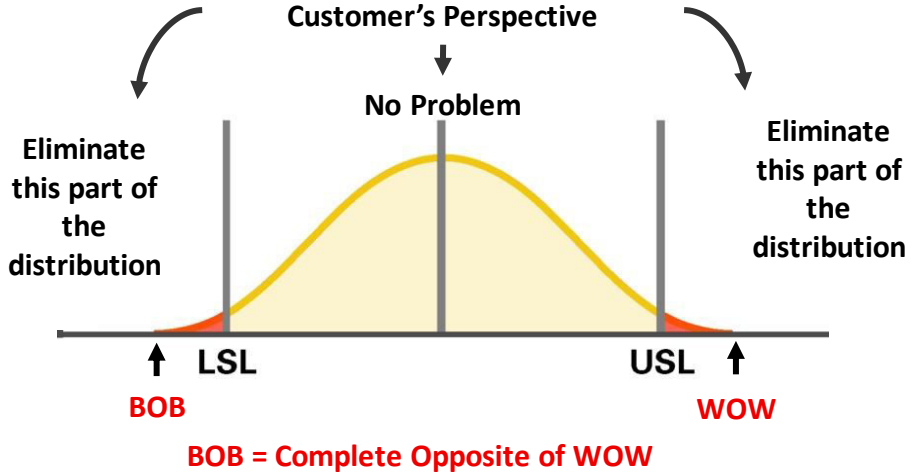


Problem solver and customer in phase.

BOB and WOW


Customer's Perspective
No Problem
Below the tolerance
Above the tolerance
Frequency
LSL
USL
WOW
BOB
Problem solver needs the extremes!
Not necessarily what defines customer enthusiasm.

Best of Best (BOB) & Worst of Worst (WOW)



BOB = Complete Opposite of WOW

- BOBs and WOWs are NOT just good and bad parts – they are the opposite tails of the distribution. Both can be unacceptable depending on the Customer Enthusiasm Limits.
- Practically speaking, both BOB and WOW represent a „window" from which the problem solver should select parts.
- BOBs and WOWs are alike in every way except for the Mone(Y) difference.
- BOB is not what is desired, unless it aligns with the barrier of a distribution or you are working with an attribute measurement system.

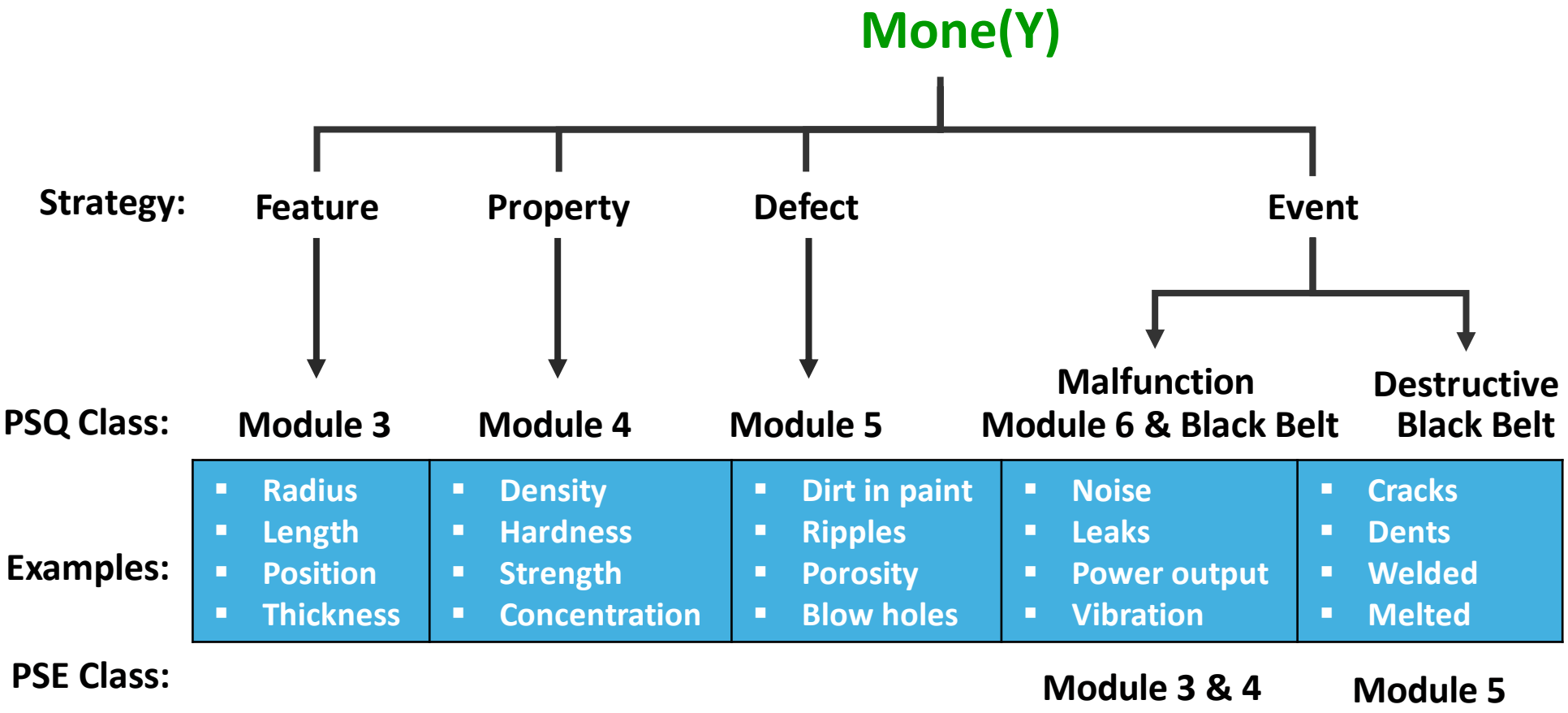
The Mone(Y)



Philosophy of the Mone(Y)

- The BIG X is the cause-effect relationship driving the largest change in the Mone(Y).
- The Mone(Y) has engineering leverage which reveals insight into the physics of the failure.
- The Mone(Y) should never be an accounting metric. Accounting metrics cannot provide insight into physics and geometry.

Different Strategies for Different Mone(Y)s



Mone(Y) - Excercise

Examples

1. Dirt in the paint
2. Vehicles pull to right during braking
3. Clicking noise during cold engine start
4. Burned bearings
5. High effort to close glove box
6. Brake pulsation
7. Shaft bore fit
8. Porosity in aluminium
9. Dented sheet metal
10. Chipped color
11. Gap between console and IP (information Panel)
12. Intermittent operation of power door lock

Defect	Property	Feature	Event

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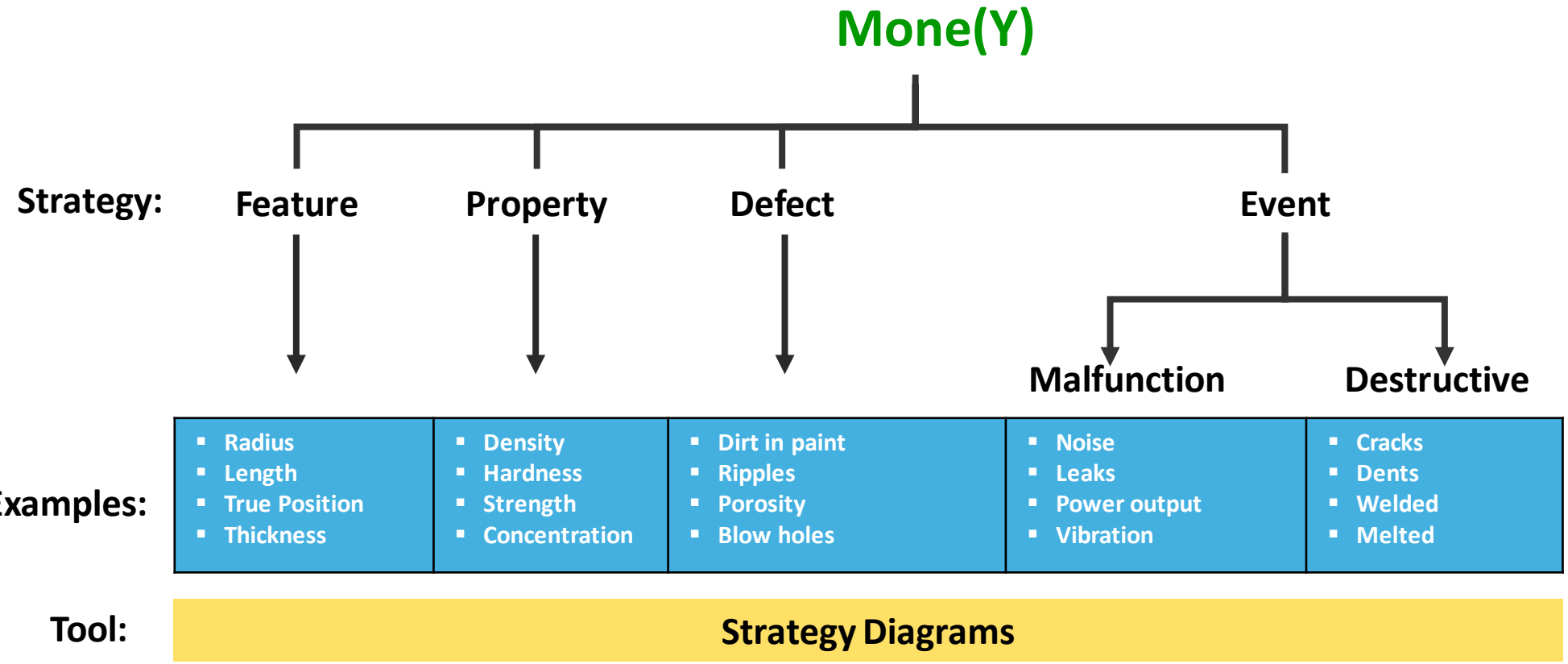
Mone(Y) - Answer

Examples

		Defect	Property	Feature	Event
1.	Dirt in the paint	X			
2.	Vehicles pull to right during braking				X
3.	Clicking noise during cold engine start				X
4.	Burned bearings				X
5.	High effort to close glove box				X
6.	Brake pulsation				X
7.	Shaft bore fit			X	
8.	Porosity in aluminium		X		
9.	Deformed sheet metal	X			
10.	Chipped paint	X			
11.	Gap between console and IP			X	
12.	Intermittent operation of power door lock				X

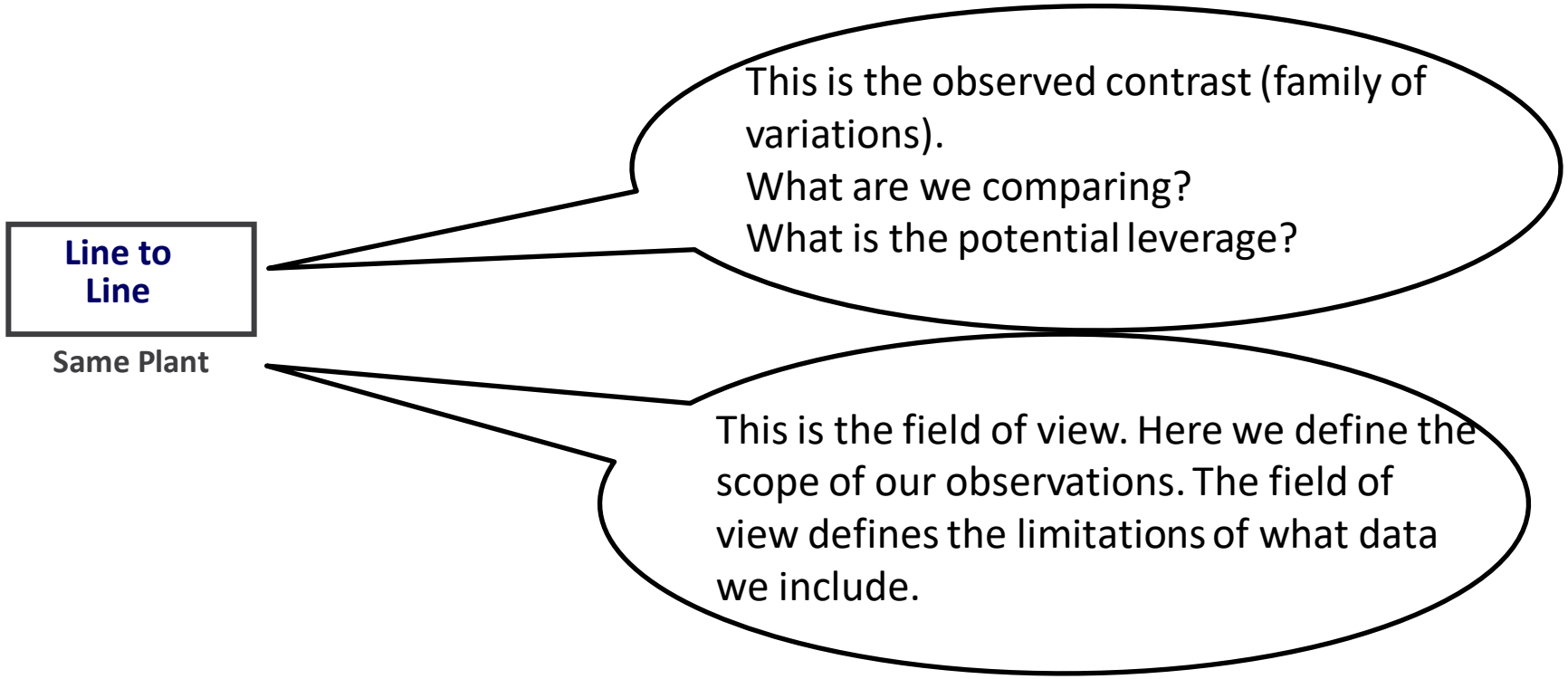
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Different Strategies for Different Mone(Y)s



Strategy Diagrams

- Strategy charts are a way to document possible contrasts for the Mone(Y) that you can use during your project.



Strategy Diagrams

Design Families

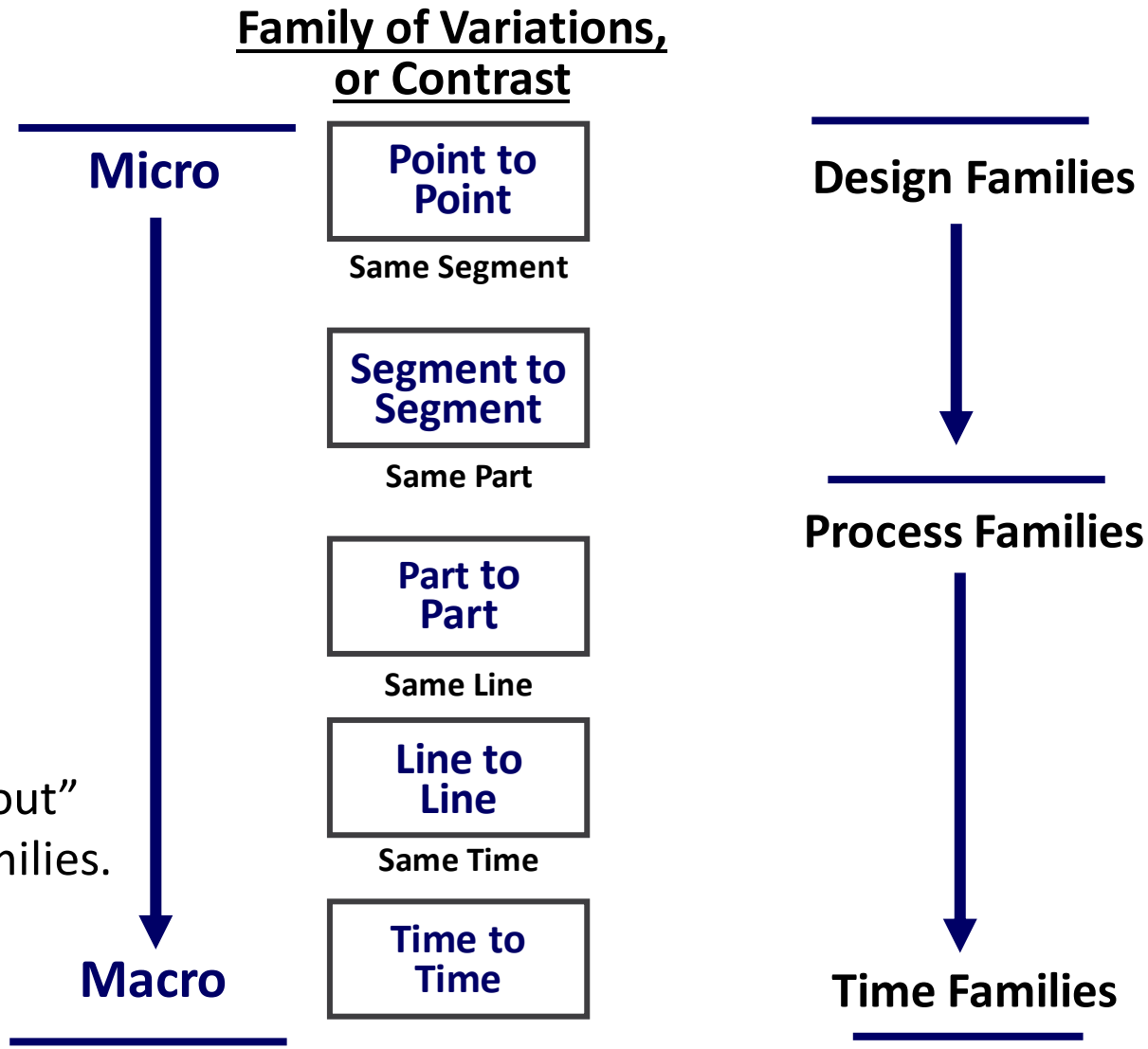
Families of variation that focus on elements of a single part are known as **Design Families**.

When creating a Strategy Diagram, we begin with Design Families and move outward towards a larger focus.

Process Families

Families of variation that focus on the manufacturing process and encompass a wide range of variables are known as **Process Families**.

We apply the same tactic of incrementally "zooming out" from each prior family when determining Process Families.



Strategy Diagram Development

1. What is the failure?
   A crack in the windshield is detected at the end of the assembly line. 5 - 15 vehicles per shift have cracks.

2. Does anyone know why?
   Lots of theories, but the problem does not disappear.



Strategy Diagram Development

First focus on the type of failure.

There are four types of failure: Feature - Property - Defect - Event

A crack in the windshield is a event.



Strategy Diagram Development

Family of Variation

Point to Point

Same Location



What does the crack look like? What kind of crack is it?

Strategy Diagram Development

Family of Variation

Location to Location

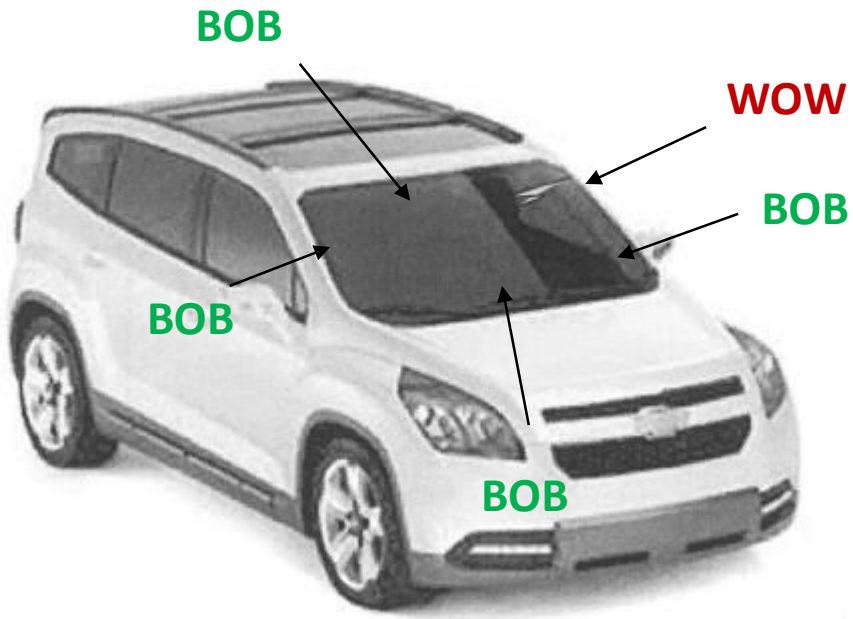
Same Edge



Are there location that do not have cracks along the same edge?

Strategy Diagram Development

Family of Variation



Edge to Edge

Same Vehicle

Are there edges that have no cracks along the same vehicle?

Strategy Diagram Development

Family of Variation



Vehicle to Vehicle

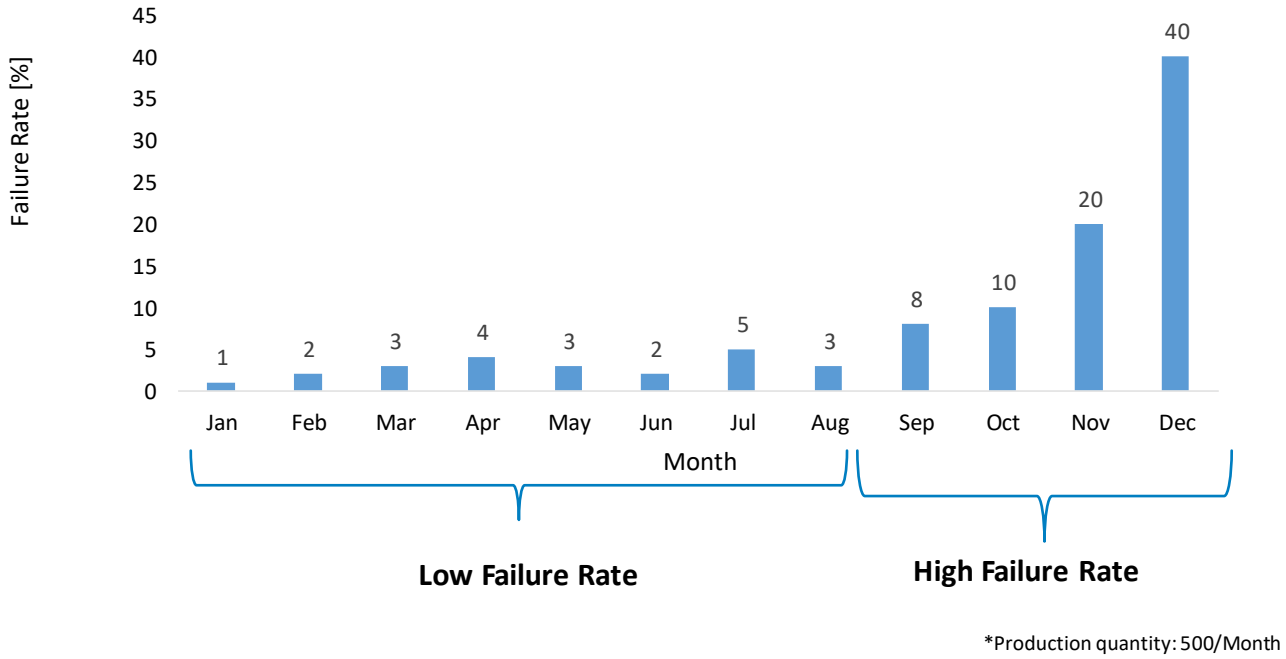
Same Time

Are there vehicles that do not have cracks at the same time?

Strategy Diagram Development

Family of Variation





*Production quantity: 500/Month

Time to Time

Are there times that have lower failure rates?

Destructive Event / Defekt Strategy Diagram

Mone(Y) = Crack in the windshield [yes/no]

Family of Variations, or Contrast

Diagram

What is Meant by Each Contrast

Micro

Point to Point

Same Location

- [ ] Large Contrast
- [ ] Small or No Contrast
- [x] Do Not Know
- [ ] N/A

There are no investigations available.

Location to Location

Same Edge

- [x] Large Contrast
- [ ] Small or No Contrast
- [ ] Do Not Know
- [ ] N/A



The crack has only been found in the upper driver side corner.

Edge to Edge

Same Vehicle

- [x] Large Contrast
- [ ] Small or No Contrast
- [ ] Do Not Know
- [ ] N/A



The crack has only been found along the driver side vertical edge.

Vehicle to Vehicle

Same Time

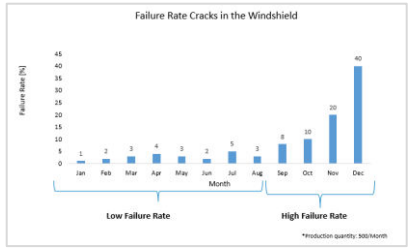
- [x] Large Contrast
- [ ] Small or No Contrast
- [ ] Do Not Know
- [ ] N/A



Some vehicles have the crack and some not.

Time to Time

- [x] Large Contrast
- [ ] Small or No Contrast
- [ ] Do Not Know
- [ ] N/A



We have see a spike in failure rate over time.

Macro

Strategy Diagram Guidelines

How to Complete the Strategy Diagram

1. Choose an appropriate template as a guide to create a Strategy Diagram(s) based on the project Mone(Y) and modify it to your specific situation.

2. For each contrast, check off “Do Not Know,” “Large Contrast,” “Small, No Contrast”, or “N/A”

3. Draw the family of variation, if appropriate label the contrast. Define nomenclature, identify orientation, features, etc. on diagram.

4. Write comments with actual data and refer to the supporting documentation.

5. If we do not know the answer, what will the answer allow us to do? We have to carefully consider whether it is worth the investment to go after obtaining the answer to some questions.
6. When there is a large contrast, do we understand the cause of the contrast? Would there be value in knowing the cause?

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Strategy Diagram Guidelines

How to Complete the Strategy Diagram

7. It is important to recognize when to leverage a contrast on the Strategy Diagram (narrow the distribution) and when to leverage physics (shift the distribution).

8. Process families are a function of the layout of manufacturing equipment at the plant. A process flow diagram is useful to document the various families to include when looking for contrast.

The importance of defining Failure Mode Types

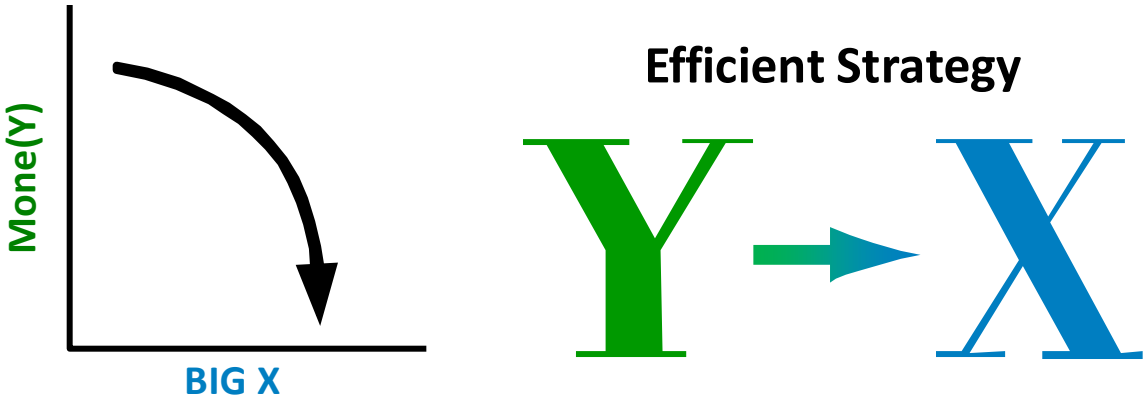
- Each Failure Mode type has a very specific “Strategy” and “BIG X Tools” associated to them.
- By classifying what Failure Mode type your **Mone(Y) is as received**, it should help you develop the right Strategy and to identify the appropriate BIG X Tools to start with clue generation.

Guidelines:

- Establish Failure Mode type
- Develop the appropriate Strategy Diagram
- Check your measurement system
- Select the appropriate BIG X Tool to proceed that support data collection for the Strategy Diagram

Note: Each project is different as is the combination of tools used.

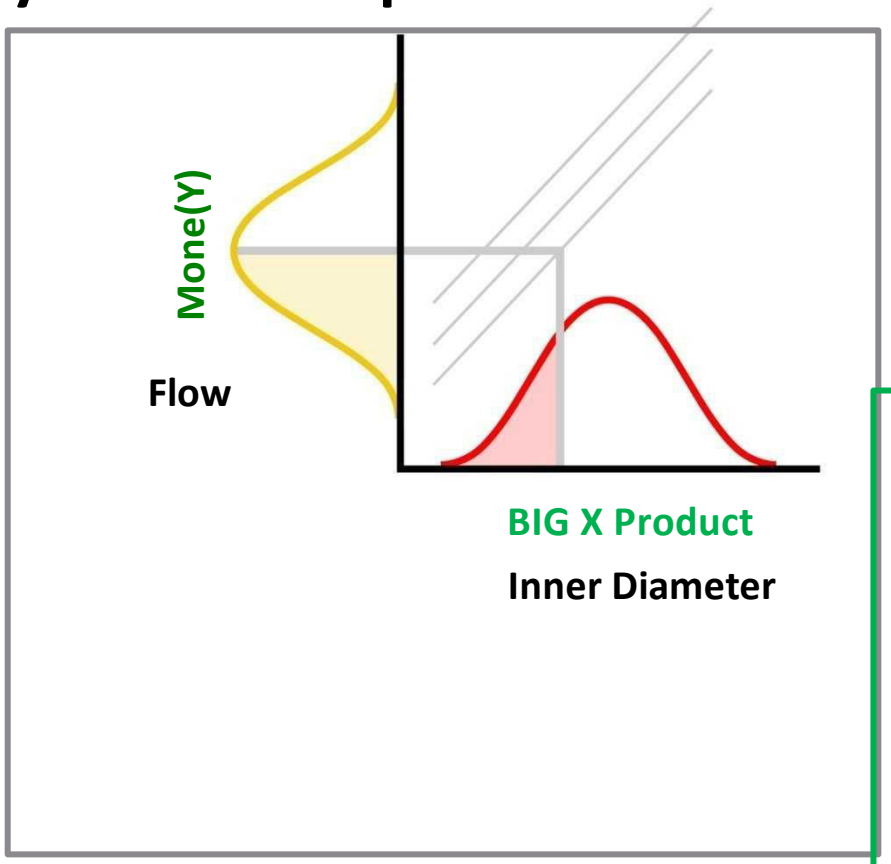
The BIG X



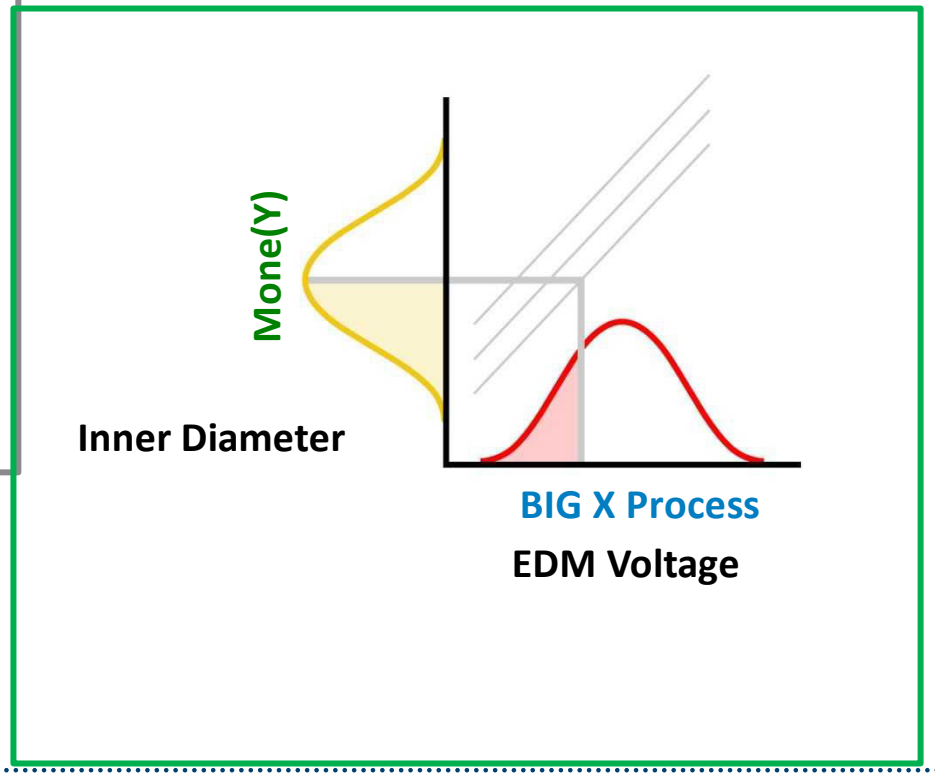
Philosophy of the BIG X

- The BIG X cause has the greatest leverage over the Mone(Y) from BOB to WOW.
- There is always a BIG X. It is often an interaction.
- The fastest route to the BIG X cause is a progressive search starting with a well defined Mone(Y) response.
- K.I.S.S.
- Nothing else matters until the BIG X is controlled.

Hierarchy Relationships



BIG X can be product features, properties or defects that affect events.



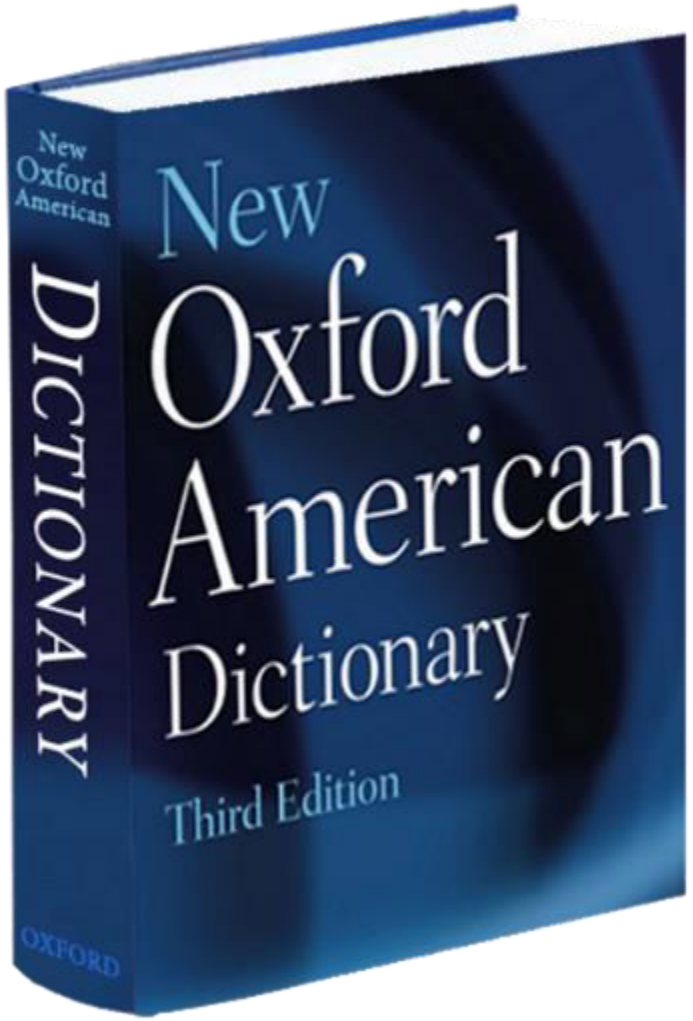
BIG X can be process parameters that affect features, properties or defects.

Elimination Strategy Concepts
Dictionary Game Thinking – The BIG X Paradigm

- What questions would you ask to find a specific word chosen from the dictionary?
- Your questions should only be answered with "Yes" and "No".
- The BIG X is the word you are looking for.
- We need effective strategies for gathering evidence to solve problems quickly!

Hints:

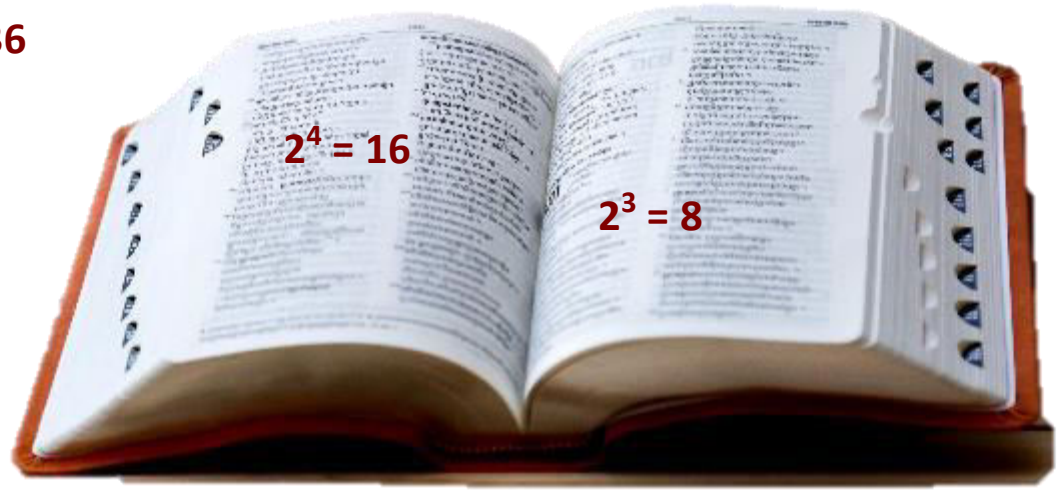
- What do I know about a dictionary?
- How can I use this to ask strategic questions?



Elimination Strategy Concepts
Rapidly Converging Power

$2^{17} = 131,072$

$2^{16} = 65,536$

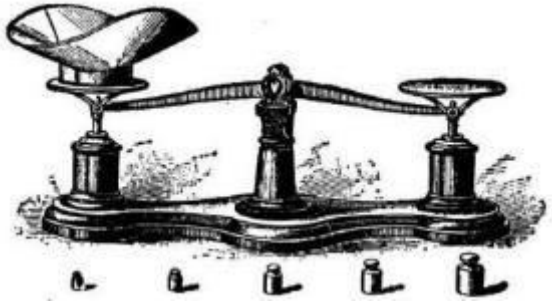


$2^1 = 2$

$2^0 = 1$

With an unabridged dictionary, find any secret word in 17 steps, but only if you know how to ask strategic questions with leveragable answers!

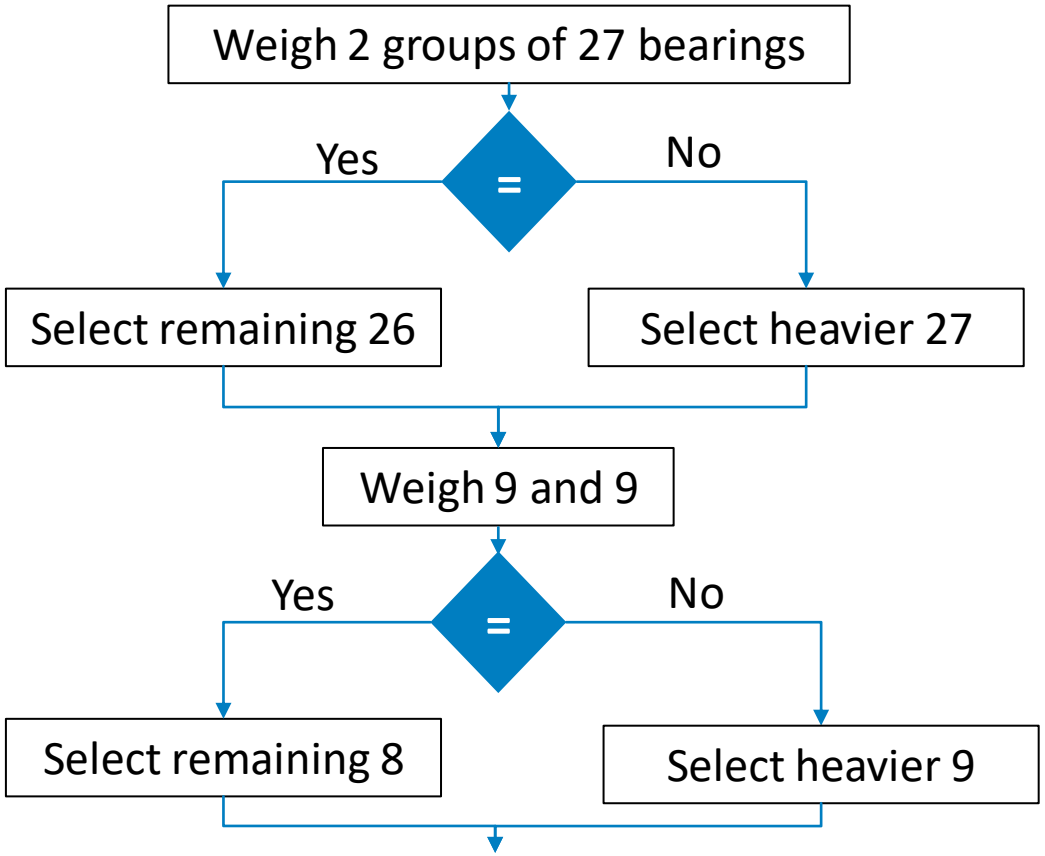
Elimination Strategy Concepts
Dictionary Game Thinking – Excercise



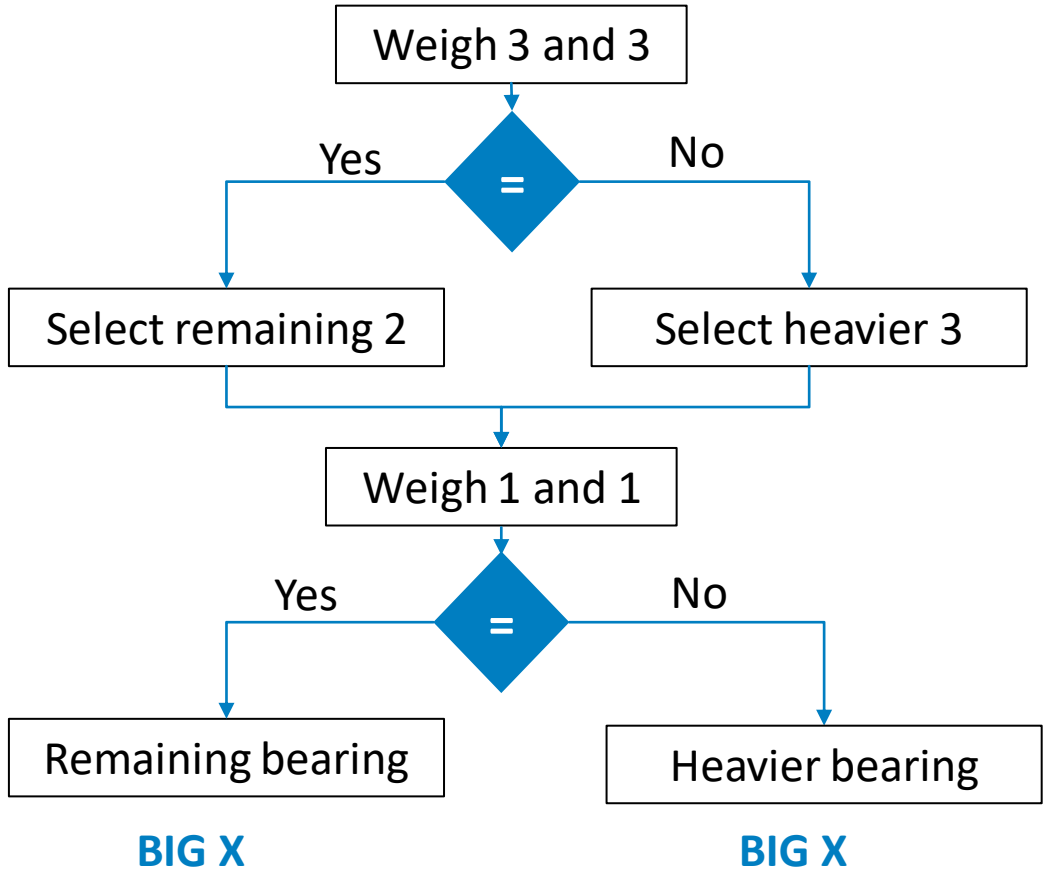
- A tray contains 80 ball bearings which are visually identical.
- 79 ball bearings weigh 20 mg each while one ball bearing weighs 19,9 mg.
- Divide into teams and develop a low cost strategy to find the 19.9 mg ball bearing.
- Each use of the precision balance scale costs € 1.000. There is no charge to use the weights.

- Break into teams.
- Appoint a spokesperson to share your strategy.
- Time allowed: 5 minutes

Elimination Strategy Concepts
Dictionary Game Thinking – Answer

		Costs
Trial 1:		€ 1.000
Trial 2:		€ 1.000
		€ 2.000

Elimination Strategy Concepts
Dictionary Game Thinking – Answer



	Costs
Trial 3:	€ 1.000
Trial 4:	€ 1.000
	€ 4.000

Total cost to find the 19.9 mg bearing.

DMAIC in PS Projects

D	**Focus**	▪ Transformation of a business case into a technical project → Problem Definition Tree (Problem Definition) → Management → Project Definition Tree (Project Statement (4Ps)) → Project Leader
M	**Approach**	▪ Development of an investigation strategy → Mone(Y) type ▪ Mone(Y) Identification and description ▪ Development of Strategy Diagrams ▪ Checking the measuring system → Isograph
A	**Converge**	→ Solution Tree ▪ Convergence to the BIG X → Dictionary game thinking ▪ Compare BOBs vs WOWs ▪ BIG X candidate identification
	Test	▪ Risk assessment → Risk assessment matrix ▪ BIG X Confirmation → B vs W or other statistical tests
I	**Understand / Apply**	▪ Mone(Y) → BIG X relationship understood (explanation of physics) ▪ Understanding of interactions ▪ Customer requirements defined within reasonable tolerance limits: E. g. Tolerance Parallelogram ▪ Corrective actions implemented and verified
C	**Apply**	▪ Procedures updated ▪ Mone(Y) Monitoring → Mone(Y) Run Chart, P Chart, ...
	Leverage	▪ Calculated savings, transfer knowledge to other areas, FMEA, ...

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Problem Definition

Continuous Improvement:

- Data Analysis
- Identify Pareto
- Select High Impact Projects

For **continuous improvement**, the **Management** is defined **for the Project Leader the Project Statement and Problem Definition Tree** is required.

Crisis:

- Customer Complaint
- Line Stoppage
- New Product Launch

For **crisis situations**, the problem is often defined for the Project Leader and **no Problem Definition Tree** is required.

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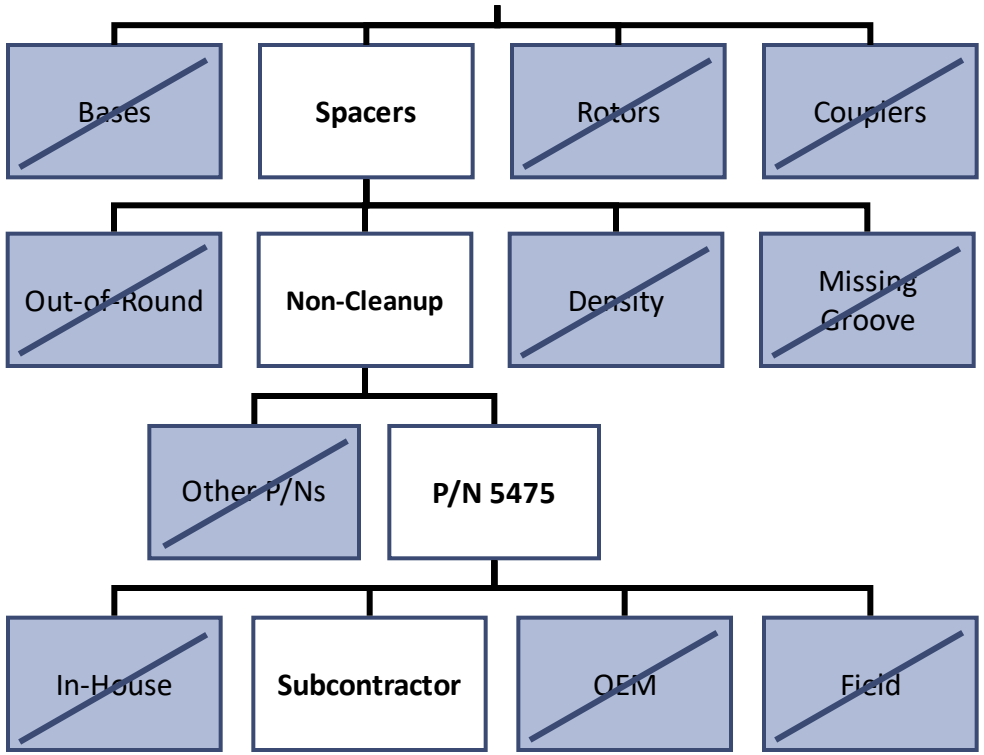
Develop a Problem Definition for Continuous Improvement

Continuous Improvement:

- Data Analysis of key business metrics
  - *Operational Metrics*
  - *Quality Metrics*
  - *Performance and field metrics*
- Identify Pareto
- Select high impact projects
- Document how the project was selected using a Problem Definition Tree

Problem Definition Tree

Problem Statement: Operating costs at BOCKA are exceeding budgeted levels.



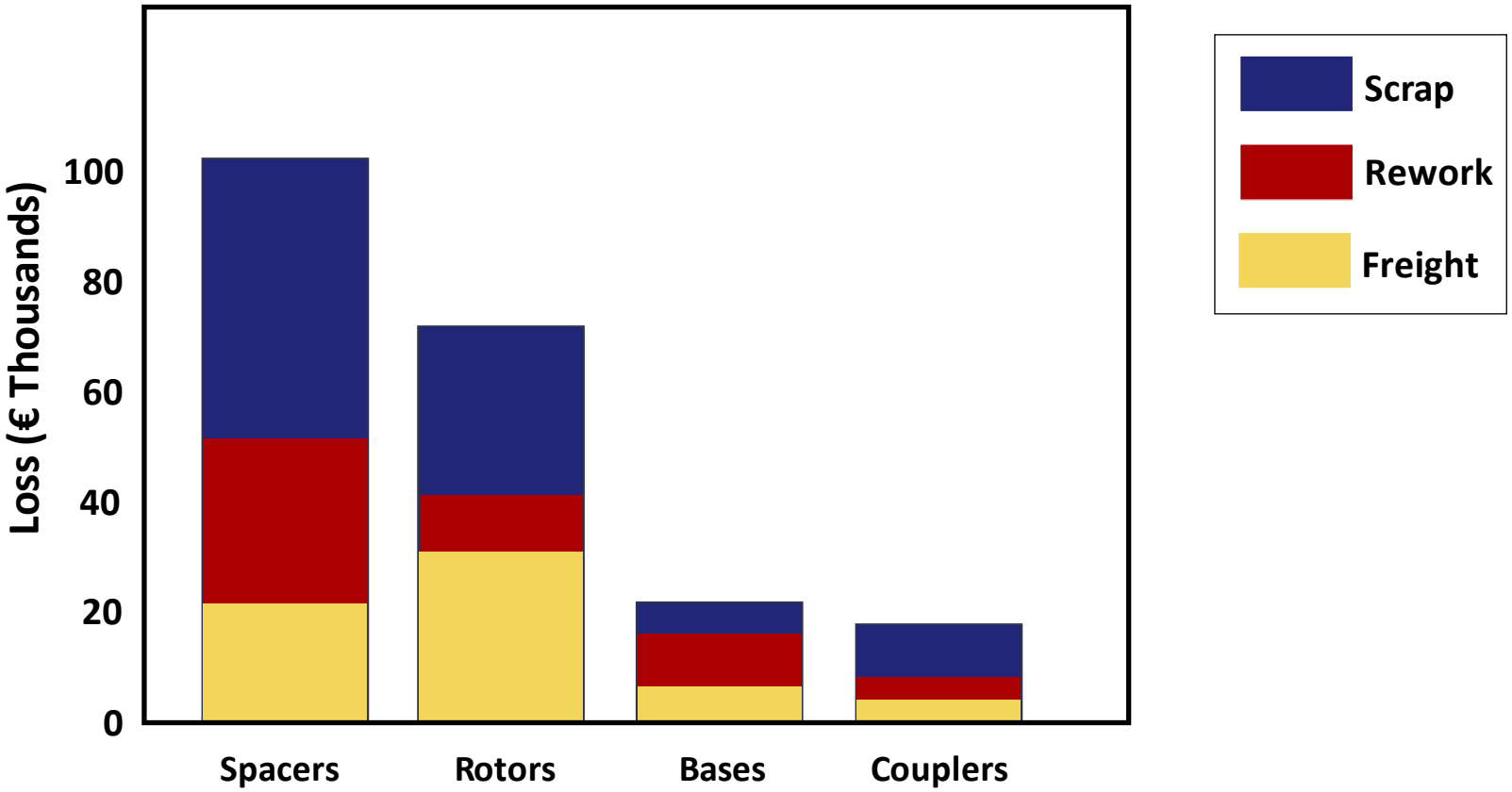
Rationale:

- Spacers product line has the highest cost of poor quality driven by scrap, unscheduled rework , and premium freight (Pareto #1).
- Problem description (Pareto #2).
- Focus split P/N 5475 has highest fallout for non-cleanup (Pareto #3)
- Problem discovery point.

Pareto #1

Strategic Business Metrics for BOCKA

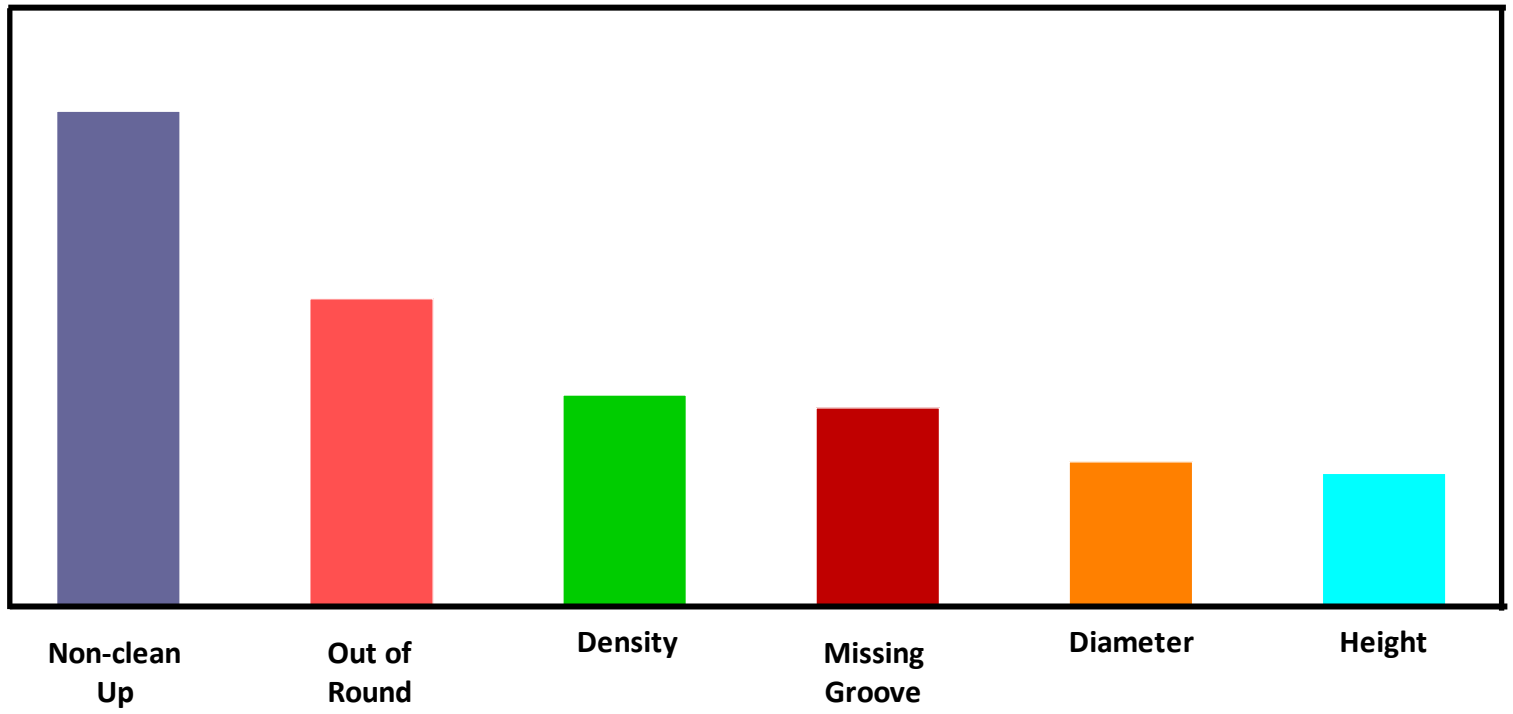
Pareto by Product Families



Pareto #2

For Spacer Family

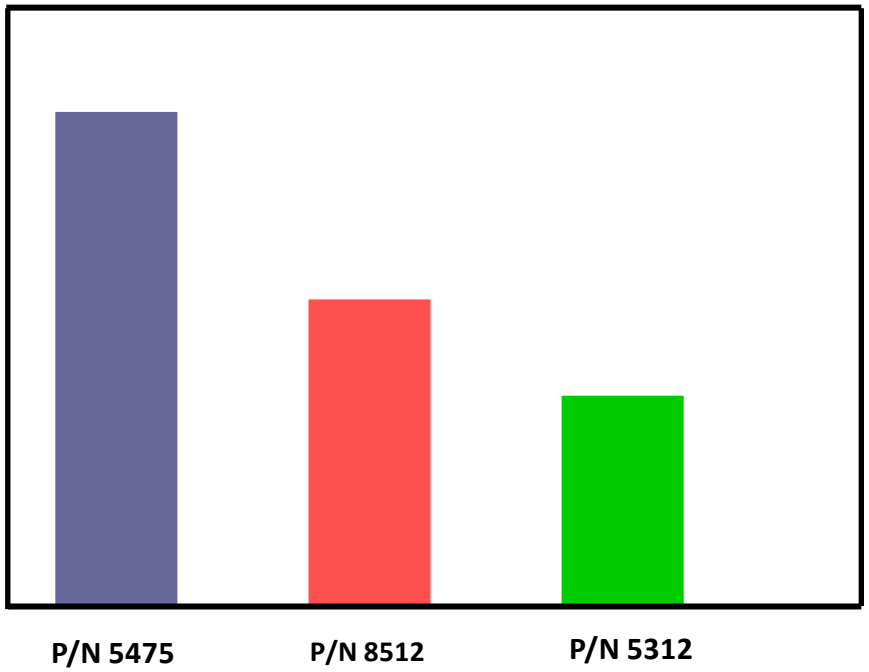
Pareto by Defect Type



Pareto #3

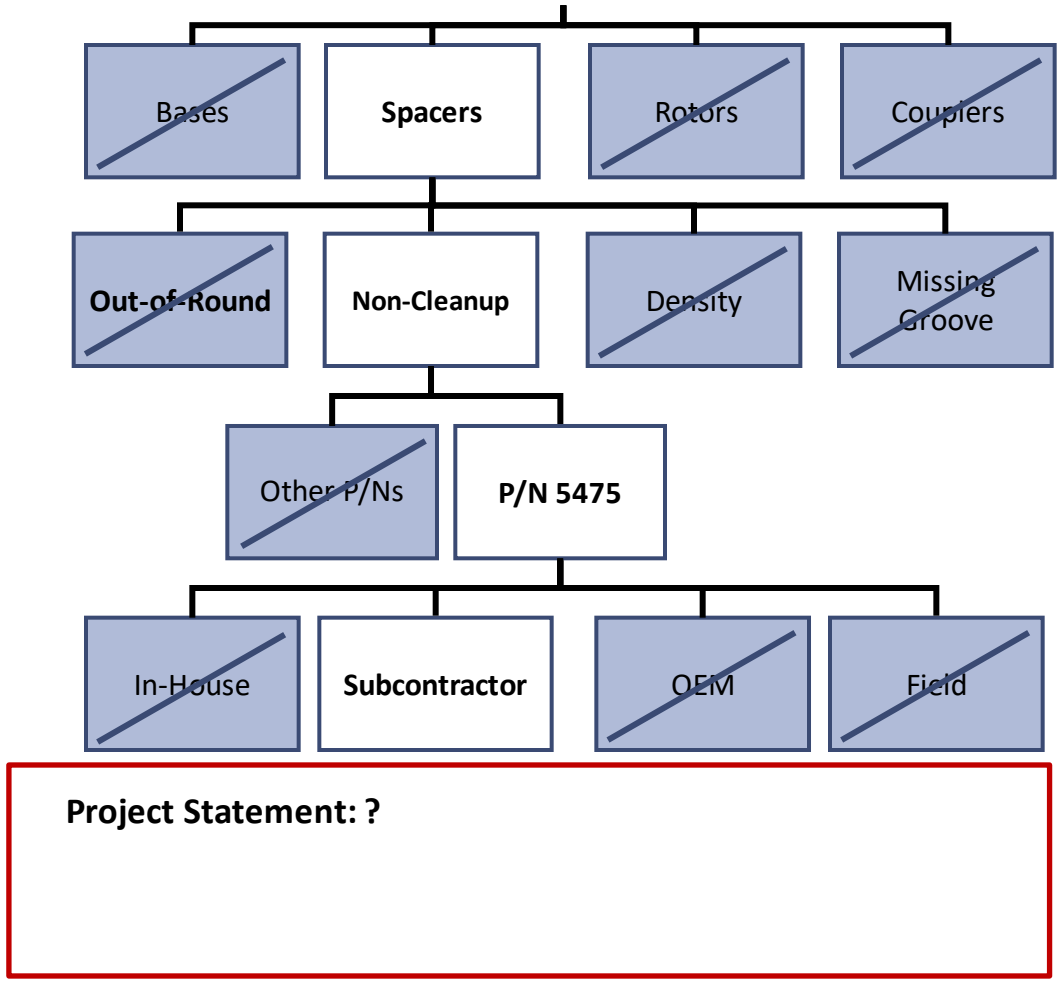
For Spacer Family

Pareto by Typ No



Problem Definition Tree

Problem Statement: Operating costs at BOCKA are exceeding budgeted levels.



Rationale:

- Spacers product line has the highest cost of poor quality driven by scrap, unscheduled rework , and premium freight (Pareto #1).
- Problem description (Pareto #2)
- Focus split P/N 5475 has highest fallout for non-cleanup (Pareto #3).
- Problem discovery point

Using 4Ps to Develop a Project Statement

Product or **P**rocess that has the problem.

Problem description

Place where the problem is discovered

How is the **P**ain measured (metric)

Project Statement:

Find and Eliminate the BIG X that drives (PRODUCT or PROCESS) (PROBLEM) discovered at (PLACE) to save or prevent (PAIN).

Project Selection
Project Statement – 4 Ps

Product or **P**rocess that has the problem. — Spacer P/N5475

Problem description — Non-clean Up

Place where the problem is discovered — After Lapping (subcontractor)

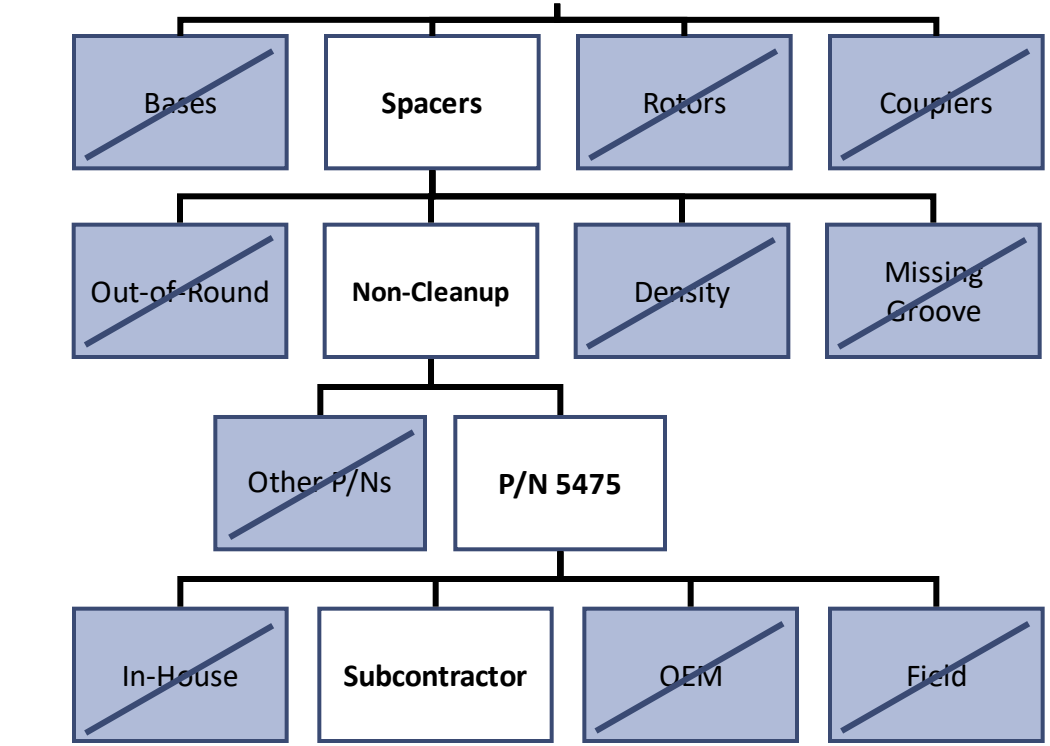
How is the **P**ain measured (metric) — containment and improve customer satisfaction and save 400 k€

Project Statement:

"Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€."

Problem Definition Tree

Problem Statement: Operating costs at BOCKA are exceeding budgeted levels.



Project Statement: Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€.

Rationale:

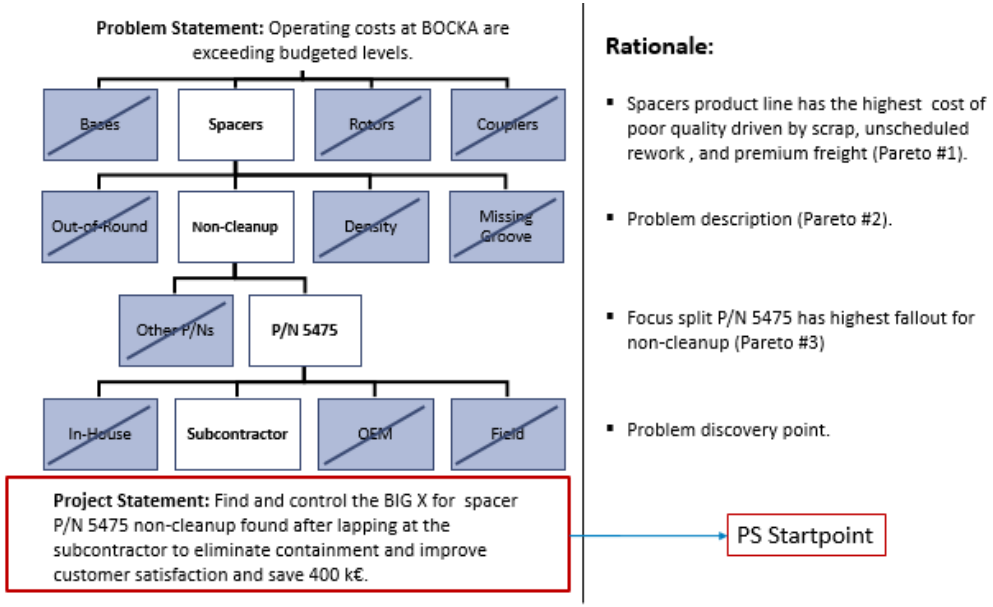
- Spacers product line has the highest cost of poor quality driven by scrap, unscheduled rework , and premium freight (Pareto #1).
- Problem description (Pareto #2).
- Focus split P/N 5475 has highest fallout for non-cleanup (Pareto #3)
- Problem discovery point.

PS Startpoint

Elements of a Problem Definition Tree

The Problem Definition Tree:

- The Management starts with a business problem statement.
- Identify the high opportunity product or process.
- Communicate the project focus (which model, failure type, etc.).
- Include rationale used to make each split shown on the right side of the tree, with supporting evidence.
- Look at multiple metrics.
- Pareto Principle is often used to find the 80/20 split.
- The tree must end with a clearly defined project statement that the problem solver can immediately work on.
- The project statement is the startpoint for the Problem Solver.

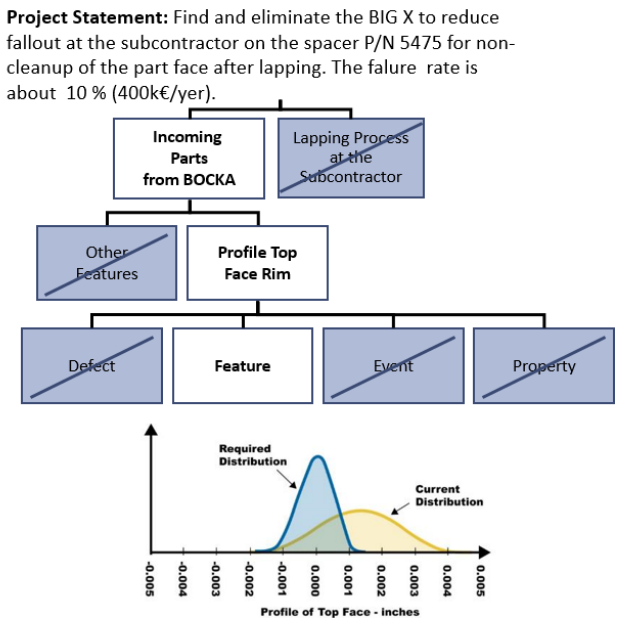


Trees are planning tools and should be constructed as the project progresses, not created at the end of the project.

Elements of a Project Definition Tree

The Project Definition Tree:

- Starts with a statement of the technical problem.
- Ends with a well defined description of the Mone(Y).
- For a malfunction event is about the characterization of the event; describe the nature of the event.
- Is used to communicate the project focus (which model, side, etc.).
- Describes the measurement system, environment and conditions used.
- Includes rationale used to make each split shown on the right side of the tree.



Rationale:

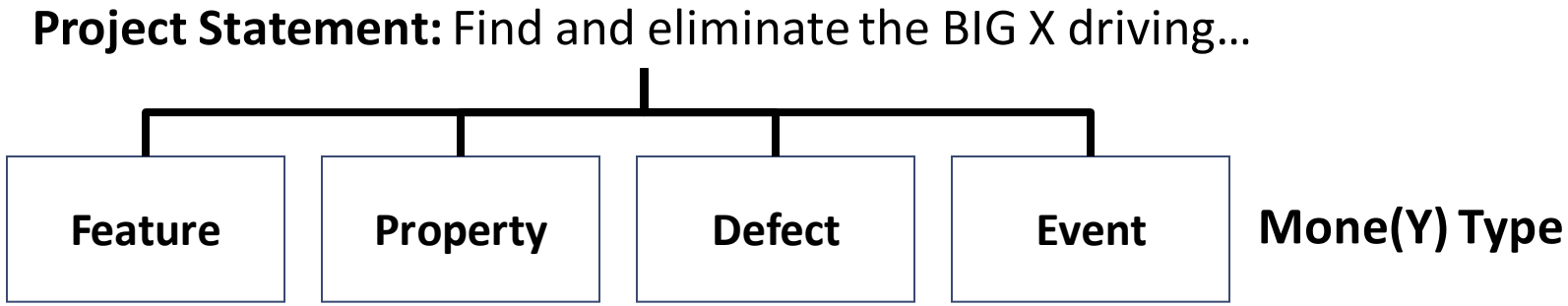
- BOCKA/subcontractor split. Mixed lot lapping.
- Group comparison.
- Strategy split.

Mone(Y) – Description = Profile of top face [inch] measured as the nominal position of points relative to the manufacturing datum.

Mone(Y) = Profile of top face [inch]

Trees are planning tools and should be constructed as the project progresses, not created at the end of the project.

Project Definition Tree

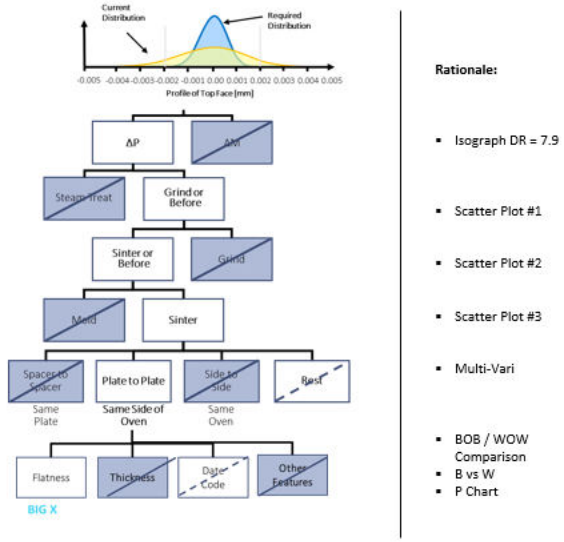


- The Project Definition Tree usually starts with a split that identifies the nature of the Mone(Y).
- The reason why we care about identifying the nature of the Mone(Y) is to help with the selection of tools and tactics to make dictionary splits.
- If the Mone(Y) cannot be expressed as a feature, property, defect, or event, then you have been assigned a problem – not a project. In that case, develop the Problem Definition Tree until the Mone(Y) can be expressed as a feature, property, defect, or event.

Elements of a Solution Tree

The Solution Tree:

- The Solution Tree is used to show the convergence to the BIG X. It is how the dictionary game was played.
- The Solution Tree starts with a well defined Mone(Y) and ends with a confirmed BIG X.
- The Solution Tree should always begin with a ΔP vs. ΔM split.
- Rationale is required for each branch, describing the logic for crossing off certain boxes and is shown on the right hand side of the tree.
- The Solution Tree must end with the identity of the confirmed BIG X.
- When the BIG X is confirmed, state in the rationale only what confirmation test (incl. risk) was applied or the rationale why is was skipped.
- Subsequent corrective actions can be included in the rationale as well.
- There can be more than one Solution Tree for a project.

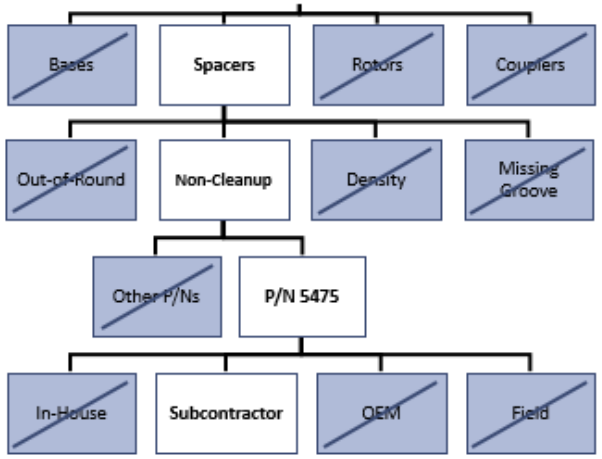


Trees are planning tools and should be constructed as the project progresses, not created at the end of the project.

All Trees Link Together

Problem Definition Tree

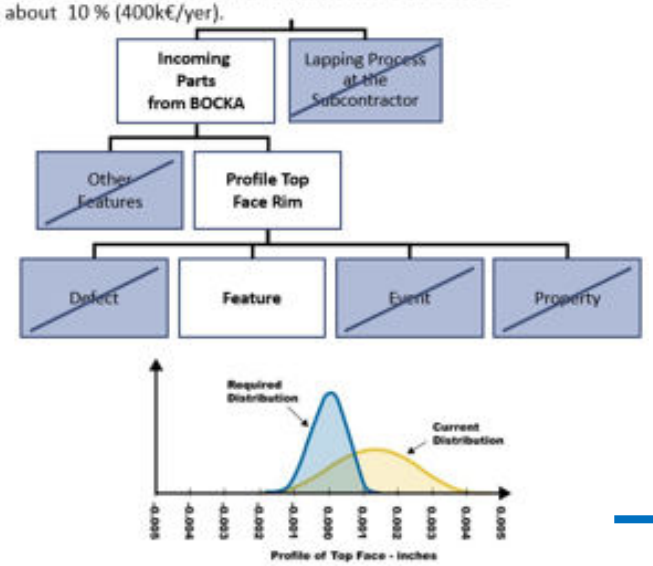
Problem Statement: Operating costs at BOCKA are exceeding budgeted levels.



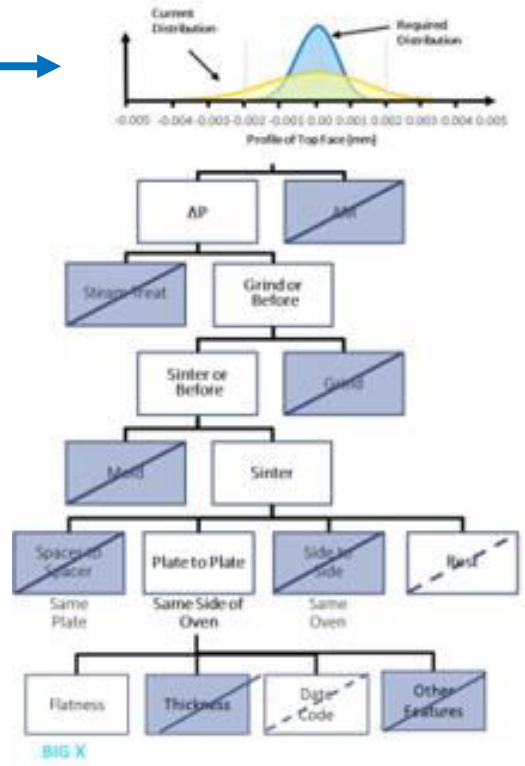
Project Statement: Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€.

Project Definition Tree

Project Statement: Find and eliminate the BIG X to reduce fallout at the subcontractor on the spacer P/N 5475 for non-cleanup of the part face after lapping. The falure rate is about 10 % (400k€/yer).



Solution Tree



Different Trees

Problem Definition Tree

Scope tree that starts with a problem as seen by management. Through a series of converging splits it focuses on a project to address the problem.

Project Definition Tree

The Project Definition Tree defining the problem begins with the problem defined for an engineer and translates it into Statistical Engineering terms. It starts with the Project Statement (4Ps) and ends with a clear Mone(Y) and description.

Solution Tree

The Solution Tree organizes the investigation after the project has been defined. The Solution Tree that represents the planning and execution of dictionary game thinking. It starts with the Mone(Y) event or feature distribution or defect statement and through a series of converging splits, focuses on the BIG X. The solution tree ends with the identification of the BIG X and a confirmation.

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Structured Problem Solving
How to Get Started



- Start with a blank sheet of paper.
- Assume you don't know anything about the problem.
- Get rid of existing "knowledge" like:
  - "This" is the cause.
  - "This" is the corrective action.
  - The measurement system is ok.
  - It is not the design.
  - Production did not change anything.
  - …

Structured Problem Solving
How to Get Started

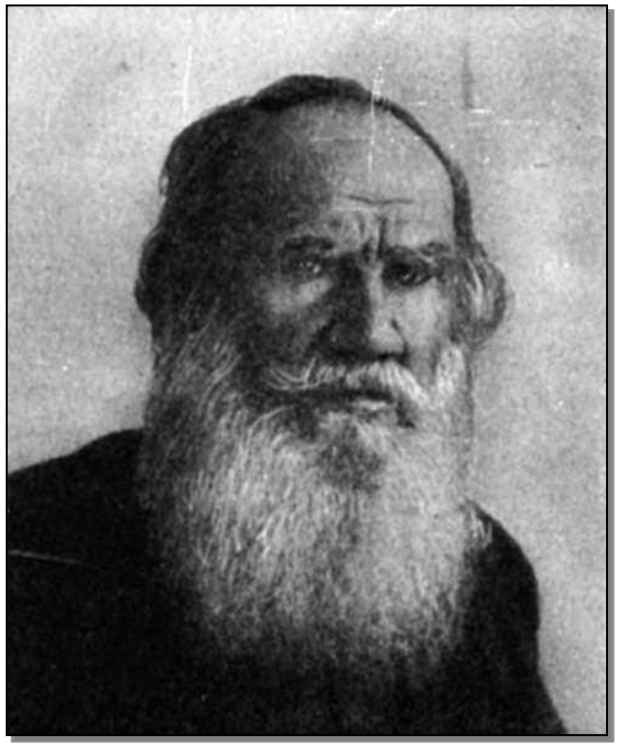


- Be wary about facts you are given regarding the problem. Are they evidence-based or has someone reached an obvious but unsupported conclusion?
- Statements about the problem may unintentionally influence you; impacting your work and results.

“It’s not what we don’t know that gets us in trouble. It’s what we know that just isn’t so.”

- Josh Billings, American Humorist

Scientific Breakthrough And the Dark Ages



Leo Tolstoy (1828 – 1910)
War and Peace

The combination of phenomena is beyond the grasp of human intellect.

Interactions are beyond our capability to understand.

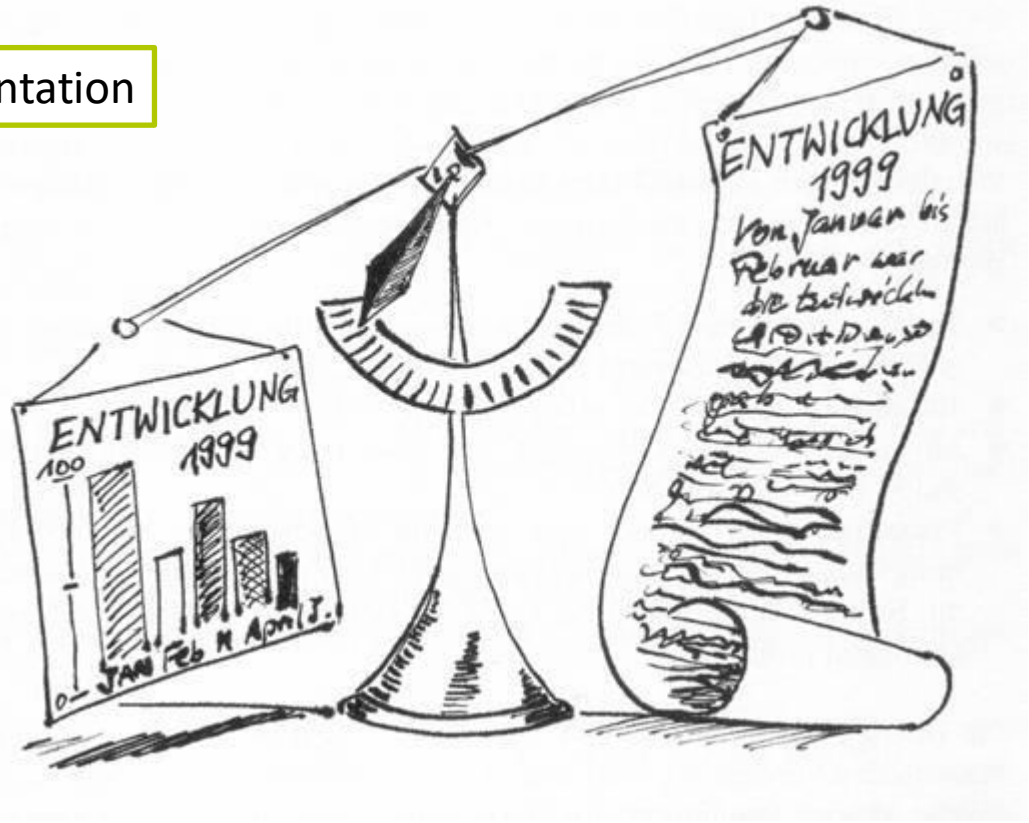
But the impulse to seek causes is innate in the soul of man. And the human intellect, with no inkling of the immense variety and complexity of circumstances conditioning a phenomena, any one of which may be separately conceived of as the cause of it, snatches at the first and most easily understood, and says here is the cause.

We will grasp at the first "obvious" cause of BIG X. Not relying on the physics and geometry.

What is Cartooning?

IS

- Graphical representation
- Pictures
- Sketch
- Reasonable data
- Good/simple graphs

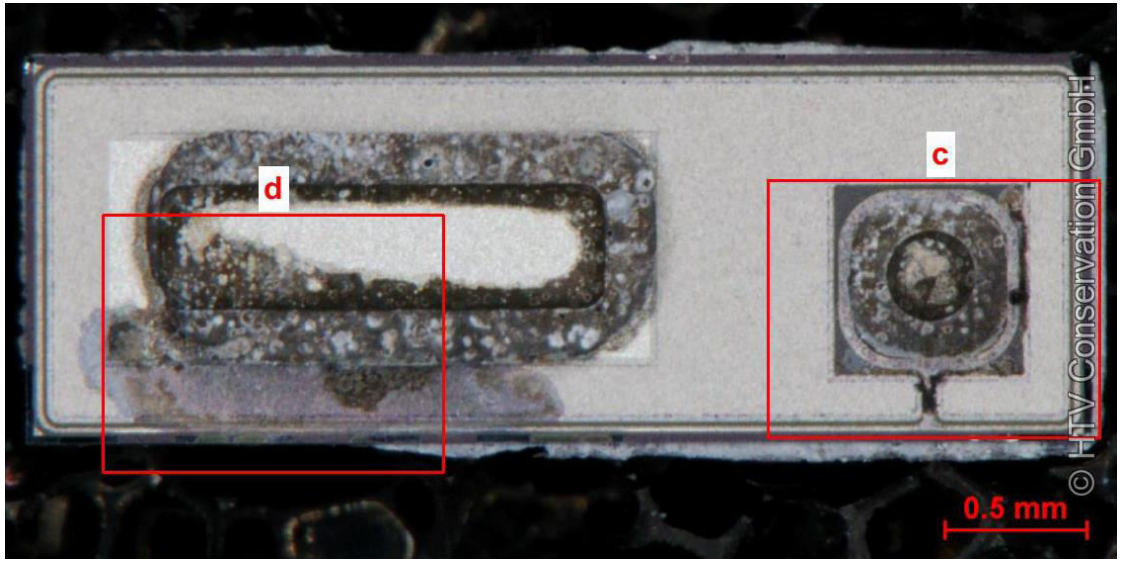


IS NOT

- Too much text
- Data tables
- Lists
- Overloaded graphs

Why Cartooning?

Used for explaining where and what the problem is



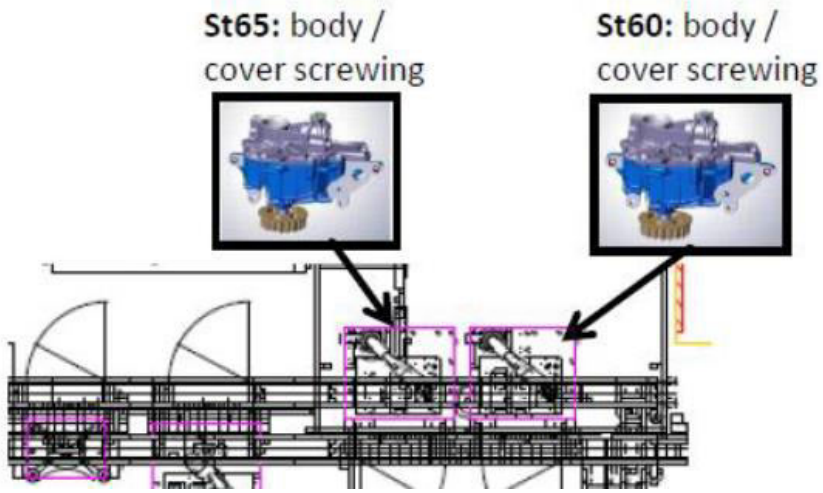
- Temperature has a significant influence on this failure mode of the MOSFETs
- Overvoltage as root cause of B6 damage can be excluded

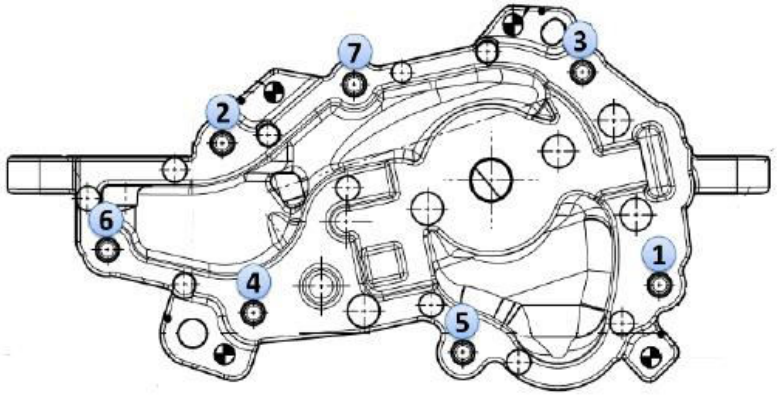
Drawings, pictures, process schematics or movie clips help the customer, the management, the coach and the problem solver to better understand and communicate where the project is focused on.

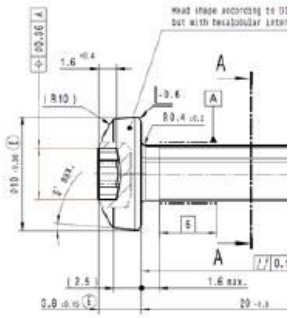
Cartooning - Examples

Screwing issue : presentation of process / design / team

Two screwing station on assy line:
Screw 1, 2, 3, 4 on ST60
Screw 5, 6, 7 on ST65



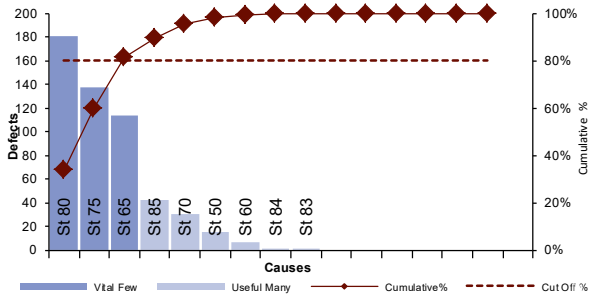


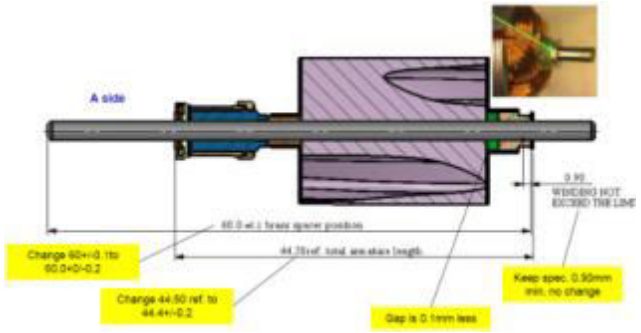




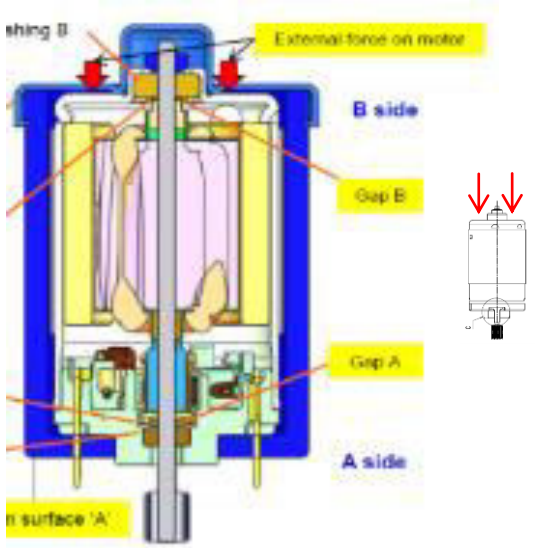


Cartooning - Examples



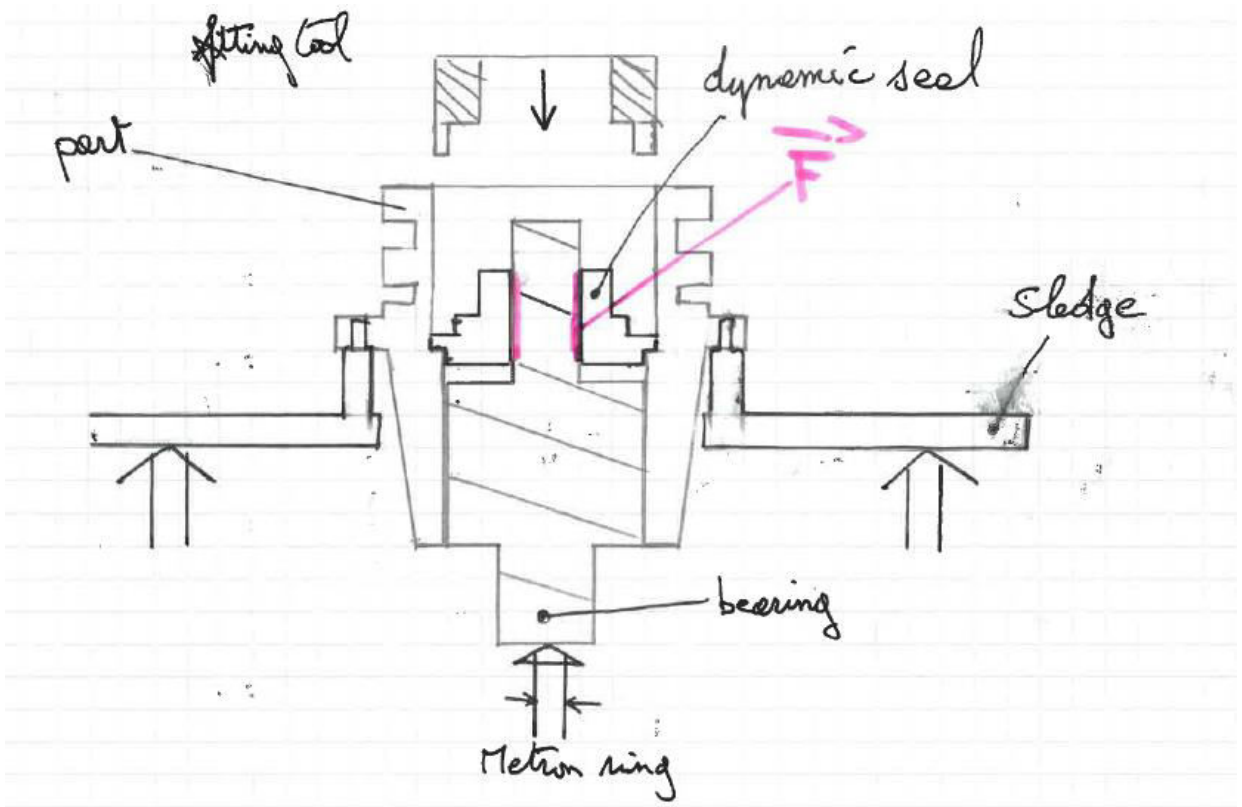


Motor current consumption problem



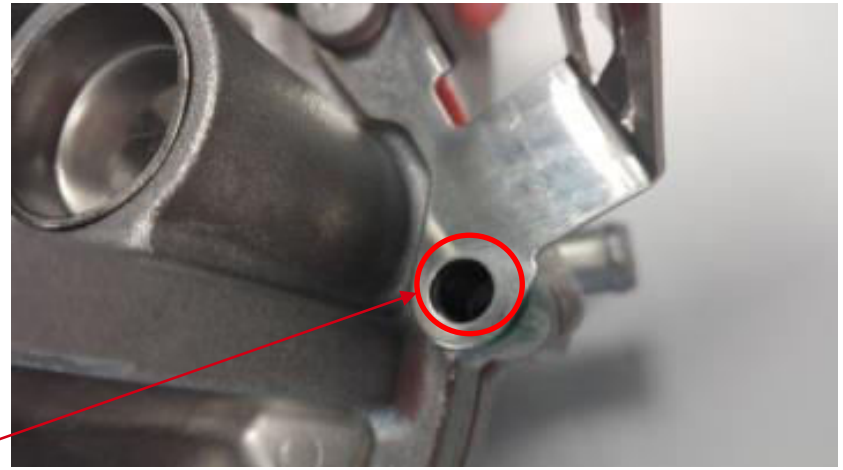
Cartooning - Examples

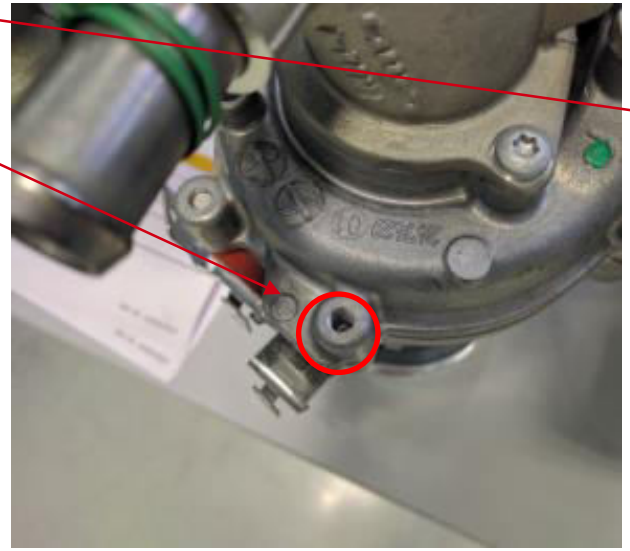
Friction/force while pressing in the dynamic seal.



Cartooning - Examples

Problem:
Missing screws

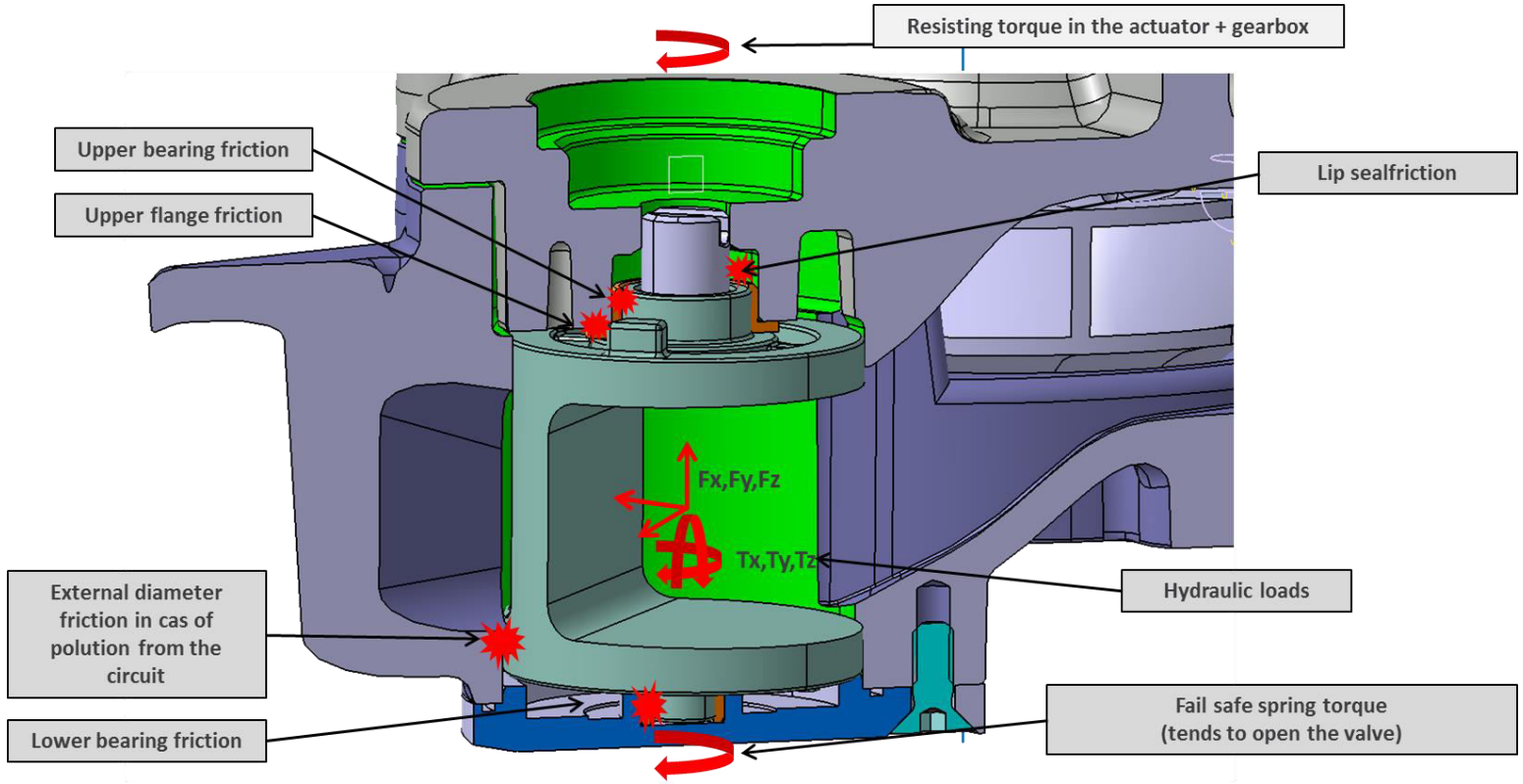






Cartooning - Examples

- Throttle partially stuck in open or closed position.
- The key measurement are the friction in the pump and actuator.



Cartooning - Examples

E_5 = Environmental Energy, liquid slugging, cavitation, **shock, vibration**

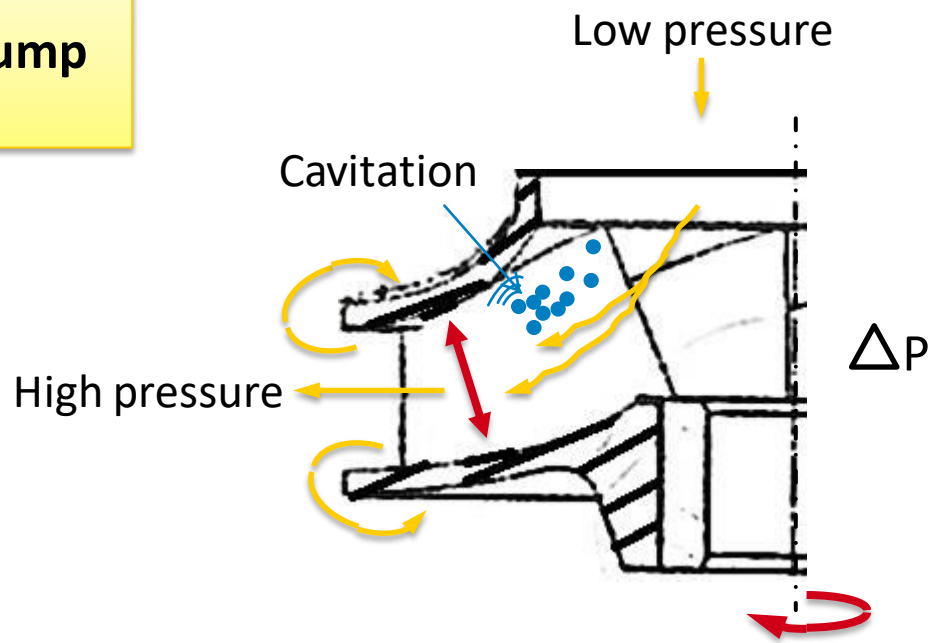
E_2 = Pressure at specific rpm;
Flow at specific rpm

E_1 = Torque, belt drive speed

Achieve Flow, Build Pressure ← ΔP ← **Pump cooling fluid** ← **Rotate Pump**

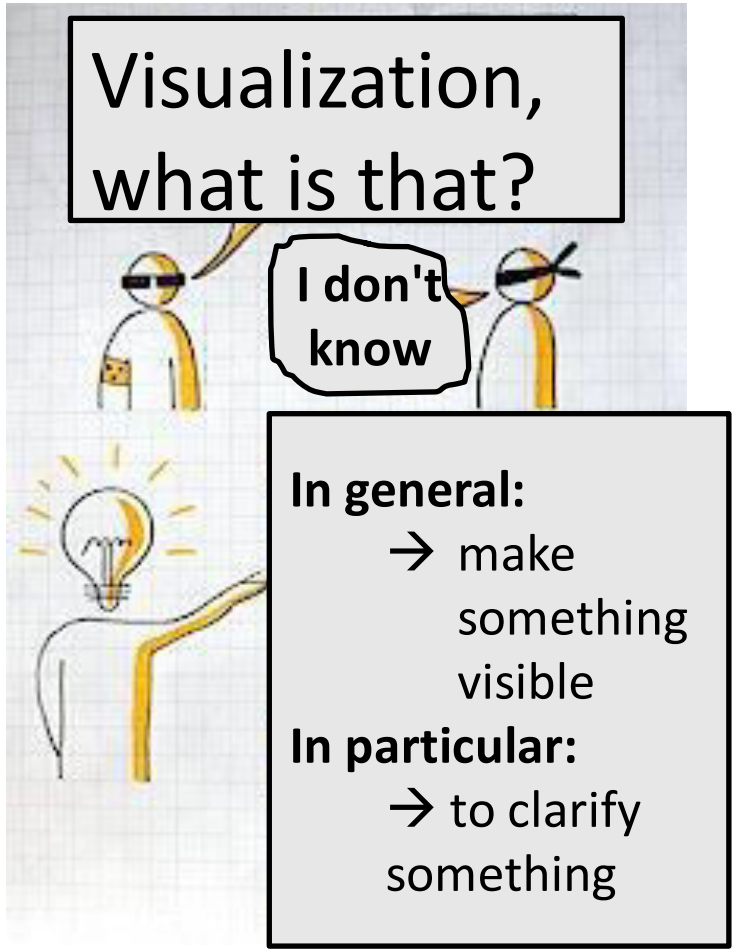
Pump cooling fluid → **Losses**

E_3 = Heat, Vibration, **pressure loss**, wear, friction, damping



Cartooning - Guidelines

- What is really your problem / project?
- If you have a problem it's essential that all involved persons have the same understanding about it.
- Cartooning helps to visualize the problem.
- It is about understanding and documenting what is happening in a process cycle and from cycle to cycle.
- It creates as common understanding and therefore focusses the problem solving team on the important topics.
- A good visualization and therefore clear problem definition almost solves the problem by 50 %.
- Use Pareto charts to narrow down to the real problem level.
- Create a sketch of your problem / part. Hand drawn is best.
- Visually explain the "physics" as you understand them.
- Keep it simple so that everybody involved can easily understand.
- If possible, combine data and sketch.



Process Flow Diagrams

Process flow diagrams pictorially represent serial and parallel processes, which, when linked together with inputs and outputs, are required to produce a component or assembly.

Serve to:

- Help people understand process steps.
- Help develop effective strategy for planning to find a BIG X.
- Assist in process characterization.

A process flow diagram has:

- Inputs, operations, and operation functions.
- Serial and parallel process steps.
- Part orientation with respect to appropriate processes.

What is a Process – Operation – Process Step?

Process

Involves multiple operations. For example, manufacturing a camshaft.



Operation

Creates a specific feature. For example, camshaft outer diameter.

Process Step

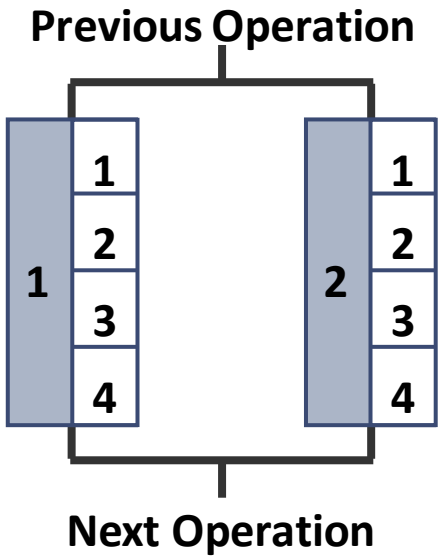
Specific task within an operation. For example, position cam lobe. If you are creating a feature – call it an operation – specific things to create that feature = step.

Difference between operations and steps is sometimes fuzzy.

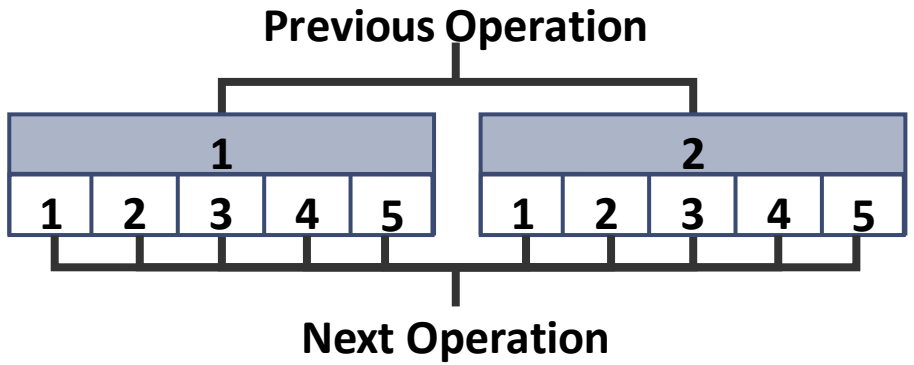
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Process Flow Diagrams

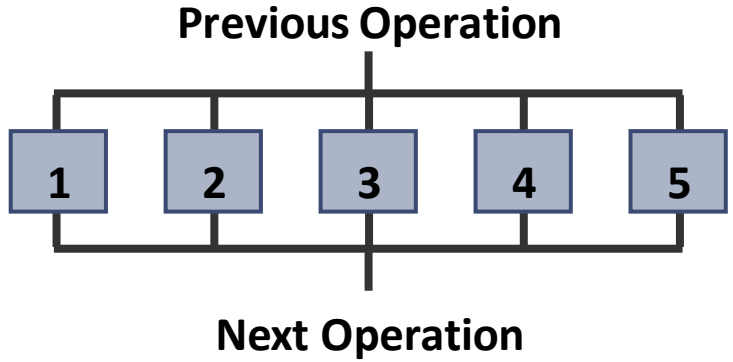
Different Illustrations / different statements



- Two machines
- Four process steps
- Each part passes across four serial stations on one of two parallel machines



- Two machines
- One process step
- Each part passes across one of five parallel stations on one of two parallel machines



- Five machines
- One process step per machine
- Each part passes across one of five parallel machines

Process Flow Diagram – Example Shaft Production

Inputs	Operation	Function
Raw Material	Material Handling	Deliver raw material to Machining Cell
Raw Material	Lathe 1 / Lathe 2 / Lathe 3 / Lathe 4 (A, B, C, D)	A=Turn Bore B=Turn Backface C=Turn Major OD D=Check Dim
Finished Machined Shafts	BBC Polishing 1 / BBC Polishing 2	Polishing the surface
Finished Machined Shafts	Inspection	100 % In-Process Inspection for Surface
Finished Machined Shafts	Packaging	Pack Acceptable Shafts for Shipment

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Process Flow Diagram – Guidelines

1. Walk the process.
2. Show each operation in the flow stream in sequential order. List the inputs, operations, and operation functions. Use verb/noun logic. Name the equipment that performs the operation.
3. Diagram all possible product flow paths within and between operations.
4. Show all flow paths, both serial and parallel. Keep track of positioning and orientation information of the parts flow between operations. When conducting investigations, we want to examine the process as it normally runs.
5. Make process schematics and/or process function models to complement a process flow diagram.

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Process Families Strategy Diagram – Template

Mone(Y) =

Family of Variation, or Contrast

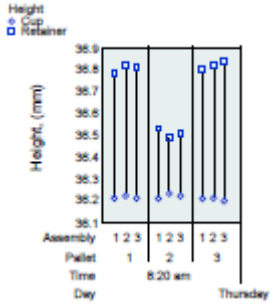
Diagram

What Is Meant by Each Contrast

Cycle to Cycle

Same Fixture

- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A



Cycle to cycle is usually the smallest process family. Do BOB and WOW units come in groups? How many parts are created with one cycle of the machine? Do BOB and WOW pieces come in groups, or are they random? Consider sampling in groups of 3 to 5 consecutive pieces. Use Multi-Vari to characterize the process.

Fixture to Fixture

Same Machine

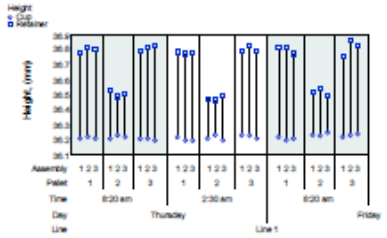
- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A

Is there contrast across fixtures? Add - or replace - Fixture to Fixture family with: Spindle, Nest, Pallet or Head as appropriate. These families represent “within machine” contrasts.

Machine to Machine

Same Line

- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A



Are there parallel machines performing the same operation without the problem?

Line to Line

Same Plant

- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A

Are there other manufacturing lines creating the same parts, or units, without the problem?

Process Families Strategy Diagram – Template

Mone(Y) =

Family of Variation, or Contrast

Diagram

What Is Meant by Each Contrast

Plant to Plant

Same Model

- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A

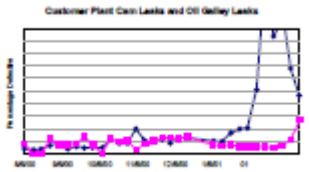
Do other plants have the same, or similar, products with different levels of quality or reliability?

Model to Model

Same Time

- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A

Are there other models which do not have the problem, or have a significantly different rate?



Time to Time

- ☐ Large Contrast
- ☐ Small or no contrast
- ☐ Do not know
- ☐ N/A

Are there time of build contrasts? Hour, shift, day, week or month may be additional time contrast worth investigating.

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Module 1 Summary

Basics

- Problem solving is based on evidence gathered by talking to the parts.
- Y → X thinking is convergent problem solving best represented by “Dictionary Game Thinking.”
- The Mone(Y) is the performance distribution or descriptive statement for either an event, a feature, a property, or a defect.
- Y → X problem solving strategies for events, features, properties, and defects are different and will be explained in the following modules.
- The BIG X is the variable or contrast with the greatest leverage over the Mone(Y) from BOB to WOW.
- The Pareto Principle tells us there is only one BIG X. Finding and controlling it is everything.
- Cartooning is used for explaining where and what the problem is.

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Module 1 Summary

Basics

- Process flow diagrams pictorially represent serial and parallel processes, which, when linked together with inputs and outputs, are required to produce a component or assembly.
- Strategy Diagrams are a way to document possible contrasts for the Mone(Y) that you can leverage during your project.

2 Effective Measurement Systems

Module 2 Objectives
Effective Measurement Systems

Understand:

- Measurement System Analysis (MSA) for features and properties

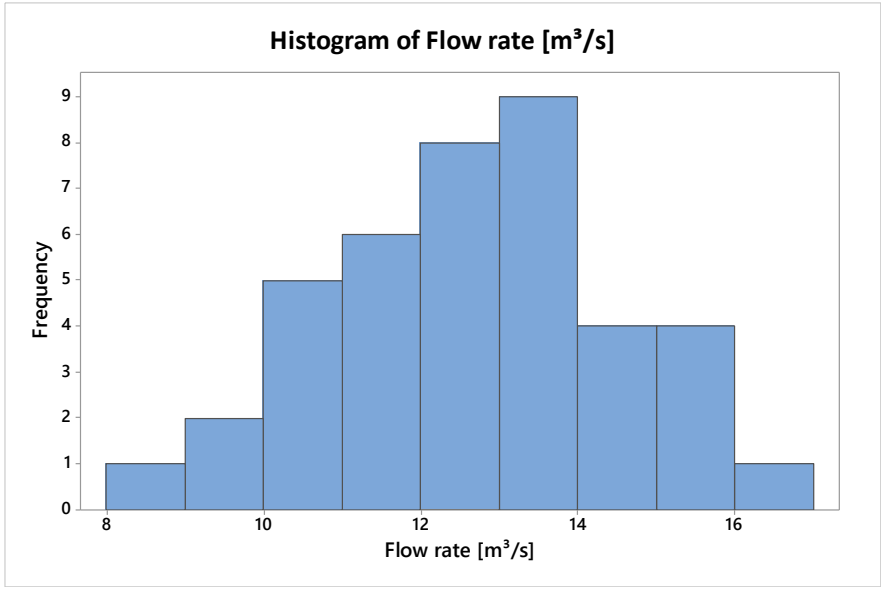
Be able to:

- Apply Isograph
- Solve measurement problem

Measurement System

Part No.	Flow [m³/s]
1	11.57
2	12.09
3	13.61
4	12.12
5	15.43
6	15.28
7	10.16
8	12.83
9	7.90
10	12.67
11	9.85
12	14.61
13	13.58
14	12.62
15	13.43
16	10.80
17	10.37
18	11.92
19	14.06
20	9.79
21	10.95
22	13.09
23	12.52
24	11.17
25	10.27
26	14.00
...	...
40	9.84

- The flow rate of 40 fuel pumps are measured once.
- The data is plotted in a run chart and a distribution is created based on the measurements of these parts.
- What does this reveal about the measurement system?

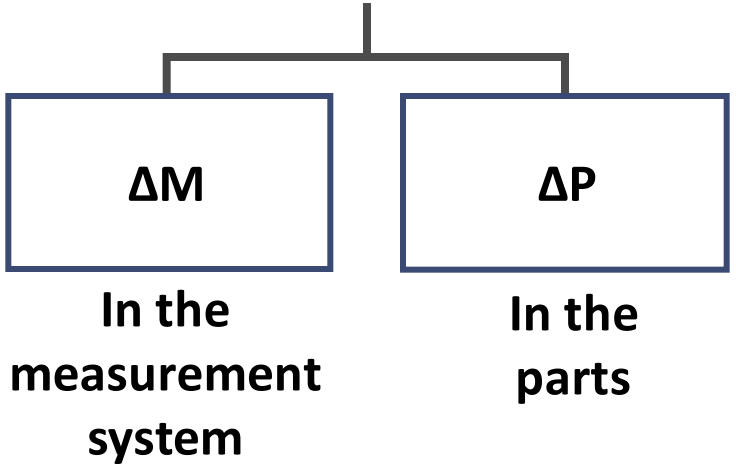




Evaluating the Measurement System

The First Split on the Solution Tree

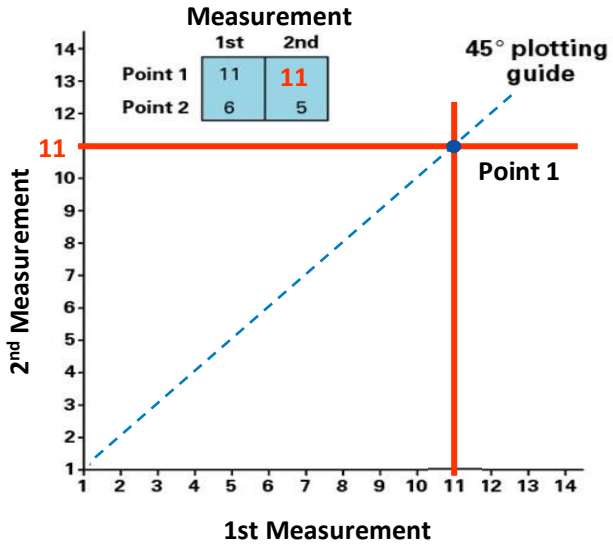
Where does the BIG X live?

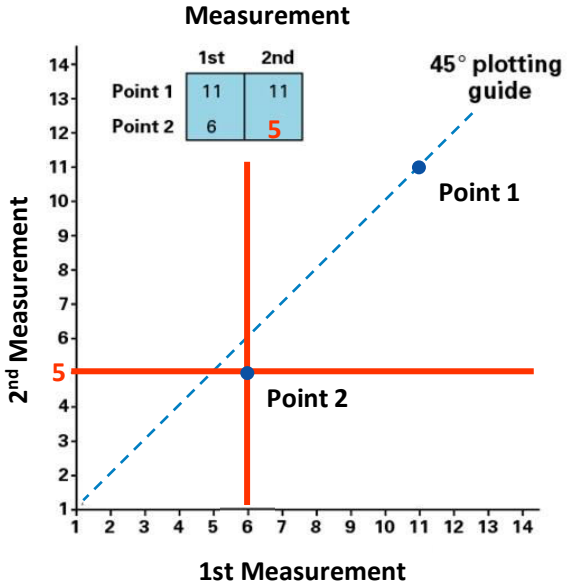


Use Isograph to evaluate the measurement system.

Isograph – Concept

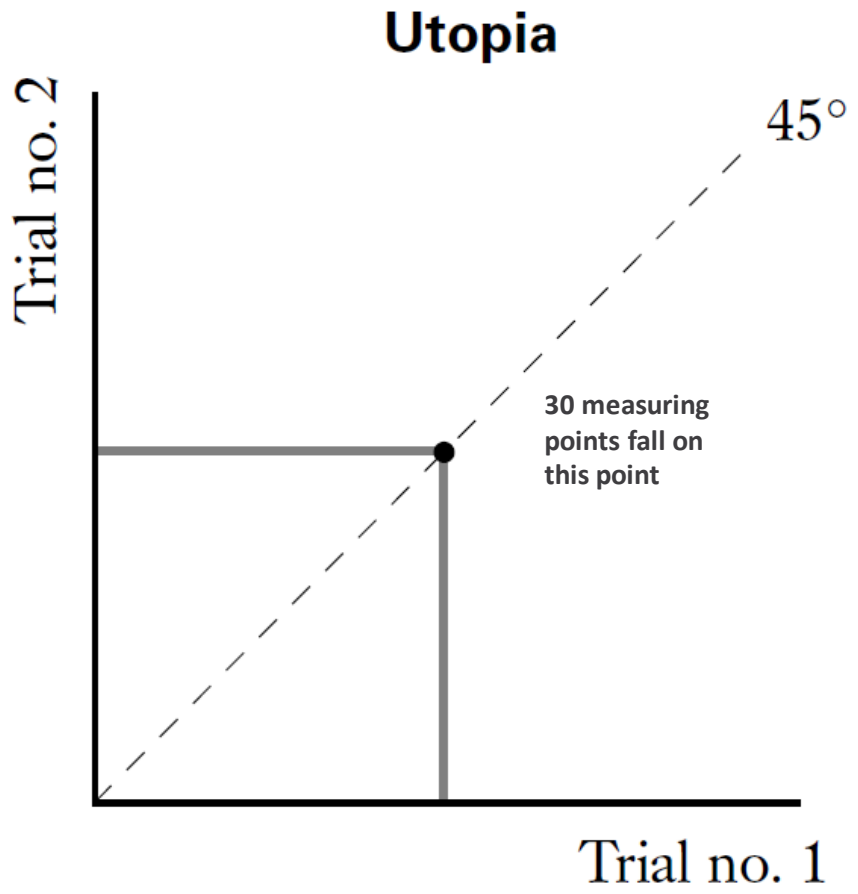
- Measure the Mone(Y) feature on the part. This is the first measurement shown on the Measurement 1 axis.
- Now, measure that same Mone(Y) feature on the same part. This is the second measurement shown on the Measurement 2 axis.
- The intersection of these two values is where the first point is plotted.





Isograph – Concept

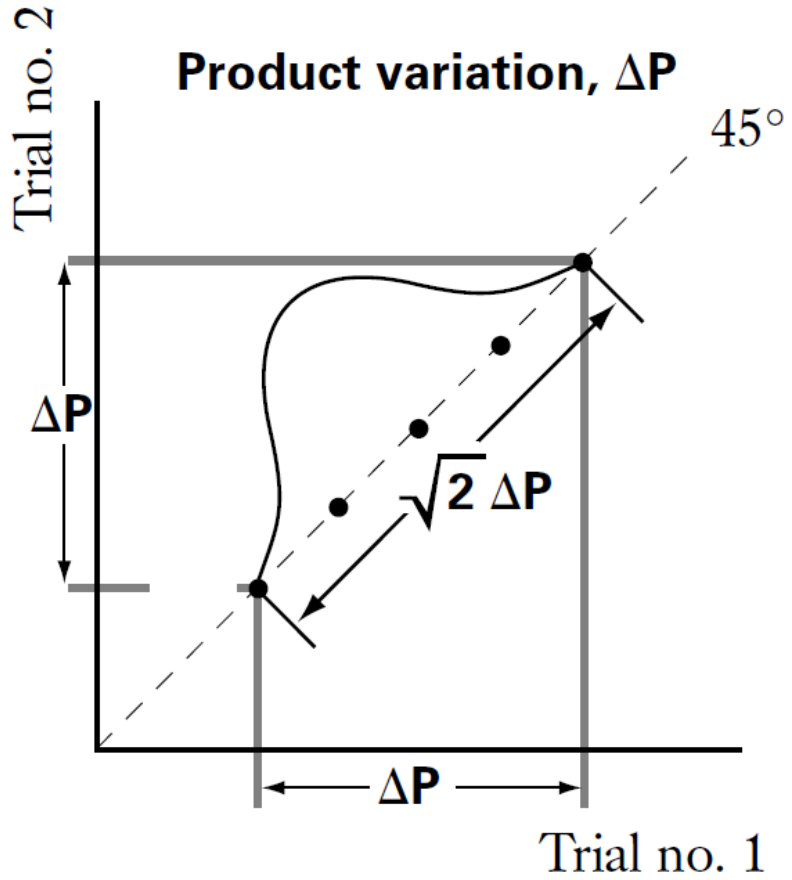
What would Utopia look like?



- A perfect manufacturing process that produces identical parts, cycle after cycle.
- The parts are measured with a perfect measuring system, which does not include ΔM.

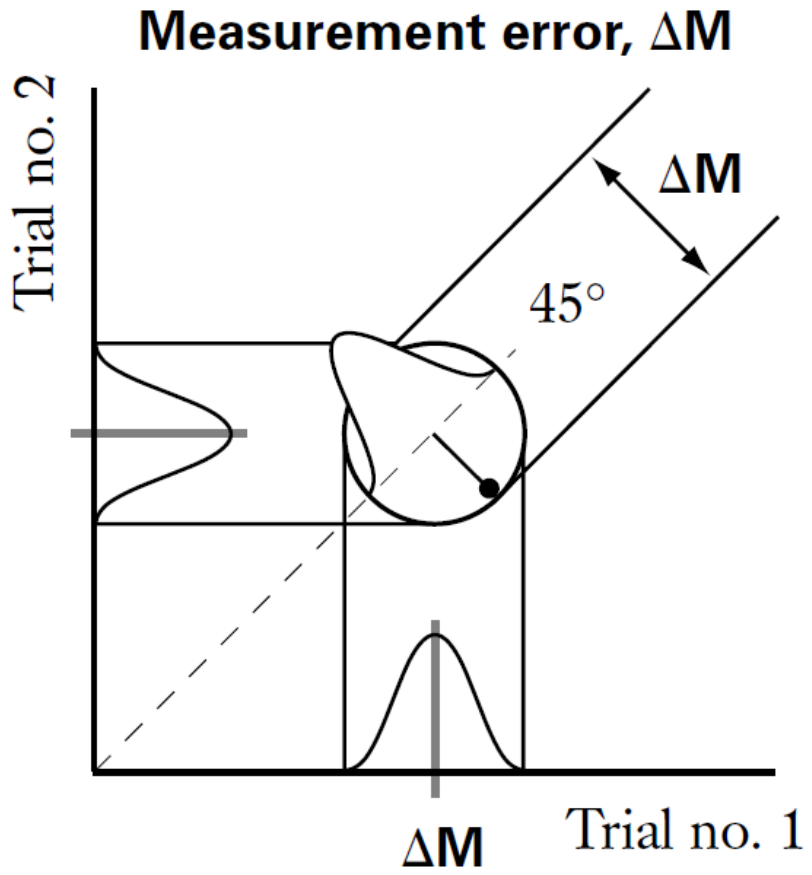
Isograph – Concept

What would change if we introduce only manufacturing variation?



Isograph – Concept

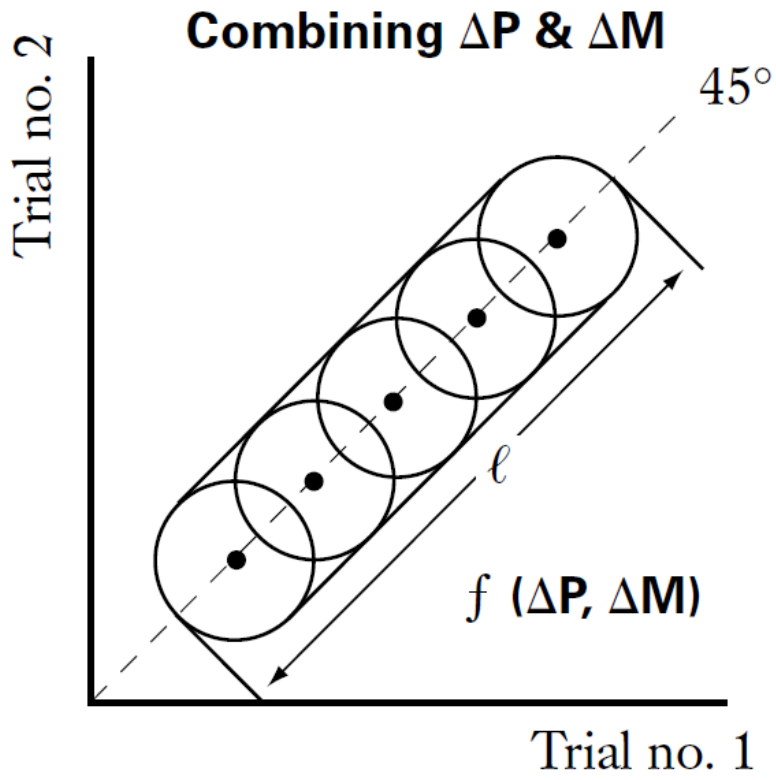
What would change if we introduce only measurement error?



Isograph – Concept

The Isograph reflect both part variation (ΔP) and measurement error (ΔM). This is the basis of Isograph.

What does the real world look like?



Isograph – Guidelines

Isograph is a simple graphical tool to verify if the measurements in your problem solving approach are OK.

- It compares the product variation (ΔP) and measurement error (ΔM) to evaluate the resolving Discrmination Ratio (DR) of a measuring system.
- The Isograph does not replace the MSA which is required for the official measurement systems in our production processes.

1. Collect Samples

- Select 30 points randomly to obtain an efficient estimate of ΔP.
- Take a random sample of 30 points from the distribution of interest.
- Thirty points may not mean 30 parts. Often one part has multiple points.
- If random selection does not provide full range of ΔP, it is appropriate to use BOB and WOW parts previously collected in the Isograph to obtain a full ΔP.

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Isograph – Guidelines

2. Mark Samples

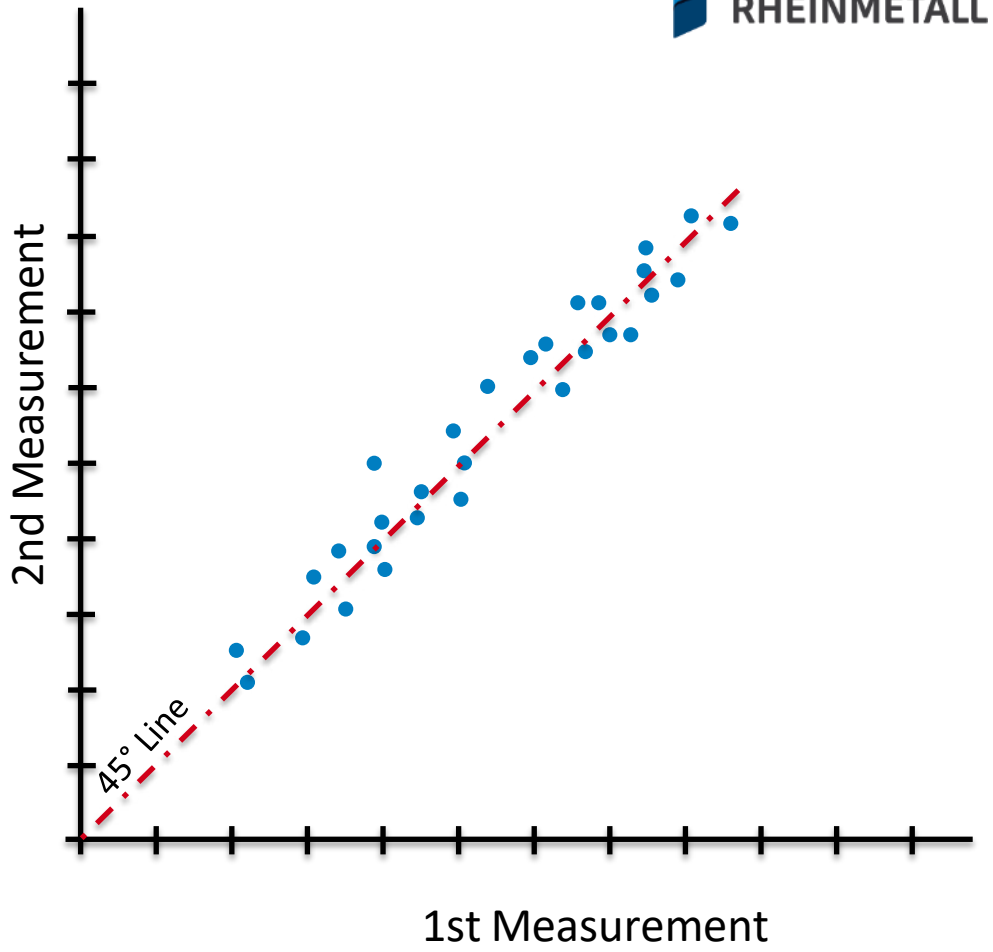
- Mark each part appropriately so it can be identified.

3. Measure Samples

- Measure each point once in a set order.
- Measure each point again in a different random order.

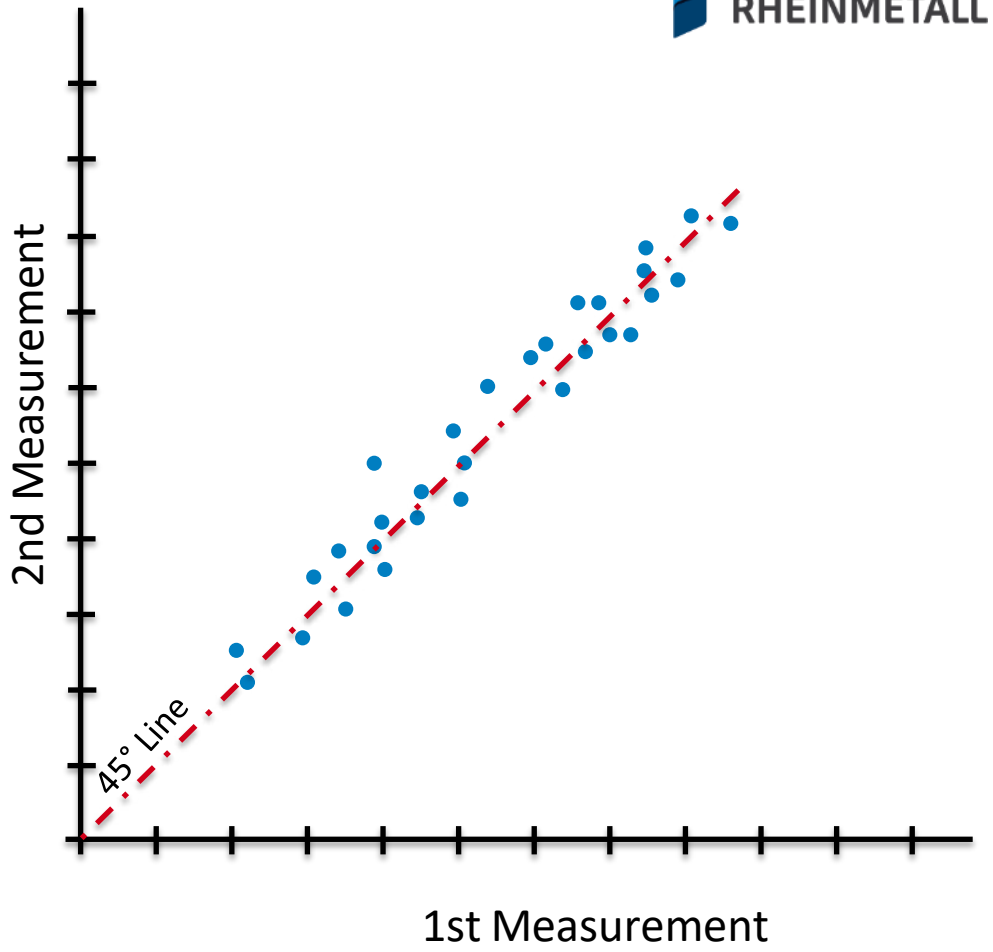
4. Plotting Data

- Use the same scale for each axis if plotting manually. Draw a 45° line for reference to avoid plotting errors.
- Plot the points. Both axes should be labeled with the name of the Mone(Y) and units.
- Check to see if there is a bias.



Isograph – Guidelines

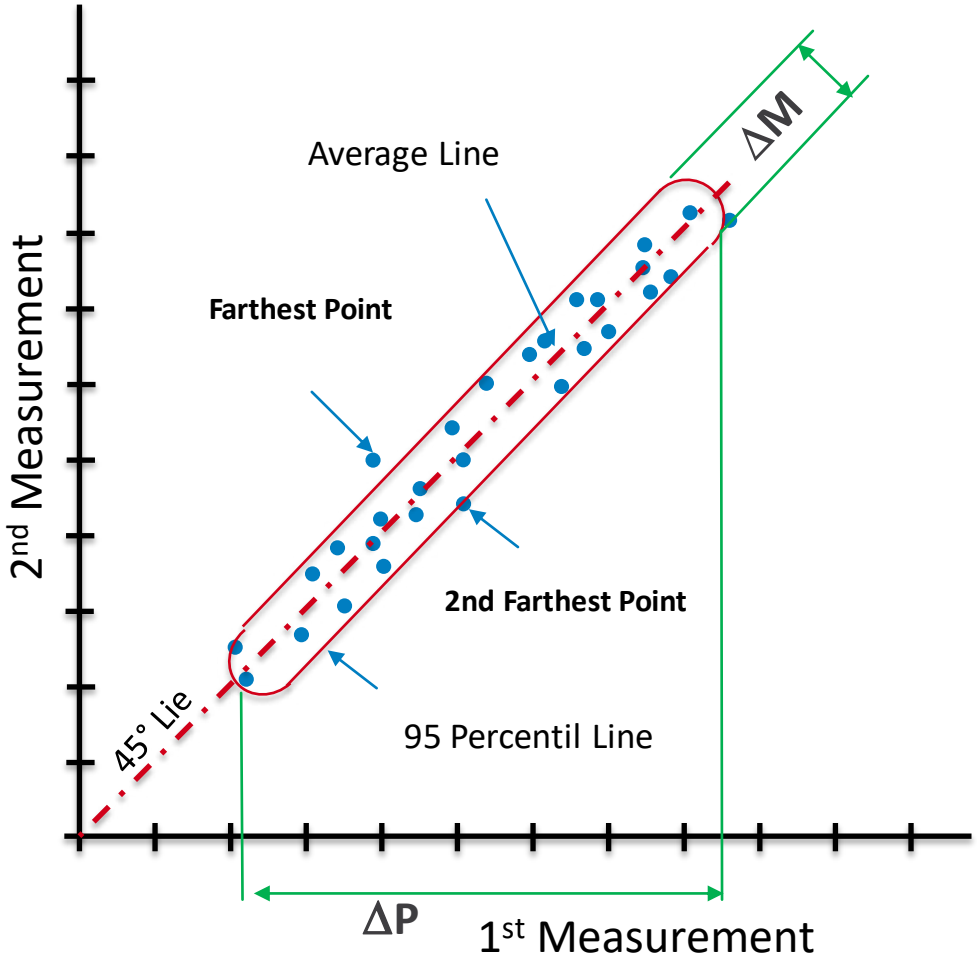
- If there is a bias, draw the average line.
- The line should go through as many points as possible.
- About ½ of the remaining points should be on either side of the line. Don't let a single outlier near the two ends of the line affect the position of the line too much.
- If there is no bias, use the 45 ° line as the average line.



Isograph – Guidelines

5. Create Isograph

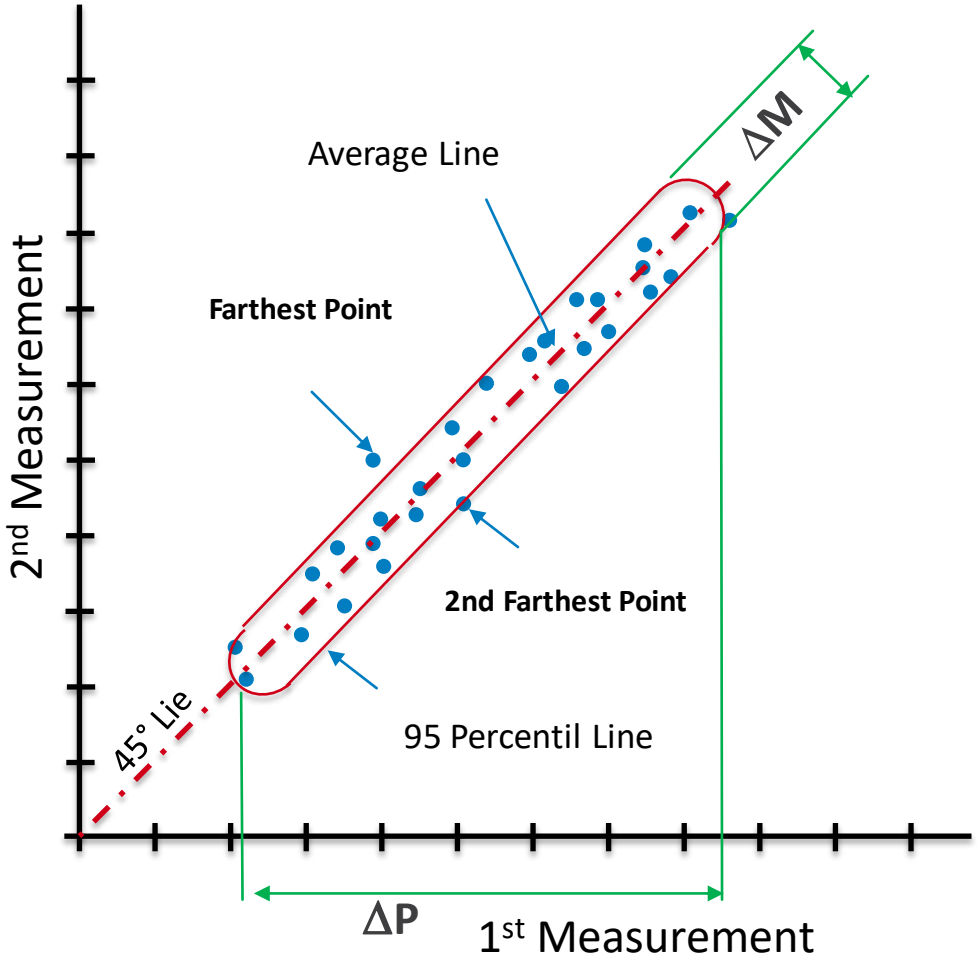
- Find the second farthest point from the 45° line, measured perpendicular to the line.
- Draw the first limit line through this point and parallel to the 45° line.
- Draw the second limit line the same distance from the 45° line as the first limit line, but on the opposite side. The range between the limit lines provides an estimate for the 95th percentile of the ΔM distribution (29 of 30 points are inside the lines 29/30 = 96,67 %).



Isograph – Guidelines

- If $\Delta P / \Delta M \geq 6.0$, you may use a "crumb scraper" to get a valid estimate of ΔP, product variation, as follows:

  ΔM defines the region of uncertainty, or blind spot. We are willing to give up 1/6 of product variation to ΔM. When $\Delta P/\Delta M$ is much less than 6 the ΔP estimate will be inflated.

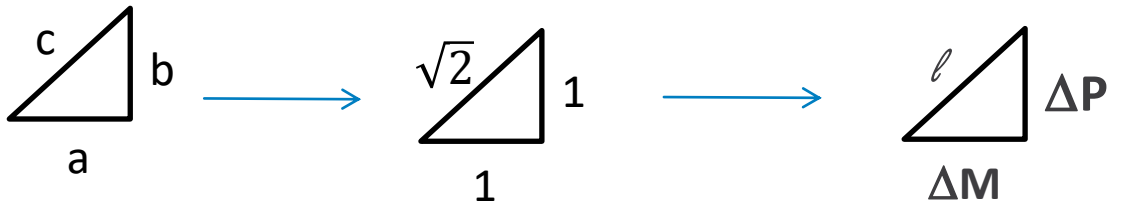


Isograph – Guidelines

If we allow ΔM to be 1/6 of ΔP, than l/ΔM = 8,5.

The ratio between l and ΔP can be calculated according to the following rules:

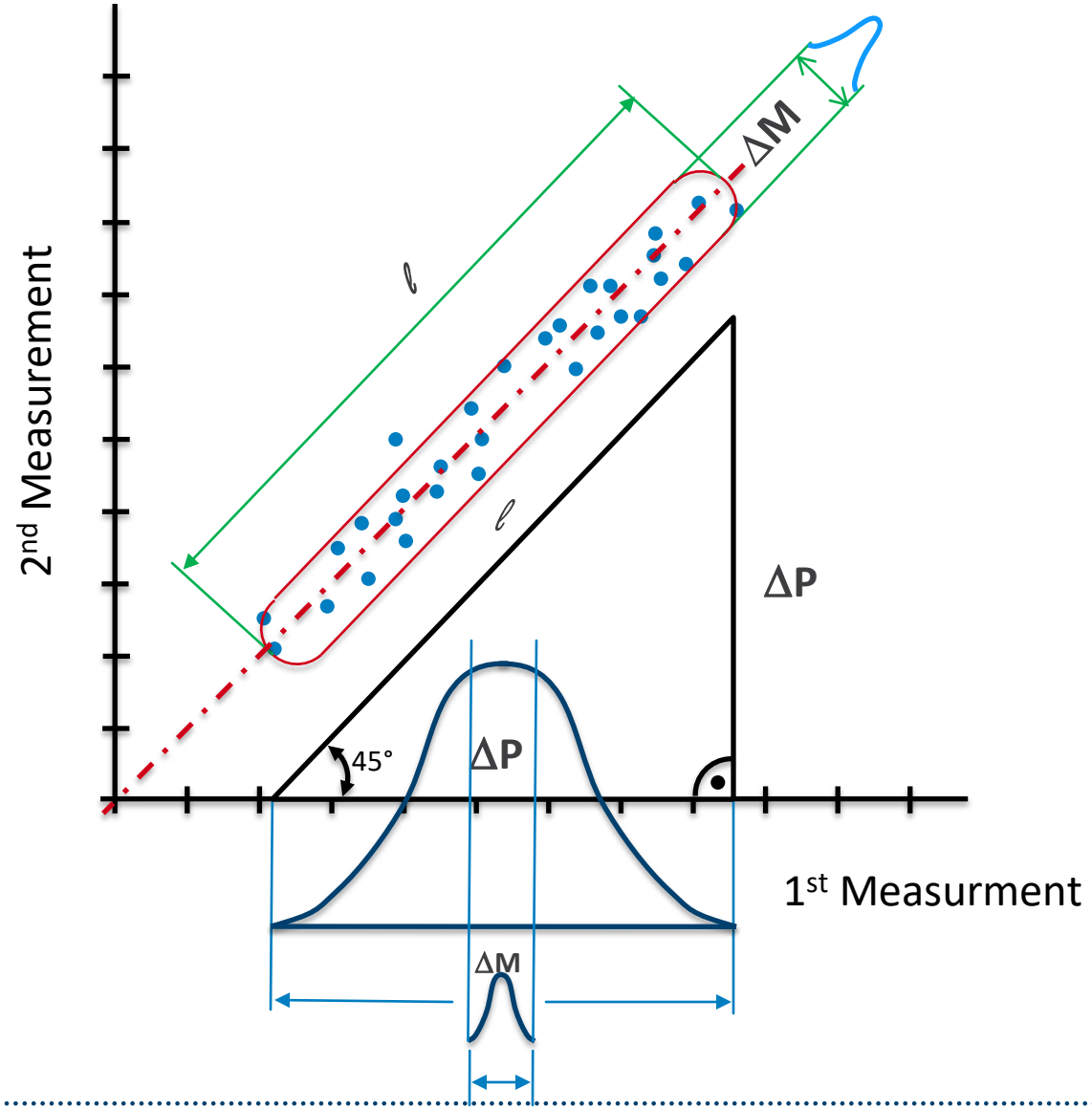
ΔP / ΔM ≥ 6, if l / ΔM ≥ 8,5, since



$c^2 = a^2 + b^2$

$l^2 = \Delta P^2 + \Delta M^2$

$$\frac{\Delta P}{\Delta M} \geq 6 \quad and \quad \frac{l}{\Delta M} \geq 8{,}5$$

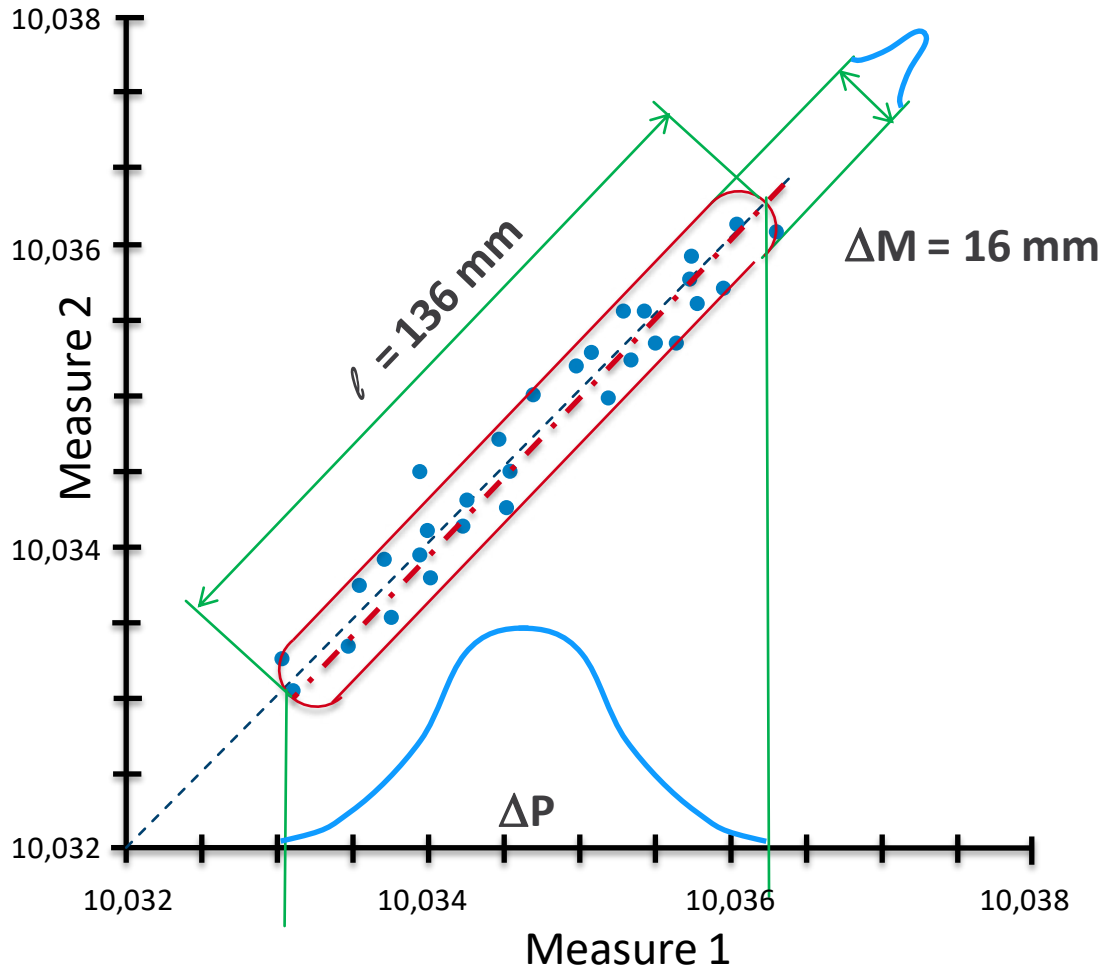


Example

ℓ = 136 mm (measured with ruler)

ΔM = 16 mm (measured with ruler)

$$\frac{l}{\Delta M} = \frac{136\ mm}{16\ mm} = 8{,}5 \text{ (good sausage)}$$



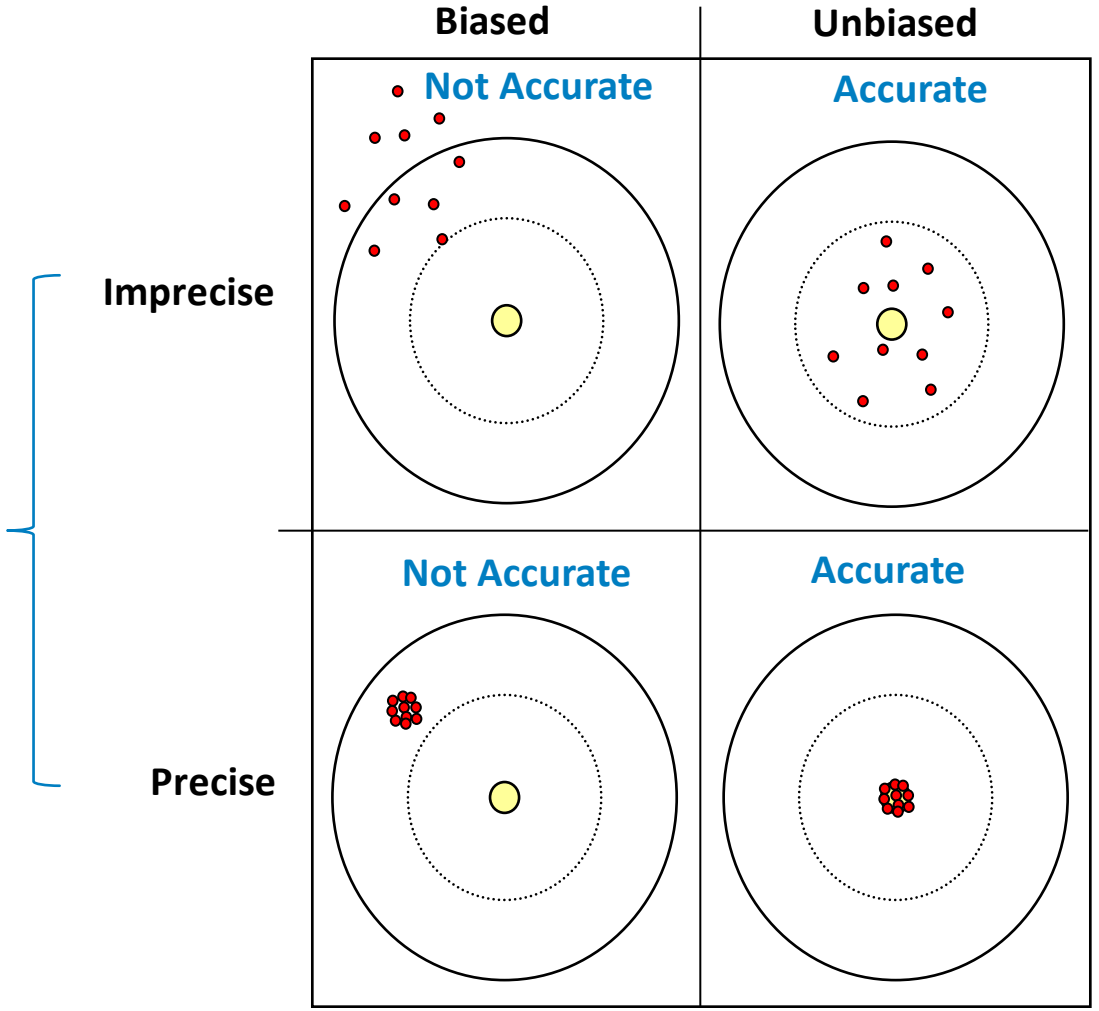
Measurement System Terminology

Understanding Measurement System Terminology:

- **Accuracy –** Closeness of the measurement to the true value.
- **Precision –** The repeatability of the measurement system.
- **Bias –** A consistent deviation from the truth.

Accuracy vs Precision vs Bias

ΔM in an Isograph represents the precision.



When accuracy is important, a calibration study must be done.

Accuracy is the combination of precision and bias.

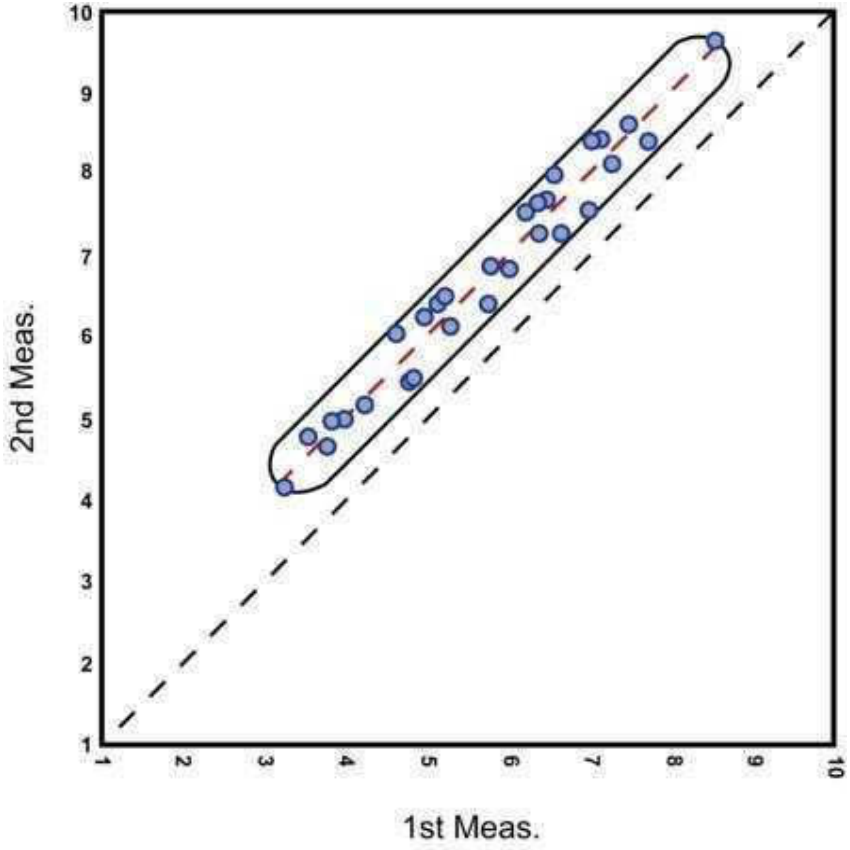
Isograph

Why is the data shifted off the 45° line?

- Constant Bias

Constant bias is a fixed offset between the first and second readings. This may be caused by:

- Temperature
- Calibration
- Measurement Procedure



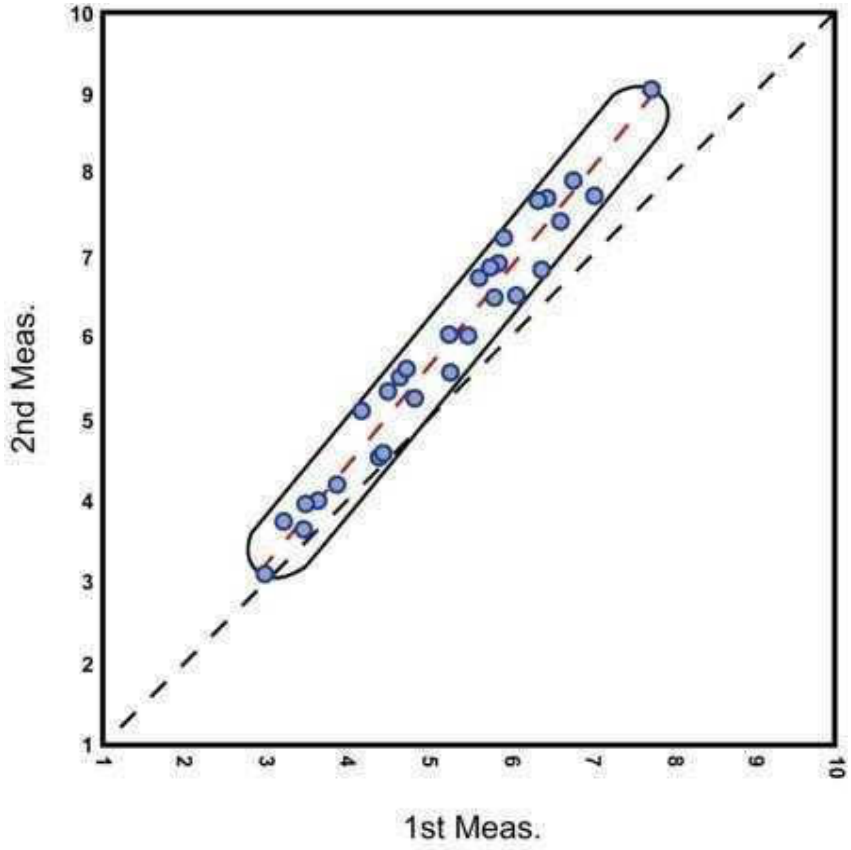
Delta P = 5.39
Delta M = 0.77
Discrimination Ratio = 7.0

Isograph

Why is the data rotated off the 45° line?

- Proportional Bias

Proportional bias is an offset where the second reading is a percentage of the first measurement value. This may be observed on an Isograph where two different measurement systems are evaluated.



Measurement System Terminology

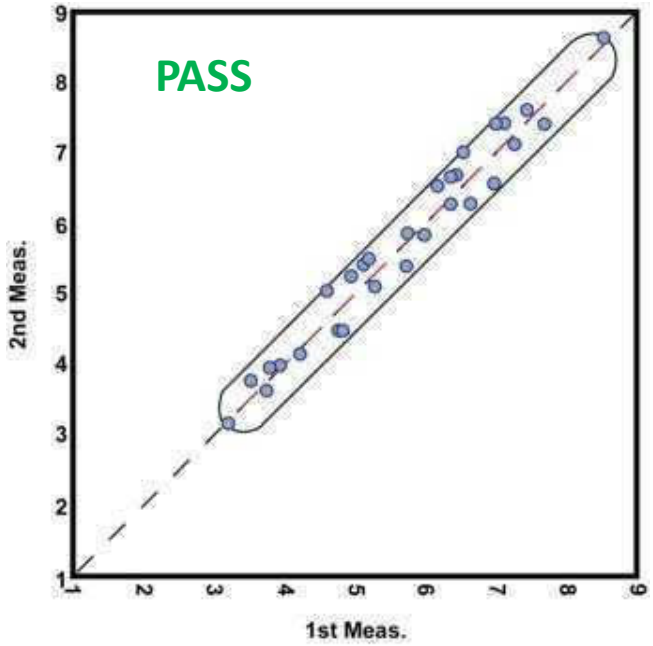
Understanding Measurement System Terminology:

- **Accuracy –** Closeness of the measurement to the true value.
- **Precision –** The repeatability of the measurement system.
- **Bias –** A consistent deviation from the truth.
- **Discrimination –** The ability to see differences.

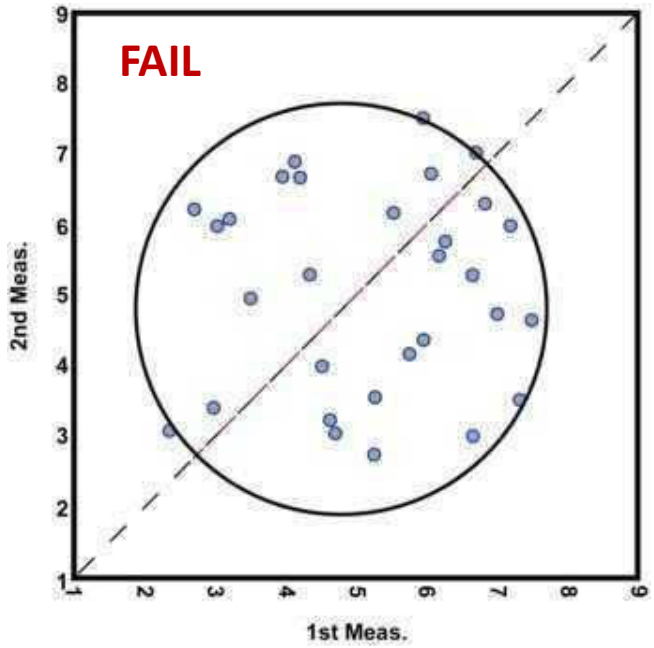
Isograph – Discrimination Ratio

The discrimination ratio is ΔP/ΔM. The standard for BIG X problem solving is six or greater.

A measurement system with good discrimination looks like a narrow "sausage."



A measurement system with poor discrimination looks like a fat sausage. This is a "meatball."



Measurement System Uses

	Timely Results	Easy to Use	Variable Output	High Accuracy	High Precision
Quality Control		✓		✓	
Process Control	✓	✓	✓	✓	✓
Problem Solving	✓		✓		✓

The choice of measurement system for:

- Quality control is based on quality requirements.
- Process control is based on production requirements.
- Problem solving is driven by the strategy.

Measurement System Terminology

Understanding Measurement System Terminology:

- **Accuracy –** Closeness of the measurement to the true value.
- **Precision –** The repeatability of the measurement system.
- **Bias –** A consistent deviation from the truth.
- **Discrimination –** The ability to see differences.
- **Resolution –** The smallest increment on the scale

Isograph – Analysis

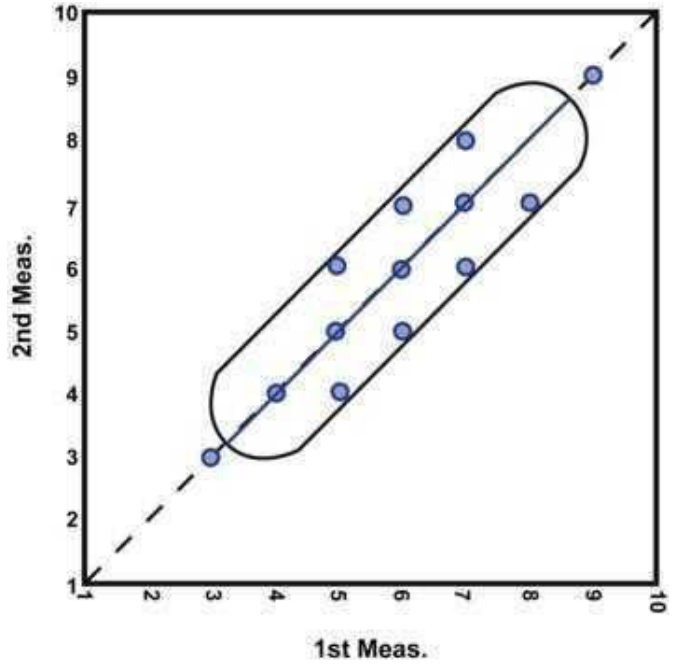
Why did the measurement system fail Isograph?

- Insufficient resolution



Resolution is the smallest increment on the measurement device.

Resolution:
X



Delta P = 5
Delta M = 2
Discrimination Ratio = 2.5

What happens if we double the resolution?

(1, 2, 3, 4 → 1.0, 1.5, 2.0, 2.5)

Isograph – Resolution

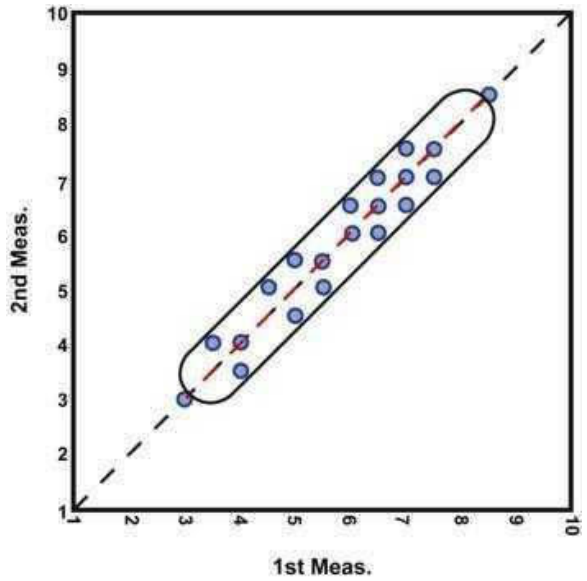
Is this enough resolution to pass?



What happens if we increase the resolution to two decimal places?

(1.0, 1.5, 2.0 → 1.XX)

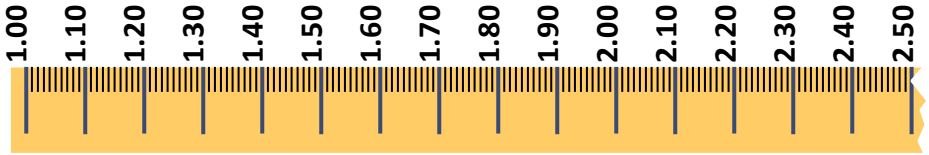
Resolution:
X.5



Delta P = 5.3
Delta M = 1.1
Discrimination Ratio = 5.0

Isograph – Resolution

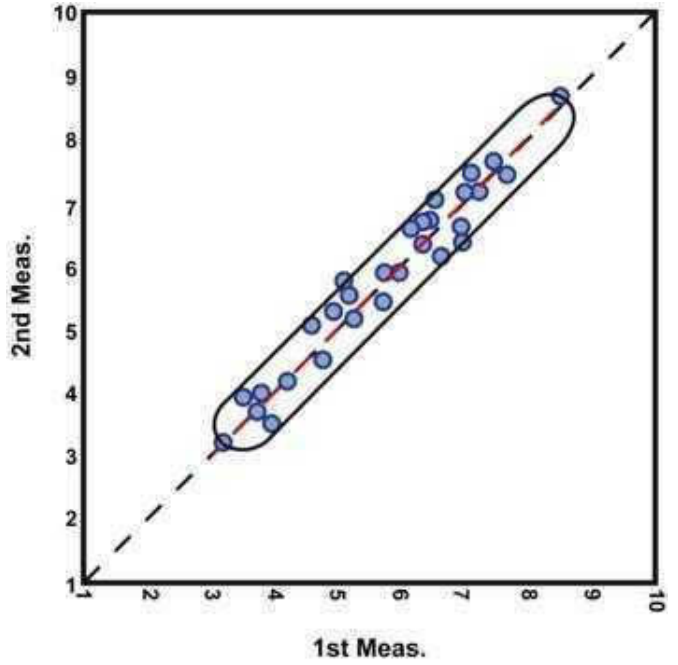
Do we need any more resolution?



What happens if the measurement system were accurate to three decimal places?

(1.XX → 1.XXX)

Resolution:
X.XX



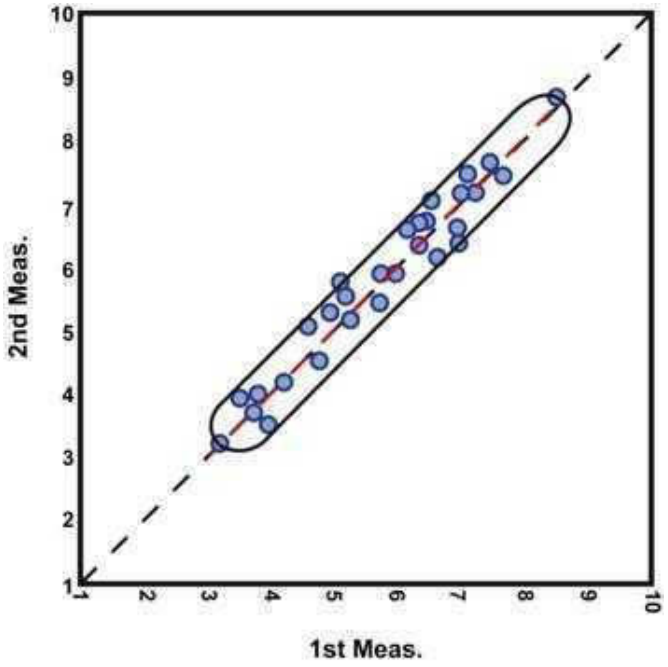
Delta P = 5.37
Delta M = 0.89
Discrimination Ratio = 6.0

Isograph – Resolution

Is the three-place measurement system more effective than the two-place?

No!

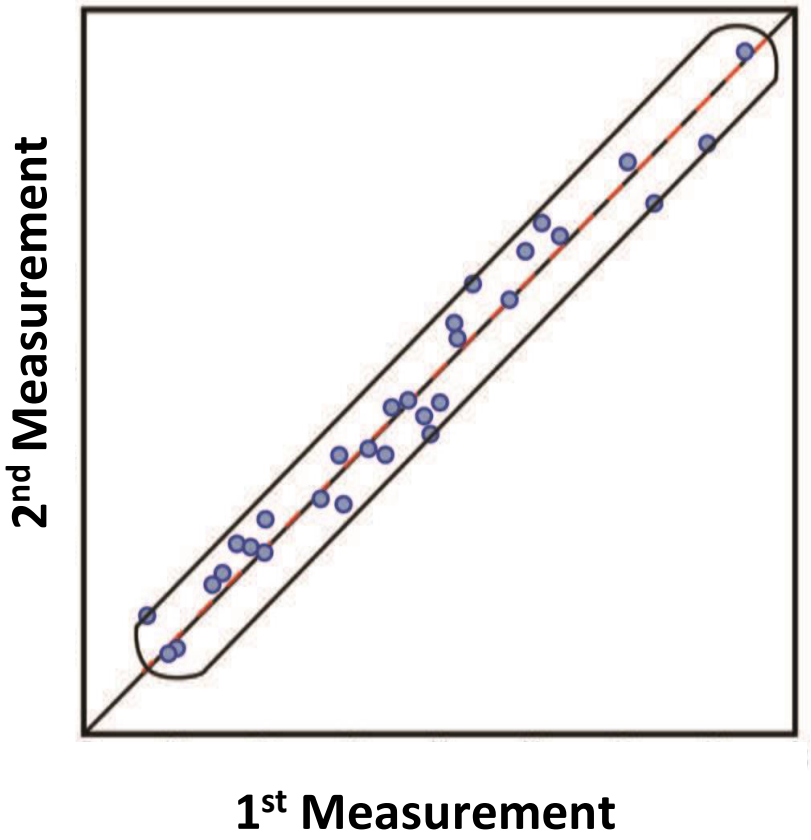
Resolution:
X.XXX



Delta P = 5.37
Delta M = 0.89
Discrimination Ratio = 6.0

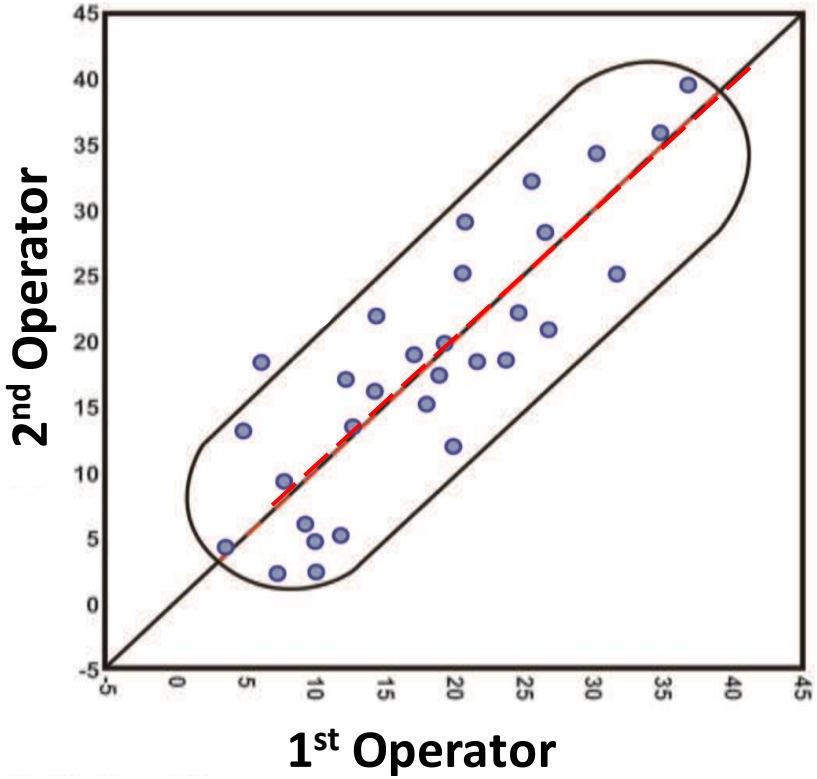
Isograph Versatility

Isograph Measurement to Measurement, Same Operator



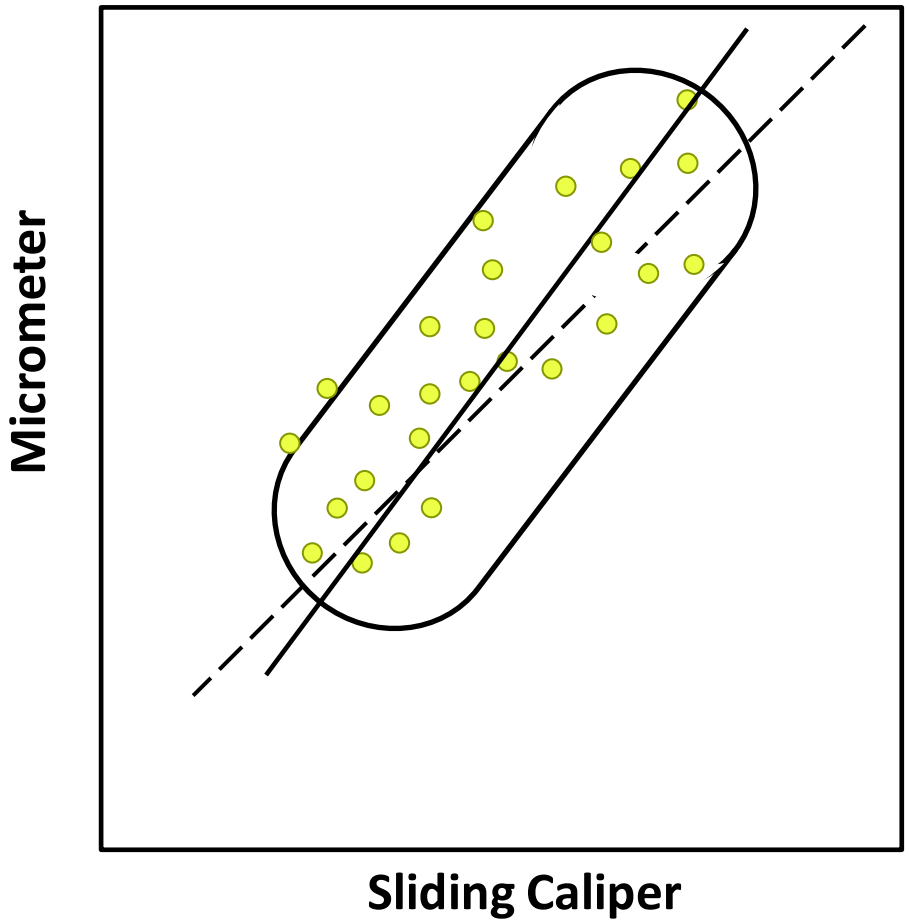
Isograph Versatility

Isograph Operator to Operator, Same Gage



Isograph Versatility

Isograph Gage to Gage, Same Time



Effective Measurement

How does measurement error affect our total variation?

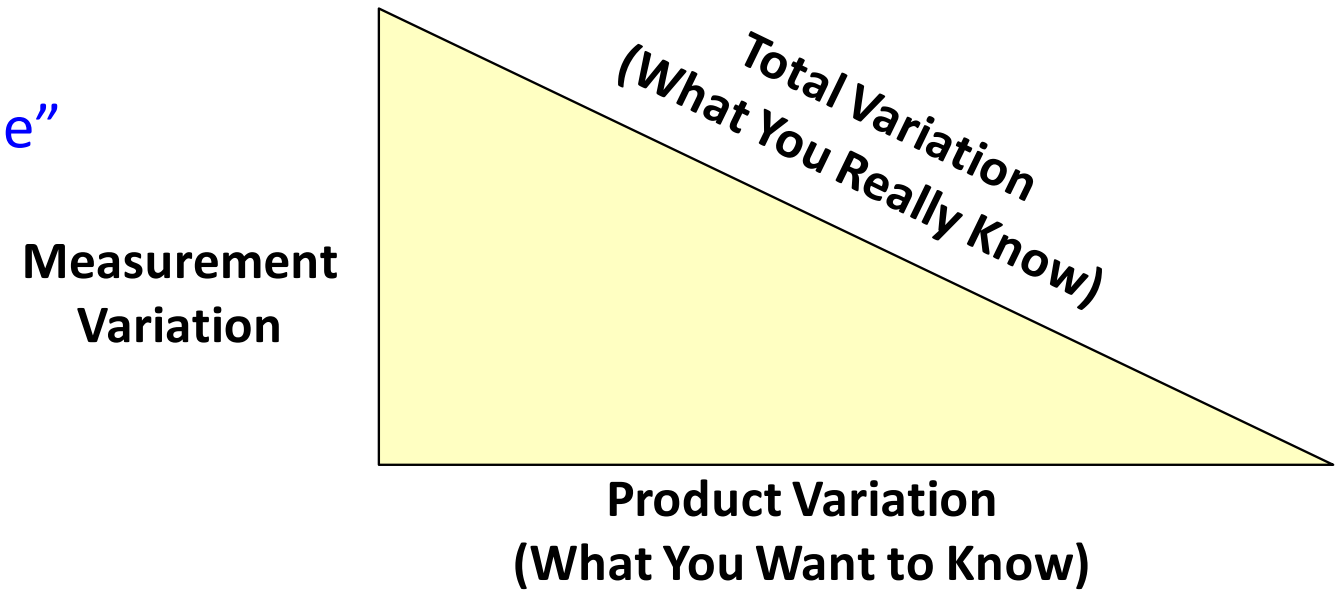
The Effect of Measurement Error

"Observed" "Actual"

Total Variation = Product Variation + Measurement Variation

$$\sigma^2_{Total} = \sigma^2_{Product} + \sigma^2_{Measurement}$$

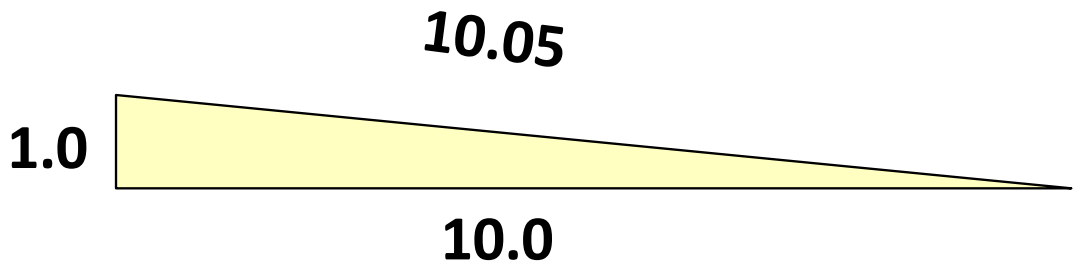
Think of a "Right Triangle"



The Effect of 10 % Measurement Error

Total Variation = Process Variation + Measurement Variation

$$10.05^2 = 10^2 + 1^2$$

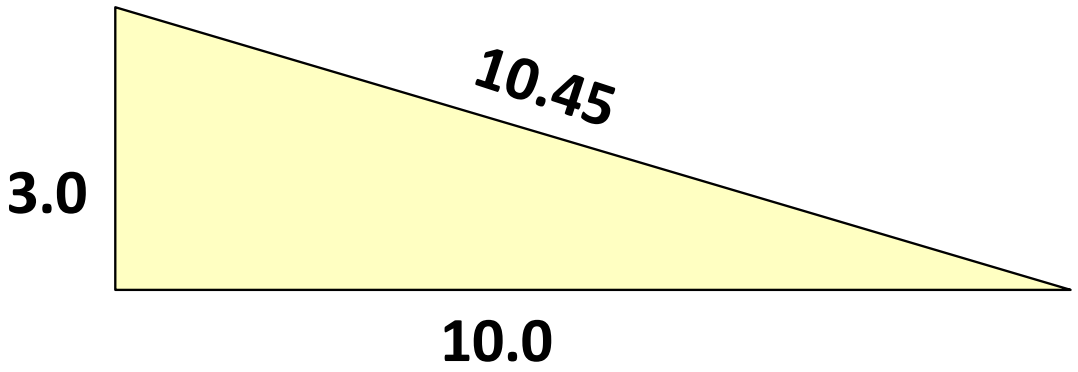


Measurement Error has little impact on the observed value

The Effect of 30 % Measurement Error

Total Variation = Product Variation + Measurement Variation

$$10.45^2 = 10^2 + 3^2$$



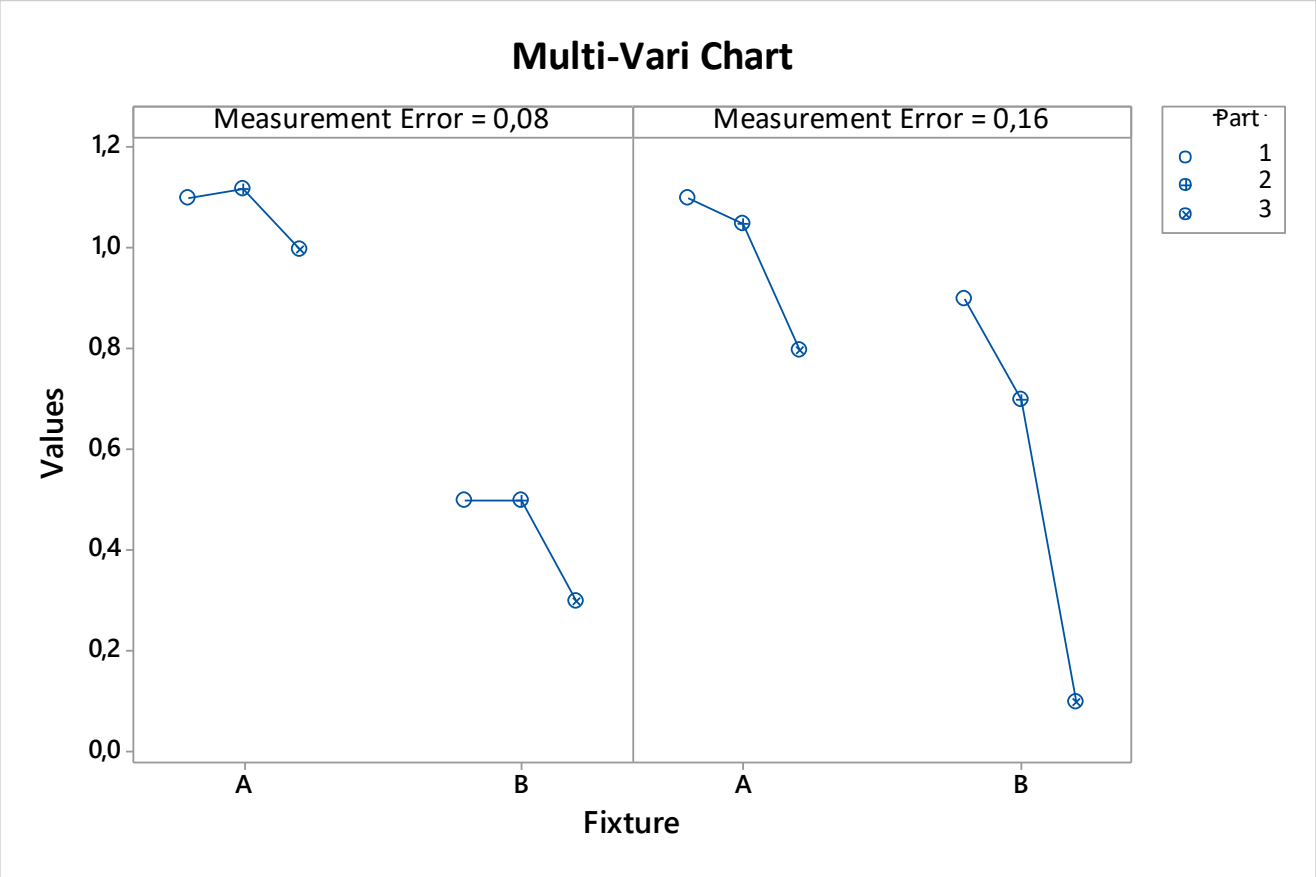
Measurement Error has larger impact on the observed value

Effective Measurement

What would happen, if our measurement system for the Mone(Y) does not give us suitable results?

Effective Measurement

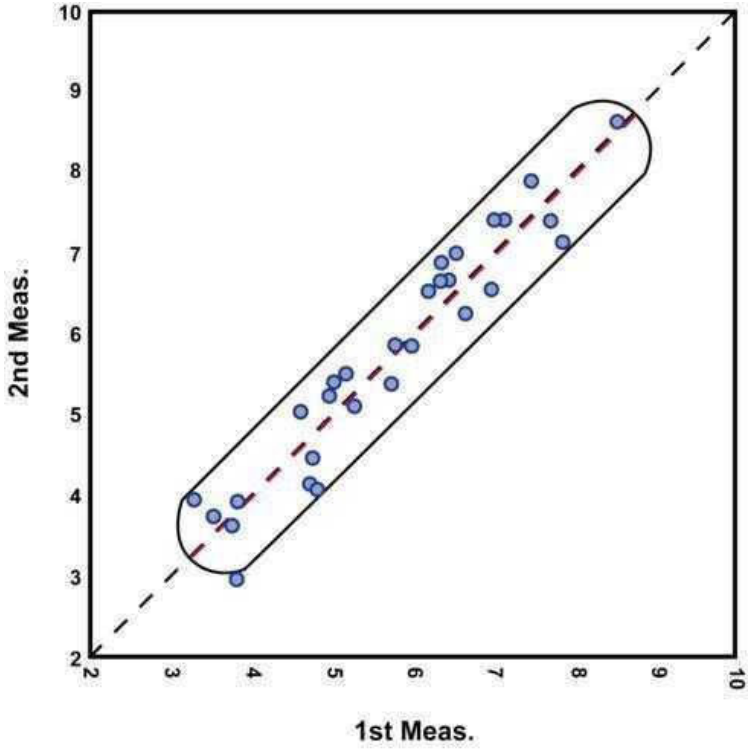
Effect of the measurement system, …



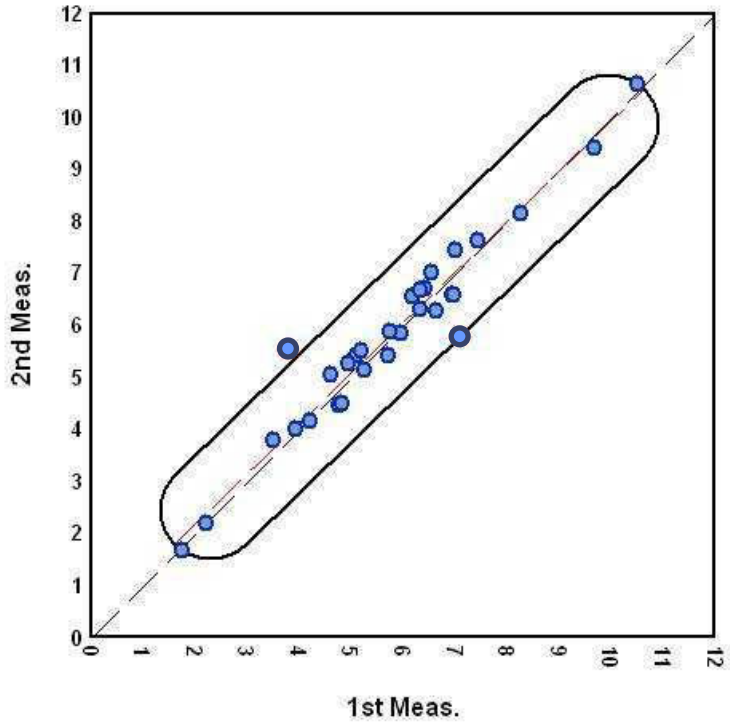
With high measurement variation, key contrasts cannot be as easily seen.

Study the Isograph For Clues

Do Not Just Look at the Discrimination Ratio



Delta P = 5.45
Delta M = 1.18
Discrimination Ratio = 4.6



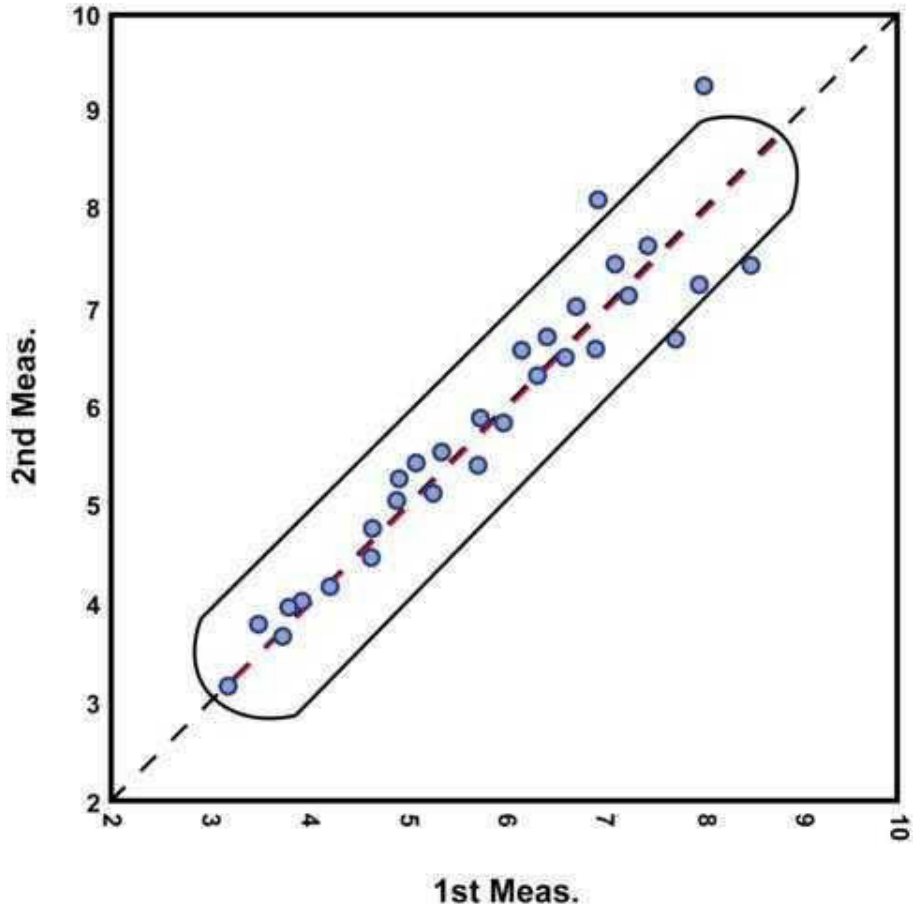
Delta P = 8.78
Delta M = 1.91
Discrimination Ratio = 4.6

What would you do in each case?

Study the Isograph For Clues

Do Not Just Look at the Discrimination Ratio

For events, it is not uncommon for WOWs to be more unstable than BOBs, resulting in a greater measurement error at one end of the sausage.



Delta P = 5.65
Delta M =1.34
Discrimination Ratio = 4.2

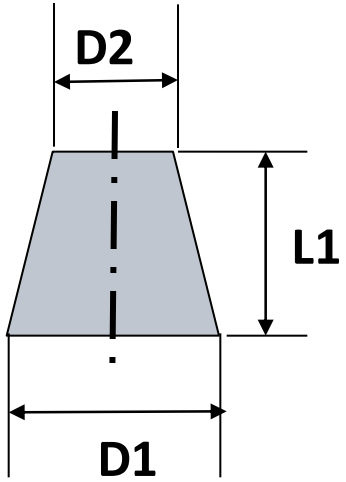
Isograph Workshop – Cylindrical Part - Excercise

- The available parts for the Isograph are listed on the next page.
- The Mone(Y) is D1 in [mm].
- For this exercise only take 30 random samples.
- Measure each sample once.
- Measure each sample a second time in a random test order.
- Create the Isograph in Excel.
- Is this a good measurement system?
- You will have 20 minutes to complete this exercise.

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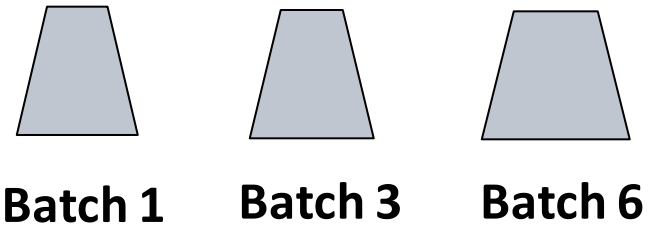
Isograph Workshop - Cylindrical Part - Excercise

Problem Description



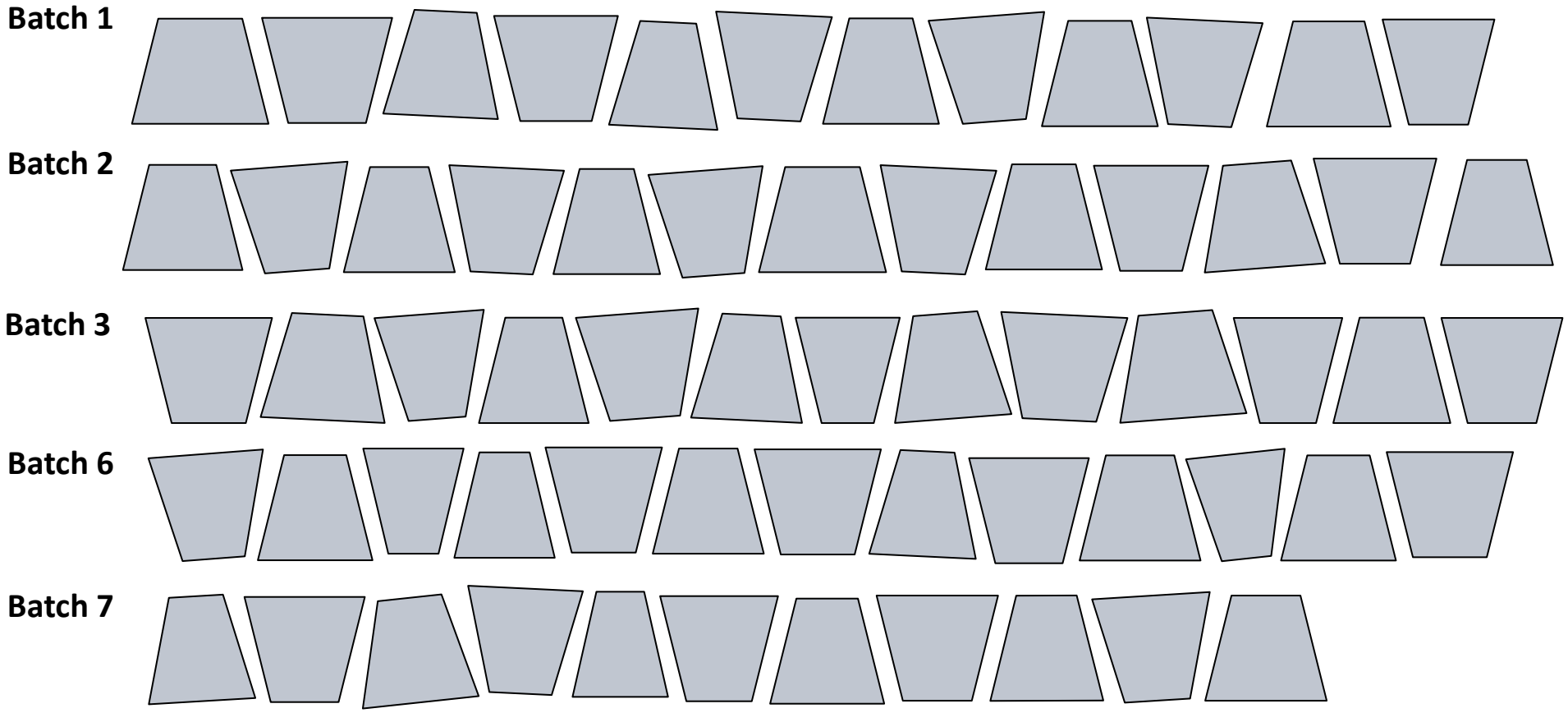
- Problem with variation in D1 of a cylindrical part. We have suspicious samples as well as some parts from existing batches.
- Before we start further analysis, the measurement system needs to be verified.
- Measurement is being done with a ruler.

Suspicious Parts

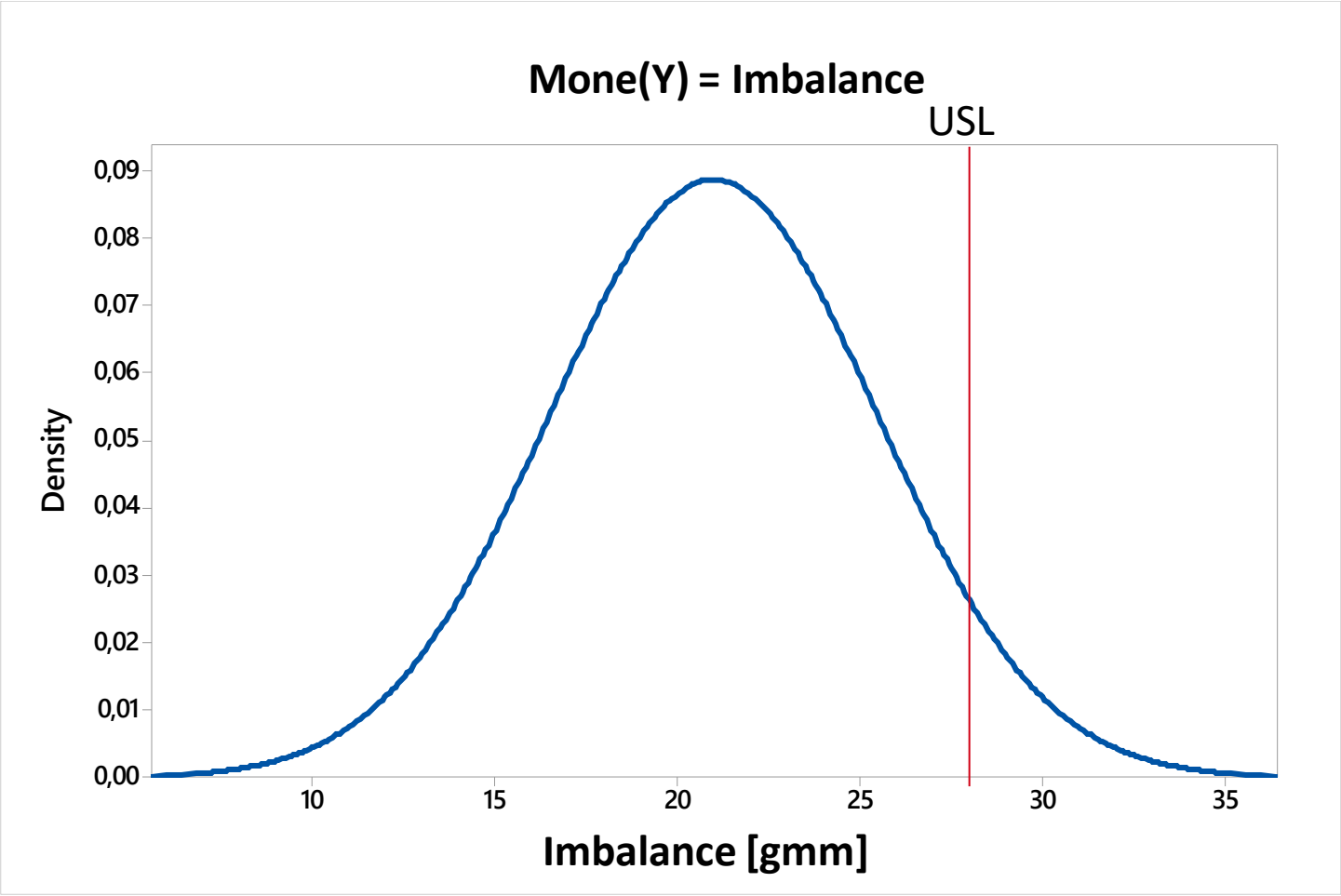


Isograph Workshop - Cylindrical Part - Excercise

Available Parts From Existing Batches



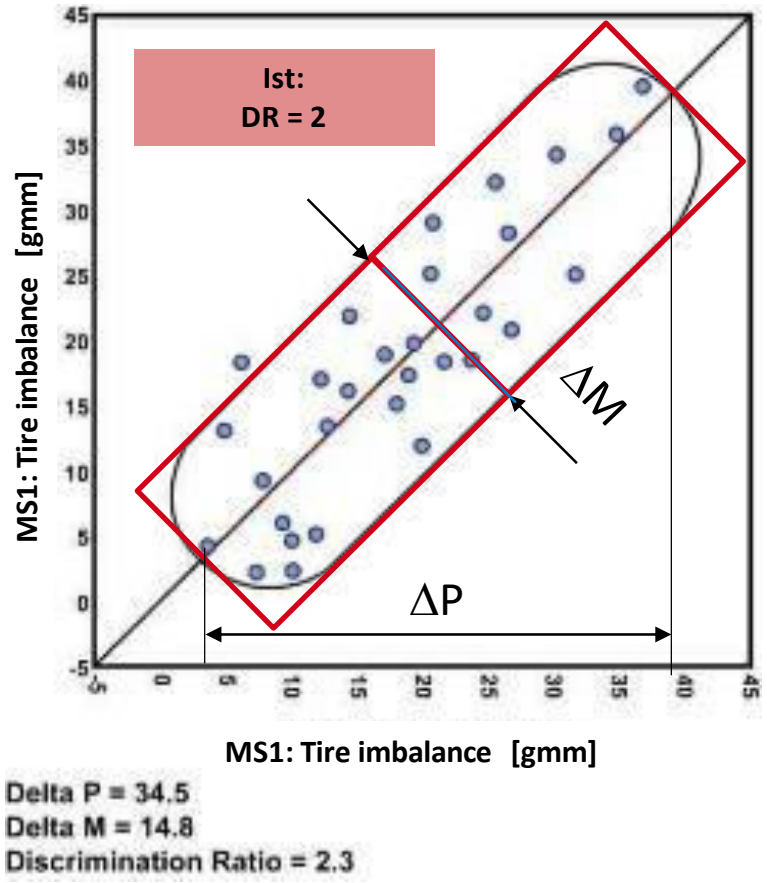
Case – Tire Imbalance



What is next?

Case – Tire Imbalance

Isograph Tire imbalance



Delta P = 34.5
Delta M = 14.8
Discrimination Ratio = 2.3

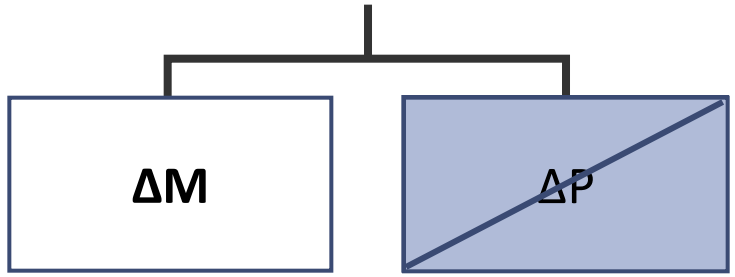
Thirty different tires measured with the original measurement system.

Failed Isograph (DR ≤ 6)

→ **BIG X** lives in the Measurement system.

Solving a ΔM Problem

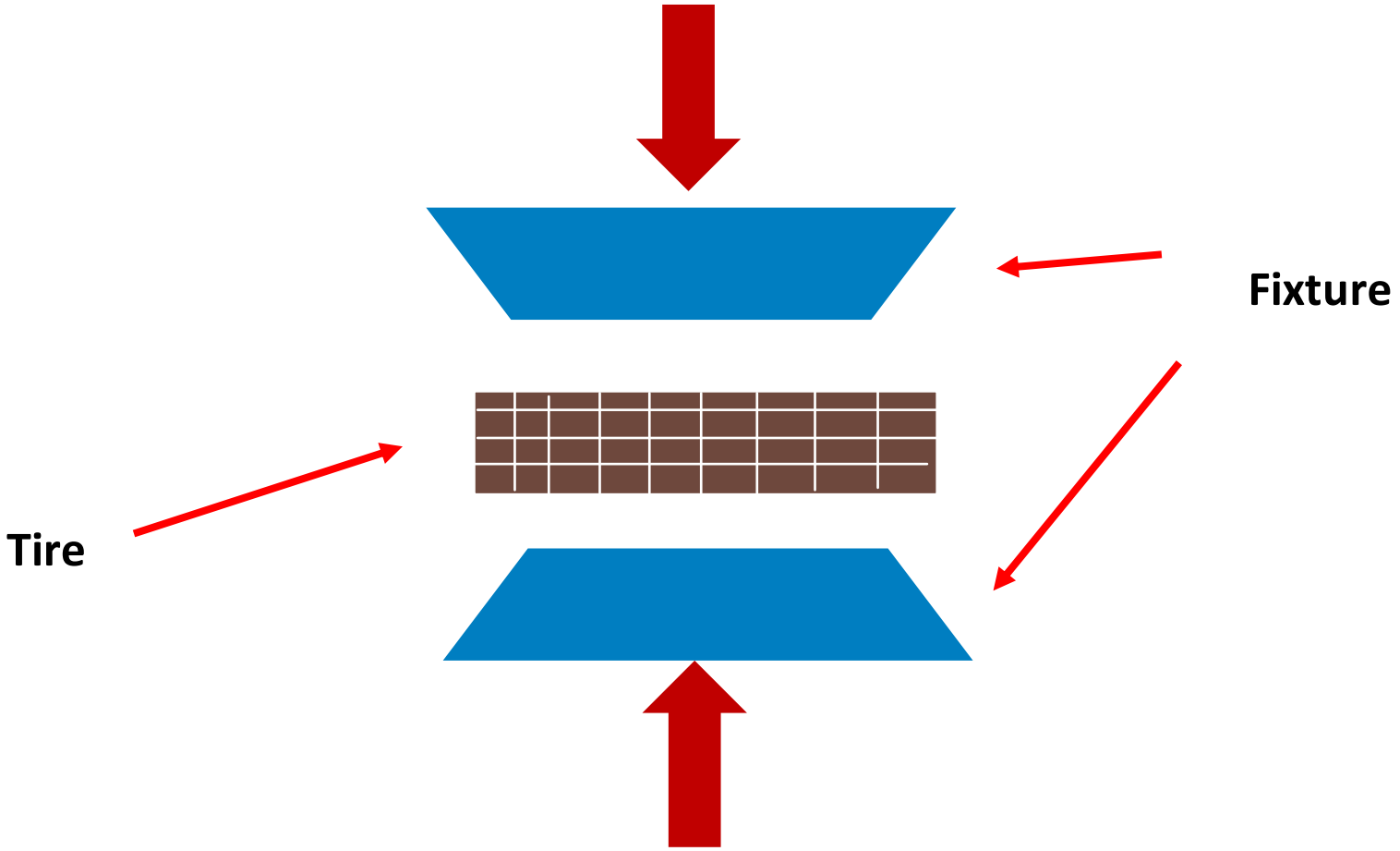
If Measurement System Does Not Pass Isograph, What do we do?



Find the BIG X in the measurement system.

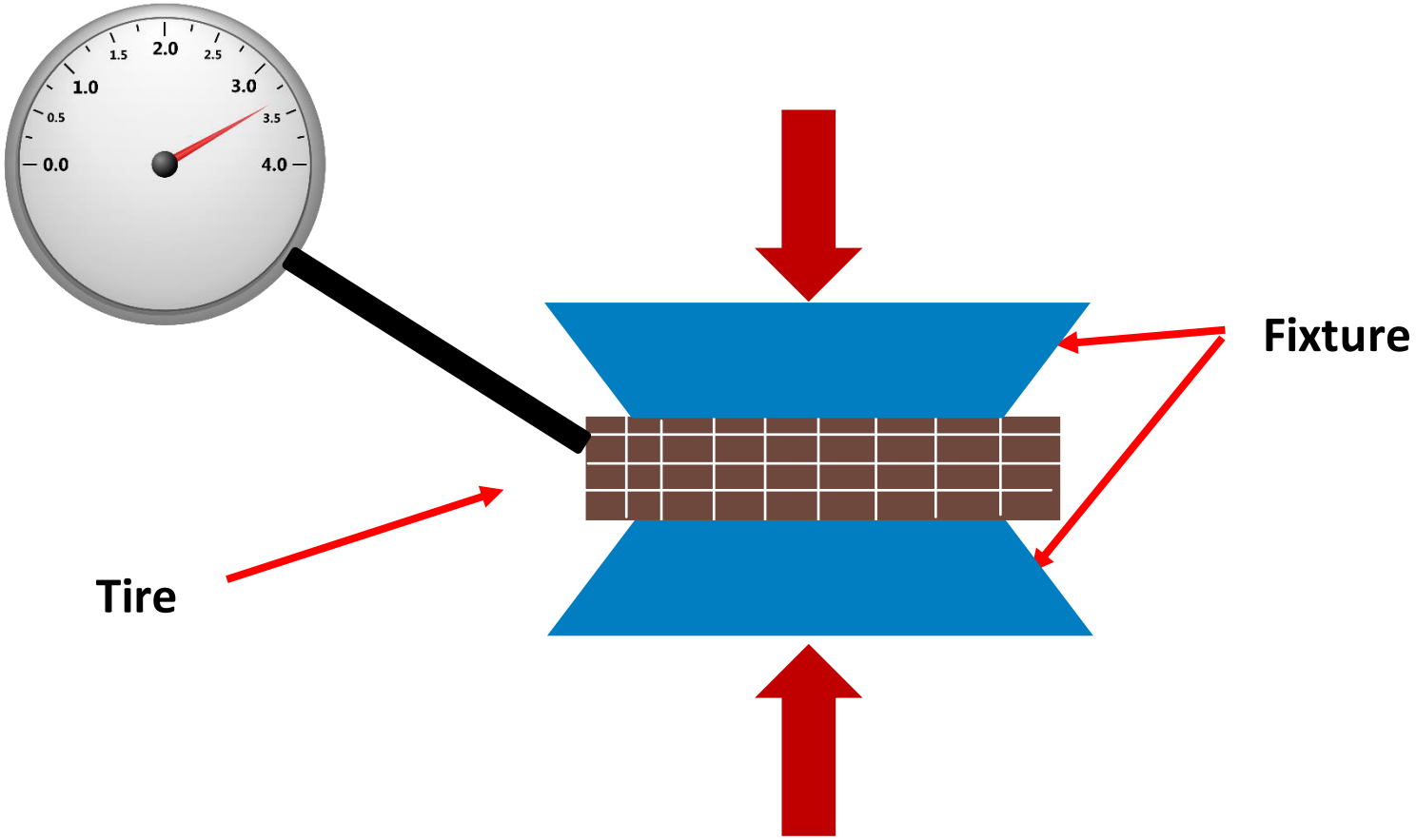
Tire - Imbalance

The tire to be tested is clamped in a fixture.



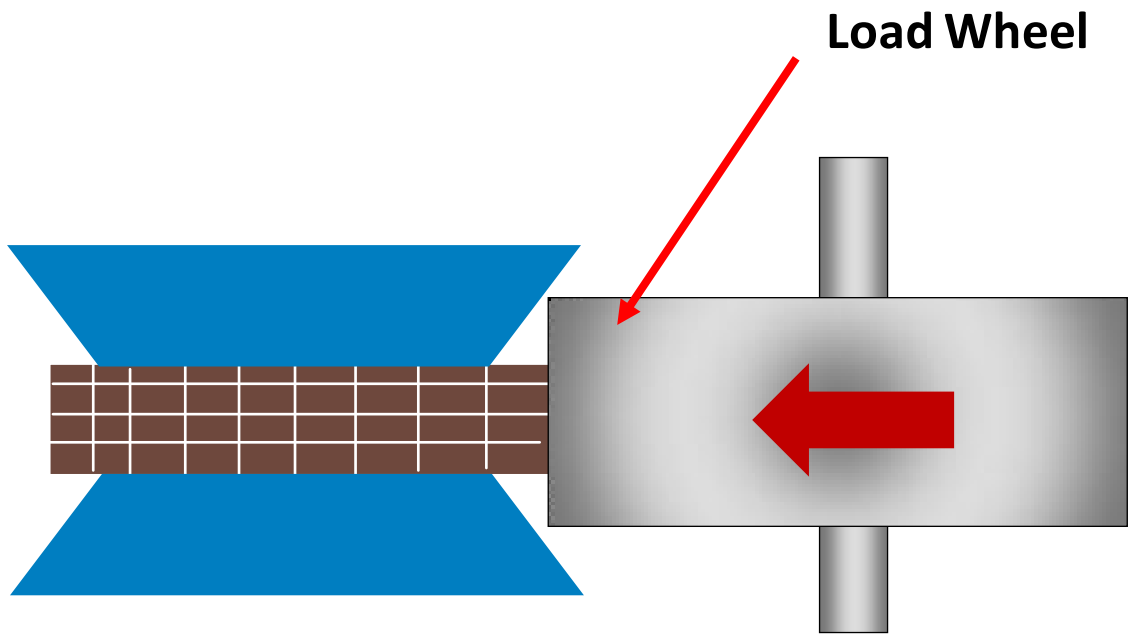
Tire - Imbalance

The tire is inflated to 3.3 bar cold.



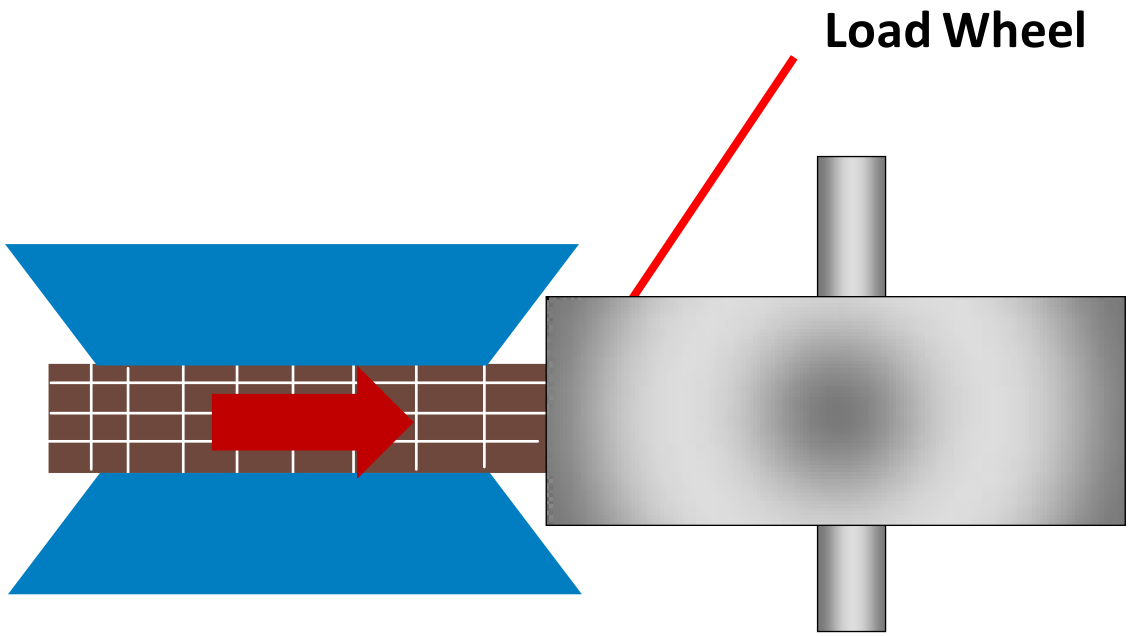
Tire - Imbalance

The load wheel is then pressed up against the tire.



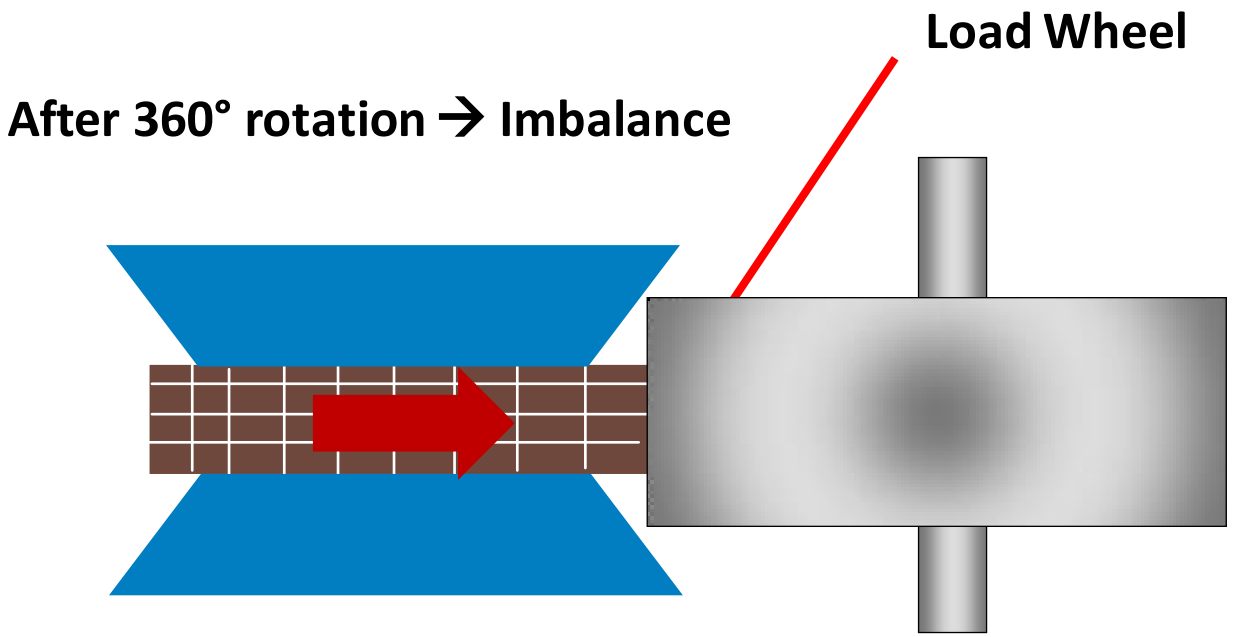
Tire - Imbalance

The tire is rotated at a speed of 30 km/h.



Tire - Imbalance

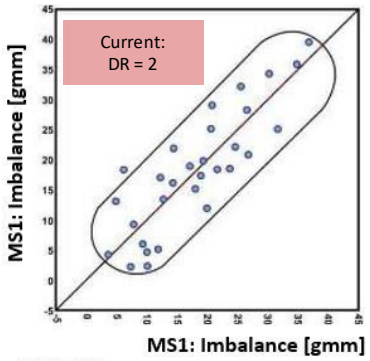
The imbalance on the load wheel measured in a full 360 degree rotation of the tire.



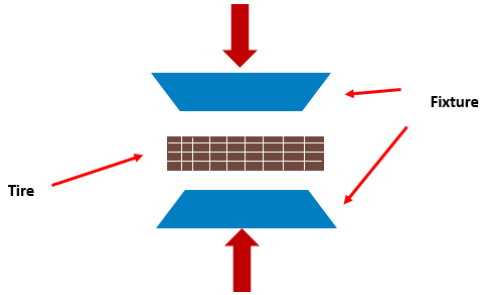
Tire - Imbalance

Technical problem: Poor repeatability in measuring the out of balance for tires.

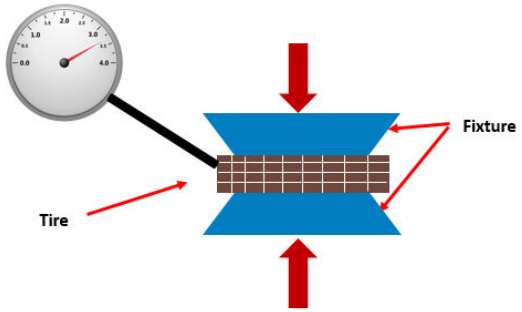
Situation:



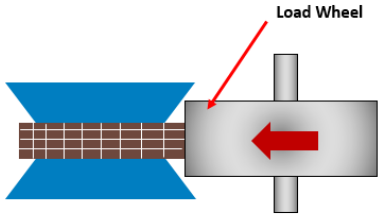
1) The tire to be tested is clamped in a fixture.



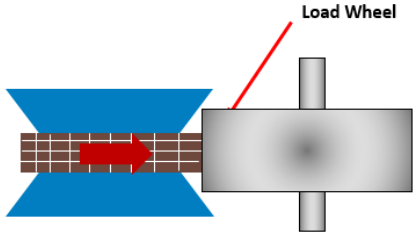
2) The tire is inflated to 3.3 bar cold.



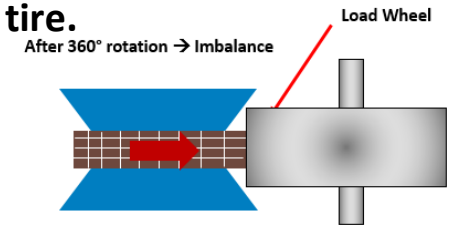
3) The load wheel is then pressed up against the tire.



4) The tire is rotated at a speed of 30 km/h.



5) The imbalance on the load wheel measured in a full 360 degree rotation of the tire.



In process step 5 the unbalance is measured.

Tire – Imbalance - Excercise

Measurement Procedure

Step:

1) Clamp tire in fixture.
2) Inflate tire.
3) Lock load wheel in place.
4) Rotate tire.
5) Read imbalance on computer screen.

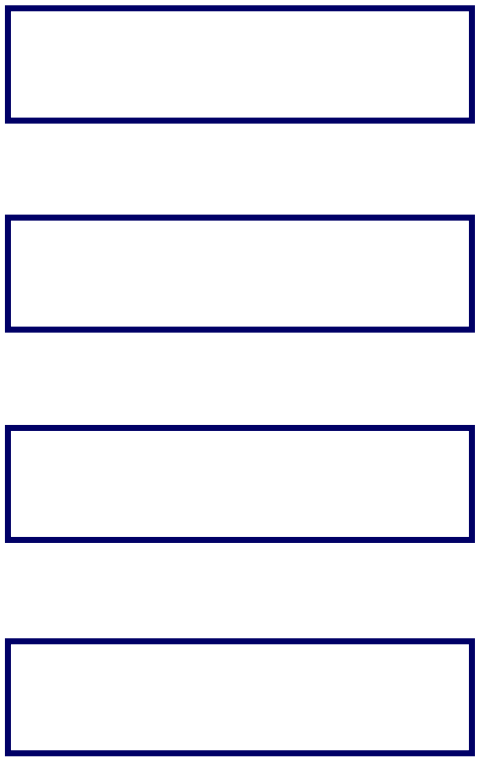
Assemble into your teams.

Develop a plan Strategy Diagram) to find the BIG X for Imbalance Measurement Error.

Capture your plan on a flip chart page.

You have 10 minutes to develop the plan.

Delta M – Plan - Excercise



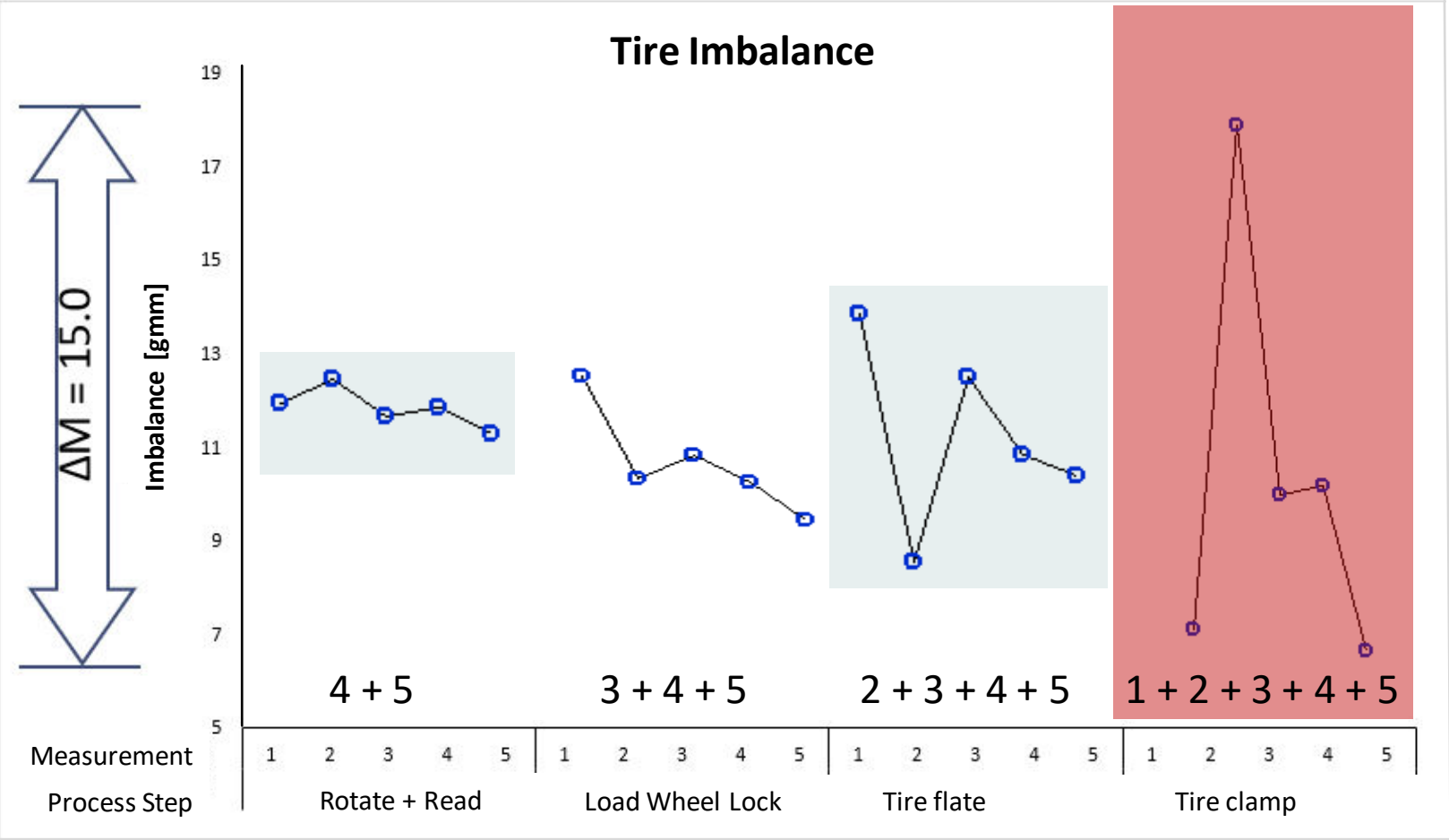
Delta M (ΔM) Multi-Vari - Answer

Strategy Diagram	Steps involved to create ΔM Multi-Vari
Reading to Reading Same Load Wheel Lock	**Repeat Steps 4 and 5 - five times.**
Load Wheel Lock to Load Wheel Lock Same Tire Inflate	**Repeat Steps 3, 4 and 5 - five times.**
Tire Inflate to Tire Inflate Same Tire Clamp	**Repeat Steps 2, 3, 4 and 5 - five times.**
Tire Clamp to Tire Clamp Same Tire	**Repeat Steps 1, 2, 3, 4 and 5 - five times.**

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Delta M (ΔM) Multi-Vari - Answer

Tire – Imbalance



Operations Search Y to X: Multi-Vari shows: Step 1) causes the largest variation in the output (Mone(Y)).

Delta M (ΔM) Multi-Vari - Answer

Strategy Diagram



Steps involved to create ΔM Multi-Vari

Minimal Contrast

Minimal Contrast

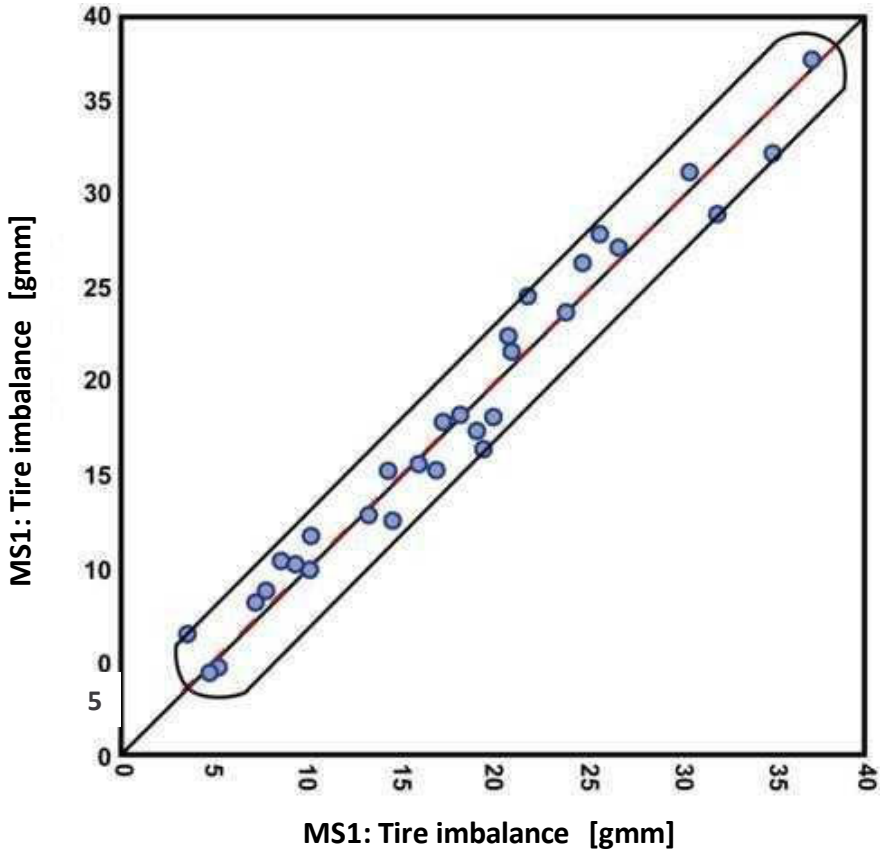
Minimal Contrast

BIG X lives in Tire Clamping

Further investigation revealed the mating halves of the clamping fixture were out of balance and not aligned. Fixture was rebalanced and aligned. A successful Isograph was completed for the measurement system.

Tire - Imbalance

New Isograph (After Improvements)



Delta P = 34.2
Delta M = 4.4
Discrimination Ratio = 7.7

Thirty different tires measured after the fixture was repaired on the original measurement system.

Passed Isograph.

Fixing ΔM for a Variable Mone(Y) - Guidelines

When a measurement system does not pass Isograph, check the following first:

- Has the full range of ΔP been captured in the Isograph?
- Does the measurement system have sufficient resolution to provide an acceptable discrimination ratio?
- Is the gage in need of calibration?

If the above steps did not improve the Isograph discrimination ratio to an acceptable level, execute solving a ΔM problem:

1. Map out the measurement process by tracing the steps starting from the last step back to the first.
2. Take a minimum of five readings for each measurement step and plot the result in a Multi-Vari.
3. Determine the measurement error in each step, realizing that the Multi-Vari shows the cumulative measurement error including the preceding steps.
4. The step with the greatest variation contains the BIG X for the measurement system.

Module 2 Summary

Effective Measurement Systems

- Isograph is a simple, graphical technique to evaluate a measurement system's ability to find the BIG X.
- Isograph provides estimates of ΔP and ΔM. ΔP represents product variation and ΔM represents measurement error.
- Isograph can be used to evaluate the repeatability of:
  - Measurement to measurement, same operator
  - Operator to operator, same gage
  - Gage to gage, same time
- Bias between the first and second readings occurs as a result of either the parts or the measurement system changing over time.
- When you fail Isograph, there are strategies to converge to the BIG X of the measurement process.

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3 Feature Strategies

Module 3 Objectives

Feature Strategies

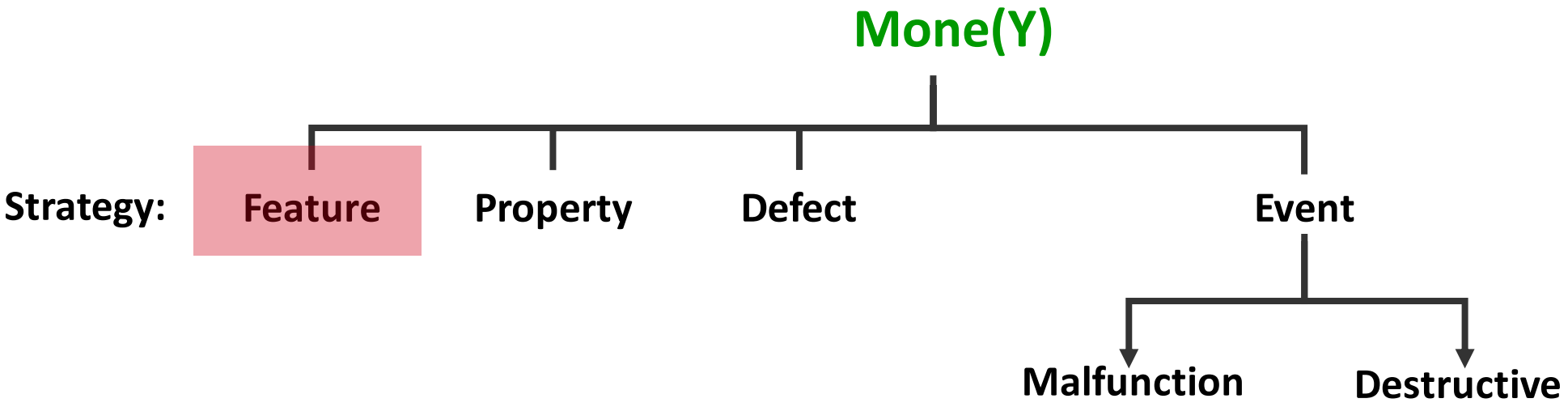
Understand:

- How to plan and execute a feature strategy
- How to recognize the structure of a system

Be able to:

- Develop a Feature Strategy Diagram
- Conduct a Multi-Vari study
- Use group comparison to identify a BIG X suspect
- Document convergence on a Solution Tree

Different Strategies for Different Mone(Y)s



Explanation: Physical dimensions of a product that describe shape; must relate to the physics of the processes that create the shape.

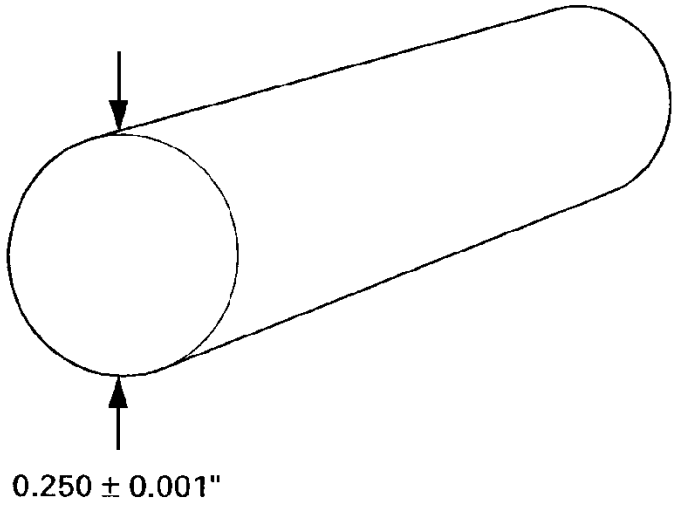
Tool: **Strategy Diagrams**

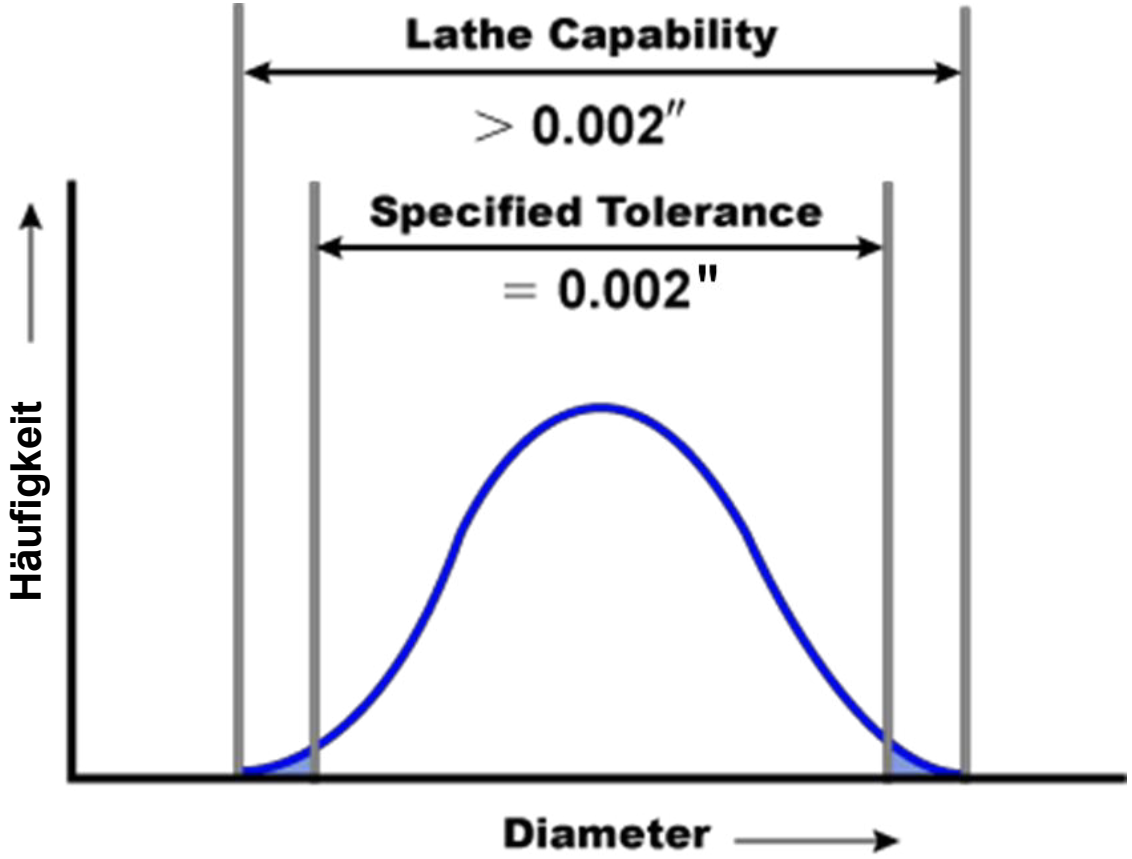
Strategy Diagrams are a way to document possible contrasts for the Mone(Y) that you can leverage during your project.

Rotor Shaft Case

The manufacturing process is not capable of holding the diameter to its required tolerance.

What would you do next?

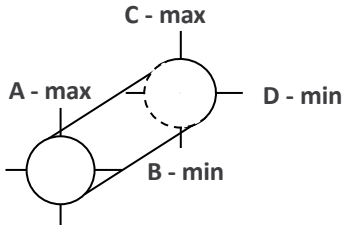


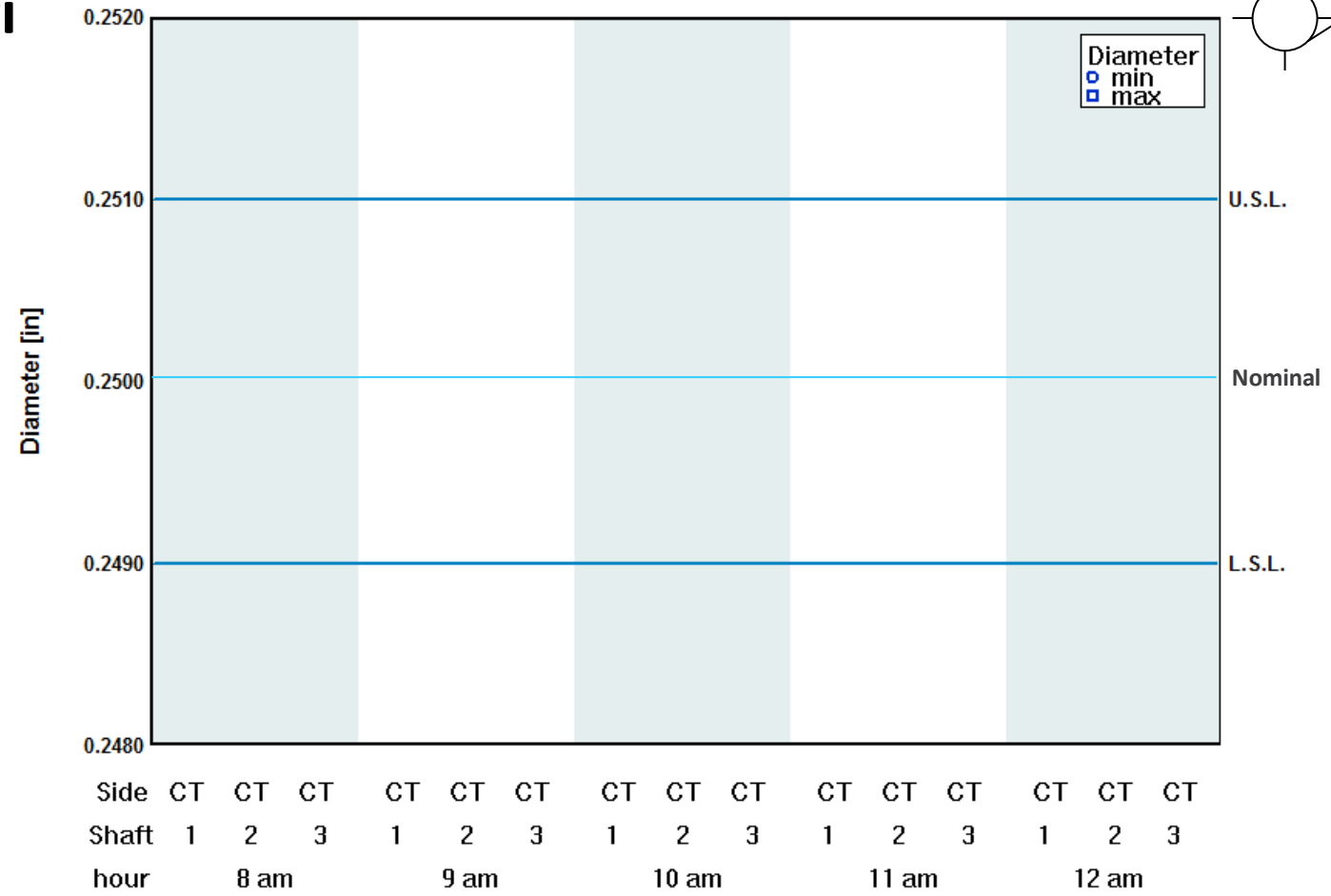


Plan of Attack

Efficiently Collecting Critical Information

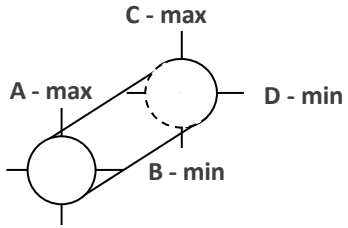
Multi-Vari

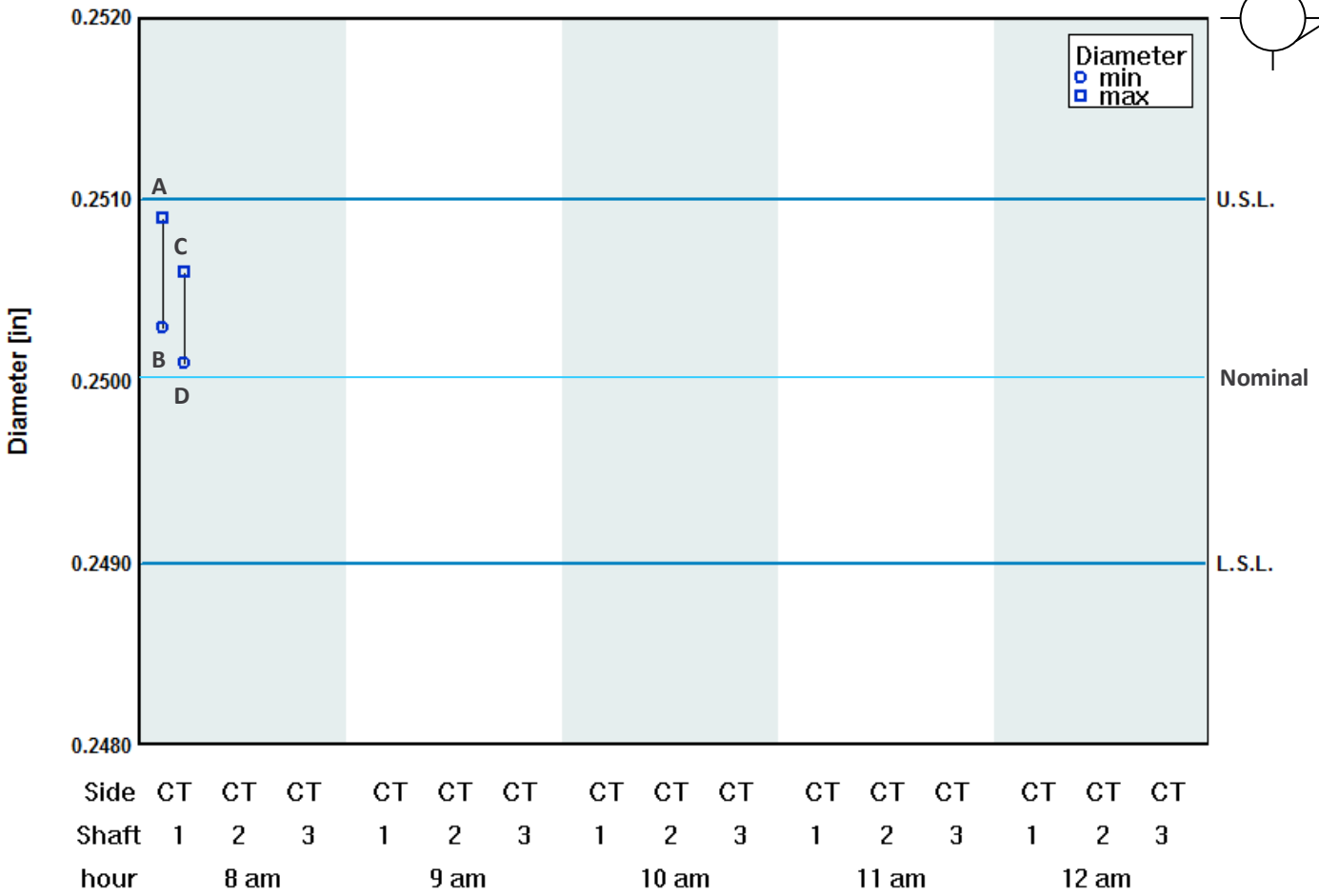




Multi-Vari

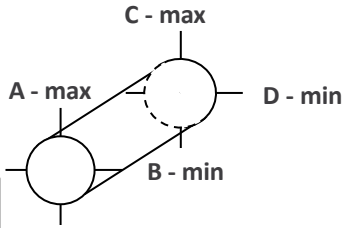
One Part, Measured Four Places

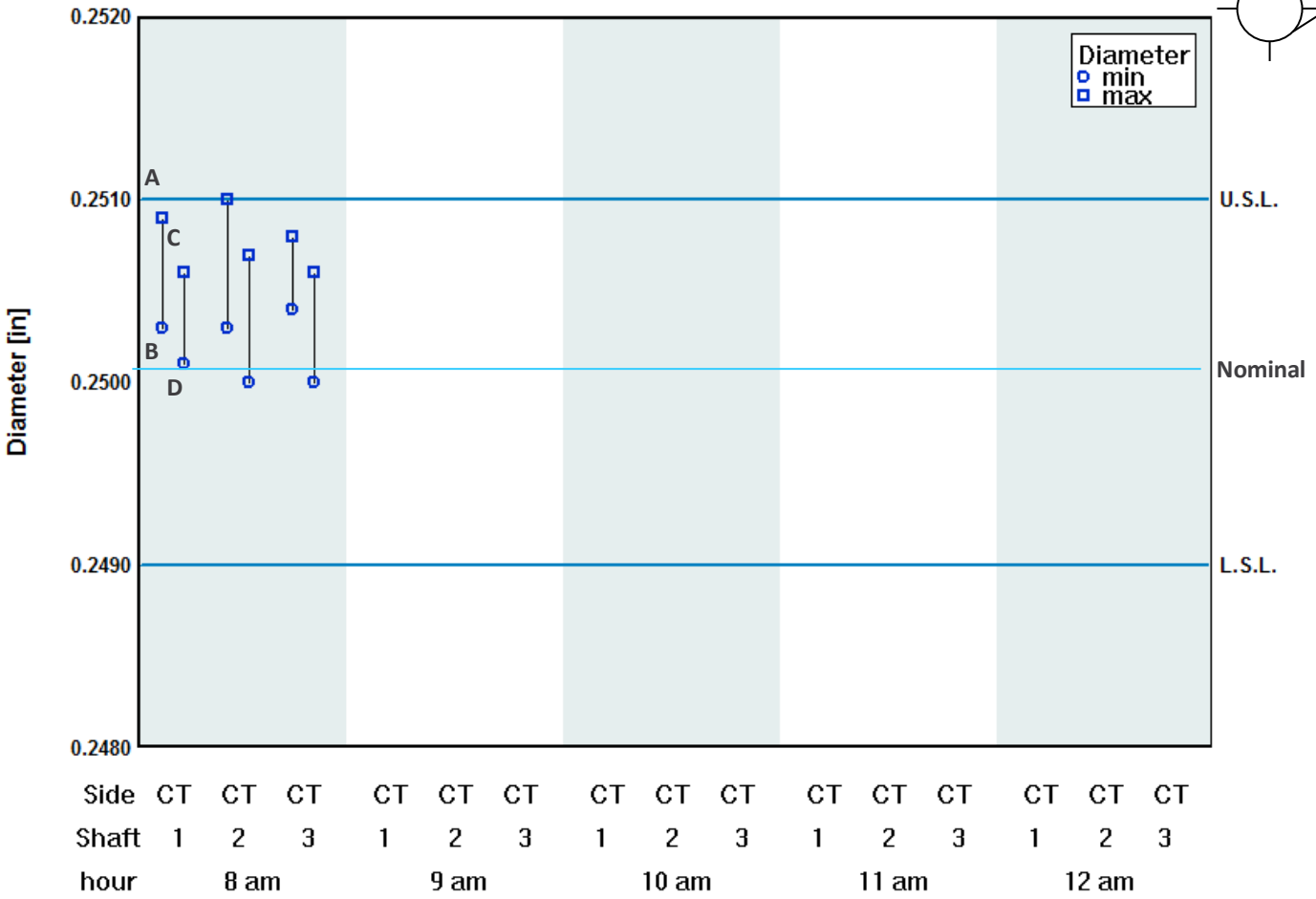




Multi-Vari

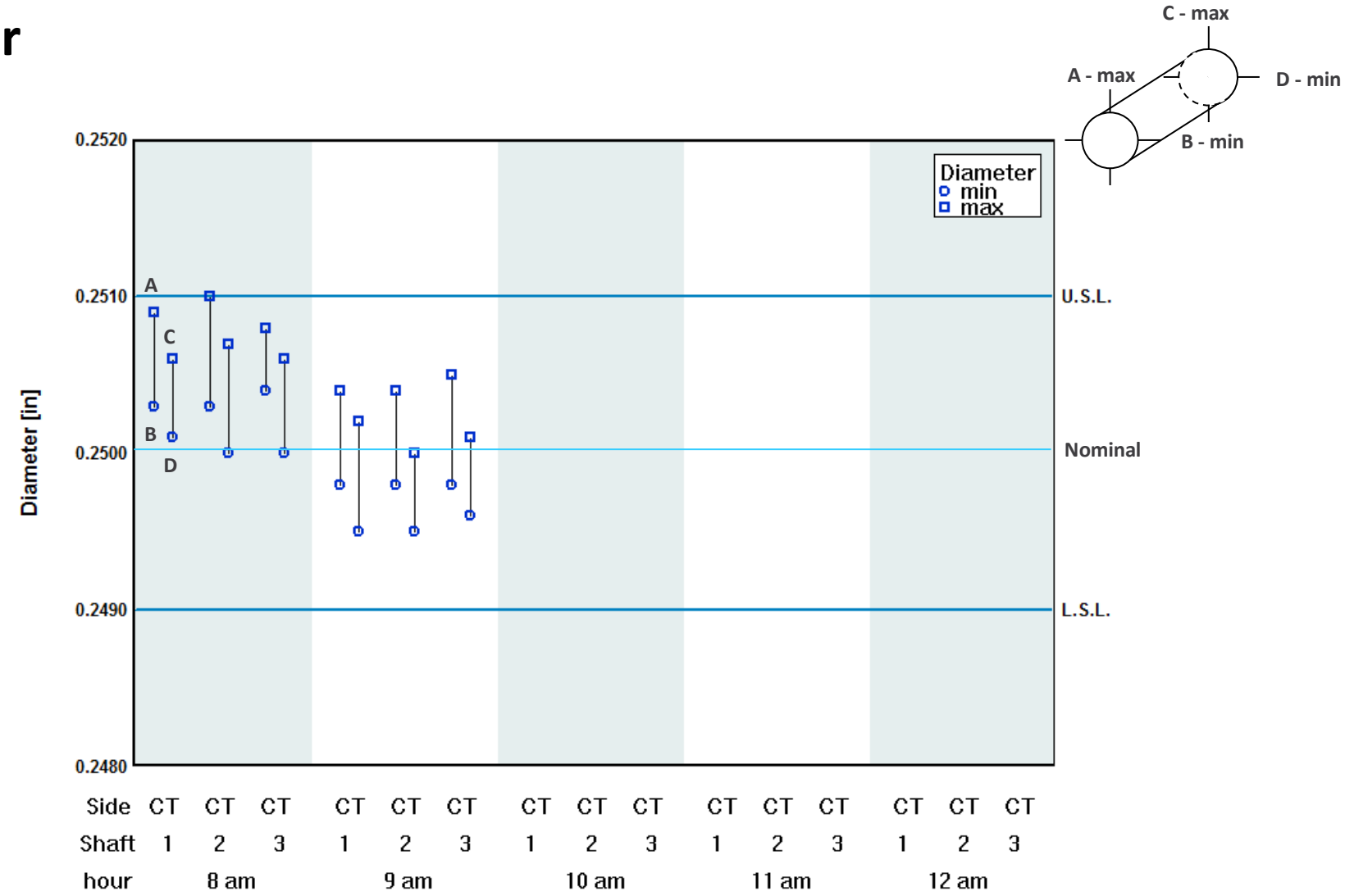
Two More Parts





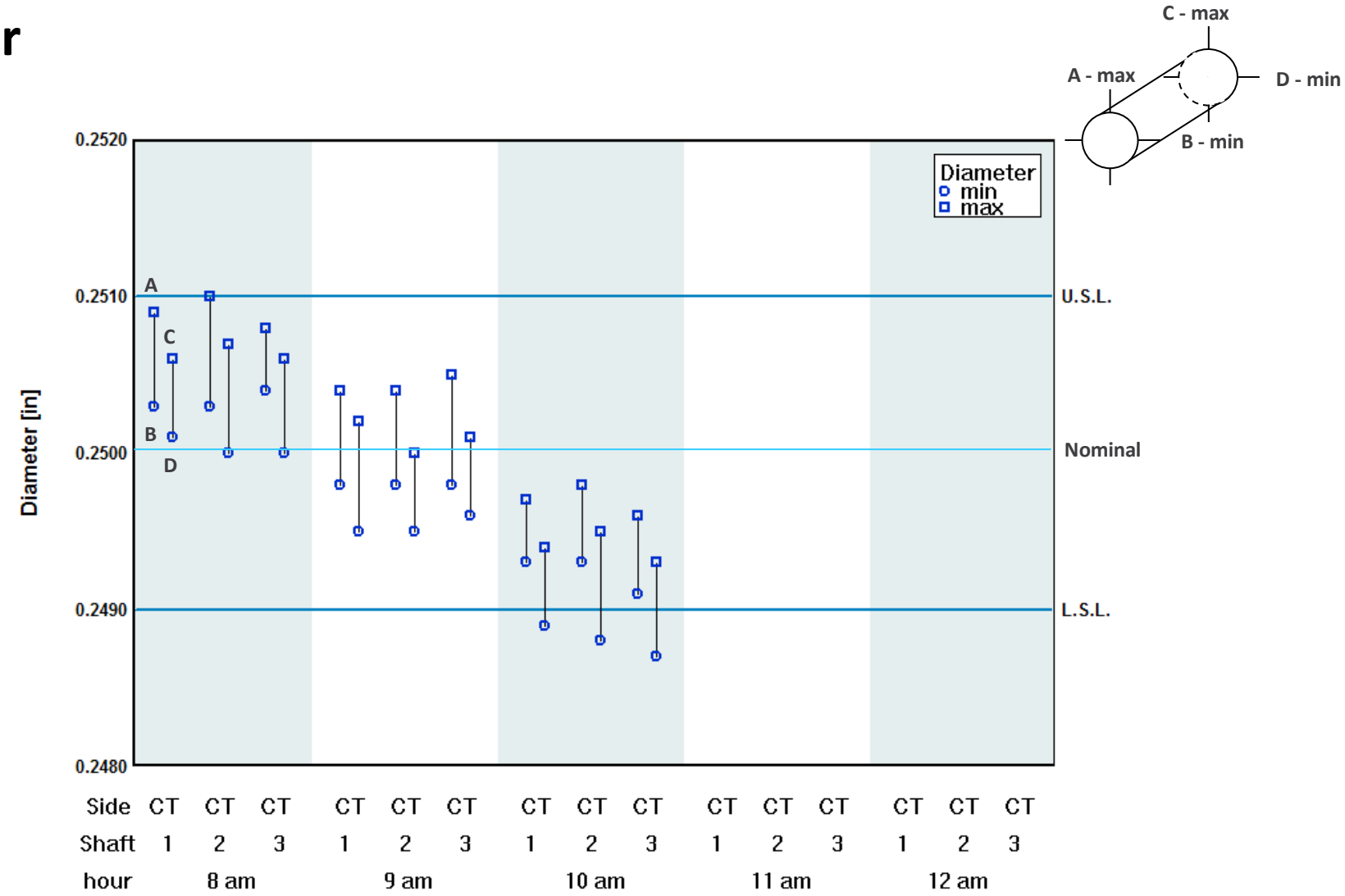
Multi-Vari

Next Hour



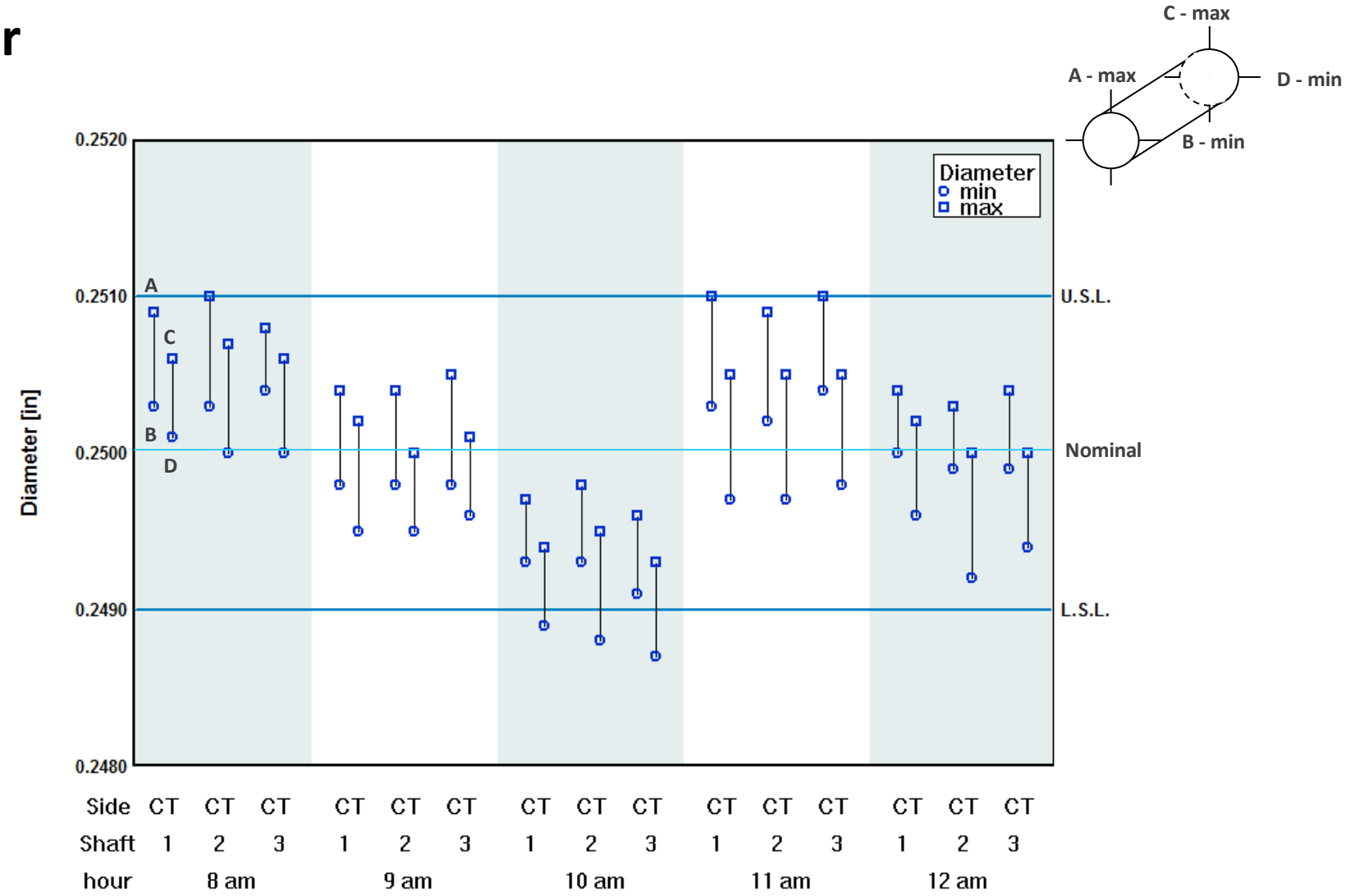
Multi-Vari

Next Hour



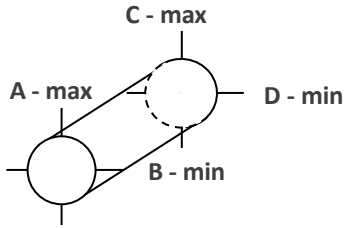
Multi-Vari

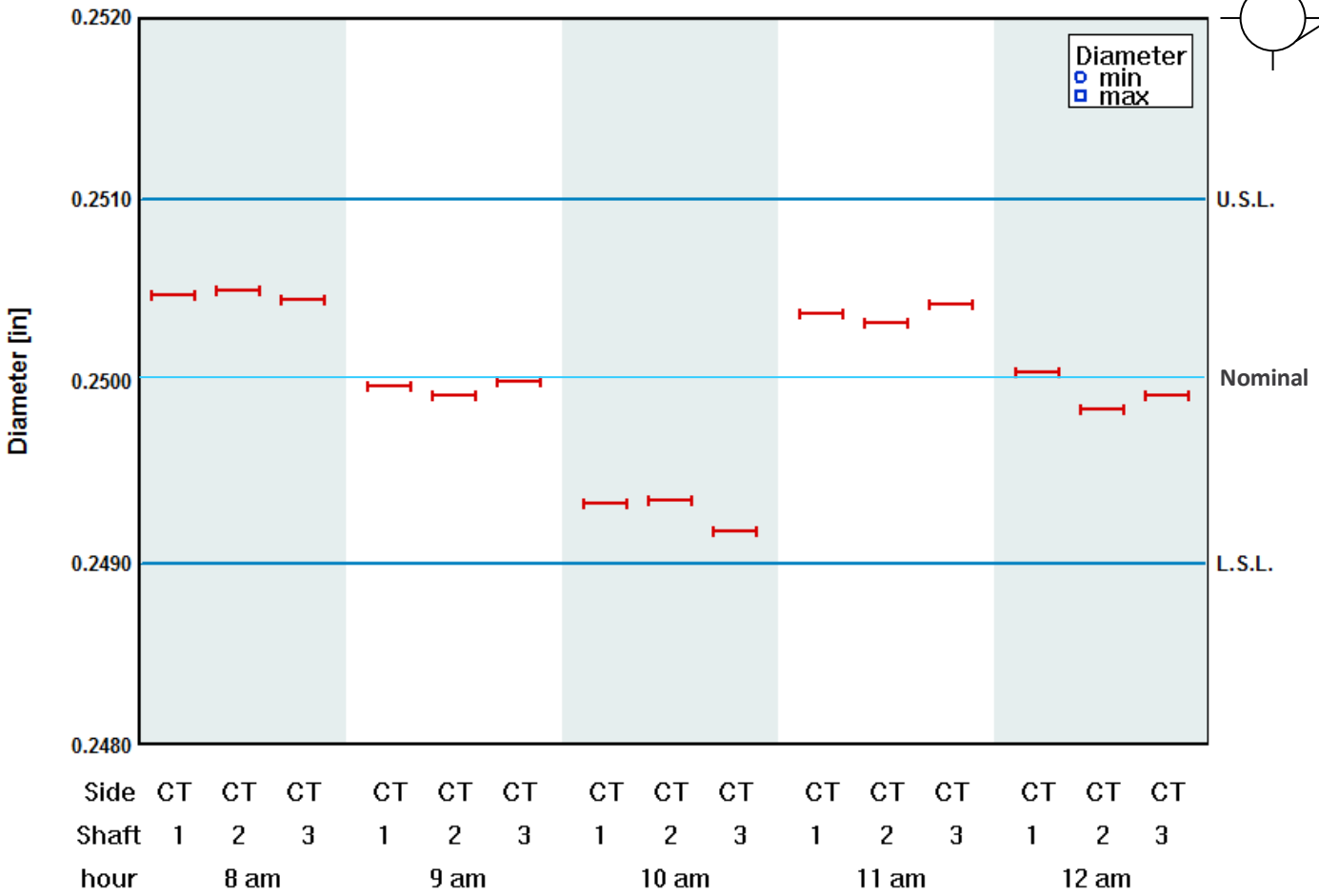
Next Hour



Multi-Vari

BIG X Leverage

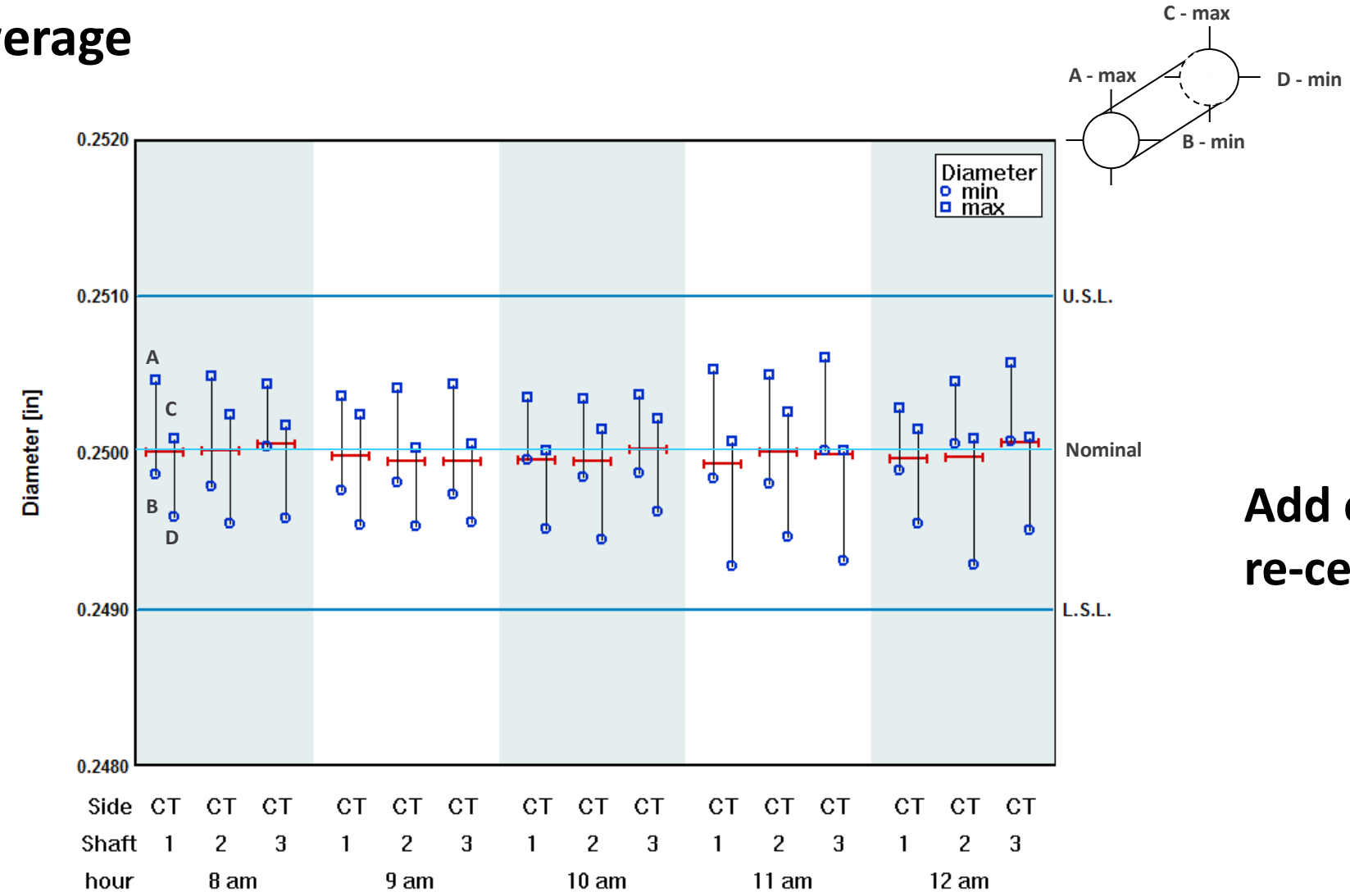




Eliminate out of round and taper.

Multi-Vari

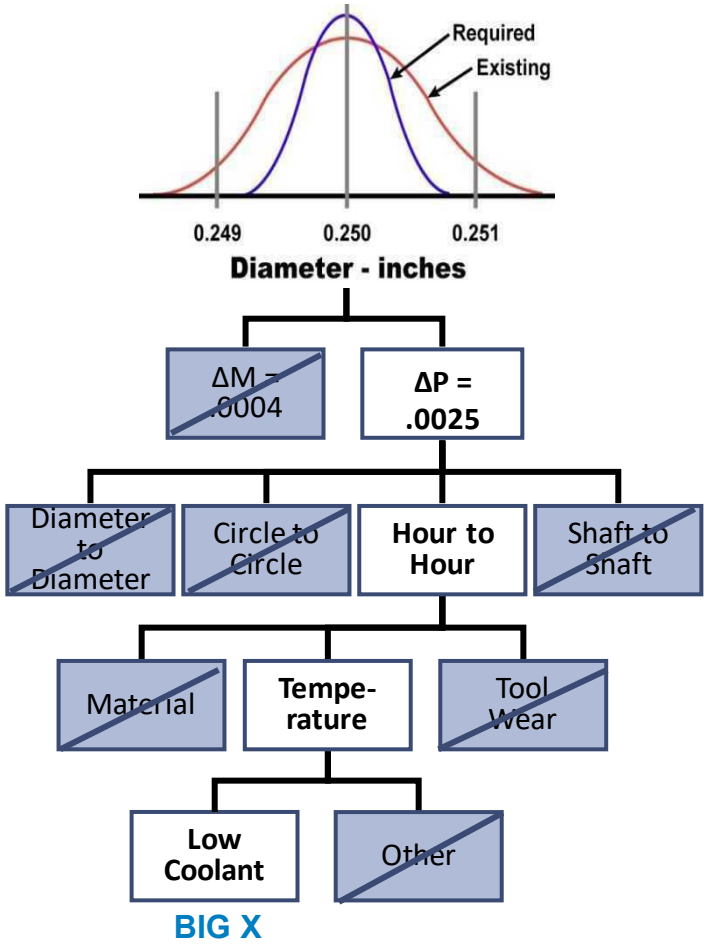
BIG X Leverage



Add coolant and re-center process.

Progressive Search

Solution Tree



Mone(Y)

BIG X

Rationale:

- Isograph
- Multi-Vari
- Trend- shift-trend pattern.
- Wrong direction for tool wear.
- Shift follows break.
- Observation.
- Filling up coolant reservoir corrects hour to hour variation.

Rotating Machinery

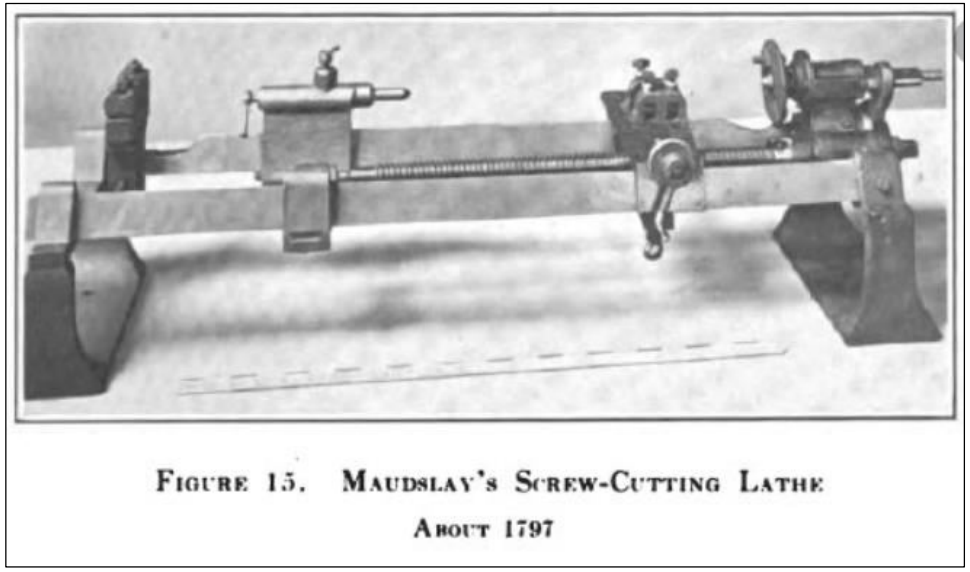
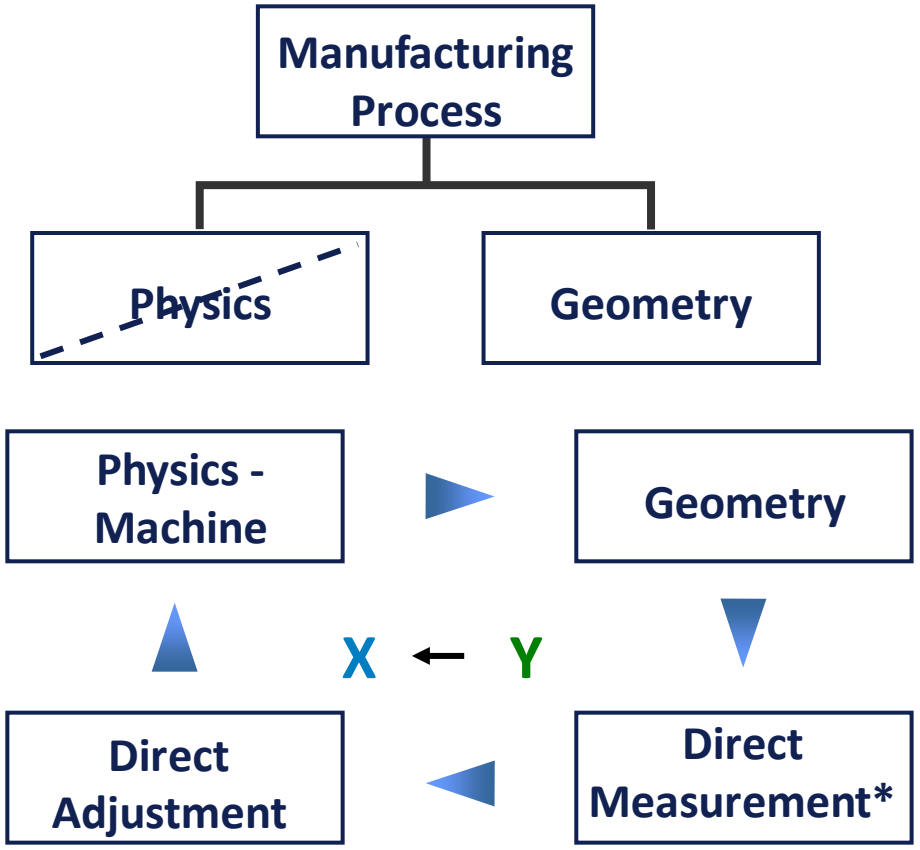


FIGURE 15. MAUDSLAY'S SCREW-CUTTING LATHE
ABOUT 1797



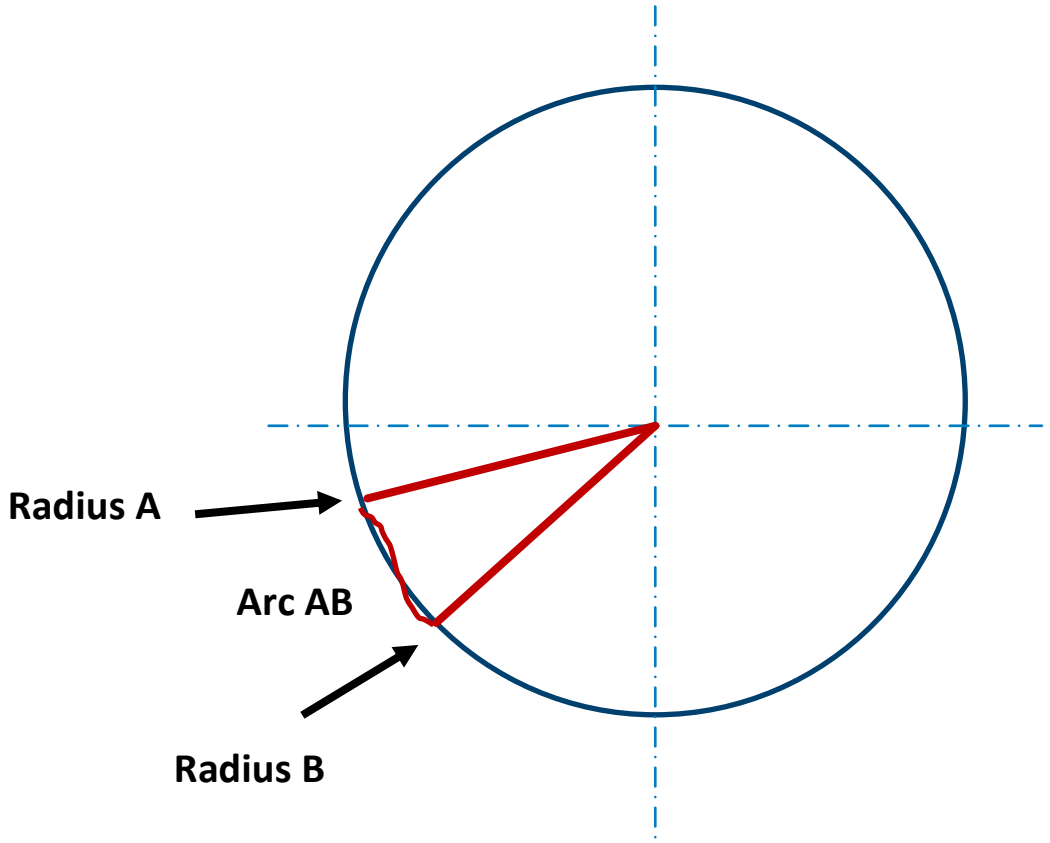
*Must be direct measurement of the physics of the machine as it creates the geometry.

Feature Strategy Diagram - Rotating Machinery

Family of Variation

Radius to Radius

Same Arc



Feature Strategy Diagram - Rotating Machinery

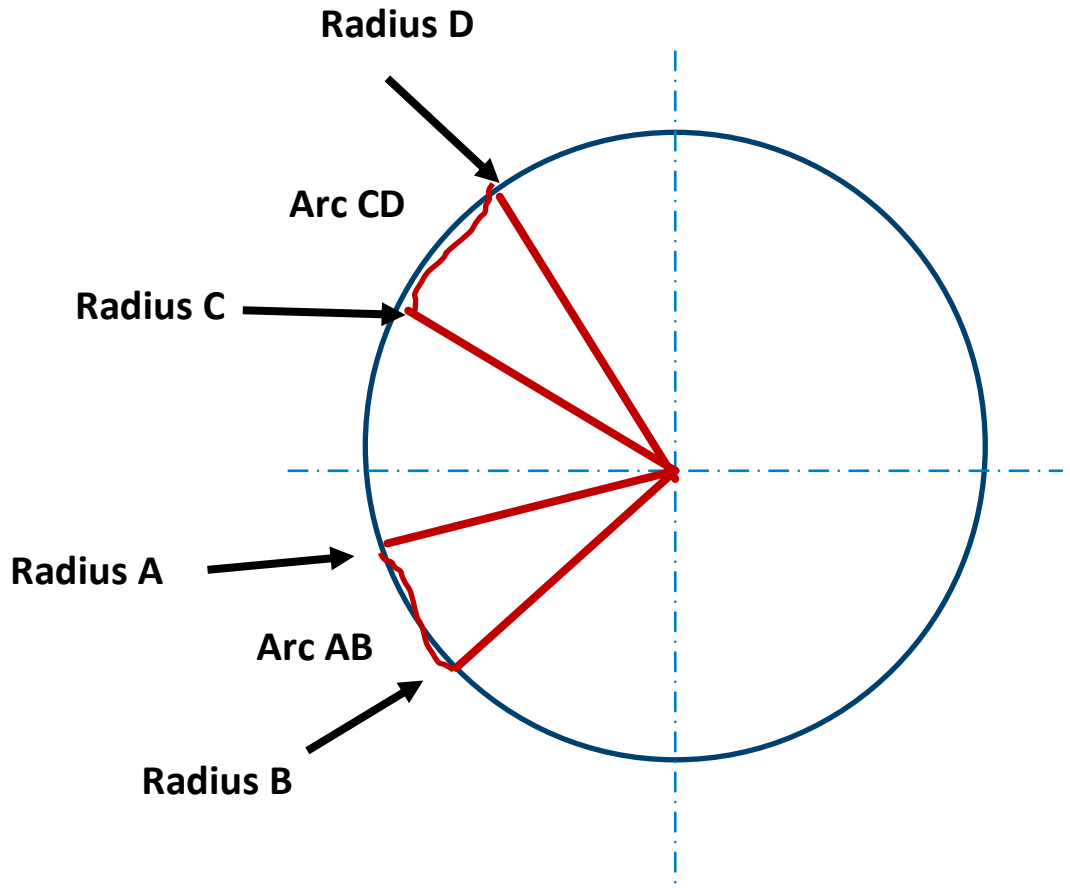
Family of Variation

Radius to Radius

Same Arc

Arc to Arc

Same Circle



Feature Strategy Diagram - Rotating Machinery

Family of Variation

Radius to Radius

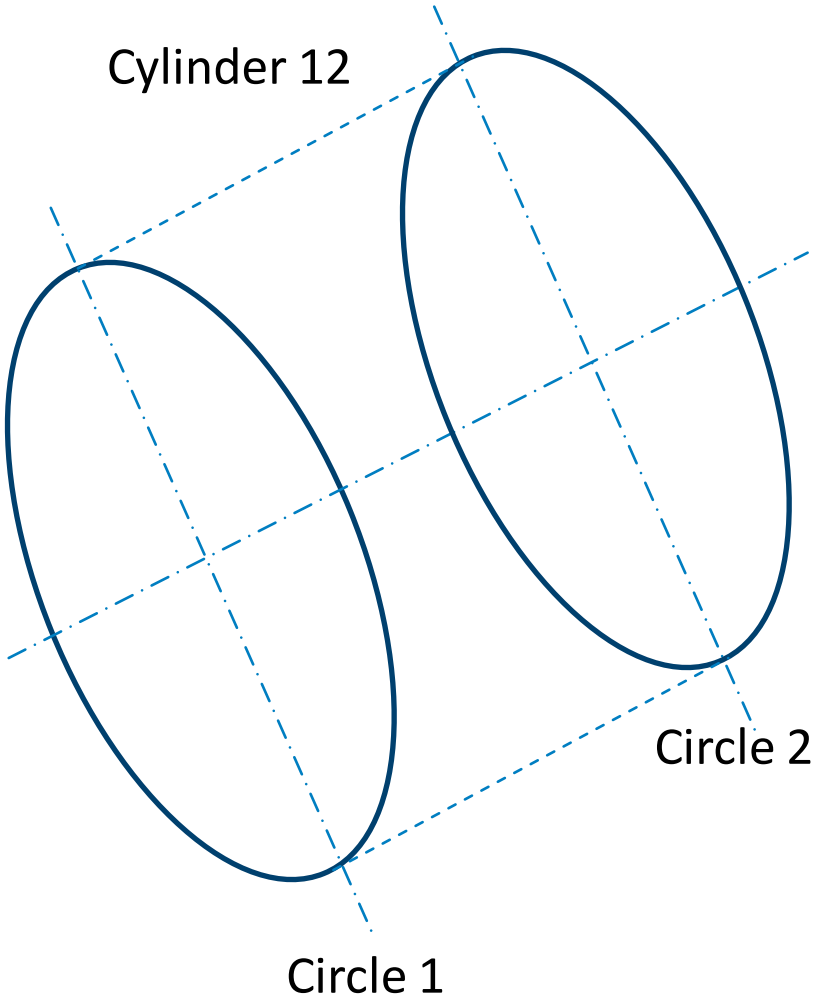
Same Arc

Arc to Arc

Same Circle

Circle to Circle

Same Cylinder



Feature Strategy Diagram - Rotating Machinery

Family of Variation

Radius to Radius

Same Arc

Arc to Arc

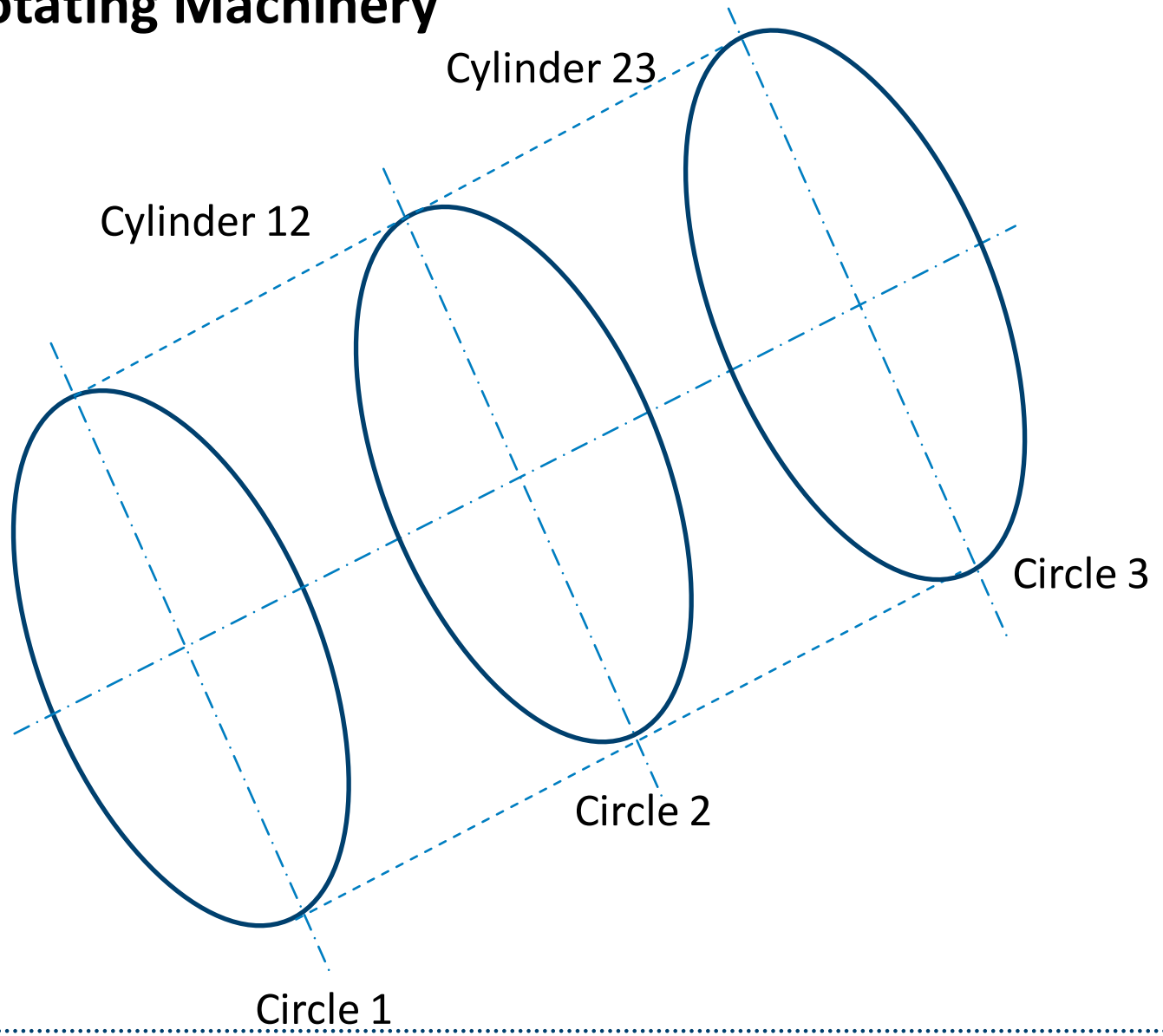
Same Circle

Circle to Circle

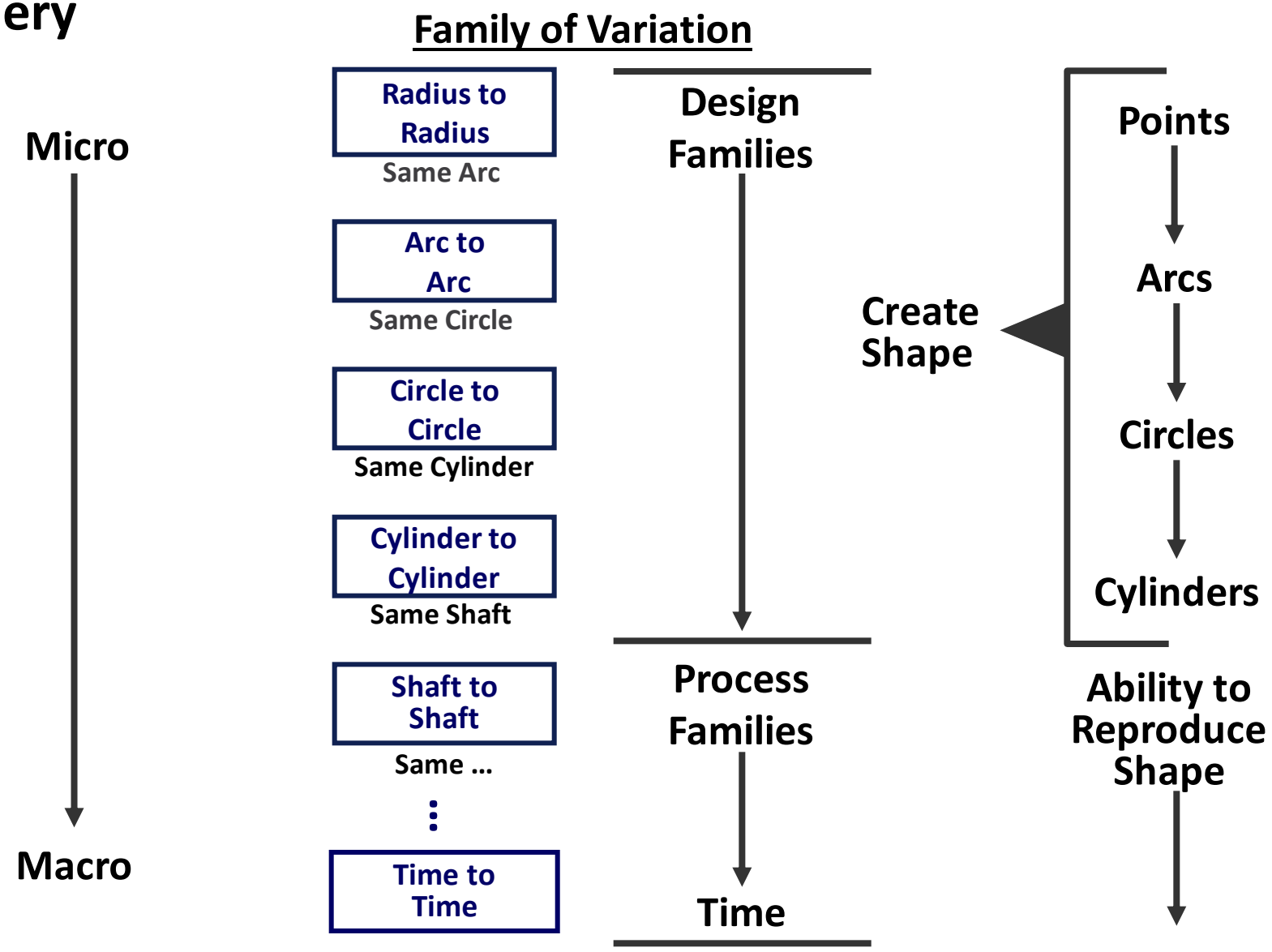
Same Cylinder

Cylinder to Cylinder

Same Shaft

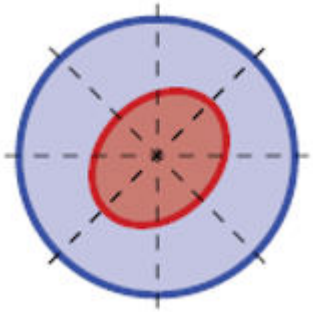
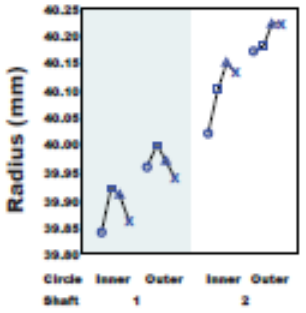


Rotating Machinery



Feature - Radius Strategy Diagram - Template

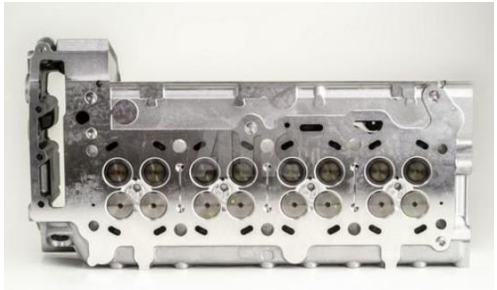
Mone(Y) = [unit]

Family of Variation, or Contrast		***Diagram***	**Design Families** ***What Is Meant by Each Contrast***
Radius to Radius Same Arc	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Rotating machinery create radii. Most of the time using diameter as a Mone(Y) is sufficient. Consider using Radius as a Mone(Y) if you get stuck.
Arc to Arc Same Circle	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Is the observed BOB / WOW contrast in the Size, or the Shape? Arc to Arc contrast would look out-of-round, or egg-shaped.
Circle to Circle Same Cylinder	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Does the bore, or shaft have taper? Is it shaped like a barrel?
Cylinder to Cylinder Same Piece	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Is there contrast bore to bore, or hole to hole, within the same piece? Do they have the same size and shape?

End of Line Test Failures

Compressor Leaks

- Compressors are failing an end of line leak test.
- A component search has isolated the BIG X to the cylinder head.
- A detailed comparison of BOB and WOW cylinder heads reveals that the BIG X feature is cylinder head flatness.
- The cylinder head surface is machined at one of several milling stations.
- A team has been assigned to find the BIG X within the machining process that controls flatness.
- Sorting costs amount to 500 k€ thousand.



4Ps for Cylinder Head Flatness Project

Product or **P**rocess that has the problem	Compressor cylinder head
Problem description	Poor flatness
Place where the problem is discovered	After milling
How is the **P**ain measured (metric)	Compressor leak failures at end of line test and 500 k€ sorting costs

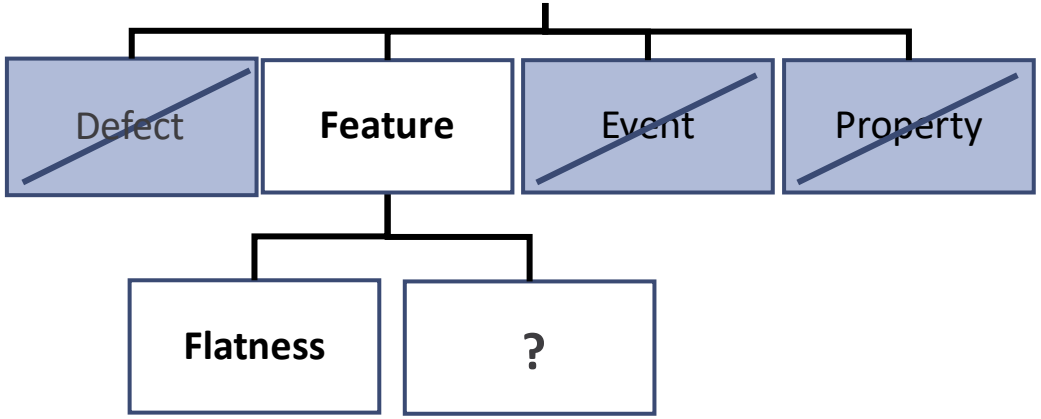
Focused Project Statement:

"Find and Control the RED X that drives (compressor cylinder head) (poor flatness) discovered (after milling) to prevent (compressor leak failures at end of line test and 500 k€ sorting costs)."

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Project Definition Tree - Cylinder Head Flatness

Project Statement: Find and control the BIG X that drives compressor cylinder head poor flatness discovered after milling to prevent compressor leak failures at end of line test and 500 k€ sorting costs.

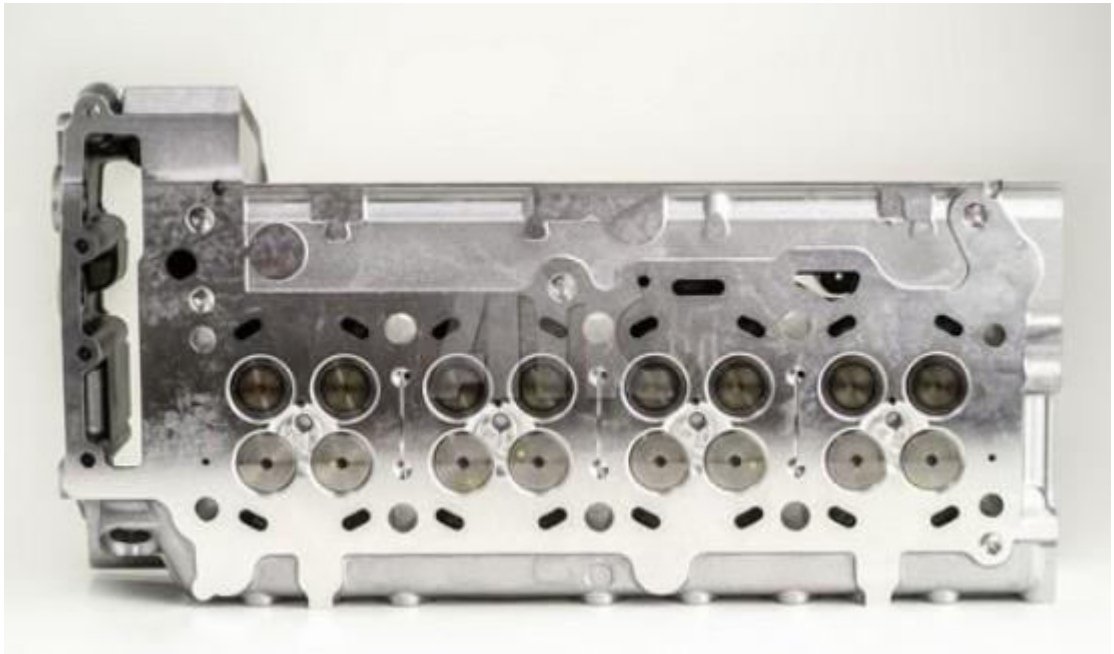


Rationale:

- Flatness is a shape characteristic.
- Will flatness reveal details of the shape?

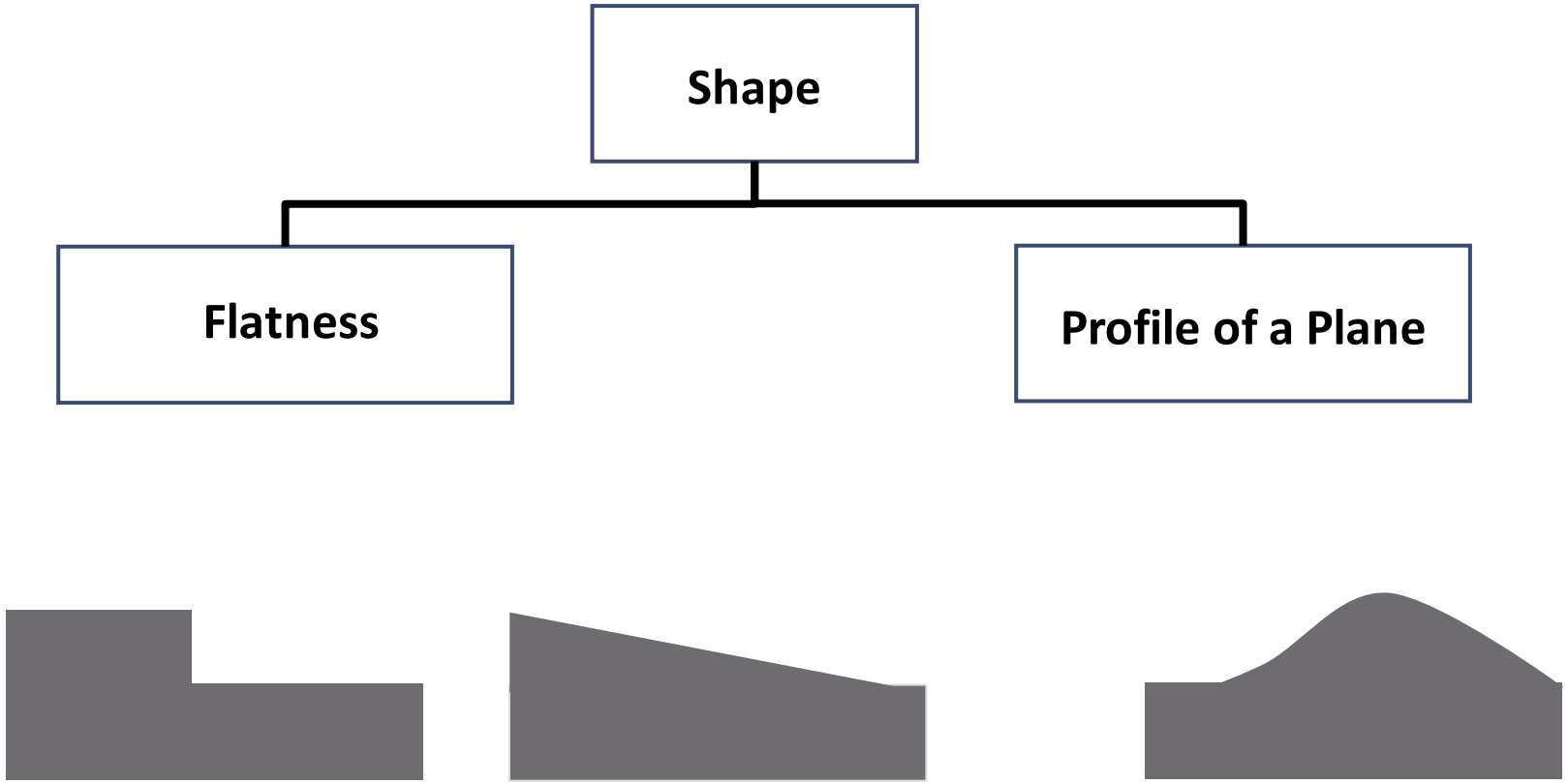
Creating Other Types of Geometry

Cylinder heads are rejected for flatness driving scrap.

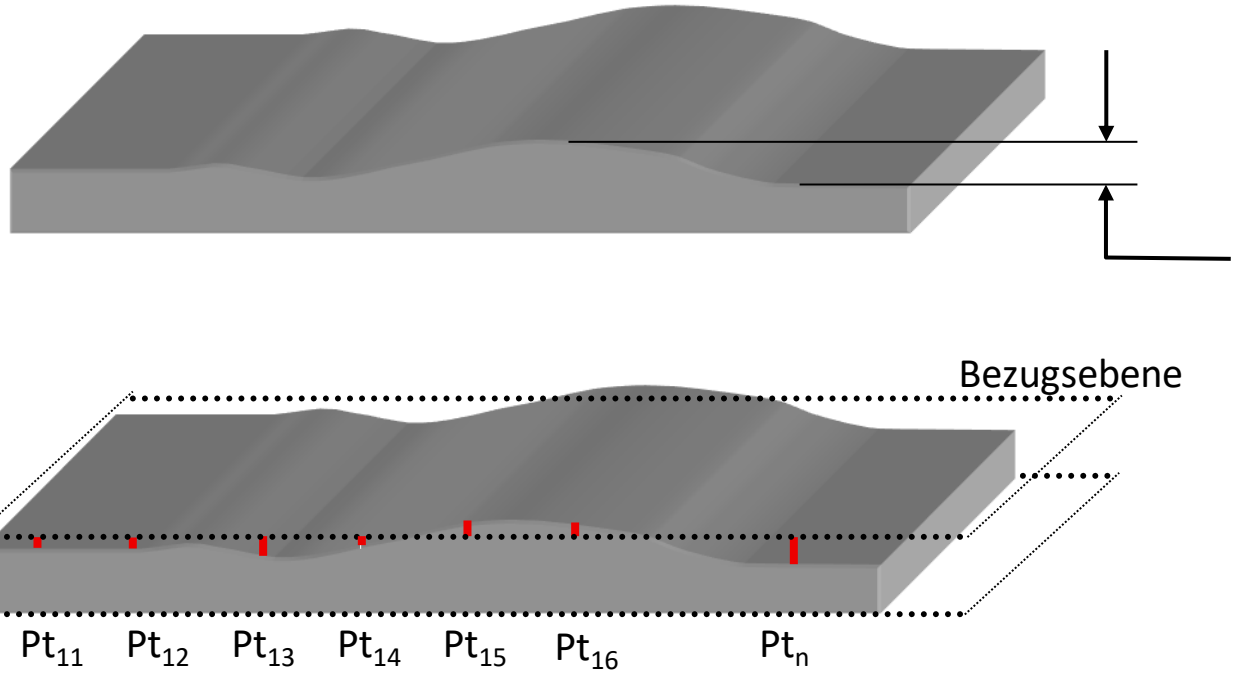


What would make a good Mone(Y) for this problem?

Profile or Flatness



Finding an Effective Mone(Y)



Flatness result is one number.

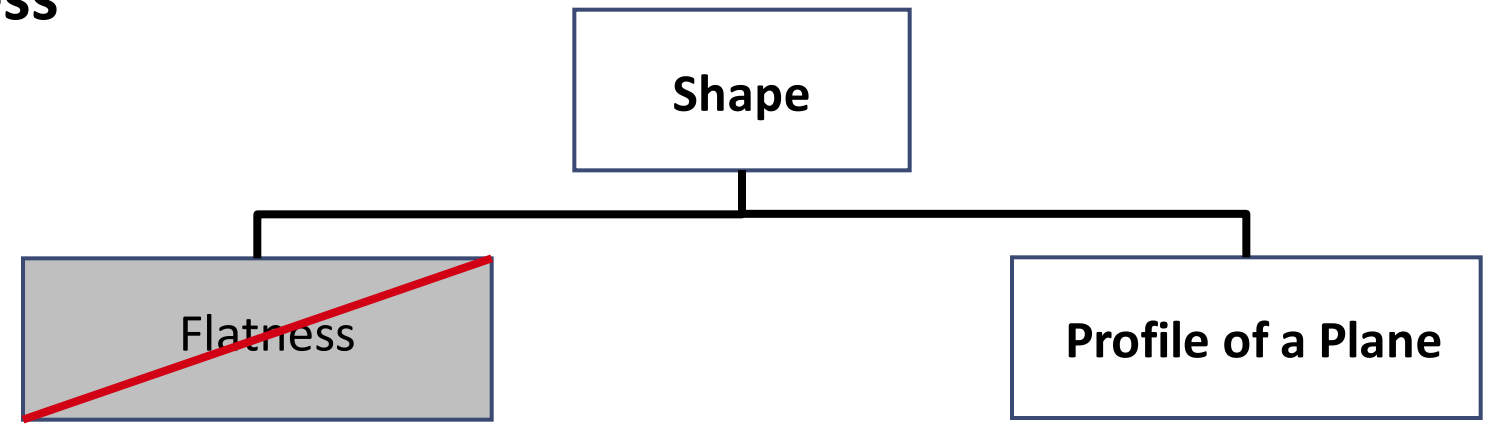
It is used for a pass/fail decision.

Height measurement from the datum describes the shape (profile).

Flatness is the range of height readings.

Profile of height with respect to a machining datum reveals the nature of the physics of creating geometry.

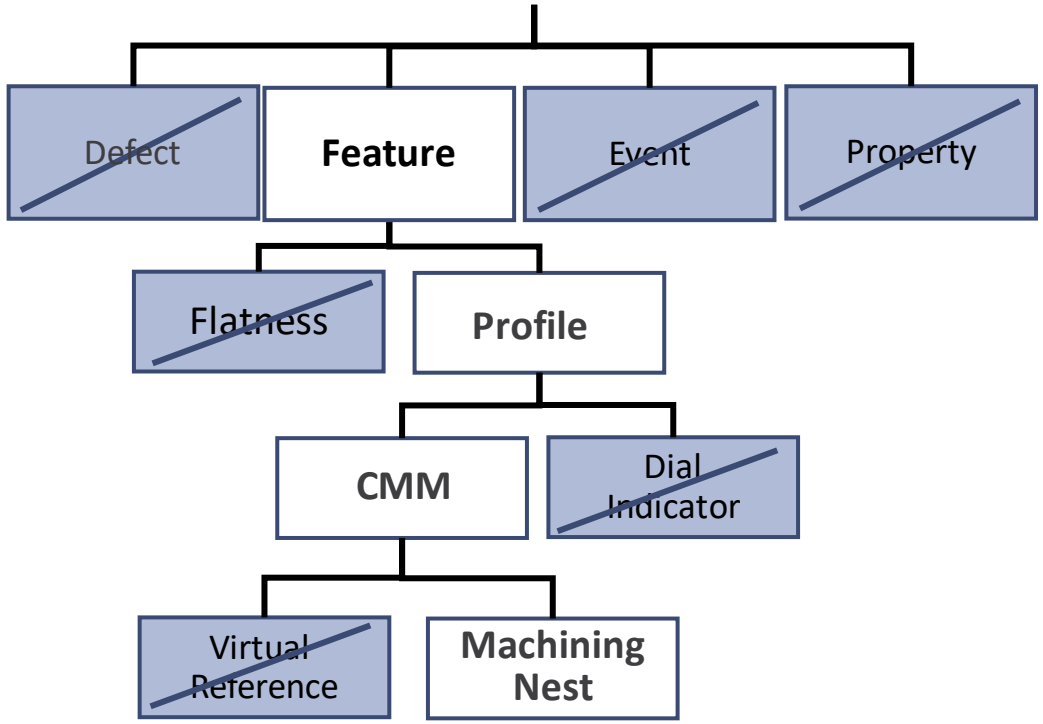
Profile or Flatness

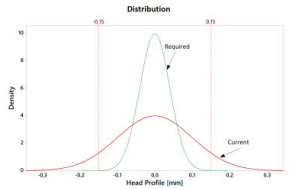




Project Definition Tree - Cylinder Head Flatness

Project Statement: Find and control the BIG X that drives compressor cylinder head poor flatness discovered after milling to prevent compressor leak failures at end of line test and 500 k€ sorting costs.



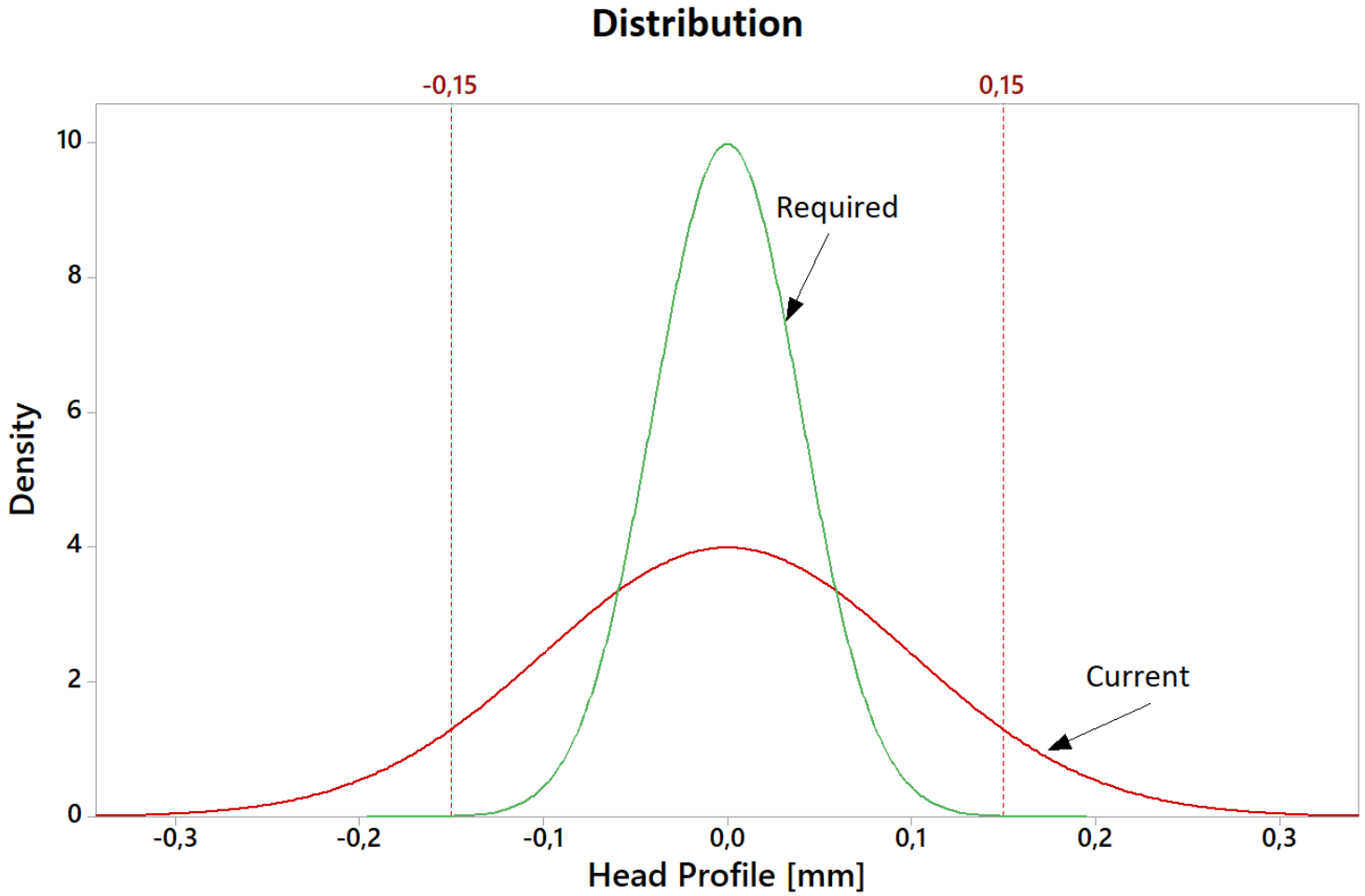


Mone(Y) = Head Profile [mm]

Rationale:

- Flatness is a shape characteristic.
- Profile will reveal details of the part shape.
- CMM provides better precision and resolution.
- Using machining station nest ensures that reference plane matches process.

Mone(Y)



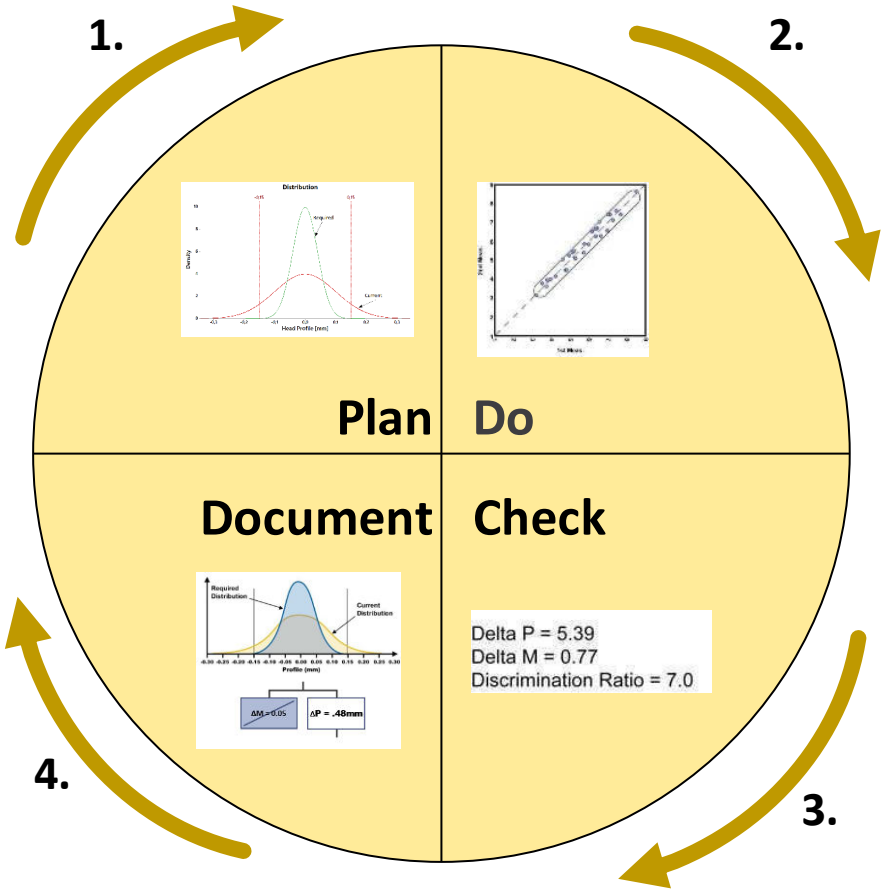
What is the next step?

*Head Profile [mm] = Distance to datum

Investigative Cycle

→ **Plan next split on the Solution Tree.**

- How can we evaluate the measurement system?
- How many part we need?



- Graph the results.
- What is the discrimination ratio?
- **Update the Solution Tree.**

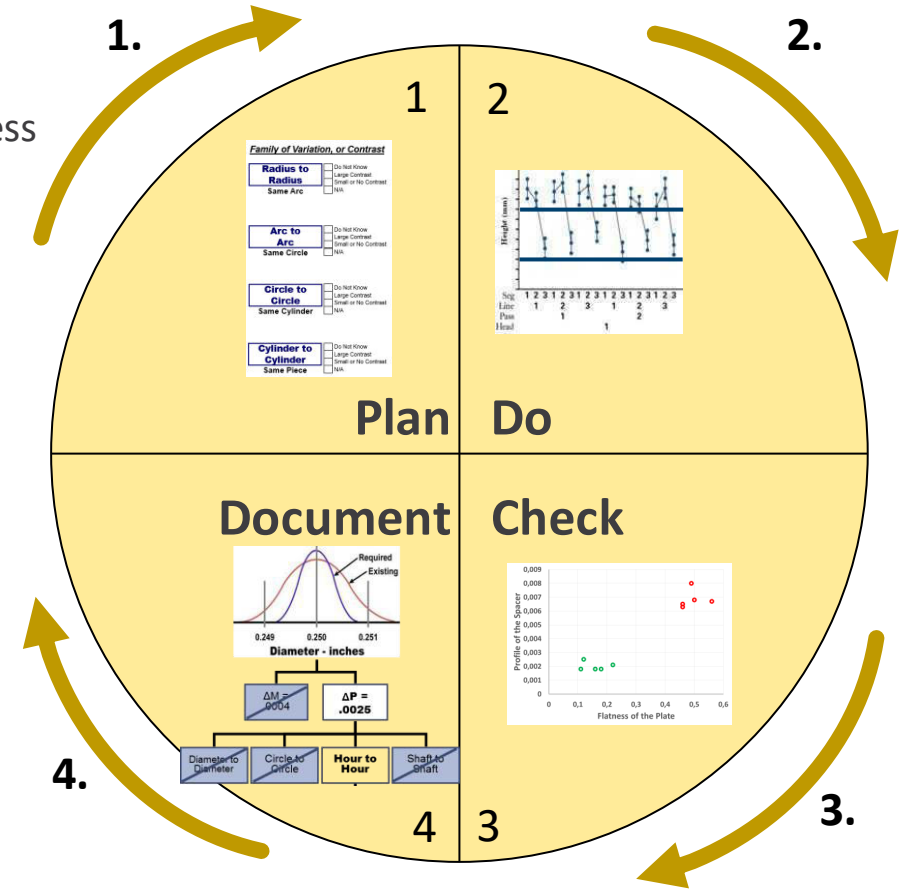
→ What is the next step?

Evolving strategies will be necessary to fill the Solution Tree level by level to identify the BIG X candidate.

Investigative Cycle

→ **Plan next split on the Solution Tree.**

- How is the geometry created?
- What are the design and process families?
- Layout Strategy Diagram.



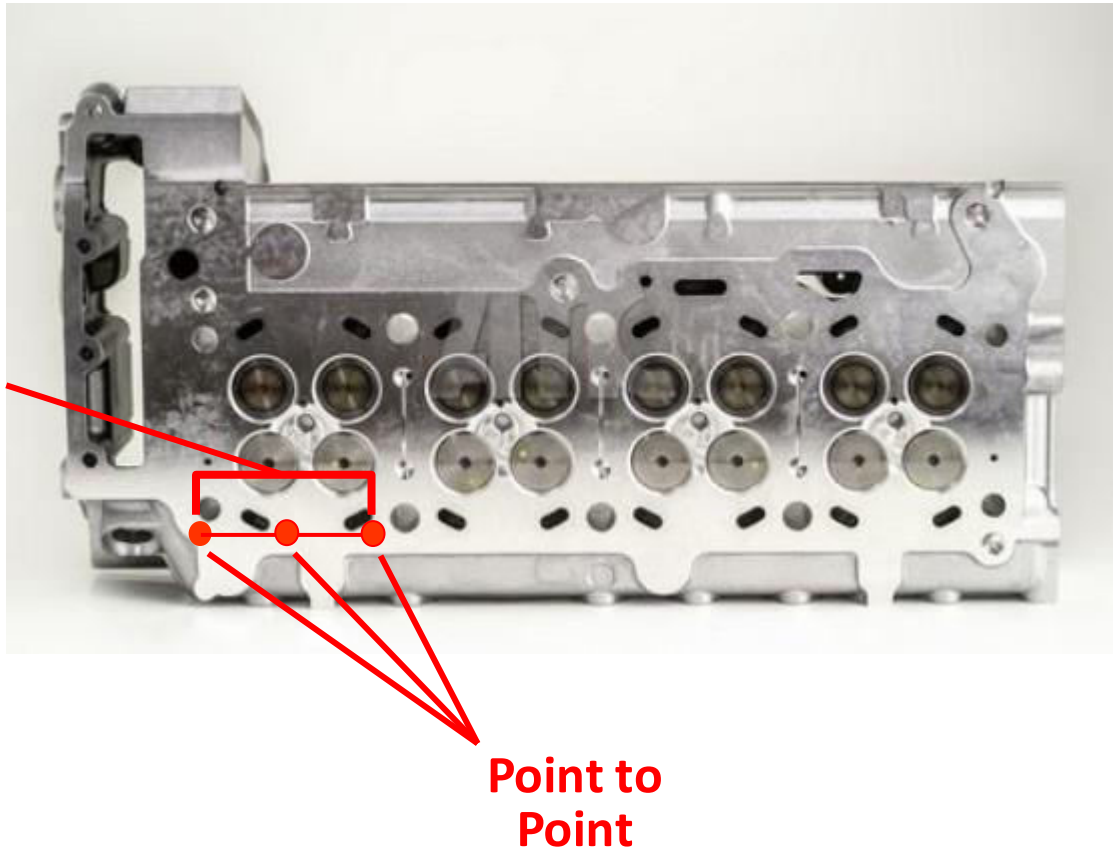
- Take stratified samples.
- Measure Mone(Y).
- Graph the results.

- Has the BOB / WOW difference been captured?
- What's the largest family of variation?

- **Update the Solution Tree.**

Evolving strategies will be necessary to fill the Solution Tree level by level to identify the BIG X candidate.

Strategy Diagrams

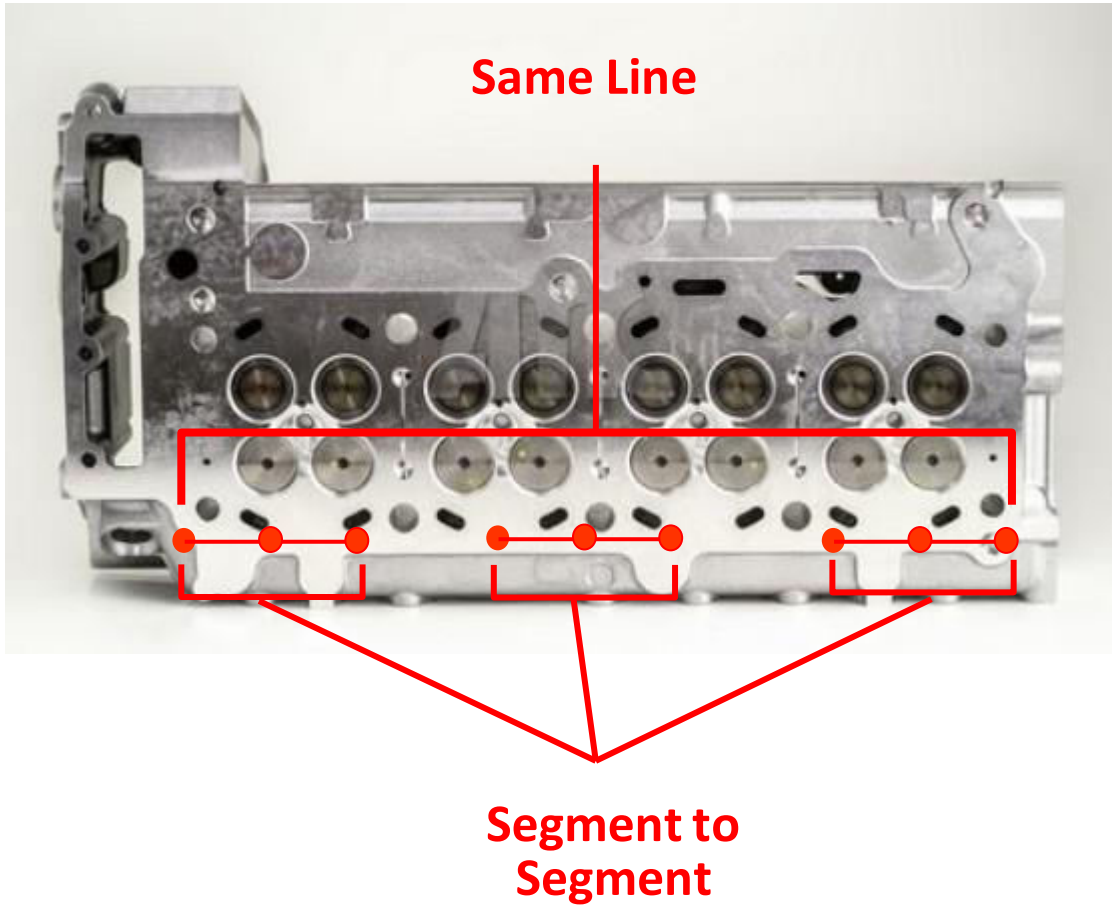


Family of Variation

Point to Point

Same Segment

Strategy Diagrams



Family of Variation

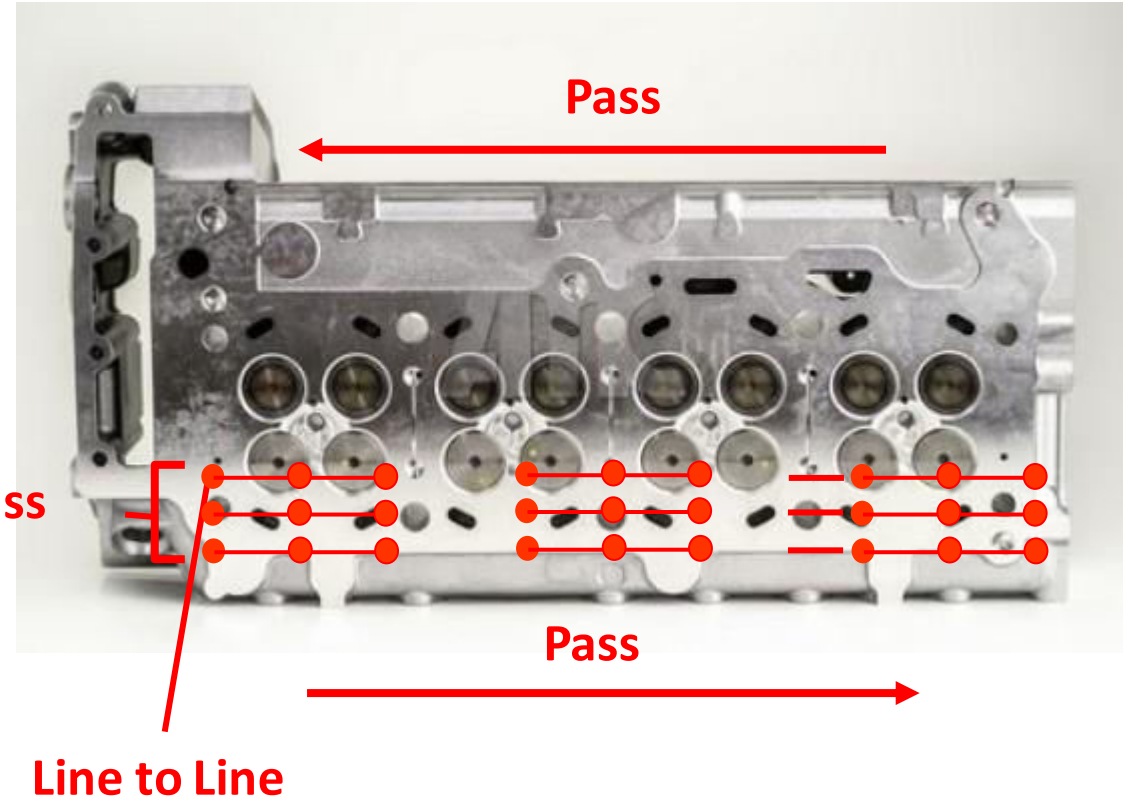
Point to Point

Same Segment

Segment to Segment

Same Line

Strategy Diagrams



Family of Variation

Point to Point
Same Segment

Segment to Segment
Same Line

Line to Line
Same Pass

Pass to Pass
Same Head

Feature Strategy Diagram

Mone(Y) = Head Profile [mm]

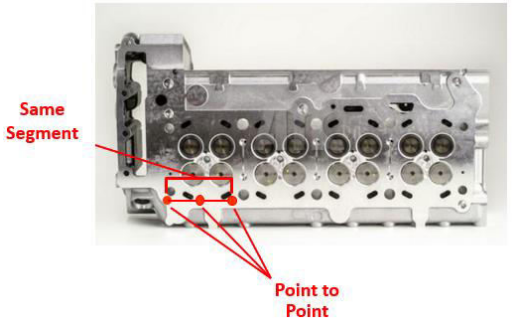
Family of Variation, or Contrast

Diagram

What Is Meant by Each Contrast

Point to Point
Same Segment

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast
- ☐ N/A



What do the profiles look like along the cutter path for milling operations?

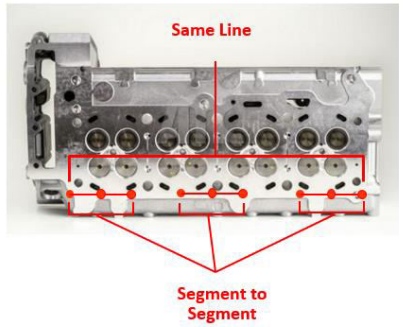
Segment to Segment
Same Line

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast
- ☐ N/A

Is there contrast in the profile cut at the start, middle and end of the cut or pass?

Line to Line
Same Pass

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast
- ☐ N/A



Cutting tools have depth. Is there contrast in the depth of the profile from one side to the other side of the cutter?

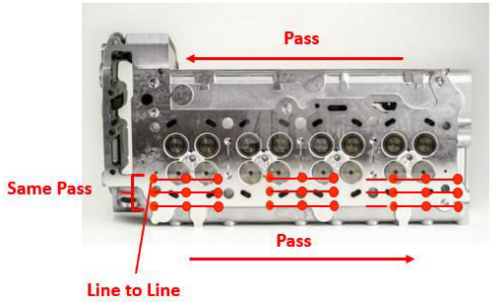
Pass to Pass
Same Cylinder Head

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast
- ☐ N/A

Milling machines often make several passes to machine a plane. Is there contrast pass to pass as the cutter moves across the same head?

Cylinder Head to Cylinder Head
Same Station

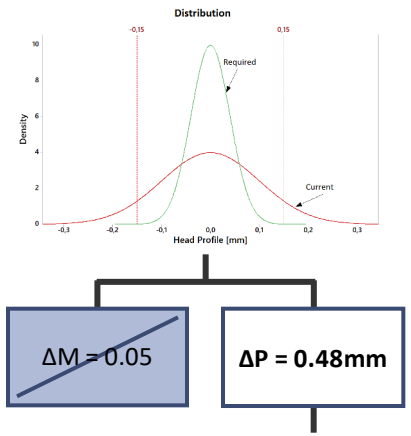
- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast
- ☐ N/A

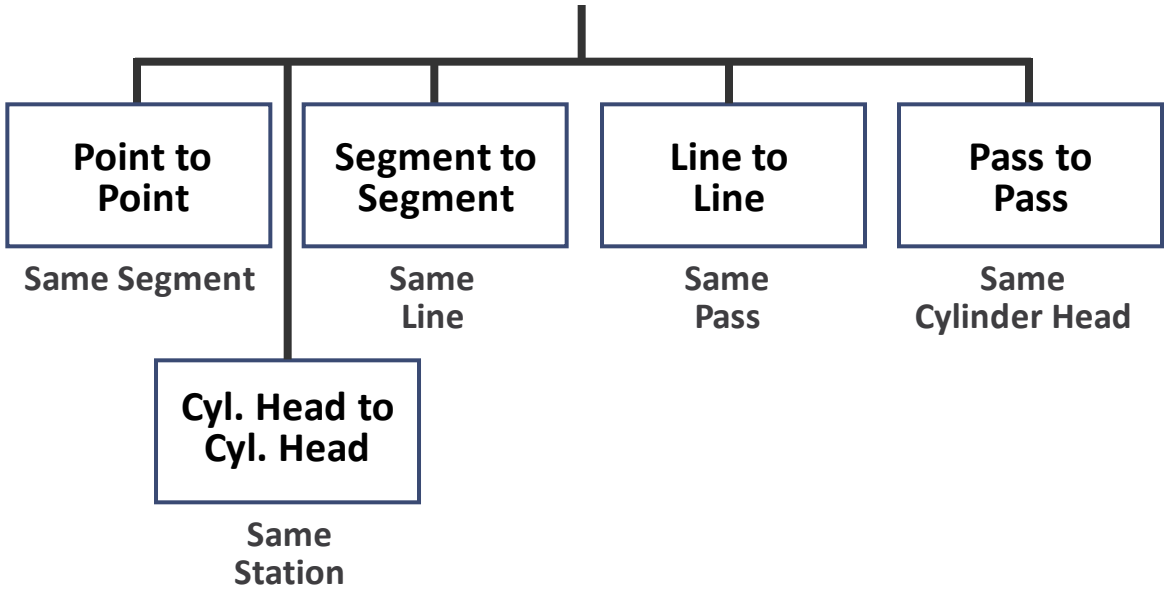


Do BOB and WOW pieces come in groups, or are they random? Consider sampling in groups of 3 to 5 consecutive pieces. Use Multi-Vari to characterize the process.

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Solution Tree

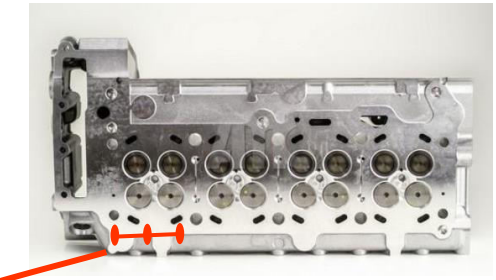


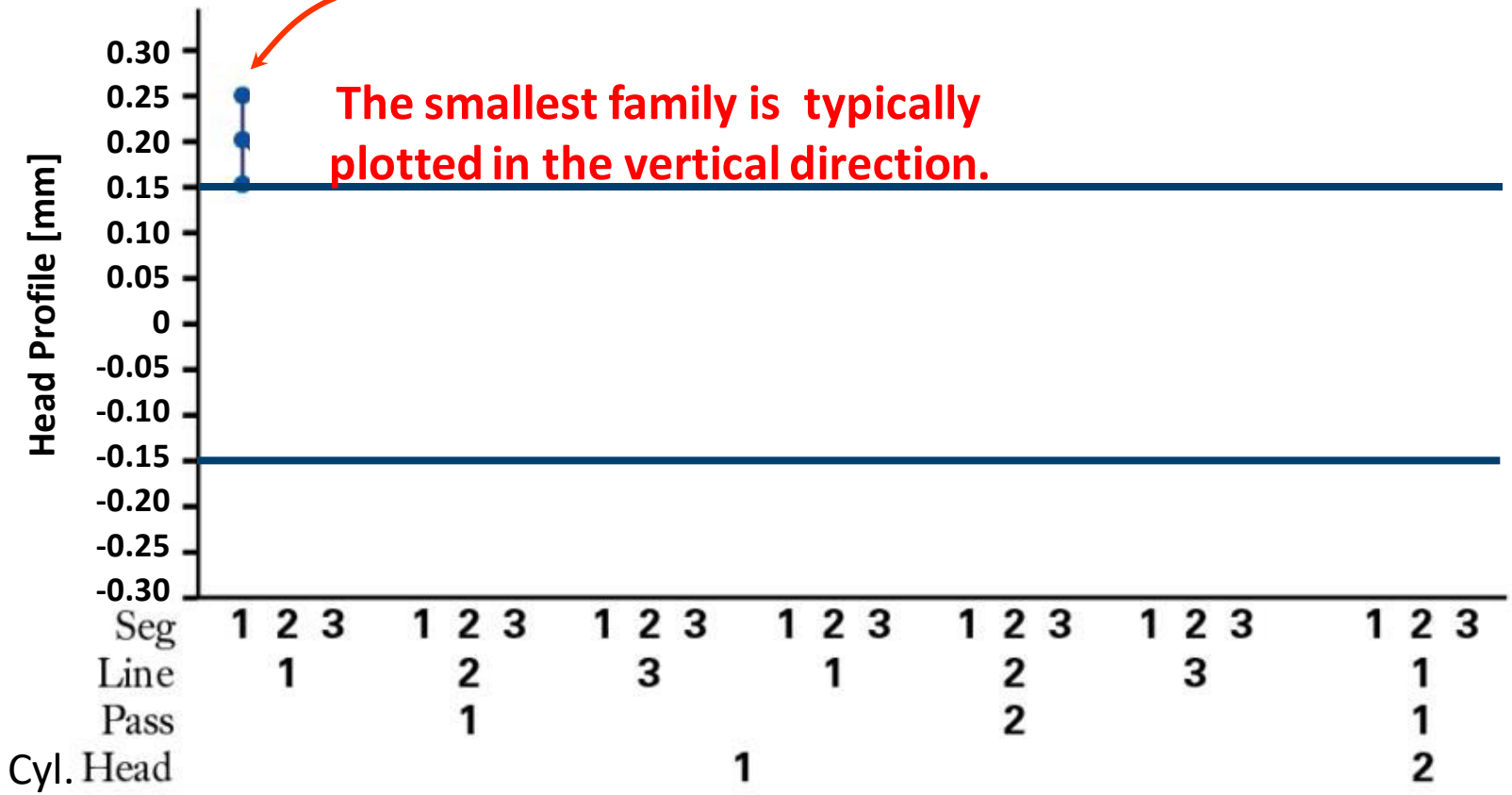


Rationale:

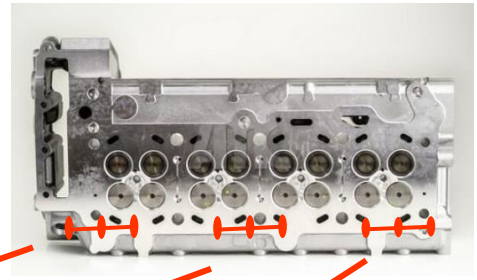
- Multi-Vari

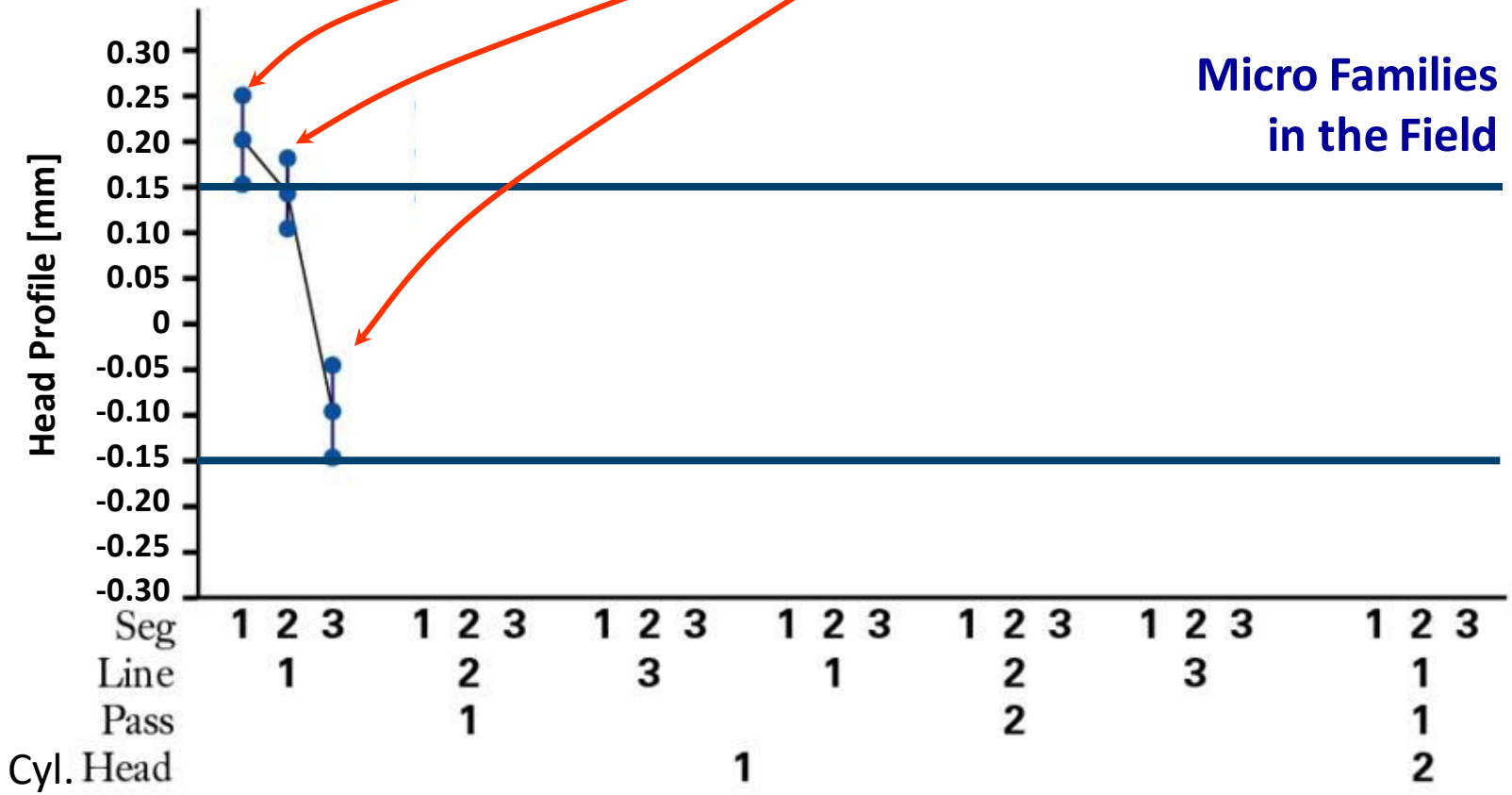
Multi-Vari



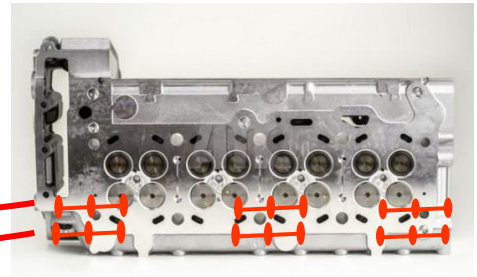

The smallest family is typically plotted in the vertical direction.
Head Profile [mm]
0.30
0.25
0.20
0.15
0.10
0.05
0
-0.05
-0.10
-0.15
-0.20
-0.25
-0.30
Seg 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3
Line 1 2 3 1 2 3 1 2 3
Pass 1 2 1 2
Cyl. Head 1 2

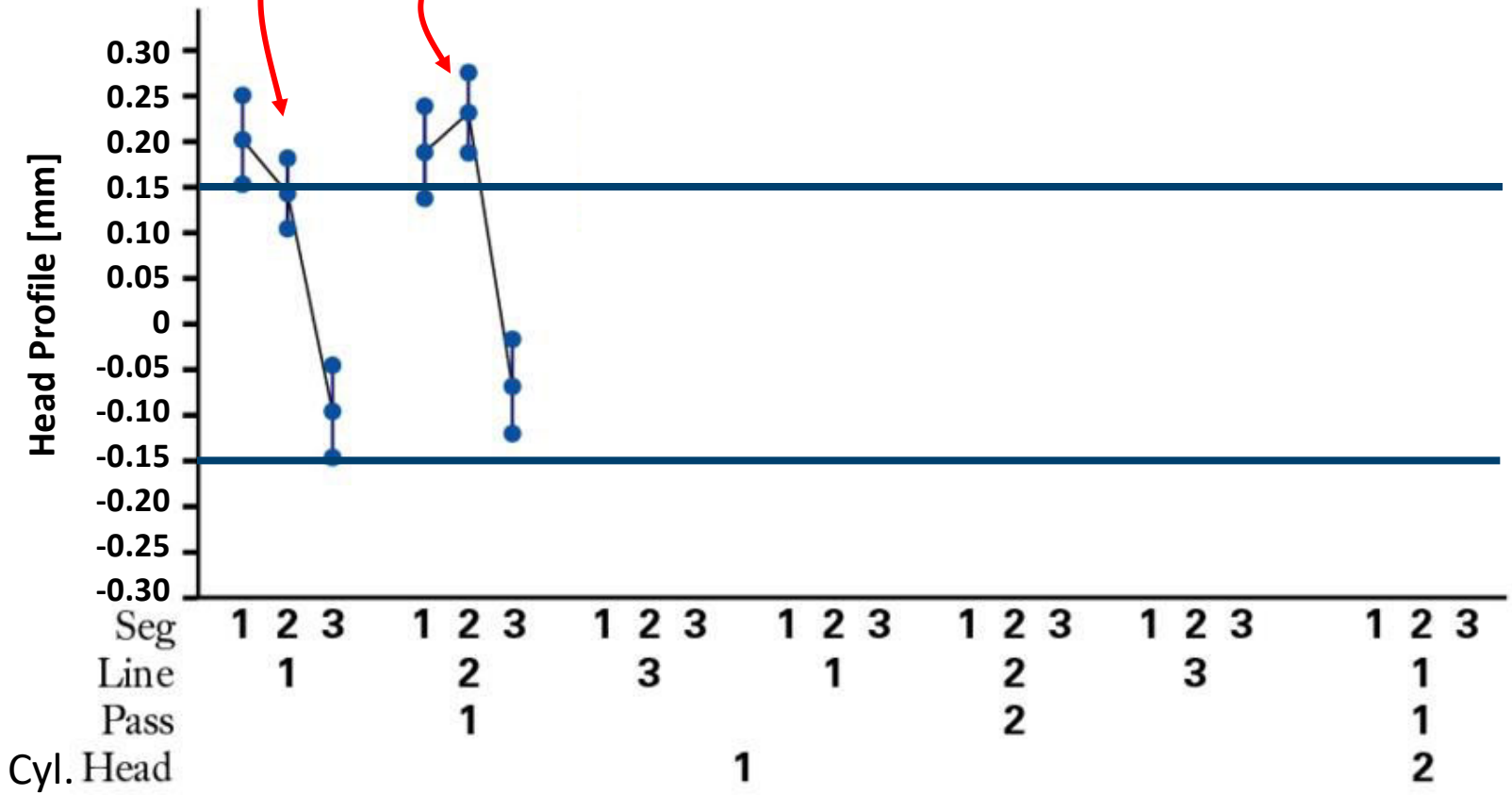
Multi-Vari



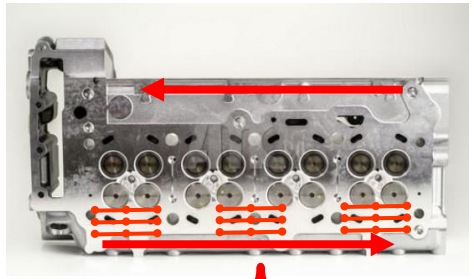

Micro Families
in the Field
Head Profile [mm]
0.30
0.25
0.20
0.15
0.10
0.05
0
-0.05
-0.10
-0.15
-0.20
-0.25
-0.30
Seg 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3
Line 1 2 3 1 2 3 1
Pass 1 1 2 2 1
Cyl. Head 1 2

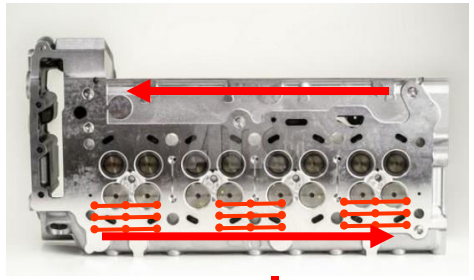
Multi-Vari

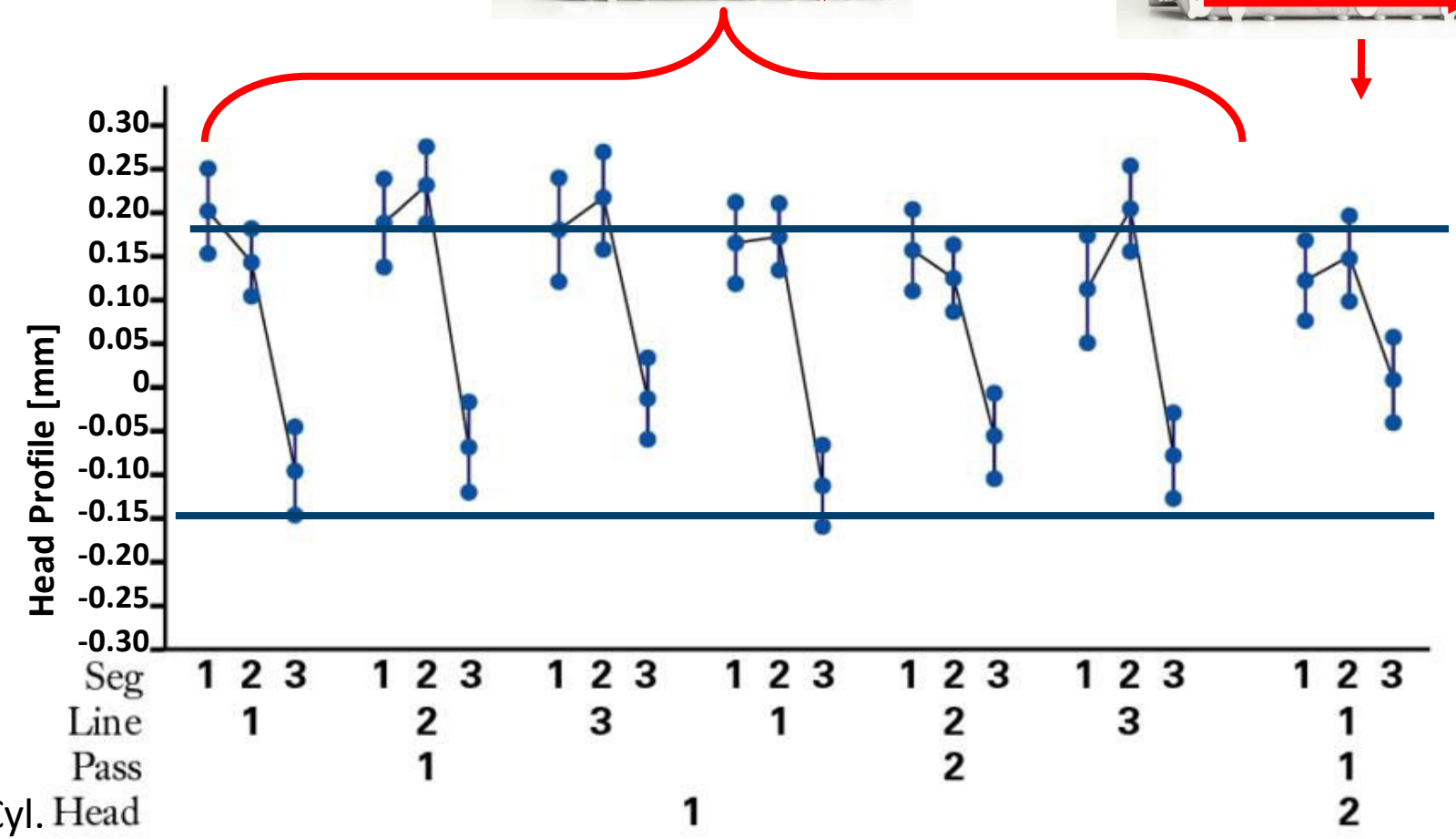



Head Profile [mm]
0.30
0.25
0.20
0.15
0.10
0.05
0
-0.05
-0.10
-0.15
-0.20
-0.25
-0.30
Seg 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3 1 2 3
Line 1 2 3 1 2 3 1
Pass 1 1 2 1 2
Cyl. Head 1 2

Multi-Vari



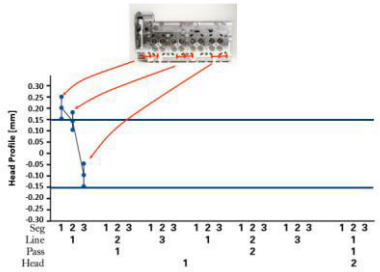
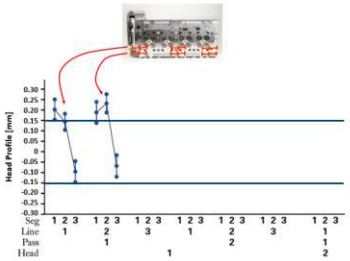
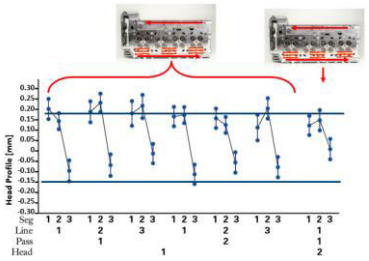




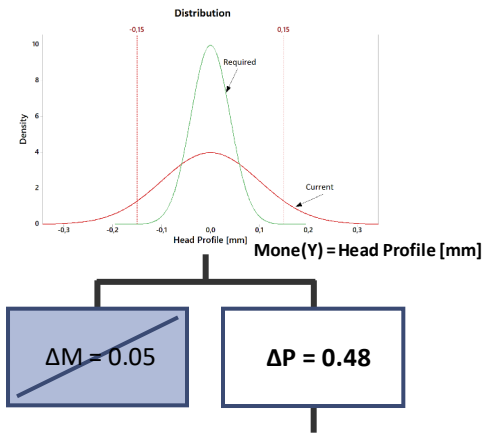
Full range of Mone(Y) variation captured in the Multi-Vari. The BIG X is active.

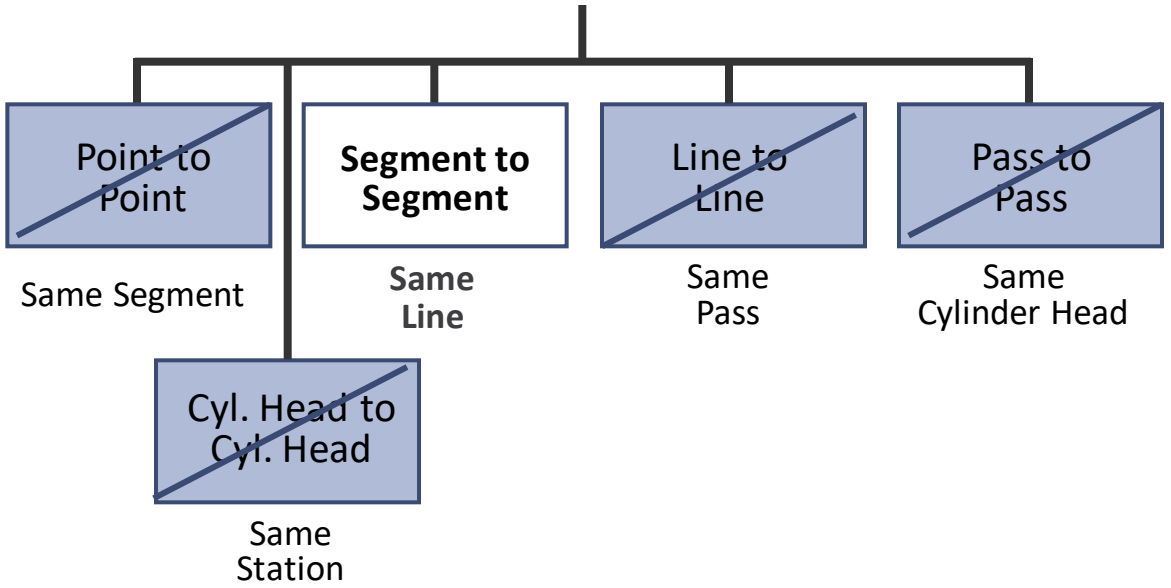
Feature Strategy Diagram

Mone(Y) = Head Profile [mm]

Family of Variation, or Contrast		***Diagram***	***What Is Meant by Each Contrast***
Point to Point Same Segment	☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast ☐ N/A		Small variation point to point. Small contrast with little leverage.
Segment to Segment Same Line	☐ Do Not Know ■ Large Contrast ☐ Small or No Contrast ☐ N/A		Segment 3 is WOW.
Line to Line Same Pass	☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast ☐ N/A		All lines show the same pattern.
Pass to Pass Same Cylinder Head	☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast ☐ N/A		Pass 1 and 2 both look similar on the Multi-Vari.
Cylinder Head to Cylinder Head Same Station	☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast ☐ N/A		Some contrast head to head.

Solution Tree

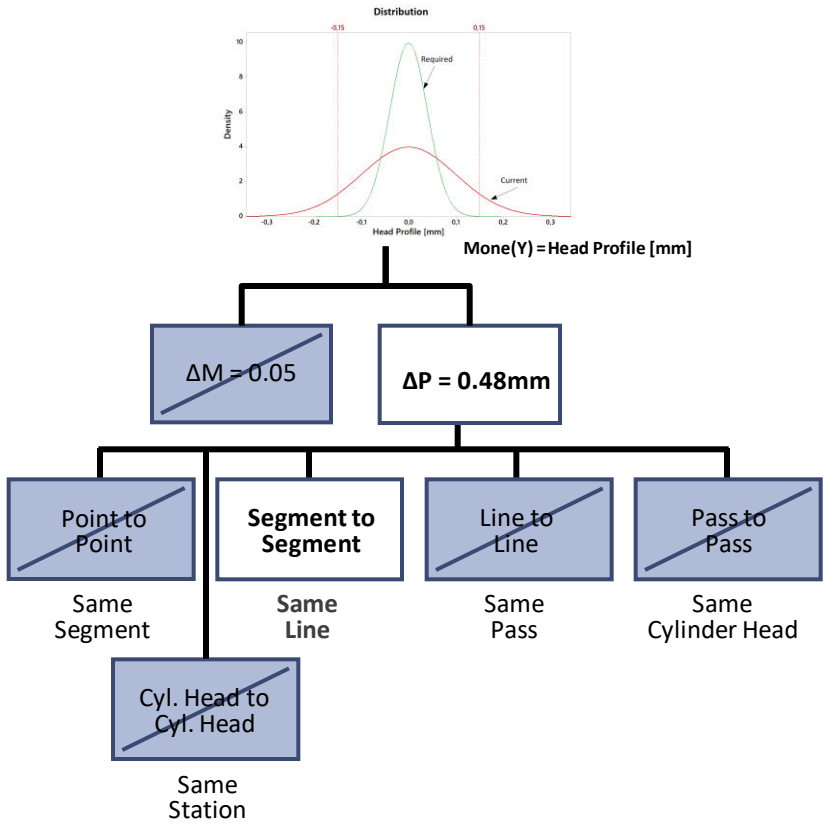




Rationale:

- Multi-Vari

Solution Tree

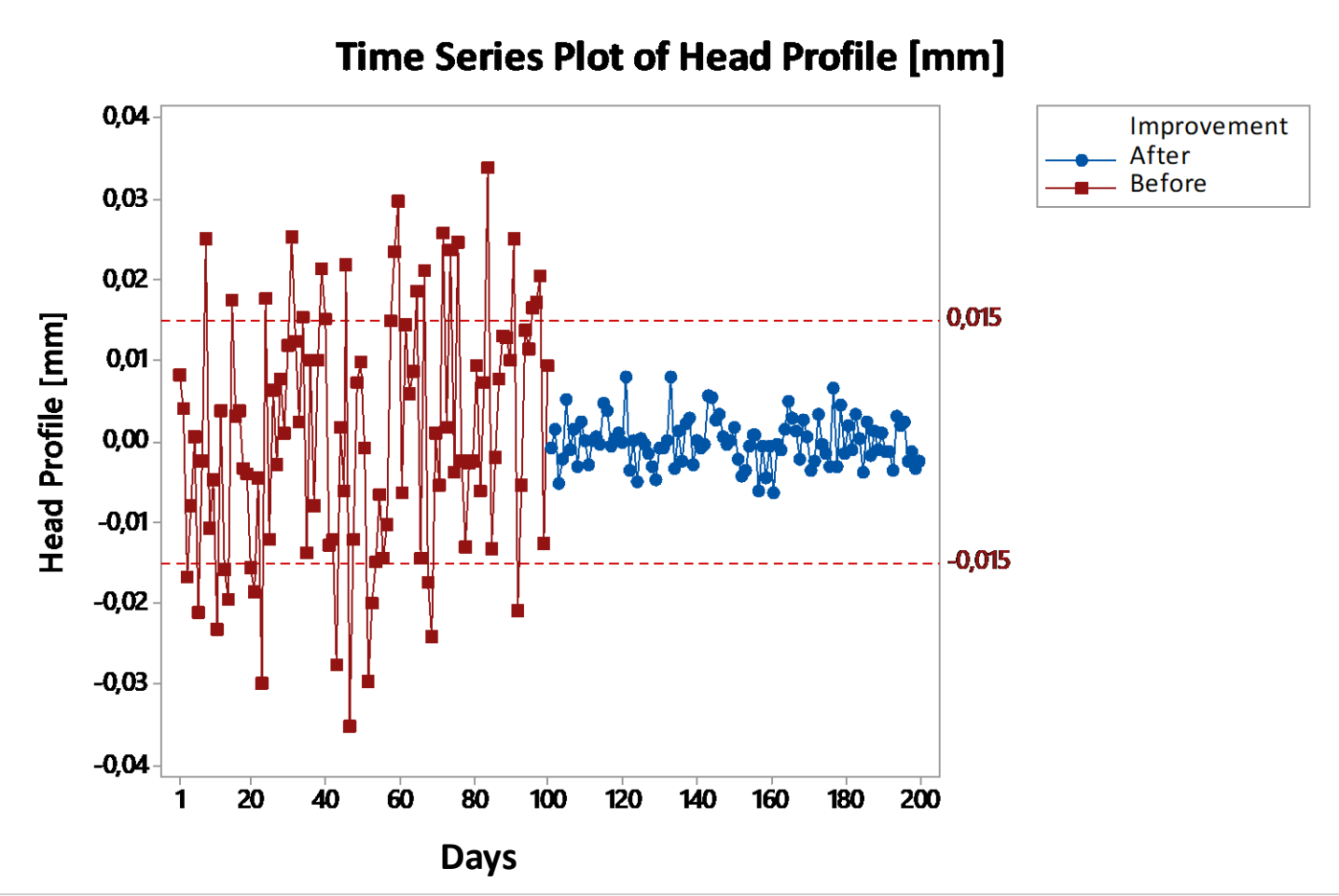


Spindle Bearing

BIG X

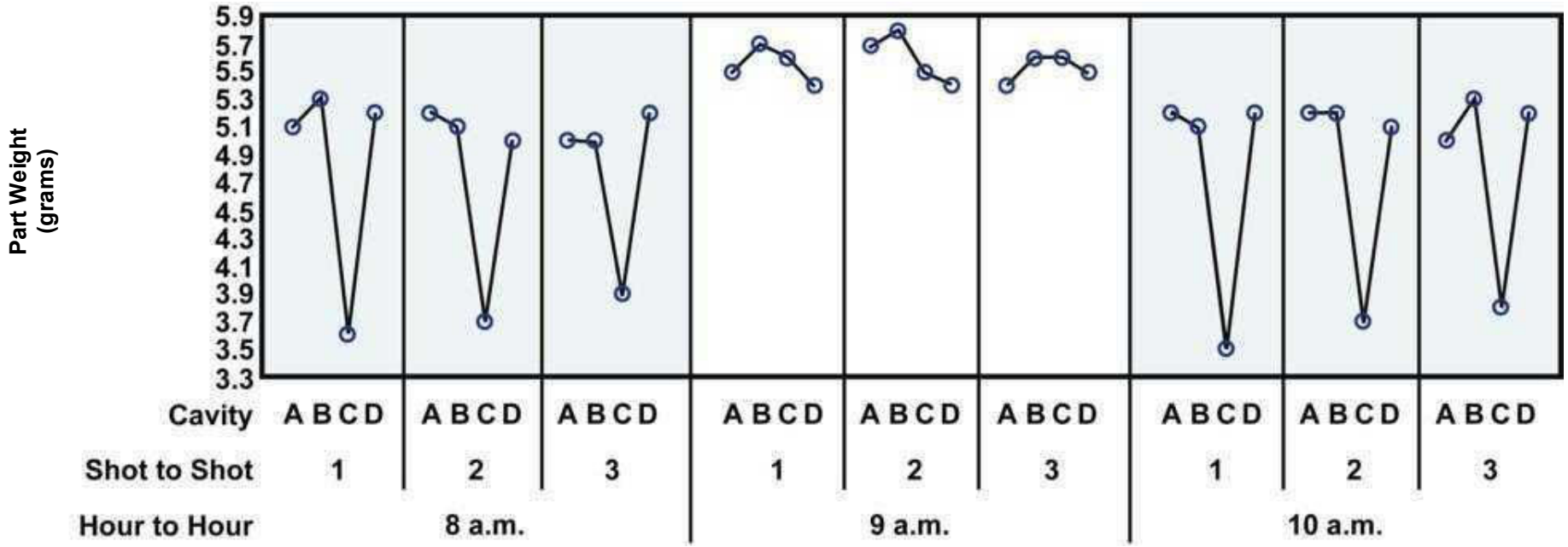
Confirmed kill with Multi-Vari of shape parameter and Mone(Y) run chart of leaks.

Head Profile – Mone(Y) Run Chart



Multi-Vari Interpretation and Observations

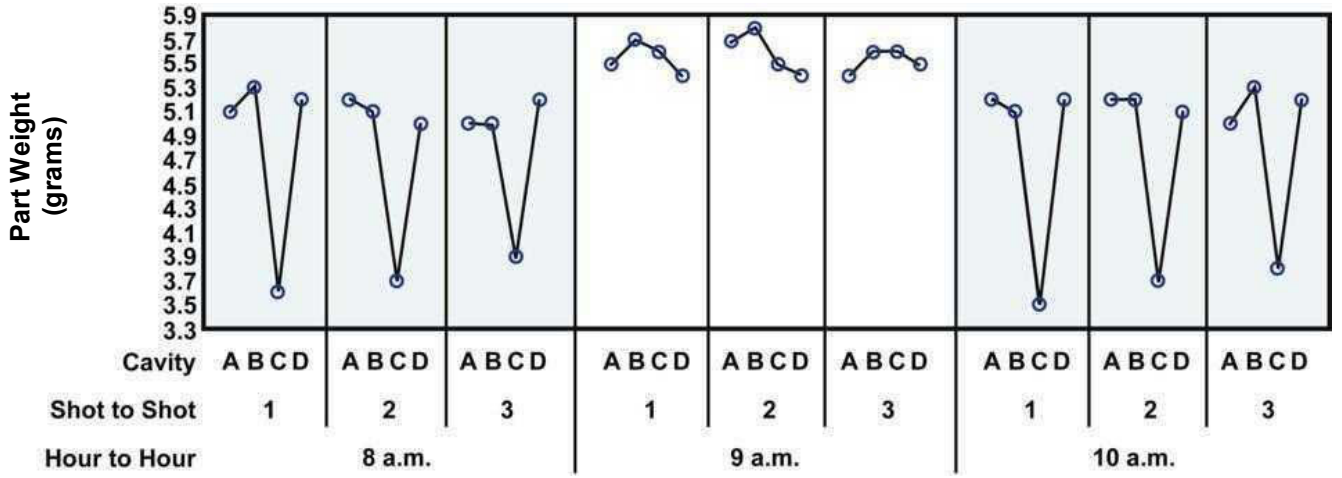
Mone(Y) = weight of injection molded part [g]



What do you see?

Multi-Vari Interpretation and Observations

Mone(Y) = weight of injection molded part [g]

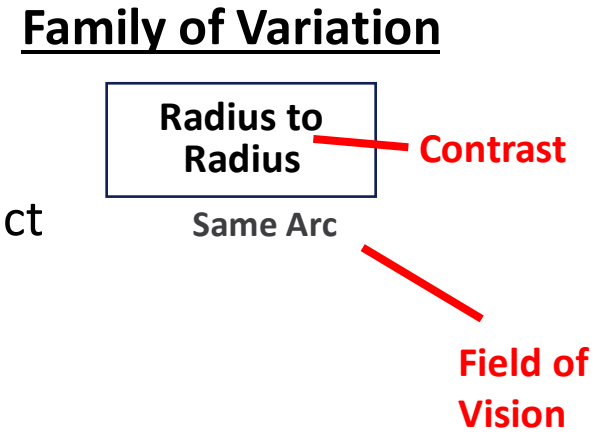


Unequal sensitivity to some unknown X

There is an interaction between the cavity/cavity and hour/hour families. They studied the cavities. When looking at the runners, they found C slightly smaller. Hour/Hour – When the operator saw a problem with the parts being too light, they increased pack pressure. After awhile, flash would increase, so they would decrease pack pressure, thus turning-on the BIG X in cavity C.

Multi-Vari - Guidelines

1. Develop a Strategy Diagram:
   - Start with the smallest logical increment of geometry.
   - Determine the appropriate field of vision.
   - Use the field of vision from the smallest order family to construct the next family.
   - Build up to the largest order, or macro family, which is usually time to time.
2. Determine an appropriate stratified sampling plan:
   - Samples are to be taken in order to capture the variation in each of the families on the Strategy Diagram.
   - Samples are taken consecutively with a minimum of three readings per family.
3. Lay out the Multi-Vari so that the first family is in the field and progressive families are shown in progressive order.
4. Run the Multi-Vari until the observed Mone(Y) variation is captured.



Multi-Vari - Guidelines

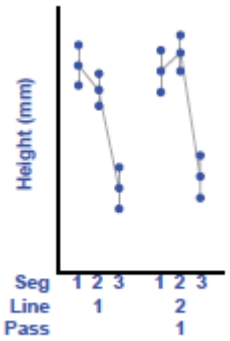
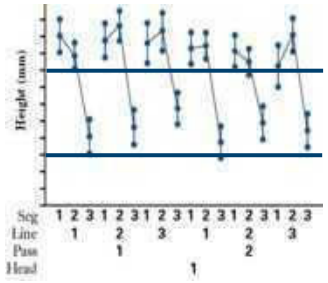
5. Identify the family which consumes Mone(Y) variation of BIG X magnitude. Focus on this family:
   - If the BIG X lives in one of the design families, look for non-random patterns that align with the orientation of the part to the physics of the machine.
   - If the BIG X lives in one of the process families (part to part), look at how the part presentation is repeated.
   - If the BIG X lives in time, look for non-random patterns associated with trends, shifts and cycles that provide clues to its nature.
6. If one or more samples exhibit unequal sensitivity, that is a clue that there is an interaction.

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Feature – Distance From Datum Strategy Diagram – Template

Mone(Y) = Distance from Datum [unit]

Design Families

Family of Variation, or Contrast		***Diagram***	***What Is Meant by Each Contrast***
Point to Point Same Segment	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		What do the profiles look like along the cutter path for milling operations?
Segment to Segment Same Line	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Is there contrast in the profile cut at the start, middle and end of the cut or pass?
Line to Line Same Pass	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Cutting tools have depth. Is there contrast in the depth of the profile from one side to the other side of the cutter?
Pass to Pass Same Plane	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Milling machines often make several passes to machine a plane. Is there contrast pass to pass as the cutter moves across the same head?
Plane to Plane Same Piece	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Are there several different surfaces machined within the same piece? If yes, is there any contrast in the shape, or profile?

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Module 3 Summary

Feature Strategies

- Problem solving is based on evidence gathered by talking to the parts.
- A Multi-Vari chart is a graphical analysis of variance (ANOVA). It reveals the largest family of variation within a manufacturing system.
- A Multi-Vari chart works best in combination with a Strategy Diagram.
- The key to developing a feature strategy is the ability to understand and recreate how the shape was formed.
- A process flow diagram describes the families within a manufacturing process and helps develop the strategy.
- An effective strategy keeps sample sizes small.
- Dictionary game thinking requires evolving strategy. Multi-Vari reveals the BIG X family; further strategy is required to identify the BIG X.

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4 Property Strategies

Module 4 Objectives
Property Strategies

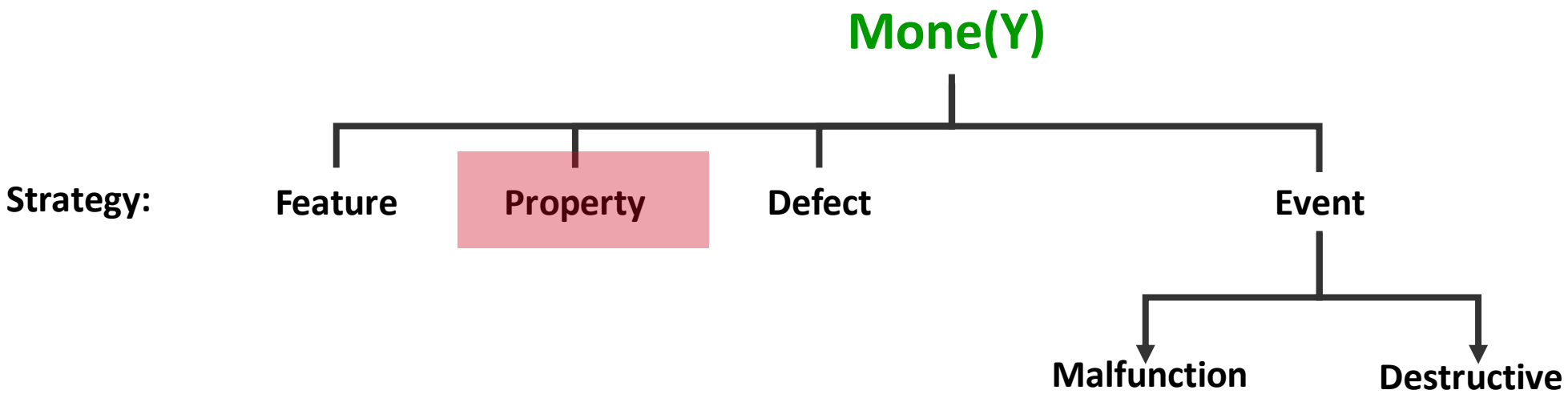
Understand:

- The principle of converging from the Mone(Y) property to the BIG X family
- The subtle differences between a property and a feature Mone(Y)

Be able to:

- Develop and execute of property strategy to identify the BIG X family

Different Strategies for Different Mone(Y)s



Explanation: A material characteristic associated with a component.

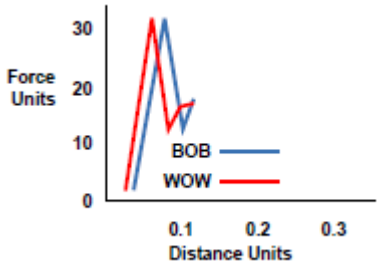
Tool: **Strategy Diagrams**

Strategy Diagrams are a way to document possible contrasts for the Mone(Y) that you can leverage during your project.

Property Strategy Diagram - Template

Mone(Y) =

Design Families

Family of Variation, or Contrast	*Diagram*	*What Is Meant by Each Contrast*
Point to Point Same Region ☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Is measuring this property destructive? If yes, try to take two readings as close as possible to each other to estimate ΔM. If readings are not destructive, use Isograph.
Region to Region Same within Piece ☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Is the property uniform across regions within the same part? Use Multi-Vari charts to characterize within part variation.
Within Piece to Within Piece Same Piece ☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		Are there contrasts within one piece (Cope side to drag side of a mold, bore to bore, boss to boss)?

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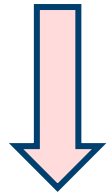
Features and Properties Comparison

Features	Properties	Clues Generation Tools
Are characterized with variable data.	Are also characterized with variable data.	Process Flow Diagram Strategy Diagram Multi-Vari Scatter Plot Operations Search
Measurements are non-destructive.	Measurements are often destructive.	
Use Isograph to calculate $\Delta P/\Delta M$.	Isograph non-destructive and destructive readings. Think “Destructive Event” for destructive readings.	
Measure features with respect to datum.	Properties have no datum scheme.	

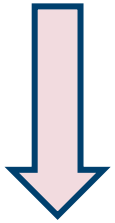
Feature and Property Strategies

- Multi-Vari charts are used in conjunction with a feature or property Strategy Diagram.
- The BIG X driving the biggest change in the feature or property must live within one of the contrasts defined in a Strategy Diagram.
- A Multi-Vari should show multiple families of variation on one chart.
- Although those families can be shown separately, it is a big advantage to have them together with the same scale for the Mone(Y).
- Patterns yield clues about the nature of the BIG X when associated with a BOB / WOW contrast.

Strategy Diagram



Multi-Vari



BIG X Family

Module 4 Summary
Property Strategies

- Property refer to the physical properties of components, both chemical and mechanical.
- Component properties are datumless.
- Measuring a component's physical properties often requires destructive measurements (i.e. energy is applied to the component to measure its properties).
- Isograph can be used where readings are non-destructive (e. g. density, intensity, field strength) or destructive (e. g. hardness).
- Properties are expressed with variable data.

a Example

Module a Objectives

Example

Understand:

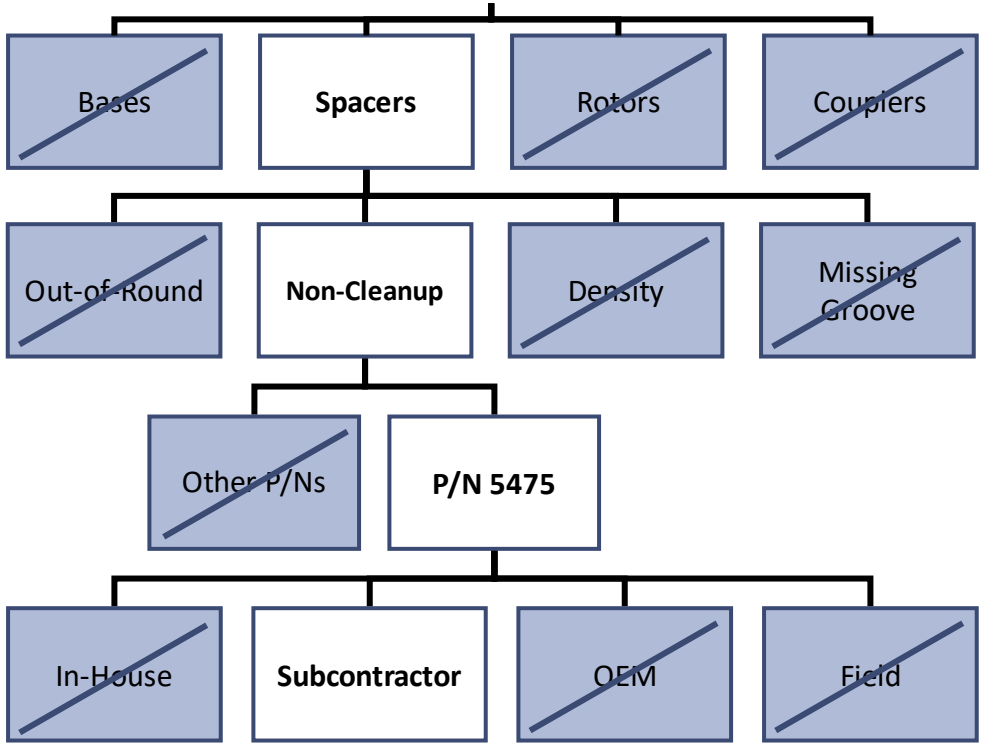
- Evolving strategies to converge from a management developed project statement to the BIG X candidate
- The relationship of Problem Definition Trees, Project Definition Trees, Solution Trees and Strategy Diagrams
- The role of scatter plots to isolate the operation that contains the BIG X

Be able to:

- Develop Feature Strategy Diagrams
- Create a Multi-Vari chart
- Complete a group comparison

Problem Definition Tree: Sintered Spacer Case

Problem Statement: Operating costs at BOCKA are exceeding budgeted levels.



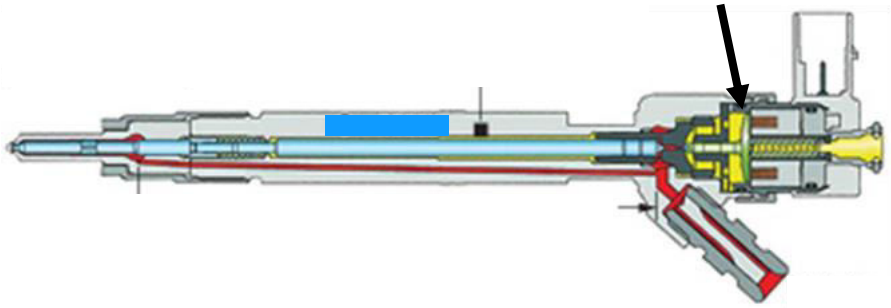
Project Statement: Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€.

Rationale:

- Spacers product line has the highest cost of poor quality driven by scrap, unscheduled rework , and premium freight (Pareto #1).
- Problem description (Pareto #2).
- Focus split P/N 5475 has highest fallout for non-cleanup (Pareto #3)
- Problem discovery point



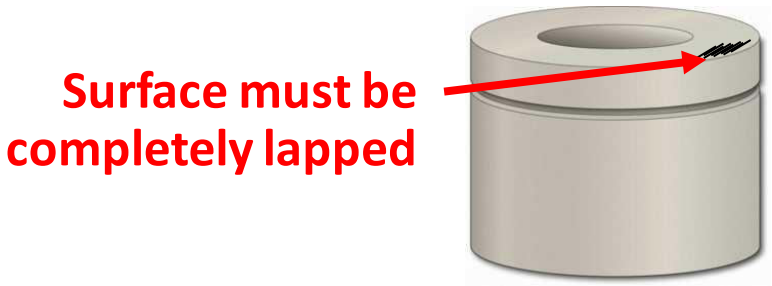
Position of the sintered spacer in a fuel injector.



Initial Situation: Sintered Spacer Case

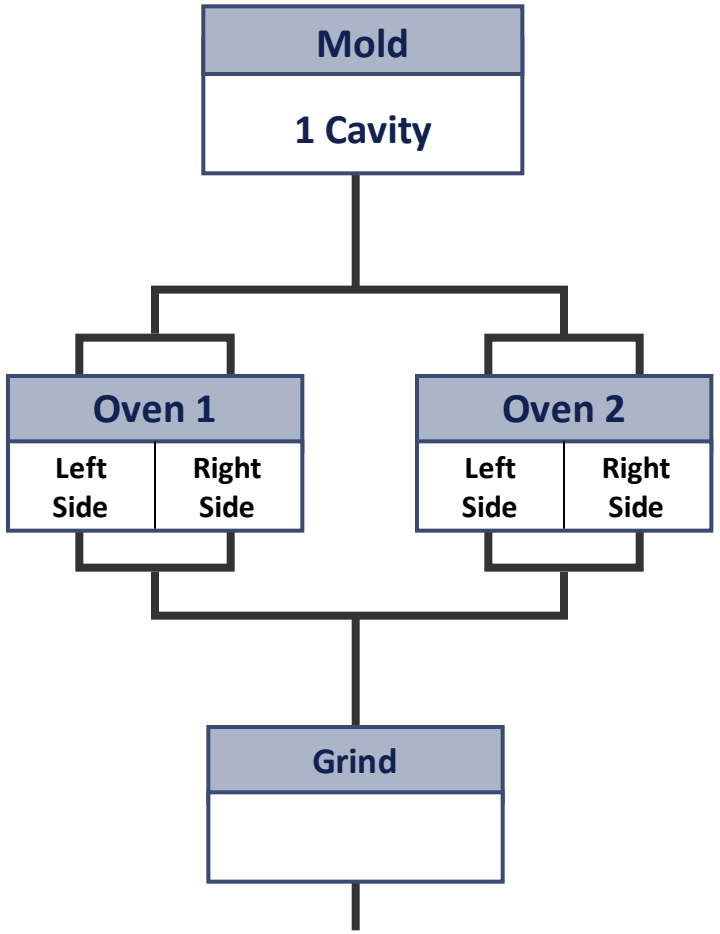
Non-clean Up Rejects

A sintered spacer is created at the manufacturer and then sent to a subcontractor, who does the final step of lapping the top face. The subcontractor is seeing high fall out of spacers at visual inspection for non-clean up (not completely lapped). There are BOB and WOW lots of product and there are approximately 5,000 parts per lot. A BOB lot after double visual inspection has less than 0,5 % fall out. A WOW lot has greater than 10 % fall out after double inspection. The potential is 400 K€.

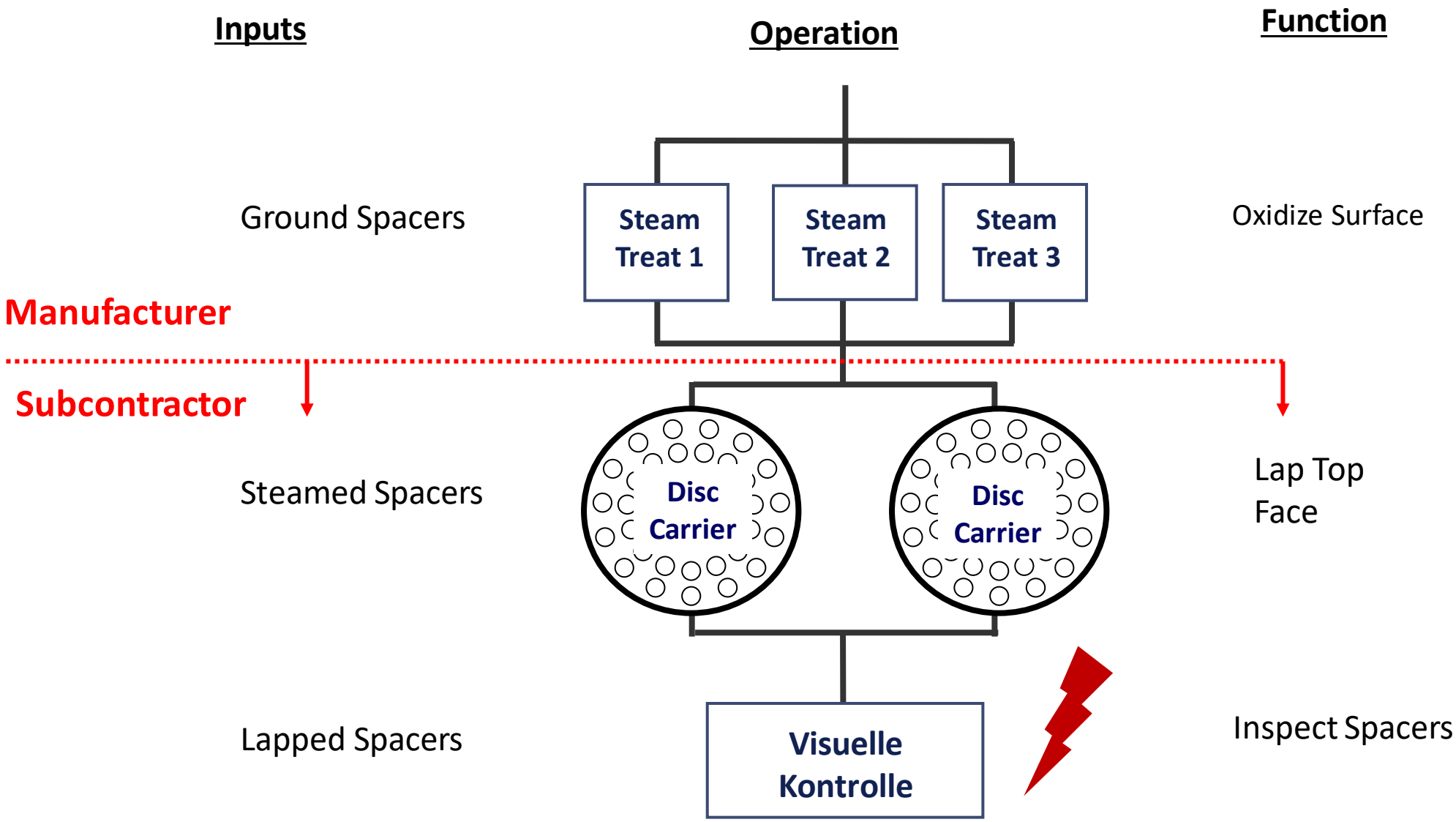


Process Flow Diagram

Inputs	Operation	Function
Blended Powdered Metal	Mold – 1 Cavity	Form Spacer in "Green" State
"Green" Spacers	Oven 1 – Left Side / Right Side; Oven 2 – Left Side / Right Side	Sinter Spacers
Hardened Spacers	Grind	Grind OD Features



Process Flow Diagram

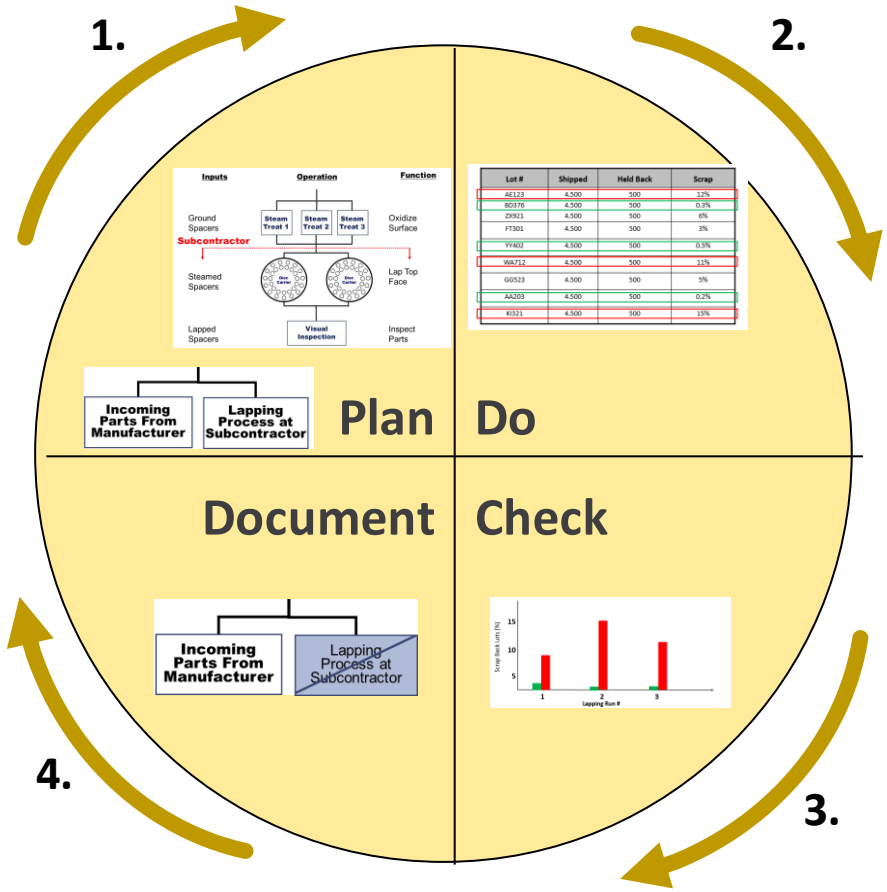


Investigative Cycle

→ **Plan next split on the Project Definition Tree**

- Defect can't be seen until after the parts have been lapped.
- Use Lot to Lot contrast to discriminate between powdered metal manufacturer and the lapping subcontractor.

→ **Update the Project Definition Tree**



- Break phase between powdered metal and lapping.
- Mixed lot lapping.

- What did we learn?

Evolving strategies will be necessary to find the BIG X.

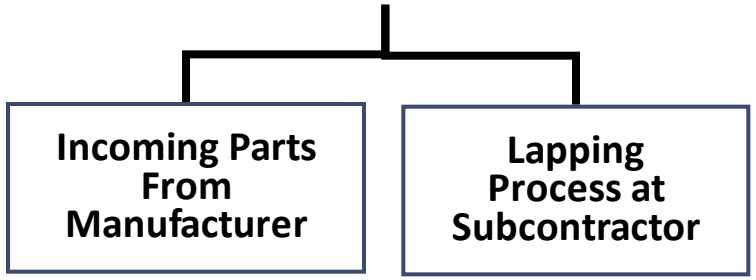
Sintered Spacer Case Dictionary Split

- Spacers are steam treated at the manufacturer and then sent to the subcontractor for the final operation of lapping.
- A dictionary split between the final step of lapping at the subcontractor and all prior operations was done by saving samples from each incoming lot from the manufacturer.
- When BOB or WOW lots were found, the samples were saved and lapped together.
- If there is clear separation between the lot samples when they are run together, the BIG X lives at the manufacturer not the subcontractor.

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Project Definition Tree

Project Statement: Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€.



Rationale:

- Mixed lot lapping.
- Breaking phase between spacer production and lapping.

Identifying BOB and WOW Lots

Lot #	Shipped	Held Back	Scrap
AE123	4.500	500	12%
BD376	4.500	500	0.3%
ZX921	4.500	500	6%
FT301	4.500	500	3%
YY402	4.500	500	0.5%
WA712	4.500	500	11%
GG523	4.500	500	5%
AA203	4.500	500	0.2%
KI321	4.500	500	15%

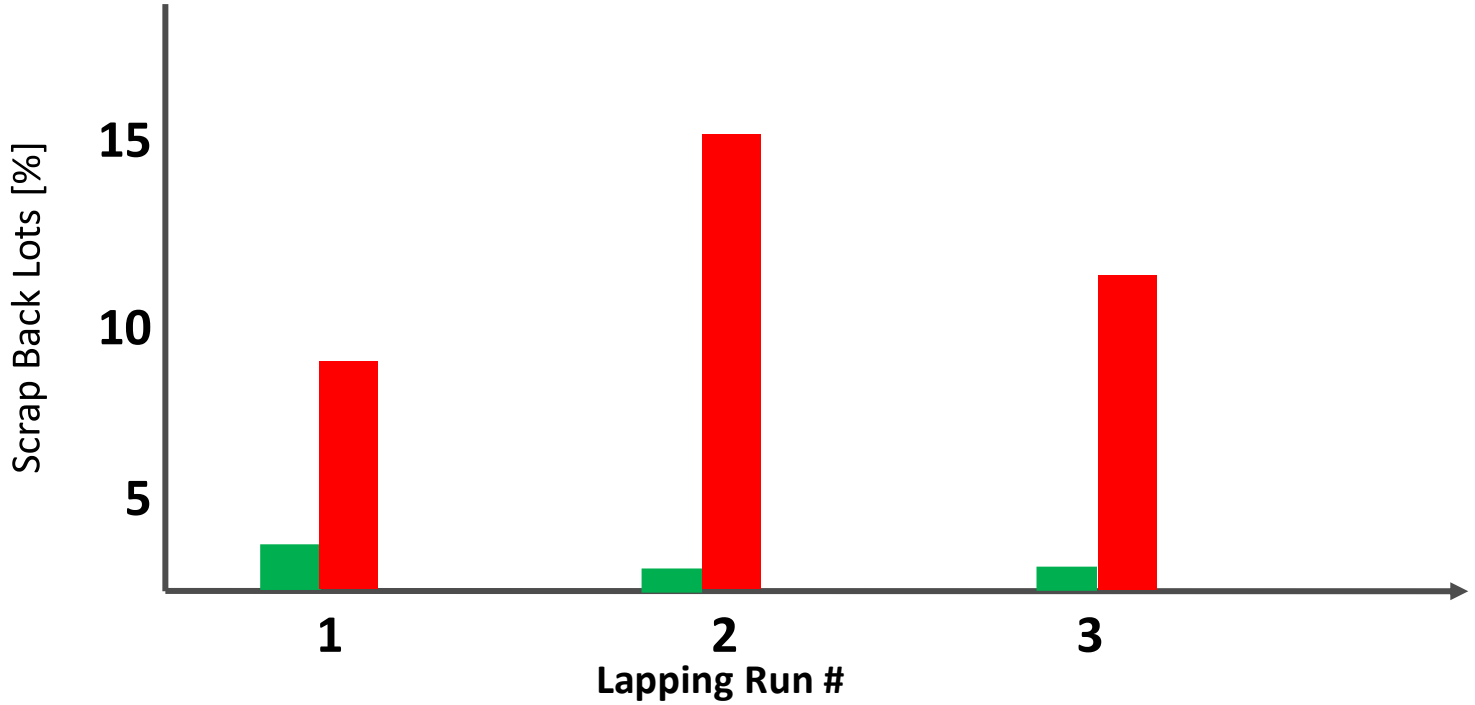
It might take a few days to get enough BOB and WOW lots.

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Dictionary Split

Represents samples from WOW lots that experienced more than a 10 % fall out at the subcontractor.

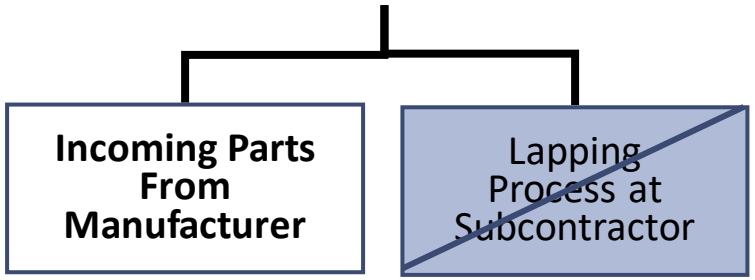
Represents samples from BOB lots that experienced less than 0.5 % fall out at the subcontractor.



Does the BIG X lives at the manufacturer or the subcontractor?

Project Definition Tree

Project Statement: Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€.



What next?

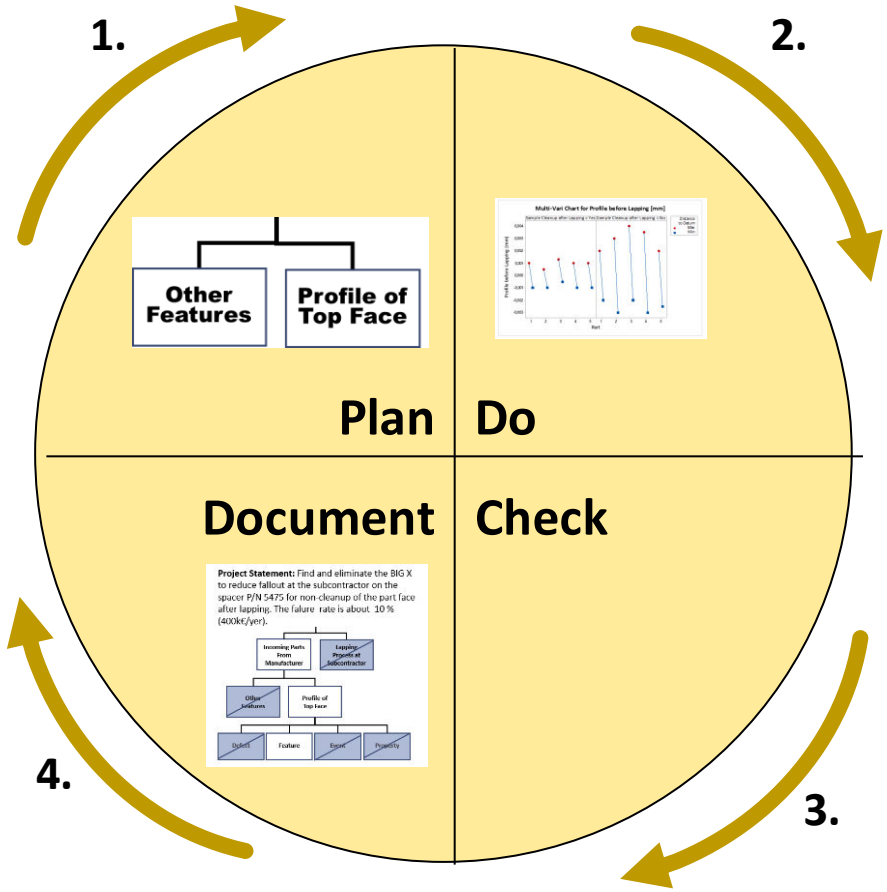
Rationale:

- Mixed lot lapping.

Investigative Cycle

→ **Plan next split on the Project Definition Tree.**

- Can we measure some characteristic in the spacers before lapping that predicts clean up results?



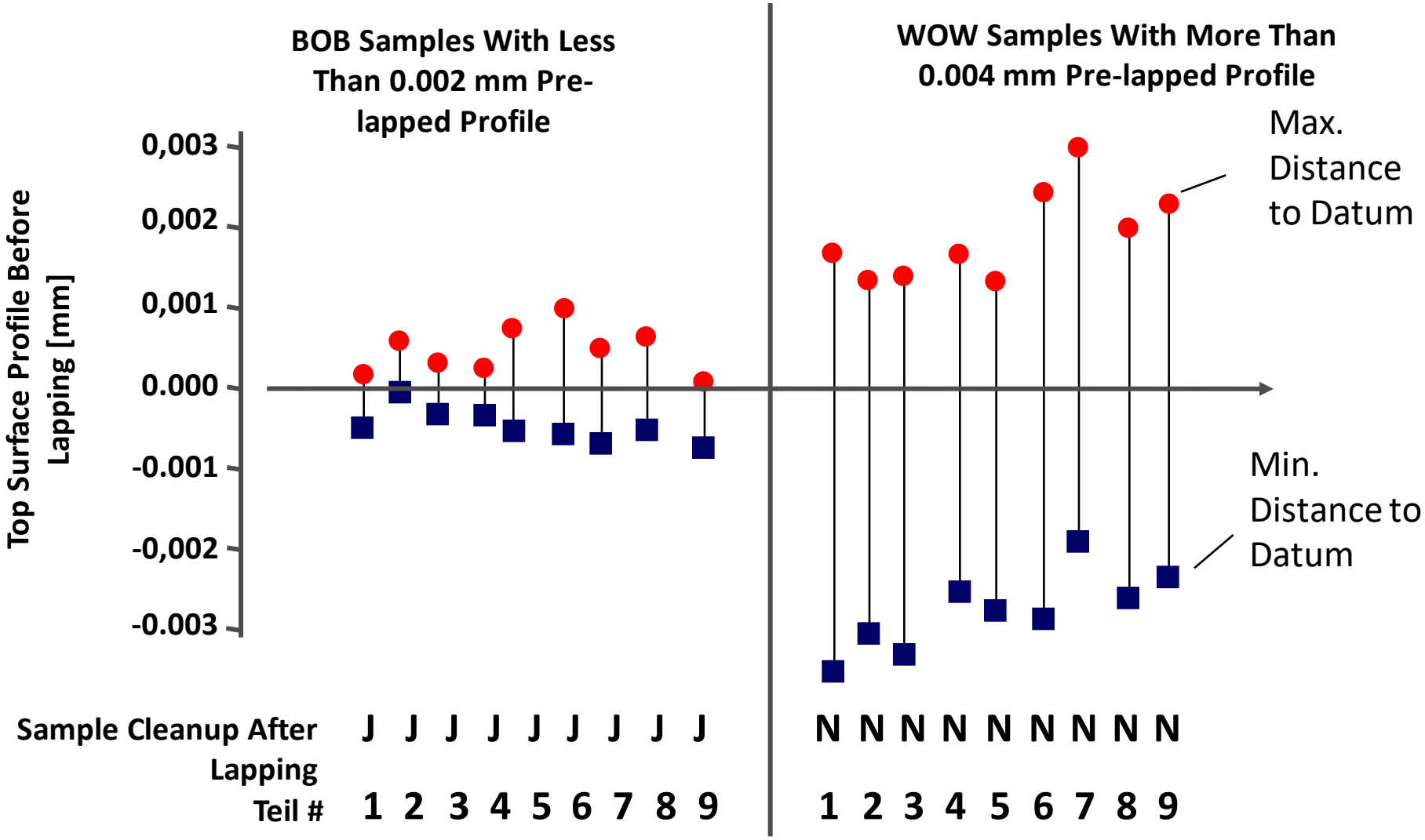
- Use group comparisons to look for lot to lot differences at the spacer level.

- What did we learn?

→ **Update the Project Definition Tree.**

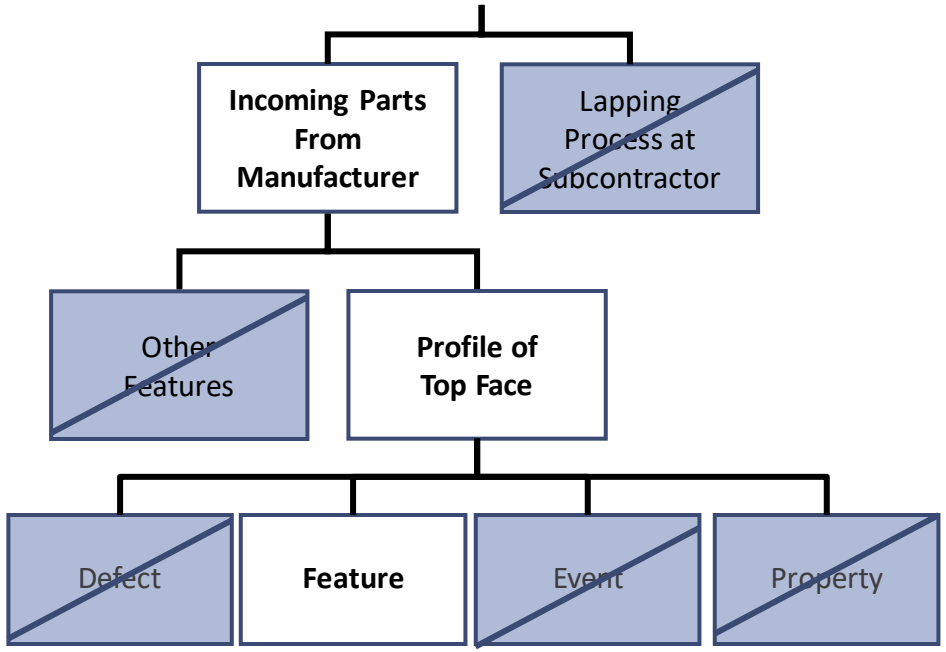
Evolving strategies will be necessary to find the BIG X.

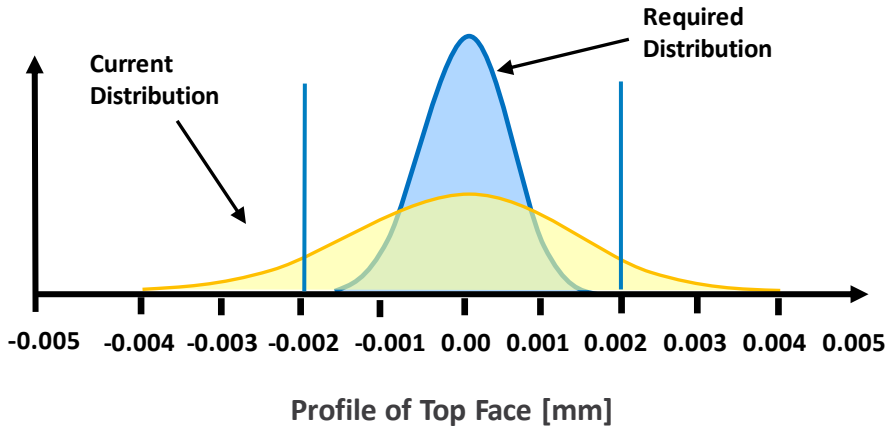
Developing a Variable Mone(Y) → Group Comparisons



Project Definition Tree

Project Statement: Find and control the BIG X for spacer P/N 5475 non-cleanup found after lapping at the subcontractor to eliminate containment and improve customer satisfaction and save 400 k€.





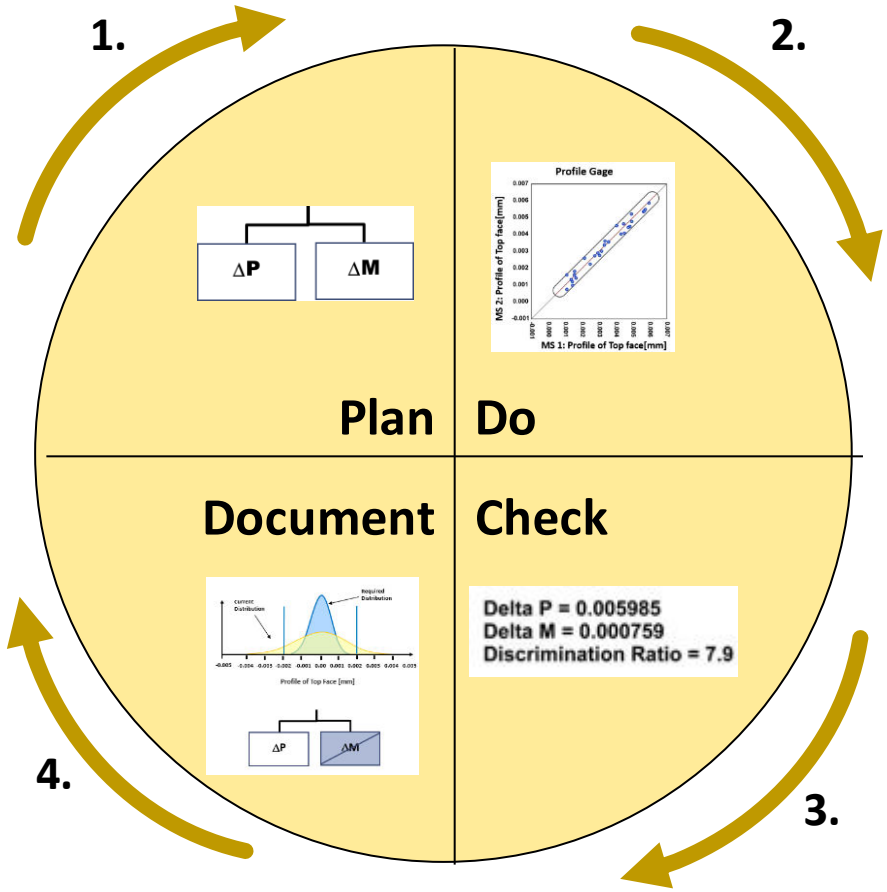
Mone(Y) description: Profile of top face measured as the nominal position of points relative to the manufacturing datum.

Mone(Y) = Profile Top Face [mm]

Investigative Cycle

→ **Plan next split on the Solution Tree.**

- Can we measure profile effectively?

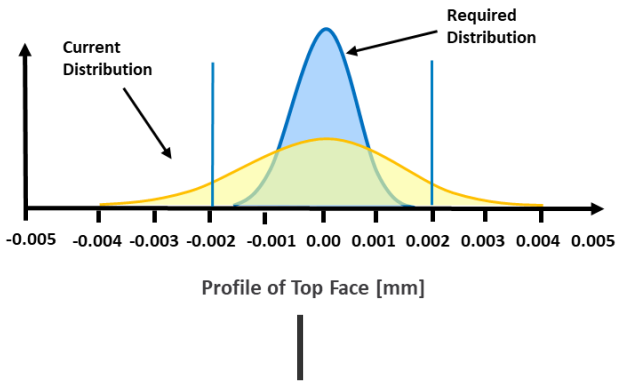


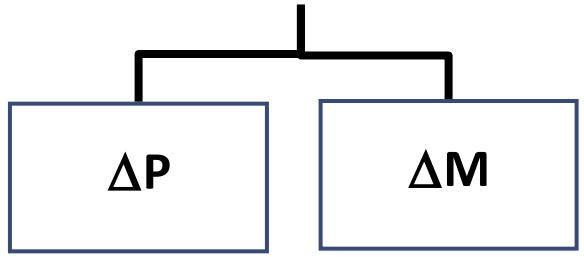
- Complete an Isograph.
- What is the discrimination ratio?

→ **Update the Solution Tree.**

Evolving strategies will be necessary to fill the Solution Tree level by level to identify the BIG X candidate.

Solution Tree

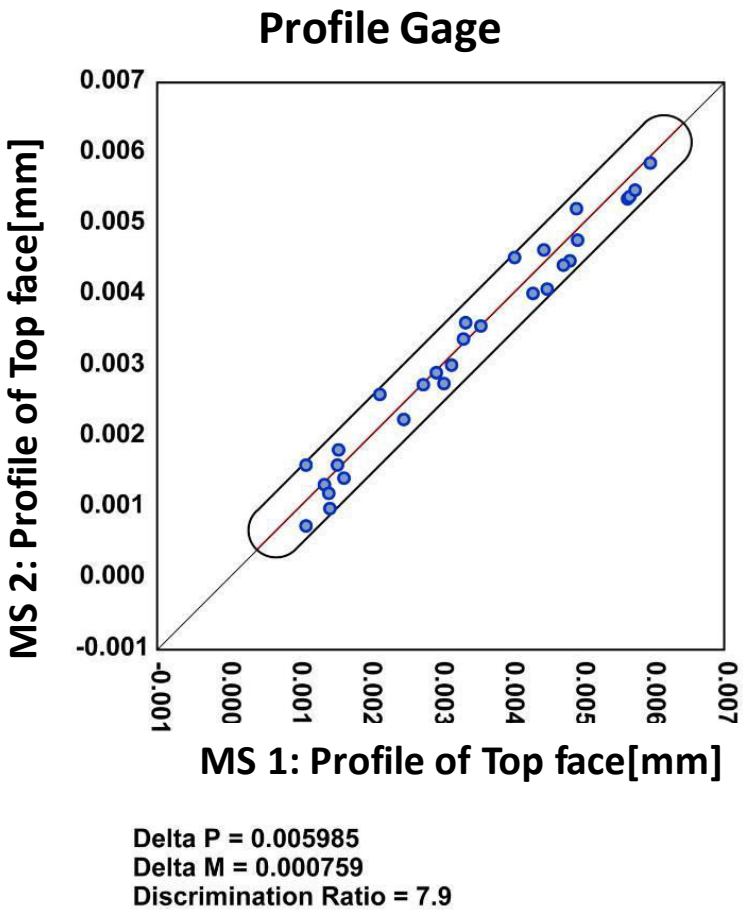




Rationale:

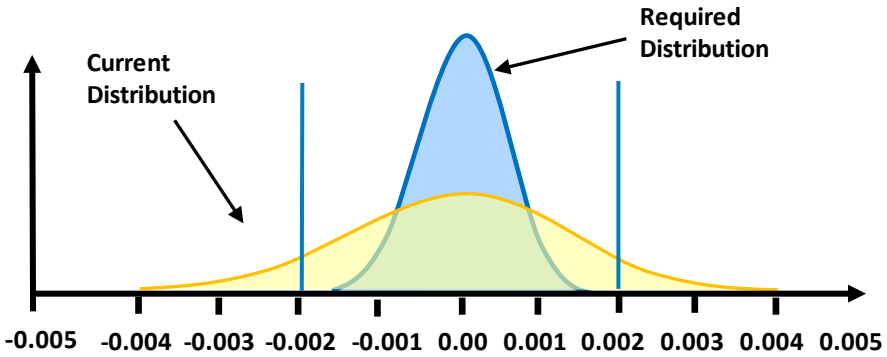
- Isograph

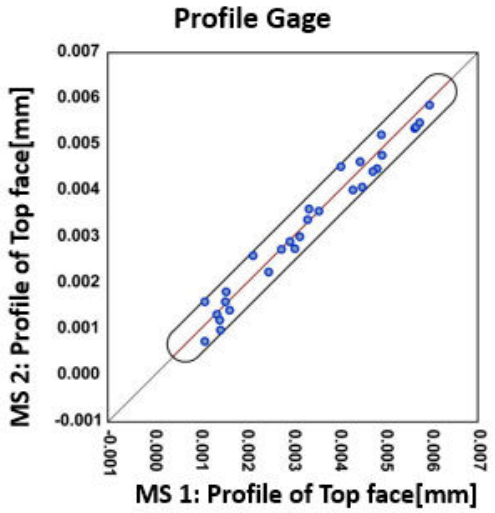
Isograph Profile Gage

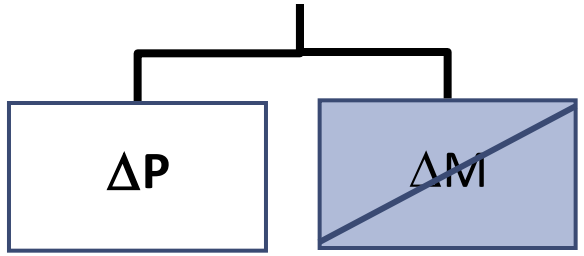


Conclusion: Profile Gage passed the Isograph, because DR ≥ 6.

Solution Tree







Rationale:

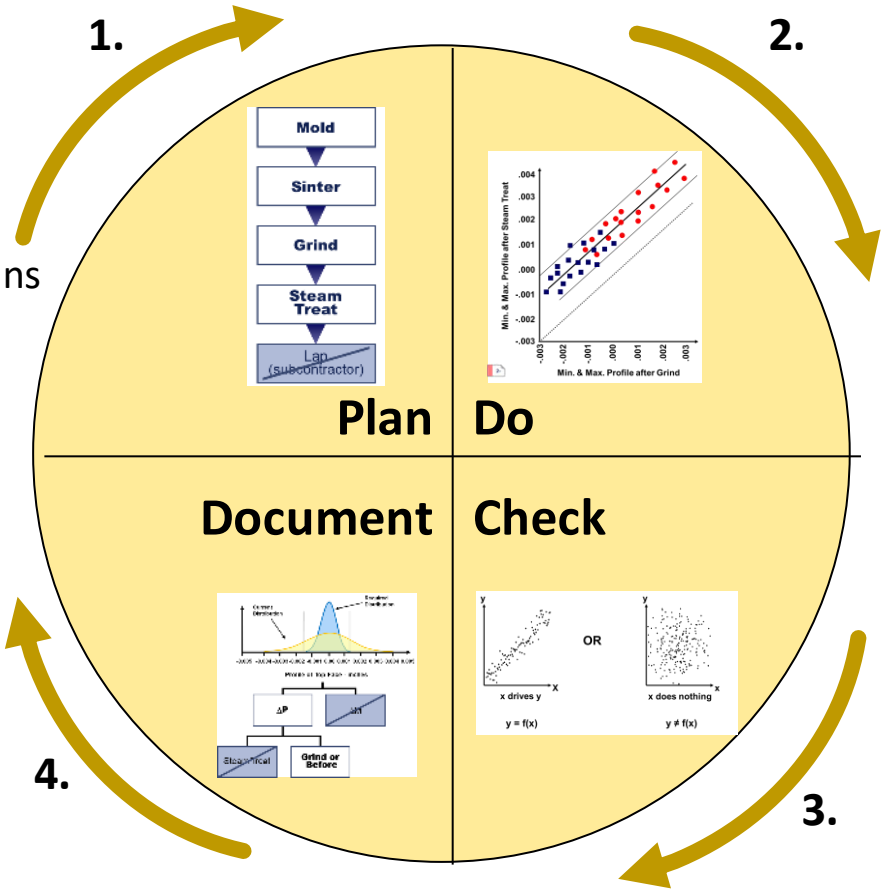
- Passed Isograph. DR = 7,9

What next?

Investigative Cycle

→ **Plan next split on the Solution Tree.**

- What are the process families?
- Can we isolate the process step that contains the BIG X?



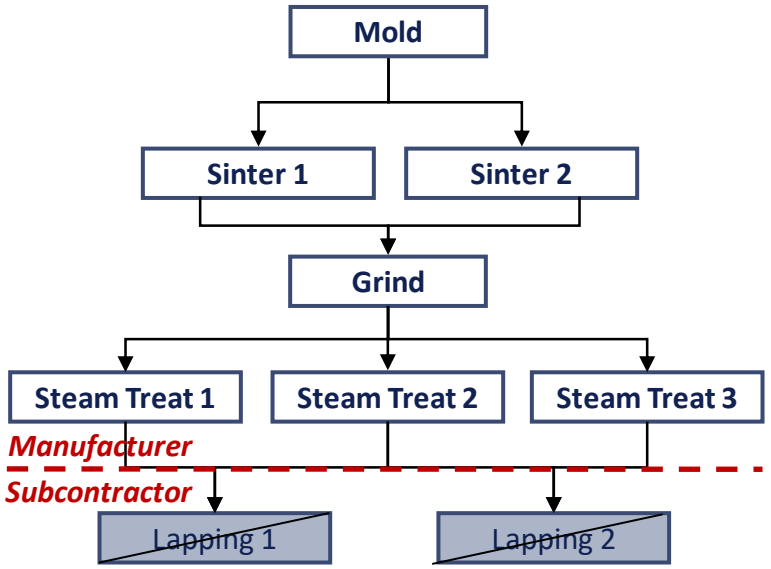
- Take stratified samples.
- Measure Mone(Y) after each process step.
- Graph the results using Y to X scatter plots.

- Where does the BIG X live?

→ **Update the Solution Tree.**

Evolving strategies will be necessary to fill the Solution Tree level by level to identify the BIG X candidate.

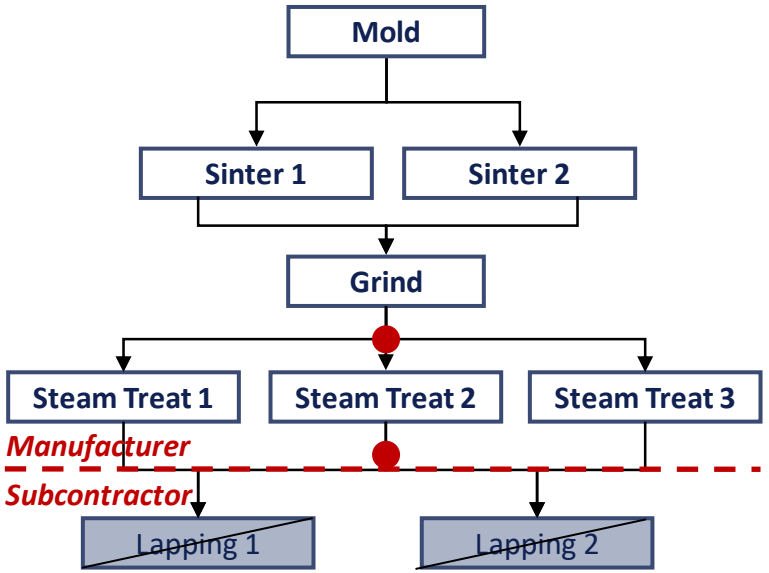
Strategy for Profile of Top Face



Strategy:

- We want to identify the process step that contains the BIG X.
- We eliminated the lapping process at the subcontractor, already.
- The profile can be measured after each process step.
- Check the profile of the top face after one step. Mark the parts and return them to the process.
- Check the profile of the top face after the next step and so on.
- Plot the results on scatter plots.

Strategy for Profile of Top Face



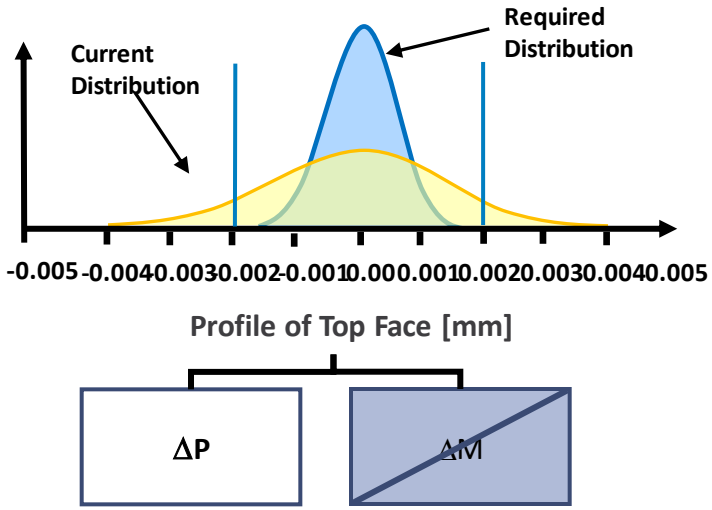
Grind vs Steam Treat

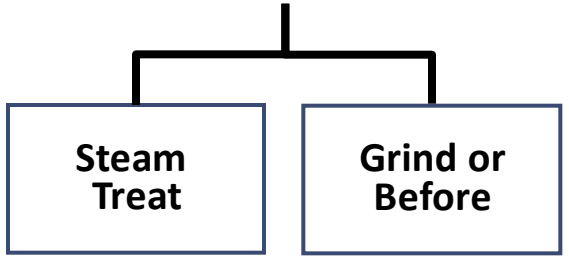
	After Grind	After Steam Treat
1	-0.0025	-0.0005
2	0.005	0.002
3	-0.001	0.0005
4	0.003	0.004
5	0.001	0.0025
…		
…		
…		
n	-0.002	-0.0005

Strategy:

Select 25 - 30 parts after “Grind.” Check profile of top face. Return the parts to the process, marking the parts. Check the profile of top face after steam treat. Plot the results on a scatter plot.

Solution Tree





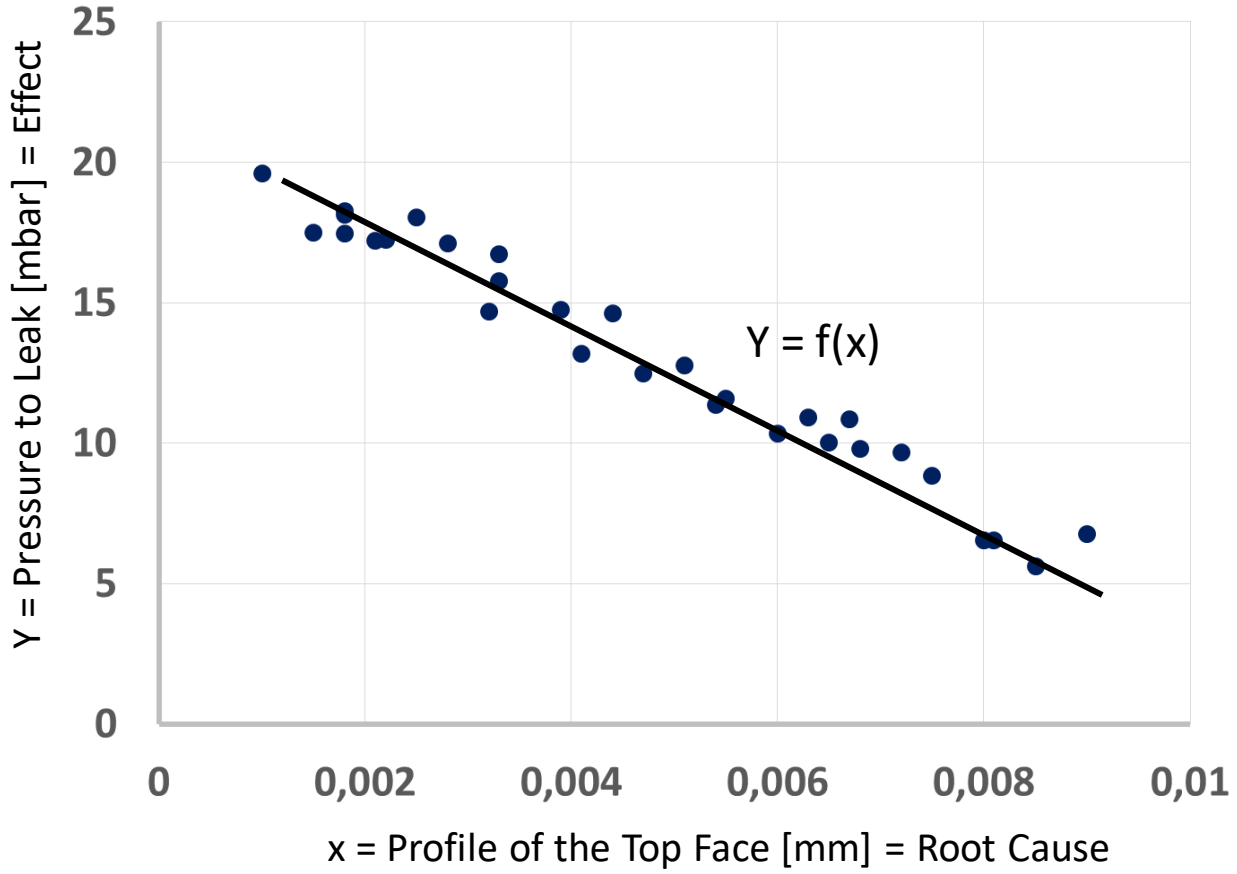
Strategy:

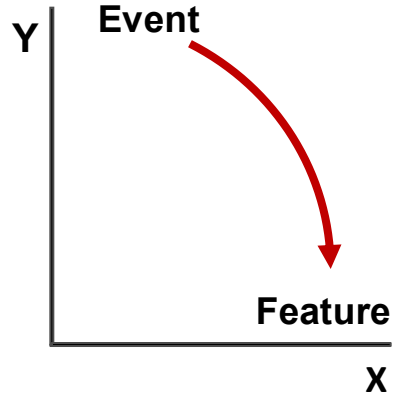
- Use Y → X scatter plots to split the dictionary.

X-Y - Scatter Pot

Why flatness matters …

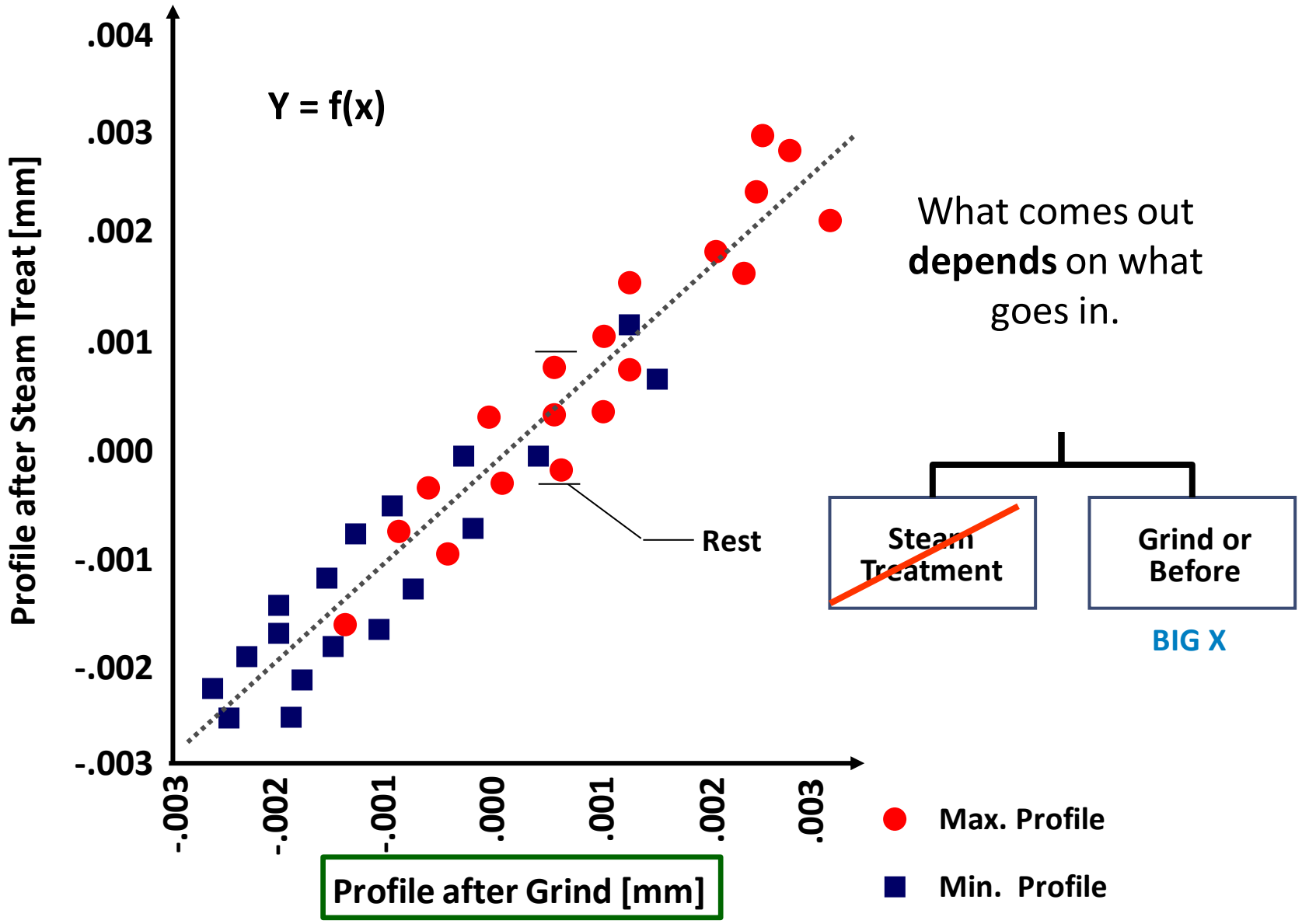
Initial Situation: The pressure to leak of the whole injector is a function of the profile of the spacer surface (flatness).



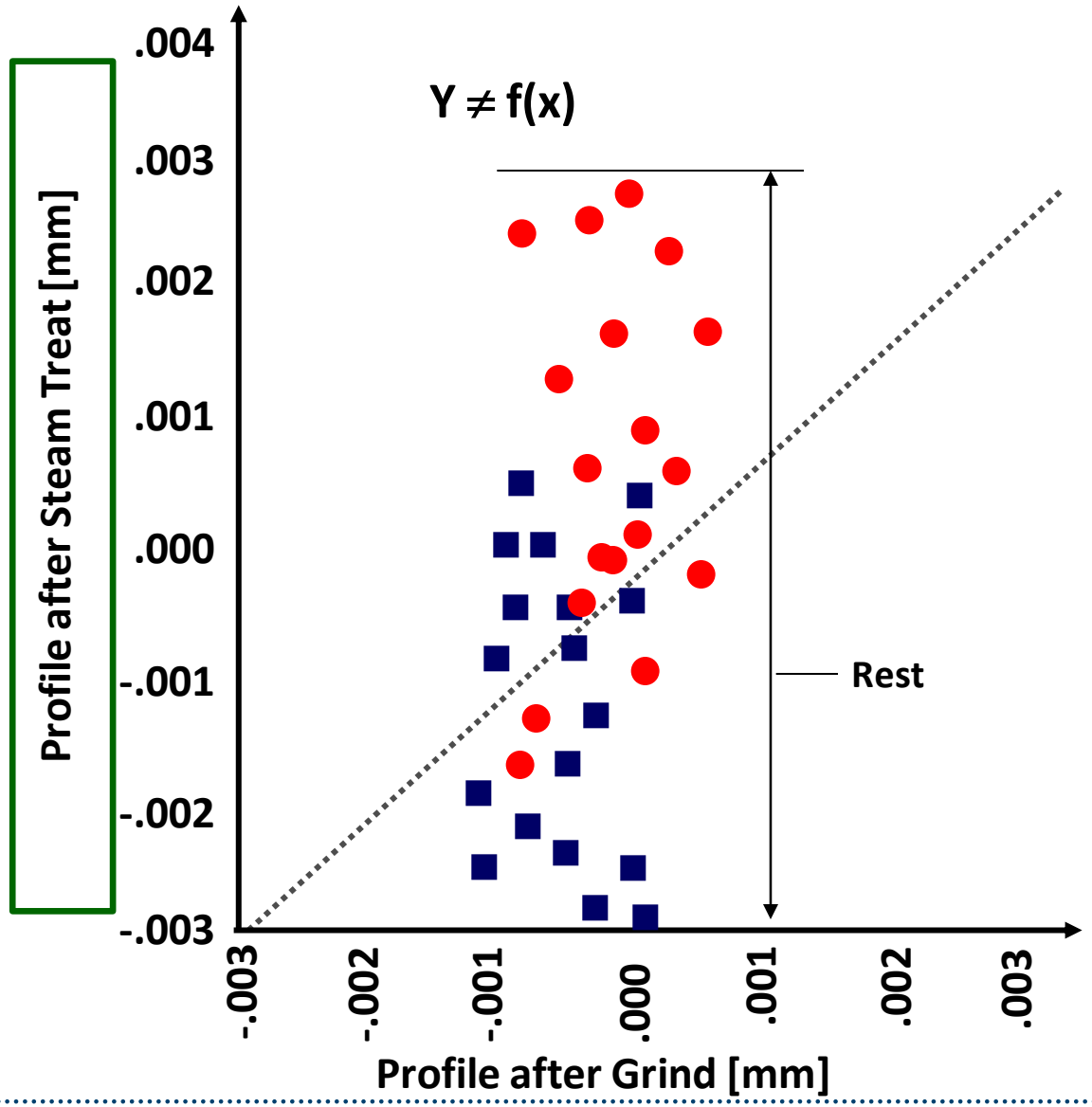




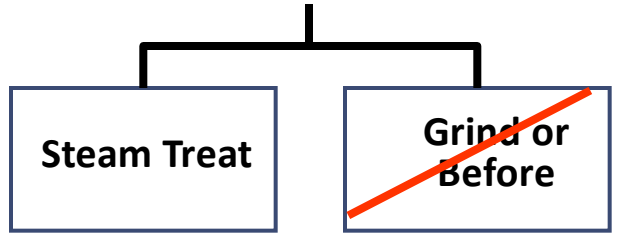
Possible Outcome 1



Possible Outcome 2



What comes out is **independent** of what goes in.

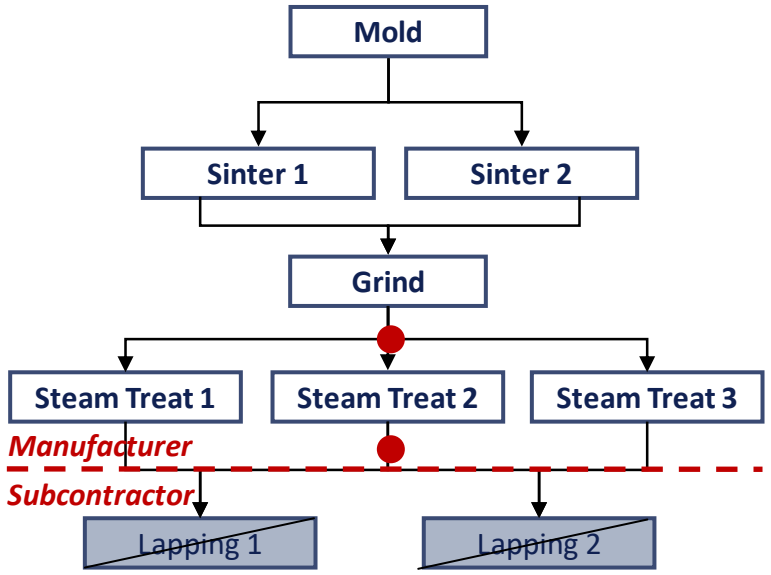


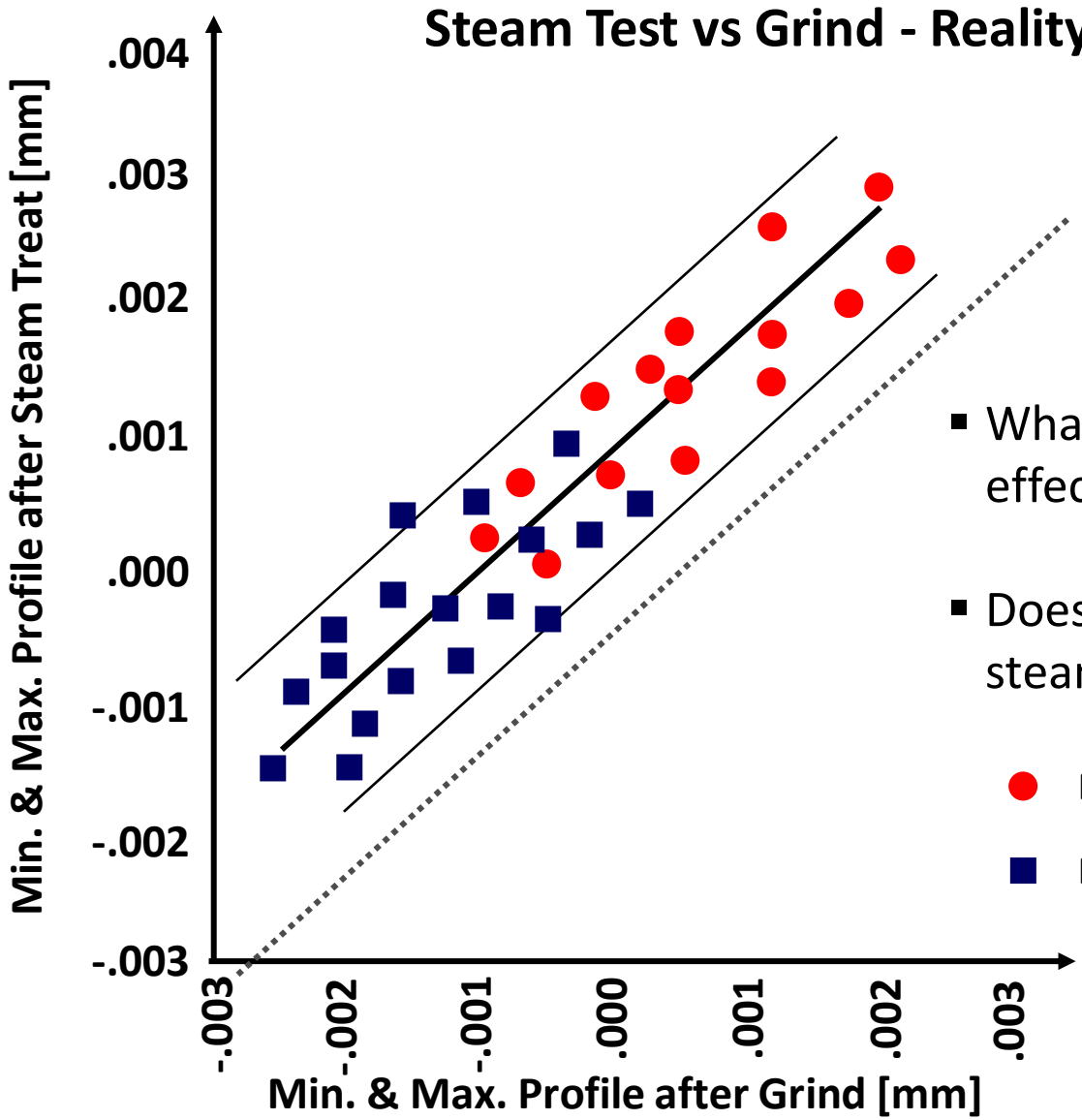
BIG X

Max. Profile

Min. Profile

Scatter Plot #1



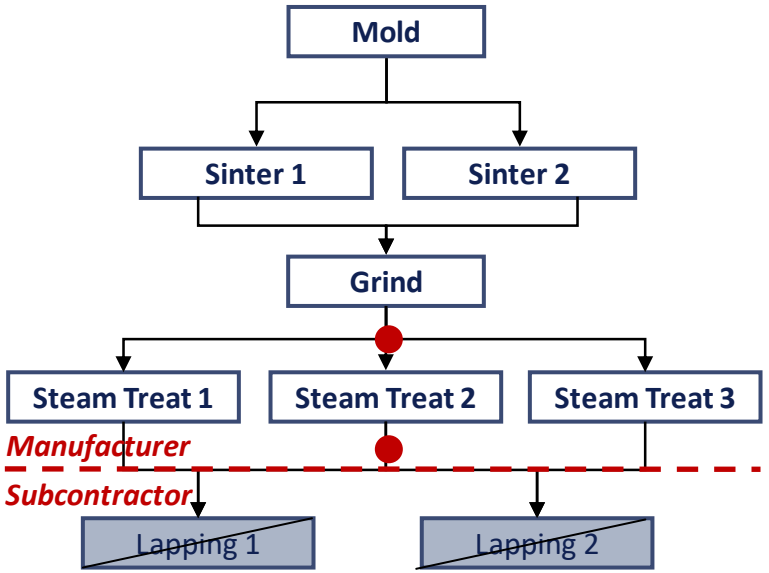


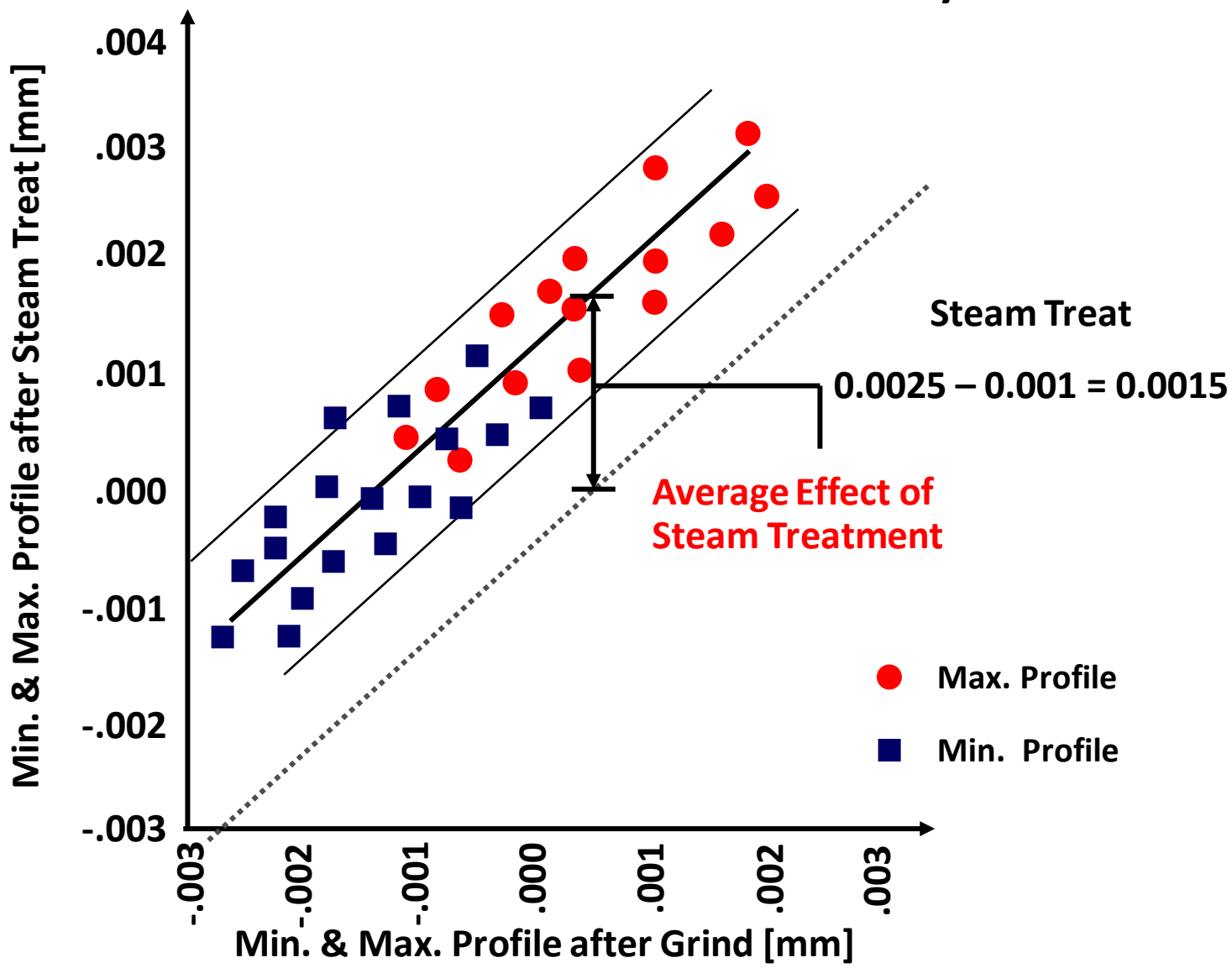
- What is the average effect of steam treat?
- Does the **BIG X** live in steam treat?

Max. Profile

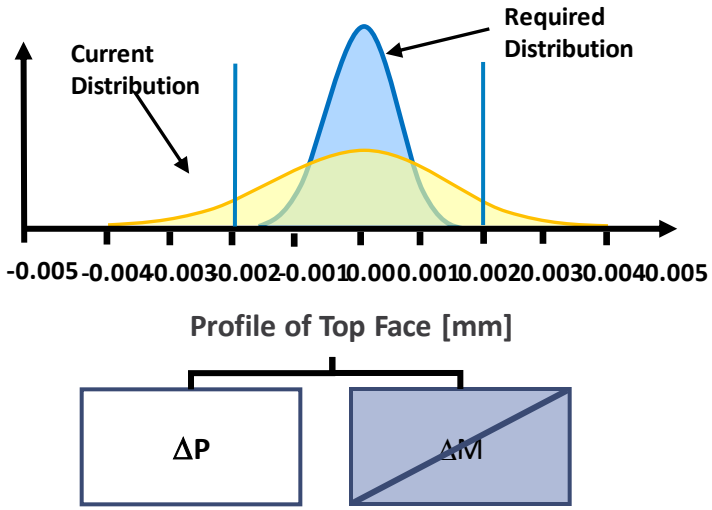
Min. Profile

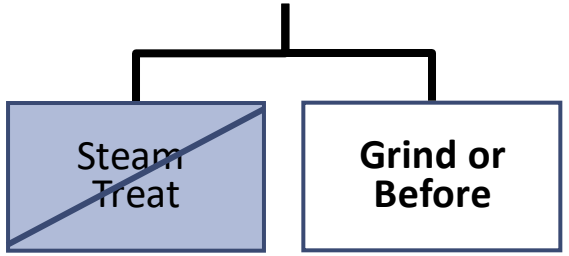
Scatter Plot #1





Solution Tree

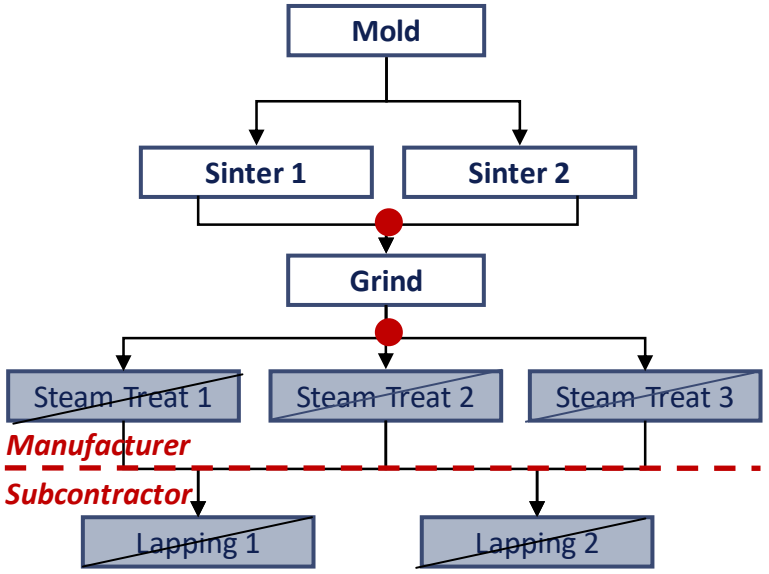




Rationale:

- Scatter plot #1

Strategy for Profile of Top Face



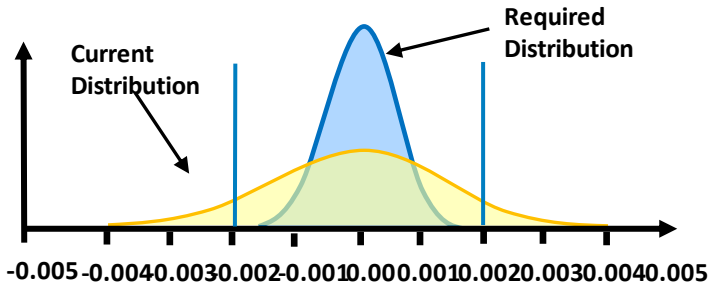
Sinter vs Grind

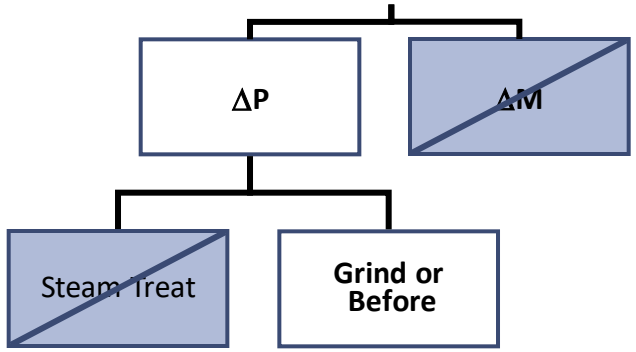
	After Sinter	After Grind
1	-0.002	-0.0025
2	0.0055	0.005
3	-0.001	-0.0005
4	0.003	0.003
5	0.001	0.0015
…		
…		
…		
n	-0.0025	-0.002

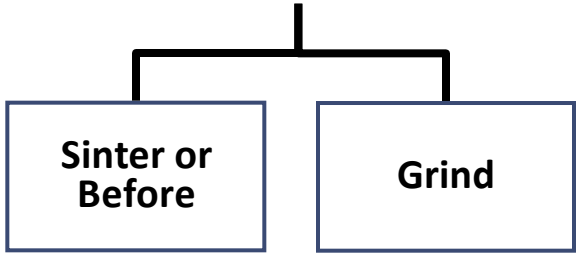
Strategy:

Select 25 - 30 parts after “Sinter.” Check profile of top face. Return the parts to the process, marking the parts. Check the profile of top face after grind. Plot the results on a scatter plot.

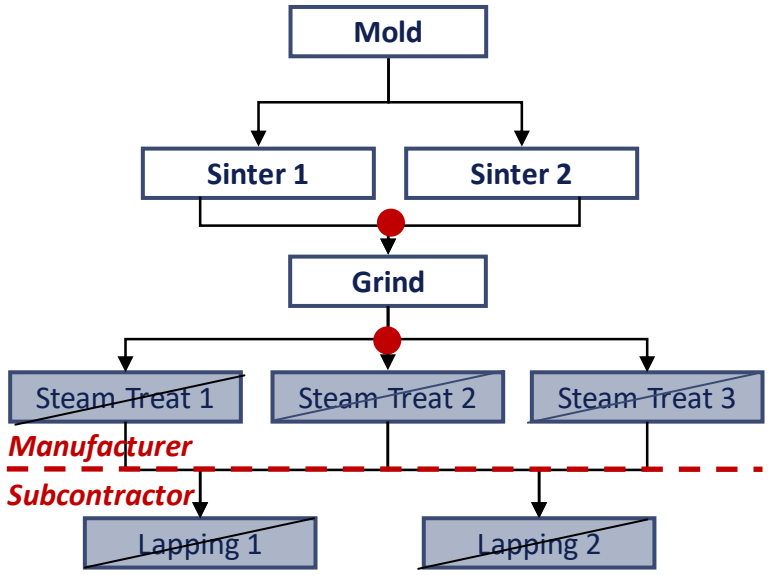
Solution Tree

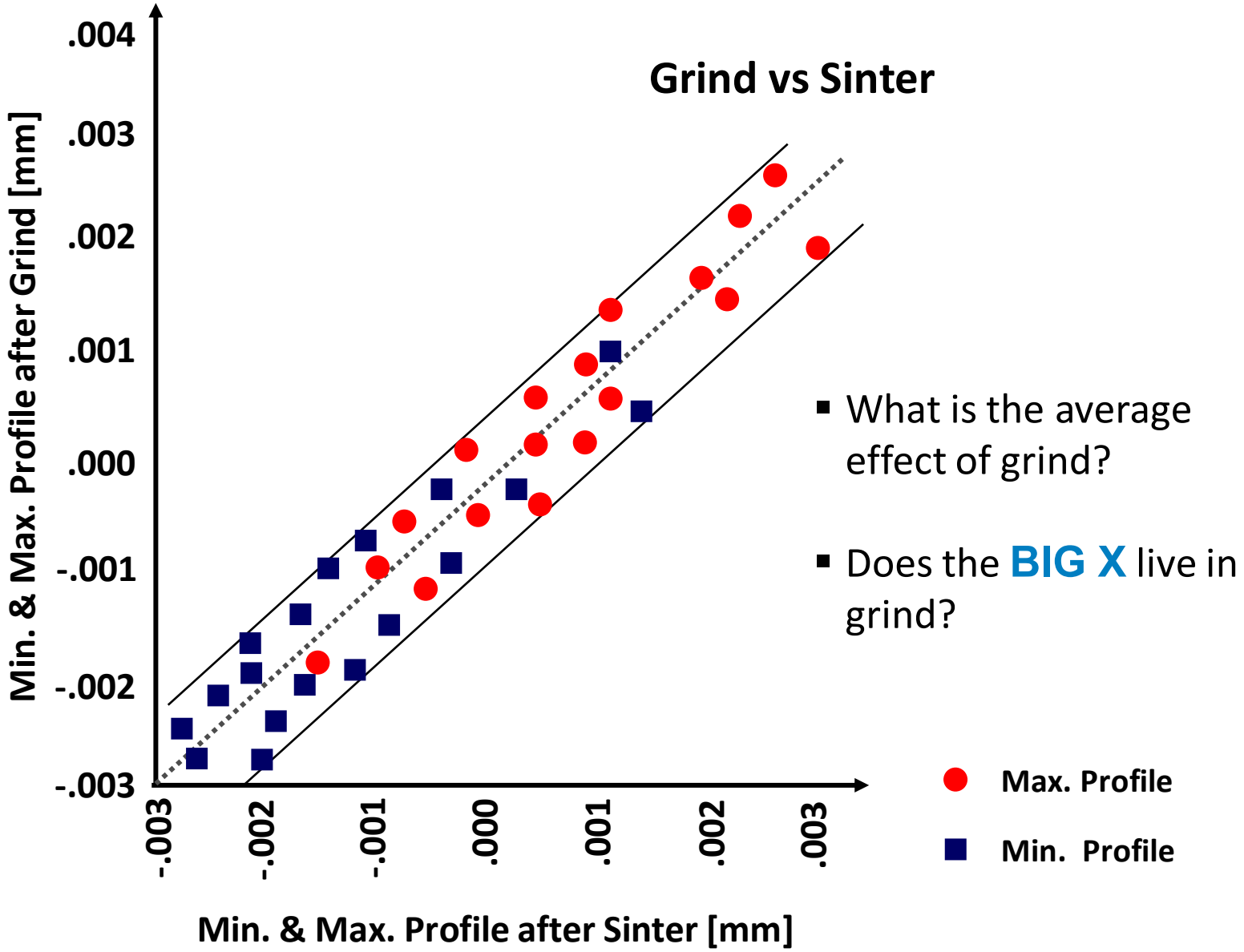






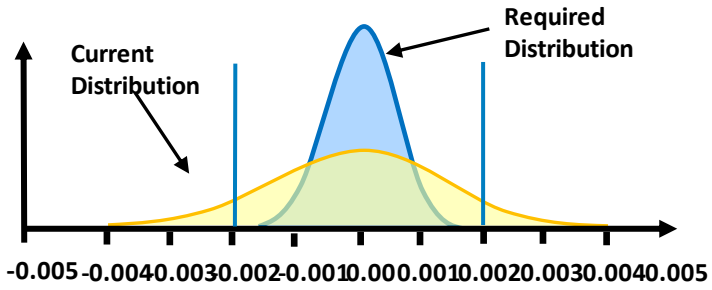
Scatter Plot #2

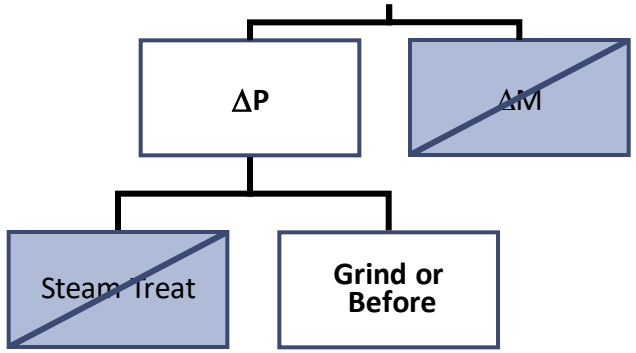


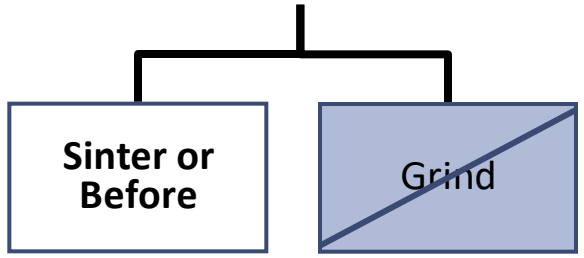


- What is the average effect of grind?
- Does the **BIG X** live in grind?

Solution Tree



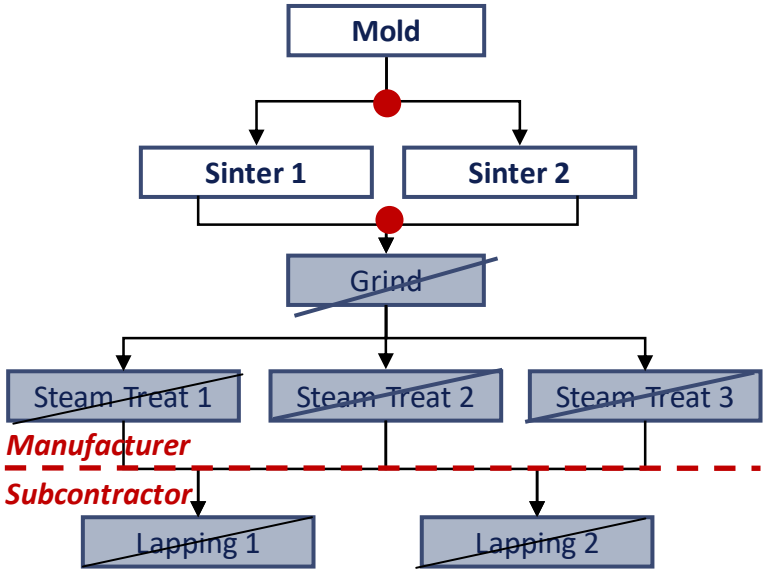




Rationale:

- Scatter plot #2

Strategy for Profile of Top Face



Mold vs Sinter

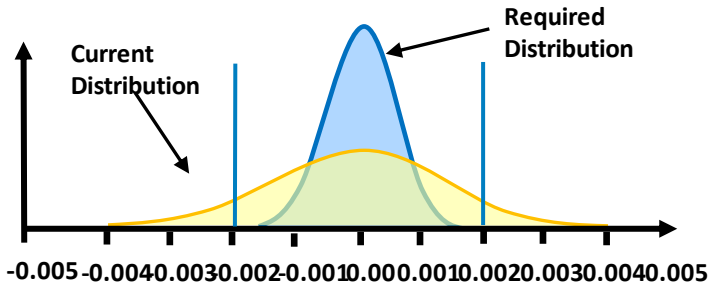
	After Mold	After Sinter
1	-0.001	-0.0025
2	0.0005	0.005
3	-0.001	-0.0005
4	0.0005	0.003
5	0.001	0.0015
…		
…		
…		
n	-0.001	-0.002

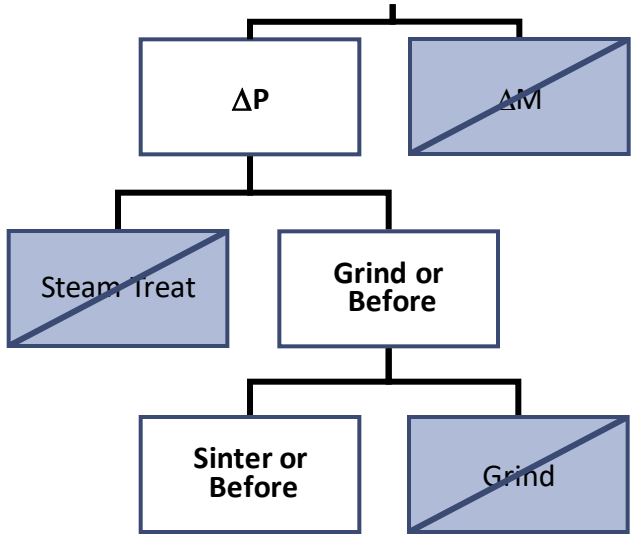
Strategy:

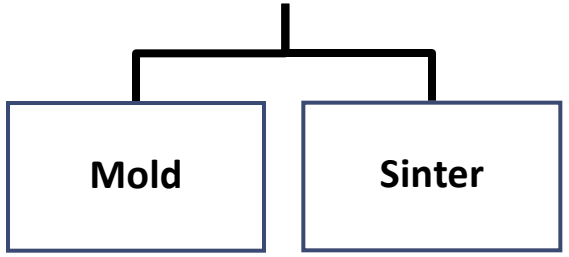
Select 25 - 30 parts after “Mold.” Check profile of top face. Return the parts to the process, marking the parts. Check the profile of top face after sinter. Plot the results on a scatter plot.

Sintered Spacer Case

Solution Tree



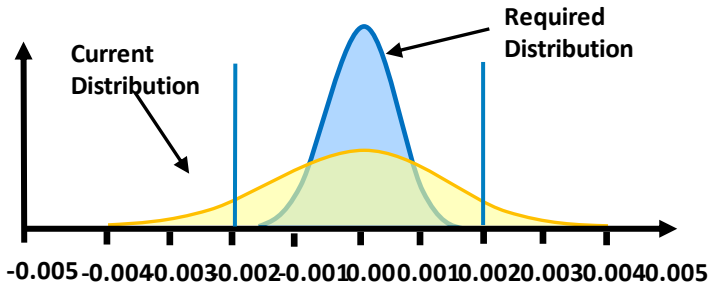


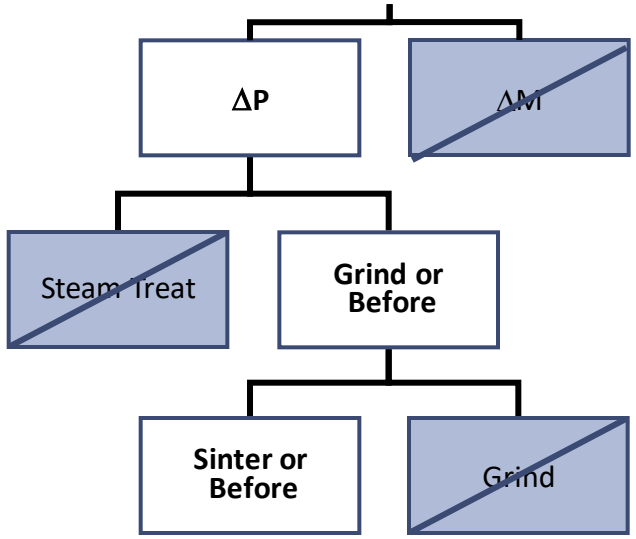


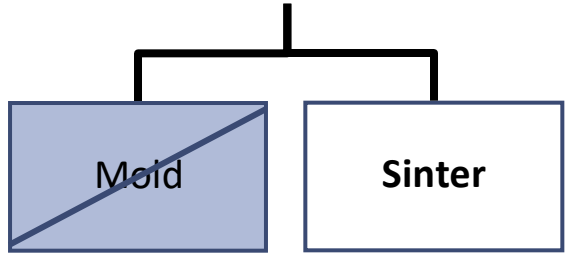
Rationale:

- Scatter plot #3

Sintered Spacer Case Solution Tree





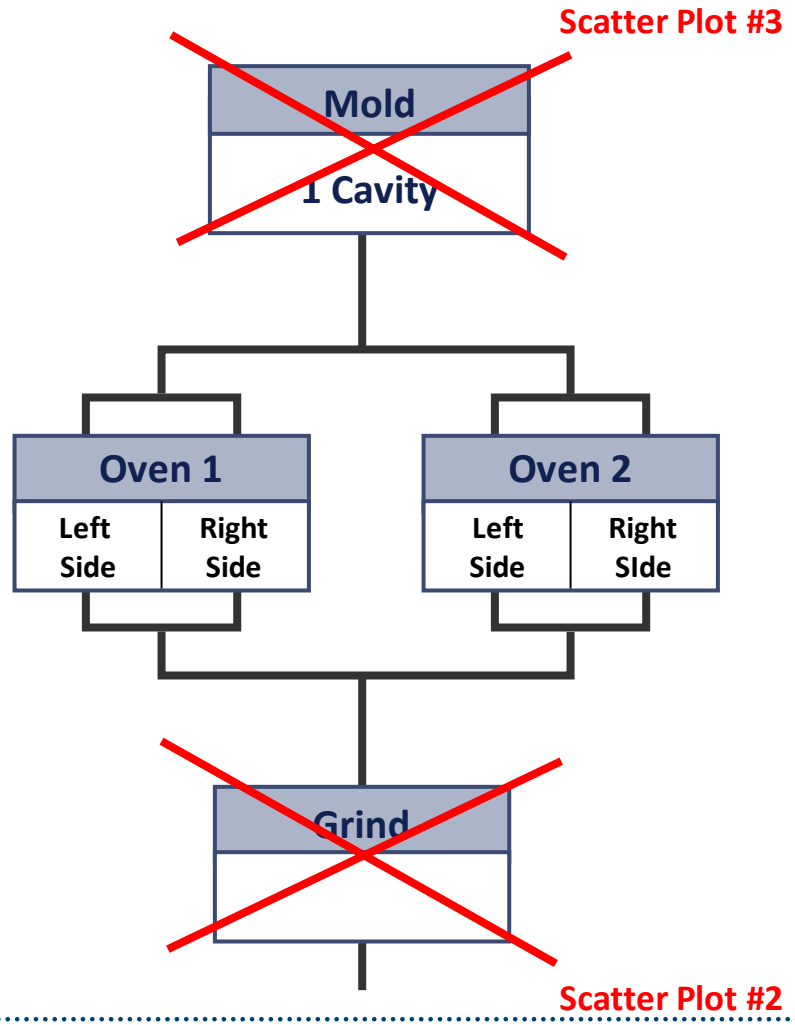


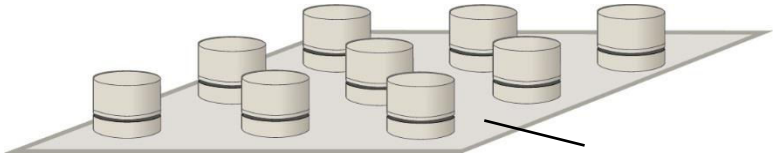
Rationale:

- Scatter plot #3

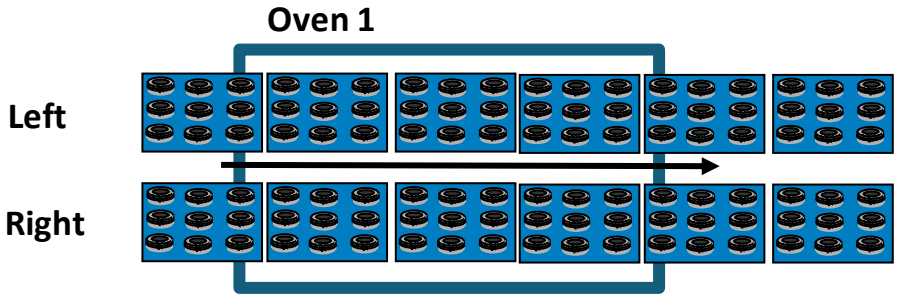
Sintered Spacer Case

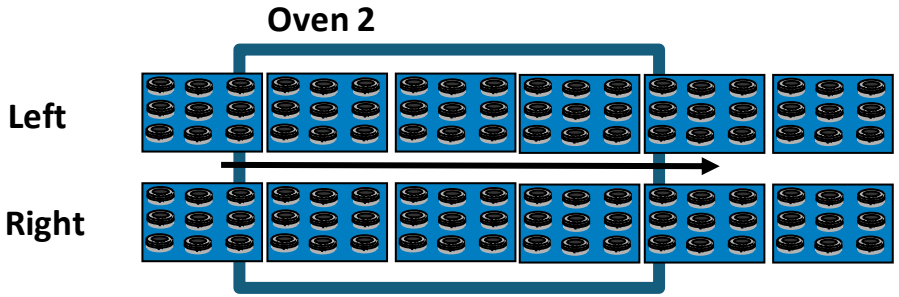
Strategy Diagram





Plate, 9 pieces per plate





Investigative Cycle

→ **Plan next split on the Solution Tree.**

- The BIG X lives in sintering.
- What are the process families?
- Layout Strategy Diagram.



- Take stratified samples.
- Measure Mone(Y).
- Graph the results.

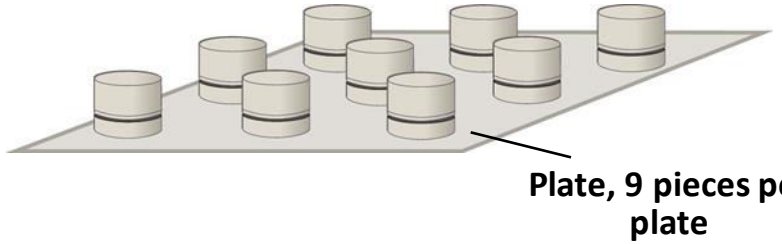
- Has the BOB / WOW difference been captured?
- What's the largest family of variation?

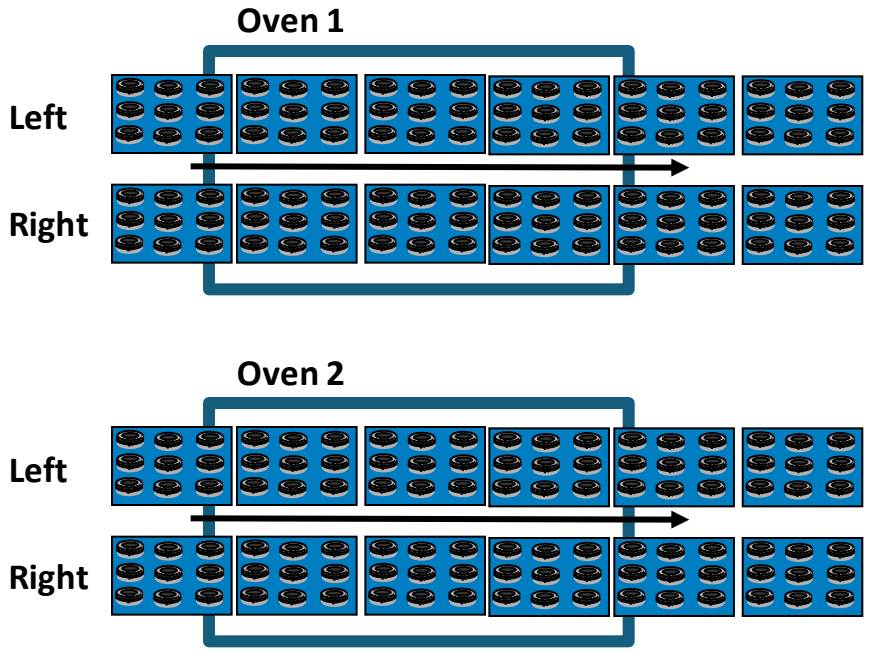
→ Update the Solution Tree.

Evolving strategies will be necessary to fill the Solution Tree level by level to identify the BIG X candidate.

Feature Strategy Diagram - Exercise

- The spacers get sintered on different plates in two continuous belt ovens with two sides each.
- The profile is measured relative to a manufacturing datum.
- Develop a strategy diagram for this case.
- Select the variation families you want to display in a Multi-Vari. Set up the Multi-Vari graph for this case. Describe where you would collect the samples from and how many.
- You will have 15 minutes to complete this exercise.





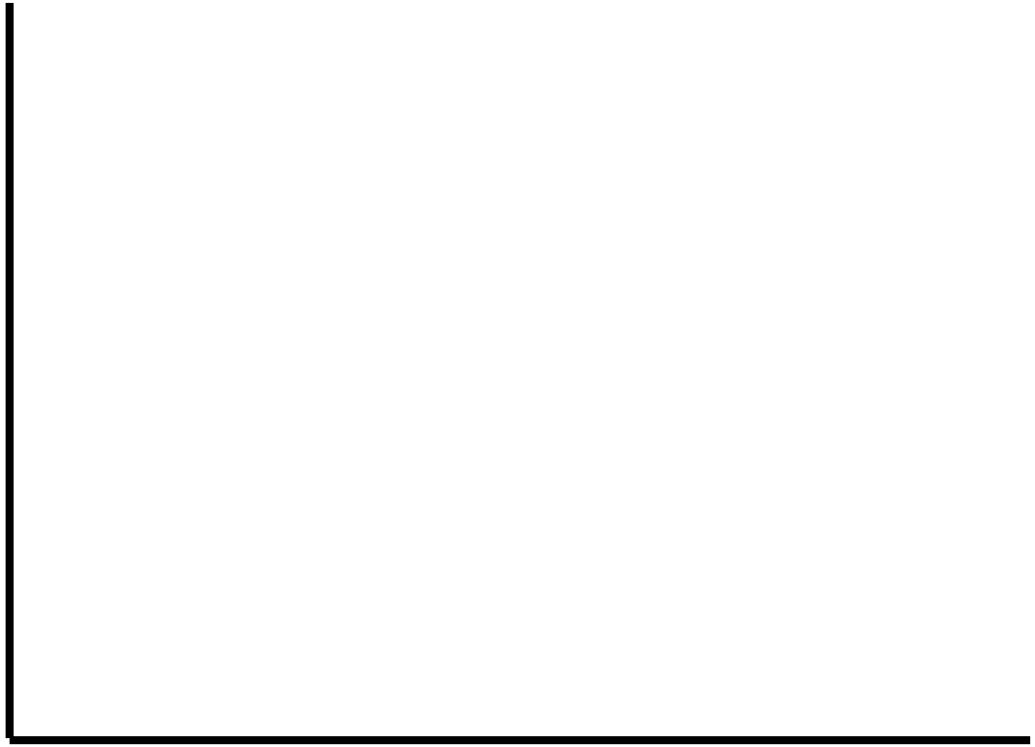
Feature Stategy Diagram - Exercise

Mone(Y) = Profile of the top surface [mm]

Family of Variations, or Contrast

	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast

Multi-Vari



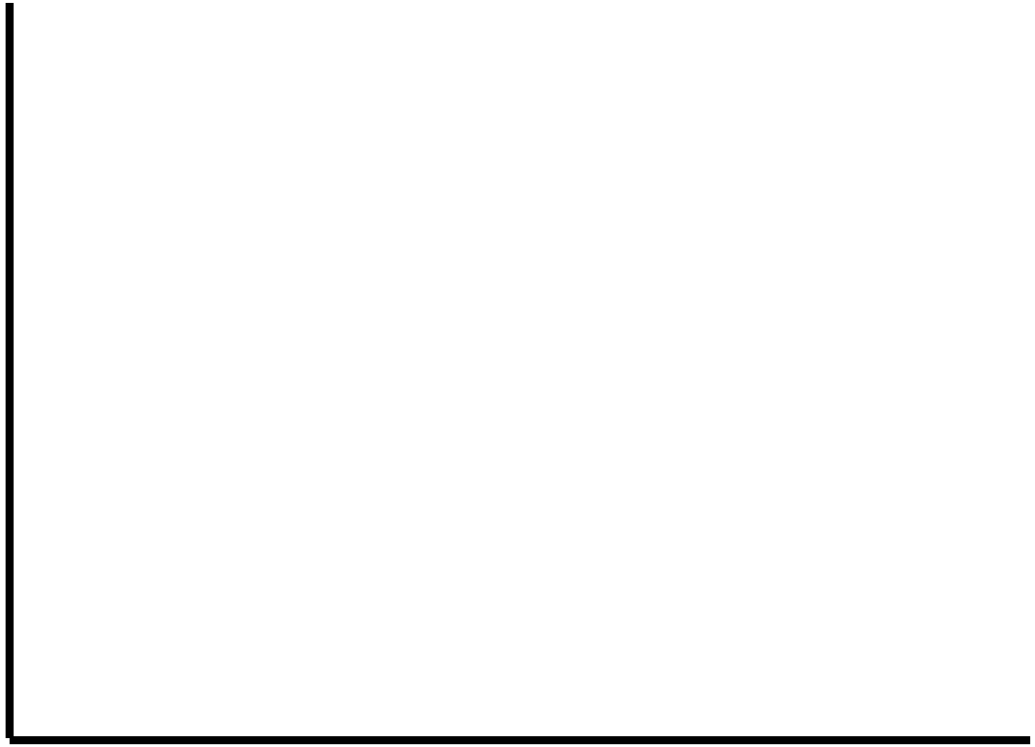
Feature Stategy Diagram - Exercise

Mone(Y) = Profile of the top surface [mm]

Family of Variations, or Contrast

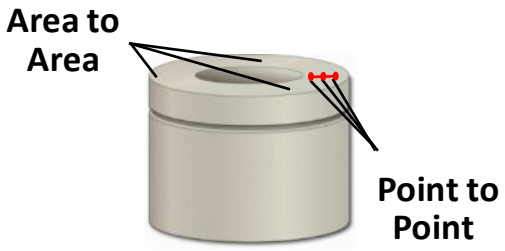
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast
	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast

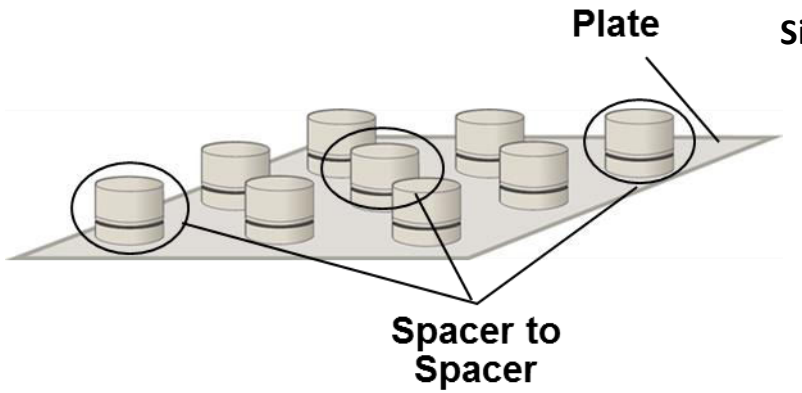
Multi-Vari

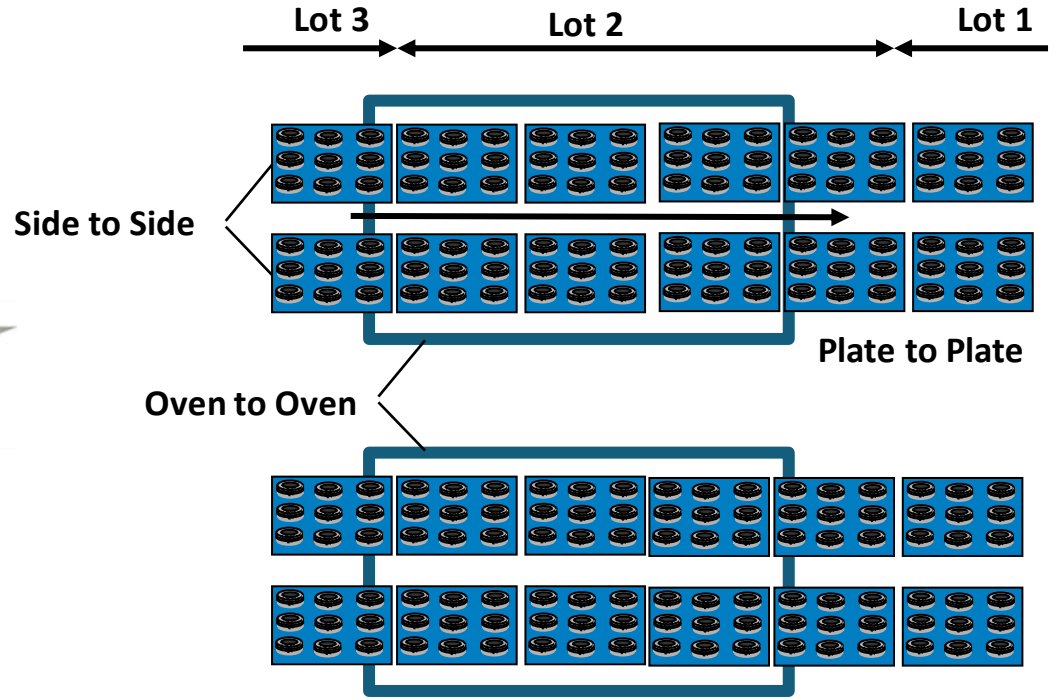


Feature Strategy Diagram - Answer

Mone(Y) = Profile of the top surface [mm]

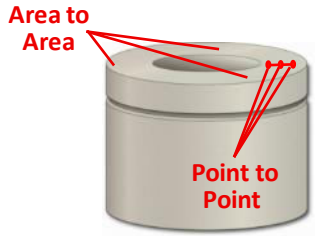






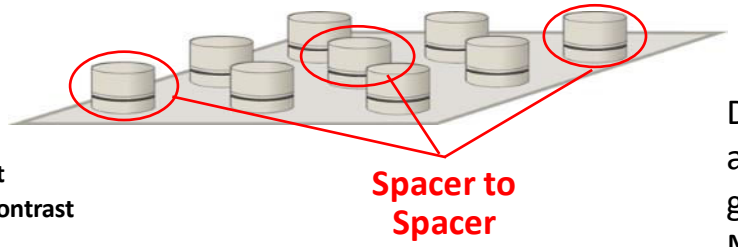
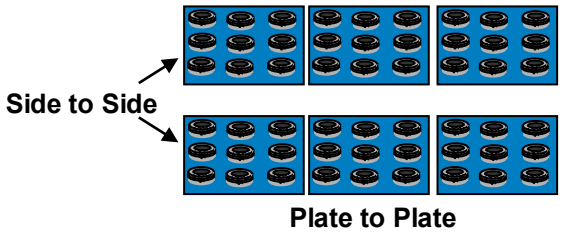
Feature Strategy Diagram – Design Families

Mone(Y) = Profile of the top surface [mm]

Family of Variation, or Contrast	*Diagram*	*What Is Meant by Each Contrast*
Point to Point (Same Area) ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		What does the profile looks like within the same area?
Area to Area (Same Spacer) ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Is there contrast in the profile cut at different areas on a spacer?

Feature Strategy Diagram – Process Families

Mone(Y) = Profile of the top surface [mm]

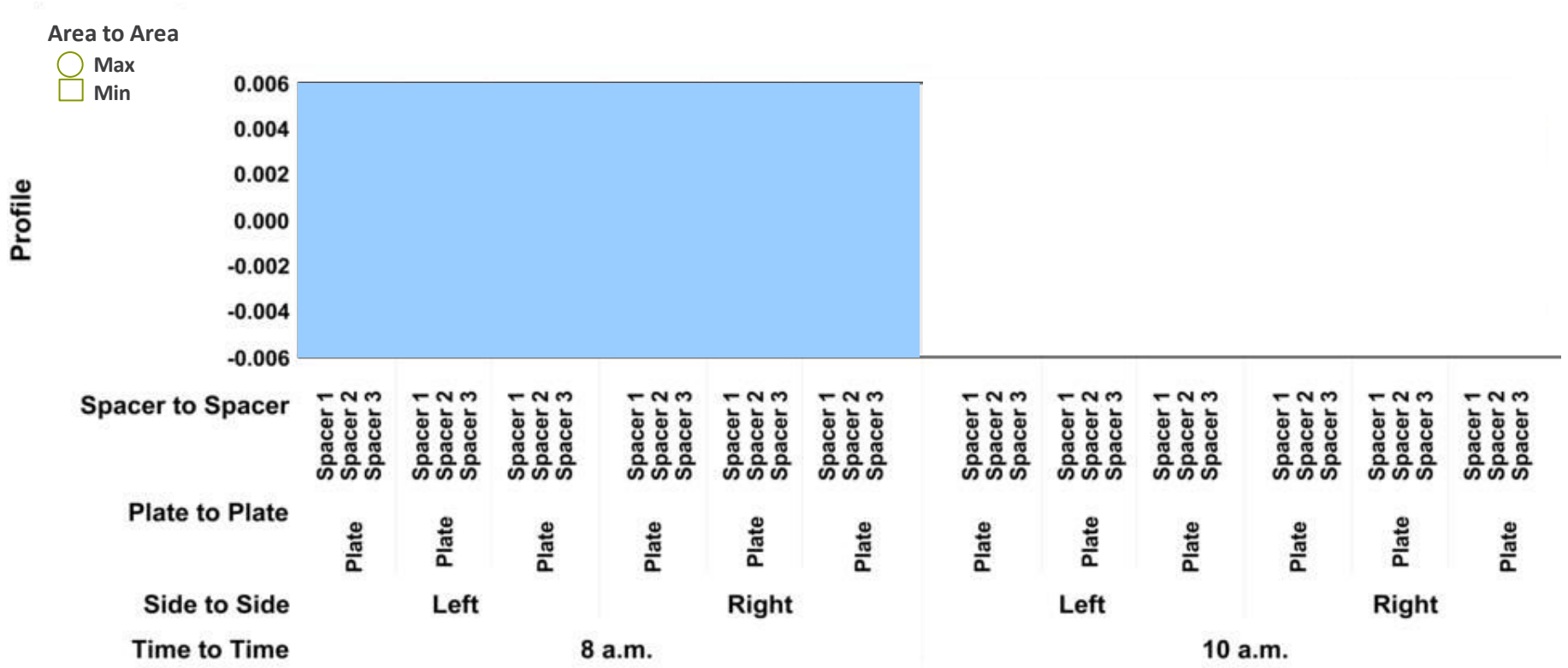
Family of Variation, or Contrast	*Diagram*	*What Is Meant by Each Contrast*
Spacer to Spacer (Same Plate) — ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Do BOB and WOW pieces come in groups, or are they random? Consider sampling in groups of 3 to 5 consecutive pieces. Use Multi-Vari to characterize the process.
Plate to Plate (Same Side) — ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast	 Plate to Plate	Is there contrast across plates?
Side to Side (Same Oven) — ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		With the oven, is there contrast side to side? What will side to side contrast tell us about the oven or sintering process at a given time?
Oven to Oven (Same Lot) — ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Are there multiple ovens performing the same operations? If so, is the oven to oven contrast?

Feature Strategy Diagram – Process Families

Mone(Y) = Profile of the top surface [mm]

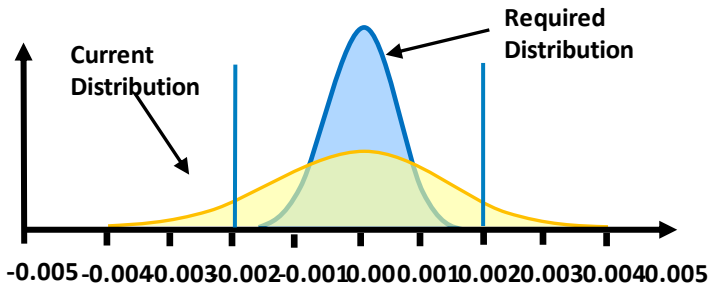
Family of Variation, or Contrast	***Diagram***	***What Is Meant by Each Contrast***
Lot to Lot ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast **Same Time**		How does the production process/scheduling affects the defect rate? Is there material lot contrast?
Time to Time ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Are there time of build contrasts? Hour, shift, day, or sintering process? Is the problem consistent over the same time period?

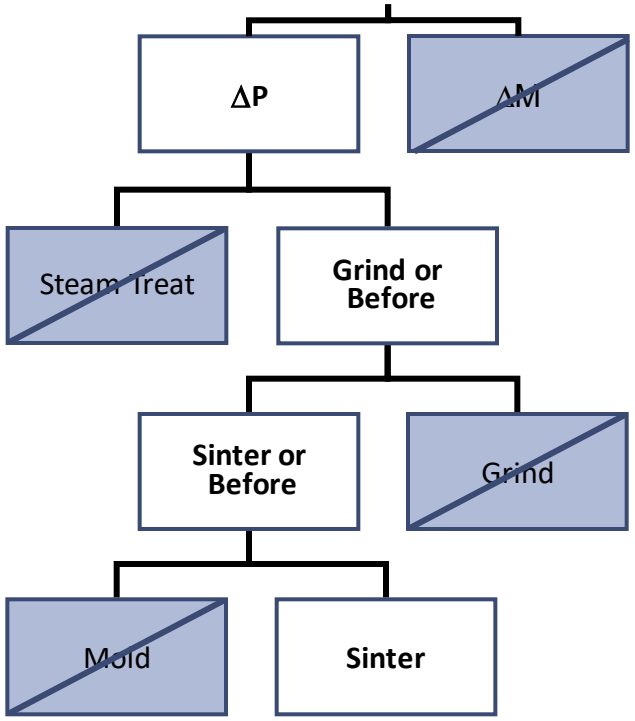
Multi Vari - Answer

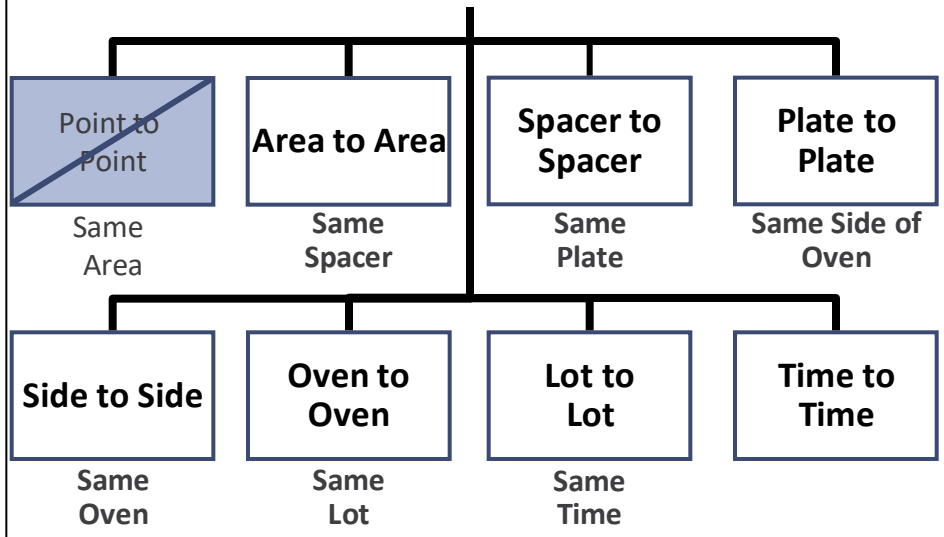


- Area to Area is represented by Min/Max values.
- Usually we take minimum three readings per family of variation.

Solution Tree







Rationale:

- Multi-Vari
- Point to Point is eliminated because of contour plot

Create a Multi-Vari - Exercise

- Create a Multi-Vari recognizing the design and process families of the Strategy Diagram.
- The Mone(Y) is the minimum and maximum value in profile depth in [mm].
- The available data for the Multi-Vari is listed on the next pages.
- Input the data in the Multi-Vari table (Excel or Minitab).
- Fill the Strategy Diagram.
- Identify the BIG X family.
- You will have 20 minutes to complete this exercise.

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Create a Multi-Vari - Exercise

Mone(Y) = Profile of the top surface [mm]

8 a.m. / Left Side

Min Max	Plate 2	Plate 5	Plate 9
Spacer 1	-4 / +4	-1 / +0.5	-1.5 / +1.5
Spacer 2	-5 / +4.5	-0.5 / +1	-2 / +2
Spacer 3	-4.5 / +5	0 / +0.5	-1.5 / +2

8 a.m. / Right Side

Min Max	Plate 2	Plate 5	Plate 9
Spacer 1	-5 / +4	0 / +1	-1.5 / +2
Spacer 2	-5.5 / +4.5	-1 / +0.5	-2 / +1.5
Spacer 3	-4.5 / +4.5	-0.5 / 0	-1.5 / +1.5

10 a.m. / Left Side

Min Max	Plate 2	Plate 5	Plate 9
Spacer 1	-5 / +5	-0.5 / +0.5	-2.5 / +2.5
Spacer 2	-4.5 / +4	0 / +0.5	-2 / +2
Spacer 3	-4 / +4.5	0 / +1	-2.5 / +3.5

10 a.m. / Right Side

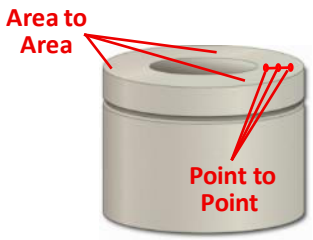
Min Max	Plate 2	Plate 5	Plate 9
Spacer 1	-4.5 / +5.5	-0.5 / 0	-2 / +2.5
Spacer 2	-5 / +4.5	0 / +1	-3 / +3
Spacer 3	-4 / +4	-0.5 / +0.5	-3 / +2.5

- For this exercise values like -0.005 are shown as -5 to make the data input faster.
- To keep this exercise manageable, the data set has been condensed.
- LSL = -1,8 / USL = 1,8

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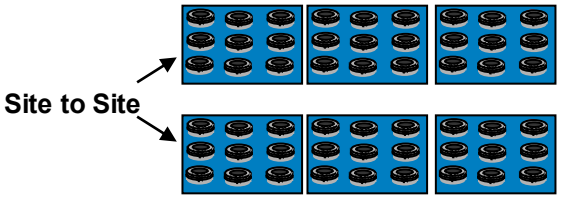
Feature Strategy Diagram – Design Families - Exercise

Mone(Y) = Profile of the top surface [mm]

Family of Variation, or Contrast	*Diagram*	*What Is Meant by Each Contrast*
Point to Point Same Area ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		What does the profile looks like within the same area?
Area to Area Same Spacer ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Is there contrast in the profile cut at different areas on a spacer?

Feature Strategy Diagram – Process Families - Exercise

Mone(Y) = Profile of the top surface [mm]

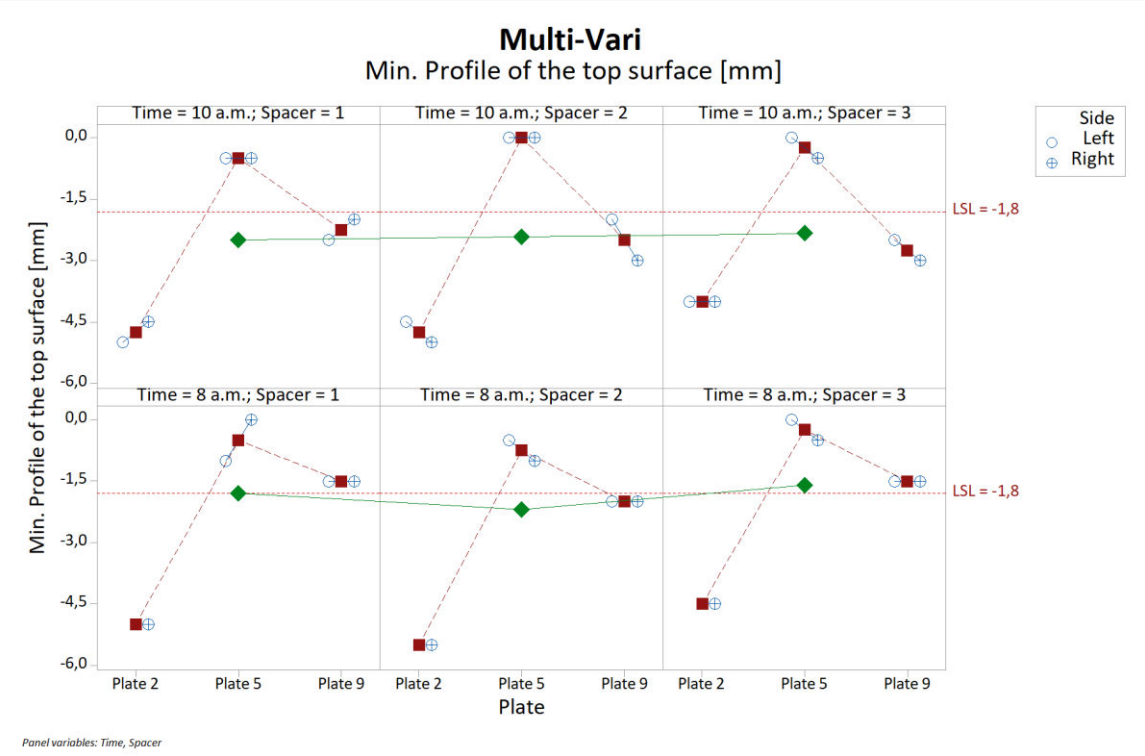
Family of Variation, or Contrast	***Diagram***	***What Is Meant by Each Contrast***
Spacer to Spacer (Same Plate) ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast	 **Spacer to Spacer**	Do BOB and WOW pieces come in groups, or are they random? Consider sampling in groups of 3 to 5 consecutive pieces. Use Multi-Vari to characterize the process.
Plate to Plate (Same Side) ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Is there contrast across plates?
Side to Side (Same Oven) ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast	 Site to Site Plate to Plate	With the oven, is there contrast side to side? What will side to side contrast tell us about the oven or sintering process at a given time?
Oven to Oven (Same Lot) ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Are there multiple ovens performing the same operations? If so, is the oven to oven contrast?

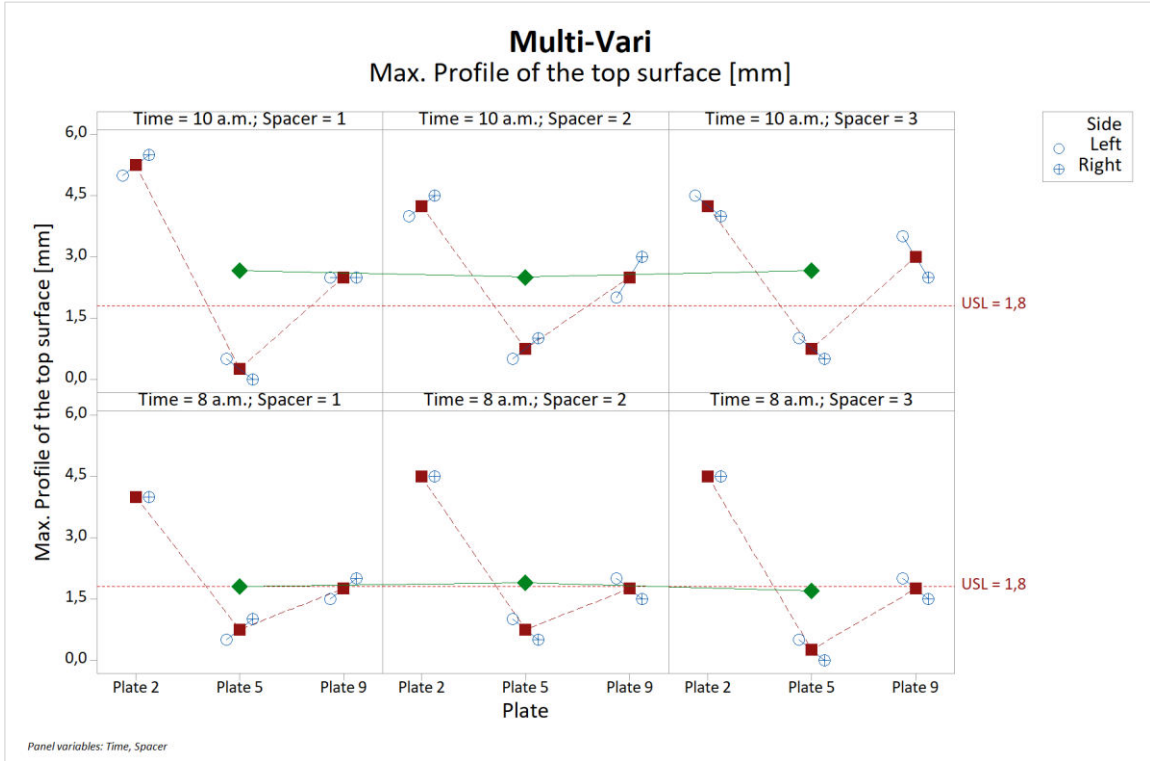
Feature Strategy Diagram – Process Families - Exercise

Mone(Y) = Profile of the top surface [mm]

Family of Variation, or Contrast	***Diagram***	***What Is Meant by Each Contrast***
Lot to Lot ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast **Same Time**		How does the production process/scheduling affects the defect rate? Is there material lot contrast?
Time to Time ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Are there time of build contrasts? Hour, shift, day, or sintering process? Is the problem consistent over the same time period?

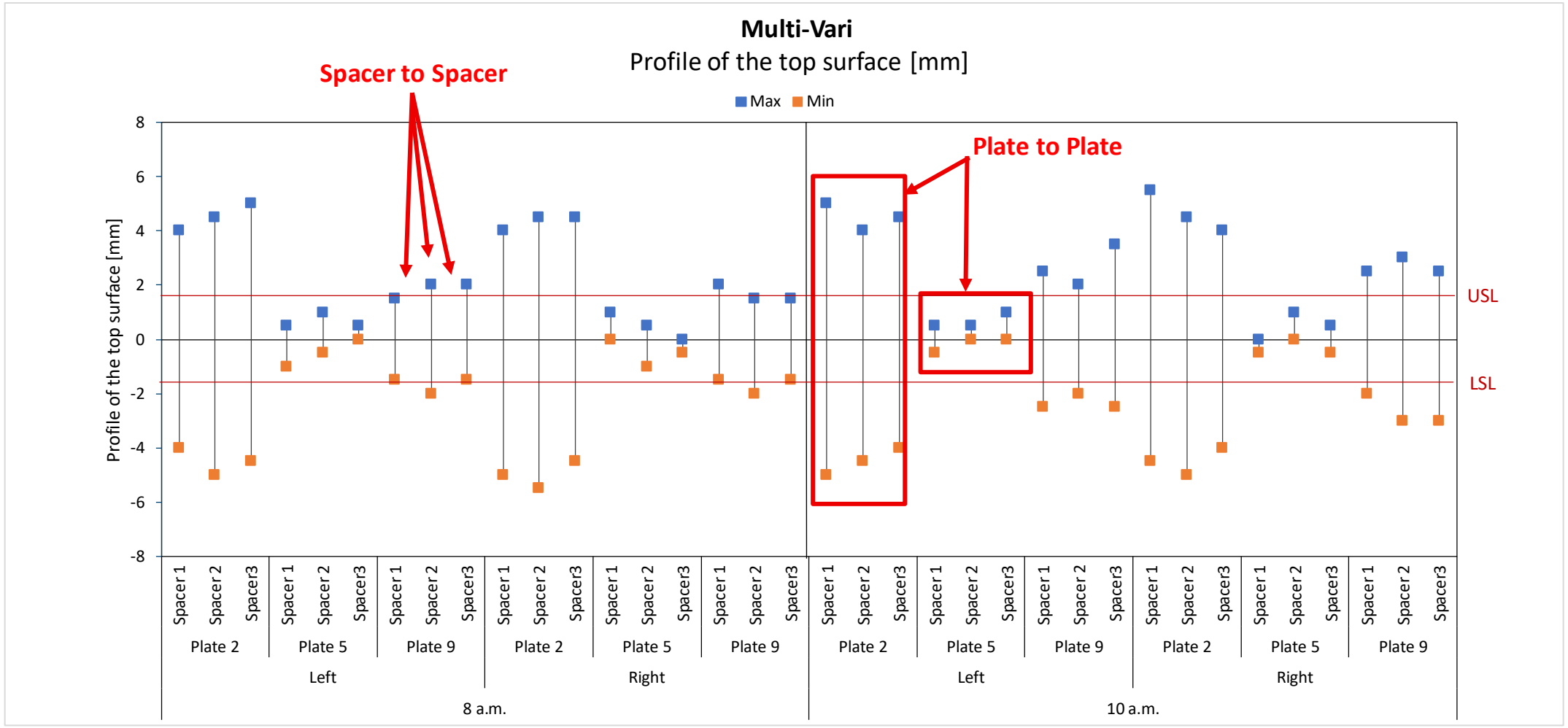
Multi-Vari - Answer





Conclusion: BIG X lives in plate to plate.

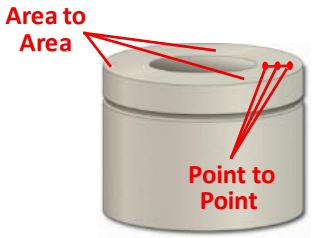
Multi-Vari - Answer



Conclusion: BIG X lives in plate to plate.

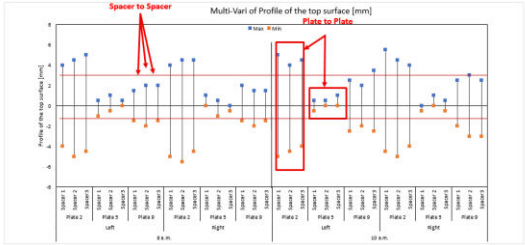
Feature Strategy Diagram – Design Families - Answer

Mone(Y) = Profile of the top surface [mm]

Family of Variation, or Contrast	*Diagram*	*What Is Meant by Each Contrast*
Point to Point Same Area ☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast		For all practical purposes, no contrast exists point to point within the same area.
Area to Area Same Spacer ☐ Do Not Know ■ Large Contrast ☐ Small or No Contrast		Some contrast does exist area to area since some of the part do clean up. This is our Mone(Y).

Feature Strategy Diagram – Design Families - Answer

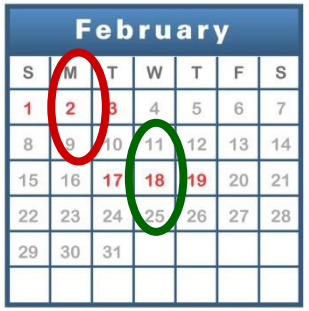
Mone(Y) = Profile of the top surface [mm]

Family of Variation, or Contrast		***Diagram***	***What Is Meant by Each Contrast***
Spacer to Spacer **Same Plate**	Do Not Know Large Contrast Small or No Contrast		Contrast exists spacer to spacer since some clean up and some do not. Parts from the same plate however show substantially less contrast.
Plate to Plate Same Side	Do Not Know Large Contrast Small or No Contrast		Large contrast seen plate to plate on the same side of the oven. Provides excellent leverage for problem solving.
Side to Side **Same Oven**	Do Not Know Large Contrast Small or No Contrast		Some contrast can be seen side to side, but substantially less than plate to plate.
Oven to Oven Same Lot	Do Not Know Large Contrast Small or No Contrast		Did not check oven to oven contrast since BIG X magnitude was seen plate to plate.

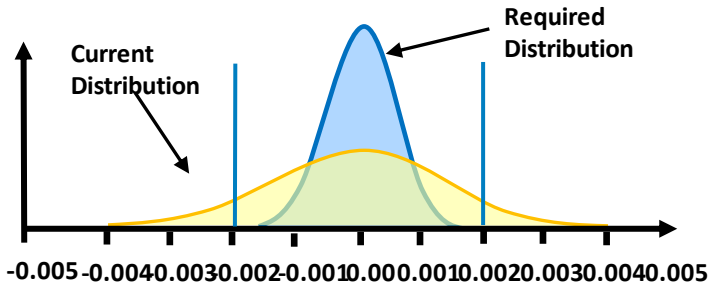
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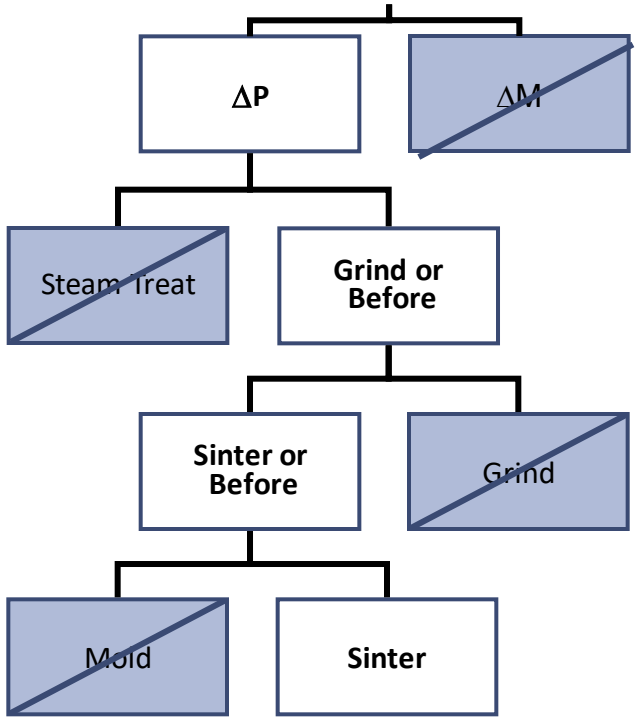
Feature Strategy Diagram – Design Families - Answer

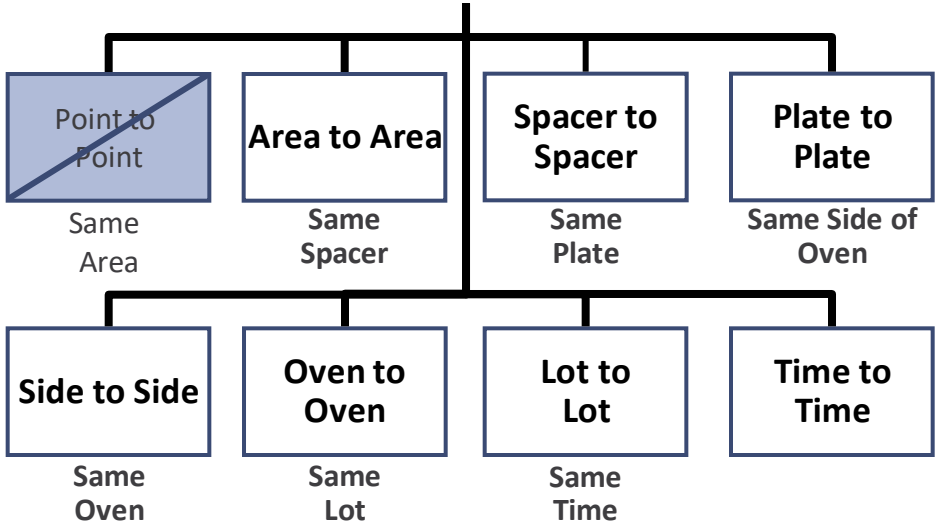
Mone(Y) = Profile of the top surface [mm]

Family of Variation, or Contrast		*Diagram*	*What Is Meant by Each Contrast*
Lot to Lot Same Time	■ Do Not Know □ Large Contrast □ Small or No Contrast		Did not check lot to lot contrast since BIG X magnitude was seen plate to plate.
Time to Time	□ Do Not Know □ Large Contrast ■ Small or No Contrast		Contrasts exits Time to time but is presently confounded with lot to lot.

Solution Tree

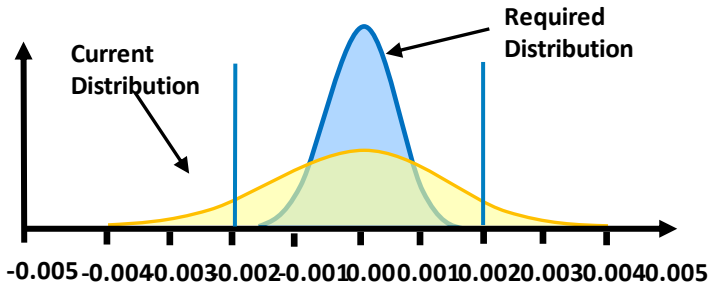


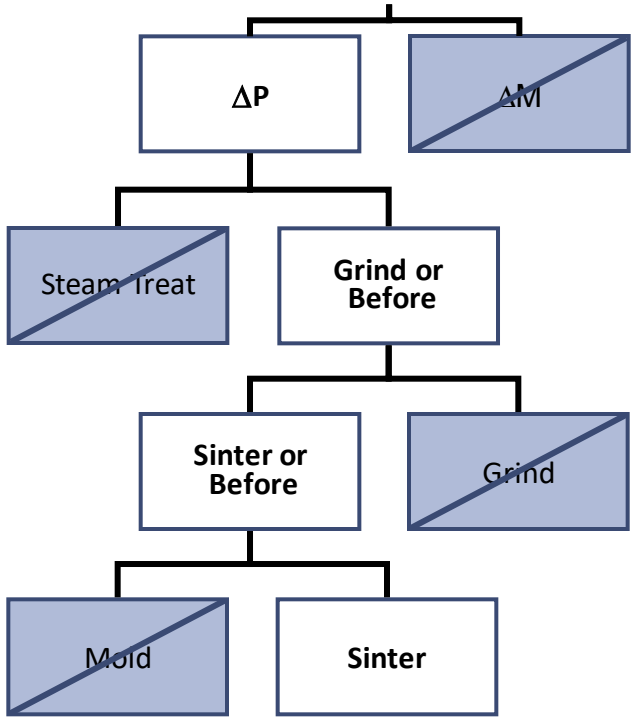


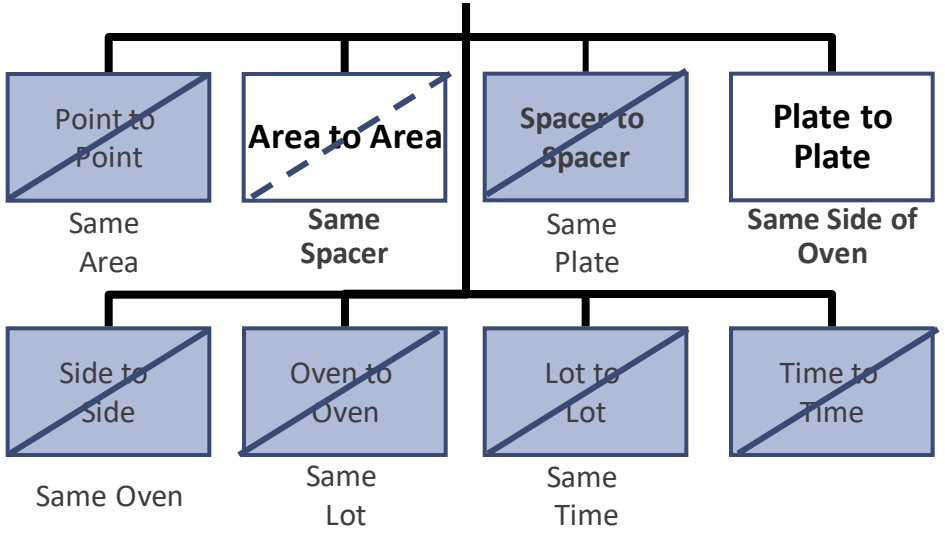


- Use the Multi-Vari for answering the current split in the Solution Tree.

Solution Tree







Rationale:

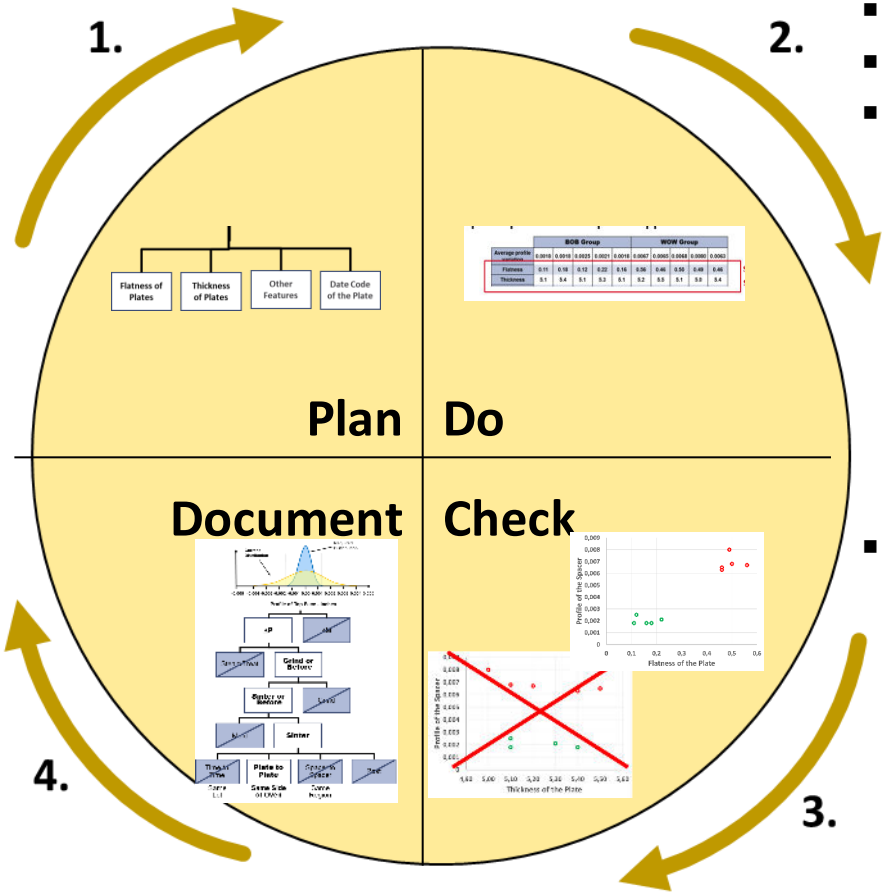
- Multi-Vari

Investigative Cycle

→ **Plan next split on the Solution Tree.**

- What's different between BOB and WOW plates?

→ **Update the Solution Tree.**

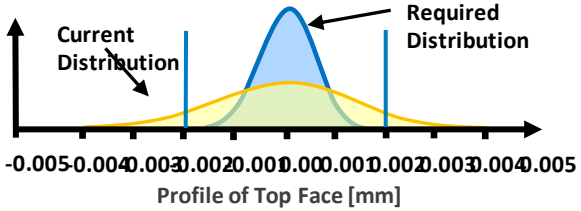


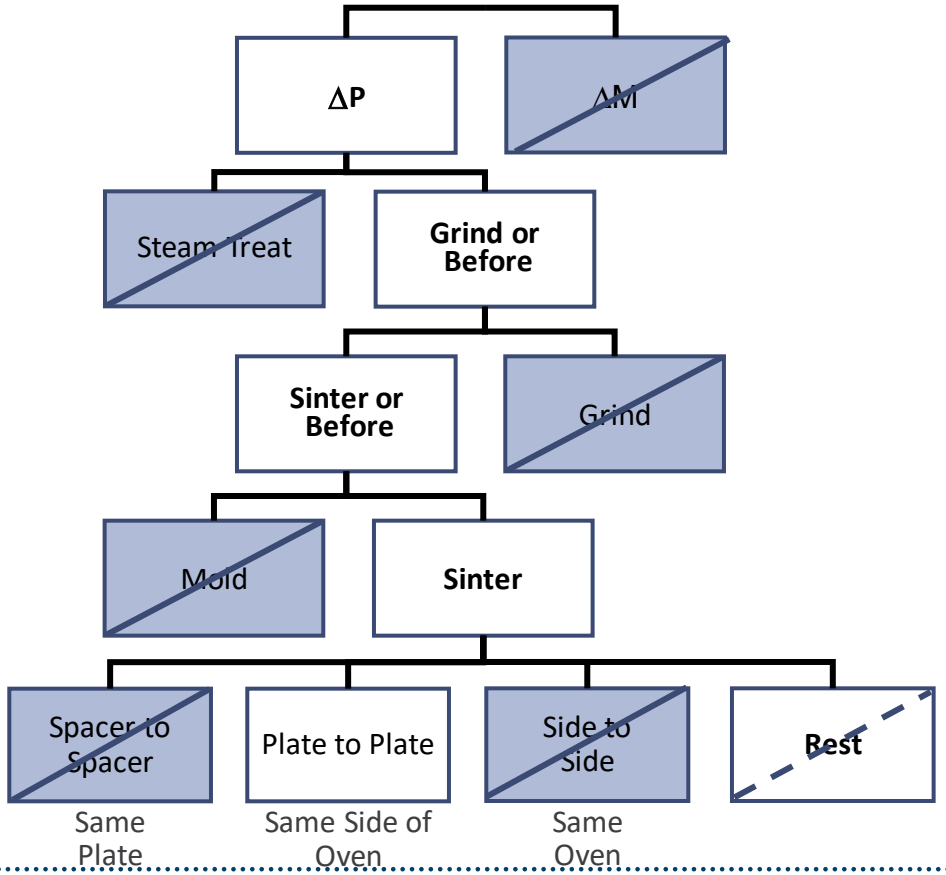
- Take stratified samples.
- Conduct group comparisons.
- Graph the results.

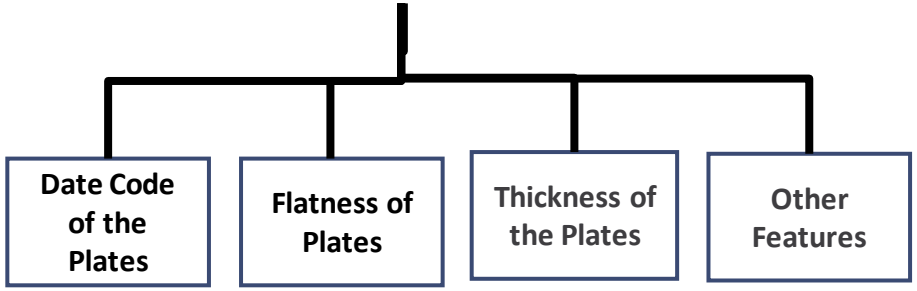
- What's the largest driver of the differences?

Evolving strategies will be necessary to fill the Solution Tree level by level to identify the BIG X candidate.

Solution Tree







Rationale:

- Group comparisons

Group Comparisons



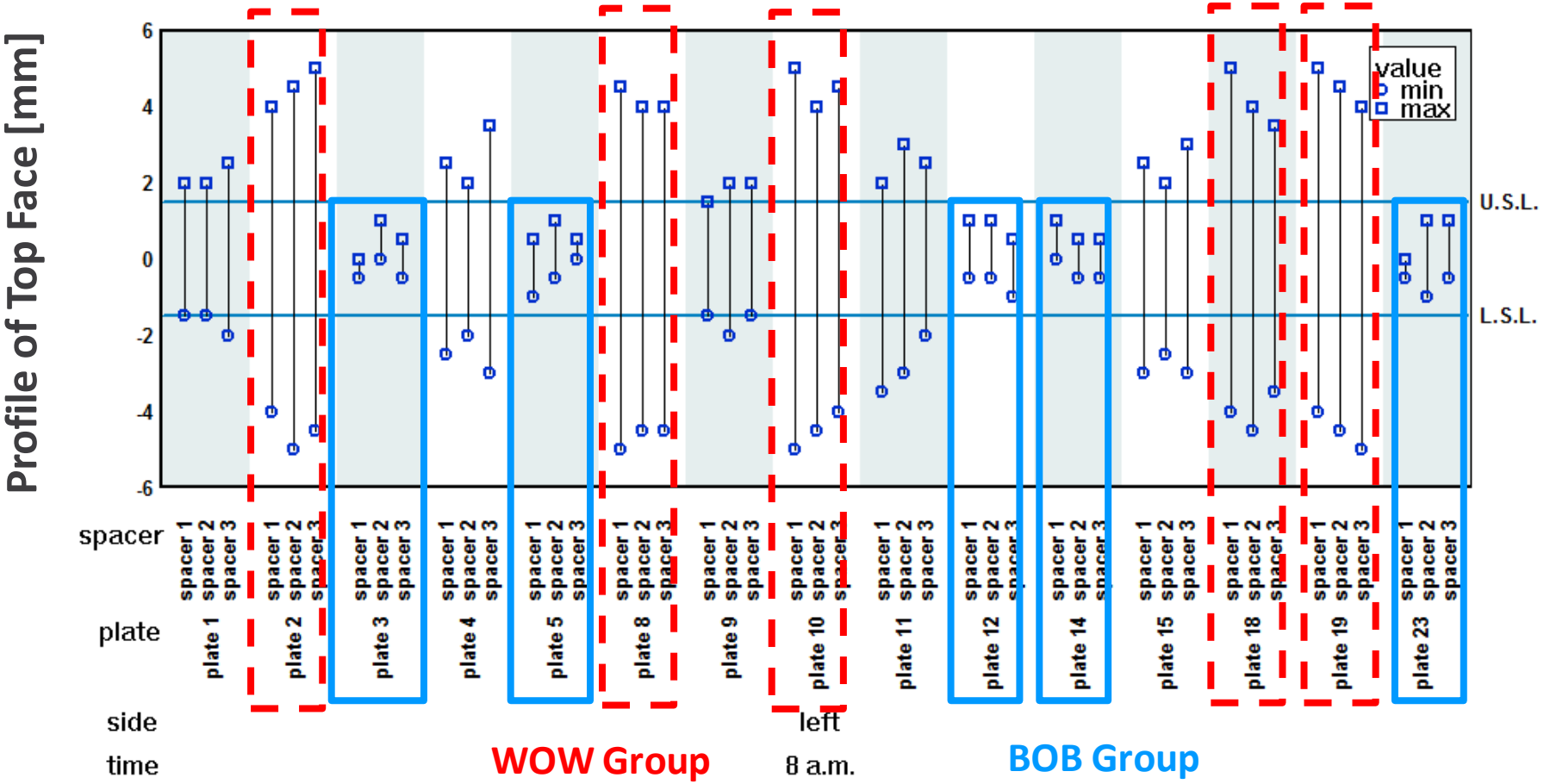
BOB Group



WOW Group

What is the difference between these two groups of shoes?

Group Comparisons - Selecting BOBs and WOWs Plates



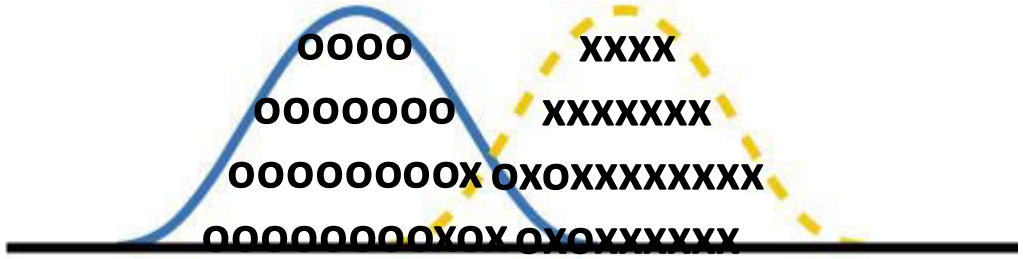
- This Multi-Vari chart shows only one time, one side but more plate variations.
- Largest family is plate to plate.
- Select BOB and WOW groups of plates for a group comparisons.

Group Comparisons

		BOB Group					WOW Group				
Mone(Y)	Average profile variation [mm]	0.0018	0.0018	0.0025	0.0021	0.0018	0.0067	0.0065	0.0068	0.0080	0.0063
Features / properties of the plate	Date code	Sep 14	Mar 14	Nov 13	Jan 14	Jul 14	Aug 13	Dec 13	Jun 13	Jan 13	Dec 13
	Flatness [mm]	0.11	0.18	0.12	0.22	0.16	0.56	0.46	0.50	0.49	0.46
	Thickness [mm]	5.1	5.4	5.1	5.3	5.1	5.2	5.5	5.1	5.0	5.4

- **In a Group Comparison all suspect variables are listed with respect to BOB and WOW group members.**
- **In the following steps we are looking for correlation between the suspects and the Mone(Y).**

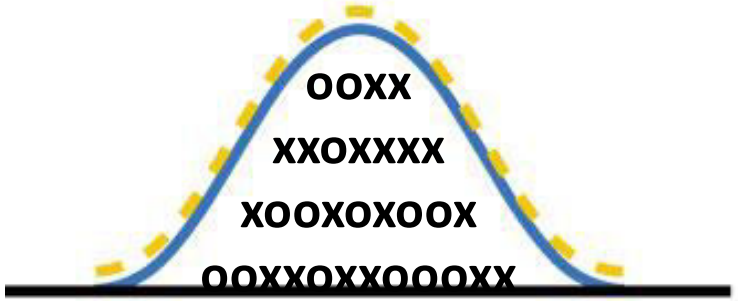
Tukey End Count



Populations

$\mu_o \neq \mu_x$

Samples oooo | xooxo | xxxxx



Populations

$\mu_o = \mu_x$

Samples oxooxoxxxooxxo

Group Comparisons – End Count Method

Date Code		Flatness [mm]		Thickness [mm]	
Sep. 14	B	0.11	B	5.0	W
July 14	B	0.12	B	5.1	B
March 14	B	0.16	B	5.1	B
Jan. 14	B	0.18	B	5.1	B
Dec. 13	W	0.22	B	5.1	W
Dec. 13	W	0.46	W	5.2	W
Nov. 13	B	0.46	W	5.3	B
Aug. 13	W	0.49	W	5.4	W
June 13	W	0.50	W	5.4	B
Jan. 13	W	0.56	W	5.5	W

4 + 3 = 7

5 + 5 = 10

= 0

EC

Group Comparisons – End Count Method

Number of Dimensions or Properties	Group Size					
	Step 2 → 5, 5	6, 6	7, 7	8, 8	9, 9	10, 10
1	5	5	5	5	5	5
Step 1 → 2	6	6	6	6	6	6
3	6	6	7	7	7	7
4	7 Step 3	7	7	7	7	7
5	7	7	7	7	7	8
6	7	7	8	8	8	8
7	8	8	8	8	8	8
8	8	8	8	8	8	8
9	8	8	8	8	8	8
10	8	8	8	8	8	9

The required end count is six or larger.

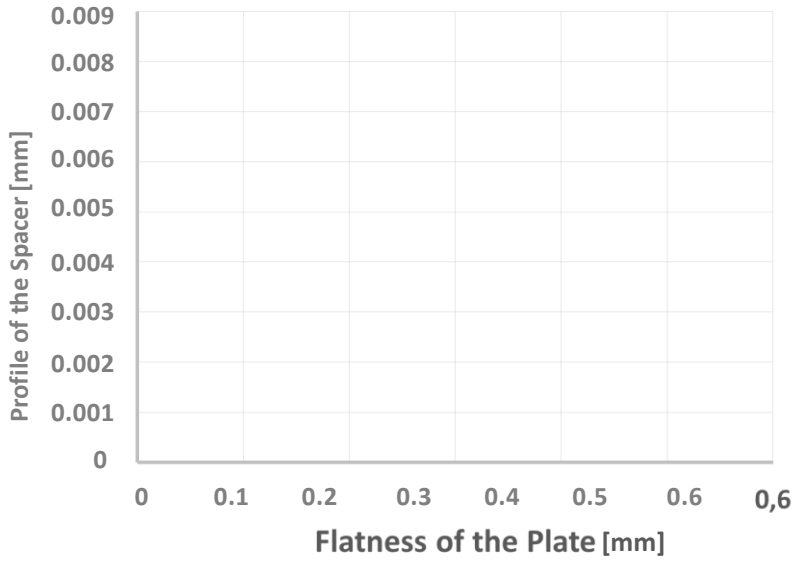
Features tested having an end count of six or less can be eliminated.

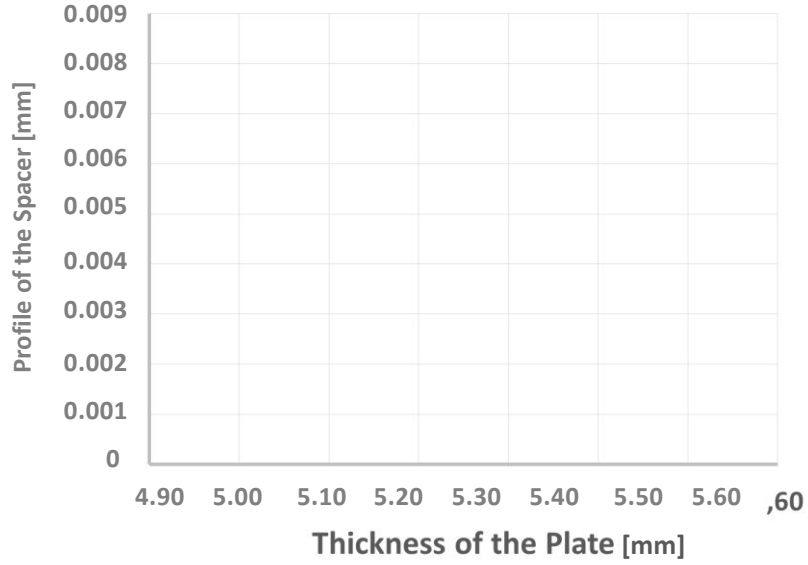
Group Comparisons – Graphical Approach - Exercise

	BOB Group					WOW Group				
Average profile variation [mm]	0.0018	0.0018	0.0025	0.0021	0.0018	0.0067	0.0065	0.0068	0.0080	0.0063
Date code	Sep 14	Mar 14	Nov 13	Jan 14	Jul 14	Aug 13	Dec 13	Jun 13	Jan 13	Dec 13
Flatness [mm]	0.11	0.18	0.12	0.22	0.16	0.56	0.46	0.50	0.49	0.46
Thickness [mm]	5.1	5.4	5.1	5.3	5.1	5.2	5.5	5.1	5.0	5.4

Step 1: Plot the Mone(Y) …

Step 2: … over the BIG X candidates

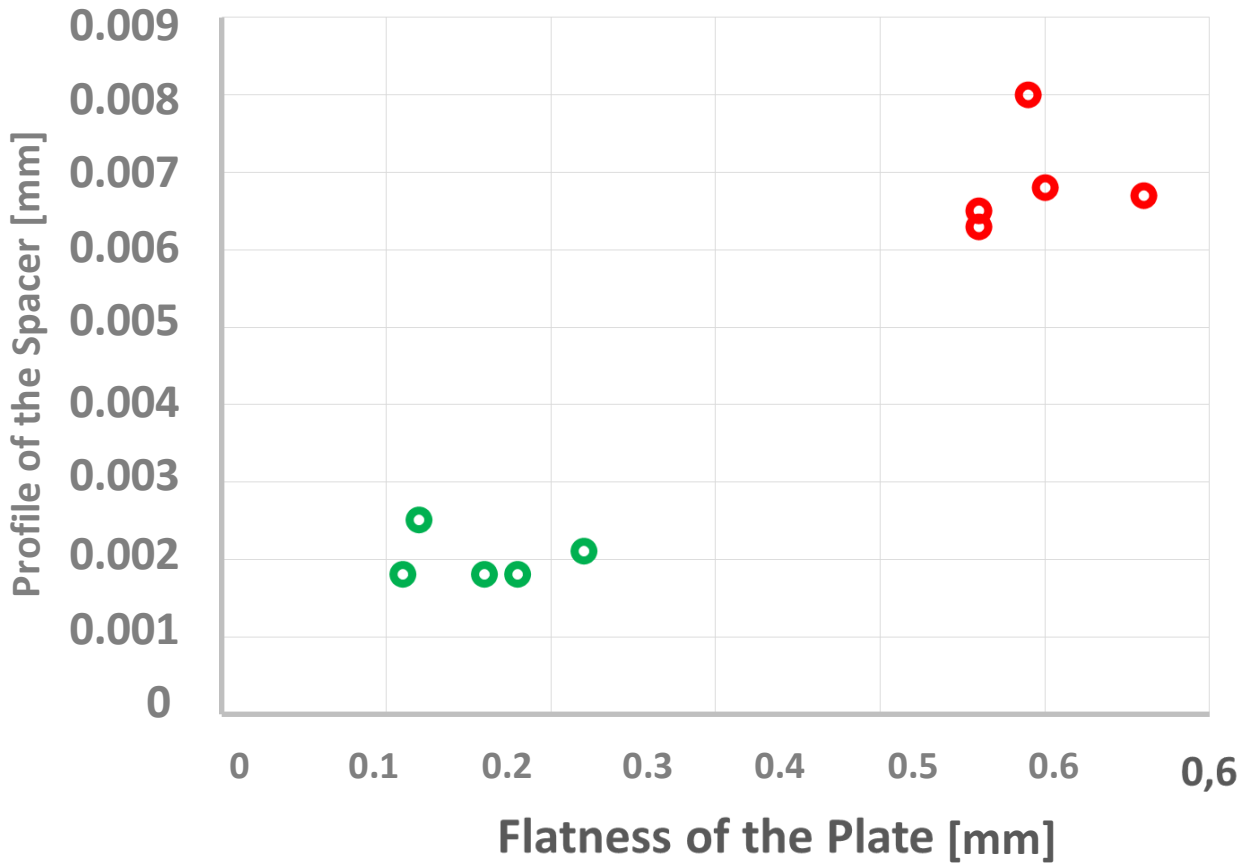


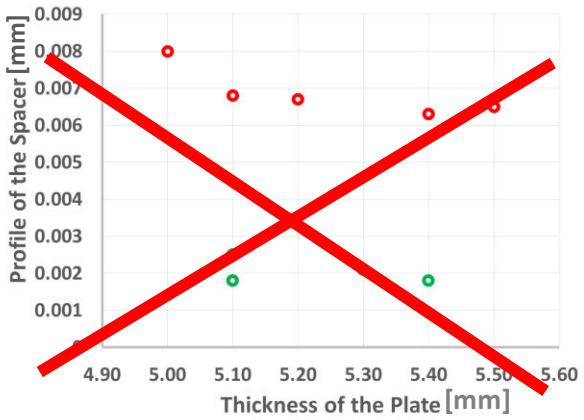


Do the graphical approach for “Flatness” and “Thickness.”

You will have 5 minutes.

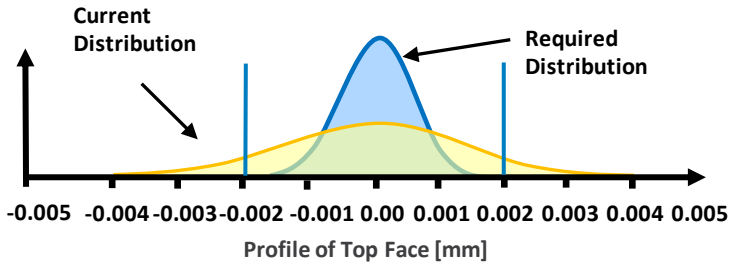
Group Comparison Graphical Approach - Answer

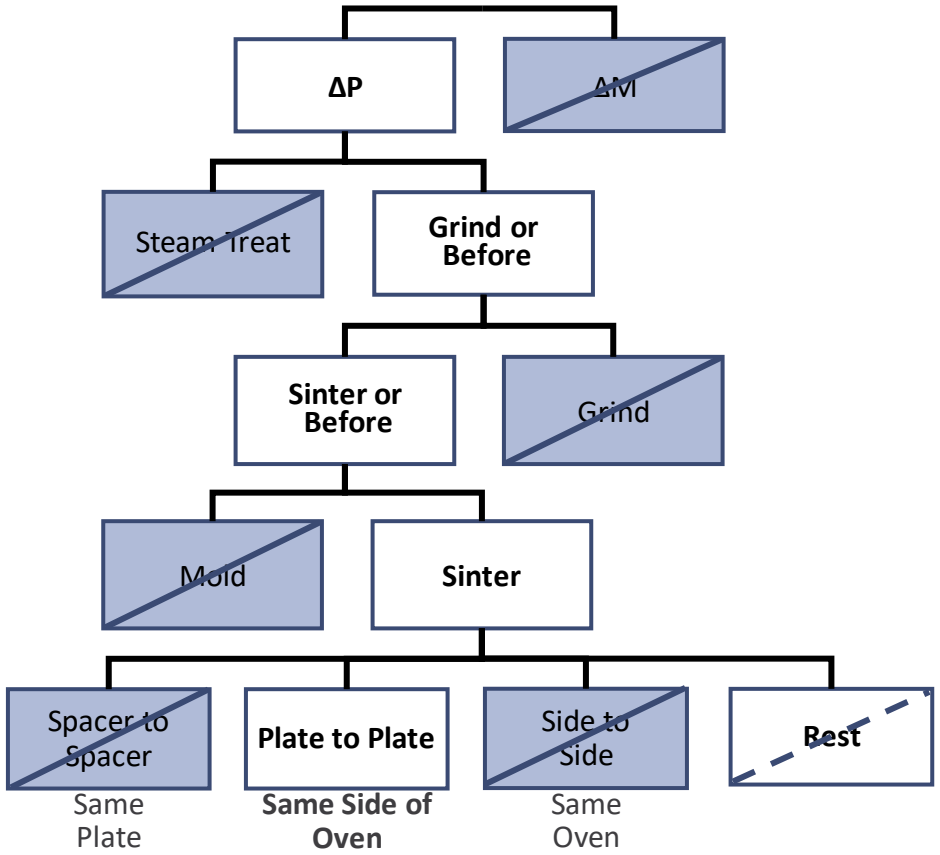


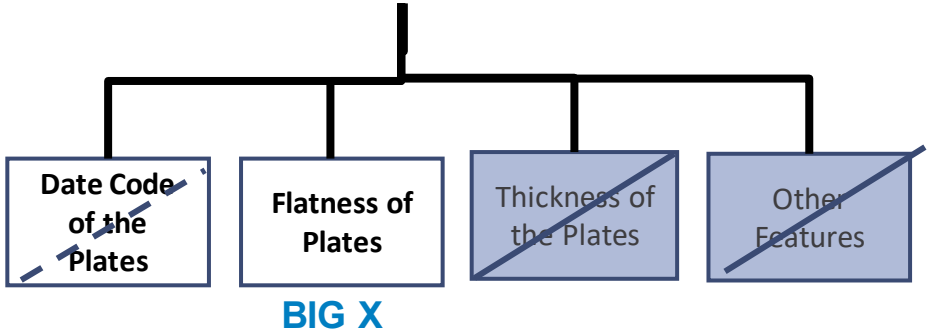


Conclusion: The BIG X candidate is the flatness of the plate.

Solution Tree







Rationale:

- Group comparisons
- BIG X proved with a p-Chart

Y → X

Problem



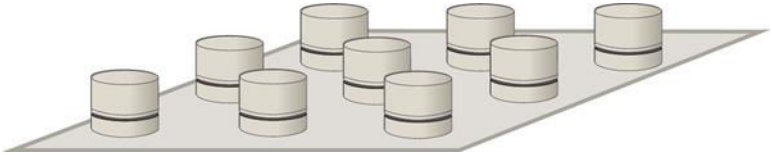
Surface must be completely lapped

Mold



“Green” Spacers

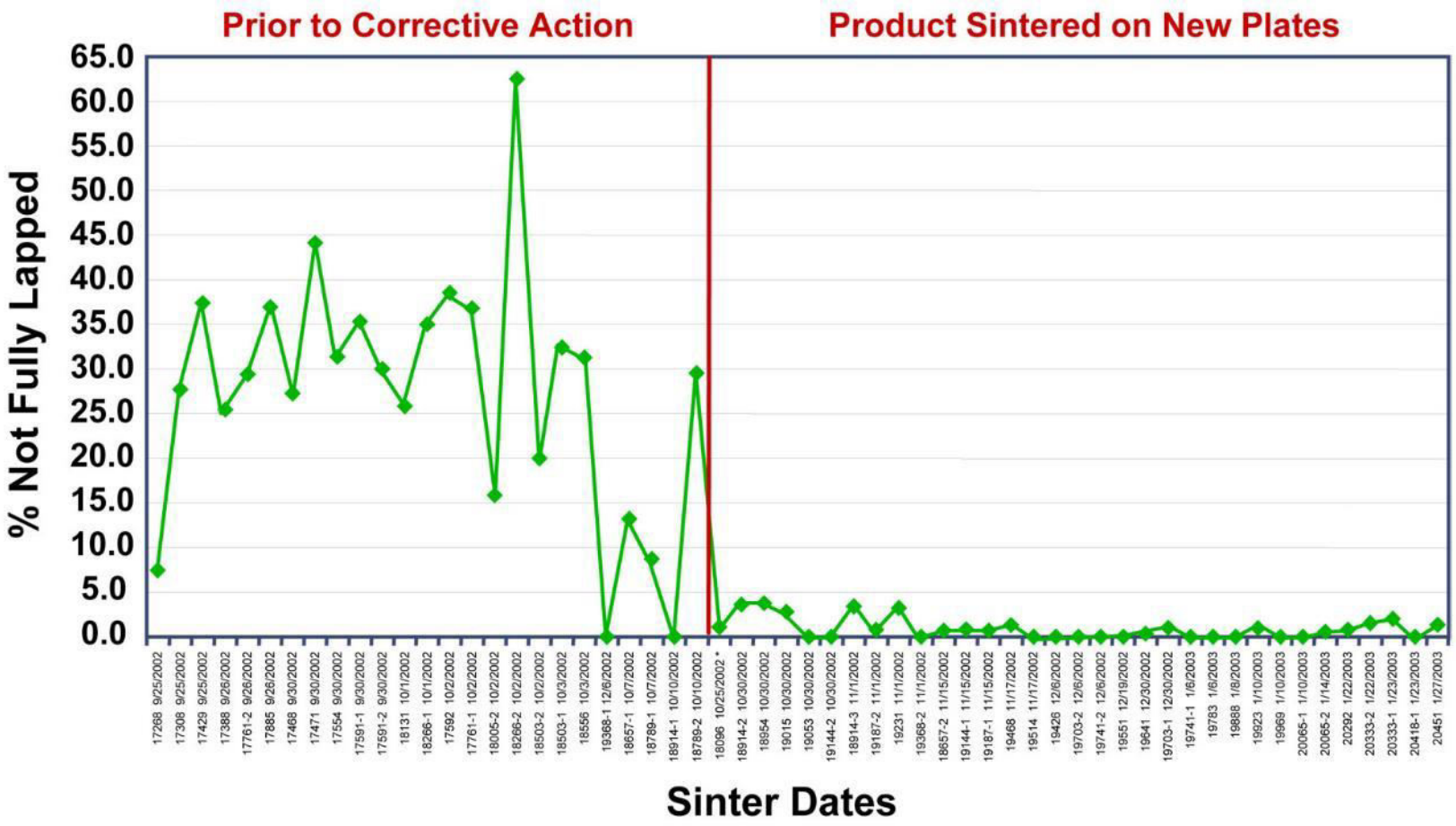
Sinter



Sinter Spacers

p-Chart

All sintering plates used on both ovens were replaced with new plates and a preventive maintenance program was put it place to remove plates before the wear became excessive. A p-chart was performed to confirm the kill.



Daily production quantity: 10.000

5 Defect Strategies

Module 5 Objectives
Defect Strategies

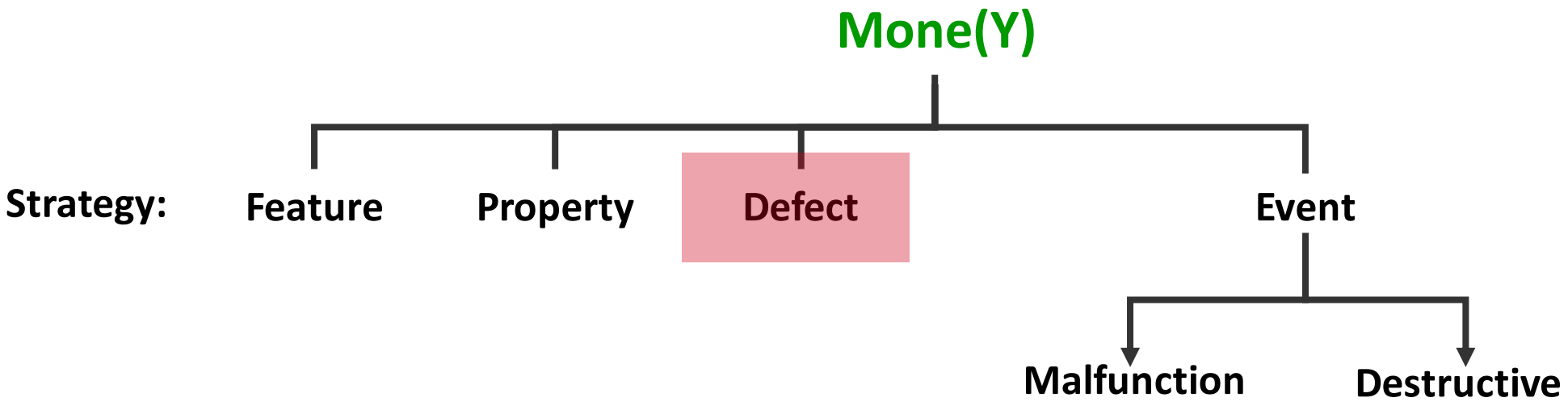
Understand:

- How to plan and execute a defect strategy

Be able to:

- Develop a Defect Strategy Diagram
- Use a concentration diagram to find the BIG X family
- Document convergence on a Solution Tree

Different Strategies for Different Mone(Y)s



Explanation: Product imperfections that occur during manufacturing. They are localized deviations. Evidence left behind after an undesirable event. One time non-repeatable events. Can be quantified as attributes (good/bad).

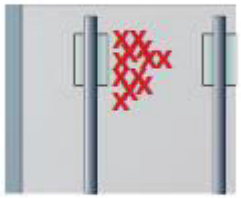
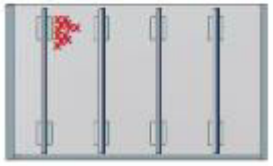
Tool: **Strategy Diagrams**

Strategy Diagrams are a way to document possible contrasts for the Mone(Y) that you can leverage during your project.

Destructive Event / Defect Strategy Diagram – Template

Mone(Y) =

Design Families

Family of Variation, or Contrast		***Diagram***
Point to Point **Same Region**	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A	
Region to Region **Within Same Unit**	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A	
Within Same Unit to Within Unit **Same Unit**	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A	
Unit to Unit	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A	

What Is Meant by Each Contrast

What does the damage or defect look like under a microscope? For damage, what do you see at the initiation site of the failure? Is there any evidence to explain why it initiated in the exact spot it did as opposed to another location? For a defect, what is it made of? Consider sending a sample to the materials lab for analysis.

Is there a pattern within the part regarding the initiation site of the damage or the position of the defect (certain corner, side, end, etc.)? Consider all three dimensions. If there is not a repeating pattern with other parts, is there evidence to explain why the failure initiated in the region it did on just this one part? Use a concentration diagram to map the failure locations.

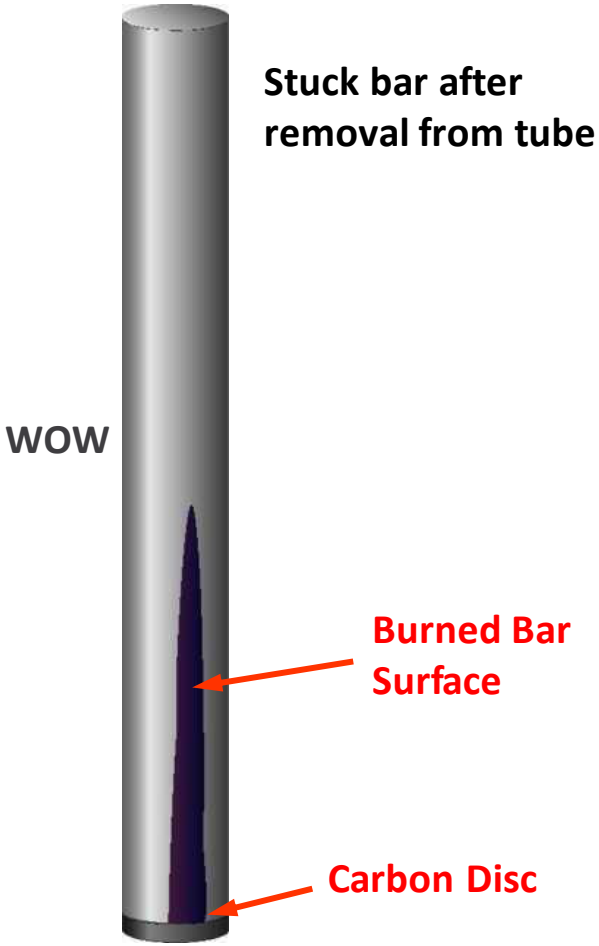
Are there contrasts within a unit (cylinder 1 damaged vs. cylinder 4, cope side to drag side of a mold, bore to bore, boss to boss)? Rename unit appropriately for your project (e.g. engine, pump, printer).

Are there BOB and WOW units? Non-failures are not necessarily BOBs? Consider the rest of the system as part of the unit since that could explain excess energy. What is different about BOB and WOW units? Consider using a Paired or Group Comparisons to leverage this contrast.

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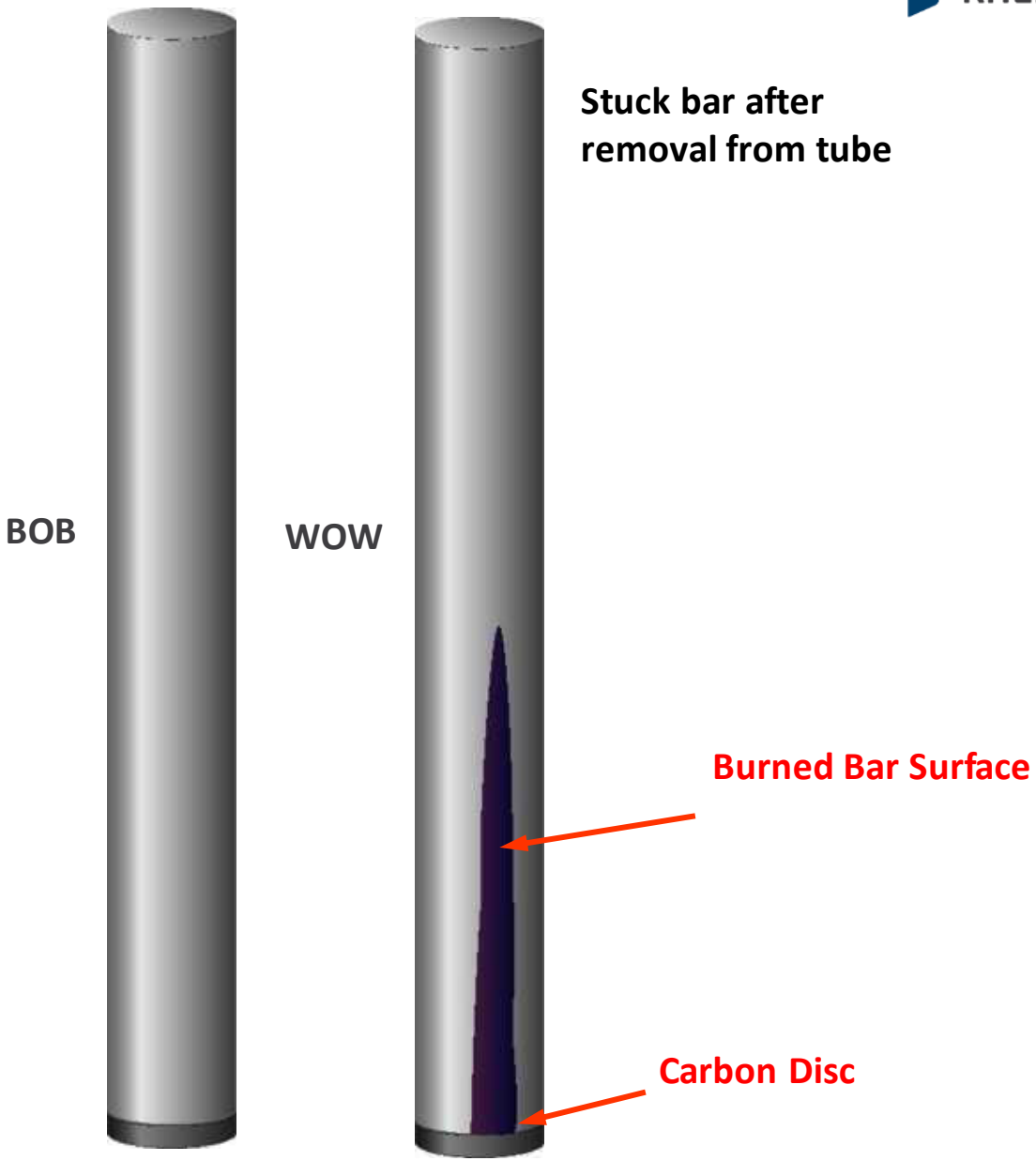
Sticking Bars

- A company manufactures bars of a high value alloy used in billet form by investment casting operations.
- Bar defects have a significant economic impact because of yield, and one defect in particular causes both yield and equipment problems – sticking bars.
- Molten alloy is poured into an upright steel tube that has a carbon disc press-fit into the base.
- Up to 60 tubes are poured in a batch. When cold, the bars are pushed out of the tubes.
- If they stick, the equipment sometimes breaks down.
- Sticking is caused by fusion of the alloy bar into the wall of the steel tube.
- The tube is scrap, and much of the bar is scrap when sticking occurs.

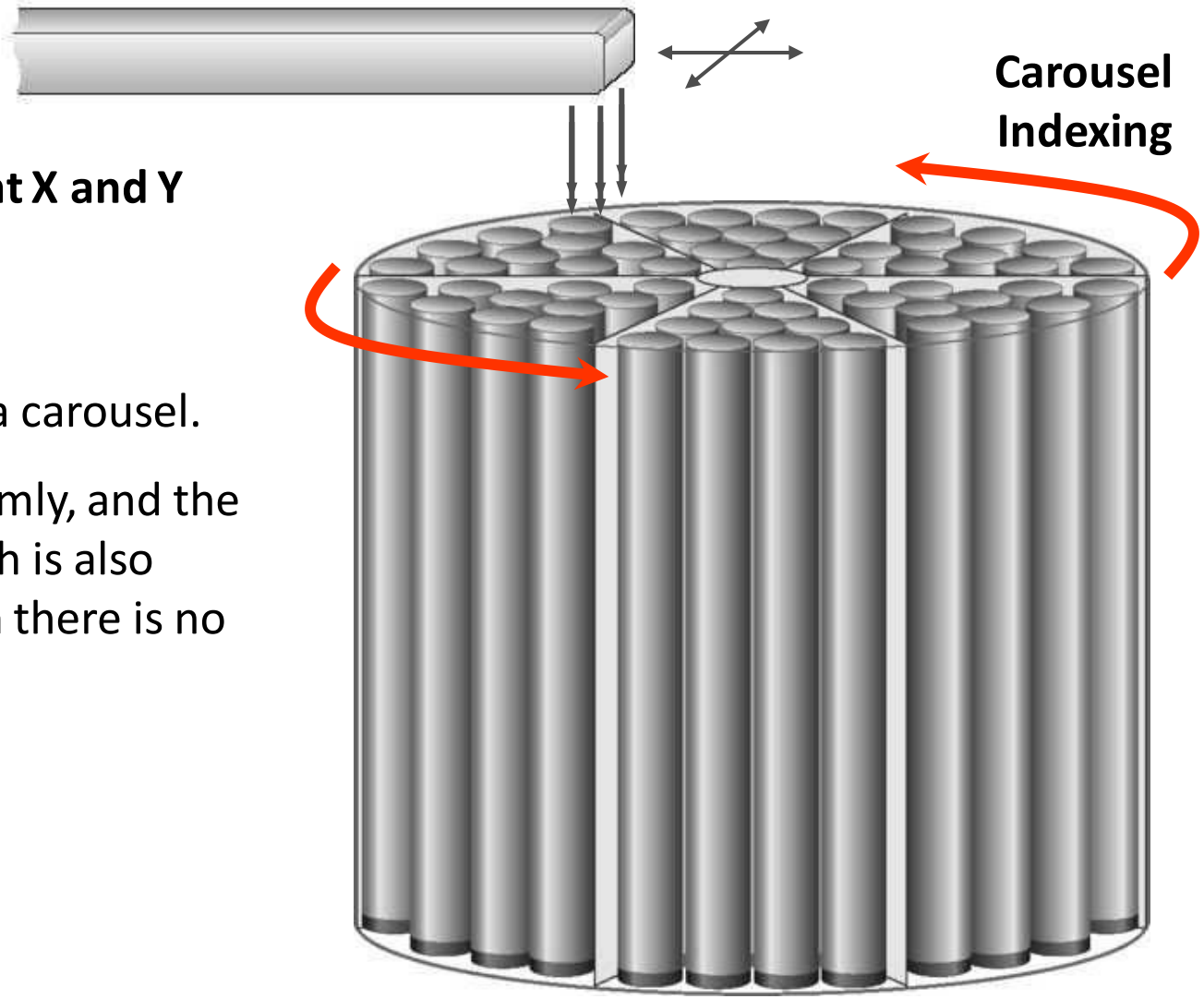


Sticking Bars

- Between 10 % and 20 % of bars typically stick.
- The company statistician has shown that smaller diameters give more problems, but these are more popular with customers too.
- Management blames the operators.



Process Schematic



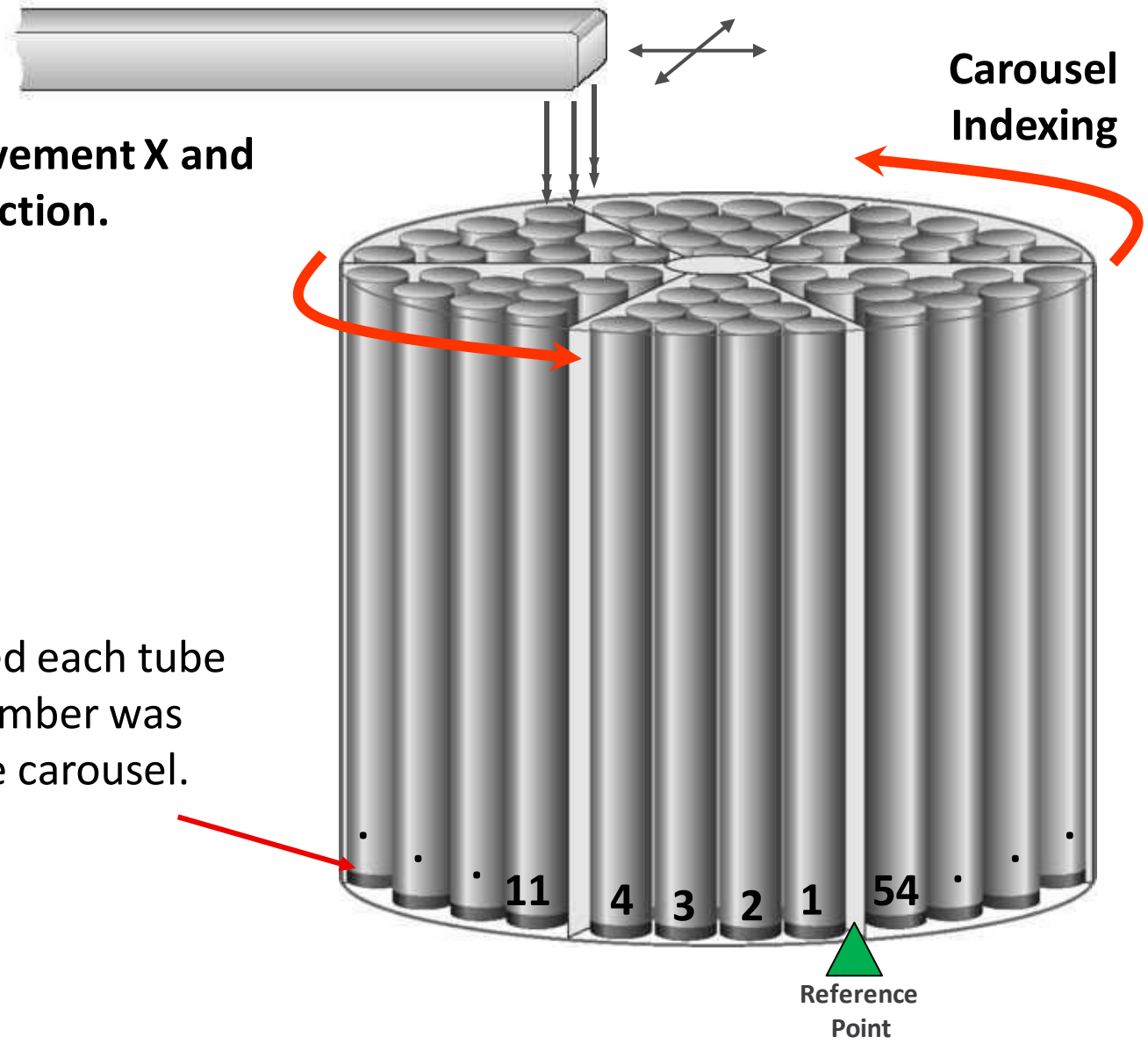
Pour cup movement X and Y direction.

Up to 60 tubes are placed on a carousel.

Operators placed tubes randomly, and the starting position for each batch is also random. In normal production there is no traceability.

Identifying Tube Position

Pour cup movement X and Y direction.



A technician numbered each tube and ensured the number was facing outside of the carousel.

Concentration Diagram - Guidelines

- A concentration diagram is a tool for identifying non-random patterns of variation within a part or process.
- A concentration diagram provides positional information which can yield critical clues to the BIG Xs identity.
- A concentration diagram should be used in conjunction with a Strategy Diagram in order to plan the search for non-random patterns.
- A concentration diagram should include multiple families of variation in order to reveal interactions.
- A concentration diagram can be used in batch processes to show whether defective parts are randomly or non-randomly dispersed within the process.

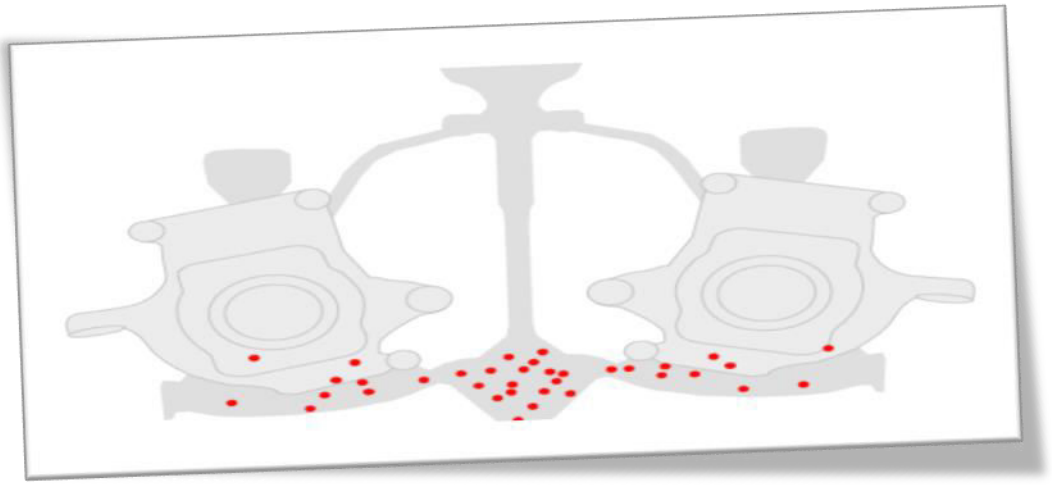
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Approach for creating a Concentration Diagram

1. Design the concentration diagram
   - Prepare a drawing of the part / the assembly / area
   - Decide about the method of marking

2. Train the people who should collect the data

3. Collect the data
   - Samples should be representative of the process
   - Collect information about the defect mode also, wherever possible
   - If more people are involved collecting the data, collate all the data
   - Sample size depends on how quick a pattern is visible
   - Mark the defect concentration diagram showing the location of each defect



Approach for creating a Concentration Diagram

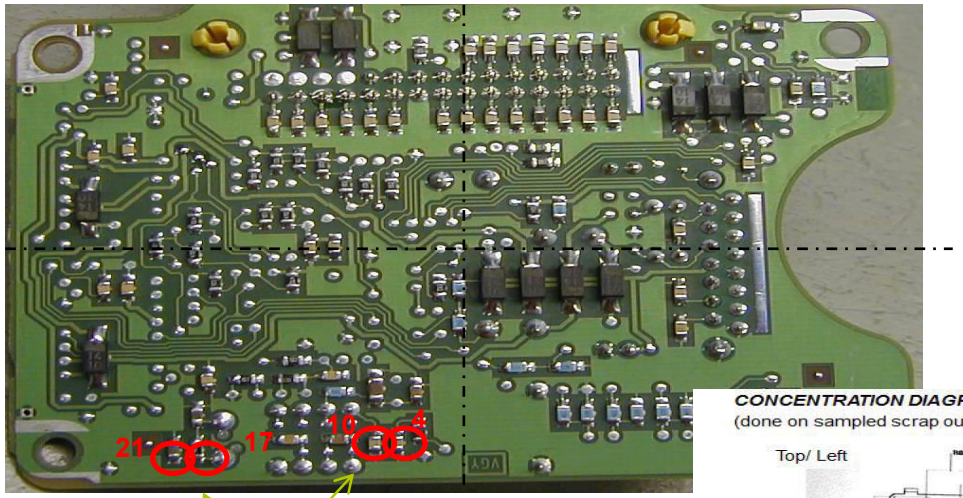
4. Simply mark every time there is an occurrence of a problem, the specific location

5. Analyze the diagram for patterns, diagnose the problems and start with improving:
   - If random, proceed with other problem solving technique
   - If not random, ask which process step exhibits that type of defect at the marked location



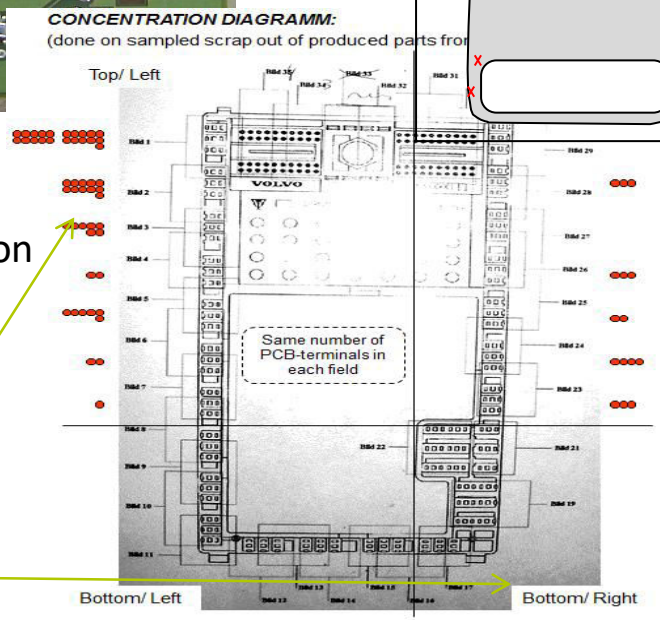
Examples and applications of a concentration diagram

PCB with concentration of cracked capacitors

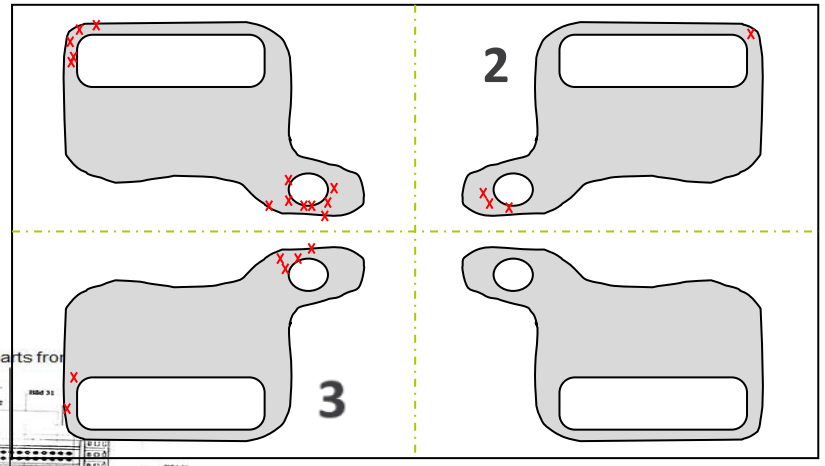


PCB with marks (circle) for defect location and amoun (number) of defects

Plastic terminal with injected metal contacts. Marked amount of defective contacts related to reference point.



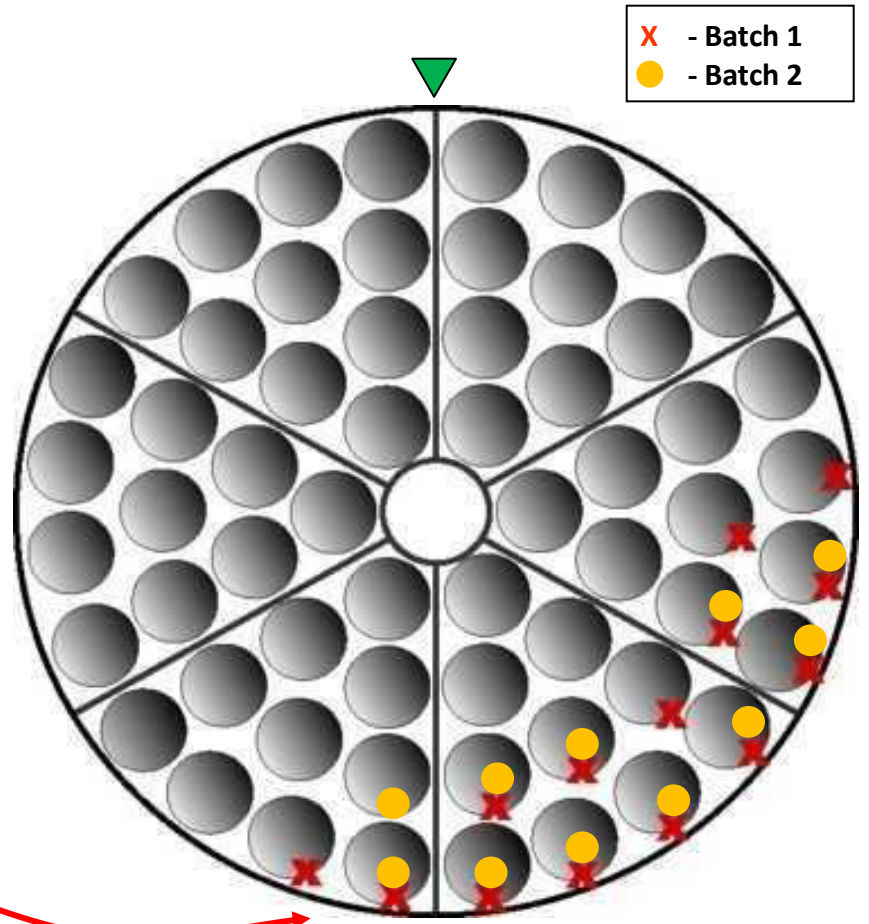
Casting mold with 4 cavities and marked defects on the components



Concentration Diagram

By observation, stuck bars are caused by the pour stream hitting the tube side and burning in. Filled tubes are removed when cool, and bars are pushed out in a completely random sequence.



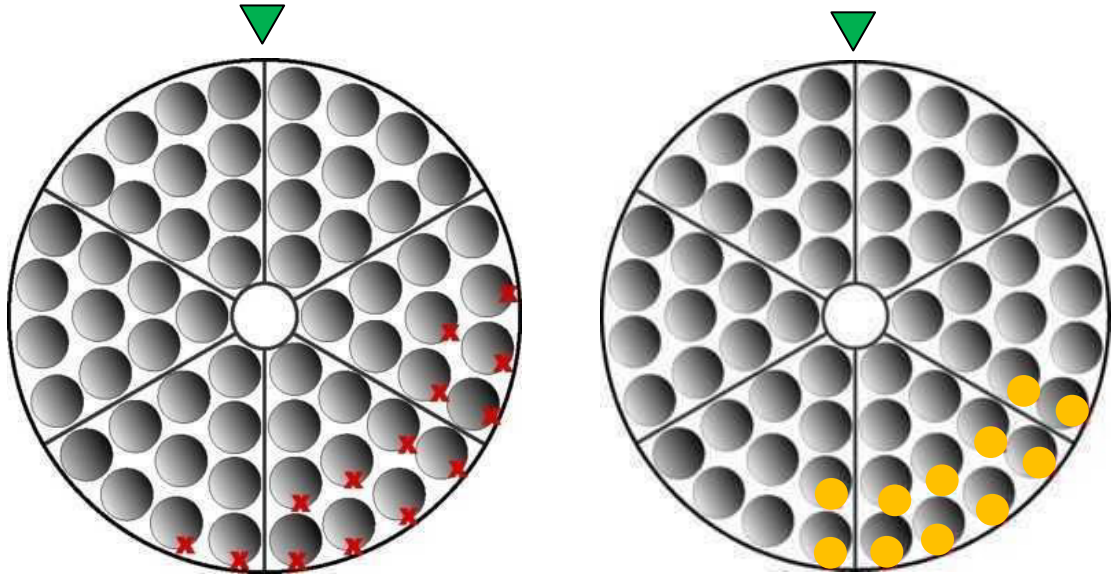


Burn in Location

Concentration Diagram – Exercise

Results from the strategy executed for this project:

- All sticking occurs closest to the carbon disc. We can show other non-random patterns in the plan view.
- Both locations with respect to the bar and the carousel can clearly be seen in this view.



Batch 1 **Batch 2**

- Complete the strategy diagram by identifying the contrasts and making observations around those contrasts.
- What conclusions do you draw?
- You will have 10 minutes to draw your conclusions.

Strategy Diagram – Design Family - Exercise

Family of Variation, or Contrast

Point to Point

Same Depth

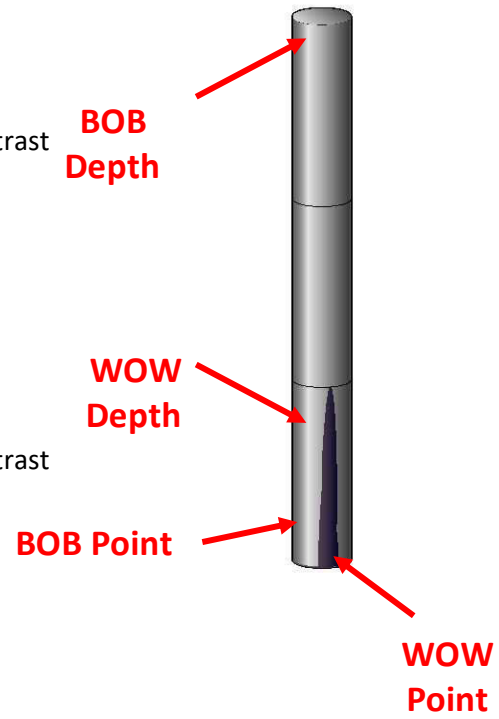
- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Depth to Depth

Same Bar

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Diagram



Observations

Strategy Diagram – Process Family - Exercise

Family of Variation, or Contrast

Diagram

Observations

Bar to Bar

Same Segment

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

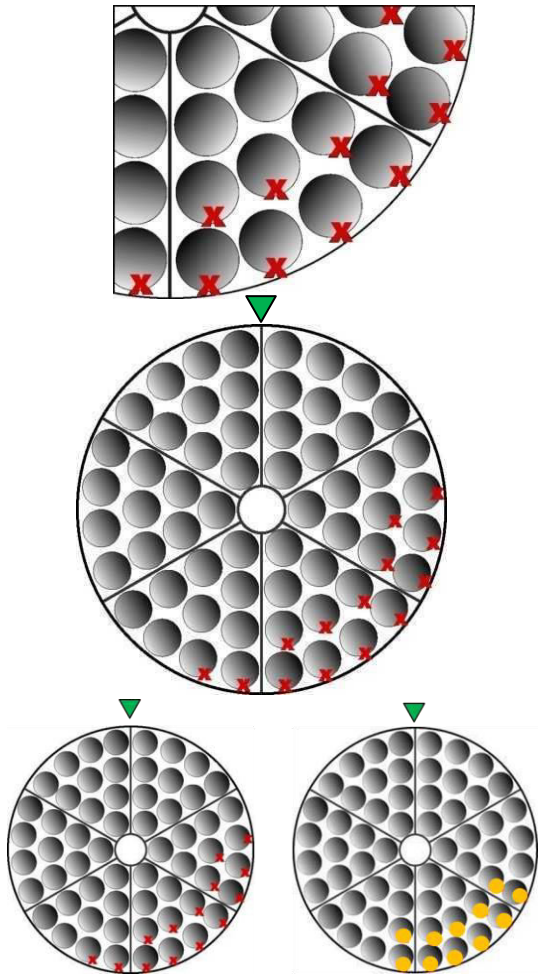
Segment to Segment

Same Batch

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Batch to Batch

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast



Strategy Diagram – Design Family - Answer

Family of Variation, or Contrast

Point to Point

Same Depth

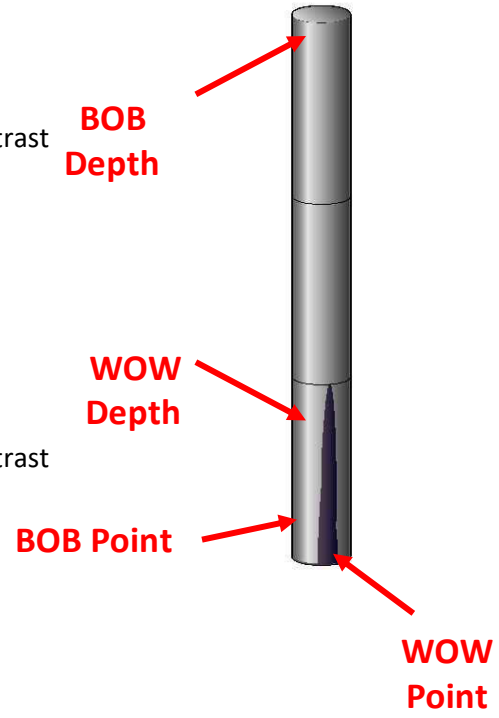
Do Not Know
Large Contrast
Small or No Contrast

Depth to Depth

Same Bar

Do Not Know
Large Contrast
Small or No Contrast

Diagram

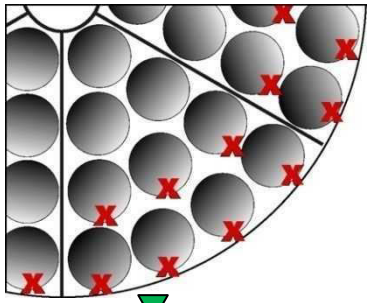
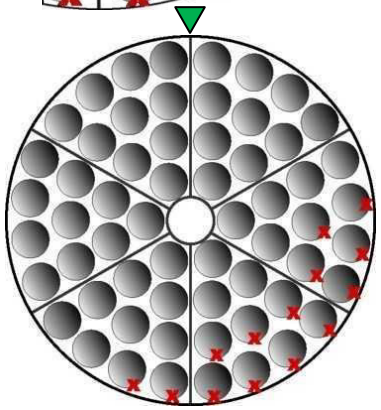
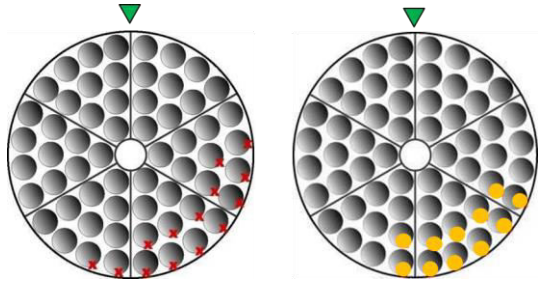


Observations

WOW point is burnt in. Can see obvious contrast between BOB and WOW points. All WOW points are oriented to the outside of the carousel.

Strong contrast with WOW depth always at the bottom. No leverage. Physics of the pour is known.

Strategy Diagram – Process Family - Answer

Family of Variation, or Contrast	***Diagram***	***Observations***
Bar to Bar **Same Segment** Do Not Know Large Contrast Small or No Contrast		Large contrast bar to bar. Bars closest to the center are BOBs and those furthest away are WOWs.
Segment to Segment **Same Batch** Do Not Know Large Contrast Small or No Contrast		Strong contrasts with segments. Have BOB and WOW segments.
Batch to Batch Do Not Know Large Contrast Small or No Contrast		No contrasts. All batches are WOWs.

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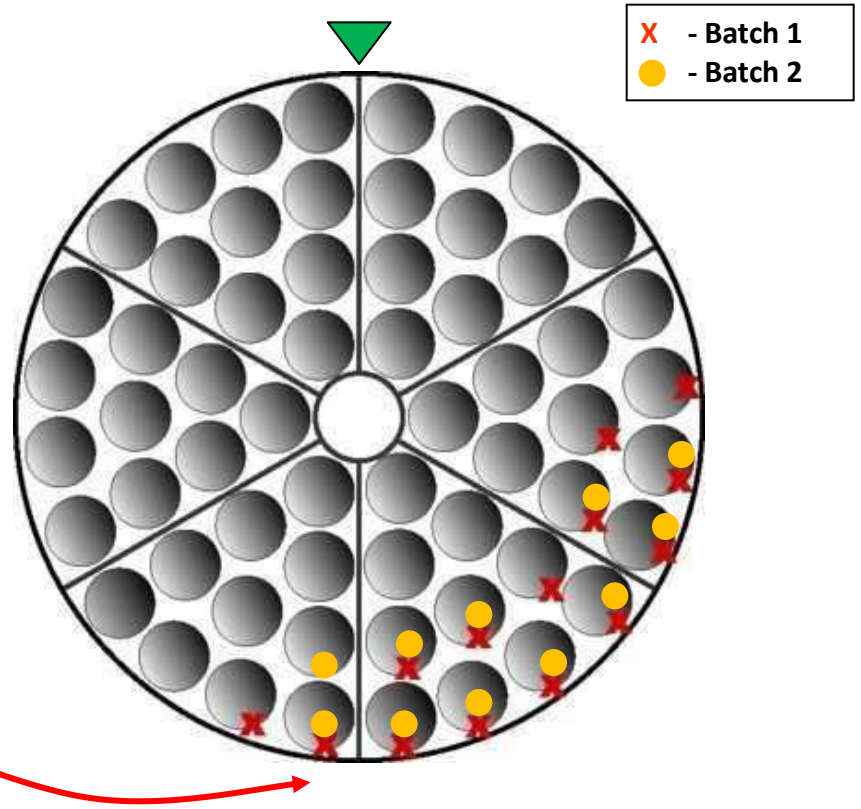
Sticking Bars

What can we eliminate as a potential BIG X?

What are some potential BIG X candidates?

What key learning points can you take away from this example?

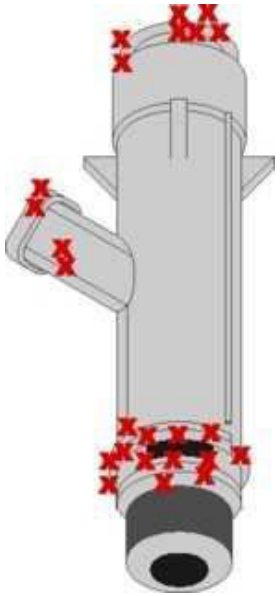




Burn in Location

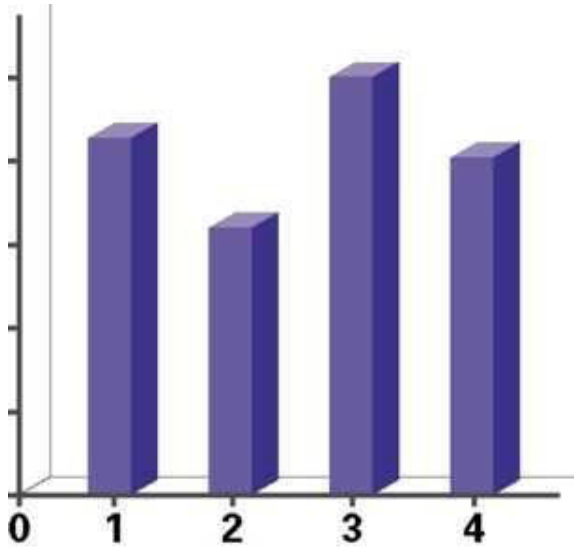
Concentration Diagram: No Strategy

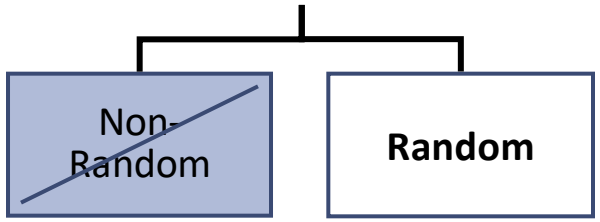
Injection Molded Part

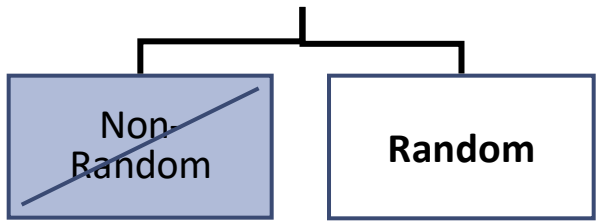


X = Defect

Defects by Cavity







The Strategy Diagram Shows the Way…

Family of Variation

Point to Point

Same Region

Region to Region

Same Part

Part to Part

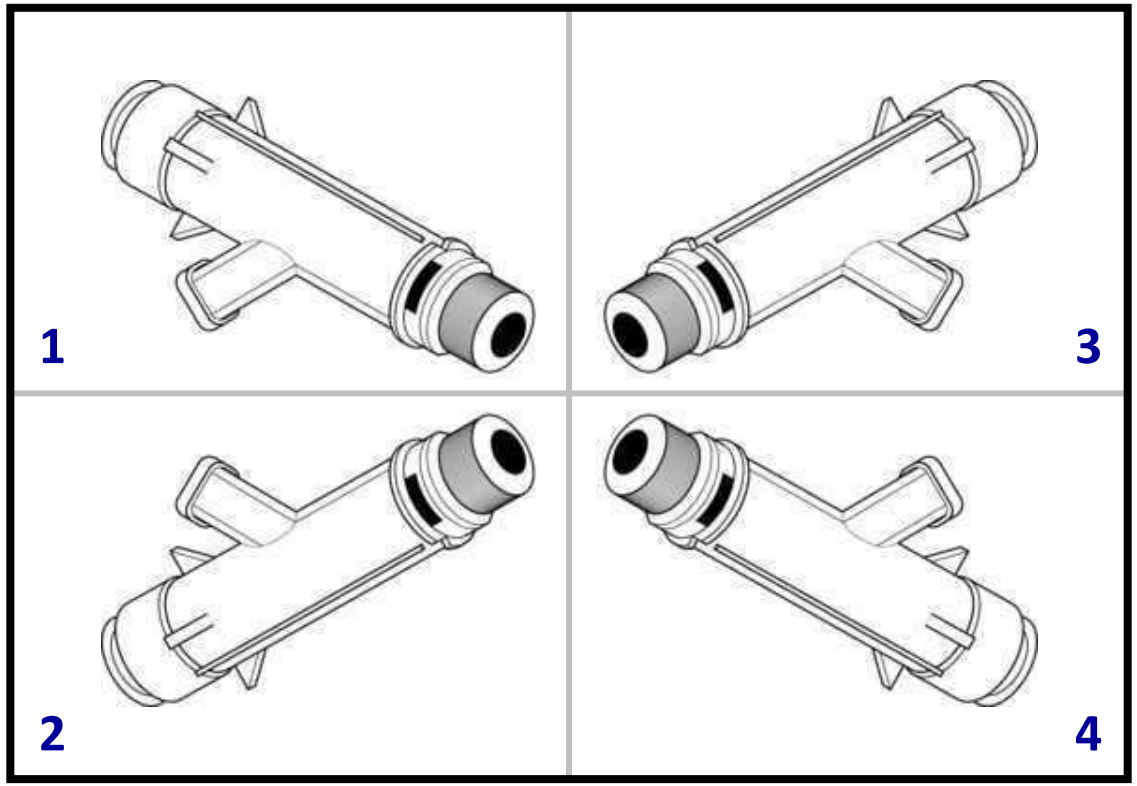
Same Cavity

Cavity to Cavity

Same Mold

Mold to Mold

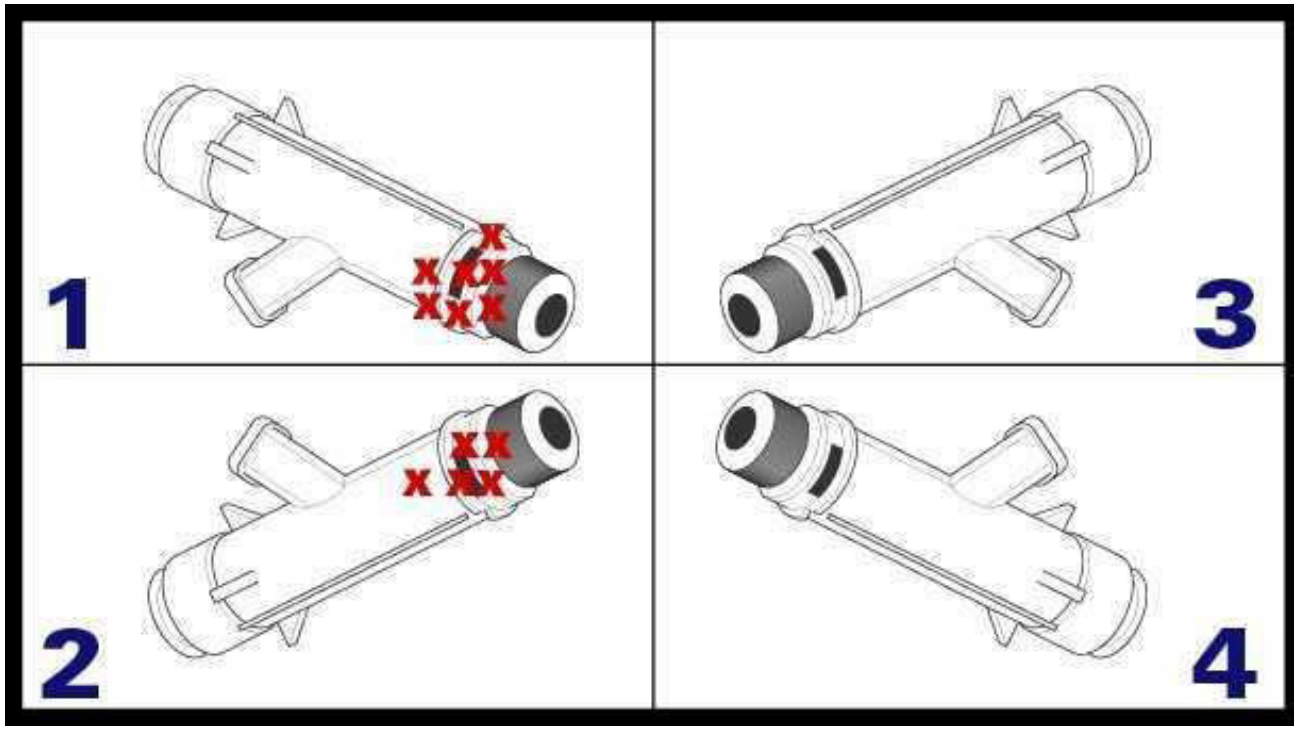
Same Time



Concentration Diagram

The identification of the defect relative to product and process reveals the truth.
The Strategy Diagram drives the concentration diagram.

Mold #1

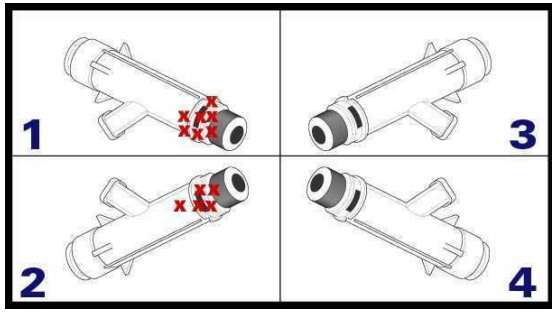


= Defect

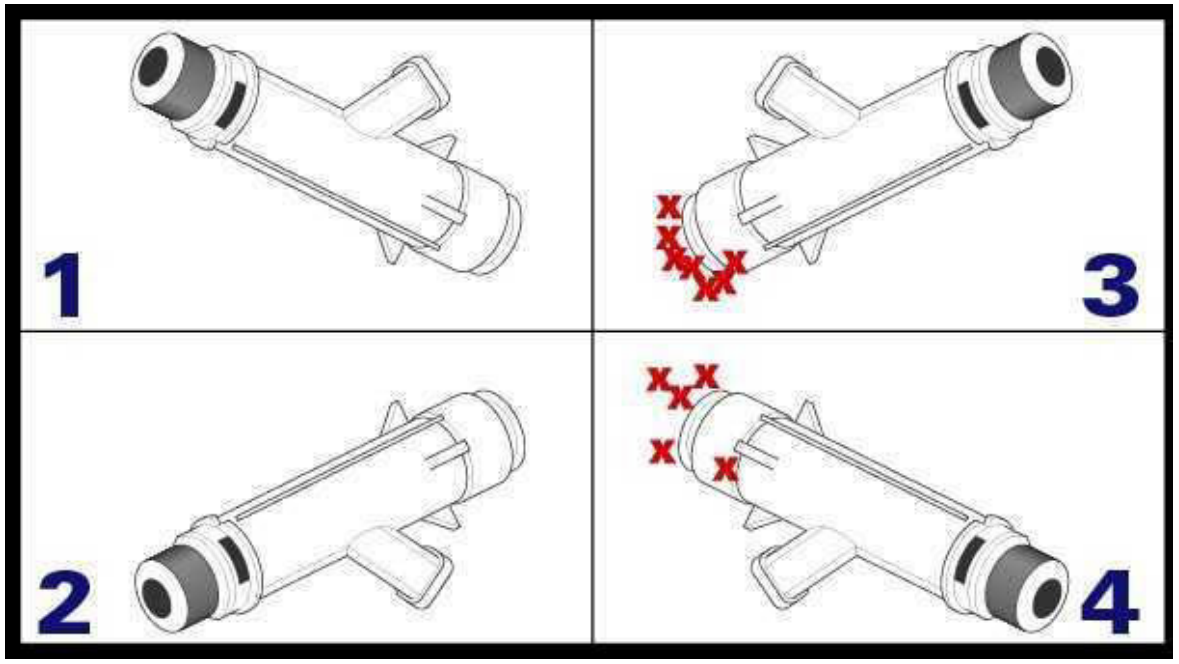
Concentration Diagram

The identification of the defect relative to product and process reveals the truth.
The Strategy Diagram drives the concentration diagram.

Mold #1



Mold #2

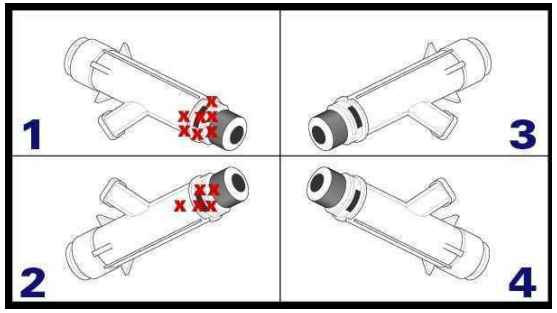


X = Defect

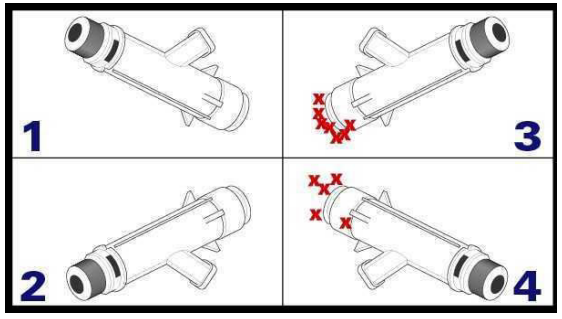
Concentration Diagram

The identification of the defect relative to product and process reveals the truth.
The Strategy Diagram drives the concentration diagram.

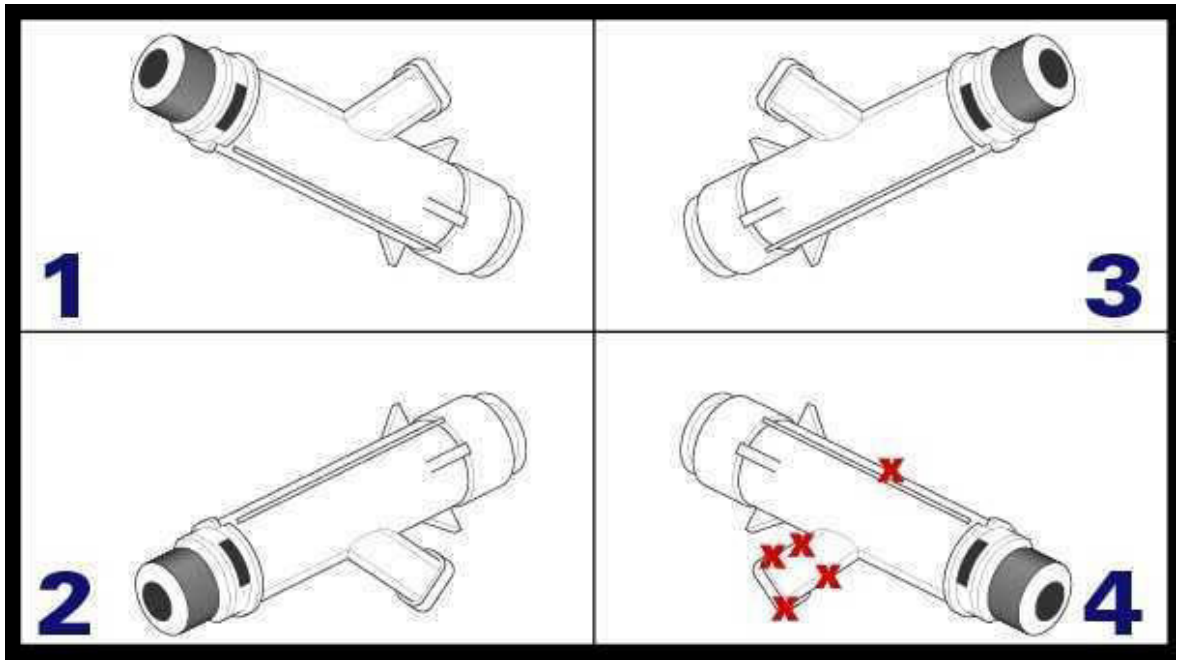
Mold #1



Mold #2



Mold #3

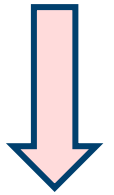


X = Defect

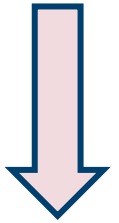
Defect Strategies

- Concentration diagrams are used in conjunction with a defect Strategy Diagram.
- BIG X that creates the defect must live within one of the contrasts defined in a Strategy Diagram.
- A concentration diagram should show the location of the defect relative to the design and process families of variation.
- The three keys to effective defect strategies are location, location and location.
- Patterns yield clues about the nature of the BIG X when associated with a BOB / WOW contrast.

Strategy Diagram



Concentration Diagram



BIG X Family

Case: Condenser Leaks

- The PS expert received a phone call late Friday afternoon.
- The client is missing shipments due to high scrap rates on condensers.
- Scrap is due to leaks at braze joints.
- At initial launch, scrap was not a problem since there was extra capacity.
- Ramp-up starting, so now scrap matters.
- The PS expert lives 75 km away, so he agrees to go in on Saturday morning.

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Racing the Clock

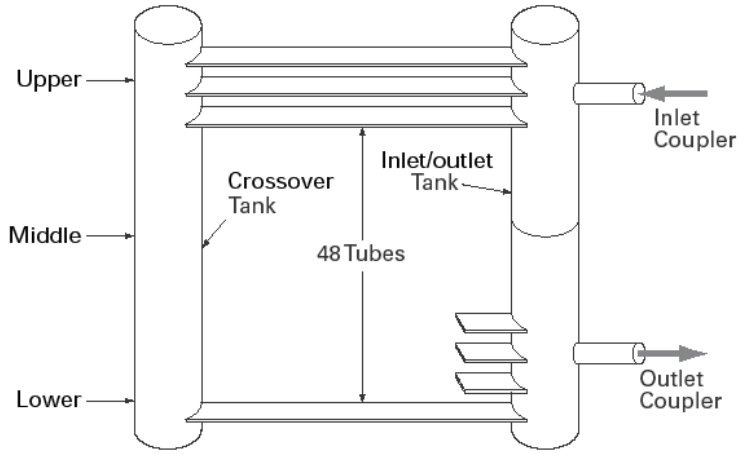
- The engineer's wife was planning a big birthday party for their daughter on Sunday afternoon with two dozen 1st graders.

- She told her husband, "If you're not back for the party, do not bother coming home."



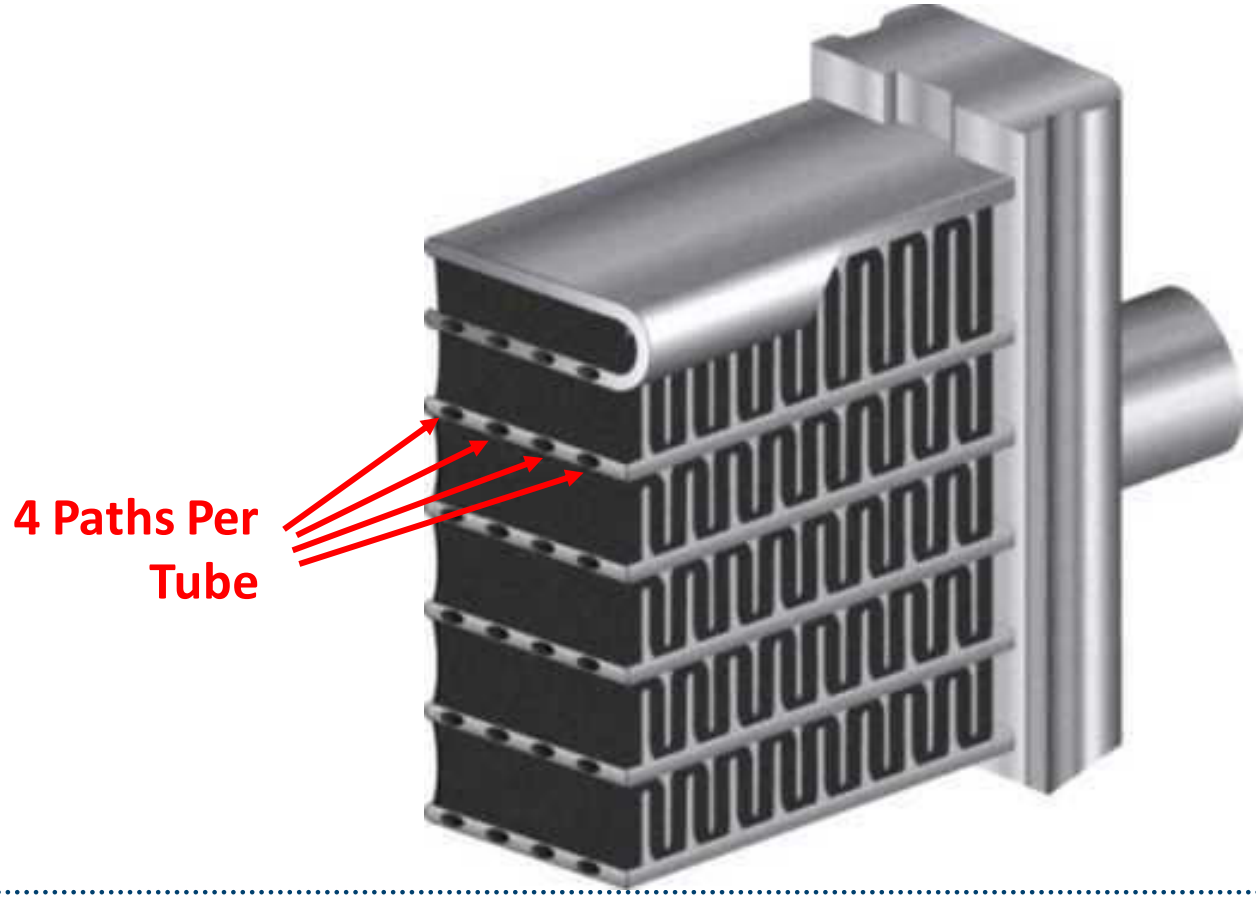
Condenser – How it Works

- Condensers are used to convert a gas to a liquid.
- Hot gas enters the inlet/outlet tank through the inlet coupler.
- The hot gas then travels through the 32 upper tubes and cools as it reaches the crossover tank.
- The cooled gas migrates to the bottom of the crossover tank and travels back to the inlet/outlet tank through the lower 16 tubes as it fully condenses to a liquid.
- The liquid exits through the outlet coupler.



Condenser – How it Works

- Each tube has four flow paths through which the fluid flows.
- Air flows around the tubes and removes heat.



Condenser – How it's Made - Exercise

- Condensers are assembled in one of three identical stacker machines:
  - Inputs to the stacker machines:
    - 48 Tubes
    - 47 Centers
    - 2 Tanks
    - Frames (support mechanism for transporting assemblies to oven)
  - Each stacker performs the same two steps:
    - Assemble Components
    - Place Frame
- All unbrazed assemblies go through one continuous belt oven with four steps:
  - Flux
  - Braze
  - Cool Down
  - Remove Frame

Condenser – How it's Made - Exercise

- Each part then goes to one of five Induction braze and test machines:
  - Inputs to the Induction braze and test machines:
    - Oven Brazed Assembly
    - 2 Couplers (1 inlet and 1 outlet)
    - 2 Flux Rings
  - Each braze and test machine performs the same two steps:
    - Induction Braze Couplers
    - Test for Leak
- If the condenser passes leak test, it is painted and then shipped.
- If the condenser fails leak test, it is held for additional leak testing to ID area of the leak.
- Create Process Flow Diagram for the condenser braze void case.
- You will have 10 minutes to complete this exercise.

Process Flow Diagram - Exercise

Inputs	Operation	Function

Process Flow Diagram - Exercise

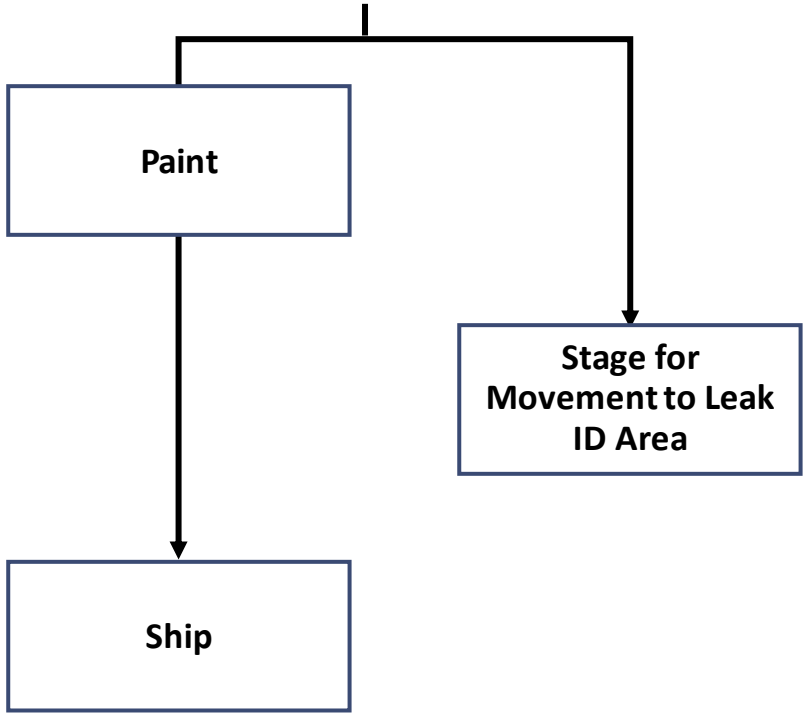
Inputs	Operation	Function

Process Flow Diagram - Answer

Inputs	Operation	Function
Tanks (2) Tubes (48) Air Centers (47) Frame (Tooling)	Stacker 1: Assemble Components; Place Frame Stacker 2: Assemble Components; Place Frame Stacker 3: Assemble Components; Place Frame	Assemble Tubes and Air Centers to the Tanks Place Frame
Unbrazed Assembly	Continuous Belt Oven: Flux; Braze; Cool Down; Remove Frame	Braze Condenser to Seal
Brazed Assembly (1) Couplers (2) Flux Rings (2)	Braze & Test 1: Braze Couplers; Leak Test Braze & Test 2: Braze Couplers; Leak Test . . . Braze & Test 5: Braze Couplers; Leak Test	Braze Input and Output Couplers Leak Test Assembly

Process Flow Diagram - Answer

Inputs	Operation	Function
Assemblies That Passed Leak Test	Paint	Apply Paint
Assemblies That Failed Leak Test	Stage for Movement to Leak ID Area	Define Leak Area
Painted Assemblies	Ship	Transport Parts



Talking to the “Experts”



Saturday Morning – 8:00 AM

- Saturday morning all the “experts” are telling how the problem lives in the braze oven.
- The processing time for each part is 12 hours.
- There is not going to be enough time to follow parts through the process, make up for missed shipments, and get home before the party.

Using 4Ps to Develop a Project Statement - Exercise

Product or **P**rocess that has the problem.

Problem description

Place where the problem is discovered

How is the **P**ain measured (metric)

Create the Project Statement:

"Find and Control the BIG X that drives (PRODUCT or PROCESS) (PROBLEM) discovered at (PLACE) to save or prevent (PAIN)."

You will have 5 minutes.

Using 4Ps to Develop a Project Statement - Answer

Product or **P**rocess that has the problem. — Condenser

Problem description — Leakage

Place where the problem is discovered — End of line tester

How is the **P**ain measured (metric) — Delivery: Missed shipments and assembly plant shutdowns.

Project Statement:

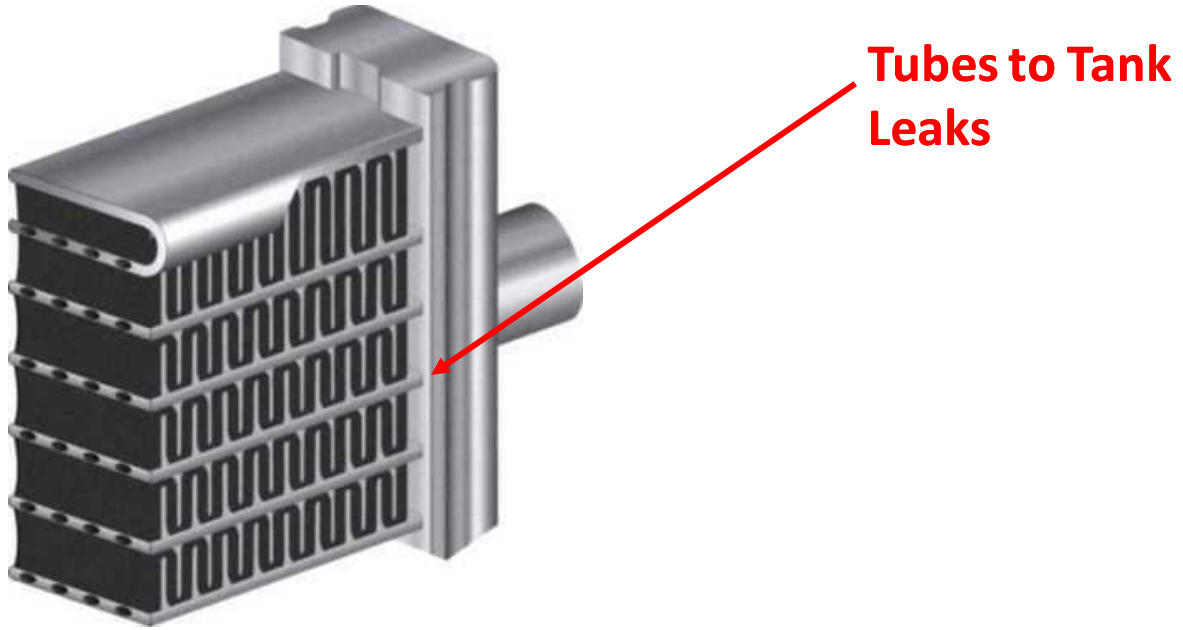
"Find and Control the BIG X that drives condenser leaks discovered at end of line tester and prevent missed shipments and assembly plant shutdowns."

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Condenser Scrap

Defect Strategy

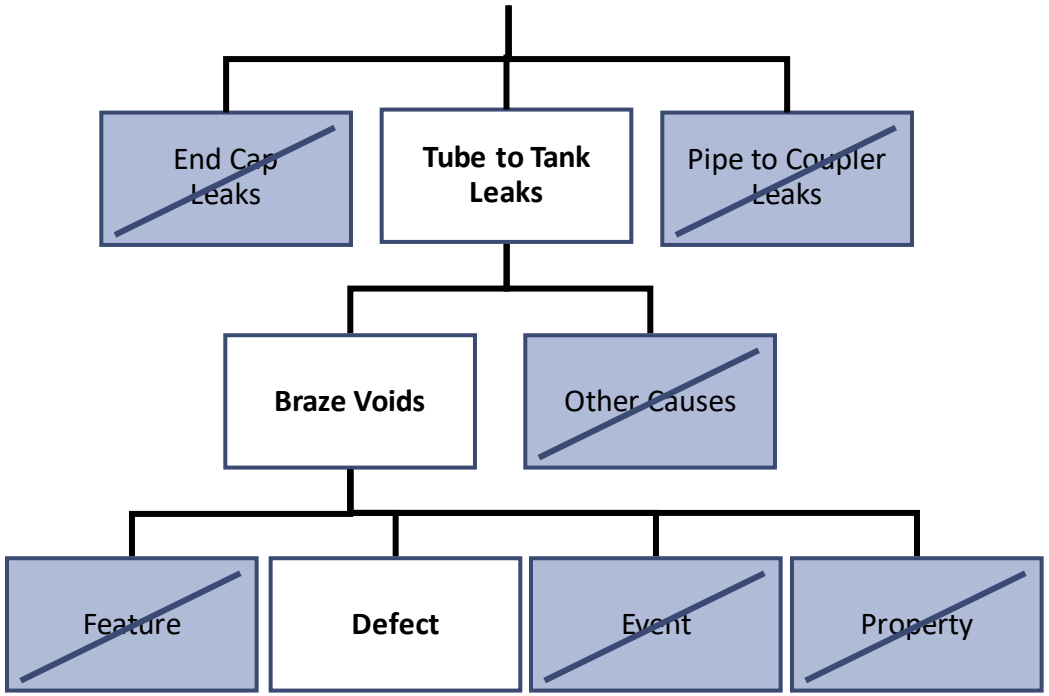
Condensers are scrapped for leaks at the end caps, the inlet/outlet coupler joints, and at the tube to tank joints. Tube to tank leaks cannot be repaired and are the largest driver of missed shipments and scrap dollars.



Condenser Scrap

Project Definition Tree

Project Statement: Find and control the BIG X that drives automobile condenser leaks discovered at end of line tester and prevent missed shipments and assembly plant shutdowns.



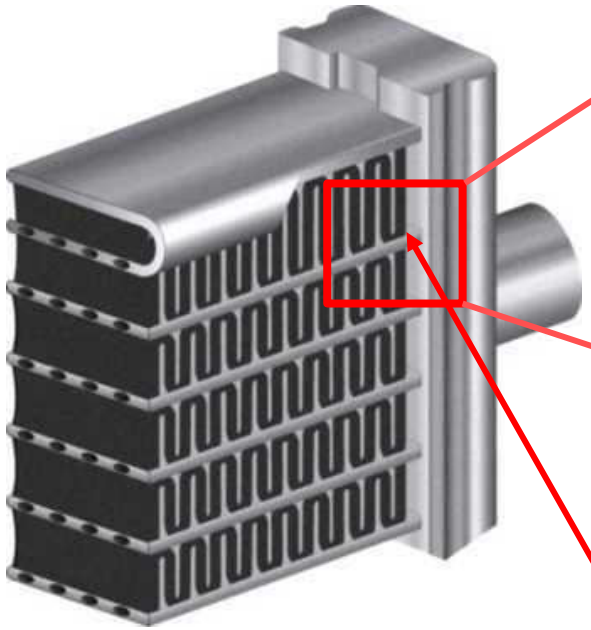
Mone(Y) Description: Tube to tank interface, leaks or does not leak.

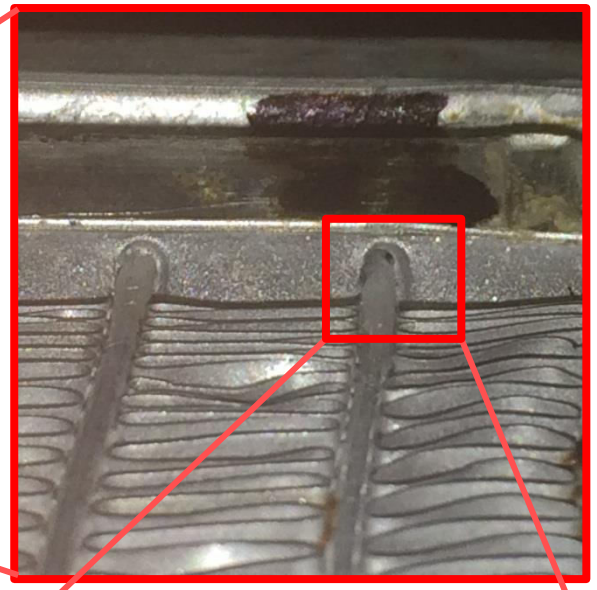
Rationale:

- End cap and pipe to coupler leaks can be repaired. Tube to tank leaks cannot be repaired and are driving scrap.
- 98 % of leakers are braze voids.
- Results of the Point to Point contrast.
- Strategy split.

Condenser Scrap

Point to Point Contrast







Tubes to Tank Leaks

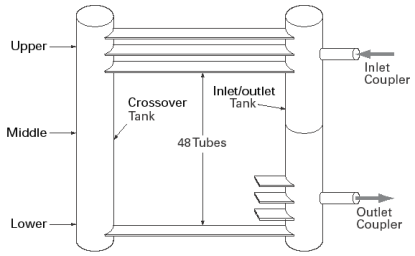
Point to Point Contrast:

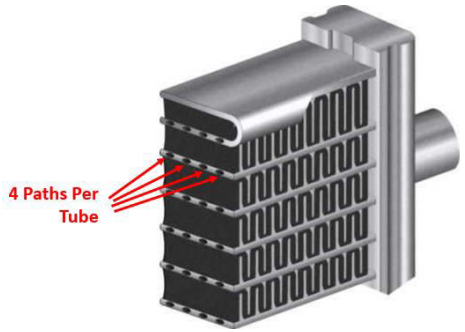
Detailed analysis of the identity of the Mone(Y).

Talking to the Parts - Exercise

Objective: Develop a Defect Strategy Diagram for Tube to Tank Leaks

- Each condenser has markings identifying which stacker and which shift it was produced on.
- It is possible to ID the joint and path that is leaking (braze void).
- There is an equal rejection rate across all five testers and all three shifts.
- Process takes too long to capture new parts and analyze them (only have until Sunday morning).
- Scrap from Friday is piled up.
- You will have 15 minutes to develop a Strategy Diagram.





Defect Strategy Diagram - Exercise

Family of Variation, or Contrast	***Diagram***	***What is Meant by Each Contrast***
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		

Defect Strategy Diagram - Exercise

Family of Variation, or Contrast	***Diagram***	***What is Meant by Each Contrast***
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		
☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		

Defect Strategy Diagram - Answer

Family of Variation, or Contrast

Diagram

What is Meant by Each Contrast

Design Families

Point to Point

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Same Path

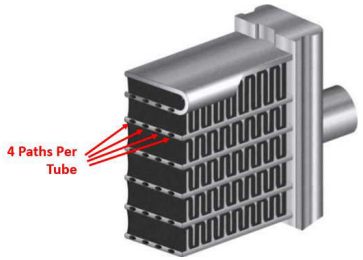


What does the defect look like? Can we explain why the defect initiates at this point?

Path to Path

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Same Tube

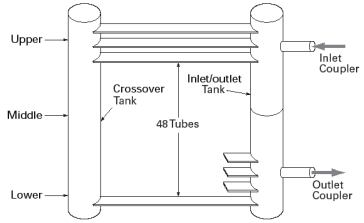


Are there more leaks from certain paths more than others? How does the location of WOW paths relate to the rest of the part?

Tube to Tube

- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Same Region



Are there BOB and WOW joints with the same region? What can be leveraged from the BOB and WOW joints?

Region to Region

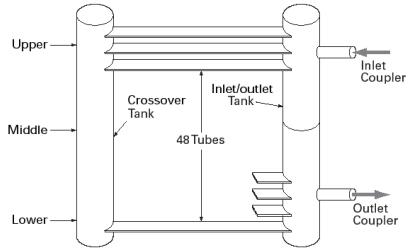
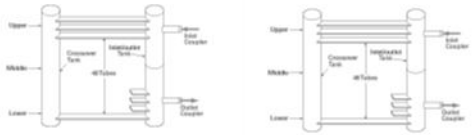
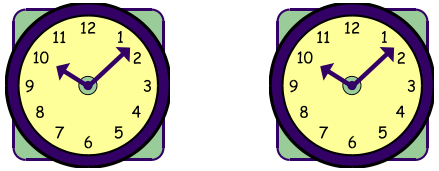
- ☐ Do Not Know
- ☐ Large Contrast
- ☐ Small or No Contrast

Same Tank

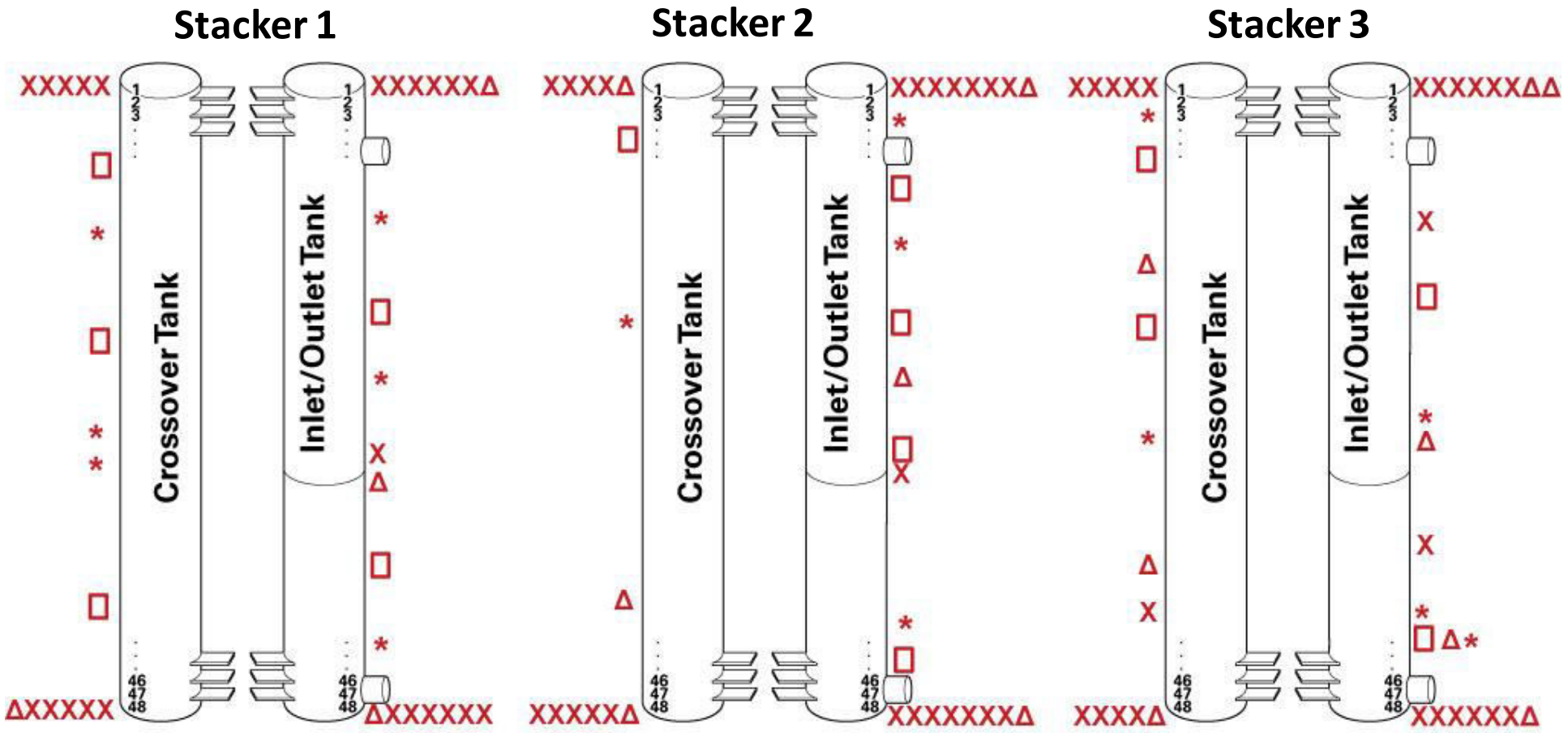
Which of the three regions; upper, middle, lower, are most of the leaks showing up?

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Defect Strategy Diagram - Answer

Family of Variation, or Contrast	***Diagram***	***What is Meant by Each Contrast***
Tank to Tank Same Condenser ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Is there a difference in the leak locations tank to tank (side to side) within a condenser? Are the leak locations the same on both tanks?
Process Families		
Condenser to Condenser Same Frame ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		What can be deduced from BOB and WOW condensers?
Frame to Frame Same Stacker ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Do we have BOB and WOW frames in our process?
Stacker to Stacker Same Shift ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Is there contrast between the three stackers?
Shift to Shift ☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Is the defect consistent over time?

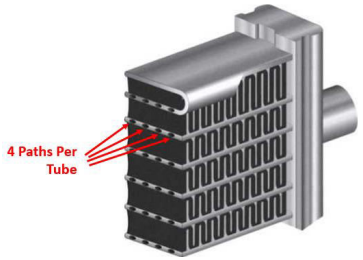
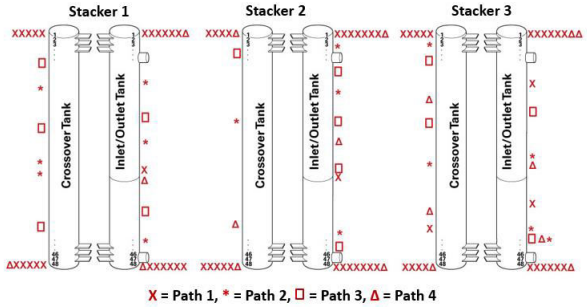
Condenser Concentration Diagram - Exercise



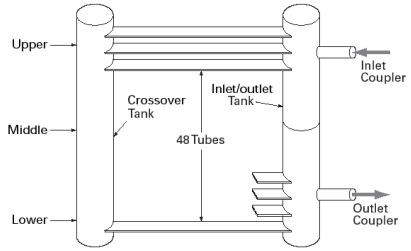
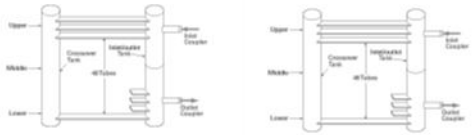
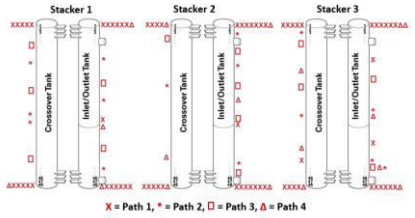
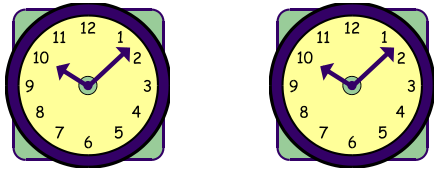
X = Path 1, * = Path 2, □ = Path 3, Δ = Path 4

Are the experts correct about temperature differences in the braze oven?

Defect Strategy Diagram - Answer

Family of Variation, or Contrast	***Diagram***	***What is Meant by Each Contrast***
Design Families		
Point to Point Do Not Know Large Contrast Small or No Contrast **Same Path**		Braze void results in visual presence of air bubbles leaking out of a hole at the tube to tank braze joint during the water bath test.
Path to Path Do Not Know Large Contrast Small or No Contrast **Same Tube**		Path 1 is worse than paths 2, 3, and 4 by far.
Tube to Tube Do Not Know Large Contrast Small or No Contrast **Same Region**		End tubes (1 and 48) are WOW, 2 and 47 provide the BOB contrast and points away from uneven heating in the braze oven.
Region to Region Do Not Know Large Contrast Small or No Contrast **Same Tank**		The middle Regions have less failure.

Defect Strategy Diagram - Answer

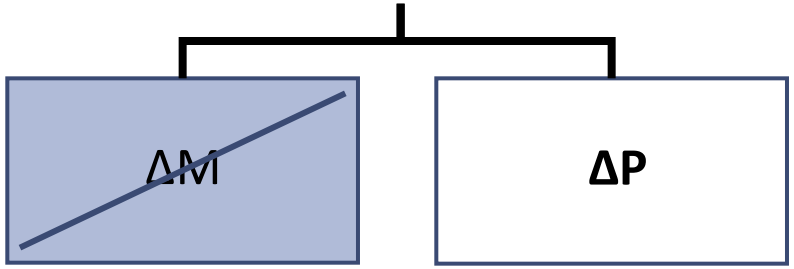
Family of Variation, or Contrast		***Diagram***	***What is Meant by Each Contrast***
Tank to Tank Same Condenser	☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast		Tanks have the same amount of defects.
Process Families			
Condenser to Condenser Same Frame	■ **Do Not Know** ☐ **Large Contrast** ☐ **Small or No Contrast**		Blind to this contrast because we only worked with scrap condensers.
Frame to Frame Same Stacker	■ Do Not Know ☐ Large Contrast ☐ Small or No Contrast		Blind to this contrast. Frames are not tracked.
Stacker to Stacker Same Shift	☐ **Do Not Know** ☐ **Large Contrast** ■ **Small or No Contrast**		All stackers make WOW condensers on end tube joints, both tanks, path 1.
Shift to Shift	☐ Do Not Know ☐ Large Contrast ■ Small or No Contrast		Shifts have the same amount of defects.

Condenser Scrap Solution Tree

The Solution Tree starts once the BOB / WOW contrast are defined.



Mone(Y) Description: Condenser leaks (braze voids) at tube to tank joints.



Rationale:

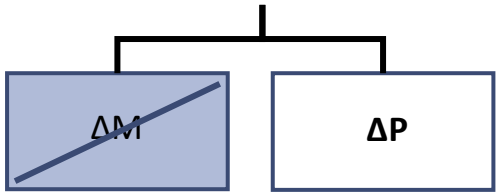
- Leaks are found on the production tester and verified on bubble tester for leak location.

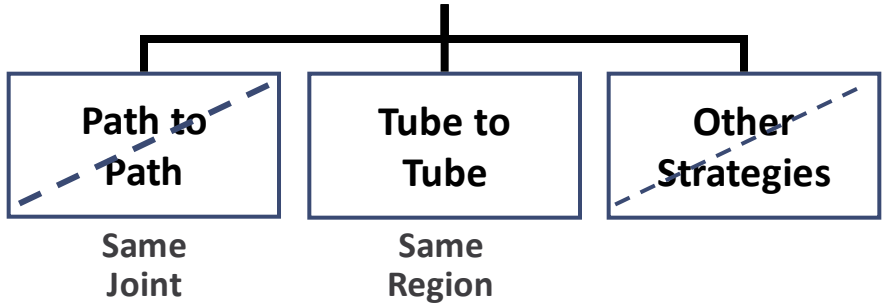
Condenser Scrap Solution Tree

Mone(Y) Description:

Condenser leaks (braze voids) at tube to tank joints.







Rationale:

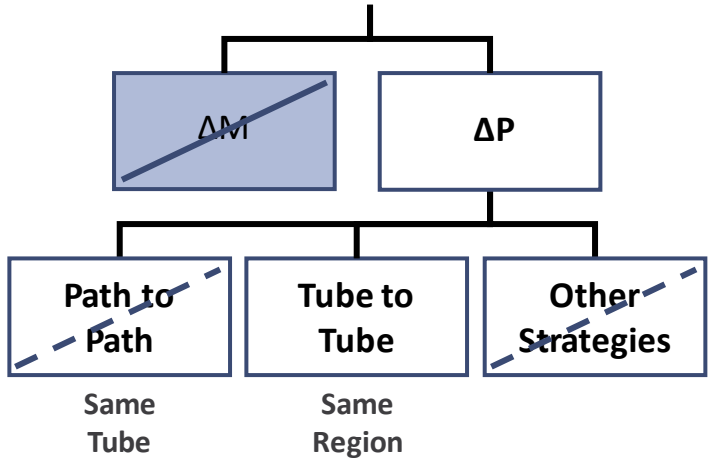
- Concentration diagram: Joint to joint provides the greatest leverage.

Condenser Scrap Solution Tree

Mone(Y) Description:

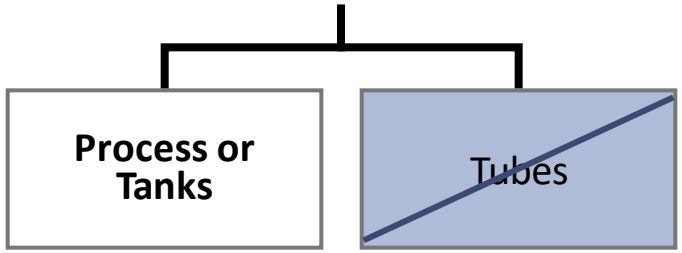
Condenser leaks (braze voids) at tube to tank joints.





Went to the line to investigate the stacker process, but the equipment moved too fast. A high-speed camera was used to see tubes 46, 47, and 48 on stacker 2 during the stacking process.

Saturday Afternoon – 4:00 p.m.



Rationale:

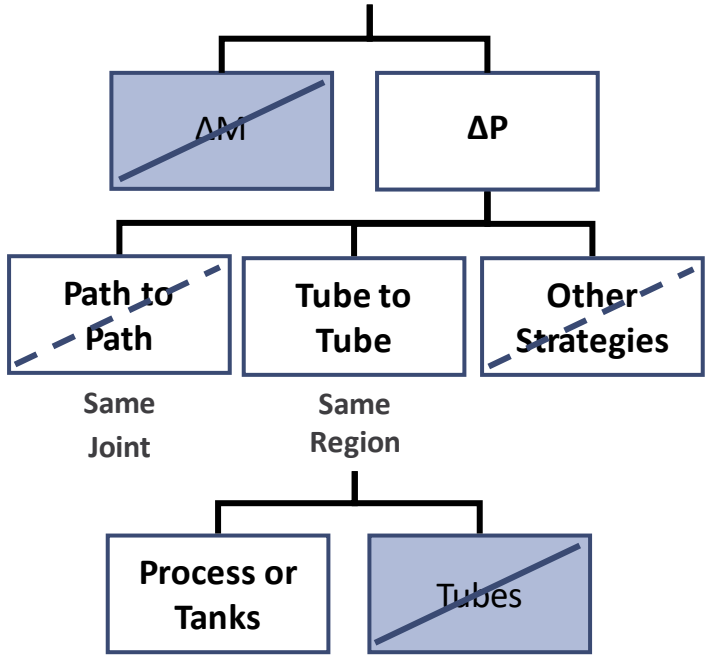
- Tube position in tank is random.

Condenser Scrap Solution Tree

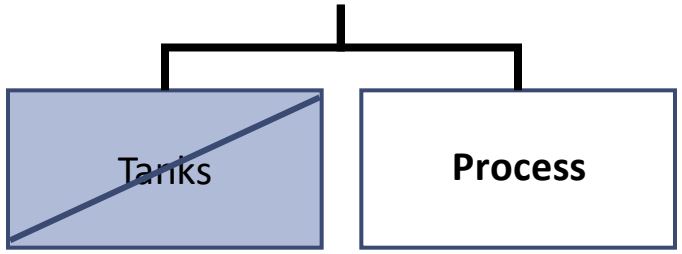
Mone(Y) Description:

Condenser leaks (braze voids) at tube to tank joints.





Saturday Afternoon – 4:00 p.m.



Rationale:

- High speed video shows tube 1 and 48 moves during assembly, 46 and 47 do not.
- Movement causes scraping of cladding at path 4 and creates a larger tube to tank gap at path 1.

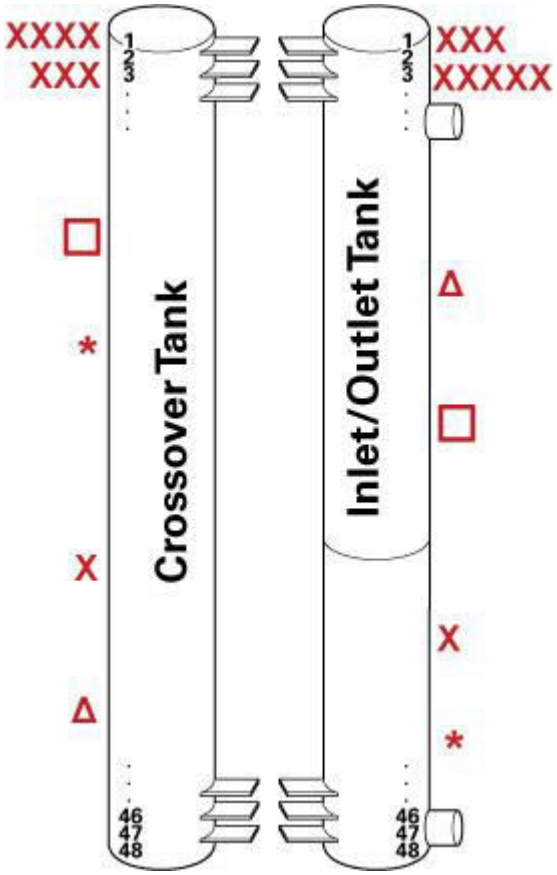
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Confirmation

Overnight the physics of failure was confirmed by moving the hold down bar to tubes 3 – 48 of stacker #2 and developing a concentration diagram of tube to tank leaks (braze voids).

No changes to stackers one and three!

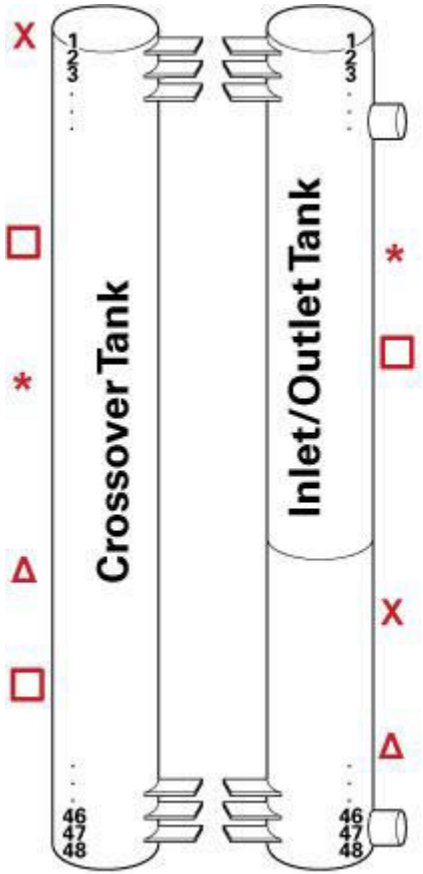
Stacker 2



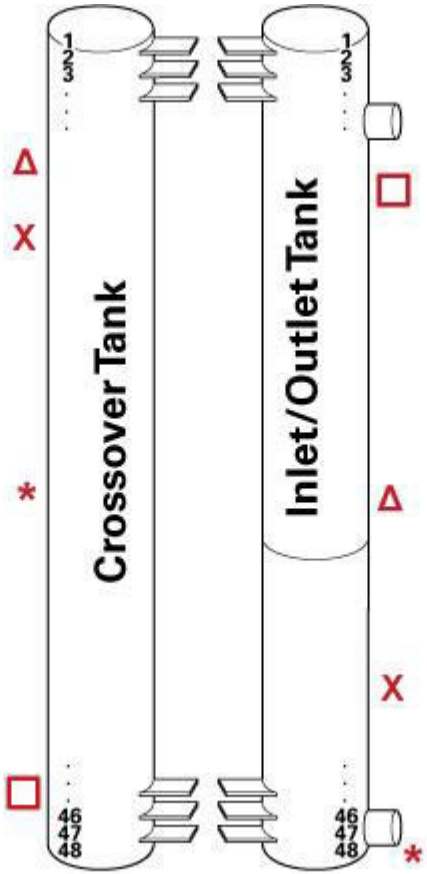
Sunday Morning – 9:00 a.m.

Killed the BIG X, Now Find the Small X

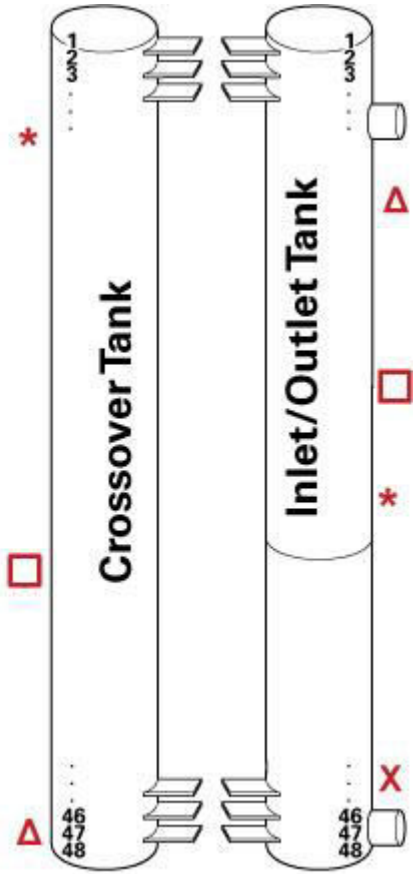
Stacker 1



Stacker 2



Stacker 3



X = Path 1, * = Path 2, □ = Path 3, Δ = Path 4

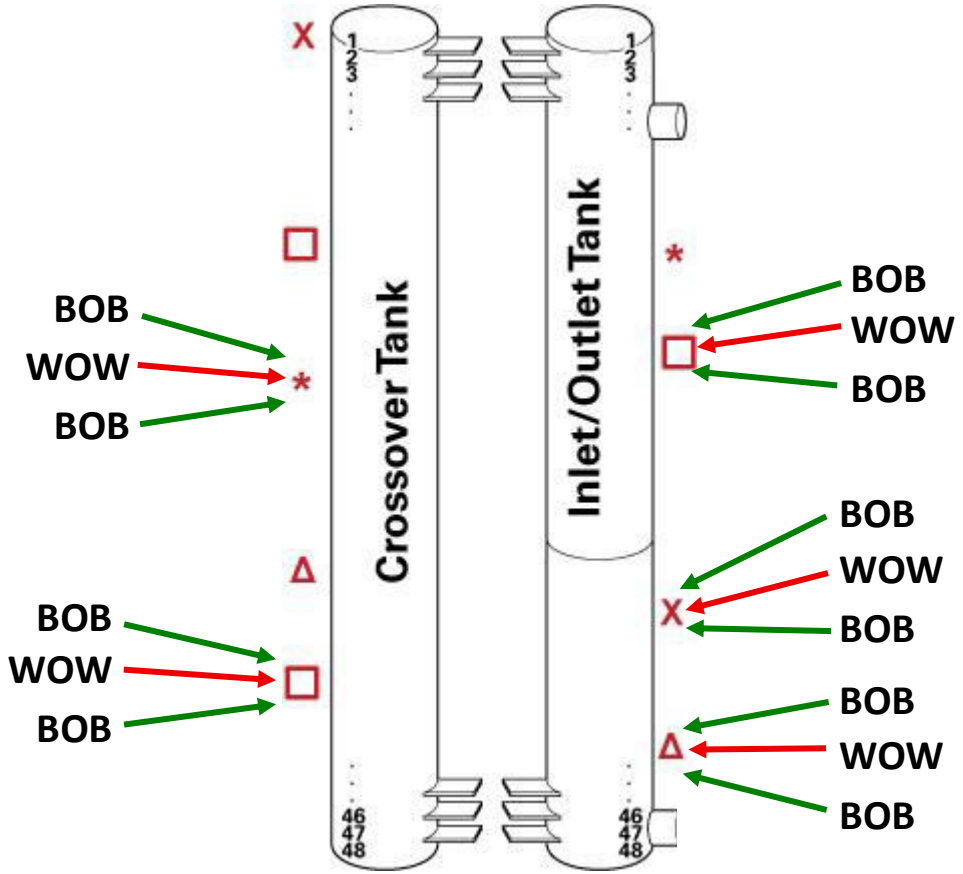
- Rejects now look random.
- Is it uneven heating?

- Does the strategy change?
- Do you need to ask new questions?

Still Leveraging BOB and WOW Joints

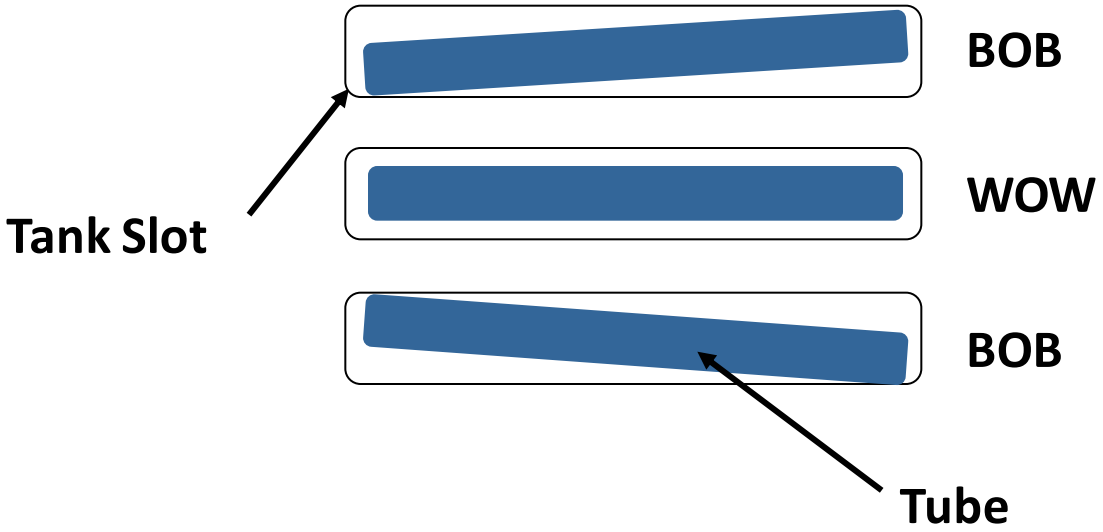
Look for differences between the BOB and WOW joints

Stacker 1



Still Leveraging BOB and WOW Joints

- Viewing cross sections of a WOW joint surrounded by two BOB joints are shown below.

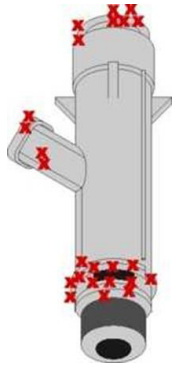


What is the Small X?

When viewing the cross section, the technician said…

“Son, cladding flows – it does not jump!”

Defects vs Destructive Events



Defect:

- Point to Point contrast
- Location, Location, Location



Destructive Events:

- Energy has permanently altered the product and has left behind a trace.
- Initial strategy is similar to Defect strategy. Start with concentration diagram
- Energy needs to be considered.

Destructive Events will be covered in detail in PSQ - Black Belt Class.

Module 5 Summary

Defect Strategies

- A Strategy Diagram for defects is a plan for how to talk to the parts and discover leveragable patterns.
- A concentration diagram reveals the location of a defect within a part and within a process.
- It is important to look at the defect up-close to make sure we are not combining different types of defects in one project.

6 Malfunction Event Strategies I

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Module 6 Objectives

Malfunction Event Strategies I

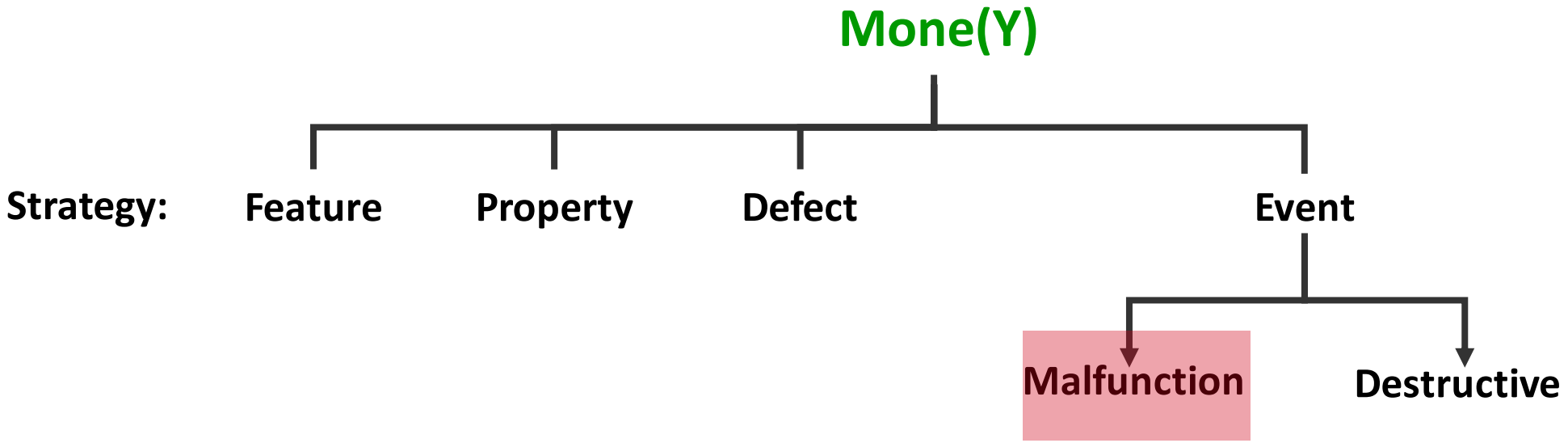
Understand:

- How to plan and execute a malfunction event strategy

Be able to:

- Develop a Malfunction Event Strategy Diagram
- Conduct a Component Search
- Use Paired Comparisons to identify a feature , property or defect BIG X suspect
- Use progressive disassembly / reassembly to isolate the BIG X step within the an assembly process
- Document convergence on a Solution Tree

Different Strategies for Different Mone(Y)s



Explanation: Product does not work at the expected level of performance. Evidence of the events disappears as energy supply is removed. Energy can be applied to attempt to reproduce the malfunction.

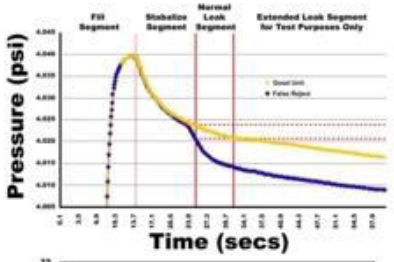
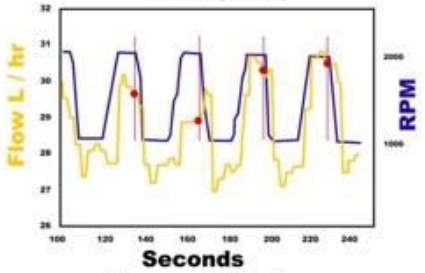
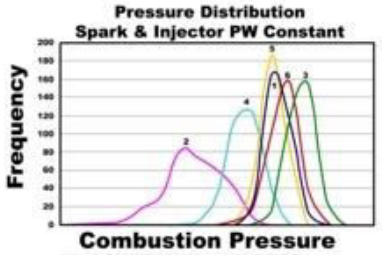
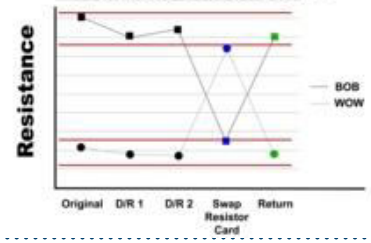
Tool: **Strategy Diagrams**

Strategy Diagrams are a way to document possible contrasts for the Mone(Y) that you can leverage during your project.

Malfunction Event Strategy Diagram - Template

Mone(Y) =

Design Families

Family of Variation, or Contrast		*Diagram*	*What Is Meant by Each Contrast*
Point (in time) to Point **Same Event**	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast ☐ N/A		What is the "signature" of the event? Look "inside" the WOW event by plotting it as it occurs over time to look for a contrast within the event. A energy model may be useful to identify alternative Mone(Y)s to characterize the event.
Event to Event **Within Same Unit**	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast ☐ N/A		Does the event repeat or is it intermittent? What other functional parameters can we measure along with the Mone(Y) which will offer clues to the physics? Does orientation of components matter?
Within Same Unit to Within Same Unit **Same Unit**	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast ☐ N/A		Are there contrasts within a unit (cylinder 1 vs. cylinder 4)? Rename unit appropriately for your project (e.g. engine, pump, printer).
Unit to Unit	☐ Do Not Know ☐ Large Contrast ☐ Small or No Contrast ☐ N/A		Are there BOB and WOW units? Rename unit appropriately for your project (e.g. engine, pump, printer).

Start at the Level of the Event

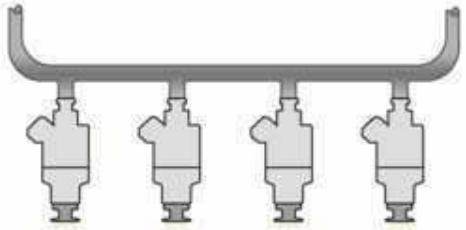


Car

System



Machining Center



Fuel injection system

Sub-system



Controller



Fuel injector

Assembly



Circuit Board



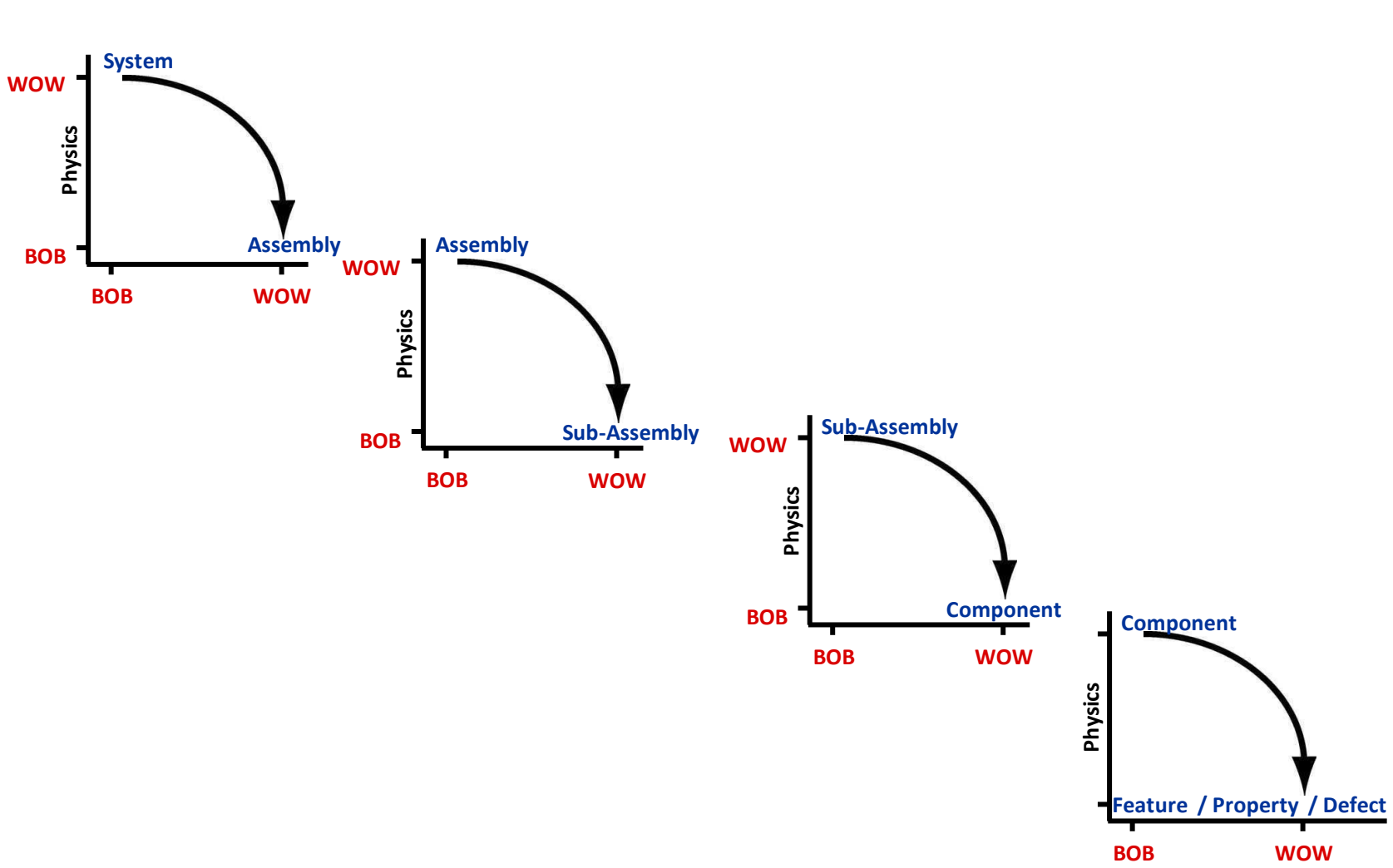
Valve

Component (Detail Part)



Capacitor

Event Strategy – Convergence Using Progressive Search



Important Linkages

Customer Experience



Product Mone(Y)



Product BIG X Feature or Property



Process Mone(Y)



Process BIG X Parameter

Strategy Diagrams are About Contrasts

If there is a noise problem with vehicle transmissions, what questions would you ask about the potential contrasts without knowing anything about the causes:

- What is the “signature” of the noise?
- Does the noise happen all the time or is it intermittent?
- Do all units of this particular model make the noise (are there BOBs)?
- Is it worse under certain conditions?
- Do other models make the same noise?
- Do only transmissions made on a specific date have the problem?
- Made on a specific line?
- After a certain time in the field?
- Etc.

Event Strategy
Example: Wiper Chater Case

An automotive OEM is experiencing a chronic warranty problem with windshield wiper chatter



Define Macro Conditions

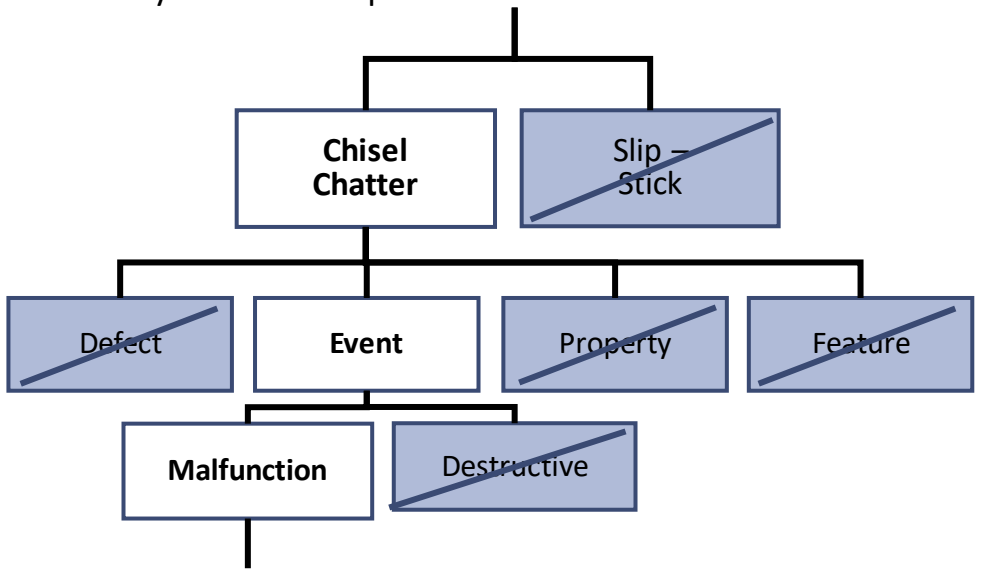
Oftentimes problems only occur under certain environmental or operating conditions. These are intermittent. Intermittent events involve interactions. Something within the system is sensitive to the change in conditions.

- Customer reclaims noisy chatter windshield wiper discovered in the field
- Warranty costs: 5 Mio. €
- The problem is intermittent.
- Warranty claims are concentrated in colder climates.
- The problem occurs at the lowest level.
- Customers note that the chatter only occurs during light misty rains.
- Customers take their vehicles into the dealer. The dealer tests for chatter and the system works fine.
- To appease customers the dealer replaces the blades. Most of the time, this resolves the problem.
- The wiper blade supplier has a test buck which is an actual windshield for this vehicle. They test return blades. No trouble found.

Example: Wiper Chater Case

Project Definiton Tree

Project Statement: Find and Control the BIG X that drives wiper system noise discovered in the field to save €5 million in warranty costs and improve customer satisfaction.

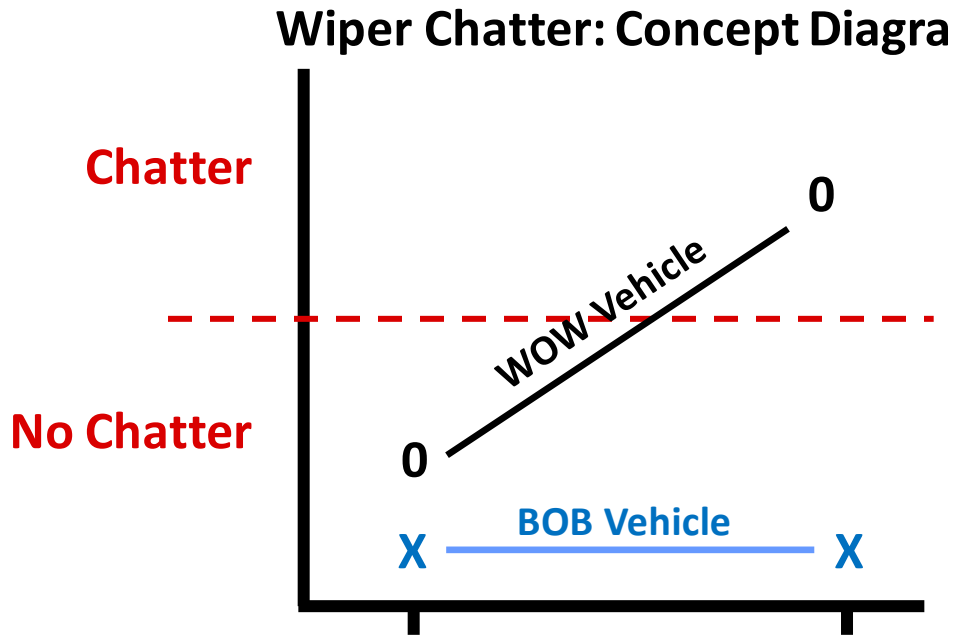


Rationale:

- Most noise complaints are for chisel chatter.
- Wiper Chisel Chatter is an event.
- Chatter is a malfunction.

Concept Diagram

Concept diagrams are an important strategy tool. They help us determine which parts to test and how to test them.

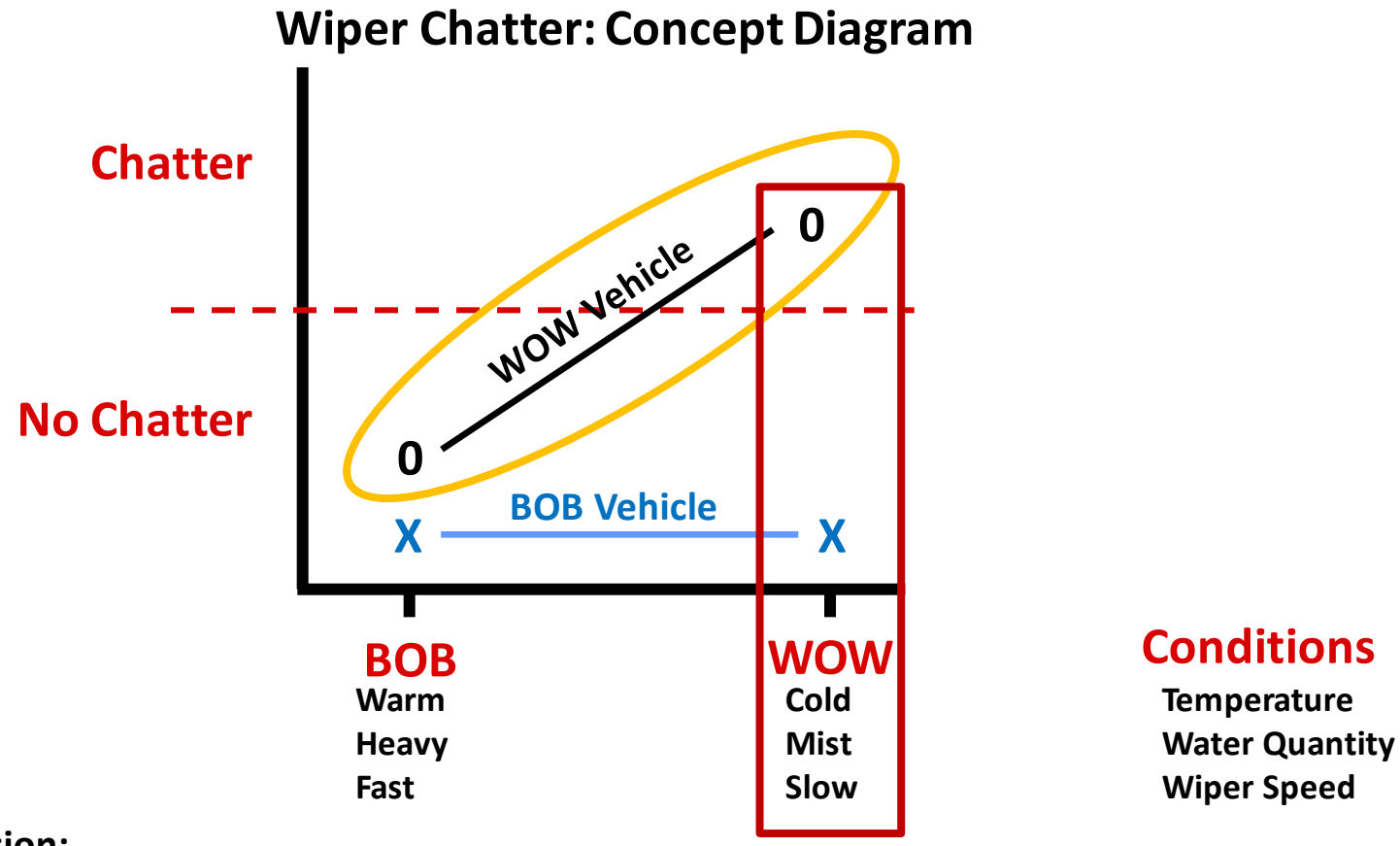


BOB	WOW	Conditions
Warm	Cold	Temperature
Heavy	Mist	Water Quantity
Fast	Slow	Wiper Speed

Conclusion: ?

Concept Diagram

Concept diagrams are an important strategy tool. They help us determine which parts to test and how to test them.



Conclusion:

- Must work with a known **WOW vehicle** to discover BOB / WOW conditions.
- Must test under known **WOW conditions** to discover what is different between BOB / WOW vehicles.

Concept Diagram - Guidelines

A concept diagram is a tool to describe relationship of two X inputs to the Mone(Y) output and is useful to convey knowledge of a partially understood event. Since this is a concept of the physical observations, actual data points may not be required to create the diagram. Often the concept diagram can be derived from information on the Strategy Diagram.

1. Define the Mone(Y). The Mone(Y) can be defined as either an attribute or variable measure.

2. Identify the two X input variables. The X inputs can be defined as either an attribute or variable measure.

3. Show graphically the Mone(Y) on the y-axis and one of the X variable on the x-axis. Use BOB and WOW to describe the variable.

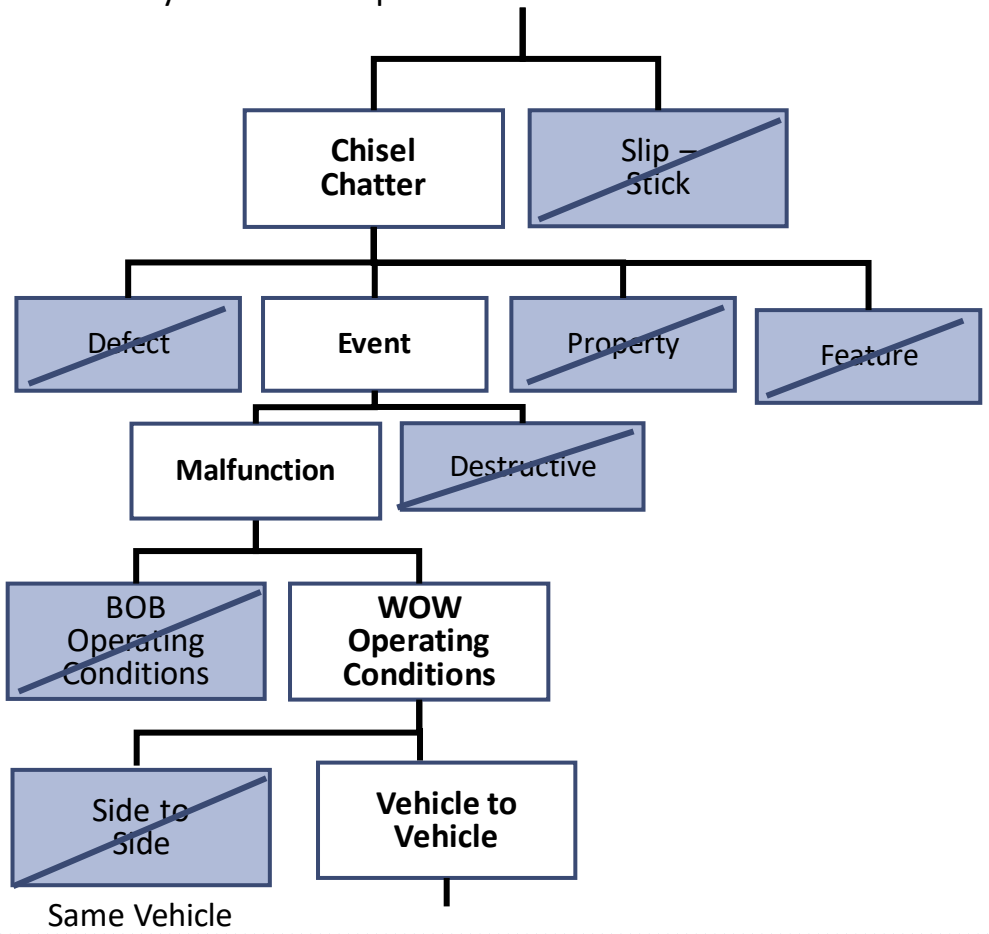
4. Show the second X variable as two lines using the BOB and WOW descriptions to describe the two lines.

5. Define what BOB and WOW means for each variable.

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Example: Wiper Chater Case
Project Definiton Tree

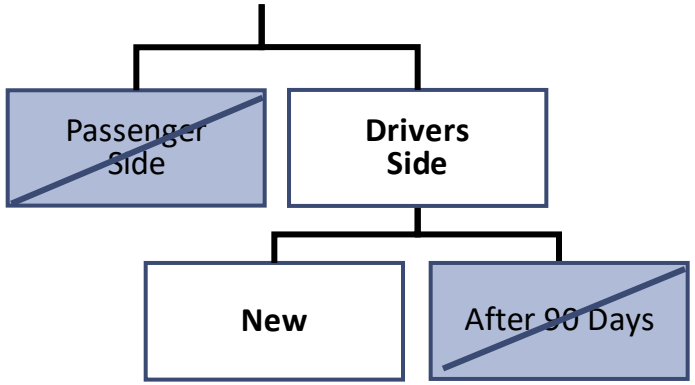
Project Statement: Find and Control the BIG X that drives wiper system noise discovered in the field to save €5 million in warranty costs and improve customer satisfaction.

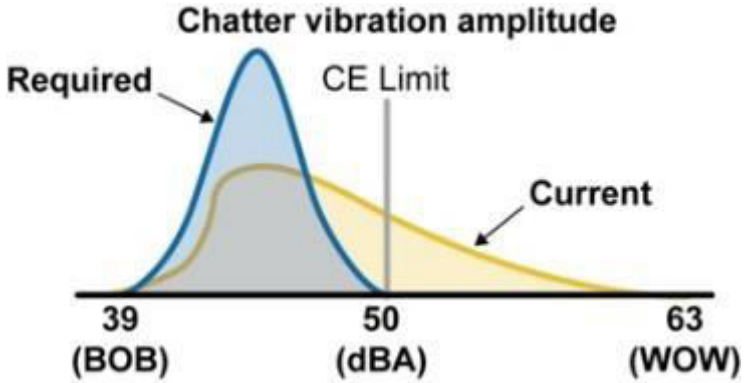


Rationale:

- Most noise complaints are for chisel chatter.
- Wiper Chisel Chatter is an event.
- Chatter is a malfunction.
- Concept diagram
- Best point of leverage. Allows the team to use Component Search.

Example: Wiper Chater Case
Project Definiton Tree





Rationale:

- Best to focus on one side only.
- Problem can be found in new vehicles. Field environments and operating conditions can be replicated in lab tests.

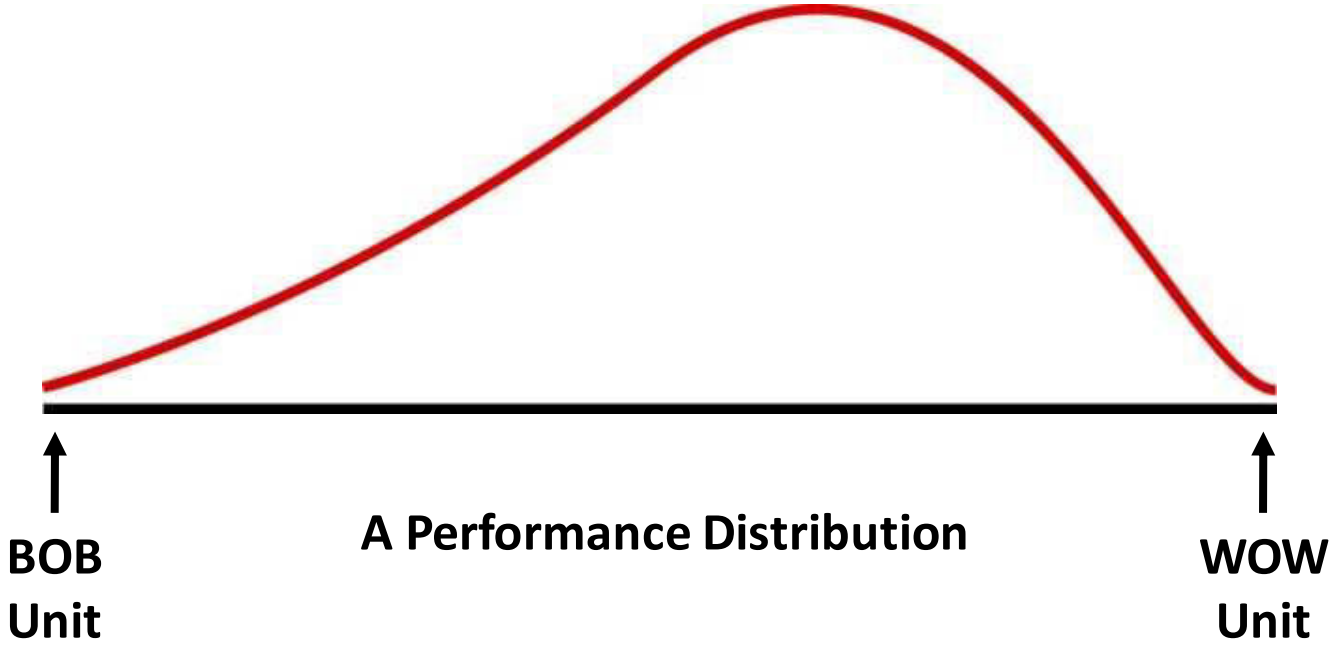
Mone(Y) Description: Chatter vibration amplitude on driver's side of new vehicles; operating the wiper system under WOW conditions.

Customer Complaint → Mone(Y) Event

Events and Component Search

Component Search is used when we have a BOB and WOW unit, and we want to learn the name of the component (part) which contains the BIG X.

Although there are exceptions, it is usually limited to units which can be disassembled and reassembled, and retested to obtain another reading.



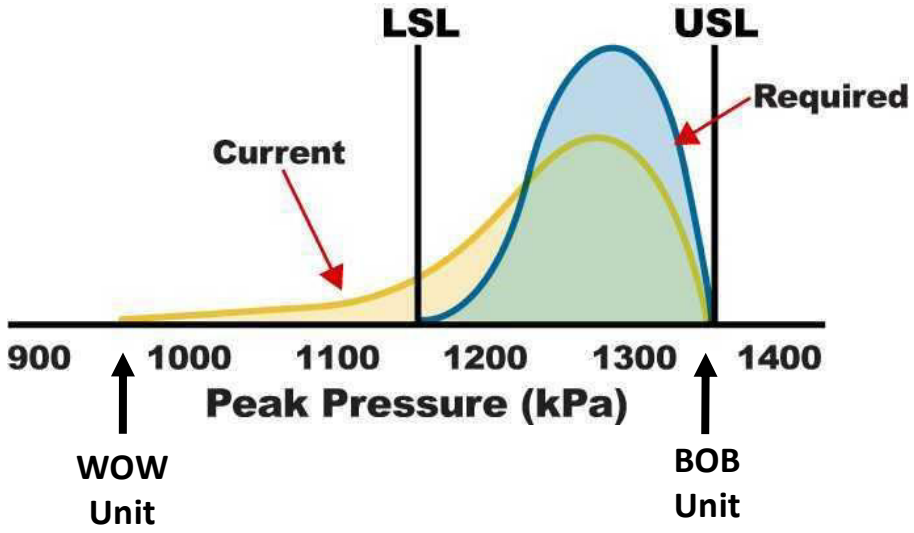
Air Conditioning Compressors Have too Much Variation in Peak Pressure

- Management has assigned a continuous improvement project to you.
- Develop malfunction event strategy diagram for the design families.
- BOB and WOW compressor can be readily found from EOL test results.
- A compressor has 6 cylinders.
- Testing options include the EOL test stand and a Lab Test.
- In the EOL test bench, the max. peak pressure recorded.



Project Statement: Find and eliminate the BIG X that drives compressor peak pressure failures found at the end of line test to reduce repairs, improve productivity and improve on time delivery.

Air Conditioning Compressors Have too Much Variation in Peak Pressure - Exercise





Strategy Diagram – Compressor Peak Pressure - Exercise

Based on what you have learned of this case, start to develop the Strategy Diagram for the design families (first page) and describe what you might do to investigate potential contrasts.

You will have 10 minutes to complete this group exercise.

Strategy Diagram – Compressor Peak Pressure - Excercise

Design Families

Mone(Y) =

Family of Variation, or Contrast

Diagram

What Is Meant by Each Contrast

	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A

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Strategy Diagram – Compressor Peak Pressure - Answer

Design Families

Mone(Y) = Peak Pressure [kPa]

Family of Variation, or Contrast

Diagram

What Is Meant by Each Contrast

Point (in time) to Point

Same Event

- ☐ Large Contrast
- ☐ Small or No Contrast
- ■ Do Not Know
- ☐ N/A

Measure pressure over time, starting at time equals zero up to steady state. How does the signature look different between the BOB and the WOW?

Event to Event

Same Cylinder

- ☐ Large Contrast
- ☐ Small or No Contrast
- ■ Do Not Know
- ☐ N/A

An MSA will show event to event combined with measurement system variation. Can we reliably see the difference between BOB and WOW?

Cylinder to Cylinder

Same Compressor

- ☐ Large Contrast
- ☐ Small or No Contrast
- ■ Do Not Know
- ☐ N/A

There are six cylinders per pump. Can we see individual cylinder performance if we graphed to pressure over time?

Compressor to Compressor

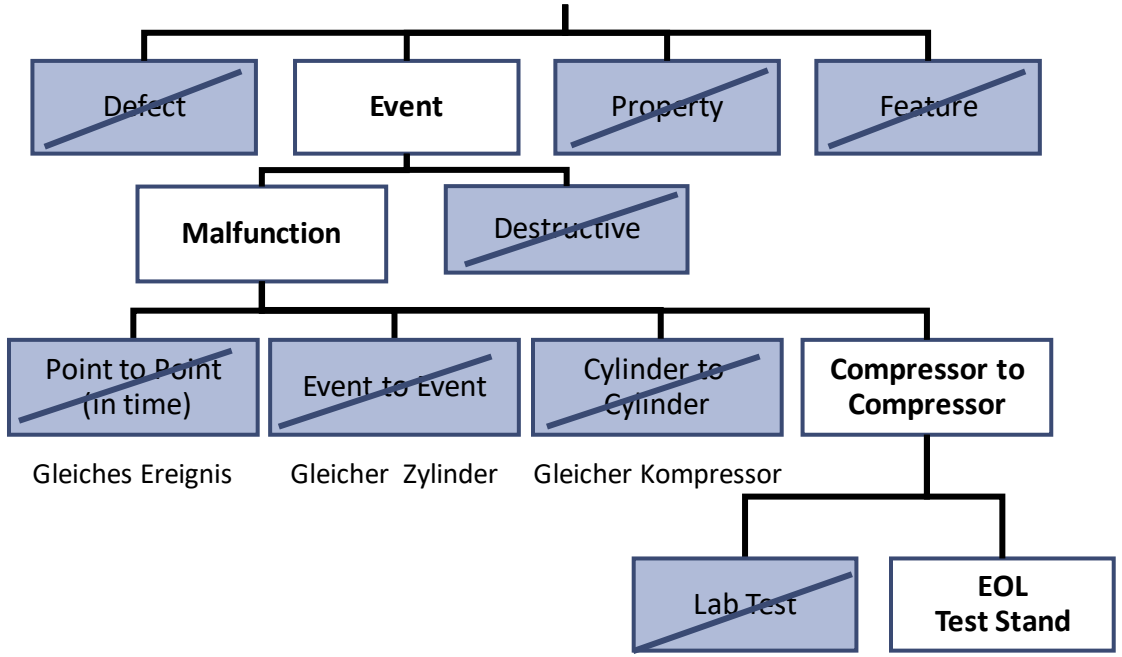
- ■ Large Contrast
- ☐ Small or No Contrast
- ☐ Do Not Know
- ☐ N/A

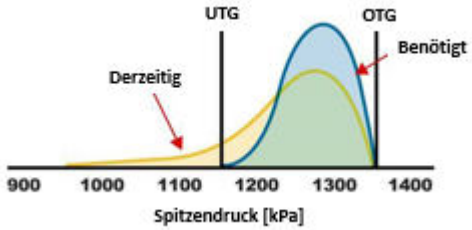
There are BOB and WOW units; use for component search.

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Project Definition Tree – Compressor

Projektstatement: Find and control the BIG X that drives compressor peak pressure failures found at the end of line test to reduce repairs, improve productivity and improve on time delivery.



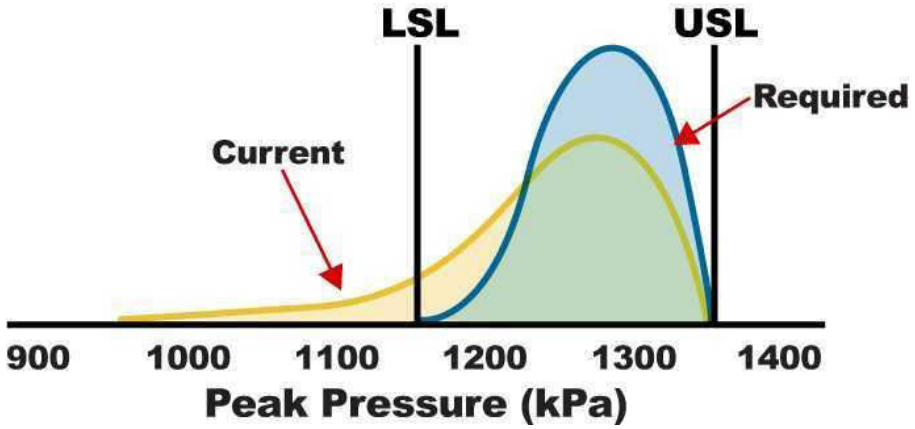


Rationale:

- EOL Compressor Test Failures
- Test measures system performance.
- Strategy Diagram
- EOL test stand is available and reflects customer requirements.

Mone(Y): Peak pressure [kPa] measured at EOL test

Required Elements of a Mone(Y) Distribution



- Draw the current distribution and include:
  - Spec limits
  - Physical barriers
- Label the horizontal axis with the name of the Mone(Y) and the units of your measuring system.
- Draw the required distribution.

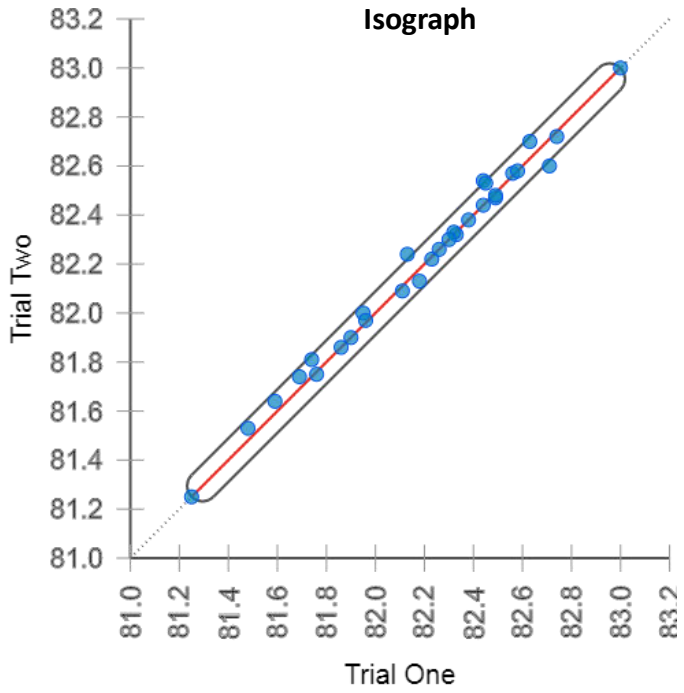
Component Search – Stage 0

An Alternative to Isograph

When measurement is quick and easy, Isograph can be used to evaluate the measurement system effectiveness and to find BOB and WOW samples.

Stage 0 is an adequate tool, when

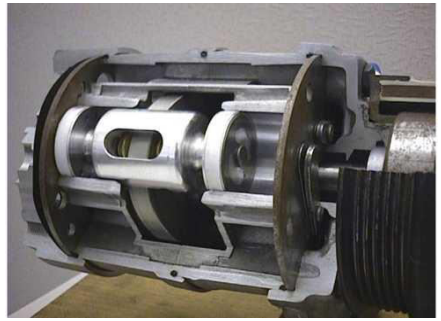
- you have limited parts,
- each measurement takes too long,
- each measurement is very expensive.



$$\frac{\Delta P}{\Delta M} = \frac{1.75}{0.13} = 13.9$$

Component Search – Stage 0

Two a/c compressors were measured for peak pressure, each in random order.



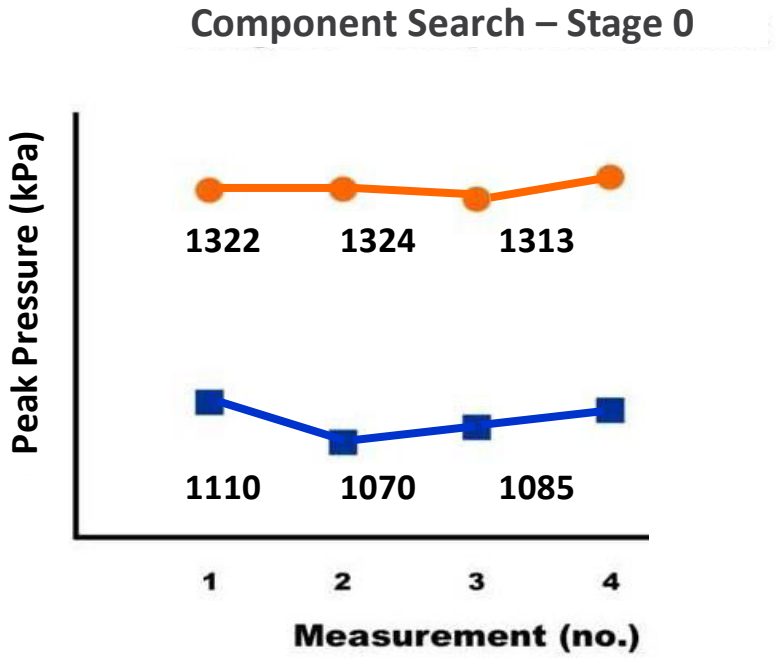
WOW
Compressor #1 – 1110 kPa
Compressor #1 – 1070 kPa
Compressor #1 – 1085 kPa
Compressor #1 – 1107 kPa

BOB
Compressor #2 – 1322 kPa
Compressor #2 – 1324 kPa
Compressor #2 – 1313 kPa
Compressor #2 – 1335 kPa

Component Search – Stage 0 - Guidelines

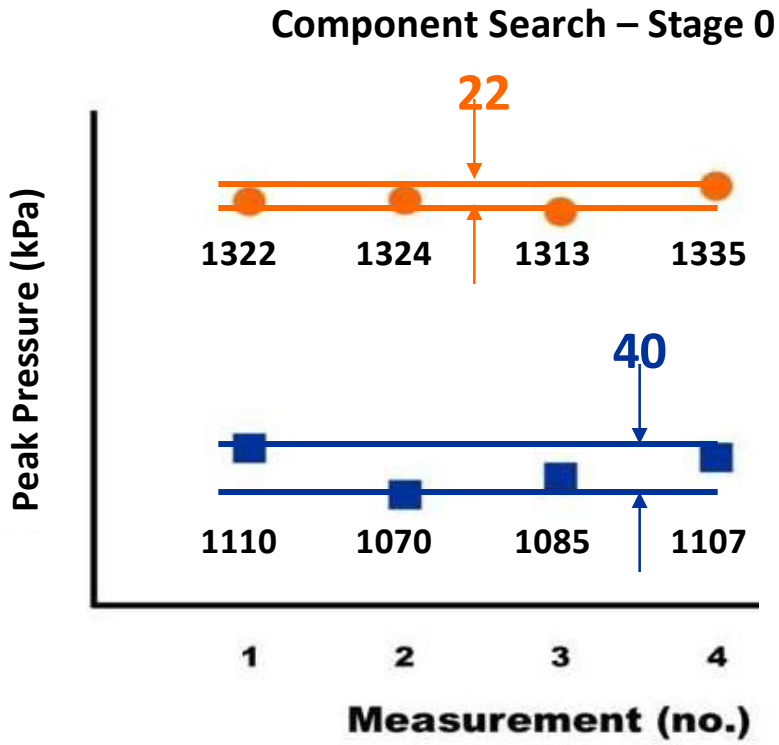
Stage 0 is a technique to evaluate the effectiveness of a measurement system to be used in a Component Search.

Step 1 – Measure each unit three to five times in a random order (includes removing and reinstalling to the measurement device) and plot:



Component Search – Stage 0 - Guidelines

Step 2 – Determine the range for the BOB and WOW. Calculate the average range.



$\text{Range}_{(\text{Compressor 1})} = 40$

$\text{Range}_{(\text{Compressor 2})} = 22$

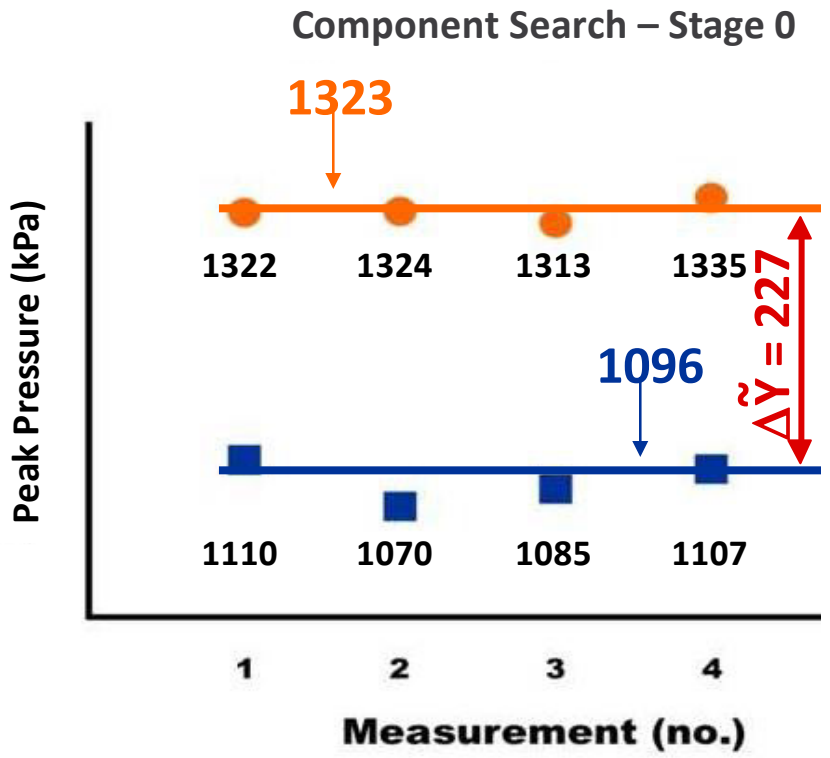
$$\overline{R_0} = \frac{40 + 22}{2} = 31$$

Component Search – Stage 0 Guidelines

Step 3 –Determine the median values for BOB and WOW. Calculate the difference in the median values.

$$| \text{Median}_{(\text{Compressor 1})} - \text{Median}_{(\text{Compressor 2})} | = \Delta\tilde{Y}_0$$

1323 – 1096 = 227



Component Search – Stage 0 Guidelines

Step 4 – Check Stage 0 Rules:

Rule 1: You must have complete separation between the BOB and WOW values. **PASS**

Rule 2: $\Delta\tilde{Y}_0 \geq V_0 \overline{R}_0$ **PASS**

$$227 \geq (0.89) \times 31$$

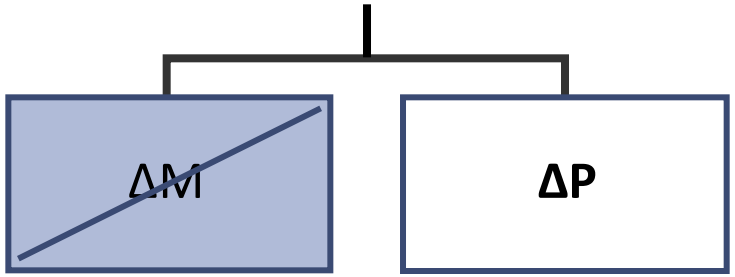
Group Size	V_0
3, 3	1.49
4, 4	0.89
5, 5	0.74

Table: V_0 for Stage 0

Where $\Delta\tilde{Y}_0$ is the difference between the median values.
$\overline{R}_0$ is the average range.
V_0 is a constant from Table for a chosen group size.

Component Search – Stage 0

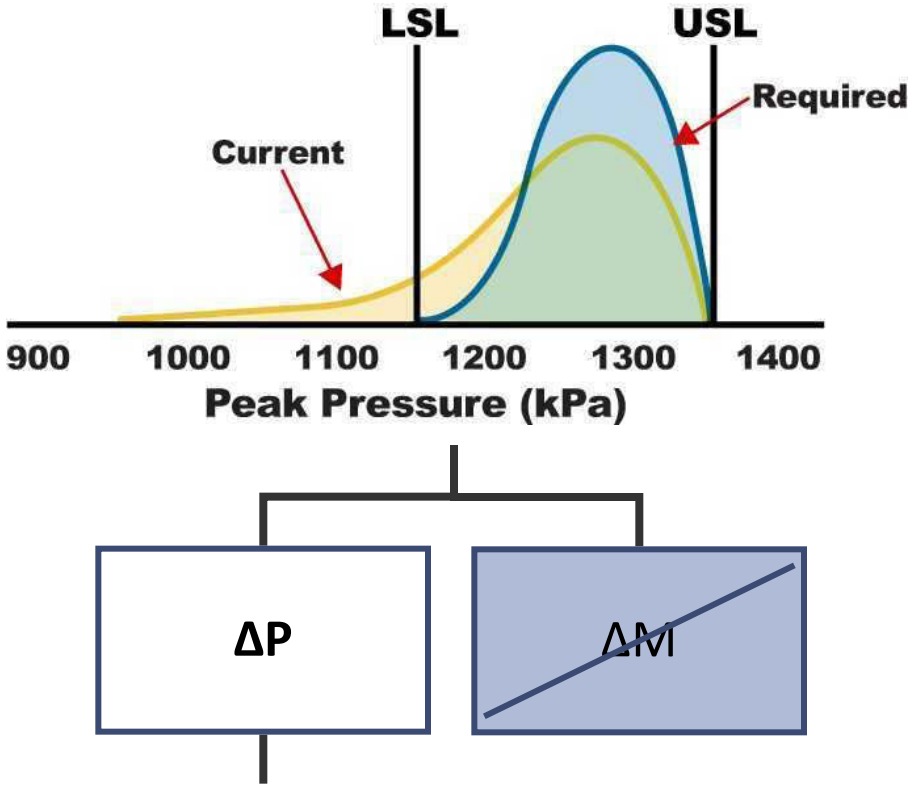
Conclusion: Measurement system for compressor peak pressure is effective.



Rationale:

- Component Search – Stage 0

First Split on a Solution Tree

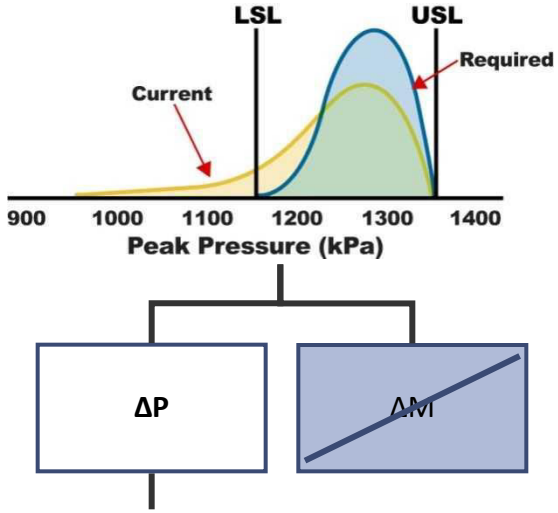


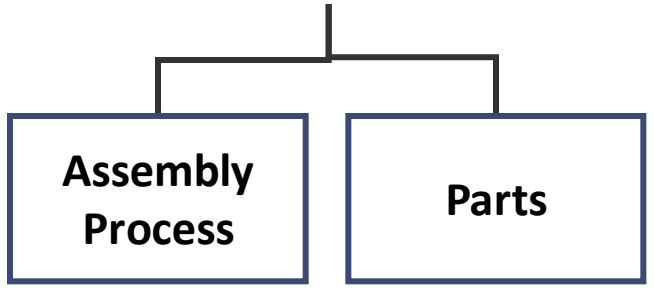
Rationale:

- Component Search – Stage 0

Component Search

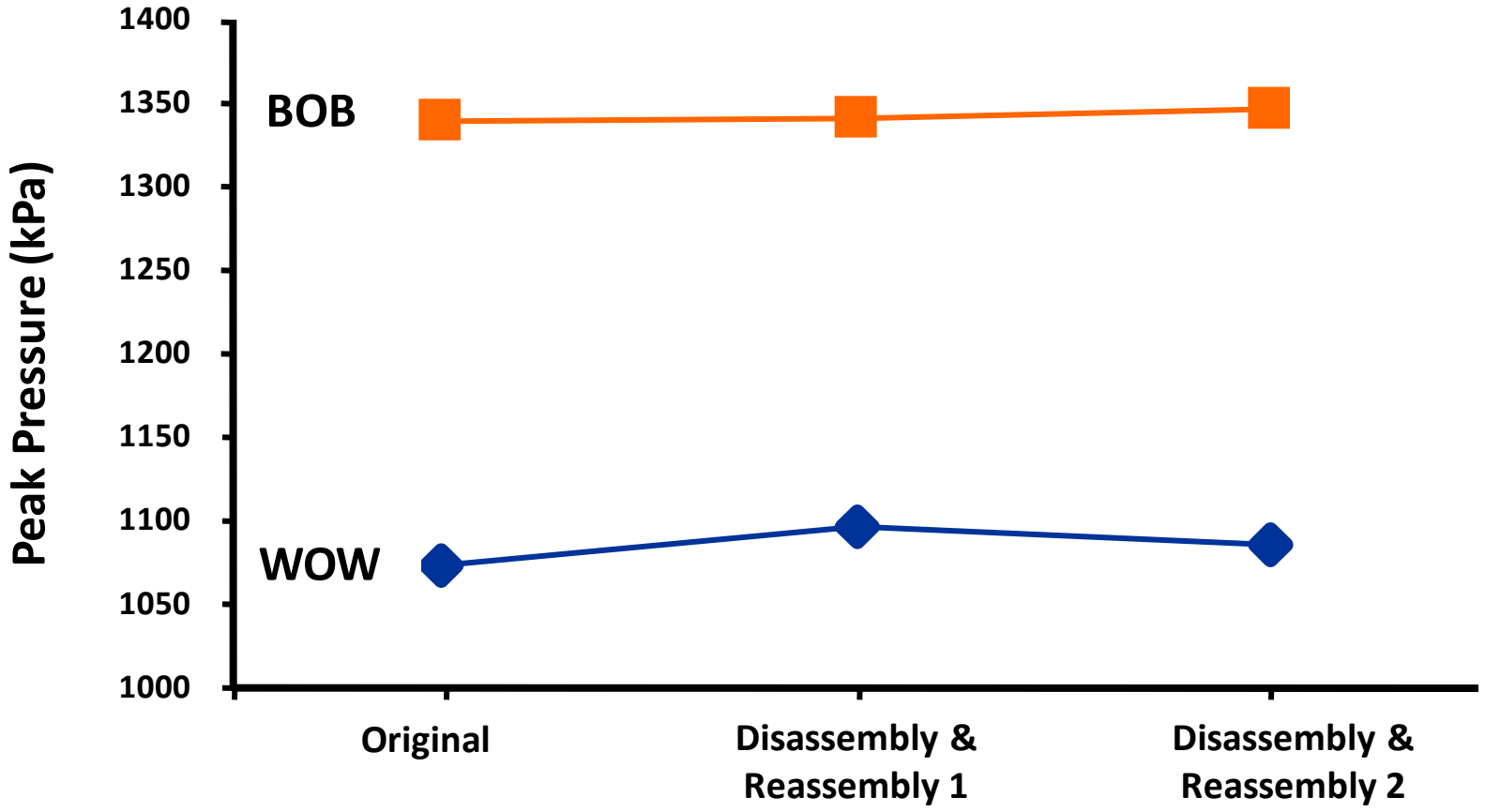
Stage 1





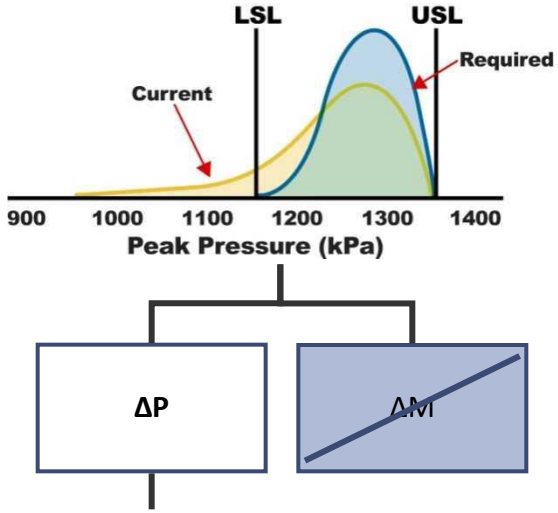
Component Search

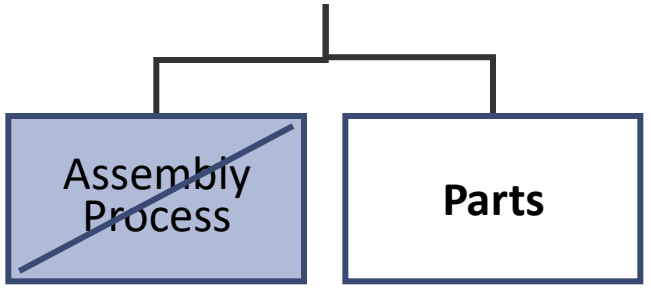
Stage 1



Component Search

Stage 1



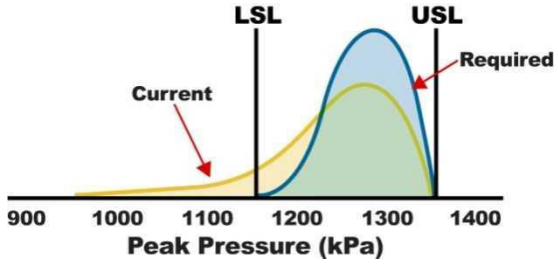


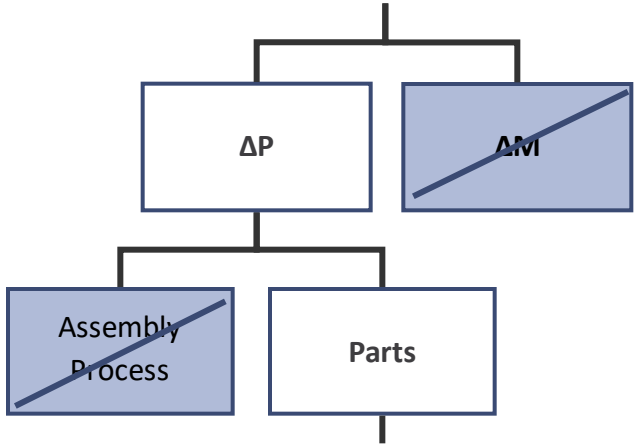
Rationale:

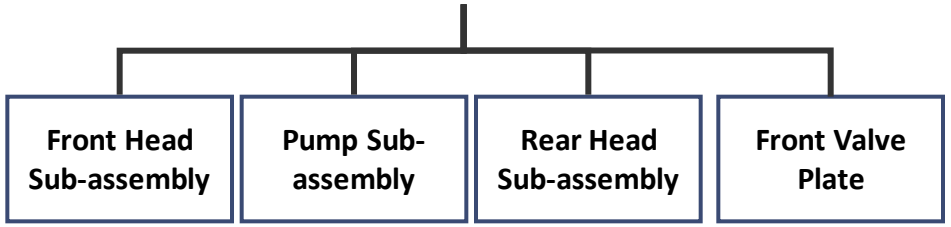
- Component Search – Stage 1

Component Search

Stage 2

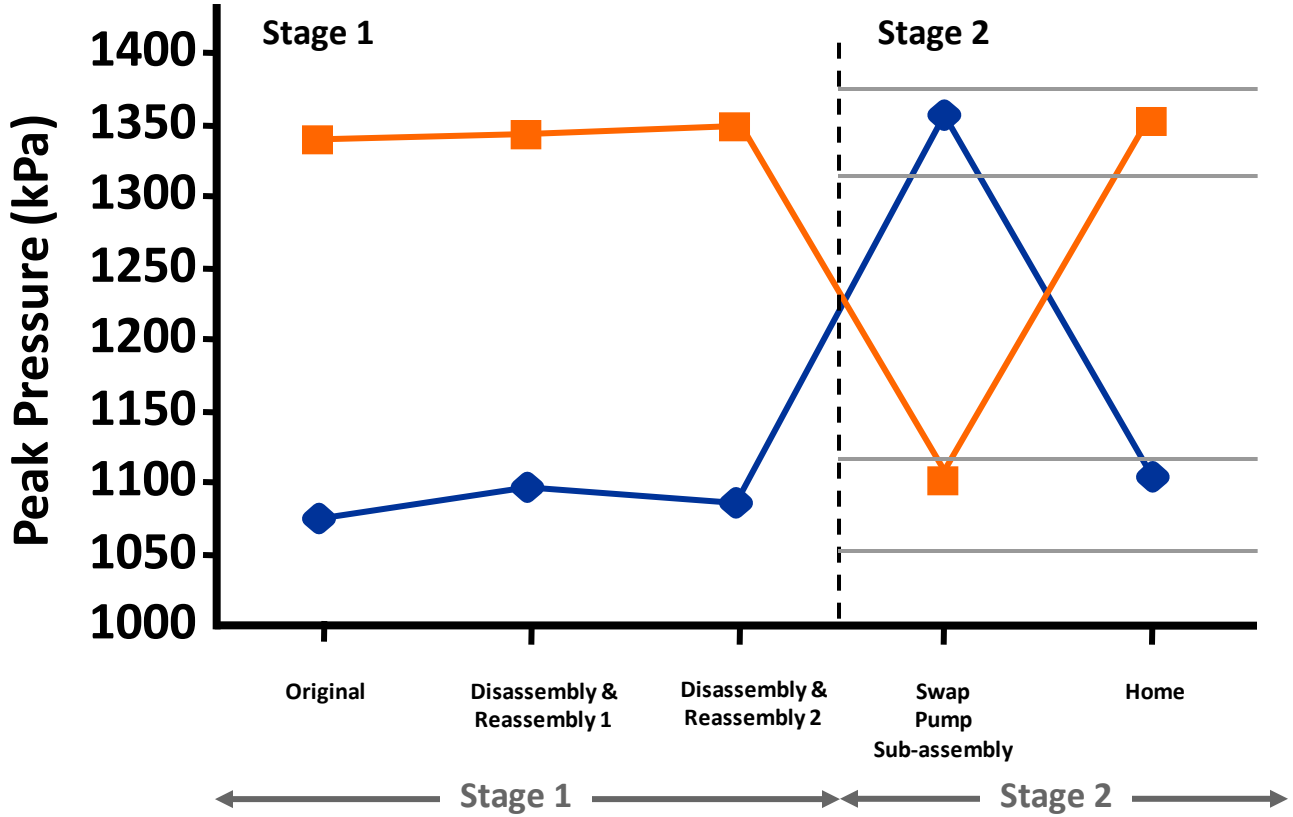






Component Search

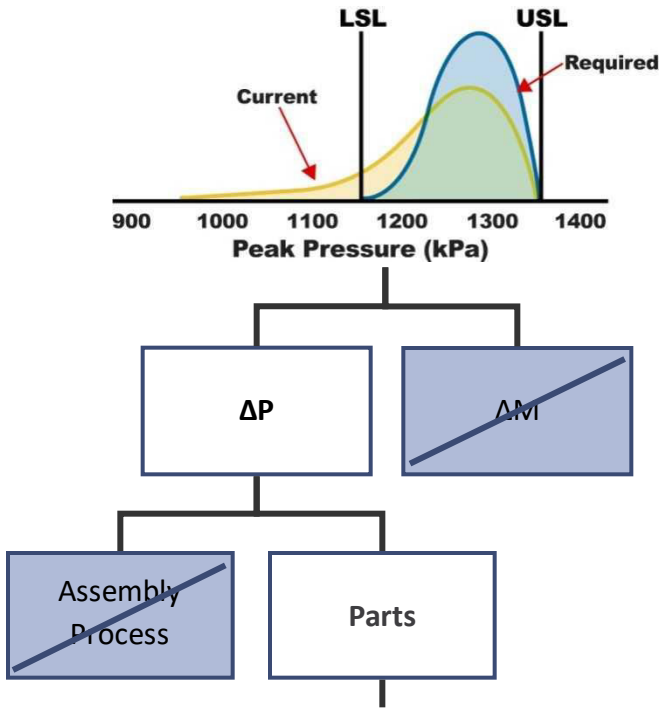
Stage 1

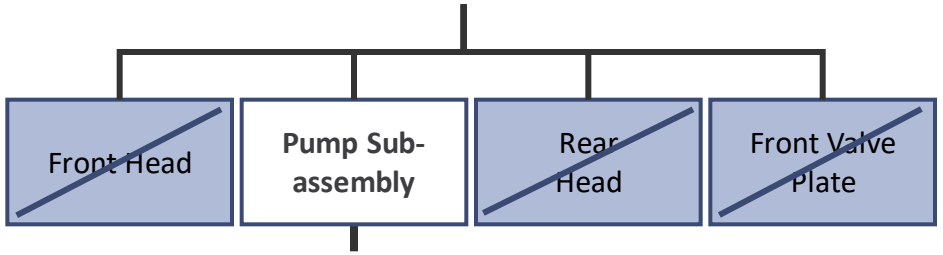


BIG X lives in the pump sub-assembly.

Component Search

Stage 1

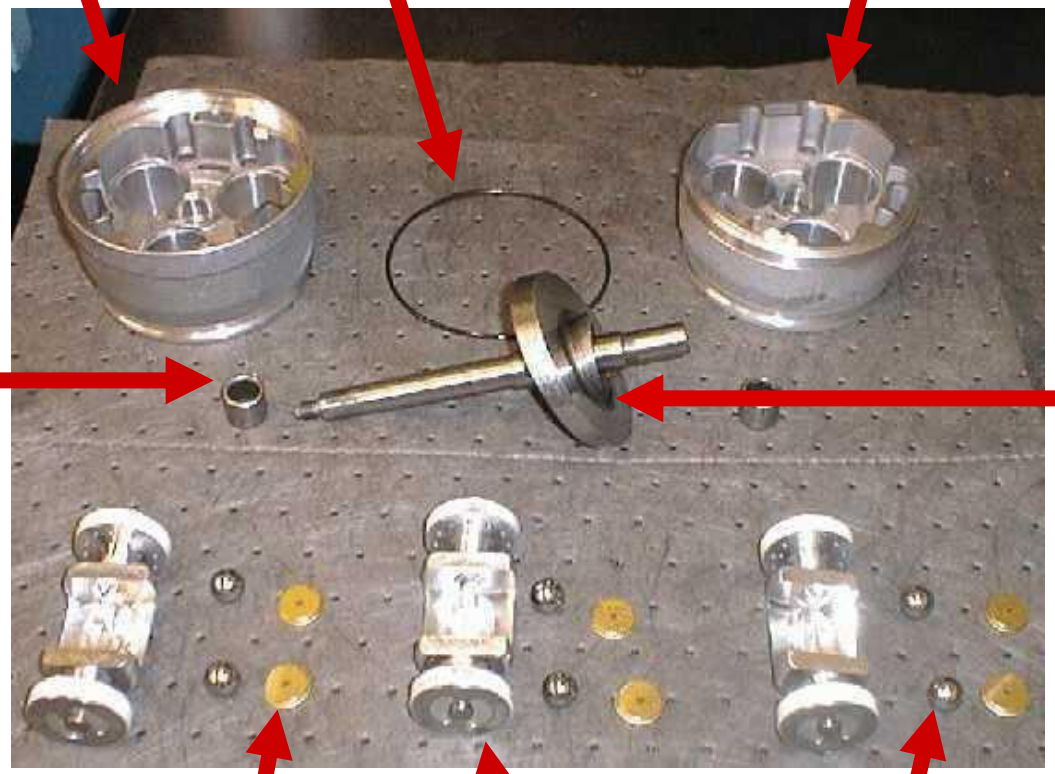




Rationale:

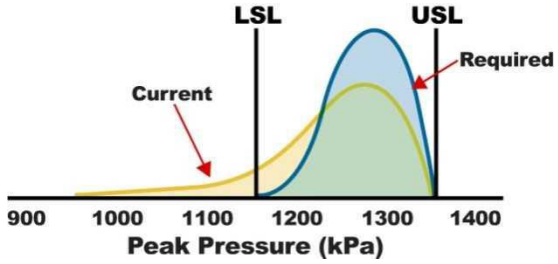
- Component Search – Stage 2

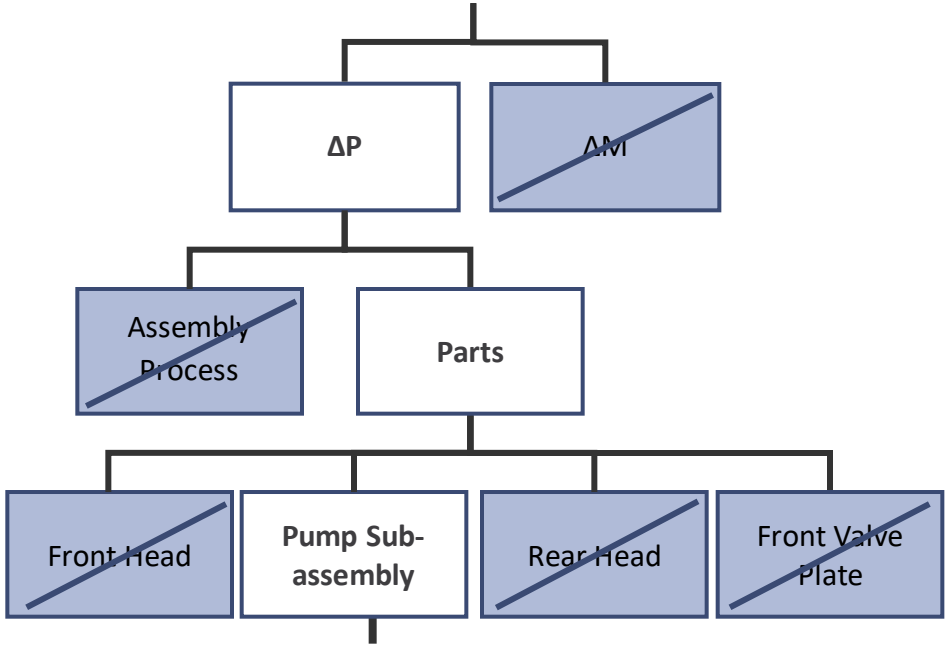
Pump Sub-assembly

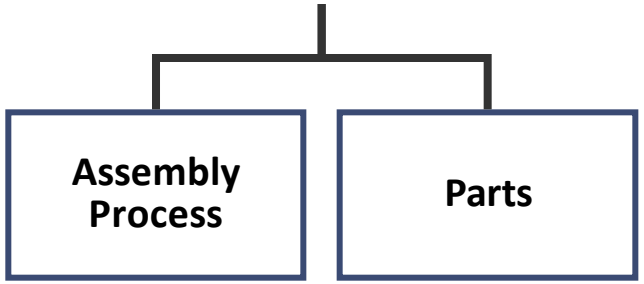


Component Search

Stage 1





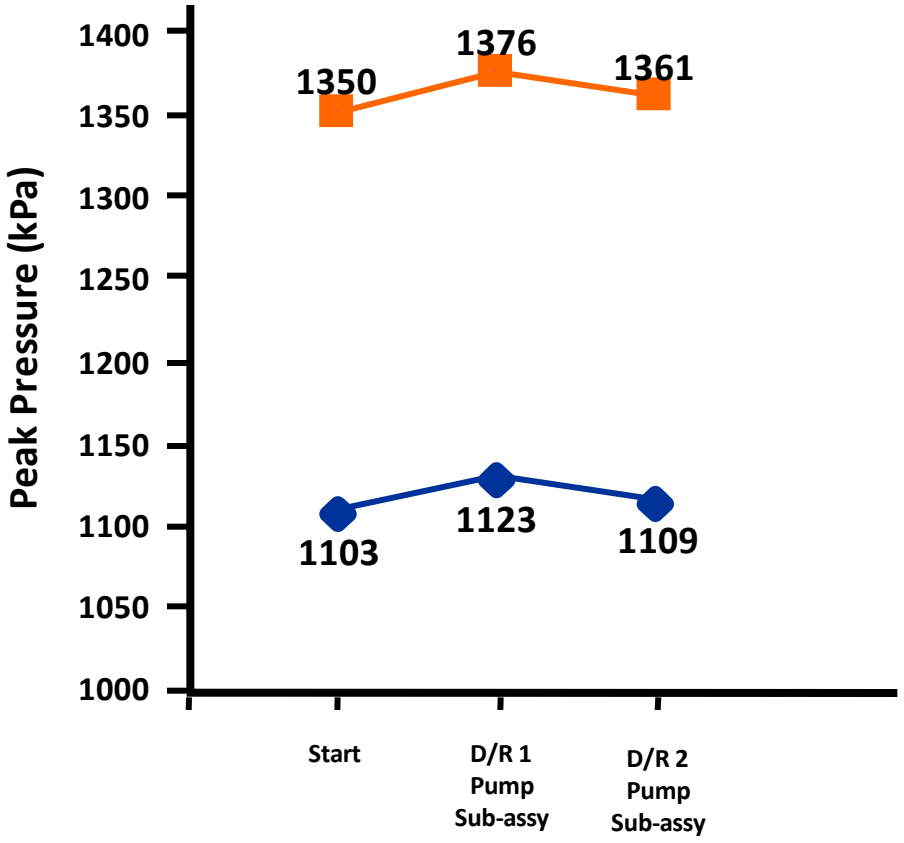


Rationale:

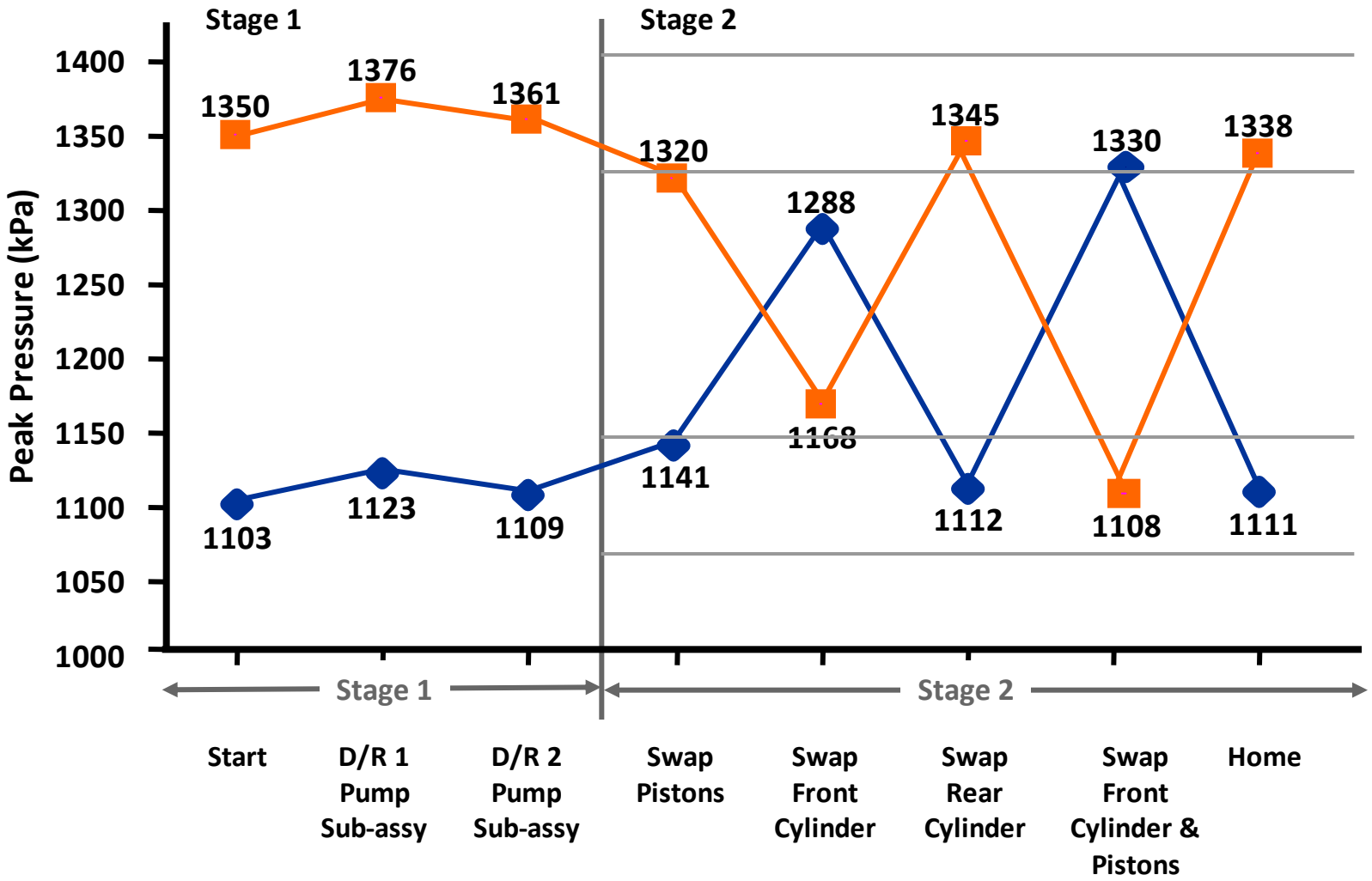
- Component Search – Stage 1 will allow us to make this split.

Component Search

Stage 1, Pump Sub-assembly



Pump Front Cylinder and Pistons Are Players

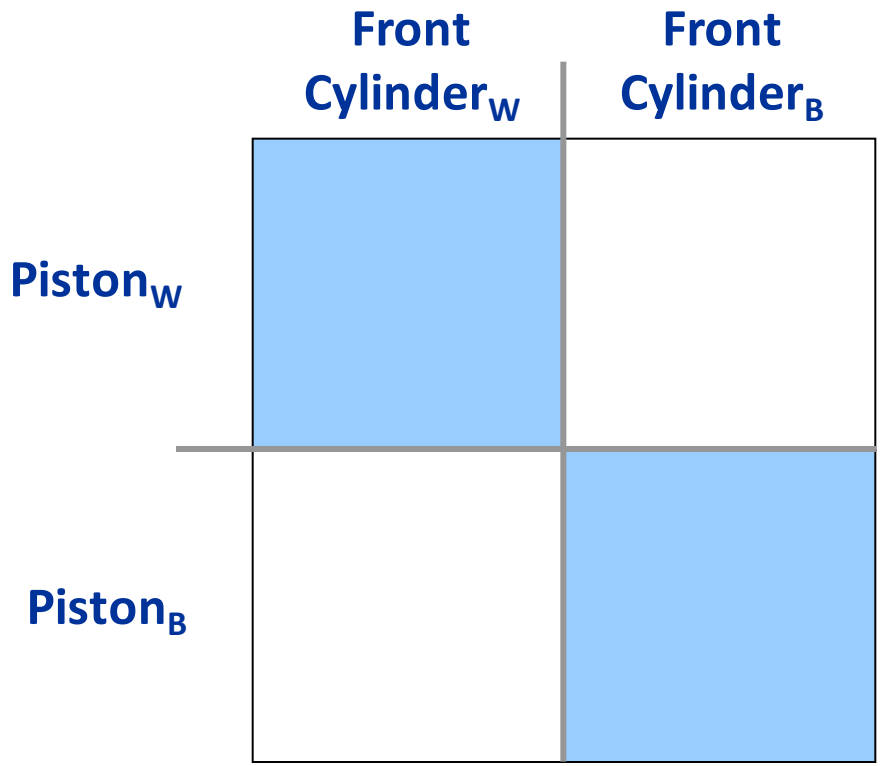


Return component to the original assembly as you are performing the swap.

Evaluation

Stage 3

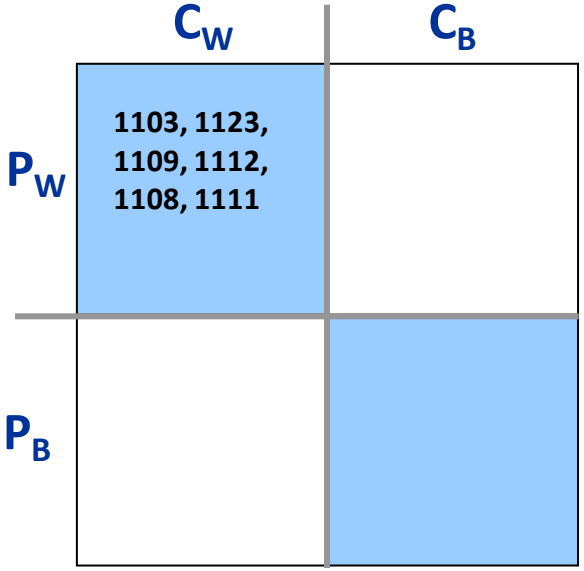
Put the data from the graph into the matrix to analyze the effects of the different components which matter (only piston and front cylinder).

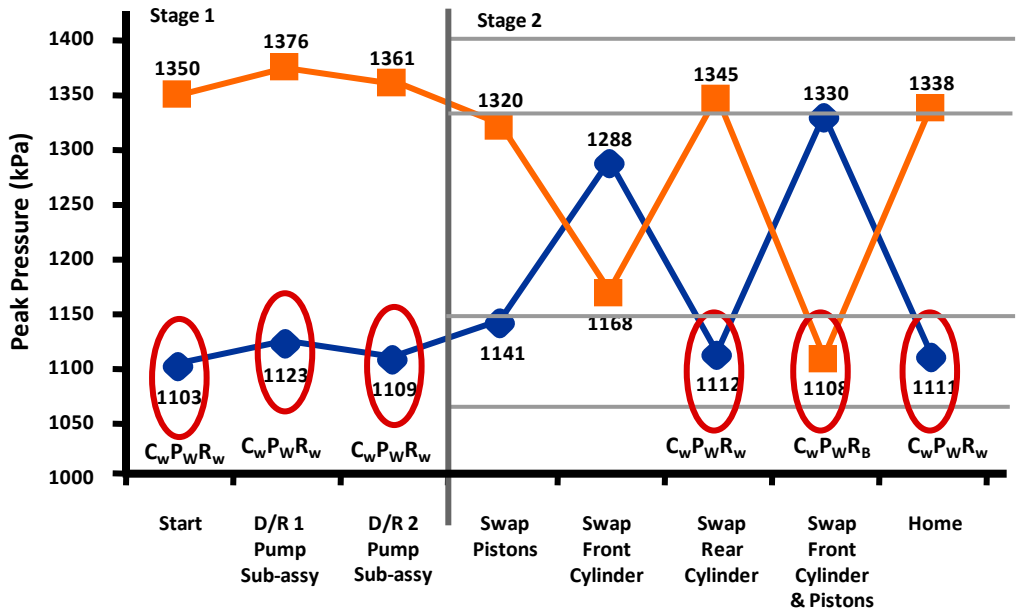


The Result

Stage 3

Start with the points where both the piston and the front cylinder were at the WOW level:





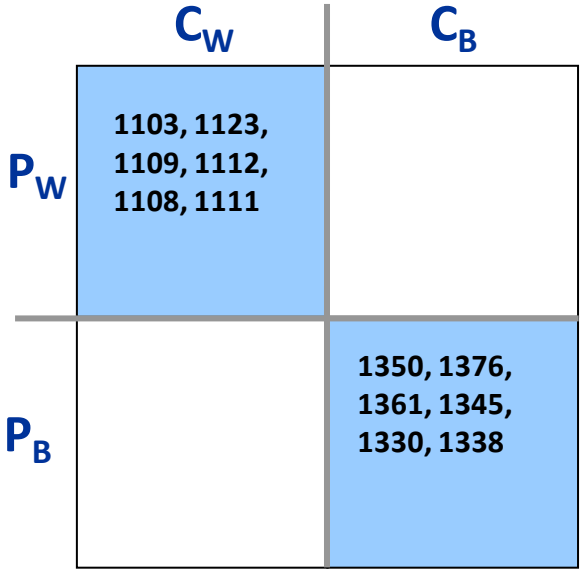
C = Front Cylinder, P = Piston, R = Rest

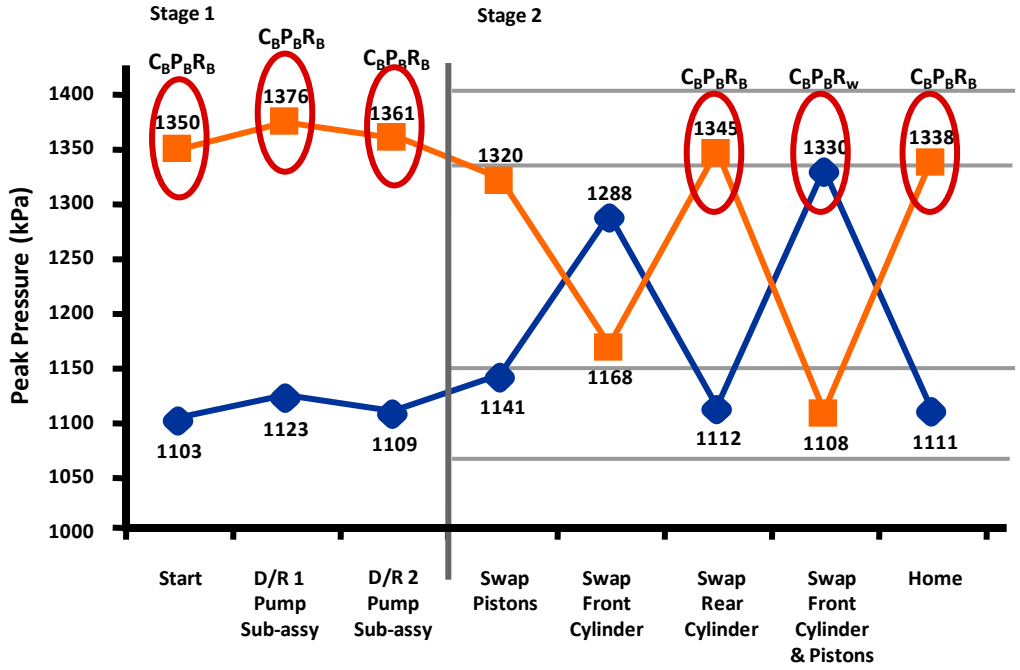
Hint: Color and symbol stay with "the rest."

The Result

Stage 3

Now move to the cell where both the piston and the front cylinder were at the BOB level:





C = Front Cylinder, P = Piston, R = Rest

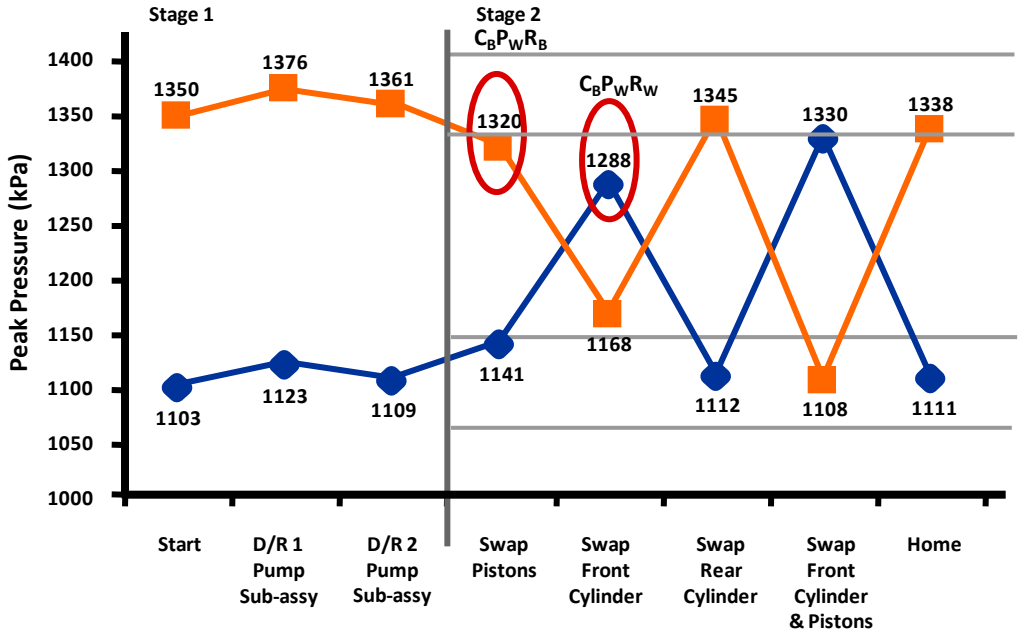
Hint: Color and symbol stay with "the rest."

The Result

Stage 3

Now move to the cell where the piston was at the WOW level and the front cylinder was at the BOB level:

	C_W	C_B
P_W	1103, 1123, 1109, 1112, 1108, 1111	1320, 1288
P_B		1350, 1376, 1361, 1345, 1330, 1338



C = Front Cylinder, P = Piston, R = Rest

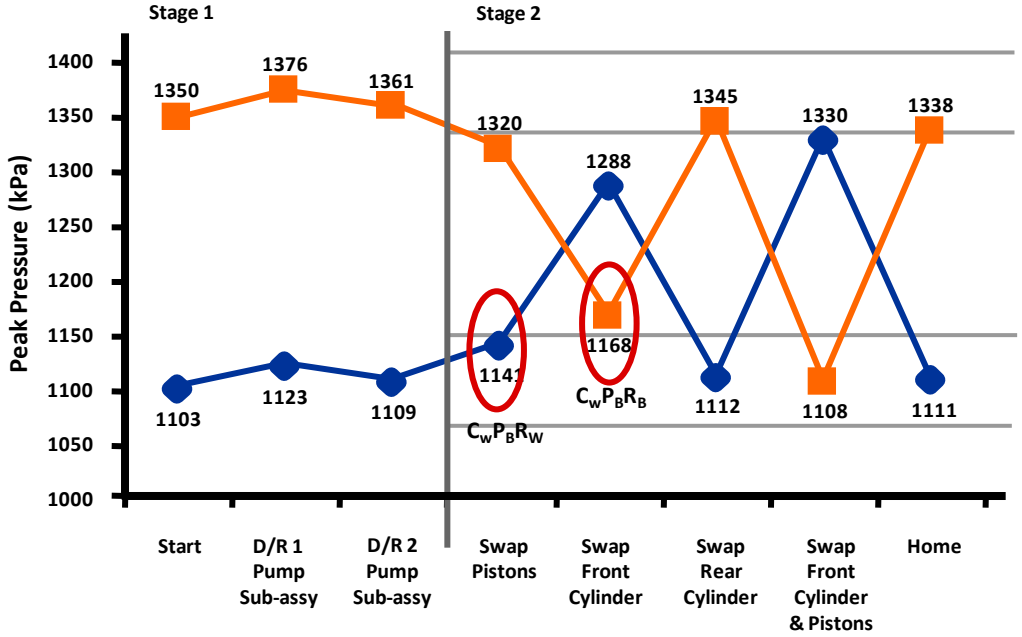
Hint: Color and symbol stay with "the rest."

The Result

Stage 3

Now move to the cell where the piston was at the BOB level and the front cylinder was at the WOW level:

	C_W	C_B
P_W	1103, 1123, 1109, 1112, 1108, 1111	1320, 1288
P_B	1141, 1168	1350, 1376, 1361, 1345, 1330, 1338



C = Front Cylinder, P = Piston, R = Rest

Hint: Color and symbol stay with "the rest."

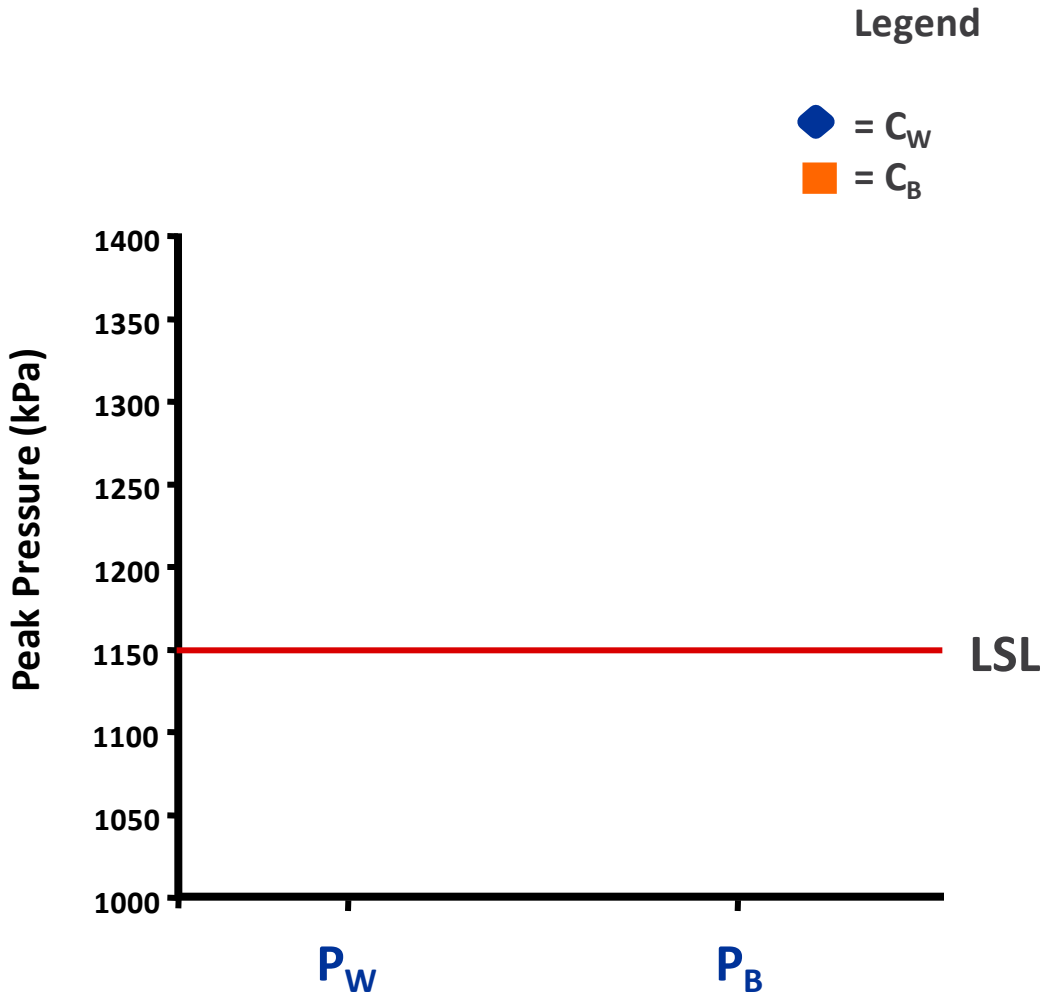
Interpreting the Results - Exercise

Stage 3

Determine the median value of each cell:

Graph the results.

	C_W	C_B
P_W	1103, 1123, 1109, 1112, 1108, 1111	1320, 1288
P_B	1141, 1168	1350, 1376, 1361, 1345, 1330, 1338



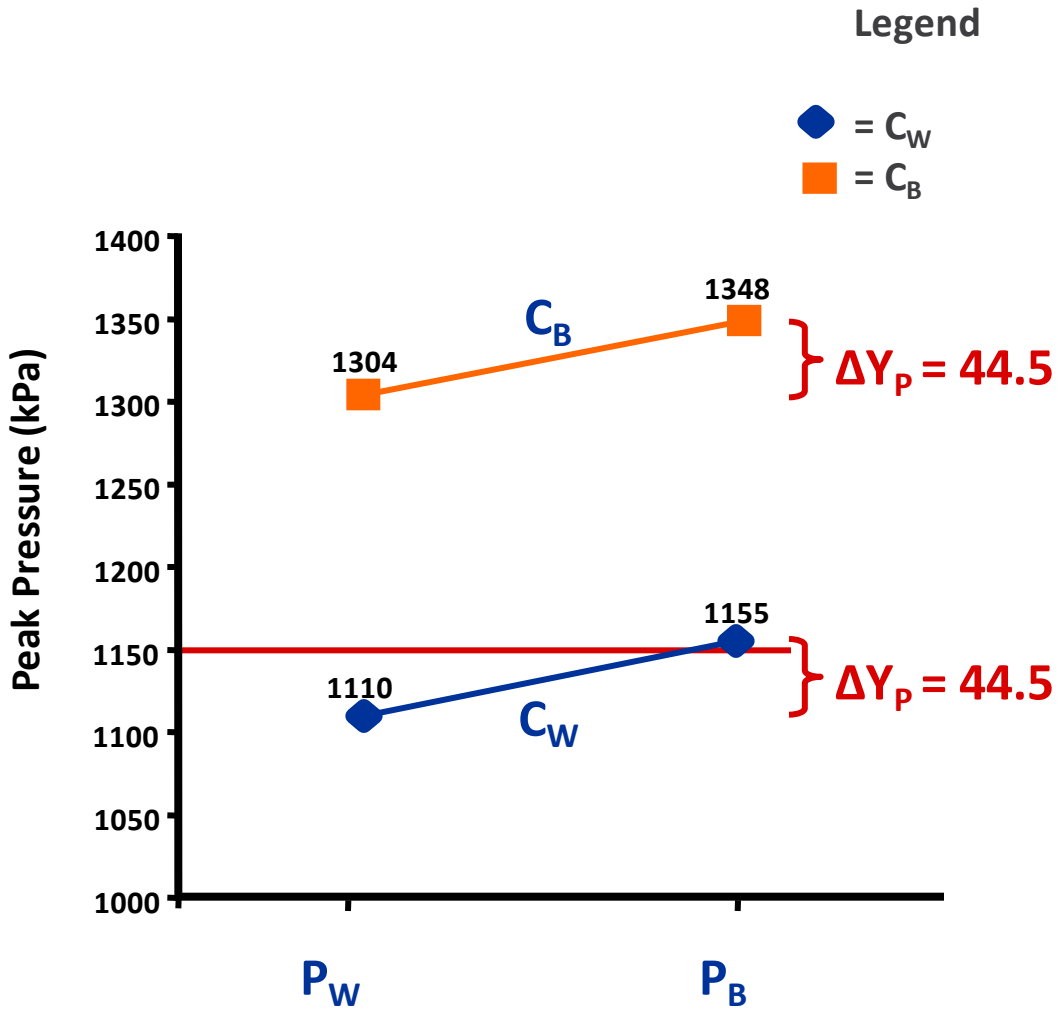
Interpreting the Results - Answer

Stage 3

How is the effect of the piston shown?

– As the slope of the lines.

	C_W	C_B
P_W	1103, 1123, 1109, 1112, 1108, 1111 **1110**	1320, 1288 **1304**
P_B	1141, 1168 **1155**	1350, 1376, 1361, 1345, 1330, 1338 **1348**



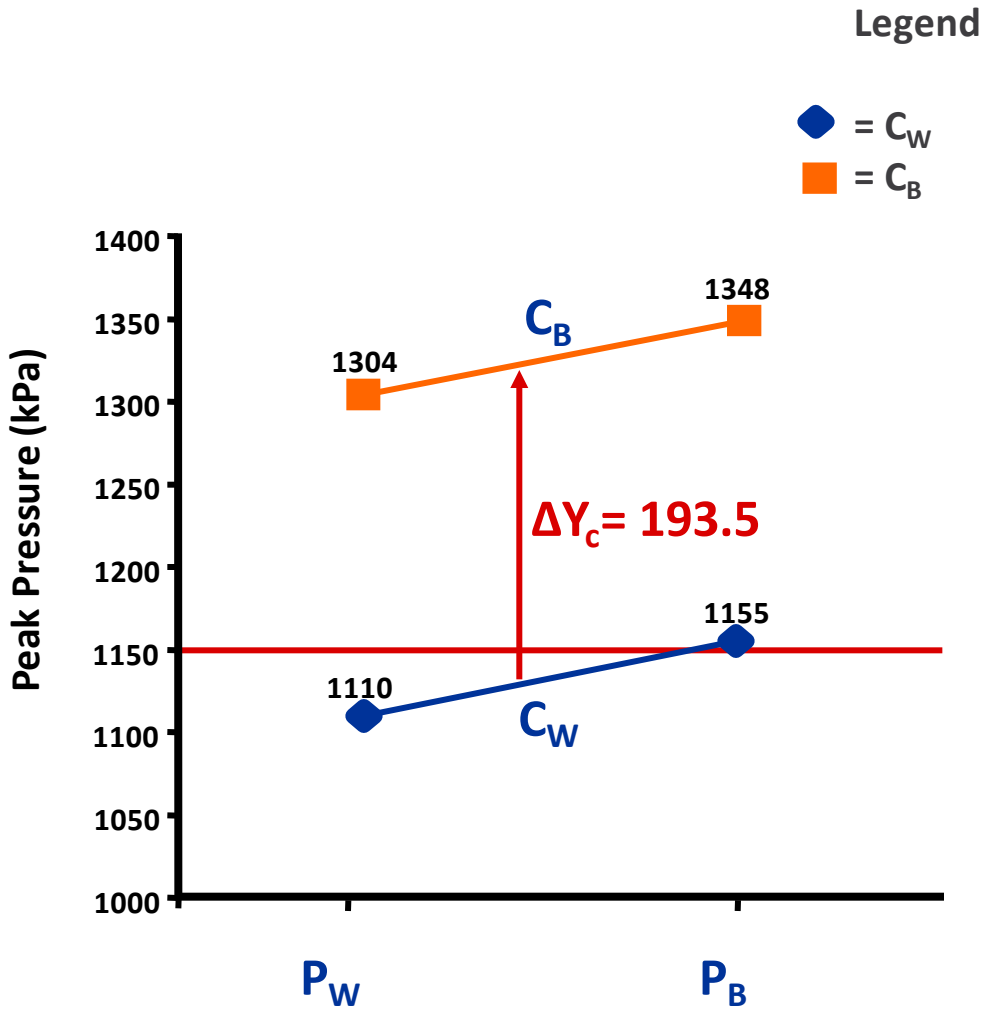
Interpreting the Results - Answer

Stage 3

How is the effect of the front cylinder shown?

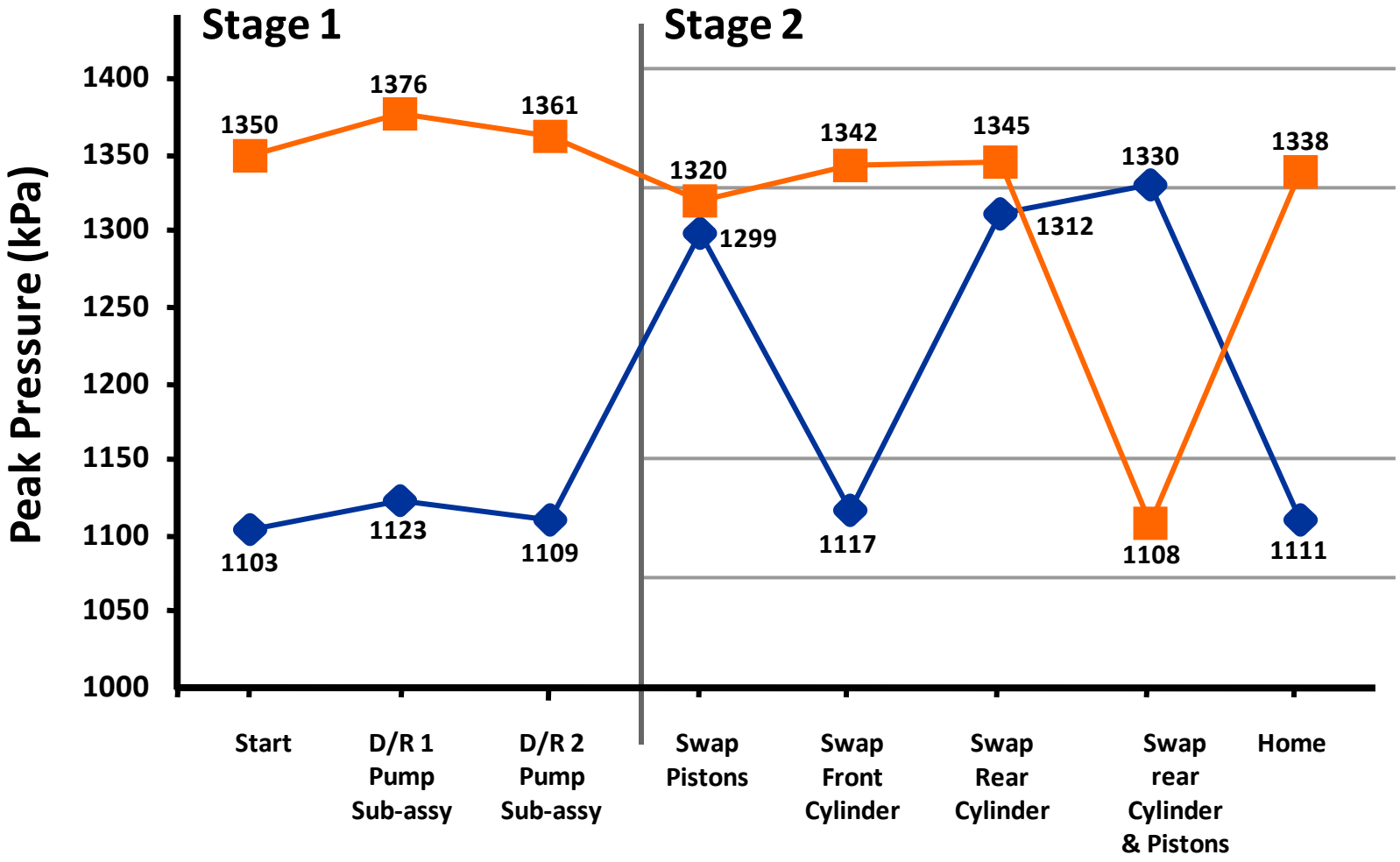
- As the distance between the lines?

	C_W	C_B
P_W	1103, 1123, 1109, 1112, 1108, 1111 **1110**	1320, 1288 **1304**
P_B	1141, 1168 **1155**	1350, 1376, 1361, 1345, 1330, 1338 **1348**



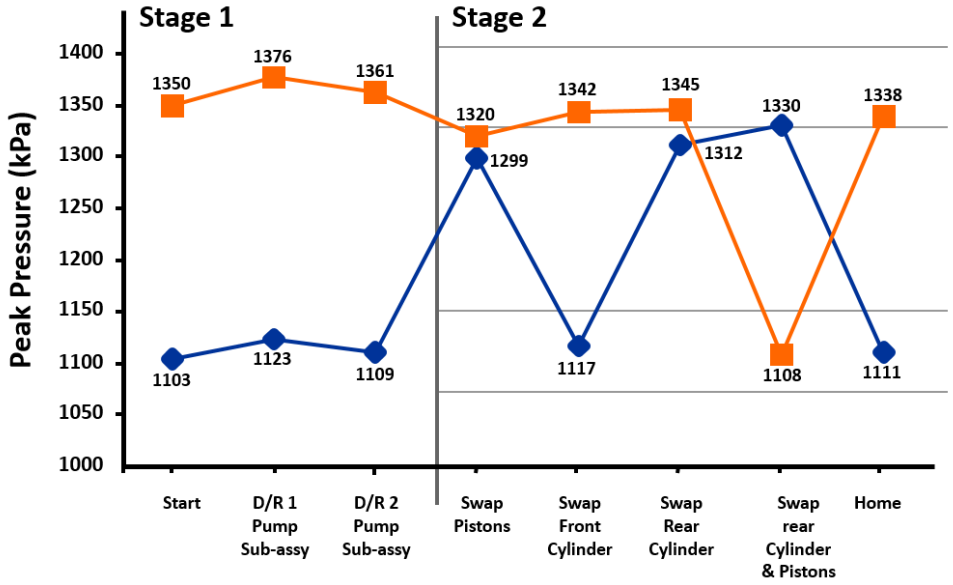
- What must you do in order to fix the pump?
- What must you do in order to maximize the peak pressure?

Another Possible Outcome to This Case: Seeing Interactions in a Component Search

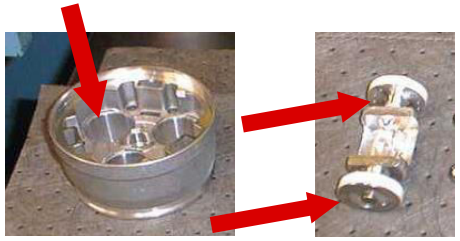


Unequal change in BOB and WOW when swapping components.

Another Possible Outcome to This Case: Seeing Interactions in a Component Search



When two components Interact, look at the features at their interface.

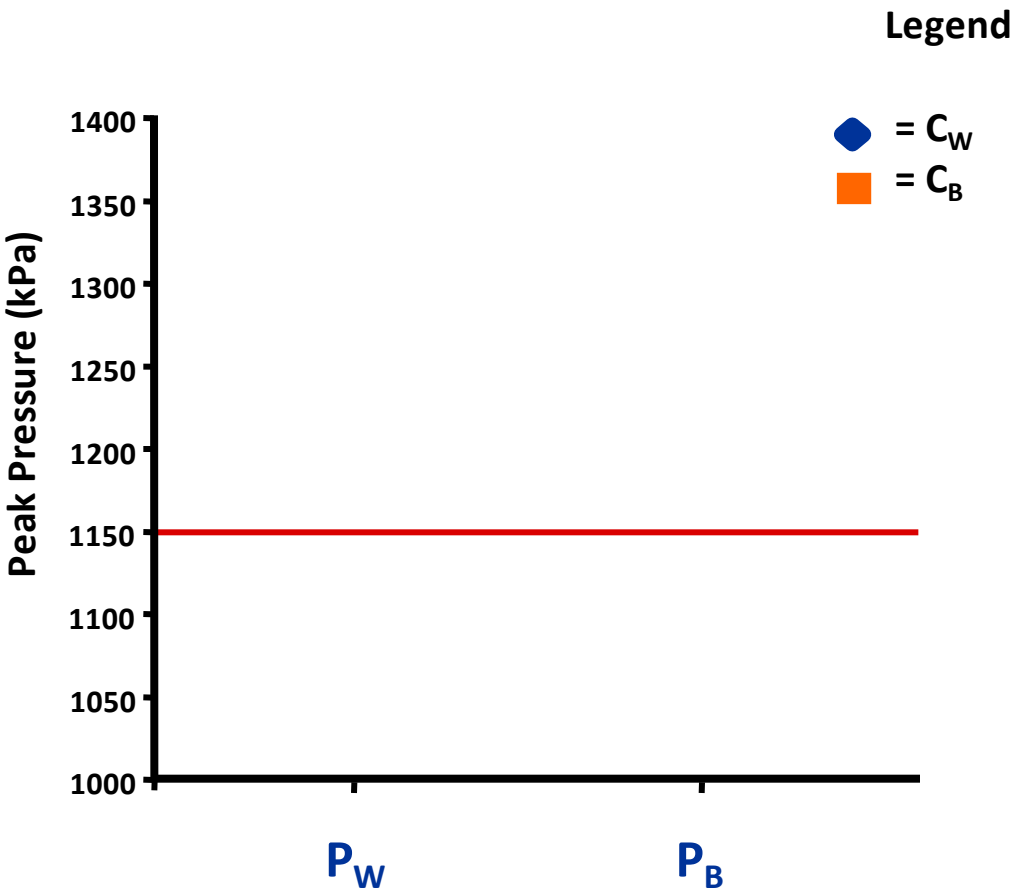


Rear Cylinder **Piston**

Interpreting Results with an Interaction

Graph the Results

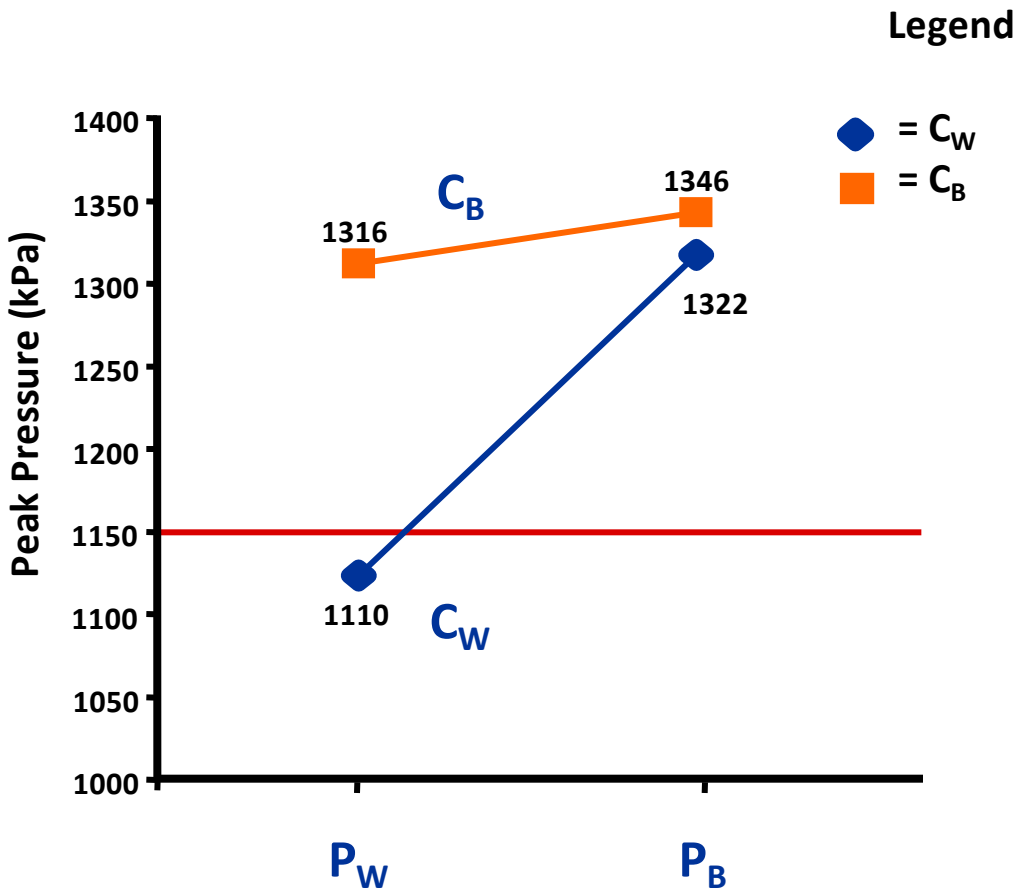
	C_W	C_B
P_W	1103, 1123, 1109, 1117, 1108, 1111	1320, 1312
P_B	1299, 1345	1350, 1376, 1361, 1342, 1330, 1338



Interpreting Results with an Interaction

Graph the Results

	C_W	C_B
P_W	1103, 1123, 1109, 1117, 1108, 1111 **1110**	1320, 1312 **1316**
P_B	1299, 1345 **1322**	1350, 1376, 1361, 1342, 1330, 1338 **1346**



What is the effect of the piston?

What must you do in order to solve the problem?

Component Search - Guidelines

0 Find BOB and WOW.

Stage 0

1 Evaluation of the effectiveness of the measurement system for BOB and WOW.

- **Rule 1:** Complete separation between BOB and WOW values.
- **Rule 2:** $\Delta\tilde{Y}_0 \geq V_0 \bar{R}_0$

Where $\Delta\tilde{Y}$ is the difference between the median values.

$\bar{R}_0$ is the average range.

V_0 is a constant from table for a chosen group size.

Group Size	V_0
3, 3	1.49
4, 4	0.89
5, 5	0.74

Table: V_0 for Stage 0

Component Search - Guidelines

Stage 1

2 Tear down and rebuild each unit two times, so the minimum group size is 3 and 3 (original plus 2 x D/R.) Check both passing rules.

- **Rule 1:** Complete separation between BOB and WOW values.
- **Rule 2:** $\Delta\tilde{Y} \geq V_1 \overline{R}_1$

  Where $\Delta\tilde{Y}$ is the difference between the median values.

  $\overline{R}_1$ is the average range.

  V_1 is a constant from table for a chosen group size.

Group Size	V_1
3, 3	1.49
4, 4	0.89
5, 5	0.74

Table: V_1 for Stage 1

3 If Stage 1 passes, proceed to Stage 2. If Stage 1 fails, the problem is in the way the parts are assembled, not in the parts themselves. Use progressive disassembly to find the BIG X.

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Component Search - Guidelines

Stage 2

4 Set decision limits for each unit.

$$DL_{BOB} = \tilde{Y}_{BOB} \pm V_2 \bar{R}_1$$

$$DL_{WOW} = \tilde{Y}_{WOW} \pm V_2 \bar{R}_1$$

Where $\tilde{Y}$ is the median value.

$\bar{R}_1$ is the average range.

V_2 is a constant from table for a chosen group.

Group Size	V_2
3, 3	1,89
4, 4	1,31
5, 5	1,10

Table: V_2 for Stage 2

5 Swap parts in their expected order of importance. Always do the mirror image. Consider swapping subassemblies; do not forget the dictionary game.

6 Any swap that results in a point outside a home decision limit means that component is contributing to the difference between units.

7 Whenever two or more significant components have been found, perform a capping run by testing all of the untested parts as a group against all of the tested parts as a group. Always test the mirror image. A capping run passes when we get a complete reversal between the units.

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Component Search - Guidelines

Stage 3

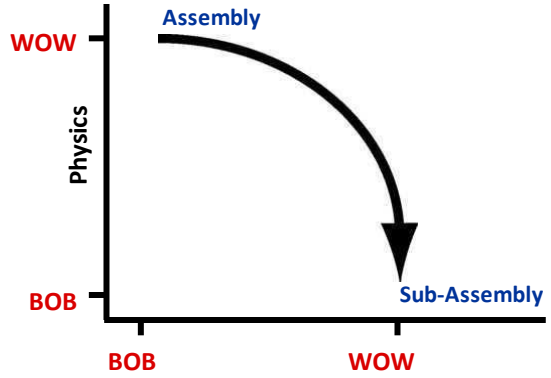
8 Make a full factorial matrix with all significant components from Stage 2 as variables. Include all data from initial selection, Stage 1 and Stage 2. Analyze the matrix using the median for each cell rather than the mean.

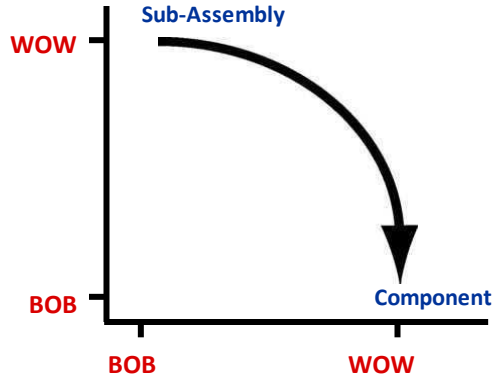
9 Draw a graph including all effects identified as significant components from Stage 2 and look at the graph to identify robust opportunities.

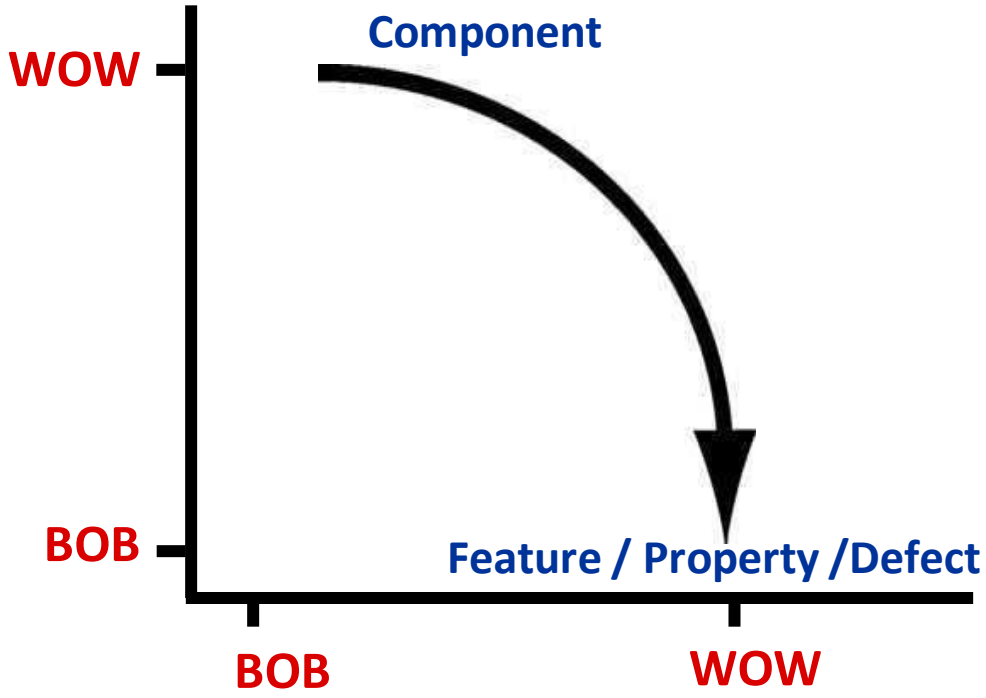
10 Once component pairs with BIG X magnitude have been captured, you will typically have to proceed to Paired Comparisons to find the name of the BIG X feature.

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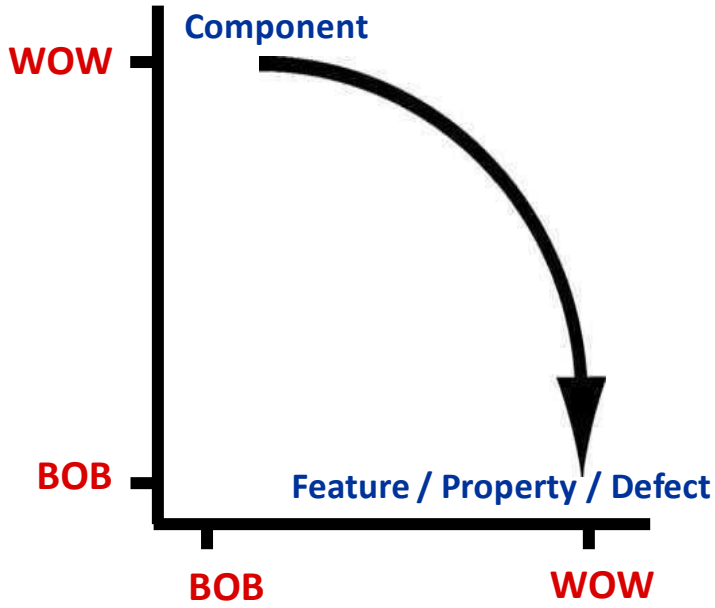
Components → Feature / Property / Defect Strategies







Paired Comparisons to Converge on Feature / Property / Defect



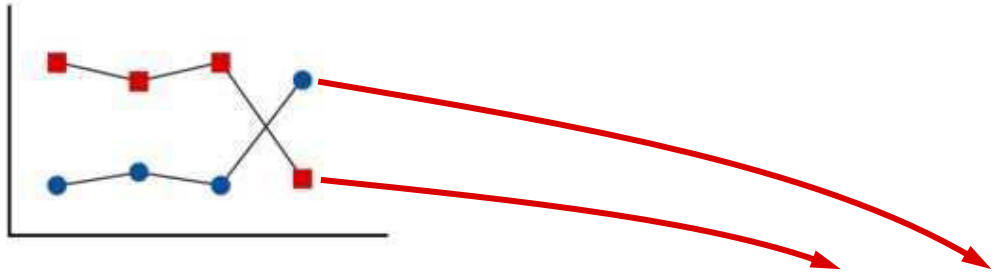
- In order to control the event, we must control the feature, property or defect driving it.
- Paired Comparisons is a powerful tool that is effective for finding the BIG X feature driving the BOB / WOW difference between components.

Analyze Logical Pairs

Compressor Low Peak Pressure – Front Cylinder

What feature on the front cylinder is causing the BOB / WOW contrast?





BOB 1 **WOW 1**

Mone(Y) →

Suspects

Pos.	Dimensions/Properties	Unit	Pair 1 BOB 1	Trend	Pair 1 WOW 1
0	Peak Pressure	kPa	1.345	↓	1.078
1	Bore Diameter	mm	38,337	↓	38,211
2	Bore Surface Finish	mm	0,170	↑	0,190
3	Bore Roundness	mm	0,952	↑	1,011
4	Bore Cylindricity	mm	1,303	↓	1,287
5	Seal Face Flatness	mm	0,020	↑	0,070
6	Seal Face Surface Finish	mm	0,120	↑	0,210
7	Interlock Face Profile	mm	0,003	↑	0,041
8	Interlock Face Surface Finish	mm	0,180	↑	0,250
9	Shaft Bore Diameter	mm	17,243	–	17,243
10	Shaft Bore Roundness	mm	0,821	↓	0,815
11	Shaft Bore Cylindricity	mm	1,340	↓	1,321
12	Shaft Bore Straightness	mm	2,658	↑	3,410

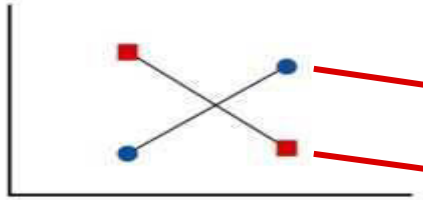
Note: The bore and interlock face dimensions were checked on all three bores and faces. The data has been condensed to keep this exercise manageable.

Three Possibilities

- **BOB > WOW:** ↓
- **BOB < WOW:** ↑
- **BOB = WOW:** –

Analyze Logical Pairs

Obtaining More BOB / WOW Pairs



Obtain a second pair of BOB and WOW front cylinders.

Mone(Y) →

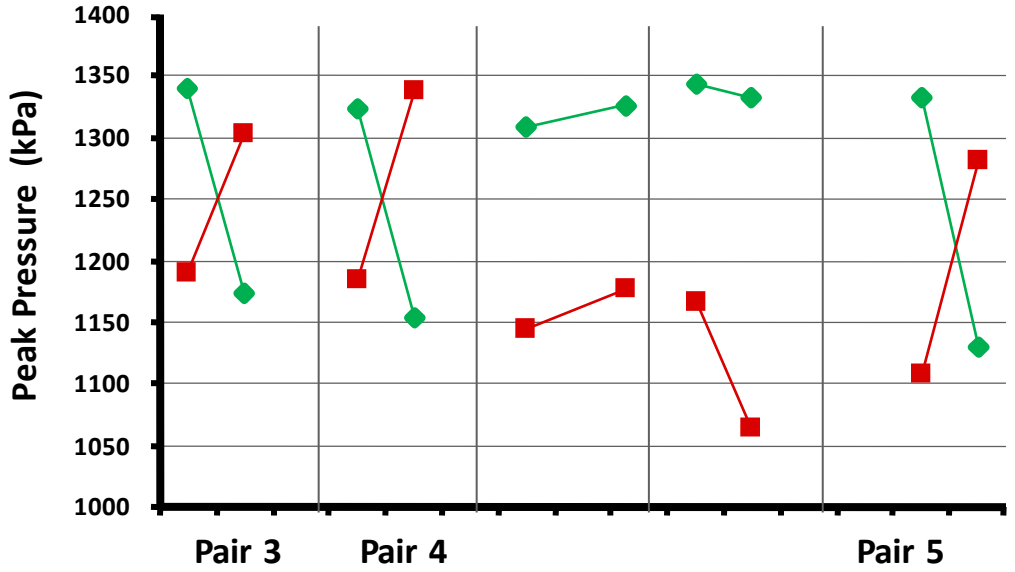
Suspects

Pos.	Dimensions/Properties	Unit	Pair 1			Pair 2		
			BOB 1	Trend	WOW 1	BOB 2	Trend	WOW 2
0	Peak Pressure	kPa	1.345	↓	1.078	1.387	↓	1.065
1	Bore Diameter	mm	38,337	↓	38,211	38,508	↓	38,451
2	Bore Surface Finish	mm	0,170	↑	0,190	0,210	↓	0,150
3	Bore Roundness	mm	0,952	↑	1,011	0,961	↓	0,958
4	Bore Cylindricity	mm	1,303	↓	1,287	1,312	↓	1,265
5	Seal Face Flatness	mm	0,020	↑	0,070	0,030	↑	0,050
6	Seal Face Surface Finish	mm	0,120	↑	0,210	0,160	↓	0,140
7	Interlock Face Profile	mm	0,003	↑	0,041	0,002	↑	0,056
8	Interlock Face Surface Finish	mm	0,180	↑	0,250	0,210	↓	0,200
9	Shaft Bore Diameter	mm	17,243	–	17,243	17,241	↑	17,243
10	Shaft Bore Roundness	mm	0,821	↓	0,815	0,812	↓	0,615
11	Shaft Bore Cylindricity	mm	1,340	↓	1,321	1,210	↑	1,450
12	Shaft Bore Straightness	mm	2,658	↑	3,410	3,413	↑	3,685

Note: The bore and interlock face dimensions were checked on all three bores and faces. The data has been condensed to keep this exercise manageable.

Obtaining More Logical Pairs

- Before we determine which one is the BIG X, we must have additional BOBs and WOWs, which also show a reversal when the front cylinder is swapped.



- Fifth and sixth BOB and WOW difference in peak pressure is caused by something other than the front cylinder.
- Put those comparisons aside and find another pair.
- Next pair 5 shows the same reversal.

Required Number of Pairs for Paired Comparisons

Number of Dimensions / Properties	Minimum Number of Pairs
1 – 3	5
4 – 7	6
8 – 10	7

Paired Comparisons – Analyze Logical Pairs - Exercise

Obtaining More BOB / WOW Pairs

Pos.	Dimensions/Properties	Unit	Pair 1			Pair 2			Pair 3			Pair 4			Pair 5		
			BOB 1	Trend	WOW 1	BOB 2	Trend	WOW 2	BOB 3	Trend	WOW 3	BOB 4	Trend	WOW 4	BOB 5	Trend	WOW 5
0	Peak Pressure	kPa	1.345	↓	1.078	1.387	↓	1.065	1.329	↓	1.066	1.366	↓	1.058	1.389	↓	1.061
1	Bore Diameter	mm	38,337	↓	38,211	38,508	↓	38,451	38,141		38,231	38,371		38,299	38,341		38,365
2	Bore Surface Finish	mm	0,170	↑	0,190	0,210	↓	0,150	0,130		0,130	0,150		0,150	0,090		0,170
3	Bore Roundness	mm	0,952	↑	1,011	0,961	↓	0,958	0,782		0,814	0,765		0,910	0,651		0,609
4	Bore Cylindricity	mm	1,303	↓	1,287	1,312	↓	1,265	1,109		1,421	1,329		0,981	1,104		1,213
5	Seal Face Flatness	mm	0,020	↑	0,070	0,030	↑	0,050	0,020		0,045	0,050		0,030	0,020		0,020
6	Seal Face Surface Finish	mm	0,120	↑	0,210	0,160	↓	0,140	0,170		0,170	0,210		0,090	0,120		0,140
7	Interlock Face Profile	mm	0,003	↑	0,041	0,002	↑	0,056	0,002		0,039	0,002		0,051	0,004		0,090
8	Interlock Face Surface Finish	mm	0,180	↑	0,250	0,210	↓	0,200	0,250		0,210	0,130		0,190	0,210		0,210
9	Shaft Bore Diameter	mm	17,243	–	17,243	17,241	↑	17,243	17,229		17,231	17,241		17,241	17,242		17,243
10	Shaft Bore Roundness	mm	0,821	↓	0,815	0,812	↓	0,615	0,614		0,730	0,650		0,640	0,780		0,910
11	Shaft Bore Cylindricity	mm	1,340	↓	1,321	1,210	↑	1,450	0,910		1,100	1,210		0,980	1,100		0,910
12	Shaft Bore Straightness	mm	2,658	↑	3,410	3,413	↑	3,685	2,412		3,710	3,540		2,160	2,080		2,850

Note: The bore and interlock face dimensions were checked on all three bores and faces. The data has been condensed to keep this exercise manageable.

- **Look for a consistent direction of change within each pair.**
- **What is the BIG X candidate(s)?**

Paired Comparisons – Analyze Logical Pairs - Answer

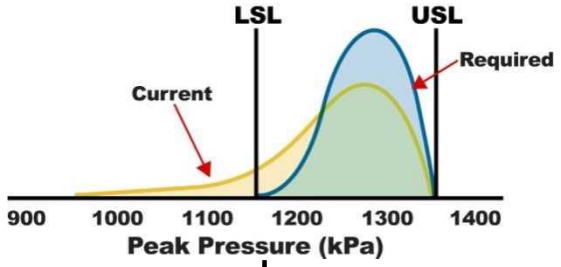
Obtaining More BOB / WOW Pairs

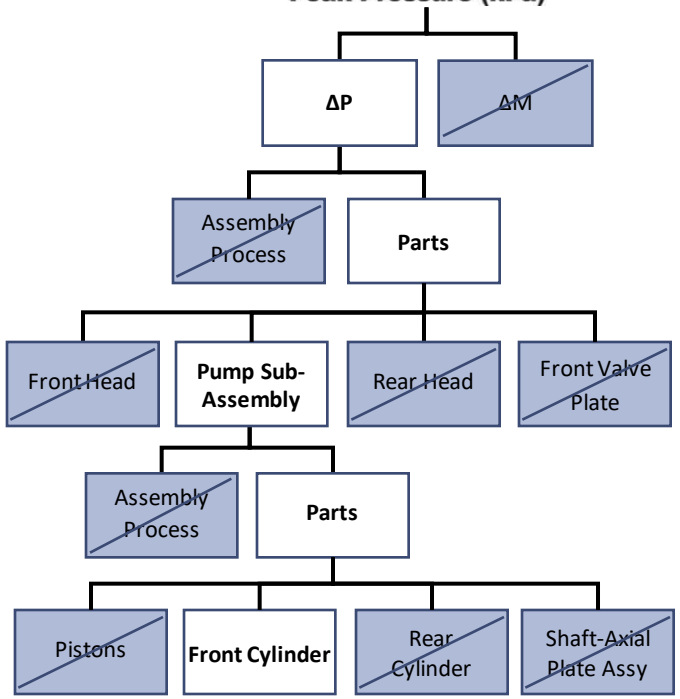
Pos.	Dimensions/Properties	Unit	Pair 1			Pair 2			Pair 3			Pair 4			Pair 5		
			BOB 1	Trend	WOW 1	BOB 2	Trend	WOW 2	BOB 3	Trend	WOW 3	BOB 4	Trend	WOW 4	BOB 5	Trend	WOW 5
0	Peak Pressure	kPa	1.345	↓	1.078	1.387	↓	1.065	1.329	↓	1.066	1.366	↓	1.058	1.389	↓	1.061
1	Bore Diameter	mm	38,337	↓	38,211	38,508	↓	38,451	38,141	↑	38,231	38,371	↓	38,299	38,341	↑	38,365
2	Bore Surface Finish	mm	0,170	↑	0,190	0,210	↓	0,150	0,130	–	0,130	0,150	–	0,150	0,090	↑	0,170
3	Bore Roundness	mm	0,952	↑	1,011	0,961	↓	0,958	0,782	↑	0,814	0,765	↑	0,910	0,651	↓	0,609
4	Bore Cylindricity	mm	1,303	↓	1,287	1,312	↓	1,265	1,109	↑	1,421	1,329	↓	0,981	1,104	↑	1,213
5	Seal Face Flatness	mm	0,020	↑	0,070	0,030	↑	0,050	0,020	↑	0,045	0,050	↓	0,030	0,020	–	0,020
6	Seal Face Surface Finish	mm	0,120	↑	0,210	0,160	↓	0,140	0,170	–	0,170	0,210	↓	0,090	0,120	↑	0,140
7	Interlock Face Profile	mm	0,003	↑	0,041	0,002	↑	0,056	0,002	↑	0,039	0,002	↑	0,051	0,004	↑	0,090
8	Interlock Face Surface Finish	mm	0,180	↑	0,250	0,210	↓	0,200	0,250	↓	0,210	0,130	↑	0,190	0,210	–	0,210
9	Shaft Bore Diameter	mm	17,243	–	17,243	17,241	↑	17,243	17,229	↑	17,231	17,241	–	17,241	17,242	↑	17,243
10	Shaft Bore Roundness	mm	0,821	↓	0,815	0,812	↓	0,615	0,614	↑	0,730	0,650	↓	0,640	0,780	↑	0,910
11	Shaft Bore Cylindricity	mm	1,340	↓	1,321	1,210	↑	1,450	0,910	↑	1,100	1,210	↓	0,980	1,100	↓	0,910
12	Shaft Bore Straightness	mm	2,658	↑	3,410	3,413	↑	3,685	2,412	↑	3,710	3,540	↓	2,160	2,080	↑	2,850

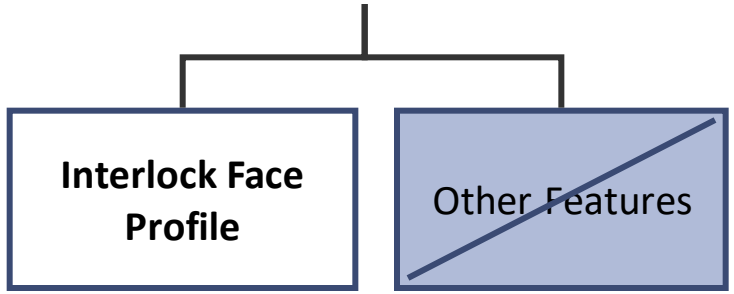
Note: The bore and interlock face dimensions were checked on all three bores and faces. The data has been condensed to keep this exercise manageable.

- **Look for a consistent direction of change within each pair.**
- **Interlock face profile is now a strong candidate for the BIG X.**
- **A confirmation test proved it.**

Solution Tree







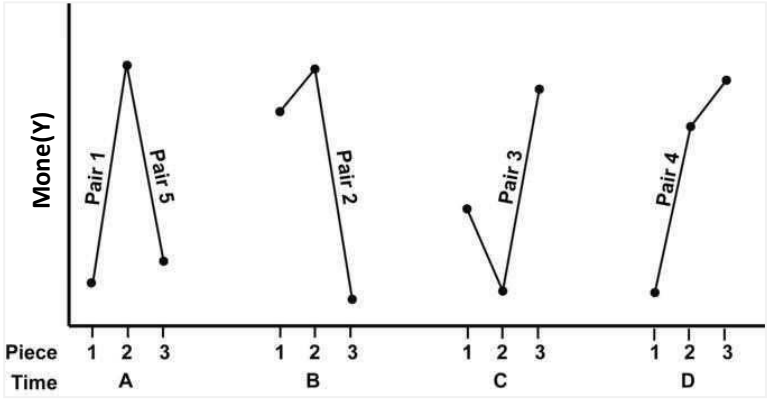
BIG X

Rationale:

- Paired Comparisons
- Confirmation test proves with 5 % risk

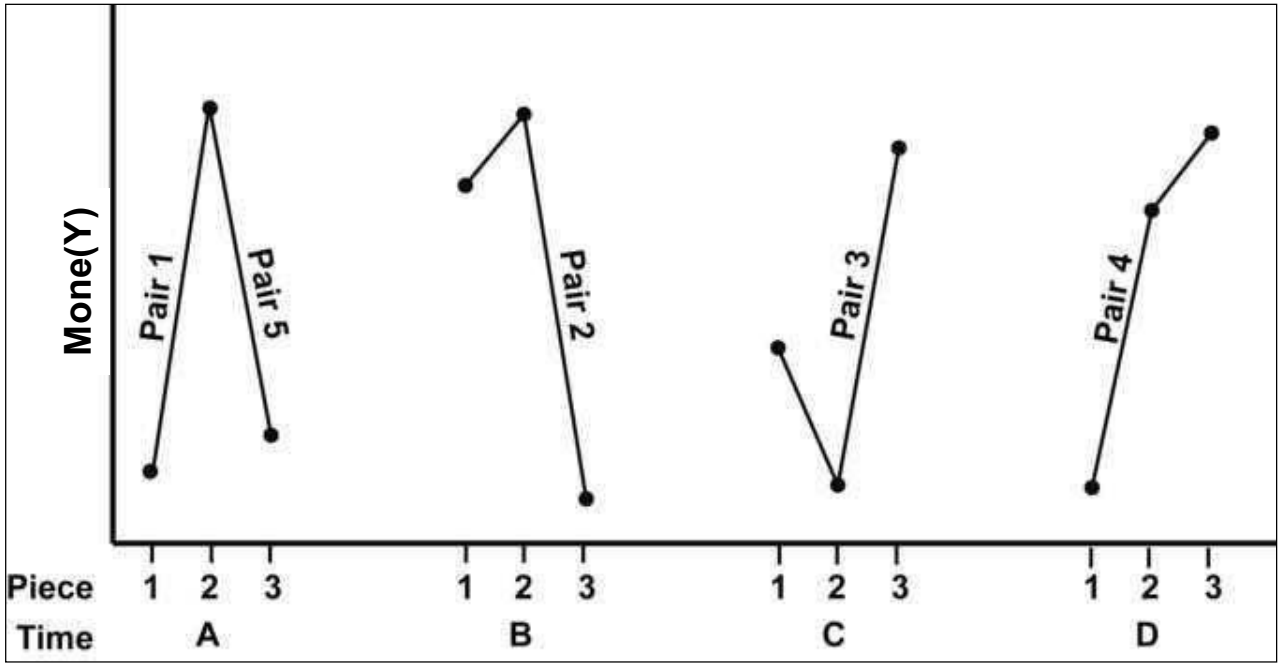
Selecting Logical Pairs Using Multi-Vari

Selecting logical pairs from a Multi-Vari depends upon the ability to read and understand the data. Identify the largest family of variation and then group pairs from that family to provide the greatest amount of leverage towards finding the BIG X. In the following Multi-Vari, the largest family of variation is piece to piece.



Logical pairs would be serial parts which capture as much Mone(Y) variation as possible. Another strategy would be to select triplets or grouped pairs of BOB / WOW / BOB parts. Triplets provide leverage by contrasting two pairs within three parts, but may be more difficult to find.

Selecting Logical Pairs Using Multi-Vari



- Identify the largest family of variation
- Select pairs from the family:
  - By restricting the selection of pairs to the BIG X family, we limit the influence of Xs from other families while capturing as much Mone(Y) difference as possible.

Paired Comparisons

BOB1 WOW1 BOB2 WOW2 BOB3 WOW3 BOB4 WOW4 BOB5 WOW5

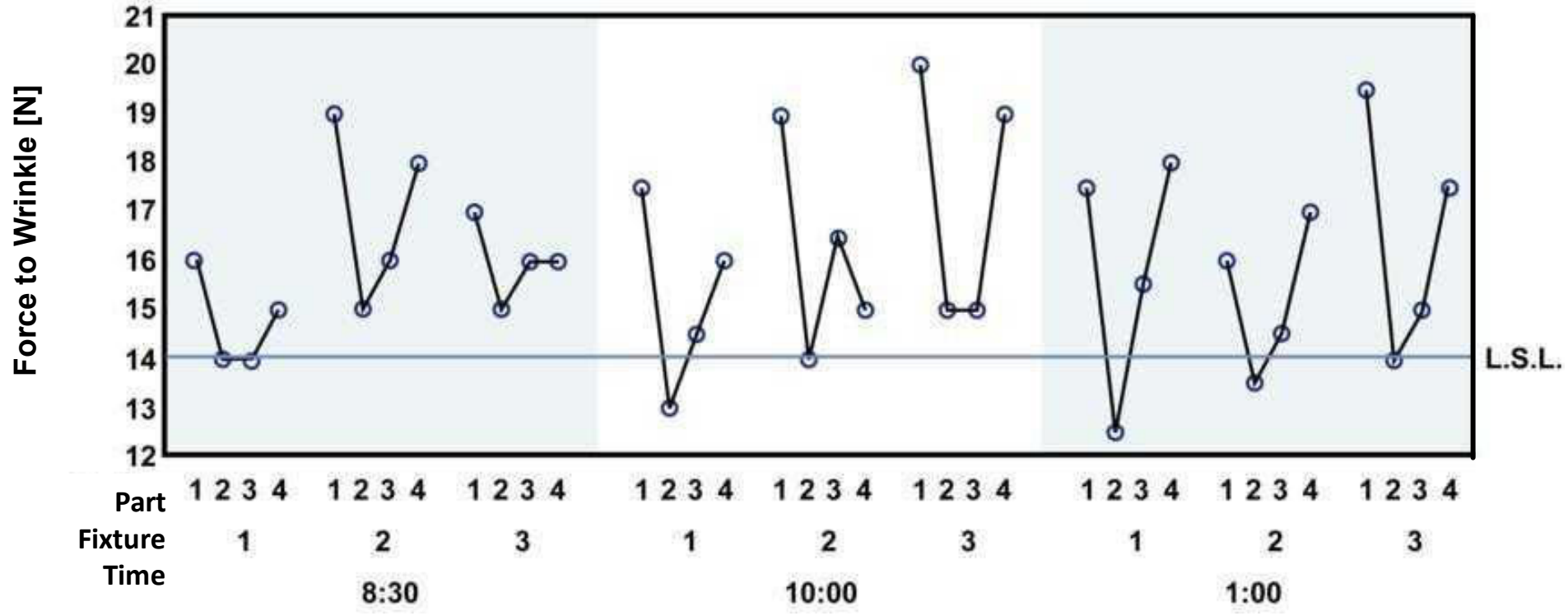


Pair 1 Pair 2 Pair 3 Pair 4 Pair 5

What difference do you see within each pair?

Selecting Logical Pairs Using Multi-Vari

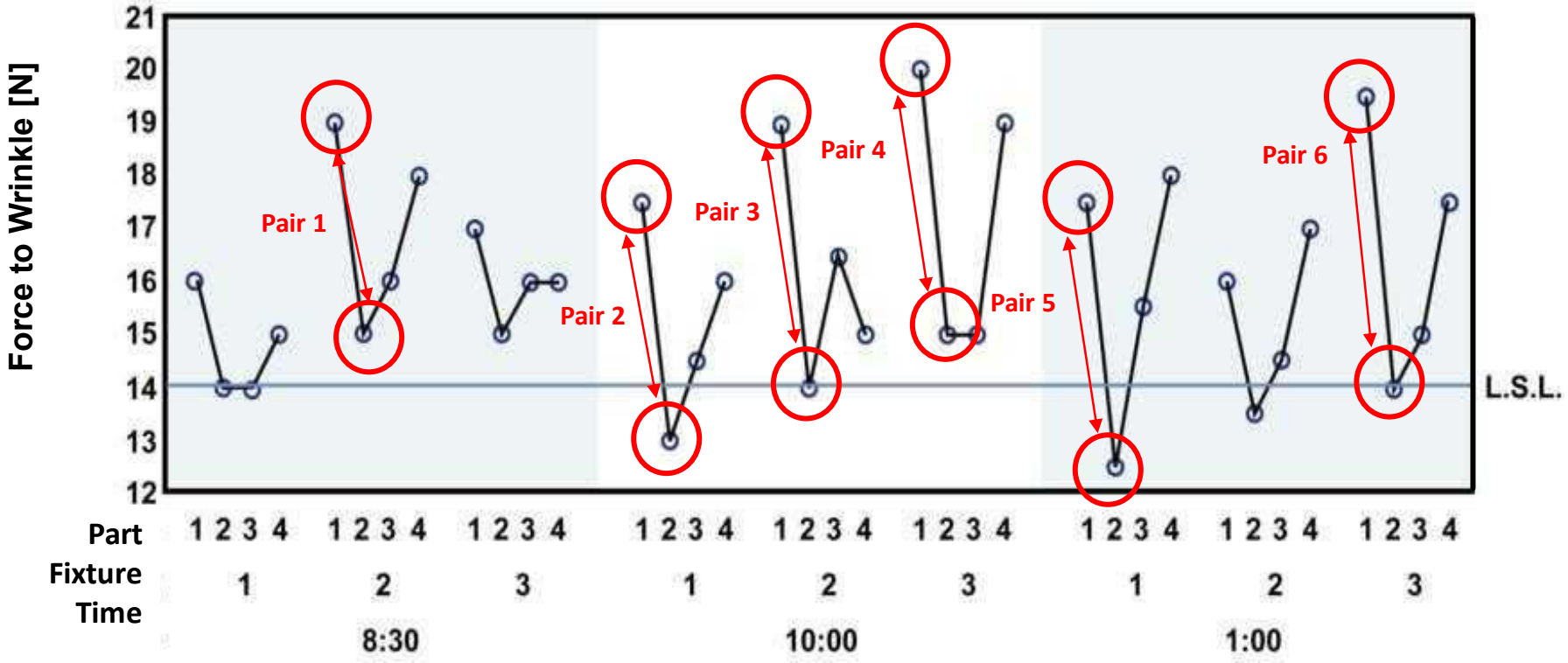
Circle what you believe to be six good logical pairs.



Selecting Logical Pairs Using Multi-Vari – Answer

Part 1 is BOB, Part 2 is WOW.

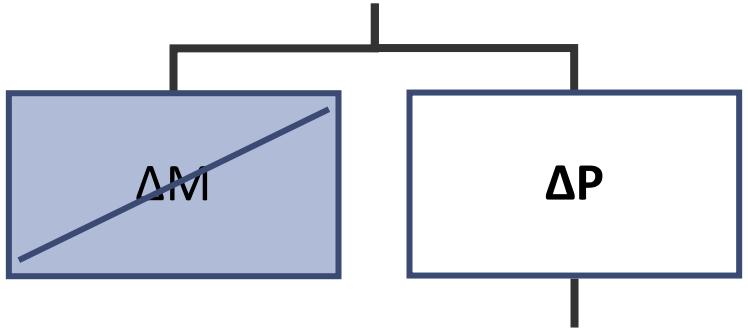
Select parts which show the greatest difference in Y.



When the BIG X Lives in the Assembly Process

Noisy latches are discovered resulting in containment for the latch supplier. There is no noise specification on the blue print, however certain latches sounded “different” (much louder) than others when the remote latch was activated.

Noise emanating from lock latches are causing customer complaints and putting the supplier in containment. (Attribute)

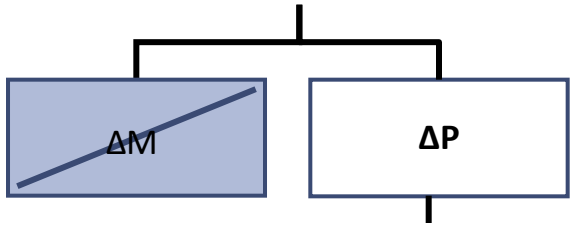


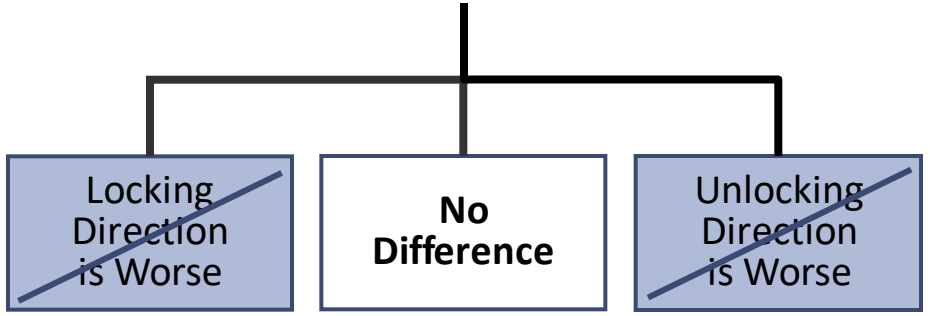
Rationale:

- Clear distinction between quiet BOBs and noisy WOWs

Solution Tree

Noise emanating from lock latches are causing customer complaints and putting the supplier in containment. (Attribute)



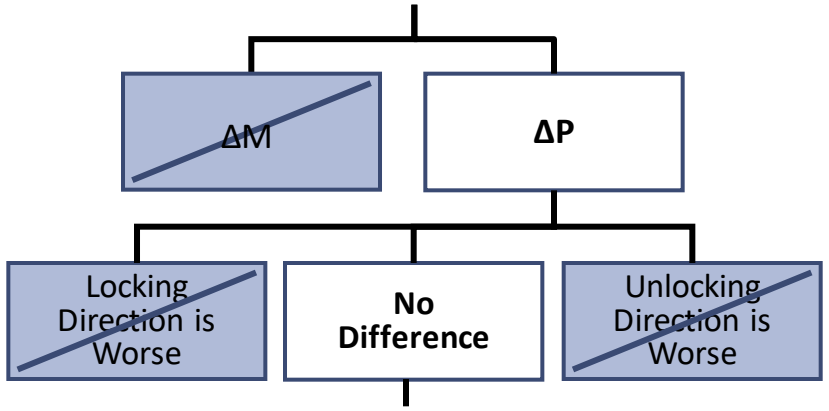


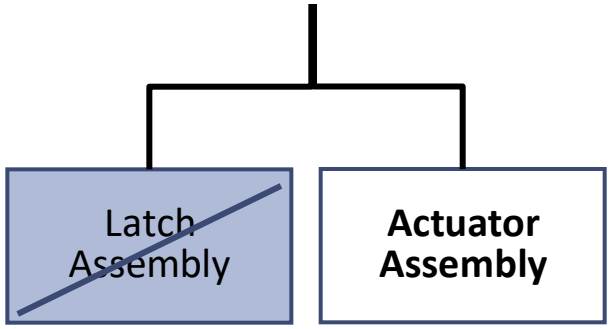
Rationale:

- Noise on WOWs are about the same in both directions.

Solution Tree

Noise emanating from lock latches are causing customer complaints and putting the supplier in containment. (Attribute)

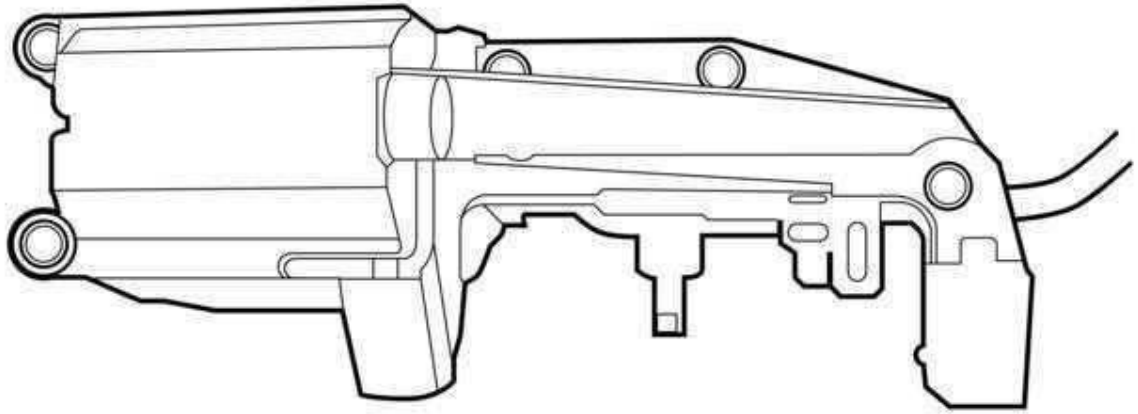




Rationale:

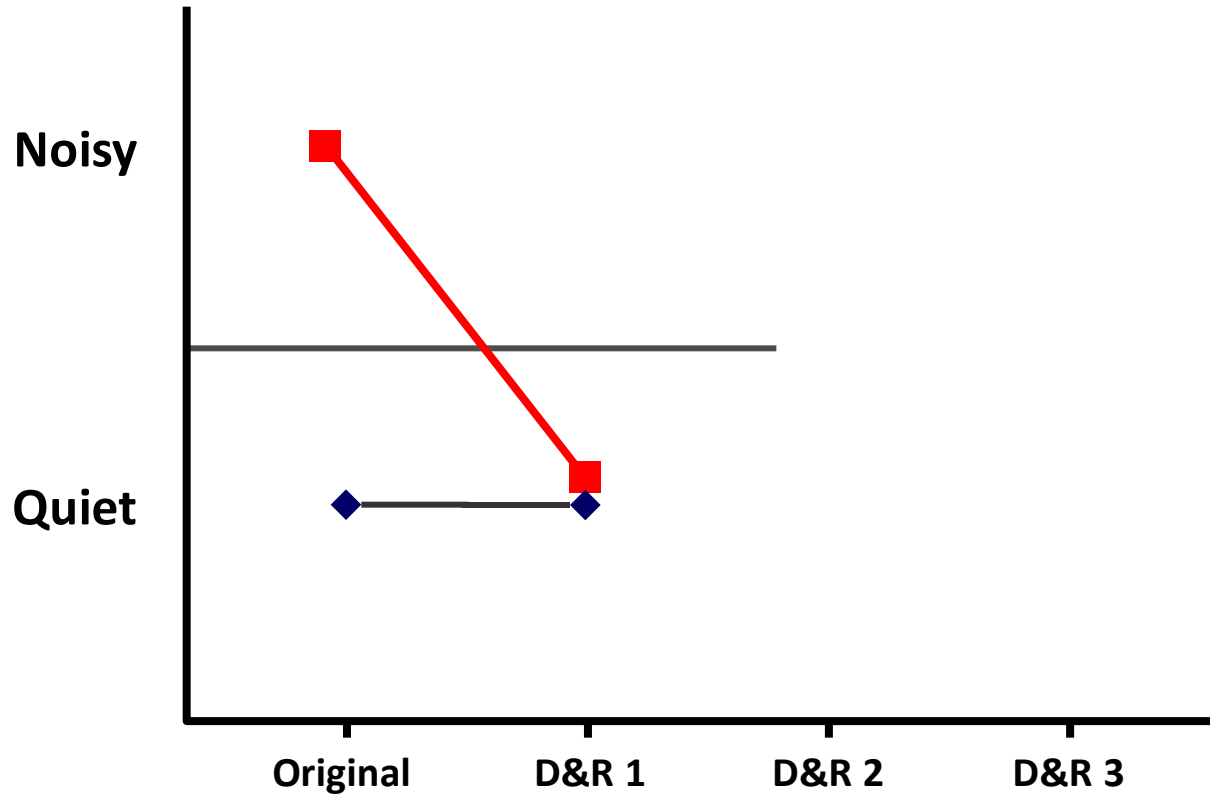
- Actuator still noisy when removed from latch.

Actuator Supplier Cannot Find Anything Wrong



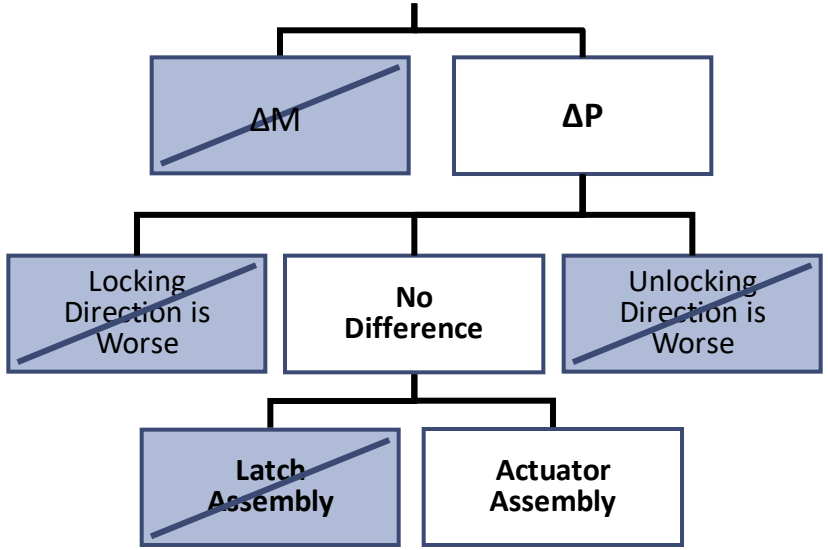
Numerous noisy actuators are returned to the actuator supplier to determine the “root cause.” The rivets holding the cover to the housing are removed and the actuators are opened. Nothing “out of the ordinary” is found with the internal components. Every actuator which is reassembled has become quiet.

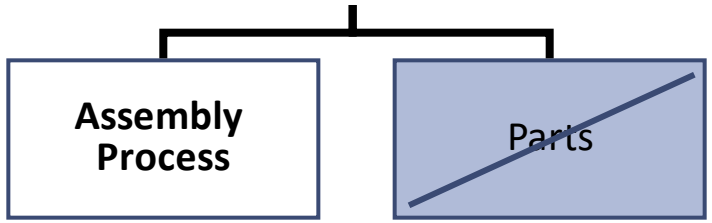
Failed Component Search - Stage 1 of WOWs Always Become BOBs



Solution Tree

Noise emanating from lock latches are causing customer complaints and putting the supplier in containment. (Attribute)

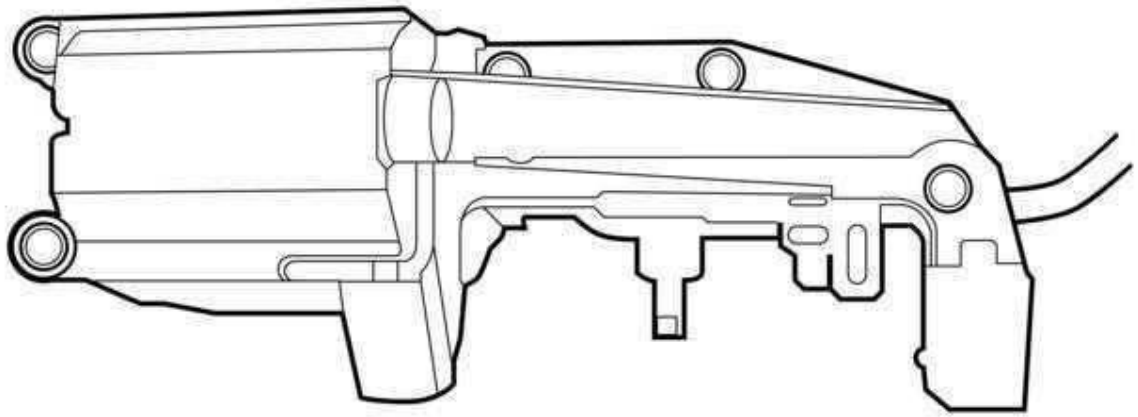




Rationale:

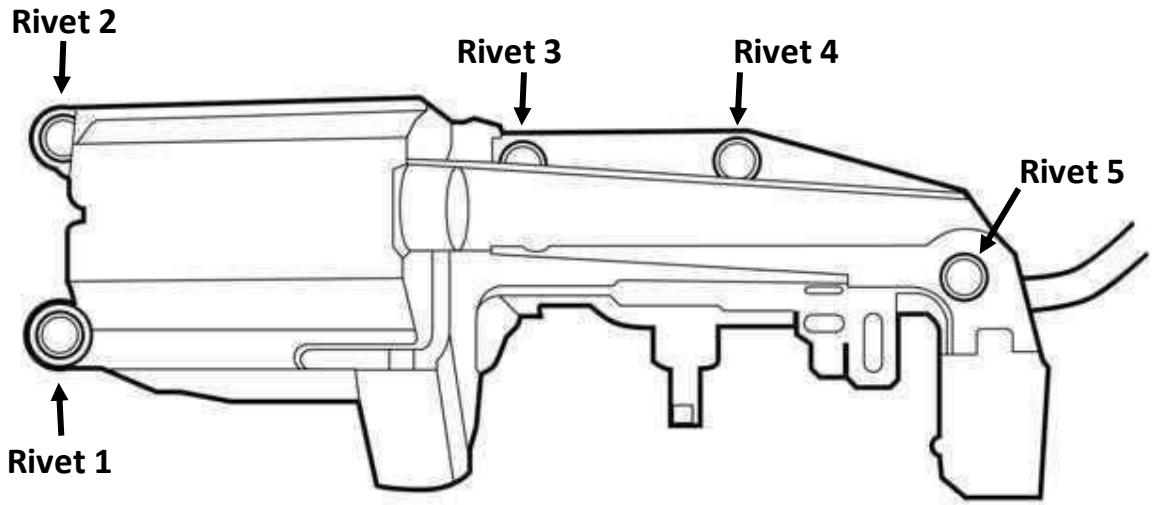
- Failed Component Search - Stage 1

New Strategy Required



The team leader requested that the next returned noisy actuators be sent directly to him. He is sent two noisy actuators. What should he do with them?

Disassembly Process



1 Remove rivets.

2 Unsnap cover and housing.

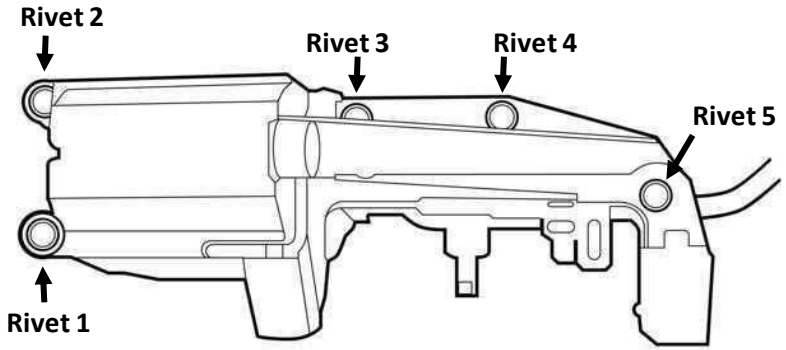
3 Remove jackscrew.

4 Remove motor and pinion gear.

We need a strategy to identify which one of the above steps turns a noisy actuator into a quiet one.

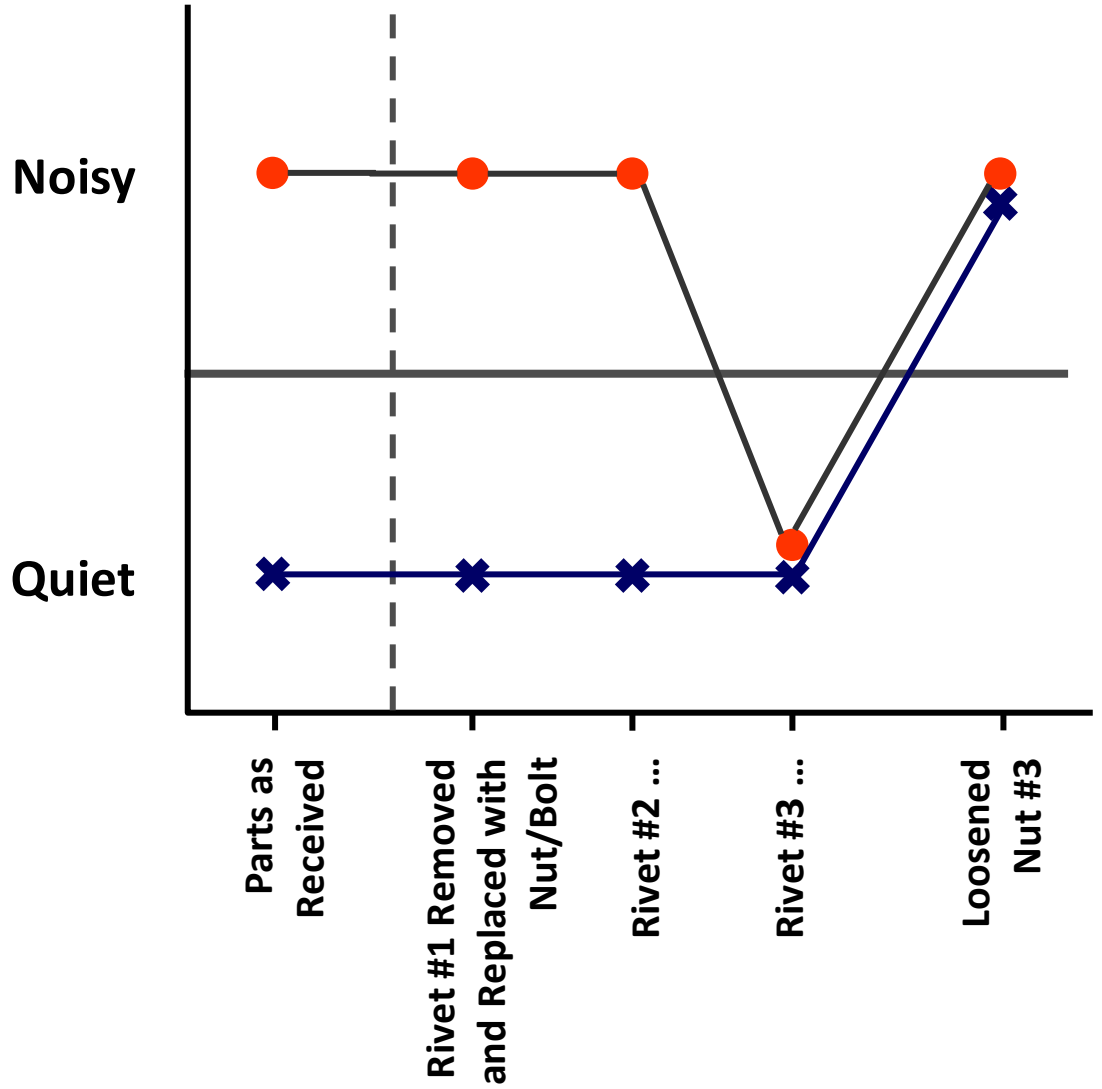
Progressive Disassembly Strategy

Using a noisy actuator, energize the motor after each one of the steps below to understand when it becomes quiet. Progressively, go further into the disassembly process after confirming the last step does not contain the BIG X.



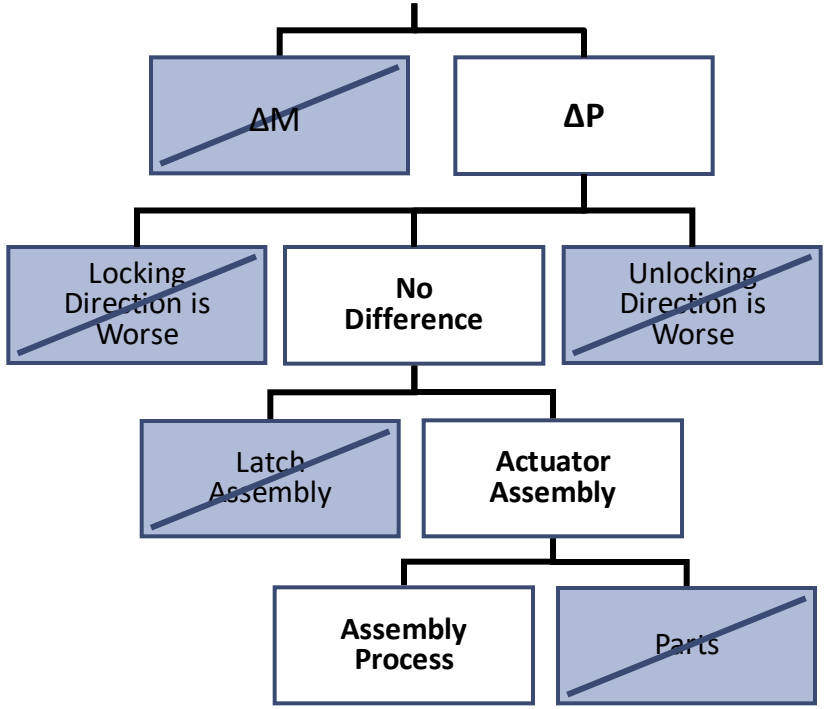
1. Remove one rivet at a time and replace with a nut and bolt so the cover can be easily attached and removed.
2. Remove all the bolts and unsnap the cover from the housing.
3. Perform step 2, and remove the jackscrew.
4. Perform step 3, and remove motor and pinion gear.

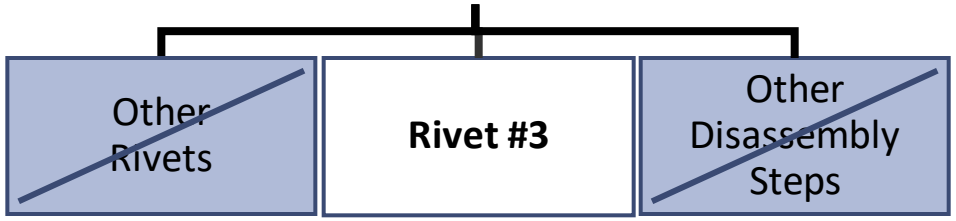
Progressive Disassembly of First WOW



Solution Tree

Noise emanating from lock latches are causing customer complaints and putting the supplier in containment. (Attribute)



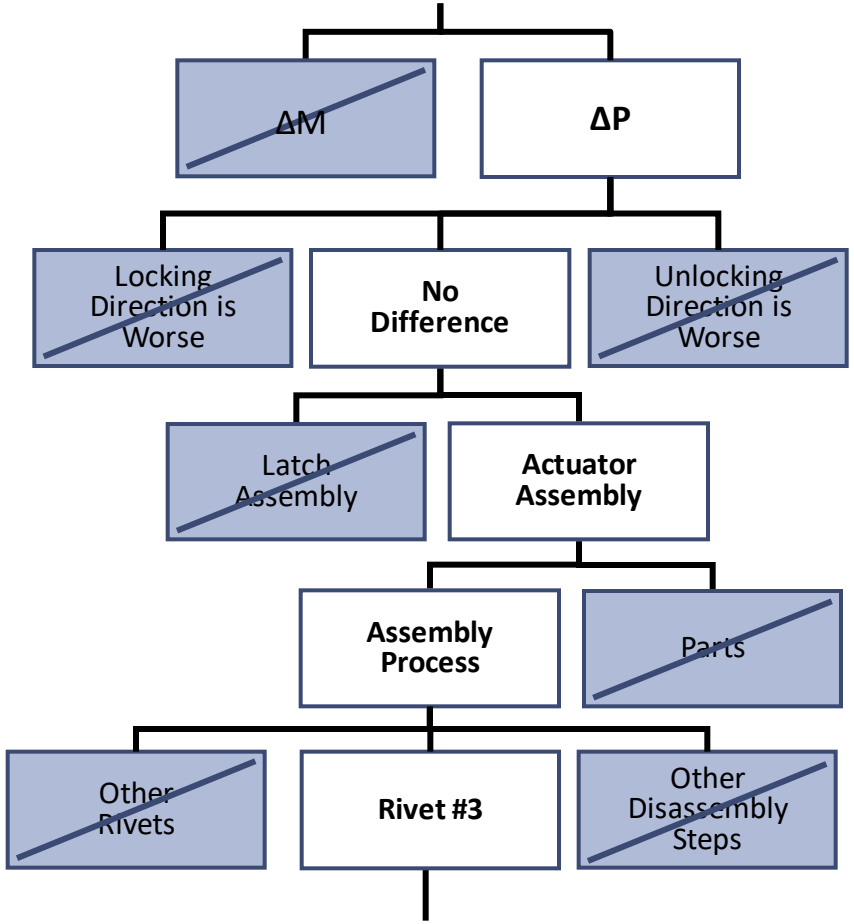


Rationale:

- Rivet #3 made both WOWs quiet when removed and replaced.

Solution Tree

Noise emanating from lock latches are causing customer complaints and putting the supplier in containment. (Attribute)



Worn Rivet Tooling

BIG X

Rationale:

- Worn rivet tooling discovered at position #3.

Workshop: Blink Problem - Exercise

Problem Description:

A leading designer and manufacturer of electromedical equipment for neonatal and pediatric care is getting customer complaints on their ventilators. When the equipment has a malfunction, a fault code is set resulting in specific lights blinking on and off.



Some customers have noticed that when a fault code is set on a unit, its lights “blink differently” than other units. While this does not impact the functionality or performance of the device, it has caught their attention and has led them to believe that there is something wrong with the ventilator.

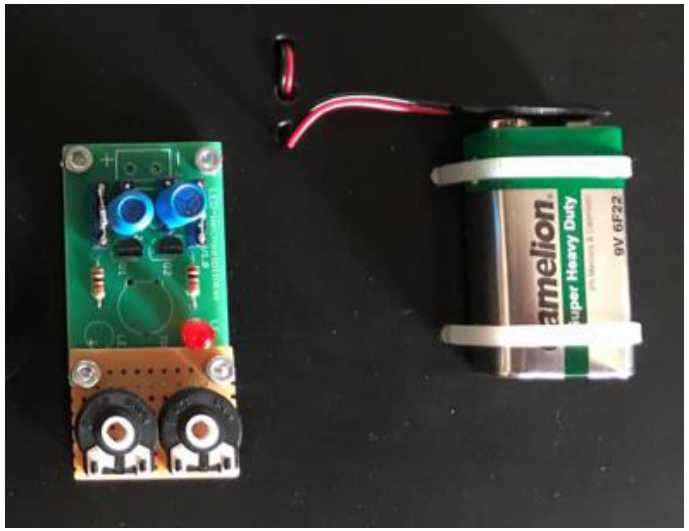
The ventilator manufacturer has traced the problem to a specific circuit board within the device, designated as 9915 after assembly on the EOL. The circuit is a supplied component, and comes from a leading manufacturer of electrical circuits used in medical devices.

Workshop: Blink Problem

Circuit Board Supplier:

Because of the problem with the 9915 circuits, the circuit board supplier has been put into containment. Each circuit board must now be tested by hooking them up to a 9 V battery and checked for the proper blinking frequency. Circuits which have a different blink frequency are rejected. The reject rate from this test is approximately 5 %. The additional testing is putting a burden on production and requires overtime.

Circuit Description:



Printed Circuit board containing:

- R1, R2: Variable Resistors
- R3, R4: Fix Resistors
- C1, C2: Variable Capacitors
- P1, P2: Potentiometer
- LED light
- Battery cables with battery

  Note: Only the variable components are allow for disassembly and reassembly.

Workshop: Blink Problem

Your Task:

Work together in a group to develop and execute a strategy to determine the BIG X, by using a BOB and WOW unit. The unit labeled BOB has the correct blink frequency of approx. 30 at 30 s.

“Clues” that have been overheard:

- This circuit is overly sensitive to input voltage. A redesign is being proposed.
- There are new workers on the assembly line that have not been properly trained.
- The problem seemed to have started when we switched to a new, lower cost component supplier.



- Definition of the project order
- Creation of the project definition tree
- Carrying out the component search
- Time: About 1 hour

Workshop: Blink Problem – Project Statemant - Exercise

Product or Process that has the problem.

Problem description

Place where the problem is discovered

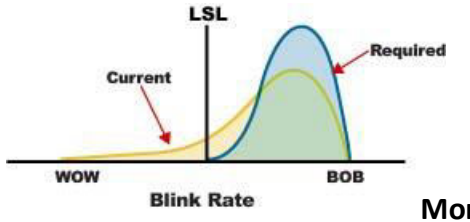
How is the Pain measured (metric)

Create the Project Statement:

“Find and Control the BIG X that drives ( ) ( ) discovered (to eliminate ( ).”

Workshop: Blink Problem – Project Definition Tree - Excercise

Project Statement: Find and Control the BIG X …

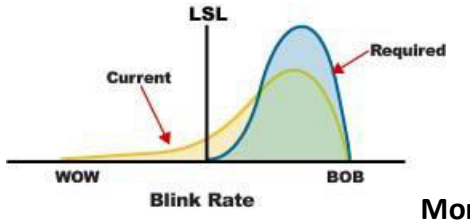


Mone(Y) = Blink rate after 30 s

Rationale:

Workshop: Blink Problem – Project Definition Tree - Excercise

Project Statement: Find and Control the BIG X …



Mone(Y) = Blink rate after 30 s

Rationale:

Malfunction Event Strategy Diagram - Circuit Bord - Exercise

Design Families

Mone(Y) = Blink rate after 30 s

Family of Variation, or Contrast

Diagram

What is Meant by Each Contrast

	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A

Processfamilies Strategy Diagram - Circuit Bord - Exercise

Mone(Y) = Blink rate after 30 s

Family of Variation, or Contrast

Diagram

What is Meant by Each Contrast

	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A
	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A

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Processfamilies Strategy Diagram - Circuit Bord - Exercise

Mone(Y) = Blink rate after 30 s

Family of Variation, or Contrast

Diagram

What is Meant by Each Contrast

- ☐ Large Contrast
- ☐ Small or No Contrast
- ☐ Do Not Know
- ☐ N/A

Workshop: Blink Problem - Exercise



Workshop: Blink Problem – Project Statemant - Answer

Product or Process that has the problem.

Assy PCB 9915

Problem description

Slow Blink Pattern

Place where the problem is discovered

After Assembly

How is the Pain measured (metric)

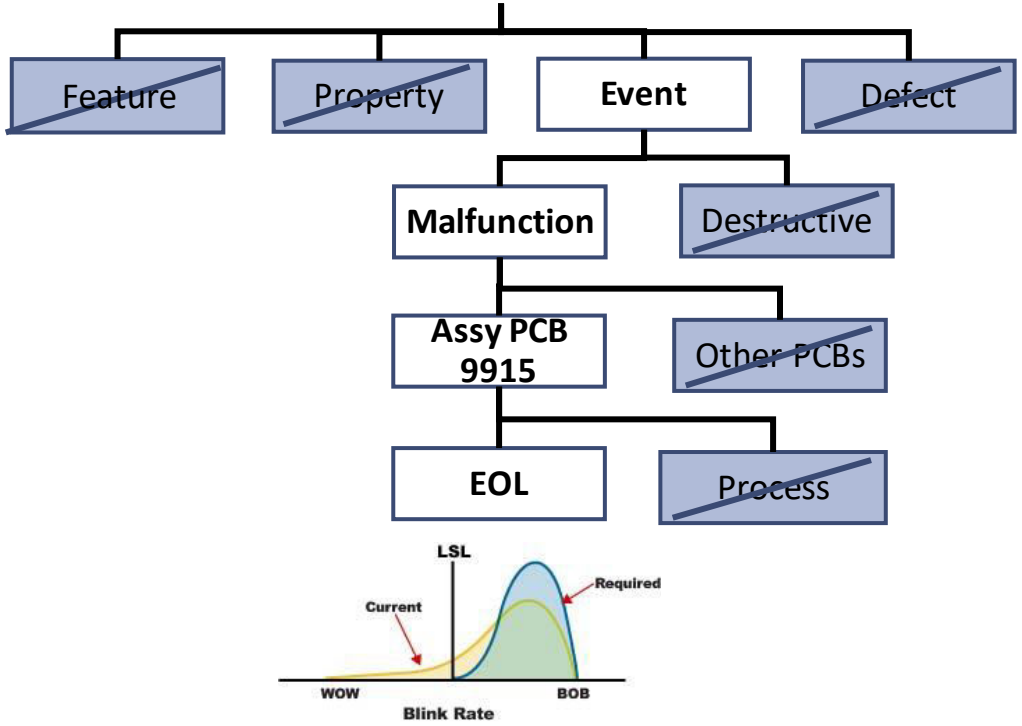
Containment costs and scrap

Project Statement:

"Find and Control the BIG X that drives (Assy PCB 9915) (slow blink pattern failures) discovered (after assembly) to eliminate (containment and scrap costs)."

Workshop: Blink Problem – Project Definiton Tree - Answer

Project Statement: Find and Control the BIG X that drives Circuit 9915 slow blink pattern failures discovered after assembly to eliminate containment and scrap costs.

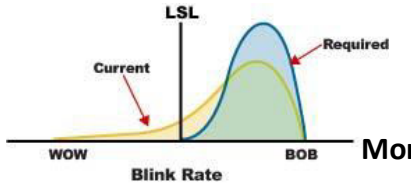


Mone(Y) = Blink rate after 30 s

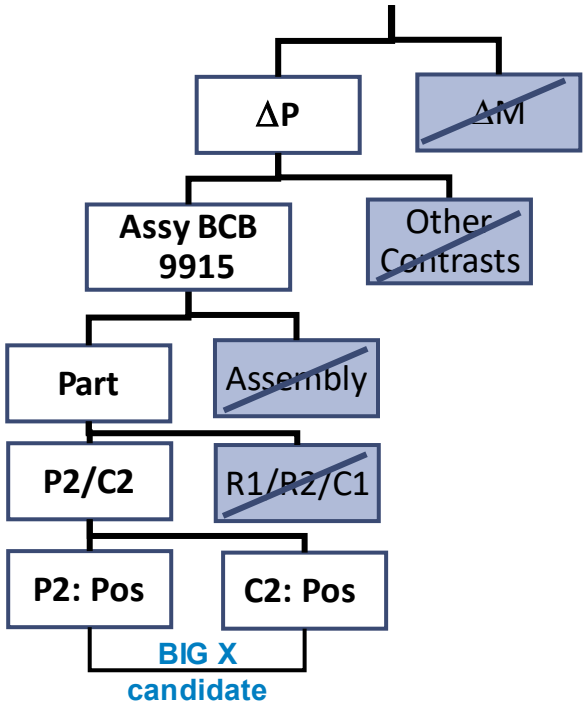
Rationale:

- Blink rate is a system performance output.
- The problem is a malfunction event.
- The problem occurs with the manufacturer of the 9915 Assy PCB.
- The problem is discovered at the EOL.

Workshop: Blink Problem – Solution Tree - Answer



Mone(Y) = Blink rate after 30 s



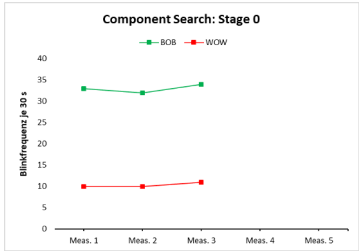
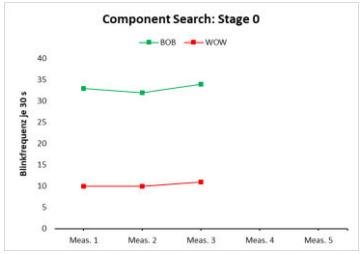
Rationale:

- Stage 0: Passed
- BOB and WOW assemblies are readily available. Able to use Component Search.
- Stage 1: Passed
- Stage 2: Passed, Interaction
- BOW vs WOW Comparison

Malfunction Event Strategy Diagram - Circuit Bord - Answer

Assy PCB 9915

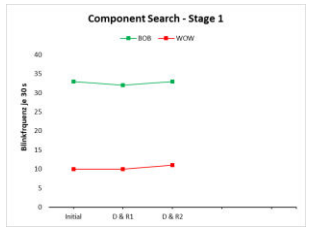
Mone(Y) = Blink rate after 30 s

Family of Variation, or Contrast		***Diagram***	***What is Meant by Each Contrast***
Point to Point **Same Event**	☐ Large Contrast ☐ Small or No Contrast ■ Do Not Know ☐ N/A		Cannot be examined with the method. This Mone(Y) prevents us from seeing into this family.
Event to Event **Within Same Assy PCB 9915**	☐ Large Contrast ■ Small or No Contrast ☐ Do Not Know ☐ N/A		Clear separation between BOB and WOW. Both remain stable.
Within Assy PCB 9915 to Within Assy BCB 9915 **Same Assy PCB 9915**	☐ Large Contrast ☐ Small or No Contrast ■ Do Not Know ☐ N/A		No findings are available.
Assy PCB 9915 to Assy PCB 9915	■ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ☐ N/A		There are BOB and WOW units.

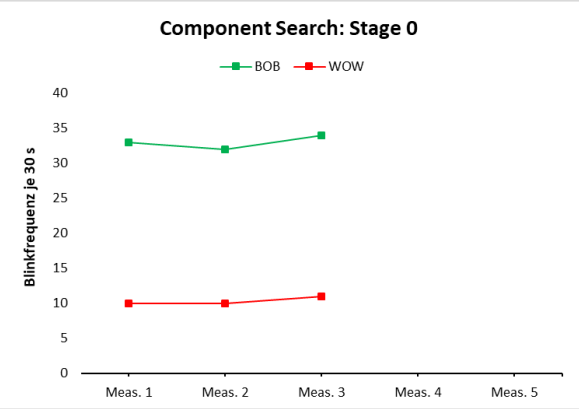
Process Families Strategy Diagram - Circuit Bord - Answer

Hand assembly Lab

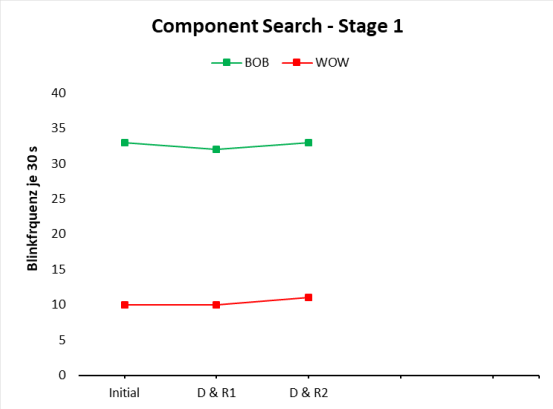
Mone(Y) = Blink rate after 30 s

Family of Variation, or Contrast		***Diagram***	***What is Meant by Each Contrast***
Component to Component **Same Fixture**	☐ Large Contrast ☐ Small or No Contrast ■ Do Not Know ☐ N/A		No documentation.
Fixture to Fixture **Same Assembly**	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ■ N/A		The assembly was carried out by a operator.
Assembly to Assembly **Same Assy PCB 9915**	☐ Large Contrast ■ Small or No Contrast ☐ Do Not Know ☐ N/A		BOB and WOW are stable.
Assy PCB 9915 to Assy PCB 9915 **Same Hour**	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ■ N/A		Assembly was done in an hour.
Hour to Hour	☐ Large Contrast ☐ Small or No Contrast ☐ Do Not Know ■ N/A		Only one type was present.

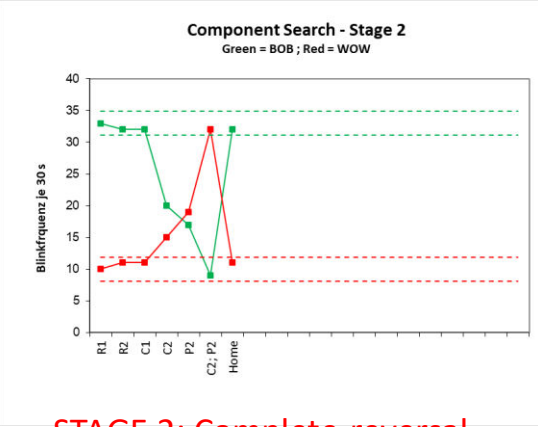
Blink Problem - Answer



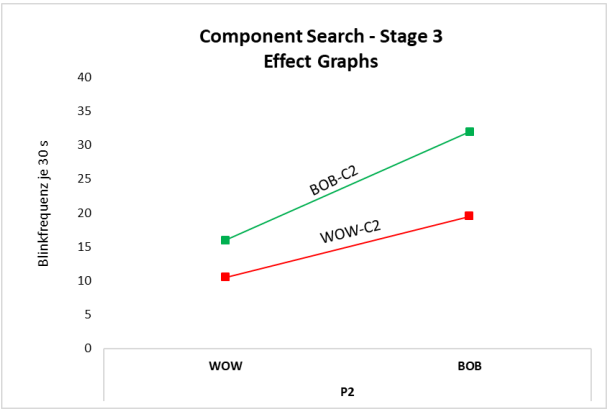
STAGE 0: Passed.
The BIG X lives not in the measurementsystem.



STAGE 1: Passed.
The BIG X lives not in the assembly.



STAGE 2: Complete reversal discovered with multiple components swapped, do a level 3 to identify its influence!

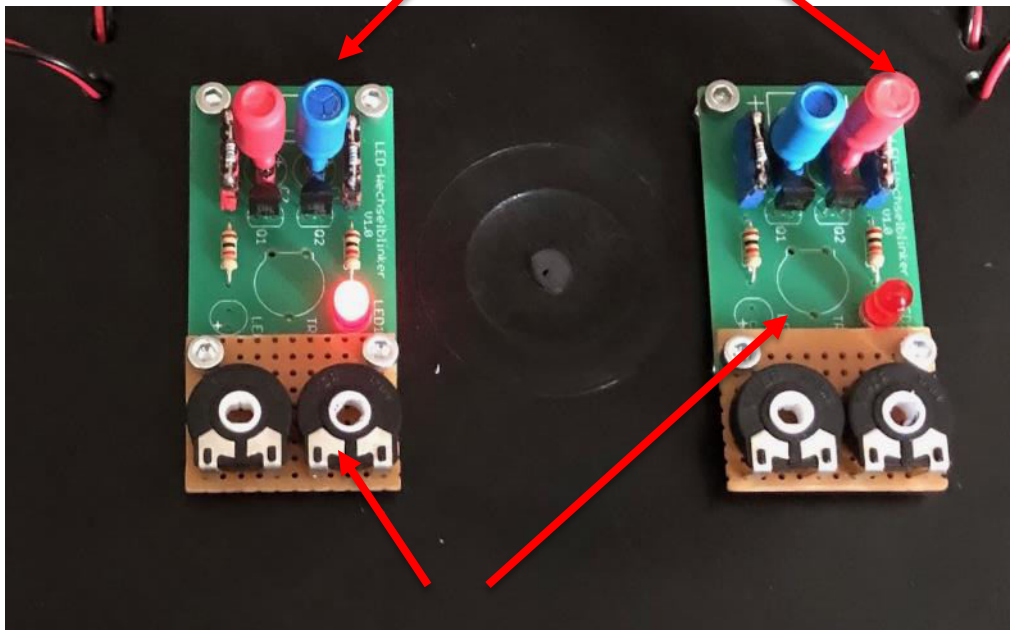


STAGE 3: BIG X is an interaction between P2 and C2.

Blink Problem - Answer

R2 Position

WOW **BOB**



Potentiometer P2

BIG X = an interaction between P2 position and Resistor R2.

Module 6 Summary

Malfunction Event Strategies I

- Projects involving events have a Mone(Y) that requires an energy supply.
- The strategy for events often requires the consideration of environments and operating conditions in order to activate the Mone(Y).
- Component Search – Stage 0 evaluates the measurement system for malfunction events.
- Component Search is often an effective tool to converge on the component that owns the BIG X.
- Paired Comparisons are often the last in a series of tools used to converge on the BIG X.
- Selecting logical pairs is key to finding the Xe that are part of an interaction, especially spike interactions.
- When logical pairs are not available, group comparisons are the proper tool.

7 Confirmation Testing I

Module 7 Objectives
Confirmation Testing I

Understand:

- The role of confirmation testing in problem solving
- The logic of risk in experimentation
- How to control risk

Be able to:

- Assess the need for a confirmation test
- Plan, execute and document an effective test

Statistical Confirmation

- After the BIG X candidate has been identified through the use of effective clue generation techniques, the next step is to statistically prove with a predetermined confidence that the BIG X candidate is truly the BIG X.
- This statistical confirmation is called a B vs W test. We will discuss in this chapter the Tukey B vs W test, too.
- We want to give the BIG X multiple chances to prove us wrong. We must demonstrate that we can produce the BOB and WOW results by only controlling the BIG X suspect. This is a prediction made before we run the test.
- If we have a misunderstanding of the largest cause-effect relationship, we want to go back to Approach and Converge.
- A failed confirmation test protects us from waste!

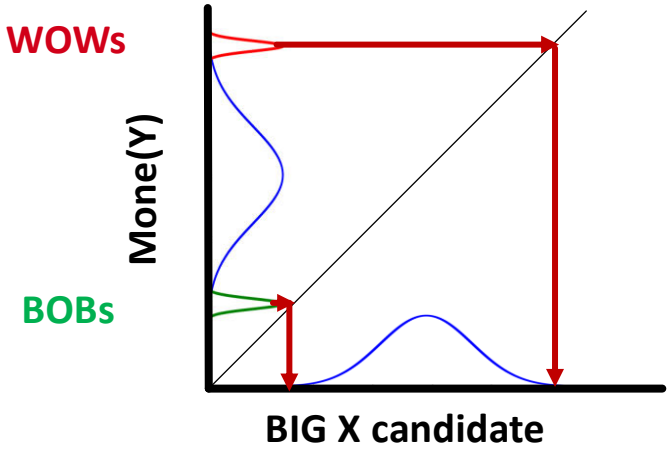


Bill, "Are you sure this rope will hold us?"

Ted, "I hope so!"

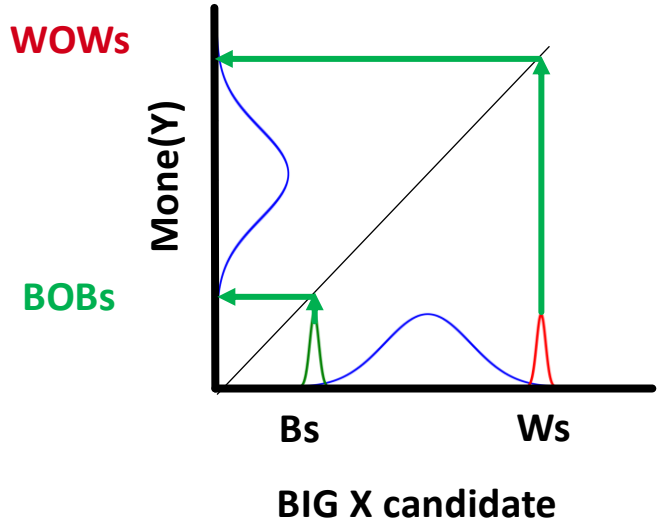
Statistical Confirmation

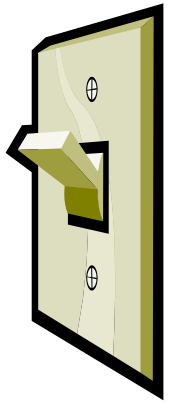
Clue Generation: Y → X



Leverage BOB and WOW contrast to expose the BIG X and to reveal it`s distribution through the use of effective clue generation.

Confirmation: X → Y





Utilize Bs and Ws to prove that you can switch on and off the Mone(Y) by only controlling the BIG X:

- B: BIG X magnitude used to predict a BOB
- W: BIG X magnitude used to predict a WOW

Cause = Effect

Steering a Car:

The Mone(Y) is the direction the car is headed.

The BIG X is the control that has the strongest effect on the Mone(Y).

How do we know that the position of the steering wheel affects the direction of travel?

You can anticipate the change in response based on the change in the input. You expect the car will go left if you turn the steering wheel to the left.



Cause ≠ Effect

Steering a car:

If the steering wheel position is changed, does the direction of the car match?

Amusement Park Ride:

In the case of an amusement park ride, the steering wheel creates the illusion of control but vehicle direction is actually determined by the track.



B vs W 6 Pack

The B vs W is used to confirm that a BIG X suspect drives the BOB / WOW range of the Mone(Y).

The 6 Pack is a B vs W test that is simple to run and confirms the BIG X with only a 5 % risk of being fooled.

Example:

If the physics are not correct, what are the odds of passing the 6 Pack test by chance alone?

1. Place six balls in a hat. Three are labeled “B” and three are labeled “W.”
2. Pick the balls out of the hat at random and note the order.
3. To pass the 6 Pack test by chance you would need to select the three “B” balls followed by the three “W” balls.



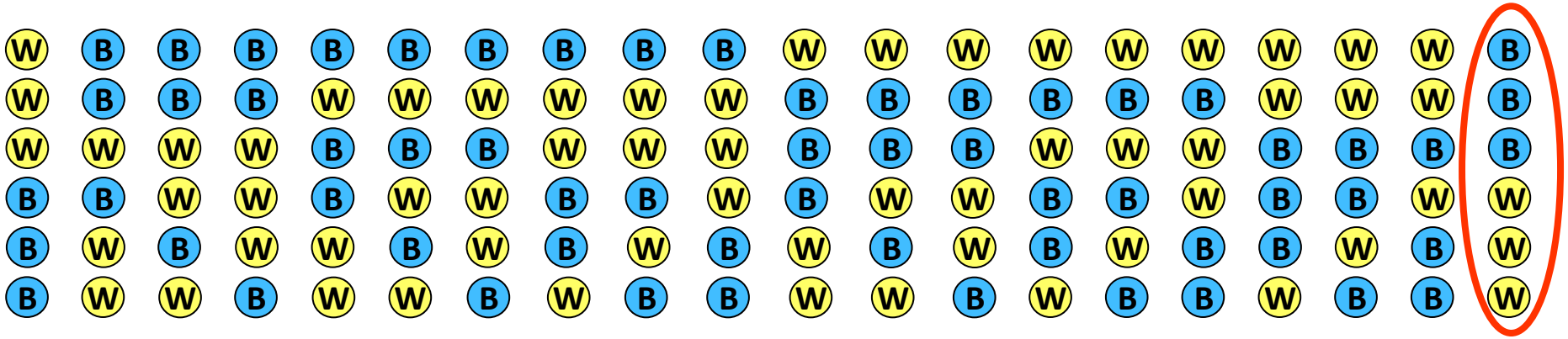
Passed



Failed

Possible Outcomes:

Possible Outcomes = $\binom{n}{r} = \frac{n!}{(n-r)! \times r!}$



Twenty possible outcomes, nineteen orders will fail the test one way to get fooled.

$$\binom{6}{3} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{(3 \times 2 \times 1) \times (3 \times 2 \times 1)} = 20$$

1/20 is a 5 % Risk.

If We Only Test X_Bs and X_Ws

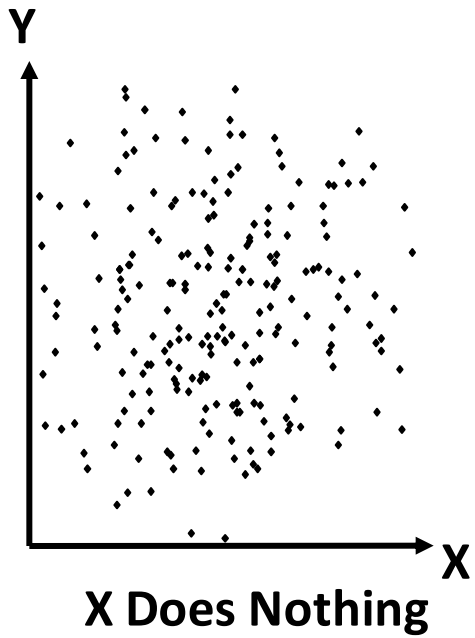
Six parts test at random, three from each of the extremes of the BIG X suspect.

Measure Mone(Y)

Rank Order

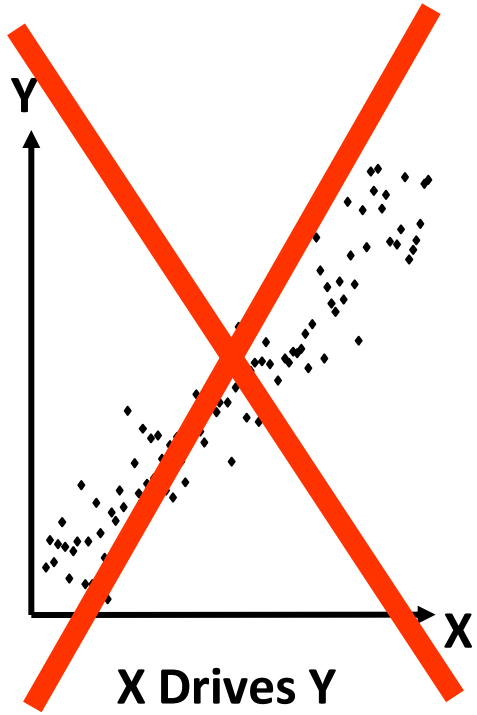
B
W
W
B
W
B

Conclusion?



X Does Nothing

OR



X Drives Y

Y ≠ f(X)

We *couldn't* prove that X controls Y at a risk of 5 %.

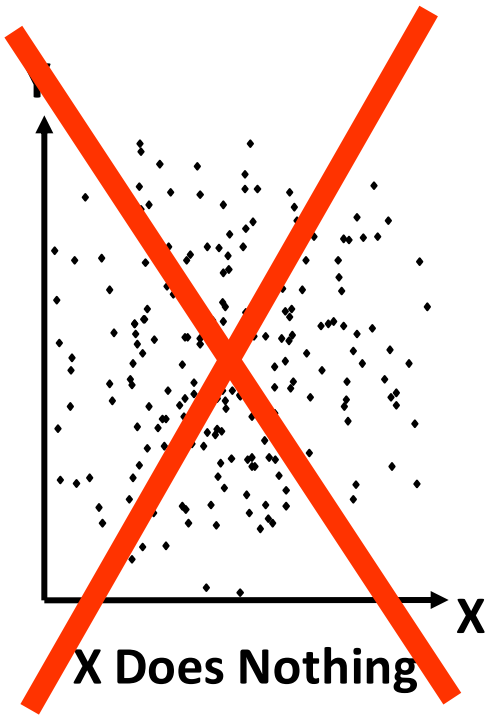
If We Only Test X_Bs and X_Ws

Six parts test at random, three from each of the extremes of the BIG X suspect.

Measure Mone(Y)

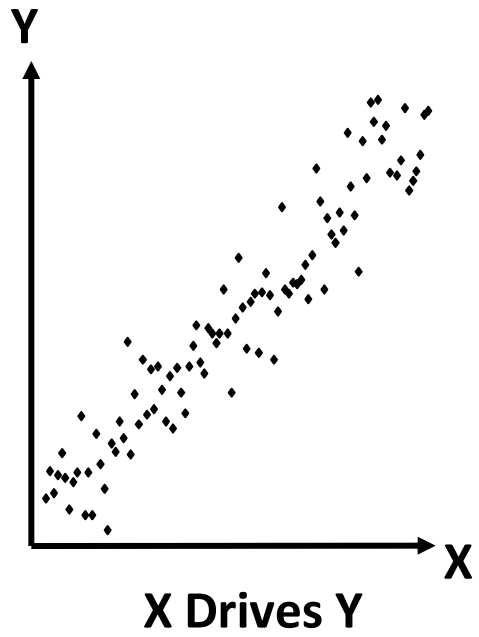
Rank Order

B
B
B
W
W
W



Conclusion?

OR



Y = f(X)

We *could* prove that X controls Y at a risk of 5 %.

B vs W Test Procedure

1. Establish test levels for the BIG X suspect.
   - B – setting expected to produce BOB results
   - W - setting expected to produce WOW results
2. Establish a basis for ranking the results. What's the Mone(Y)?
3. Determine allowable risk.
4. Select a random build and test order that minimizes risk.
5. Rank results on the Mone(Y) and evaluate outcome. *No overlap* of ranks is *permitted* by this test.
6. Results from B vs W must show the full range of variation from your product distribution.
7. Document conclusions.

B vs W Test Procedure

1. **Establish test levels for the BIG X suspect.**
   - **B – setting expected to produce BOB results**
   - **W - setting expected to produce WOW results**
2. **Establish a basis for ranking the results. What's the Mone(Y)?**
3. Determine allowable risk.
4. Select a random build and test order that minimizes risk.
5. Rank results on the Mone(Y) and evaluate outcome. *No overlap* of ranks is *permitted* by this test.
6. Results from B vs W must show the full range of variation from product distribution.
7. Document conclusions.

Establish test levels for the BIG X suspect.

Specification Limit:

- A little above top specification limit and below bottom specification limit? **No**

- A little above bottom specification limit and below top specification limit ? **No**

Equipment Limit:

- Just below the top limit and just above the bottom limit of the current process equipment? **No**

Current Process:

- Input levels from BOB and WOW samples? **Yes!**

B vs W Test Procedure

1. Establish test levels for the BIG X suspect.
   - B – setting expected to produce BOB results
   - W - setting expected to produce WOW results
2. Establish a basis for ranking the results. What's the Mone(Y)?
3. **Determine allowable risk.**
4. Select a random build and test order that minimizes risk.
5. Rank results on the Mone(Y) and evaluate outcome. *No overlap* of ranks is *permitted* by this test.
6. Results from B vs W must show the full range of variation from your product distribution.
7. Document conclusions.

Risk Assessment

Assume that the suspect is not the BIG X. What are the consequences of moving forward while believing that it is the BIG X?

If:

- the cost of implementing a corrective action is low, and
- the time to implement is short, and
- the change is reversible, and
- the improved performance can be tracked quickly and easily,

then execute and demonstrate improved performance.

However, if any of these is moderate or high then plan and execute a statistical confirmation test.

Risk Assessment Matrix

▪ **The proposed change is:** ▪ **Date of assessment:** ▪ **What leader was involved in this discussion?:**	**Consequences of a wrong decision**			
	Details	**Low**	**Med**	**High**
1. Will it ***cost*** a lot to implement the change?	?	Insignificant	Moderate	Significant
2. Will it take a long ***time to implement*** the change?	?	Quickly	Moderate	Significant
3. Will it be easy to go **reverse *the change***?	?	Completely Reversible	Partly Reversible	Non Reversible
4. Will it take a long **time find out** it wasn't effective?	?	Insignificant	Moderate	Significant
5. On going cost (€ or other losses) if the change **does not fix** the problem.	?	No Risks	Internal Risk	Customer Risk
6. Do we **understand the physics** of the problem?	?	Fully Understand	Somewhat	Not Clear
7. Is there a risk of **health damage**?	?	No	Somewhat	Significant
Overall Assessment of Risk:		Low	Med	High

Decision Made

Acceptable % of ´risk selected: 0.1 %, 1 %, 5 %, 10 % oder 50 %

(If 50%, no confirmation test will be conducted because all risks judged LOW. This is called "Run Chart Confirmation").

Write some short information in place of the Question Mark. Write out your final decision at the bottom of what kind of Confirmation and associated Risk of being fooled.

Table for B vs W and B vs C One Tailed Test

Consequences of a Wrong Decision		Number of Randomized Samples	
Desired Confidence	α-Risk	X_B	X_W
0.999	0.001	3	16
Super Critical		4	10
		5	8
		6	6
0.99	0.01	2	13
Critical		3	7
		4	5
		5	4
0.95	0.05	1	19
Important		2	5
		3	3
0.90	0.10	1	9
Moderate		2	3

(X_B → BOB and X_W → WOW, not just different)

No overlap of ranks is permitted by this test.

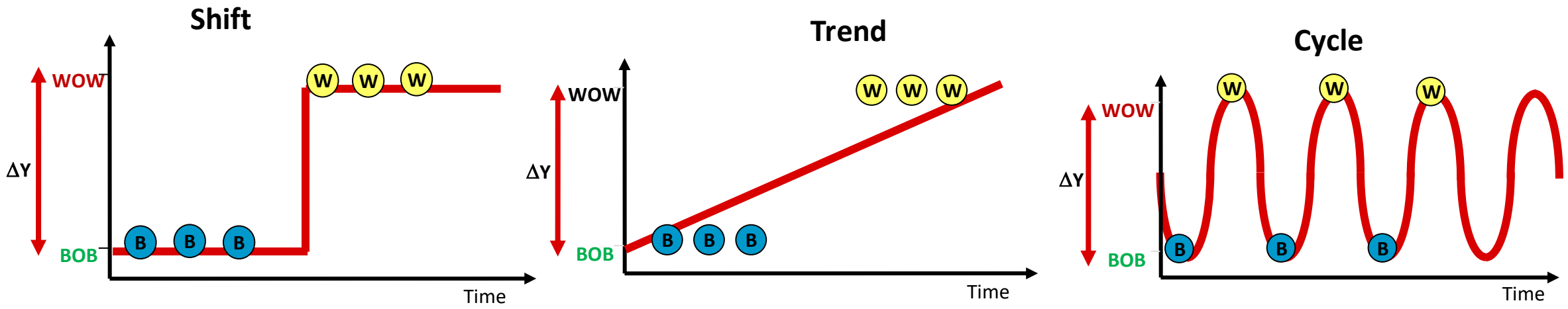
6 Pack Test

B vs W Test Procedure

1. Establish test levels for the BIG X suspect.
   - B – setting expected to produce BOB results
   - W - setting expected to produce WOW results
2. Establish a basis for ranking the results. What's the Mone(Y)?
3. Determine allowable risk.
4. **Select a random build and test order that minimizes risk.**
5. Rank results on the Mone(Y) and evaluate outcome. *No overlap* of ranks is *permitted* by this test.
6. Results from B vs W must show the full range of variation from your product distribution.
7. Document conclusions.

Select a build and test order that minimizes risk.

Without randomization the effect of the BIG X candidate may not be able to be from separated noise factors, such as:



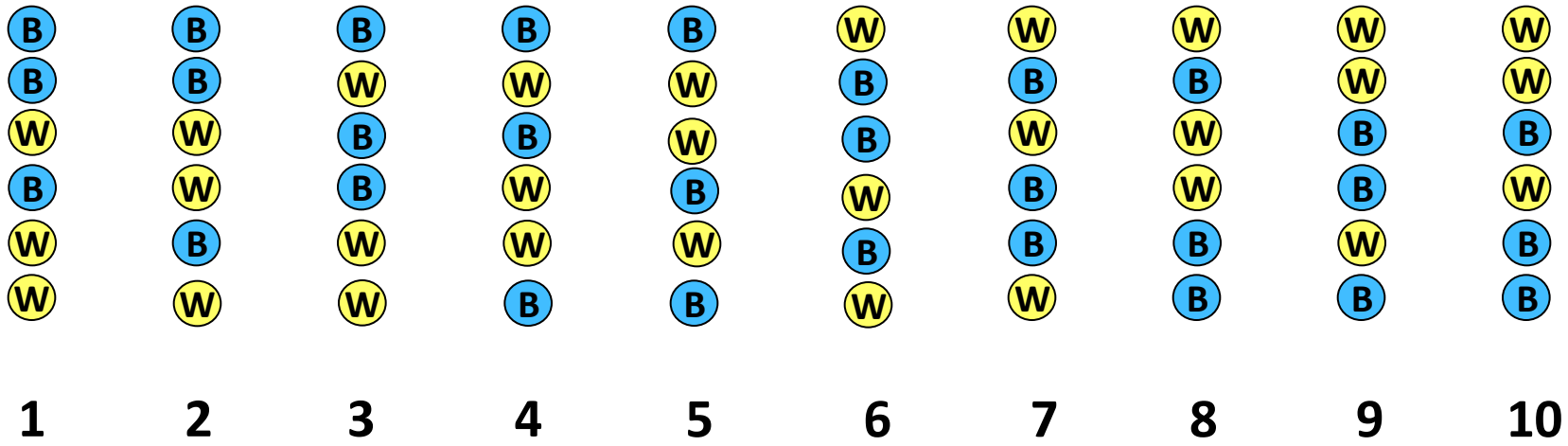
Randomization breaks the phase relationships so we can tell whether the effect is due to the BIG X candidates or noise factors.

Remember, the logic is to assume the BIG X candidate does nothing!

Select a build and test order that minimizes risk.

Minimizing risk of a false conviction

Ten Useful Test Patterns



10 of them Avoid a Trends, Shifts & Cycles.

B vs W Test Procedure

1. Establish test levels for the BIG X suspect.
   - B – setting expected to produce BOB results
   - W - setting expected to produce WOW results
2. Establish a basis for ranking the results. What's the Mone(Y)?
3. Determine allowable risk.
4. Select a random build and test order that minimizes risk.
5. **Rank results on the Mone(Y) and evaluate outcome. *No overlap* of ranks is *permitted* by this test.**
6. **Results from B vs W must show the full range of variation from your product distribution.**
7. **Document conclusions.**

Rank results and evaluate outcome. Document conclusions.

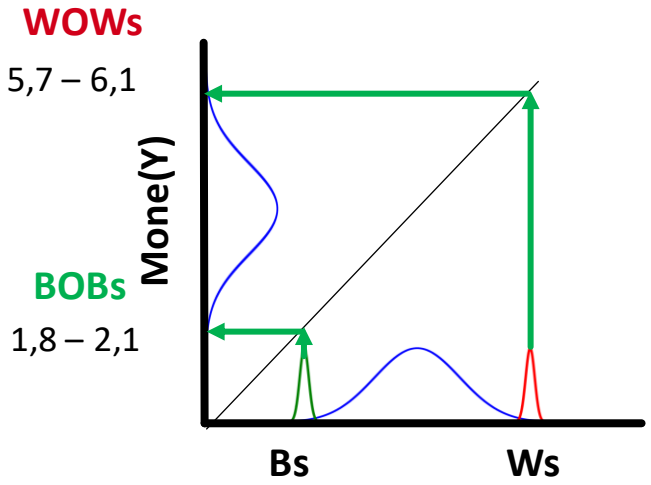
Example:

Results in Rank Order

X Condition	Y
B	1.8
B	1.9
B	2.1
W	5.7
W	5.8
W	6.1

BOB

WOW



Conclude that the suspect X is the BIG X with 5 % risk.

Reasons you fail your test

Some of the most common reasons B vs W tests fail:

1. Did not identify the BIG X.
2. Not picking true BOBs and WOWs from the tails of the product variation during clue generation.
3. Not selecting true BOBs and WOWs for the B vs W test.
4. Possible Small X present (second unknown variable).
5. Interaction present that was missed during clue generation.

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When B vs W Tests Fail

- That does not mean that the BIG X Candidate is not the BIG X.
- You simply “Failed to Confirm BIG X Candidate”.
- Using the results of the B vs W test, return to Clue Generation and evaluate what other X`s matter.

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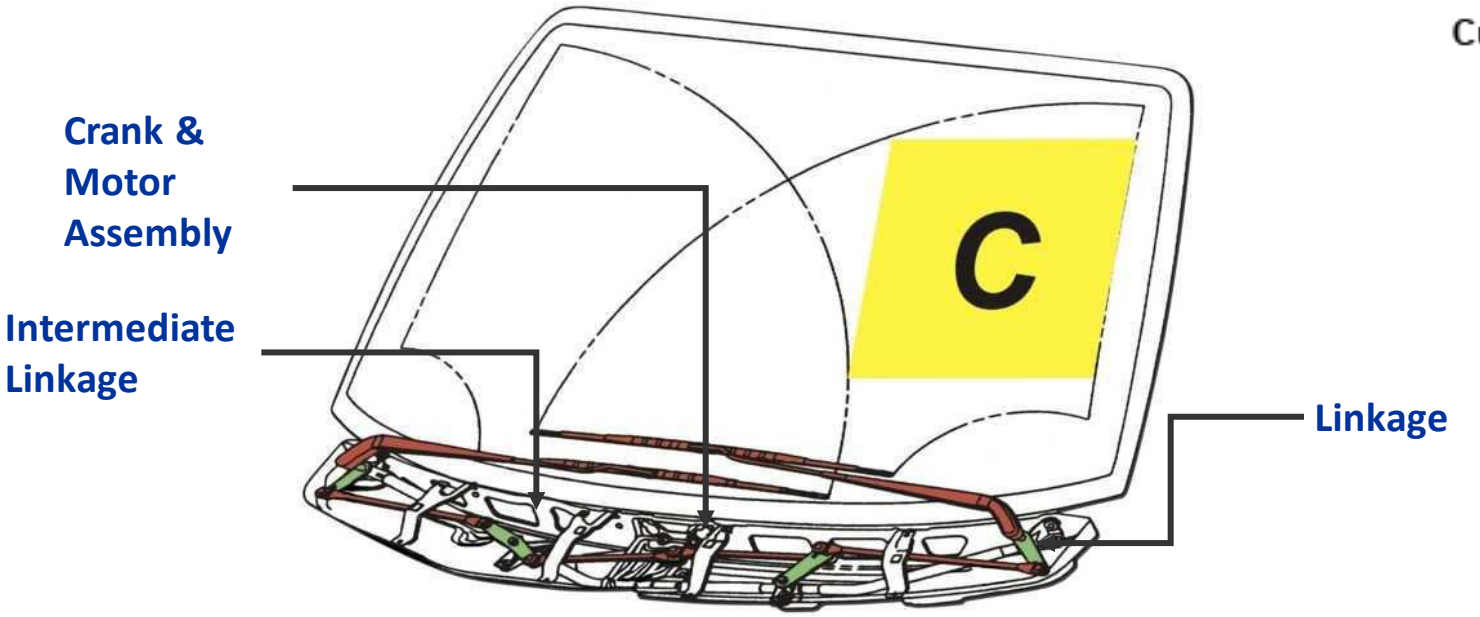
Confirmation of incoming parts

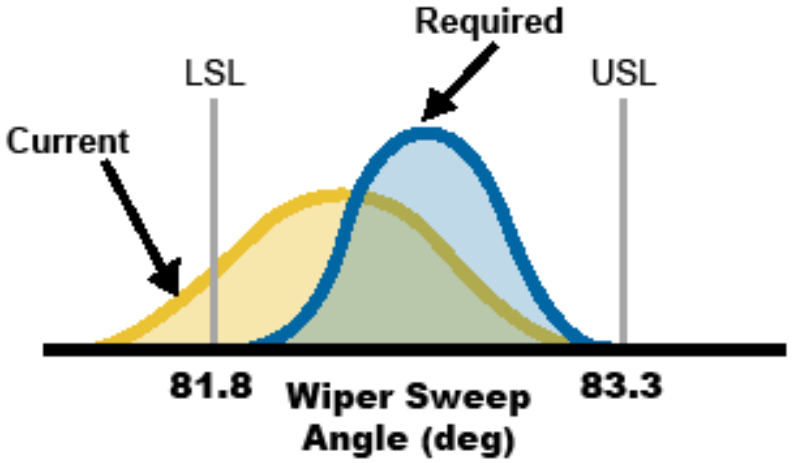
Intermediate Linkage Length and Shaft features for wiper sweep angle

- A check of incoming parts has intermediate linkage lengths and the shaft features (run out, knurl concentricity, knurl angle) that are outside of the specification. Management has focused resources on fixing the supplier's process. All other efforts have stopped because the root cause has been found.
- A test could have been used to confirm that intermediate linkage length was the BIG X before expending resources at the supplier and putting other investigative efforts on hold. If the confirmation test fails, the shaft features (run out, knurl concentricity, knurl angle) must be analyzed with the Paired Comparisons. A confirmation test must then be performed.

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Windshield Wiper: Magnification of Sweep Angle Variation



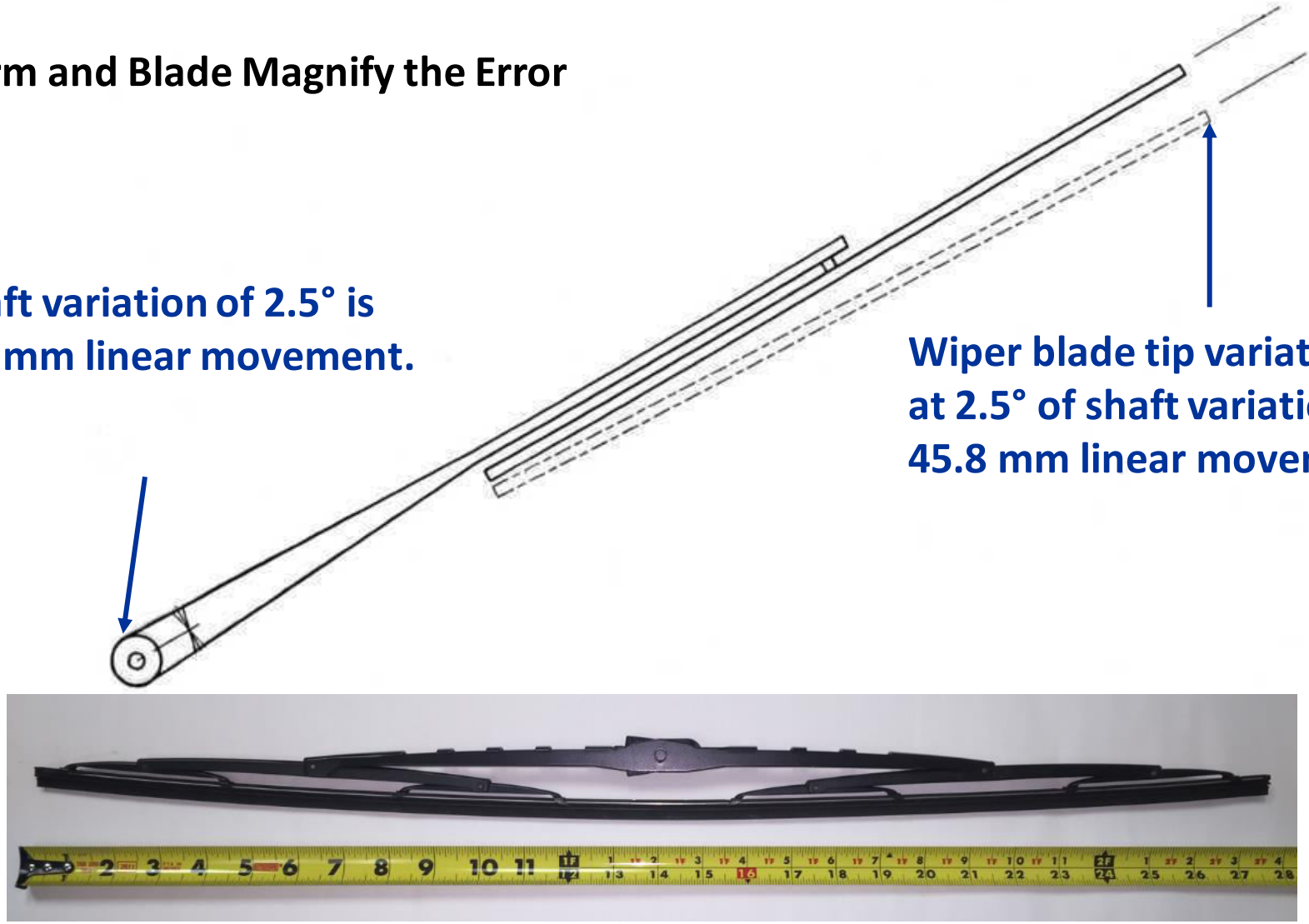


Windshield Wiper: Magnification of Sweep Angle Variation

The Longer Arm and Blade Magnify the Error

Shaft variation of 2.5° is 0.3 mm linear movement.

Wiper blade tip variation at 2.5° of shaft variation is 45.8 mm linear movement.



B vs W Test – Intermediate linkage

1 X_B: Intermediate linkages: 269.2 mm -269.4 mm

X_w: Intermediate linkages: 267,8 - 268.0 mm

2 **Response:** Sweep Angle [°]

3 **Risk:** 5 %

Required Result: All B samples must rank above the W samples.

Run Order

4

Sample	Length [mm]	Sweep Angle [°]
B1	269.2	82.44
W1	267.9	81.22
B2	269.3	82.69
W2	268	81.20
W3	267.8	82.39
B3	269.4	81.58

Rank Order

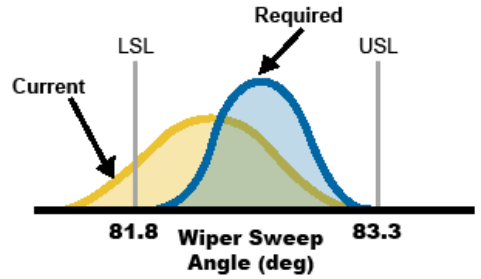
5

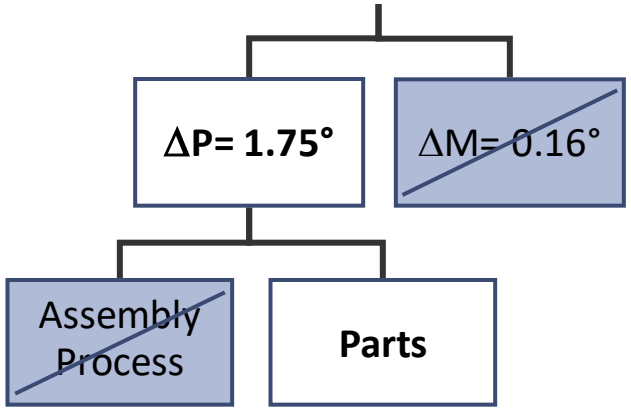
Sample	Length [mm]	Sweep Angle [°]
B2	269.3	82.69
B1	269.2	82.44
W3	267.8	82.39
B3	269.4	81.58
W1	267.9	81.22
W2	268	81.20

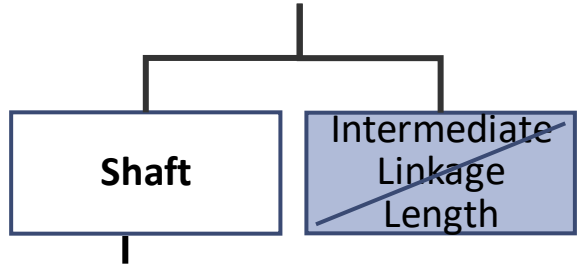
6 **Conclusion:** Failed to prove that intermediate linkage length is the BIG X for sweep angle. There is no more than a 5 % risk that we have been fooled by these data.

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Wiper Angle – Solution Tree







Rationale:

- Paired Comparisons

Paired Comparisons - Exercise

Wiper Sweep Angle Case

	Pair 1		Pair 2		Pair 3		Pair 4		Pair 5	
	BOB	WOW	BOB	WOW	BOB	WOW	BOB	WOW	BOB	WOW
Wiper Angle[°]	82,38 ▼	81,35	82,66 ▼	81,30	82,35 ▼	81,34	82,45 ▼	81,29	82,37 ▼	81,39
Run Out [mm]	0,06 ▲	0,08	0,08 ▼	0,05	0,07 ▲	0,09	0,08 ▼	0.05	0,09 ▼	0,05
Knurl Con. [mm]	0,03 ▲	0,07	0,03 ▲	0,09	0,01 ▲	0,1	0,02 ▲	0.13	0,03 ▲	0,12
Knurl Angle [°]	18,5 ▼	18,0	19,0 ▲	19,5	18,5 ▲	20,0	18,0 ▲	18.5	19,0 ▲	19,5

What is the candidate for BIG X feature?

Design a B vs W Test for the Wiper Angle Case - Exercise

X_B:

X_w:

Response:

Risk:

Required Result:

- Fill the test document.
- Consider the limits for BOBs and WOWs.
- Complete the Rank Order Table.
- Draw your conclusion.
- You have 5 minutes to complete.

Run Order

Sample		Wiper Angle [°]
B1		82.21
W1		81.58
W2		81.28
B2		82.48
W3		81.29
B3		82.66

Rank Order

Sample		Wiper Angle [°]
B1		
W1		
W2		
B2		
W3		
B3		

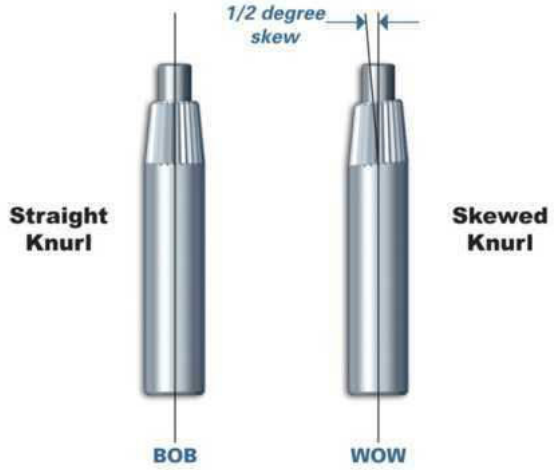
Conclusion:

Paired Comparisons - Answer

Wiper Sweep Angle Case

	PAIR 1		PAIR 2		PAIR 3		PAIR 4		PAIR 5	
	BOB	WOW	BOB	WOW	BOB	WOW	BOB	WOW	BOB	WOW
Wiper Angle [°]	106.60	105.50	106.30	105.20	106.80	105.10	106.70	105.00	106.50	105.10
Run Out [mm]	0.06	0.08	0.08	0.05	0.07	0.09	0.08	0.05	0.09	0.05
Knurl Con. [mm]	0.03	0.07	0.03	0.09	0.01	0.10	0.02	0.13	0.03	0.12
Knurl Angle [°]	18.50	18.00	19.00	19.50	18.50	20.00	18.00	18.50	19.00	19.50

Knurl concentricity is candidate for BIG X feature.



Wiper Angle Confirmation – Answer

X_B: Knurl concentricity: 0,01 - 0,03 mm.

X_w: Knurl concentricity: 0,07 – 0,13 mm

Response: Wiper angle [°].

Risk: 5 %

Required Result: All B samples must rank above the W samples.

Run Order

Sample	Knurl Conc. [mm]	Wiper Angle [°]
B1	0.01	82.21
W1	0.09	81.58
W2	0.11	81.28
B2	0.03	82.48
W3	0.1	81.29
B3	0.03	82.66

Samples found by sorting parts.

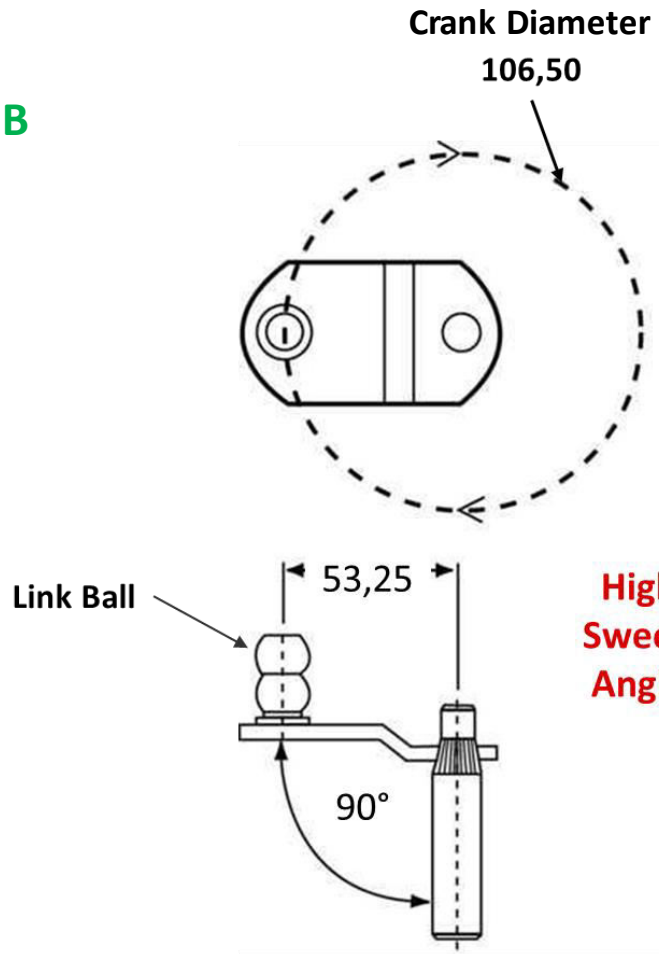
Rank Order

Sample	Knurl Conc. [mm]	Wiper Angle [°]
B3	0.03	82.66
B2	0.03	82.48
B1	0.01	82.21
W1	0.09	81.58
W3	0.1	81.29
W2	0.11	81.28

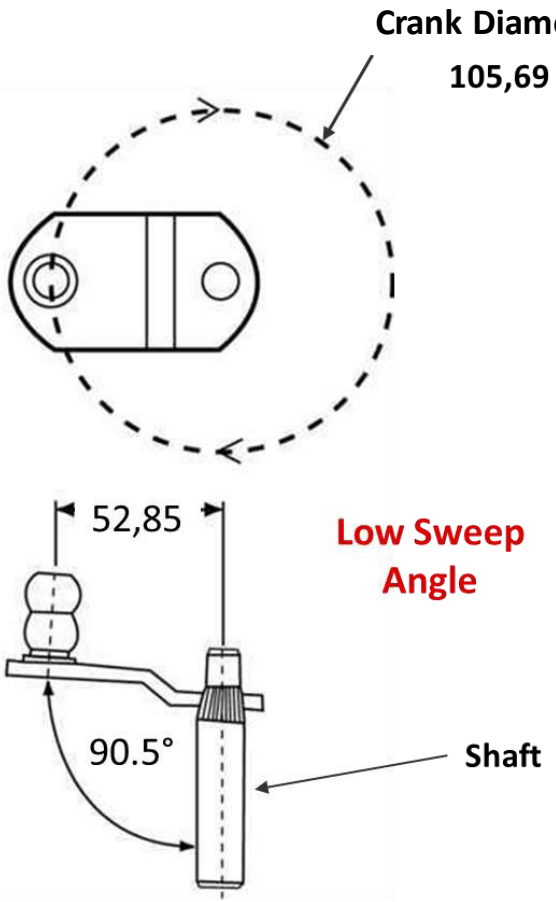
Conclusion: Test passed. Knurl concentricity is the BIG X for wiper sweep angle. There is a 5% risk that we have been fooled by these data.

Understand Y → X Relationship - Answer

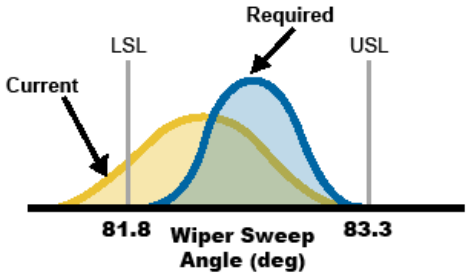
BOB

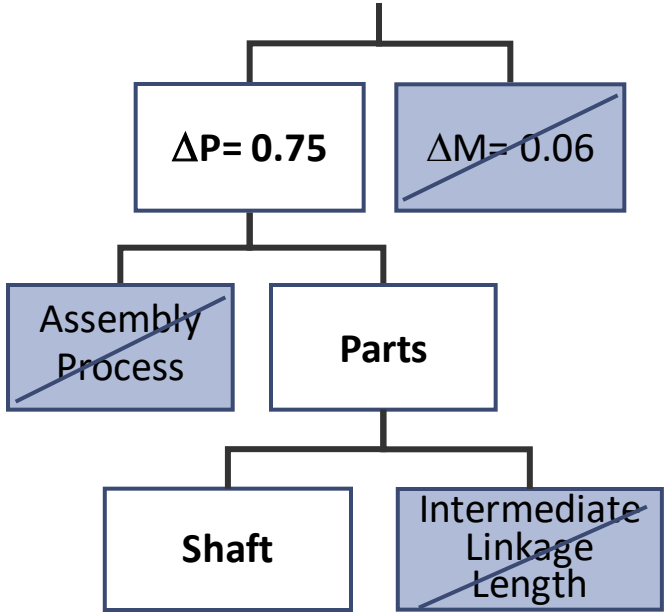


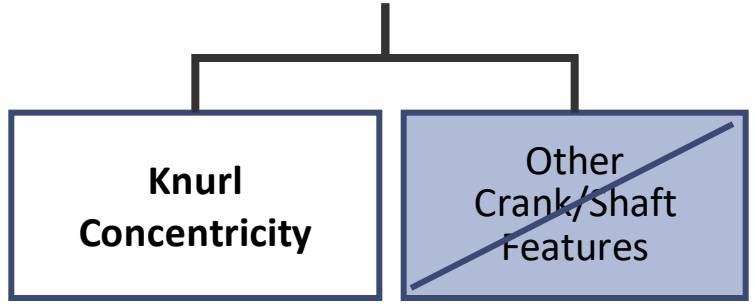
WOW



Wiper Angle – Solution Tree - Answer







BIG X

Rationale:

- Paired Comparisons
- B vs W 6 Pack test

Tukey B vs W Test

- This test is performed when clue generation has identified a single BIG X candidate , and a potential Small X candidate is present. This test only confirms the **BIG X candidate**.
- The Small X candidate could be another variable, measurement variation, or an unknown source.
- The Tukey B vs W test confirms whether or not the BIG X candidate is strong enough to ovecome any of the other candidates.

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Tukey B vs W Test Procedure

1. Establish test levels for the BIG X suspect.
   - B – setting expected to produce BOB results
   - W - setting expected to produce WOW results
2. Establish a basis for ranking the results. What's the Mone(Y)?
3. Determine allowable risk.
4. Select a random build and test order that minimizes risk.
5. Rank results on the Mone(Y) and evaluate outcome. *Overlap* of ranks is *permitted* by this test.
6. Results from B vs W must show the full range of variation from your product distribution.
7. Document conclusions.

Tukey B vs W Test

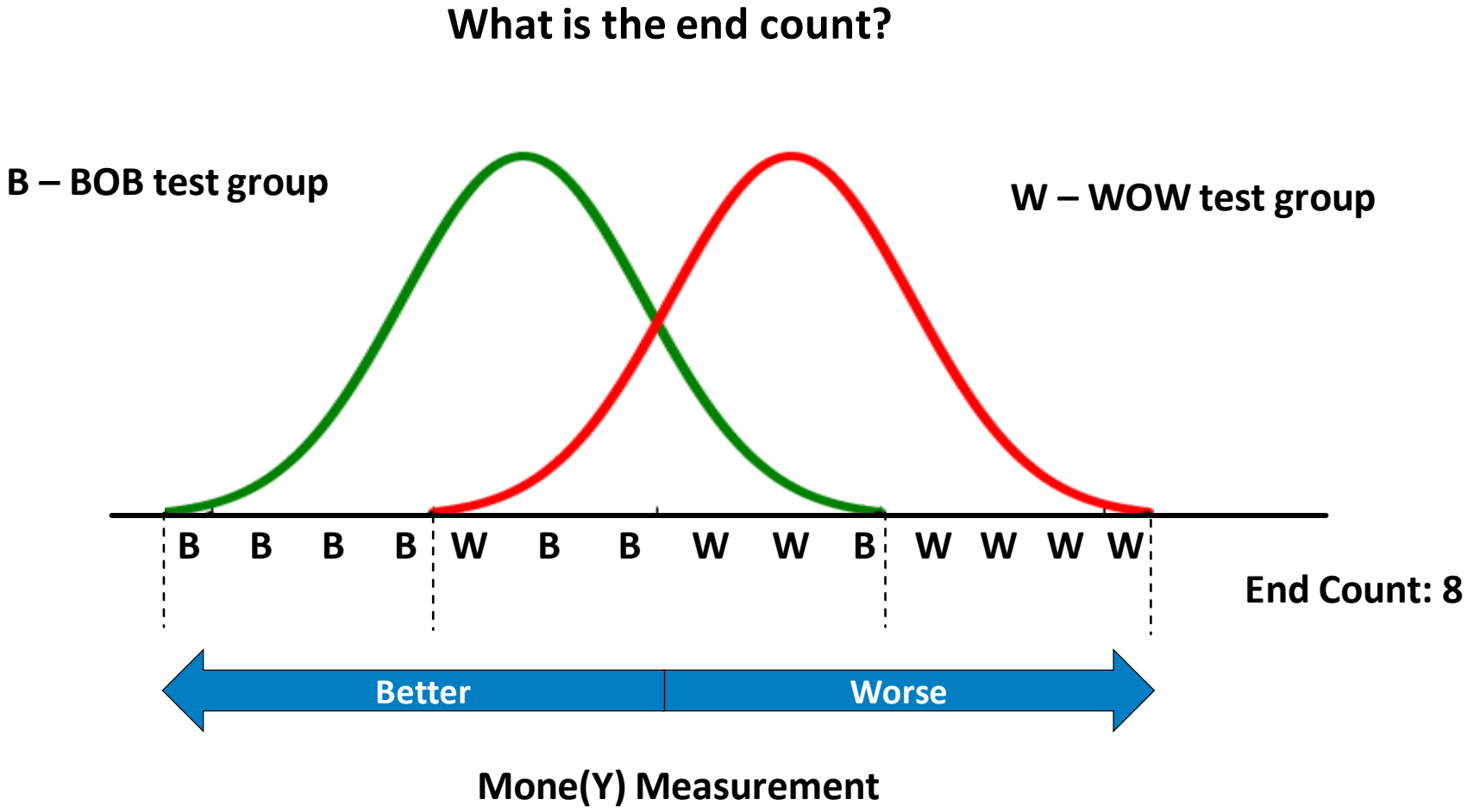


Table for Tukey B vs W and B vs C Two Tailed Test

Desired Confidence	α-Risk	Minimum Sample Size (n) for B & W*	Min. End Count
0.999	0.001	n ≥ 5	13
Super Critical			
0.99	0.01		10
Critical			
0.95	0.05		7
Important			
0.90	0.10		6
Moderate			

*Ratio of B: W or W: B is no worse than 3:4

(X_B → BOB and X_W → WOW, not just different)

Overlap of ranks is permitted by this test.

Tukey B vs W Test – Pipe Length

X_B: Pipe length : 306 – 309.3 mm

X_w: Pipe length: 322.7 – 326 mm

Response: Fuel sender noise [db]

Risk: 5 %

Required End Count: 7

Conclusion: Test passed. Shorter pipe length is the BIG X for fuel sender noise. There is no more than a 5 % risk that we have been fooled by these data.

Run Order

Sample	Pipe length [mm]	Fuel sender noise [dB]
B1	306.3	28
B3	306.0	37
W1	327.7	36
B5	307.6	29
W2	324.7	44
B2	307.2	32
W5	326.3	45
W3	323.4	43
W4	325.0	35
B4	306.9	29

Rank Order

Sample	Crank Dm [mm]	Fuel sender noise [dB]
B1	306.3	28
B5	307.6	29
B4	306.9	29
B2	307.2	32
W4	325.0	35
W1	327.7	36
B3	306.0	37
W3	323.4	43
W2	324.7	44
W5	326.3	45

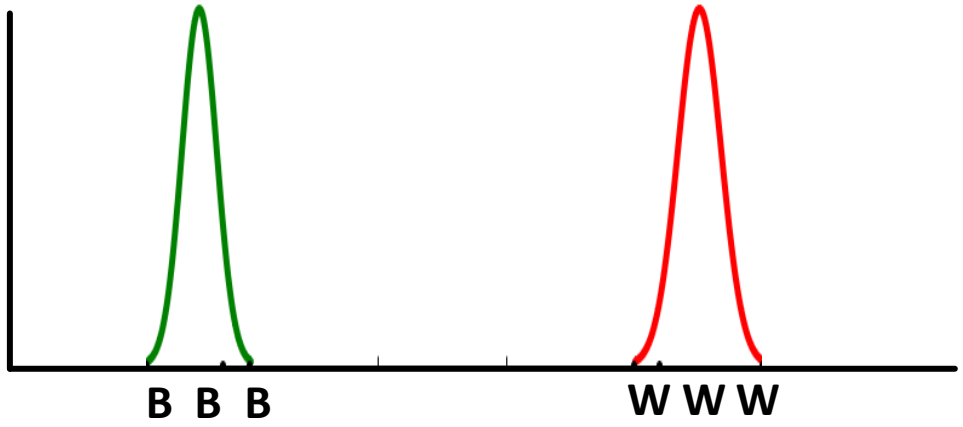
End Count: 7

Additional notes to remember when performing a B vs W Test, Tukey B vs W, …

- Parts used in Group/Paired Comparisons define BIG X distributions. However, these parts can not be used for the B vs W test.
- BOBs und WOWs are new parts (“virgin”) that have never been assembled before.
- The only thing that you know about the Bs and Ws is the BIG X magnitude.
- Always rank test results based on the Mone(Y) from BOB to WOW.
- Use normal production process.
- Requirement to pass:
  - Test must meet or exceed the required minimum end count
  - Test must have full range of product variation

Determining Sample Size (n)

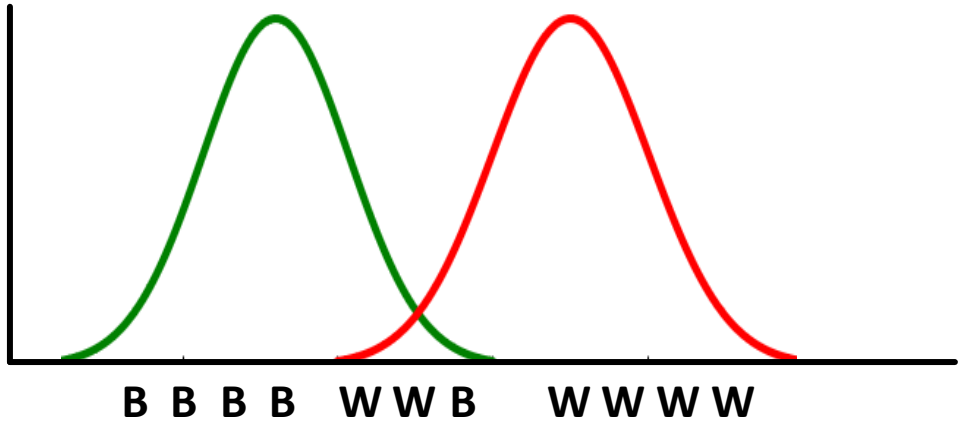
Strong – BIG X Candidate



Six Pack B vs W

→ BIG X is active
Small Sample Size of BOBs and WOWs

Weaker – BIG X Candidate



Tukey B vs W

→ Small X is active
Use larger Sample Size of Bs & Ws

Single Factor Experimental Designs

- **B vs W 6 Pack:** Used to confirm a BIG X candidate drives the Mone(Y) from the BOB to WOW limit. Need three samples each at the B and W settings. No overlap allowed.
- **B vs C 6 Pack:** Used to determine if an *alternate design or process* produces an improved Mone(Y) distribution. Need three samples each at the B and C designs. No overlap allowed.
- **Tukey B vs W 6 Pack:** This test is performed when clue generation has identified a single BIG X candidate, and a potential Small X candidate is present. This test only confirms the BIG X candidate. Need three samples each at the B and W settings. Overlap allowed.
- **Tukey B vs C 6 Pack:** This test is performed when clue generation has identified a single BIG X candidate, and a potential Small X candidate is present. This test only confirms the BIG X candidate, if an *alternate design or process* produces an improved Mone(Y) distribution. Need three samples each at the B and W settings. No overlap allowed.

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BIG X Confirmation

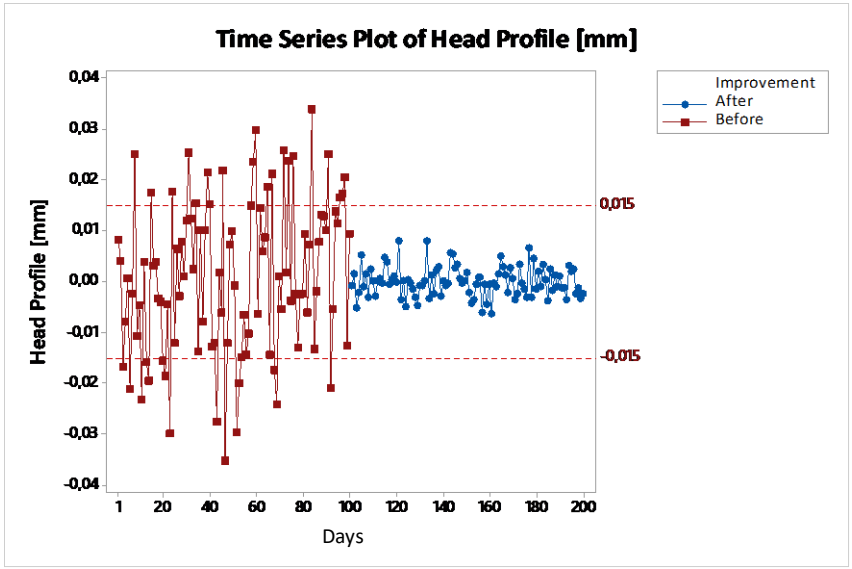
Options for confirming a BIG X suspect:

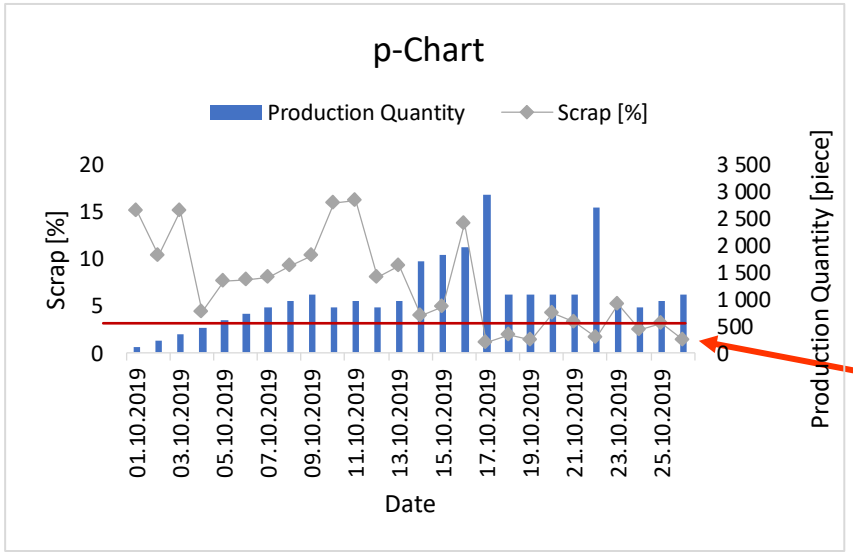
- We must confirm the BIG X suspect.
- Statistical tests assess the risk that we've been fooled.
- A 6 Pack test confirms the BIG X candidate with a 5 % risk.
- The allowed risks should depend upon the consequences of being fooled.

The effectiveness of the corrective action(s) over time for the BIG X candidate may also i. a.:

- Mone (Y) Run Chart
- p-Chart

Proving the Corrective Action





Mone(Y) Run Chart …

… shows the performance of the Mone(Y) that we used in our project.

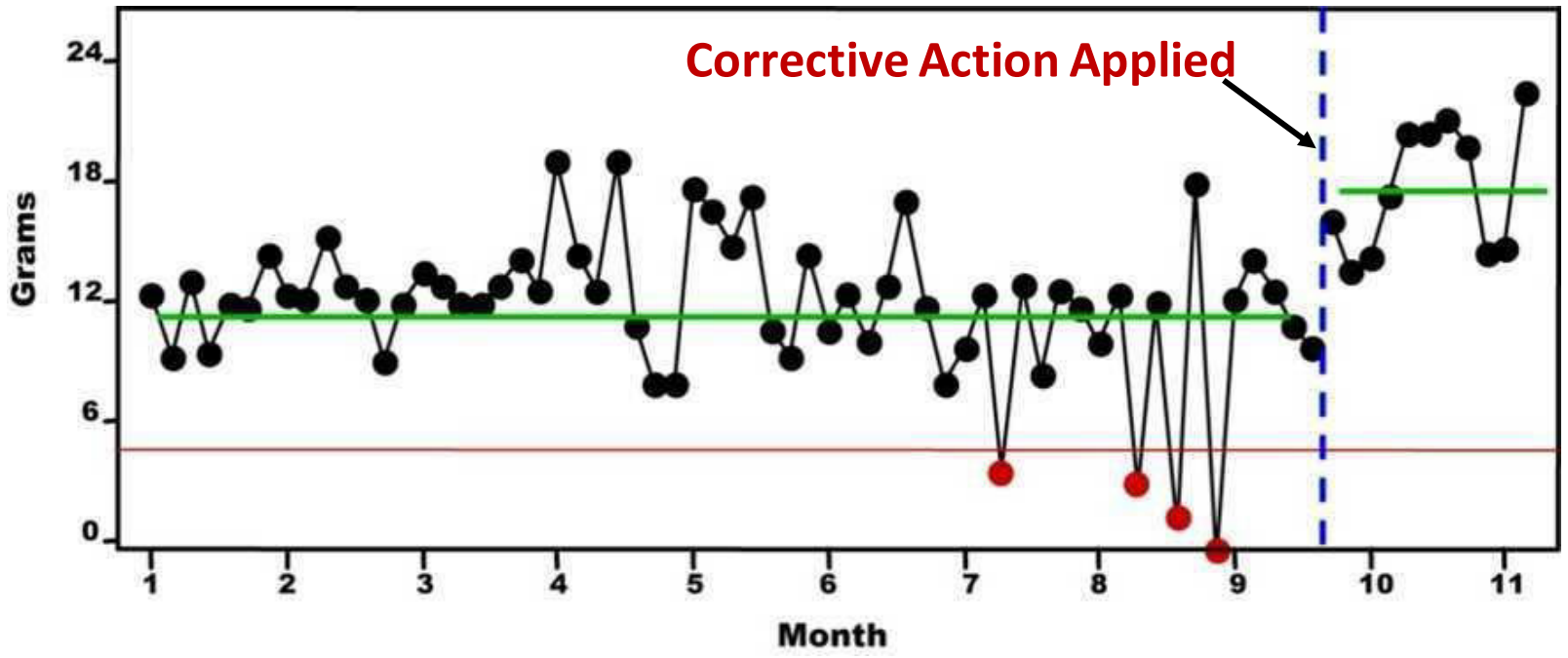
p-Chart…

…shows the failure rate over time.

There might be another cause for the leakage (2. Root cause – Small X_1)

Proving the Corrective Action

Mone(Y) Run Chart - Example 1



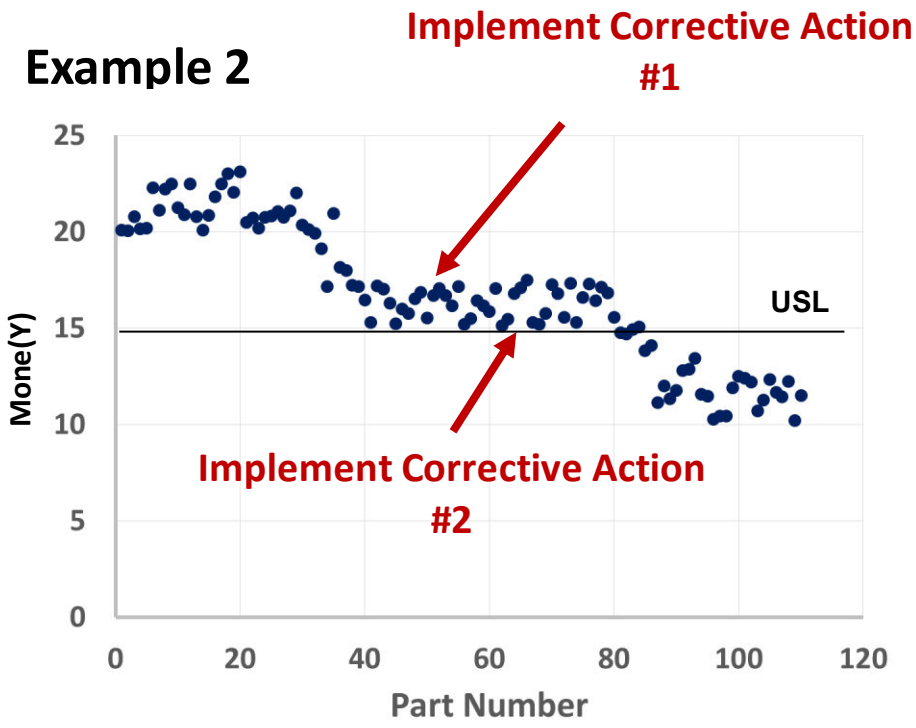
Are the improvements working?

Conclusion: Increased part strength and reduced variation. The change was effective.

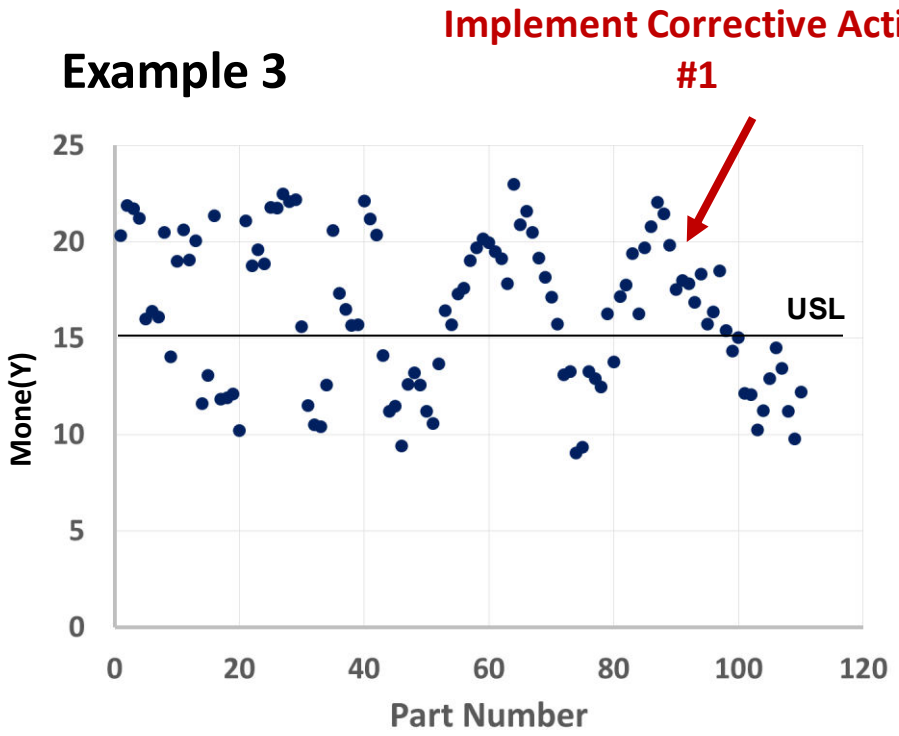
Proving the Corrective Action

Mone(Y) Run Chart – More Examples

Example 2



Example 3



Are the improvements working?

Example 2: No, improvements in the Mone(Y) do not correlate with corrective actions.

Example 3: No, we had “good times” like that before.

Module 7 Summary

Confirmation Testing I

- Confirmation is based upon correctly predicting, a number of times, that extremes in X cause BOB and WOW results in Y.
- The number of correct predictions required is only a function of the consequences of the confirmation being false.
- The correct settings for the extremes of X are critical to the conclusions - actual values, not possible or designed values, must be used.
- A lack of randomization, which fails to ensure uncontrolled Xs vary naturally, is a major source of false confirmations.

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8 Tolerancing / Finishing Projects I

Module 8 Objectives
Tolerancing / Finishing Projects I

Understand:

- How to describe the BIG X - Mone(Y) relationship
- How to develop and apply a corrective action
- How to leverage the insights gained from a successful project

Be able to:

- Determine an appropriate tolerance for the BIG X using Tolerance Parallelogram
- Demonstrate improved system performance
- Document project value
- Leverage the knowledge gained to other applications and organizational development

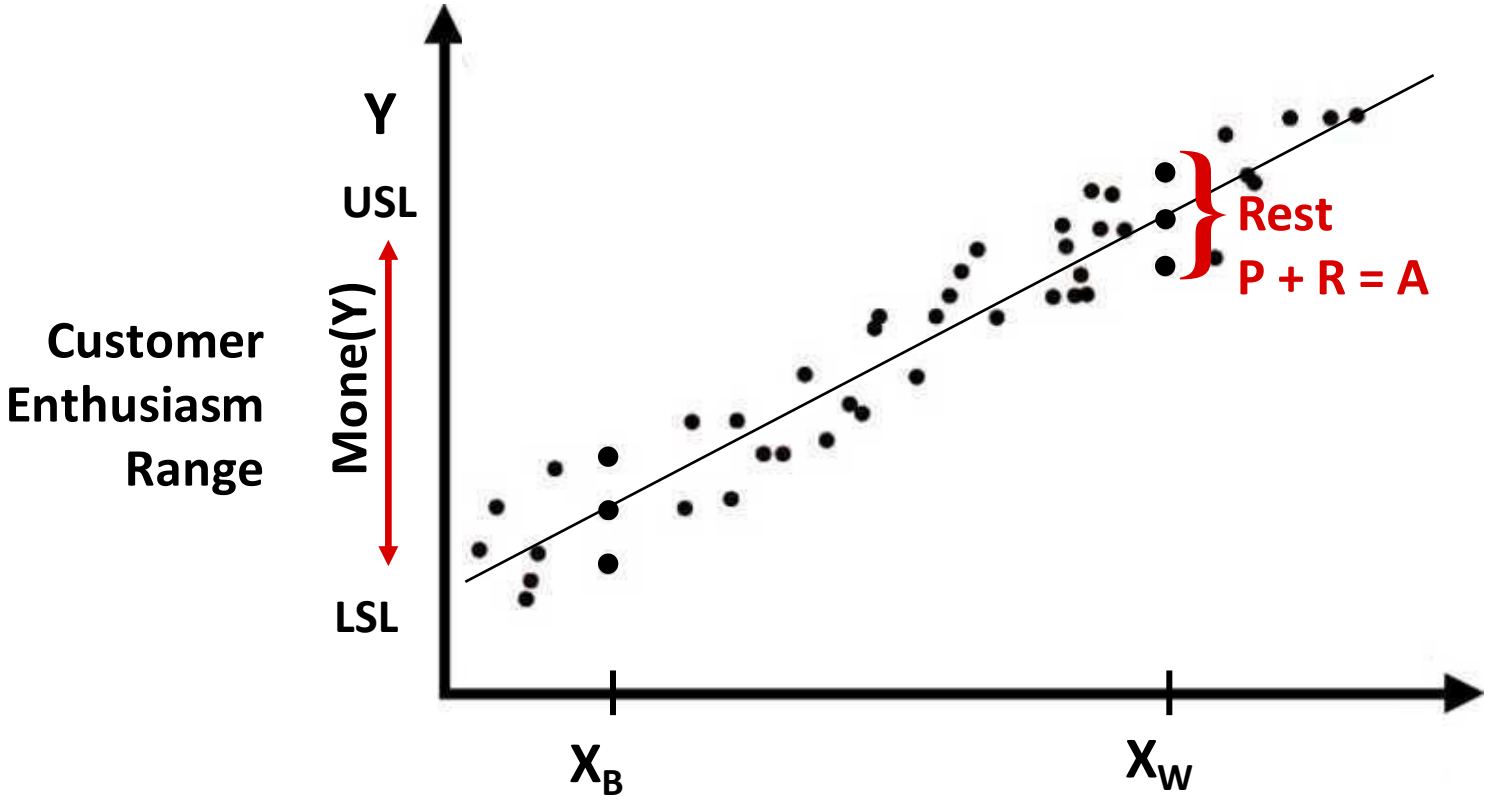
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Tolerancing

- To control the Mone(Y) we must control the BIG X.
- The purpose of a Tolerance Parallelogram is to establish the tolerance of a single BIG X, which will produce Mone(Y) values that are consistent with customer enthusiasm.
- The Mone(Y) is the output which has been the focus of the investigation. The customer enthusiasm range is the range of allowable variation within which the customer perceives no meaningful difference in product performance.

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Scatter Plot



Product BIG X Feature

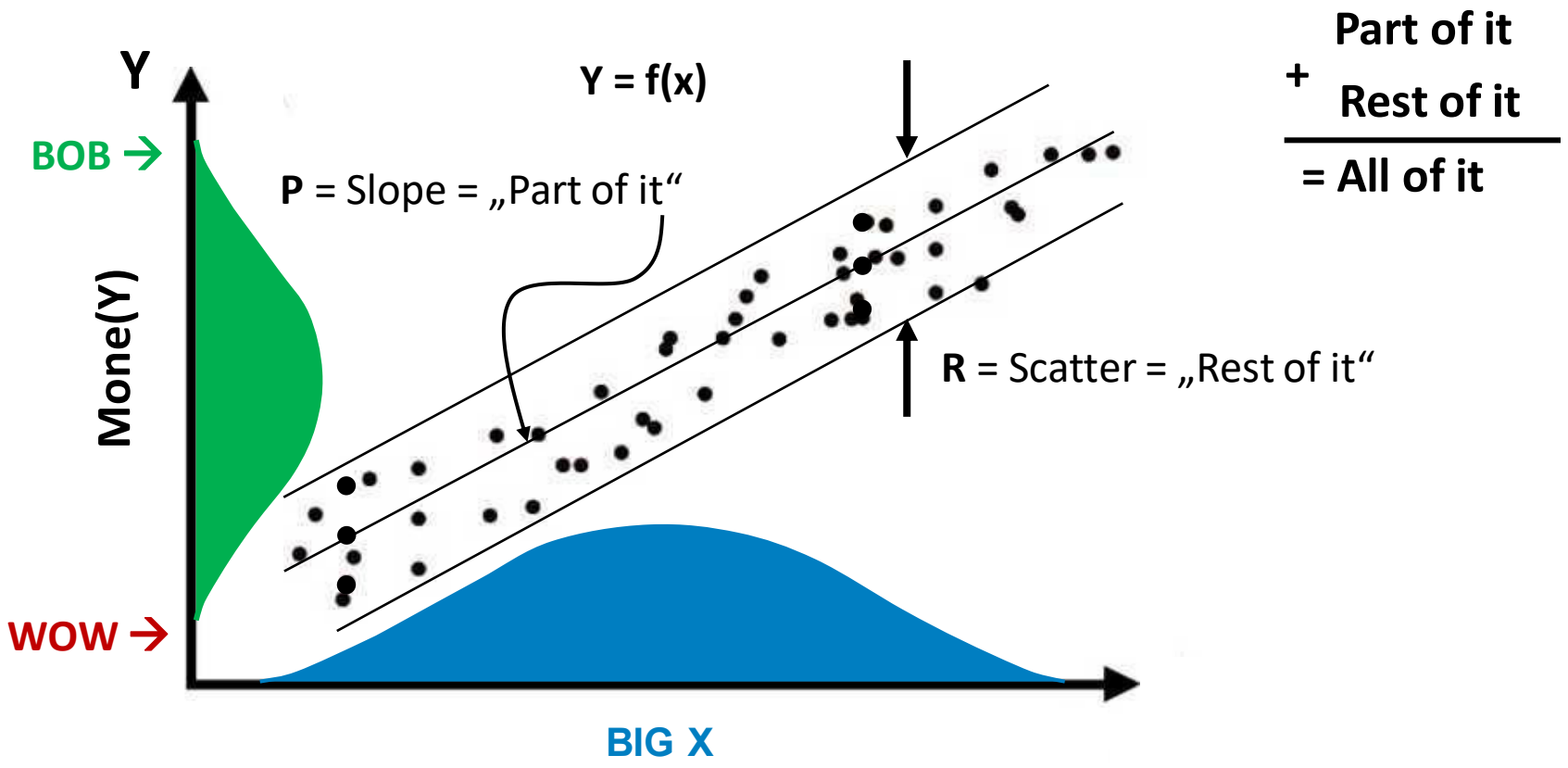
Does Y = f(X)?

What is the tolerance on X?

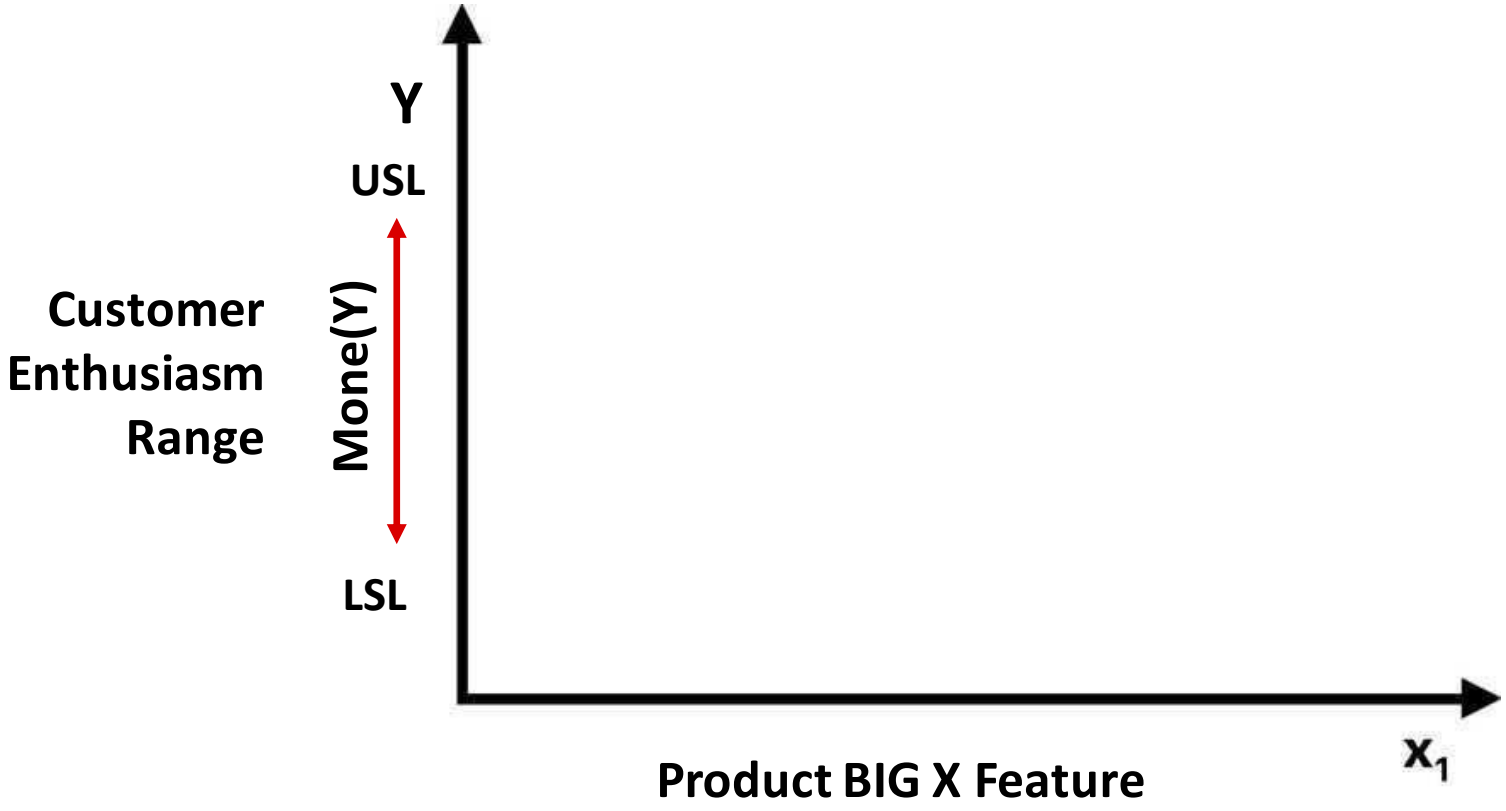
X-Y - Scatter Plot: P + R = A

Slope: The slope, shown as **P**, shows the effect the X has on the Y.

Scatter: The vertical scatter, shown as **R**, is the effect of everything other than that which is on the axis.

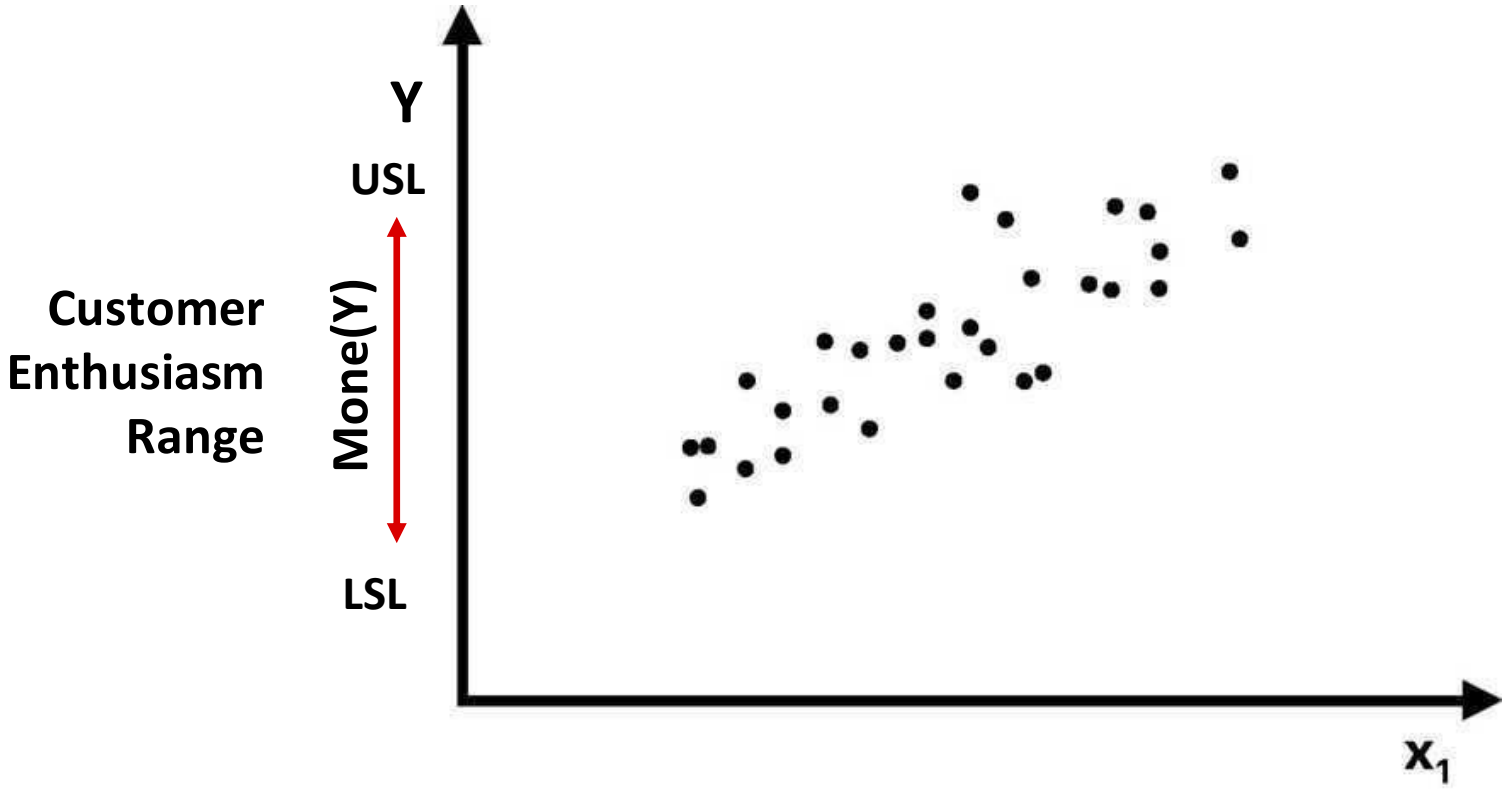


Establish Mone(Y) and BIG X Axis Using the Range of Values From the B vs W



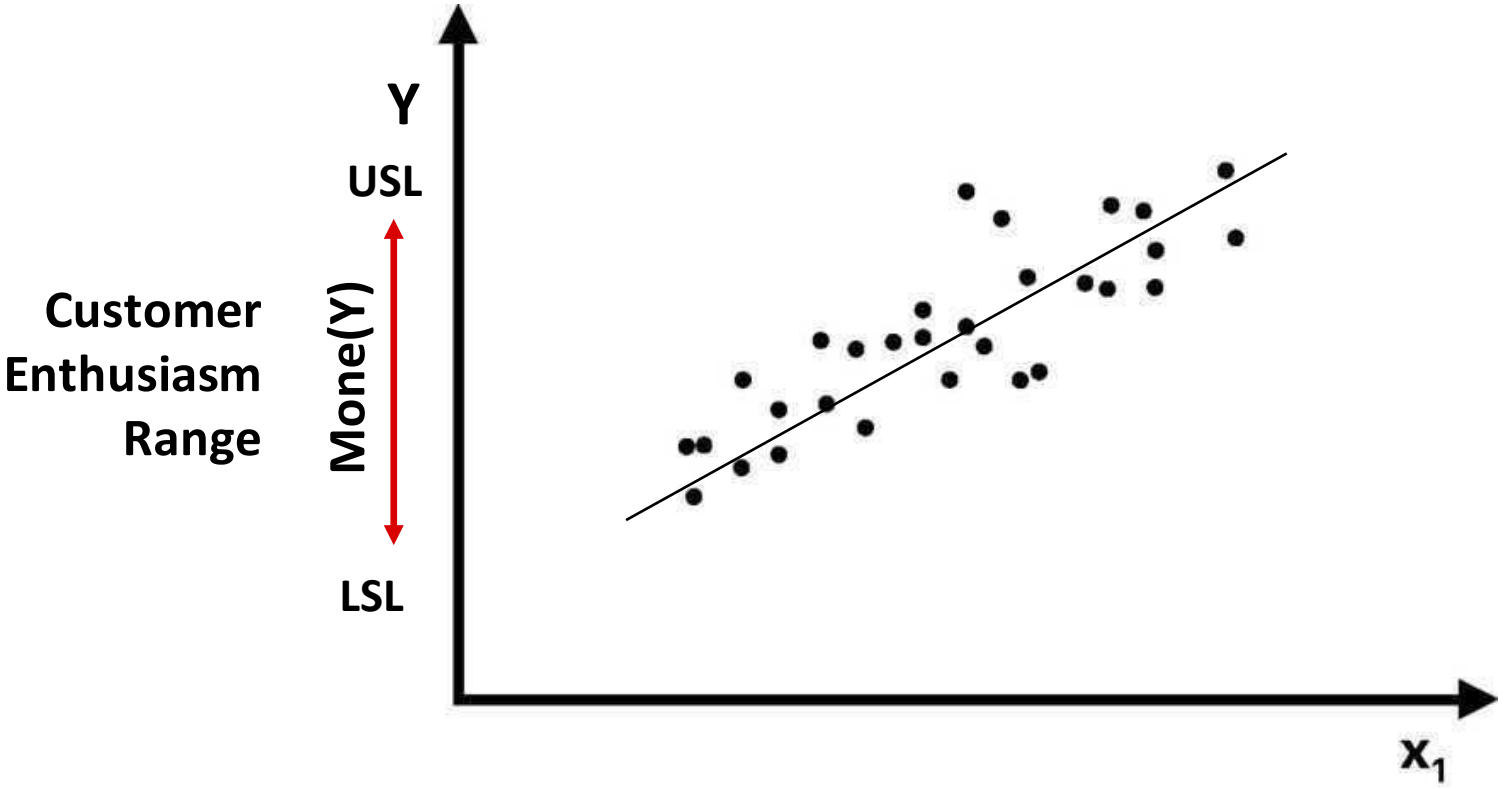
Tolerance Parallelogram

Plot 30 Random Samples* of Parts Made to the Current Process Limits As Defined by the Levels of X From the B vs W.



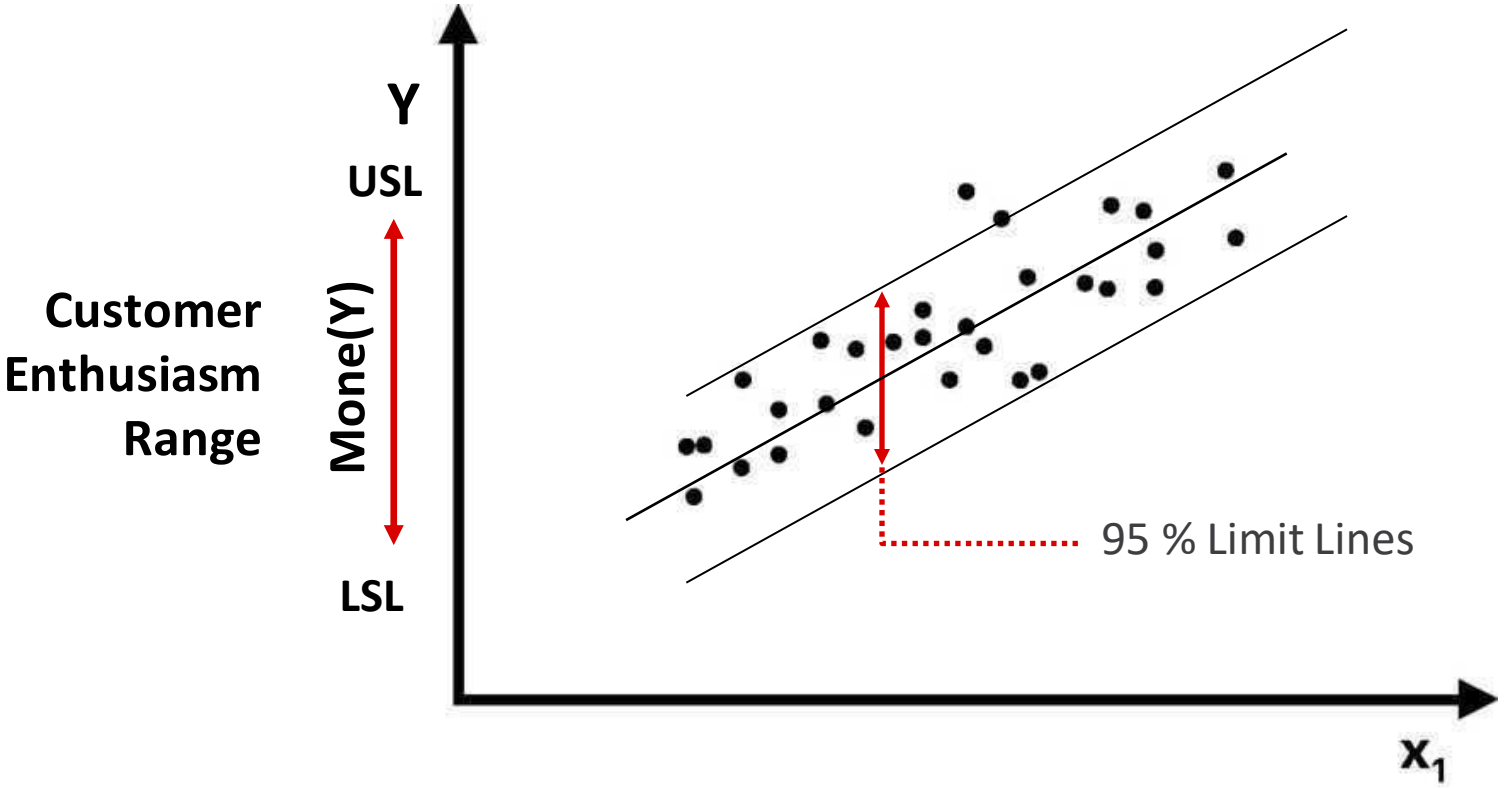
*** Or 24 random samples, plus 6 from a B vs W.**

Draw the Regression Line



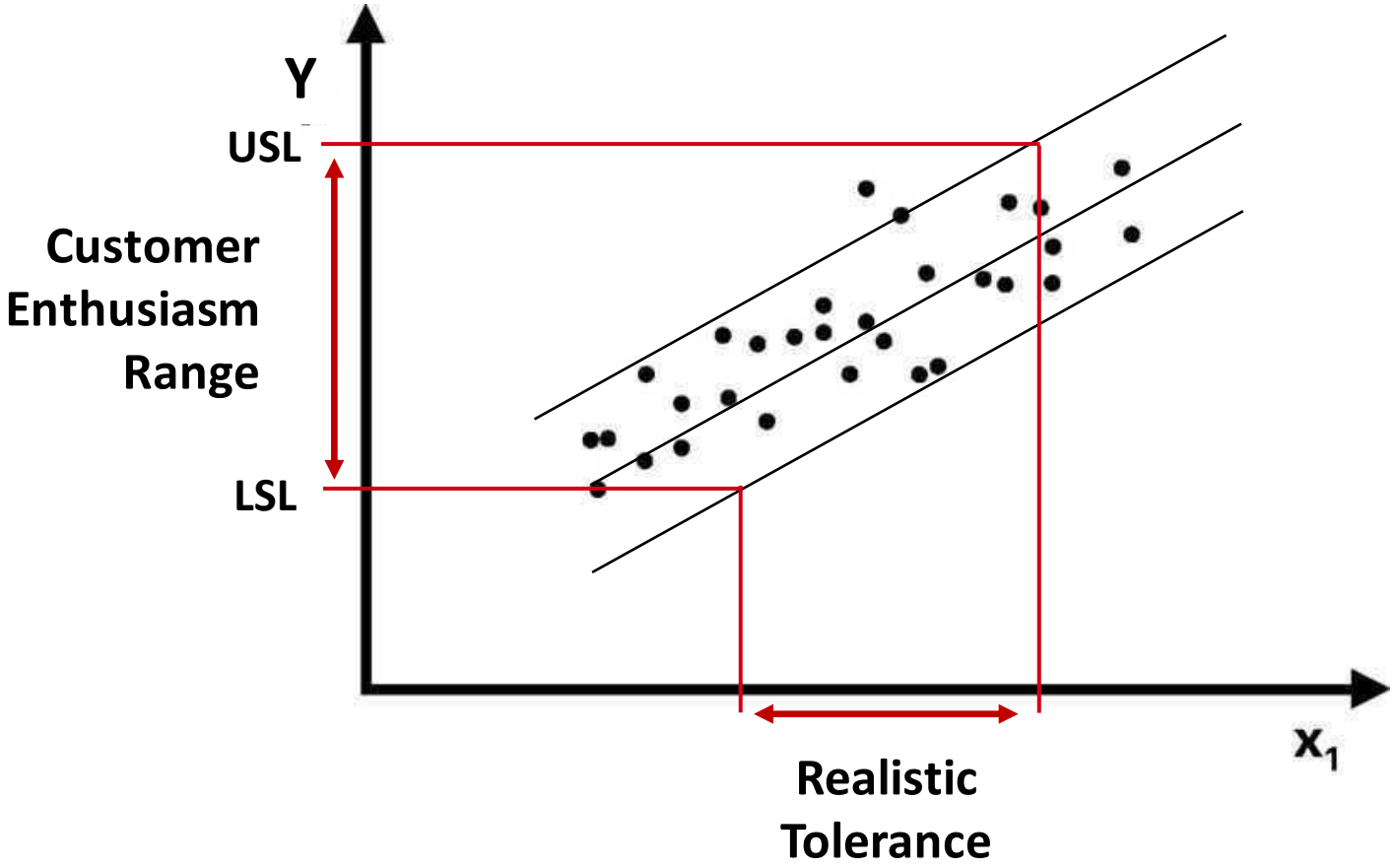
Create Limit Lines

Draw limit lines to encompass 95 % of total effects of factors other than X.



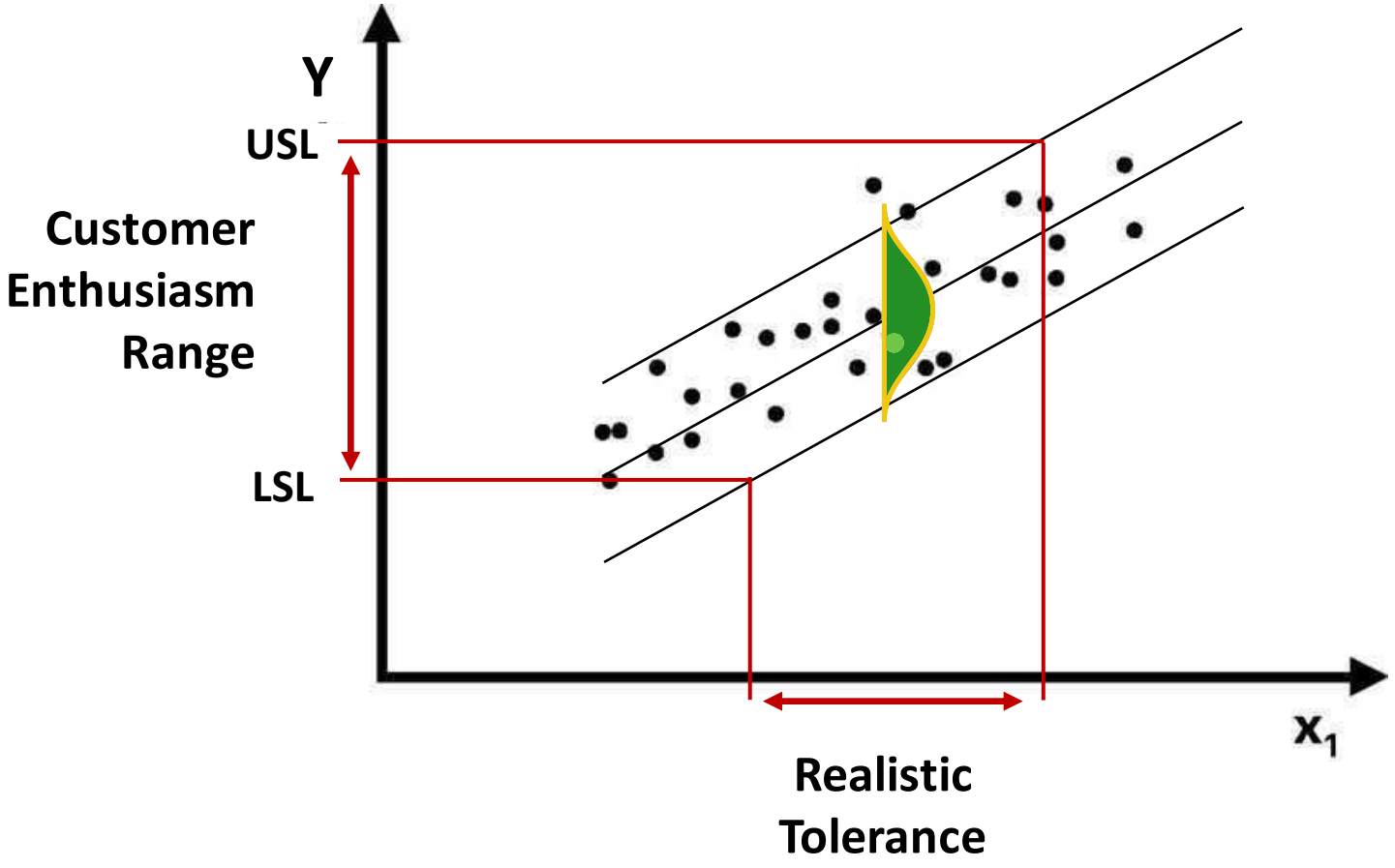
Mark Customer Enthusiasm Range

Draw Vertical Lines From Intersection Points to X Axis to Establish Realistic Tolerances.



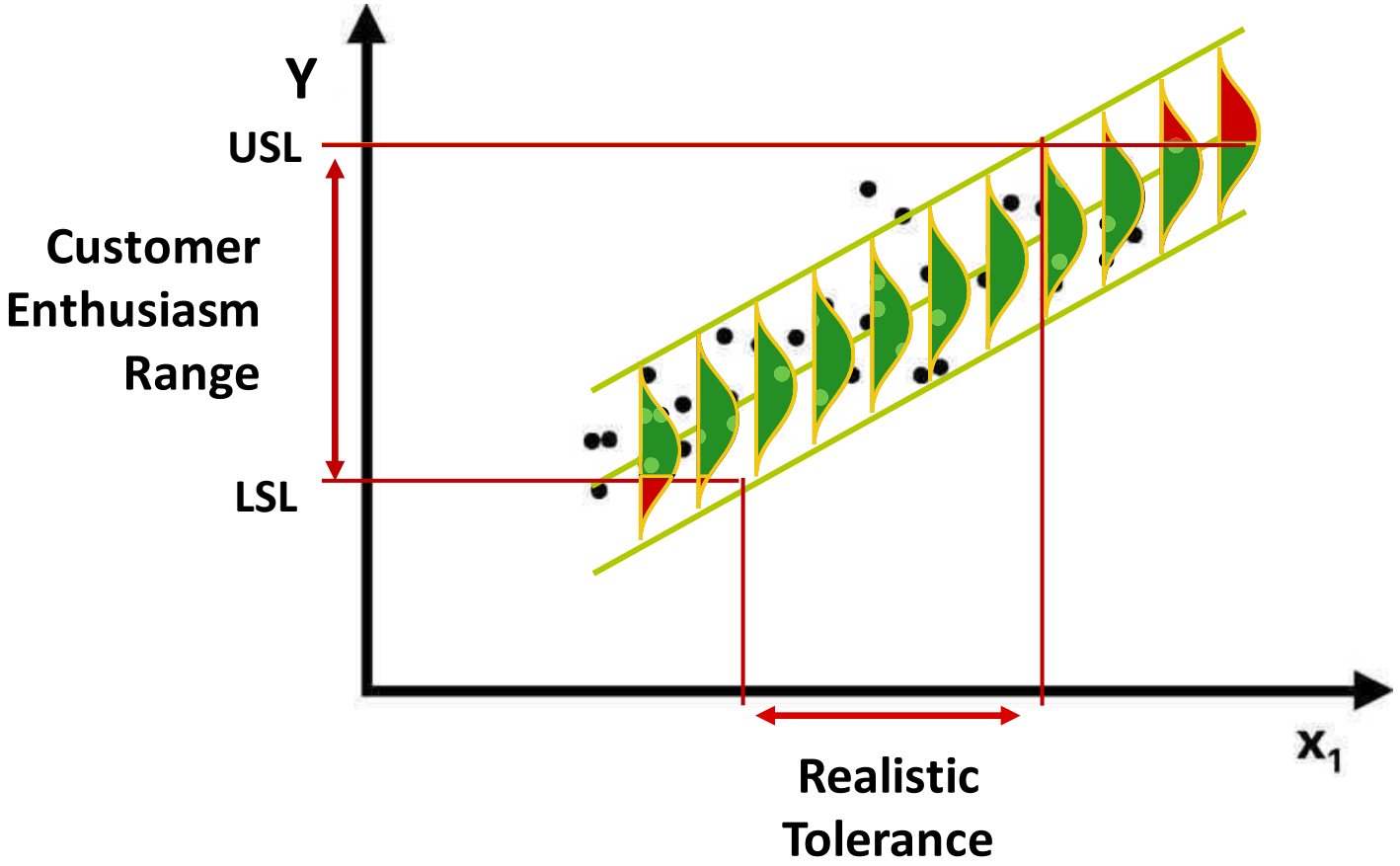
Mark Customer Enthusiasm Range

Draw Vertical Lines From Intersection Points to X Axis to Establish Realistic Tolerances.



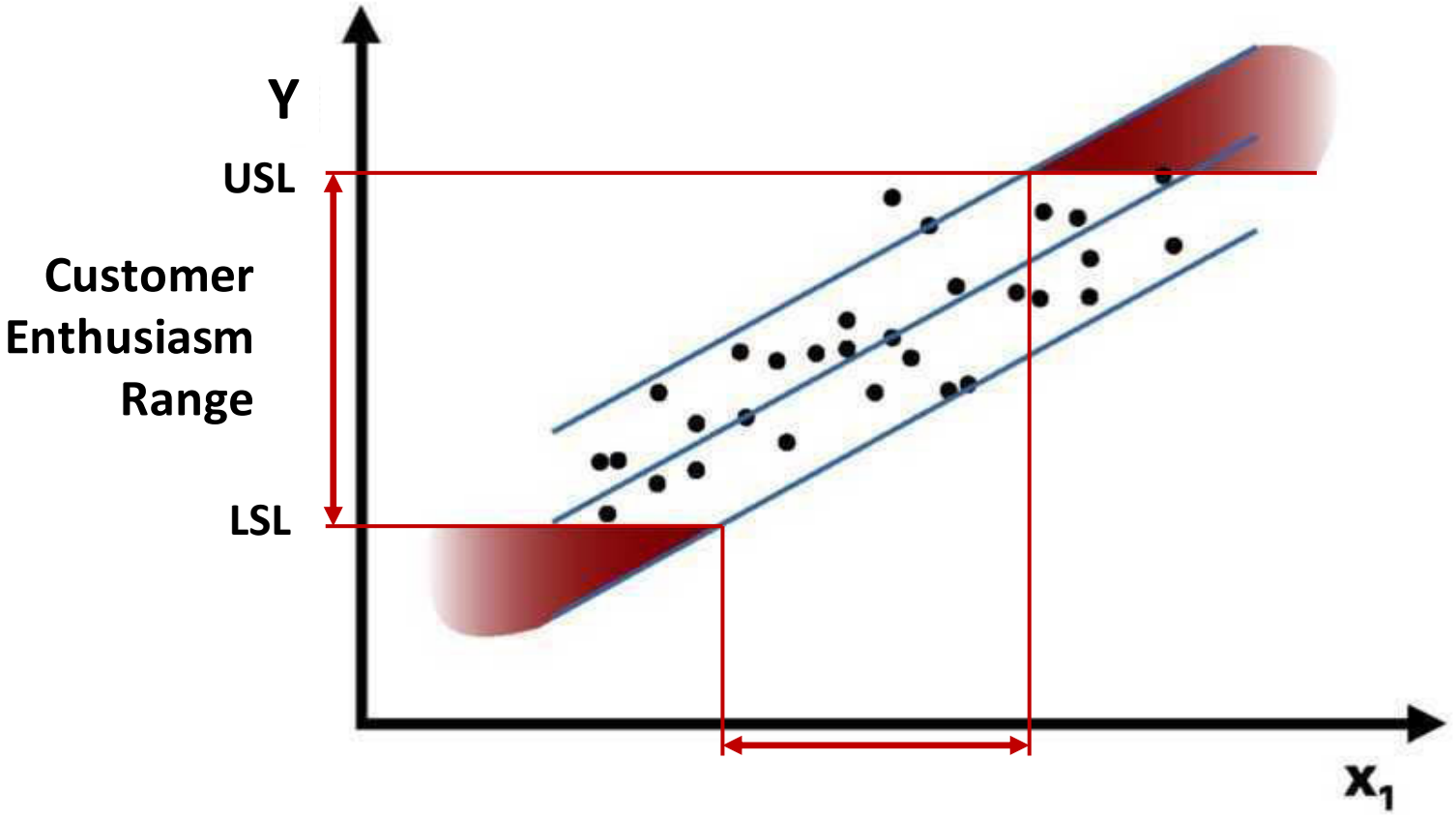
Mark Customer Enthusiasm Range

Draw Vertical Lines From Intersection Points to X Axis to Establish Realistic Tolerances.



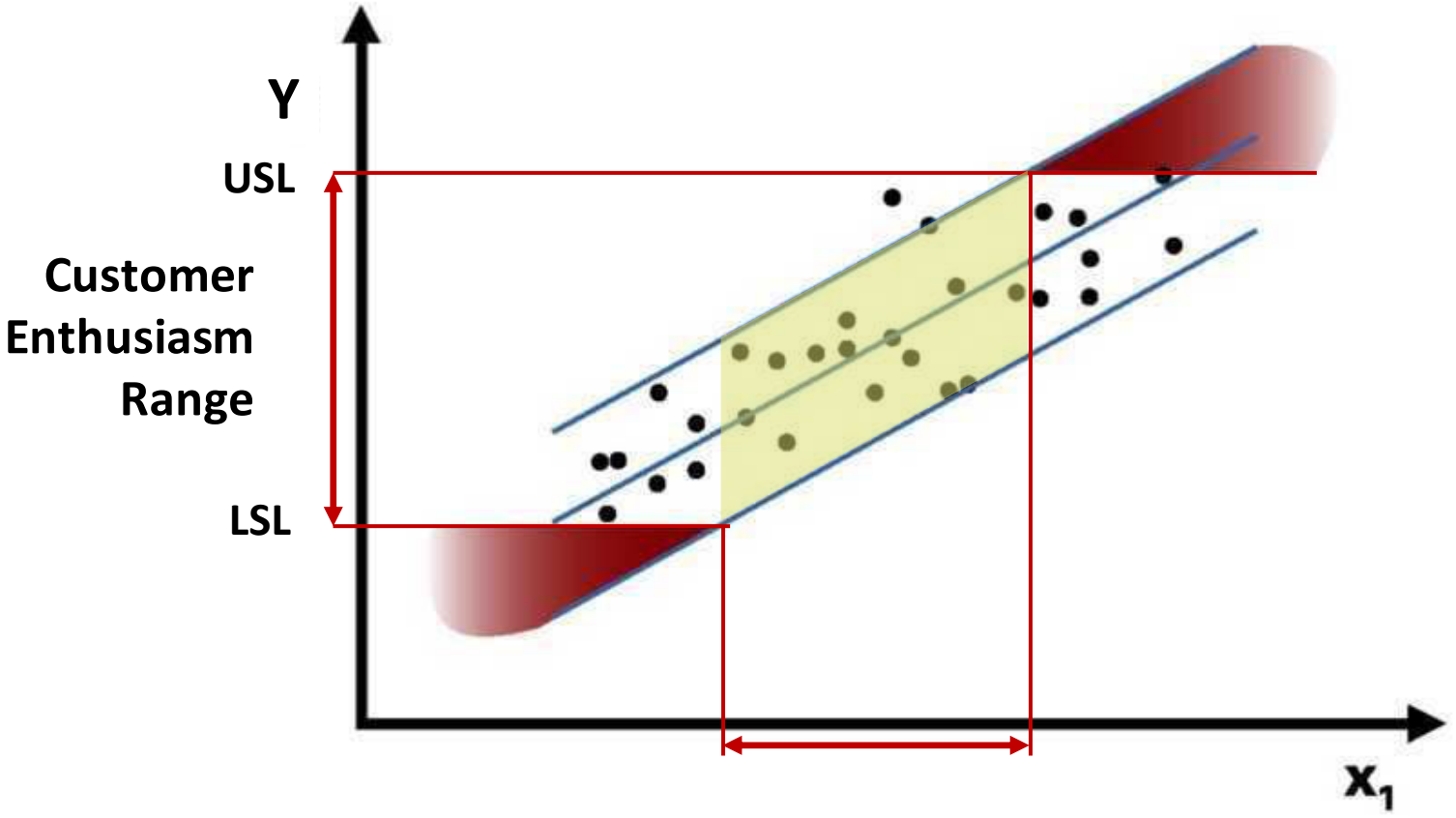
Determine Realistic Tolerances

Draw Vertical Lines From Intersection Points to X Axis to Establish Realistic Tolerances.

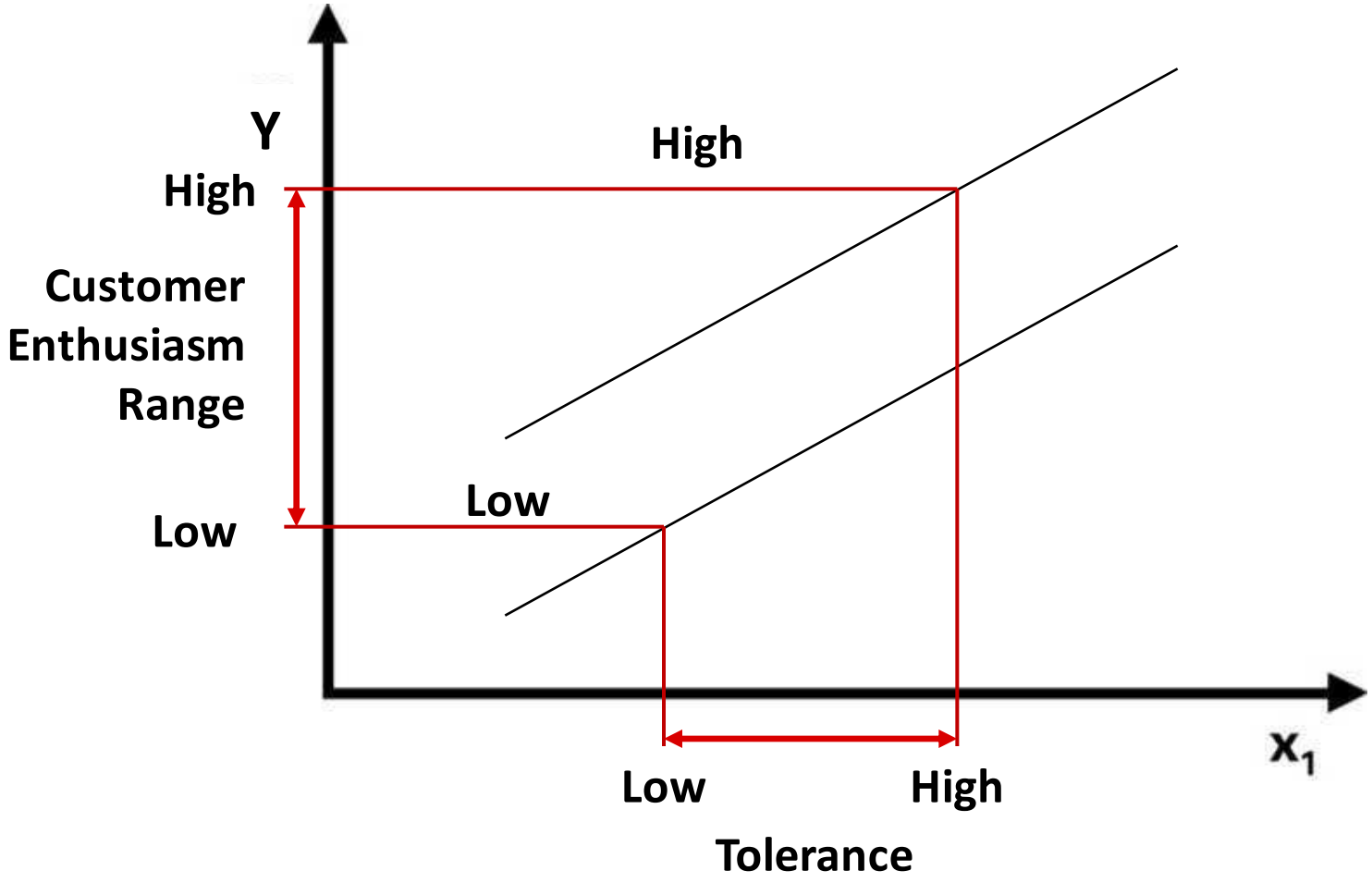


Determine Realistic Tolerances

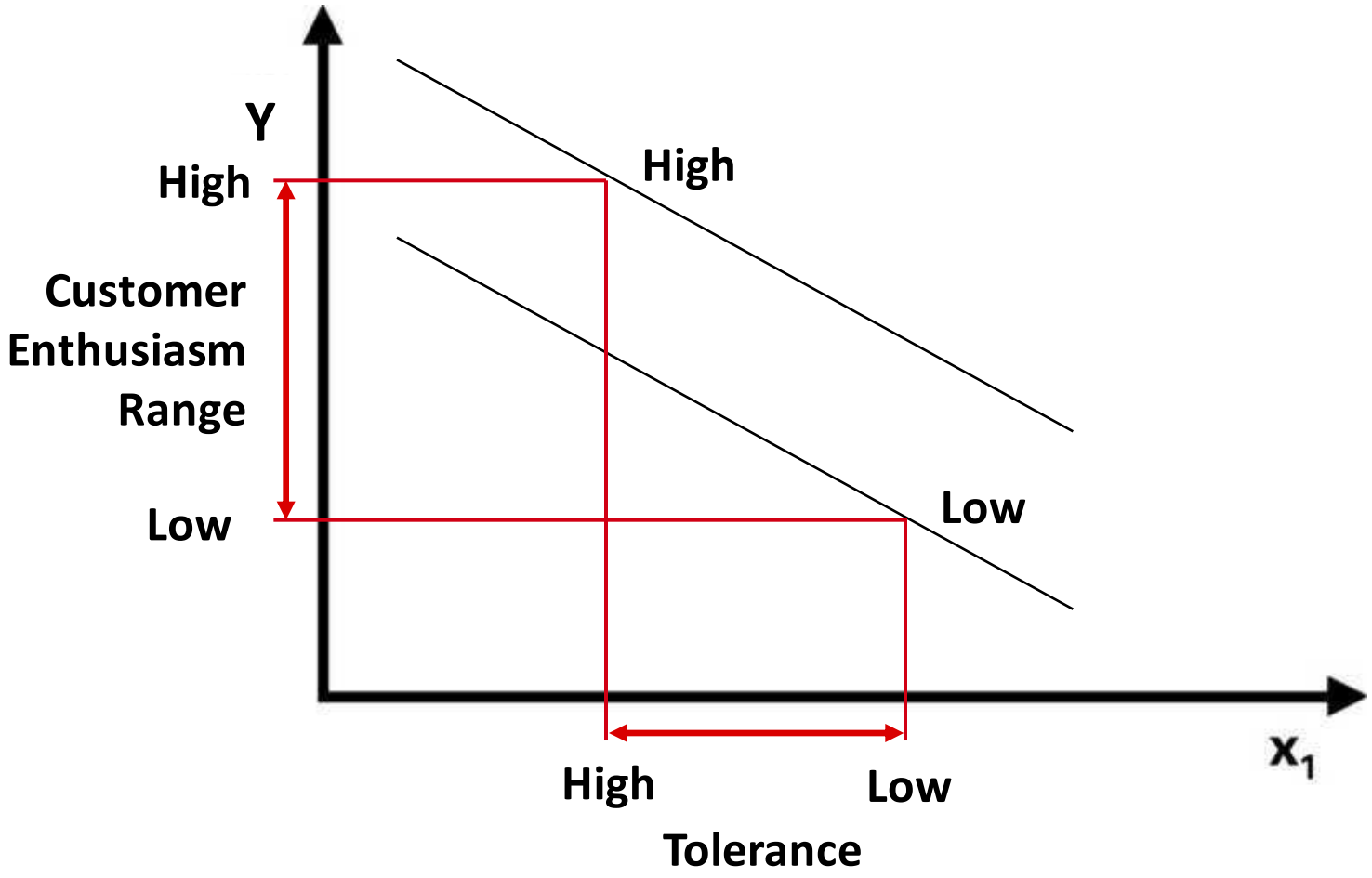
Draw Vertical Lines From Intersection Points to X Axis to Establish Realistic Tolerances.



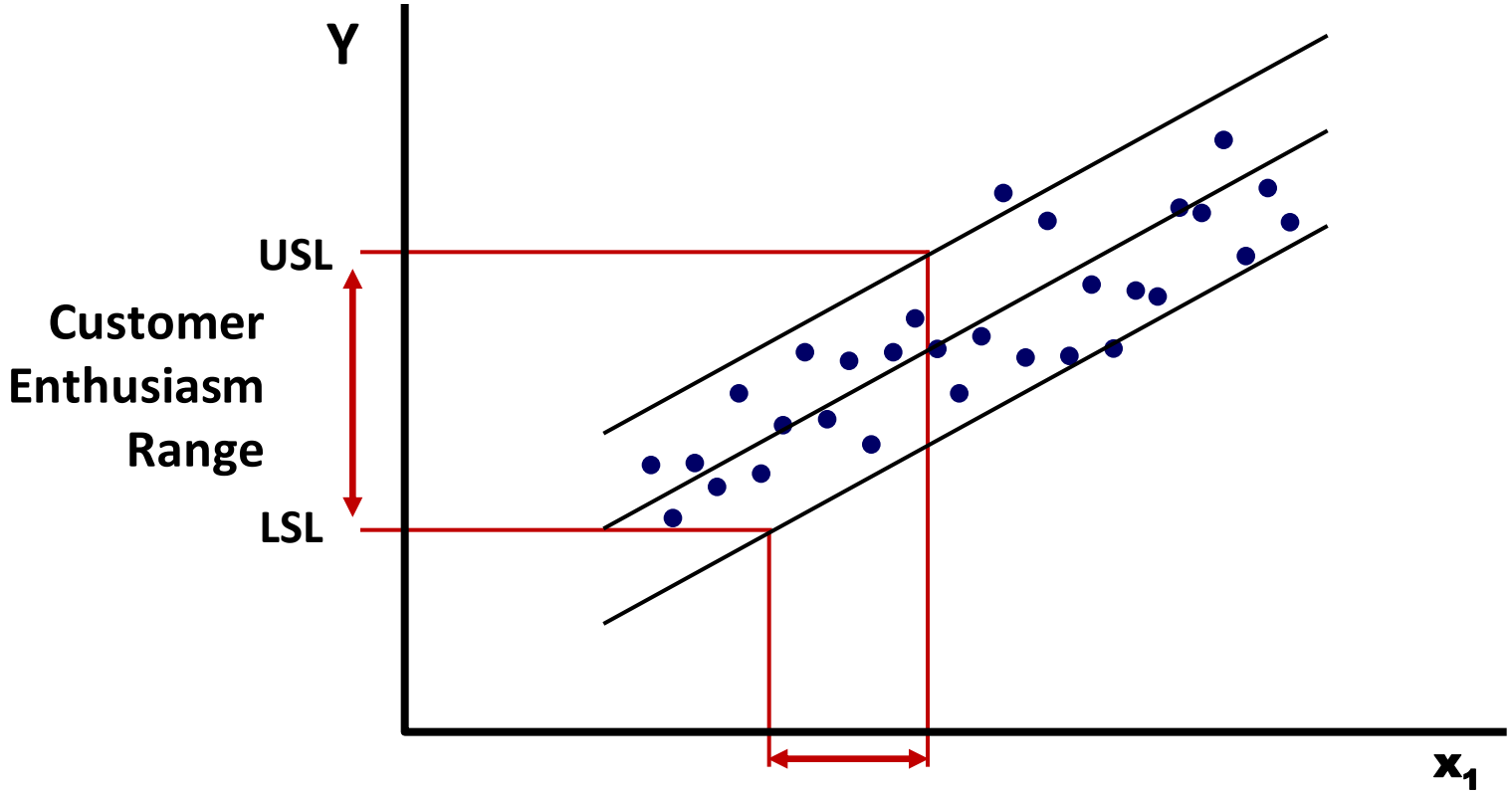
Tolerance Parallelogram



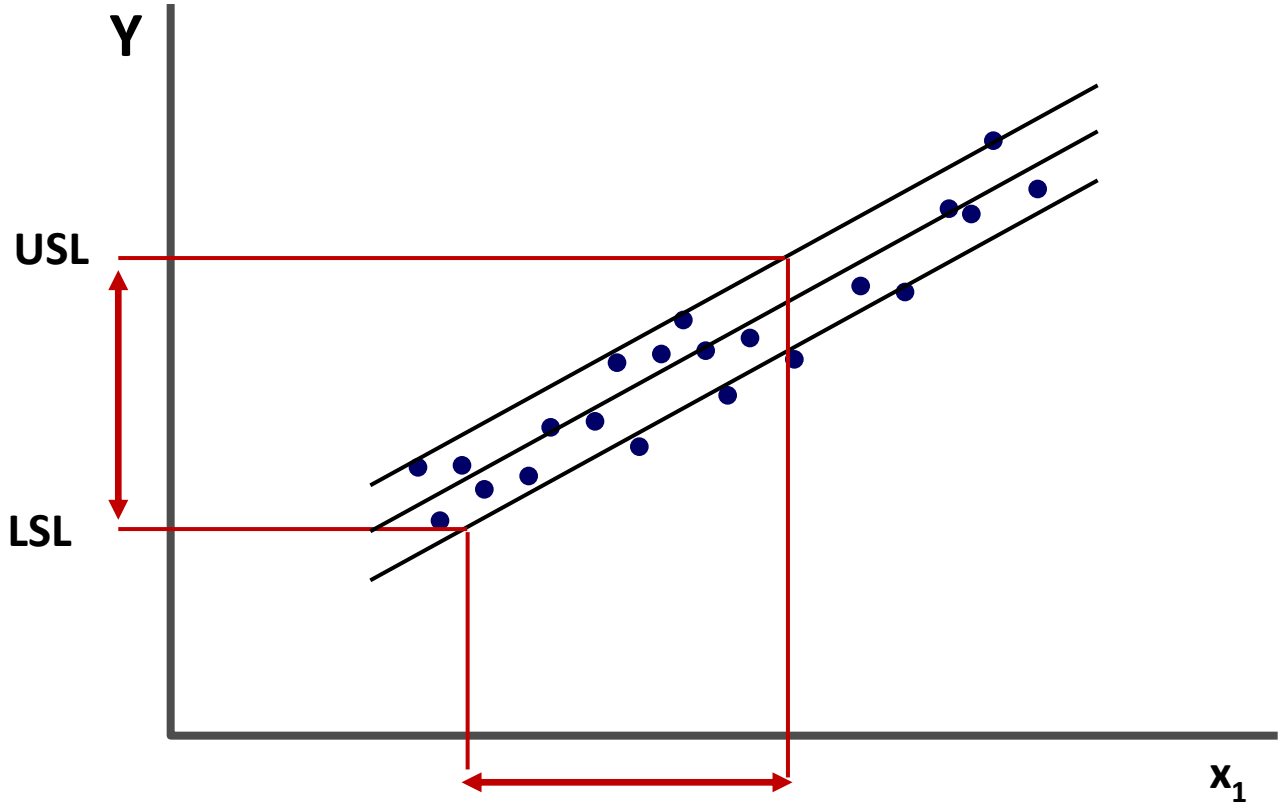
Tolerance Parallelogram



Too Much Scatter Yields Too Little Tolerance



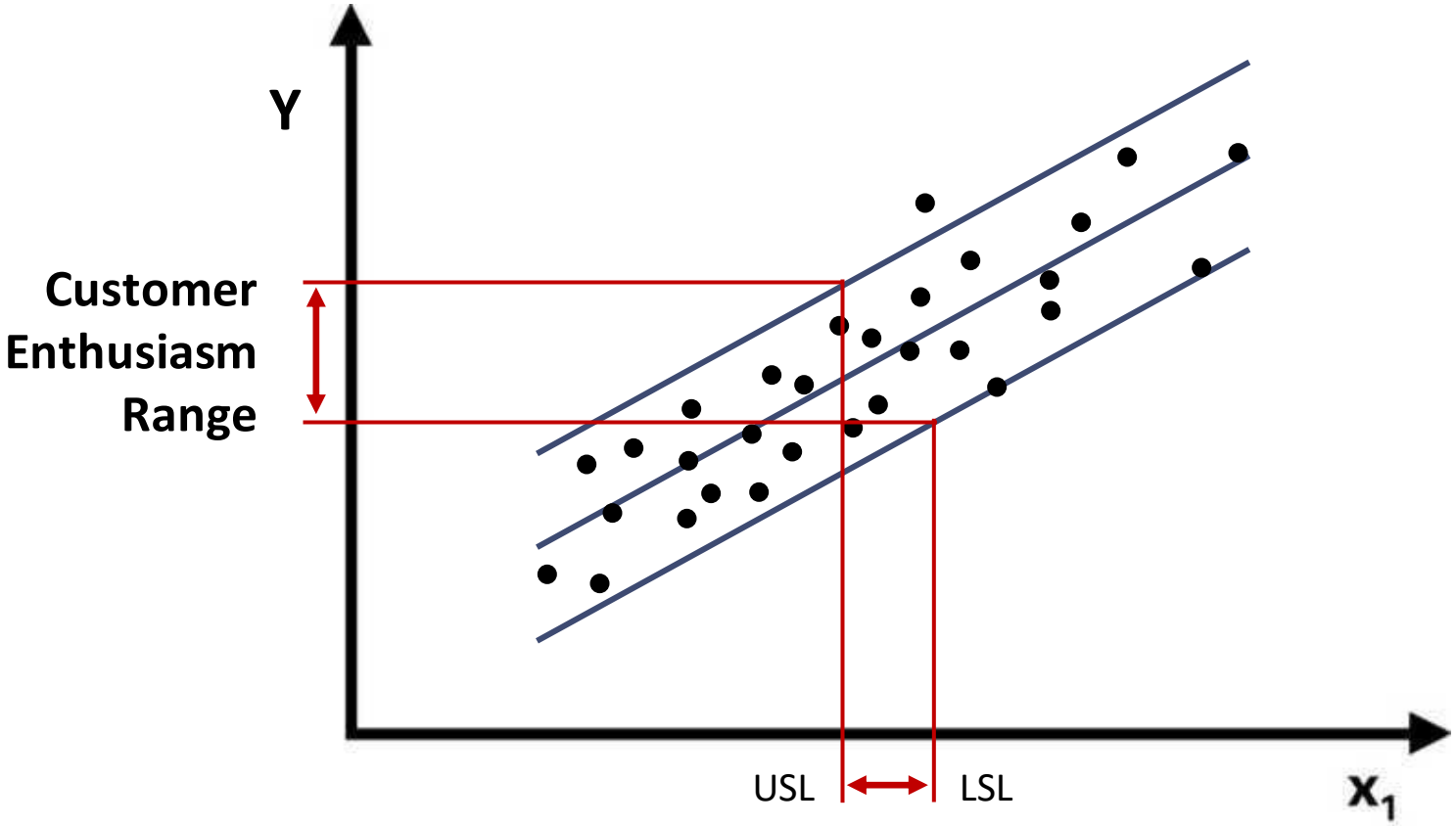
Too Much Scatter Yields Too Little Tolerance



X under Control

… or to no Tolerance

USL < LSL!



"Realistic"
Tolerance

Tolerance Parallelogram

Purpose:

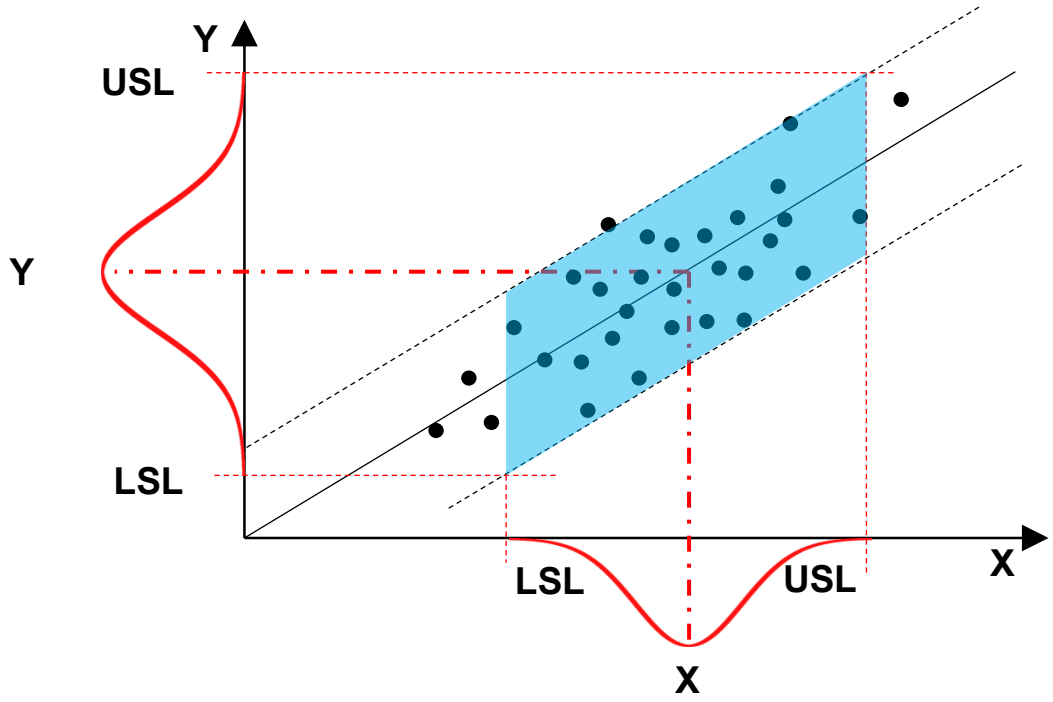
The purpose of a Tolerance Parallelogram is to establish the tolerance of a single BIG X, which will produce Mone(Y) values that are consistent with customer enthusiasm. The Mone(Y) is the output which has been the focus of the investigation. The customer enthusiasm range is the range of allowable variation within which the customer perceives no meaningful difference in product performance.

Why is it used?

- To illustrate the influence of tolerances of individual components / process steps on the tolerances of the final product and to understand.
- To determine meaningful tolerances for the individual components.

When is it used?

- In the Improve Phase to set the tolerances of the influencing components BIG X to reach the target value Mone(Y).



How to create a Tolerance Parallelogram

1. Prior to starting the Tolerance Parallelogram, measurement systems for both the Mone(Y) and BIG X must pass Isograph ($\Delta P/\Delta M \geq 6$).
2. First, the tolerances of the final product are specified. Subsequently, the tolerances of the subassemblies and components are determined.
3. For this the actual values (BIG X) of the subassemblies are measured. Take 30 data sets. Then the final product is assembled out of these 30 sets and the desired output Mone(Y) is measured.
4. This results in a correlation between the input and output variable. The correlation is shown graphically. In the diagram of the final product then the tolerances (LSL, USL for Mone(Y)) are entered on the Y-axis by drawing two lines parallel to the X-axis.
5. Draw the center line of your dot cloud.
6. Two parallel lines with equal distance to the center line are drawn to border the scatter of data excluding the data point with biggest distance to the center line (black dotted lines).
7. The points of intersection of this border lines with the two lines of the control limits results in the lower (LSL) and upper (USL) tolerance limit of the BIG X.

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Tolerance Parallelogram - Exercise

Sweep Angle Case

Concentricity [mm]	Wiper Angle [°]	Concentricity [mm]	Wiper Angle [°]
0,003	83,10	0,121	81,65
0,000	83,00	0,102	81,70
0,006	83,15	0,100	81,75
0,026	83,00	0,096	82,00
0,031	82,90	0,084	82,05
0,035	82,80	0,097	82,05
0,035	82,68	0,053	82,52
0,066	82,60	0,076	82,05
0,032	82,60	0,101	81,95
0,048	82,60	0,059	82,30
0,035	82,60	0,060	82,30
0,049	82,55	0,069	82,40
0,130	81,40	0,055	82,40
0,137	81,30	0,067	82,50
0,048	82,55	0,059	82,41

Create a Tolerance Parallelogram

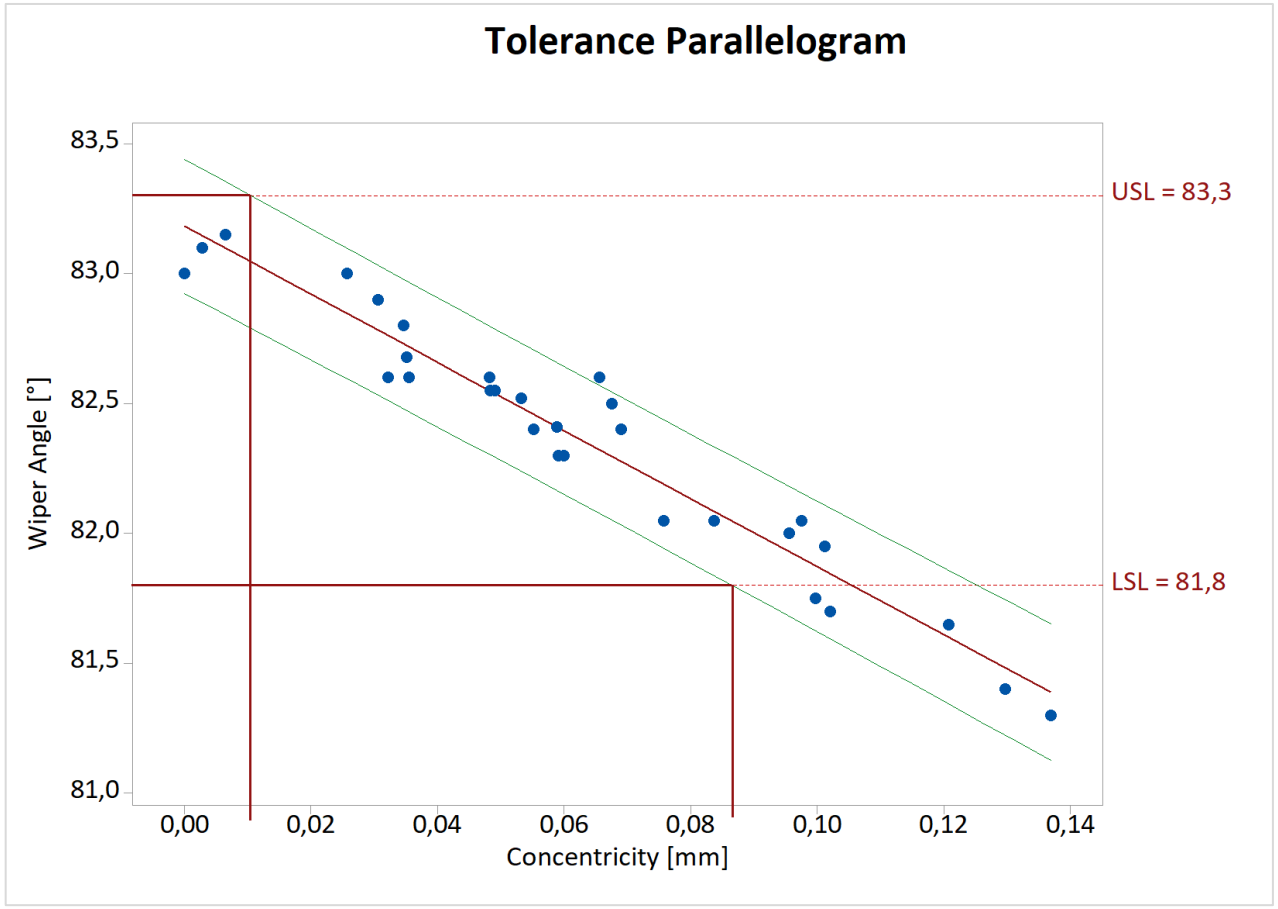
- The Mone(Y) is the wiper angle. The BIG X is the concentricity.
- The customer enthusiasm limits are:

  81.8° – 83.3°

- Input the data in the Tolerance Parallelogram table.
- What is the realistic tolerance for concentricity?
- Use Minitab.
- You will have 10 minutes to complete this exercise.

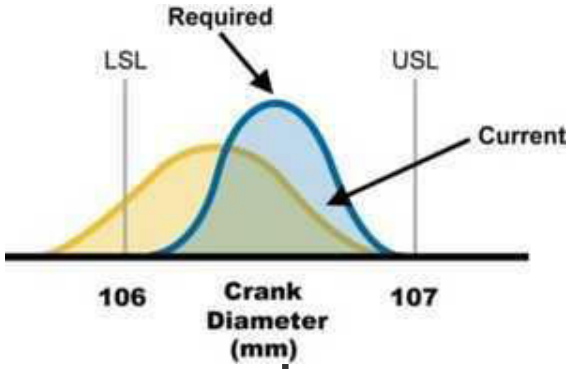
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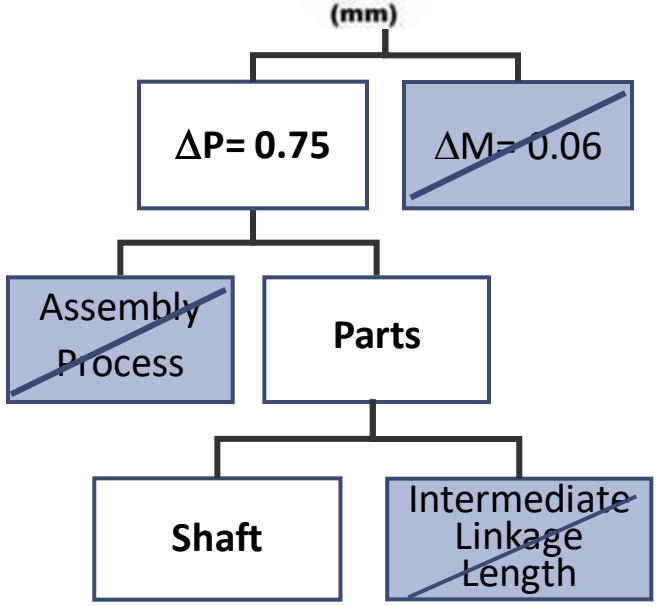
Tolerance Parallelogram Case – Answer

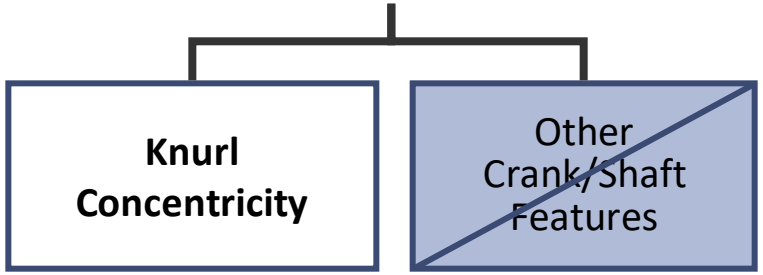


Resulting tolerance for the **BIG X**: 0.011 mm to 0.087 mm

Solution Tree







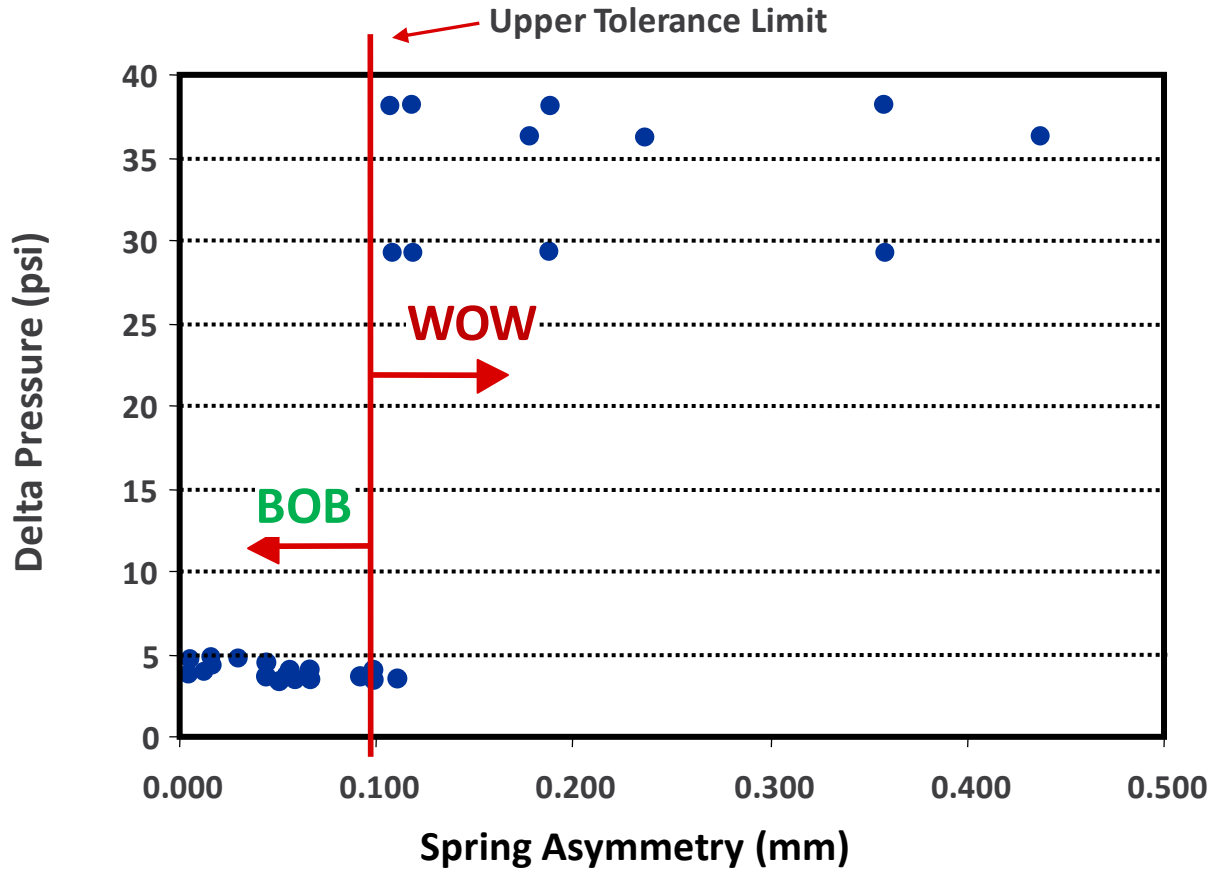
BIG X

Rationale:

- Paired Comparisons
- B vs W 6 Pack test
- Tolerance Parallelogram
  (Knurl Concentricity: 0.011 – 0.087 mm)
- Mone(Y) Run Chart

Non-linear Tolerance

Sometimes the X → Y relationship is not linear, but looks more like a cliff. We can still use this relationship to set realistic tolerances on the X to stay out of trouble on the Y.





Conclusion: Asymmetry of spring legs must be less than 0.1 mm.

Document the Bottom Line Impact

- Consider the following hard factors for money saved:
  - Scrap
  - Rework / labor time
  - Cost of containment
  - Warranty costs
- Capture the savings and quantify the value.
- Saved costs should be calculated for the next year.
- List soft factors:
  - Improved delivery performance
  - Customer satisfaction
  - Capacity increase

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Document the Bottom Line Impact

Hard Factors	Failure Rate	Value Per Part	Parts Per Year	Total
Scrap Costs	5 %	€ 56.00	2,500	€ 7,000
Rework / Labor Time	5 %	€ 13.00		€ 1,625
Measurement and Logistics	100 %	€ 8.00		€ 20,000
New Machine				€ 65,000
Savings in the Next Year				€ 93,625

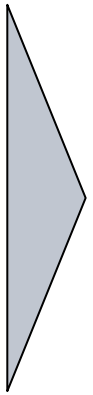
Soft factors:

- Smoother production process
- On time delivery for the next process step (in-house customer satisfaction)

Spread the Knowledge in the Company…

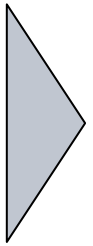
… that was gained in the initial project.

- Short term:
  - Other model
  - Other line / plant
- Long term:
  - New model
  - New line / plant



- Complete project documentation.
- Distribute the summary.
- Update procedures.
- Update FMEA material.

- Lessons learned:
  - For the project
  - For the problem solving approach



- Evaluate what went well and why?
- Apply the same strategy to similar projects.

Management needs to get active to generate more value of the performed project work!

Finishing Projects
DMAIC Documentation → IMS PREMIUM

		Project Leader:	Last Updated:
	Champion/Owner:	Coach:	Plant/department:

RHEINMETALL

Start 11.04.16	Define 28.04.16	Measure 25.05.16	Analyze 19.09.16	Improve 16.03.17	Control 31.11.17	End

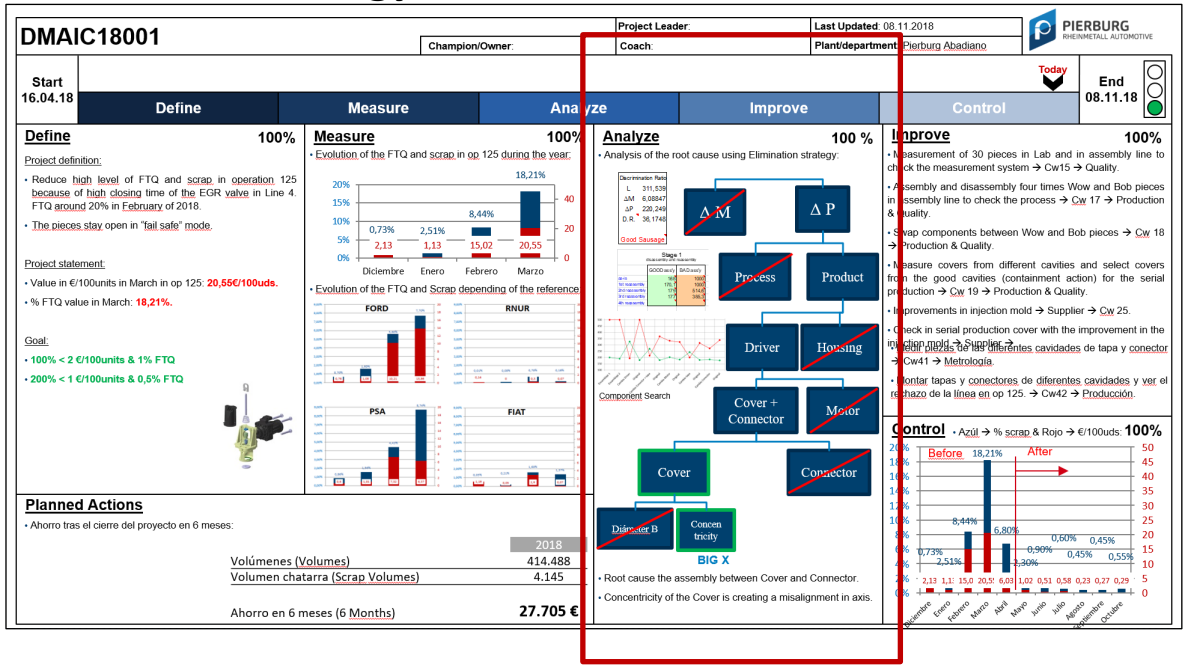
Today

Define 100%	Measure 100%	Analyze 100%	Improve 100%
			Control 100%
Planned Actions			Savings: €

Finishing Projects

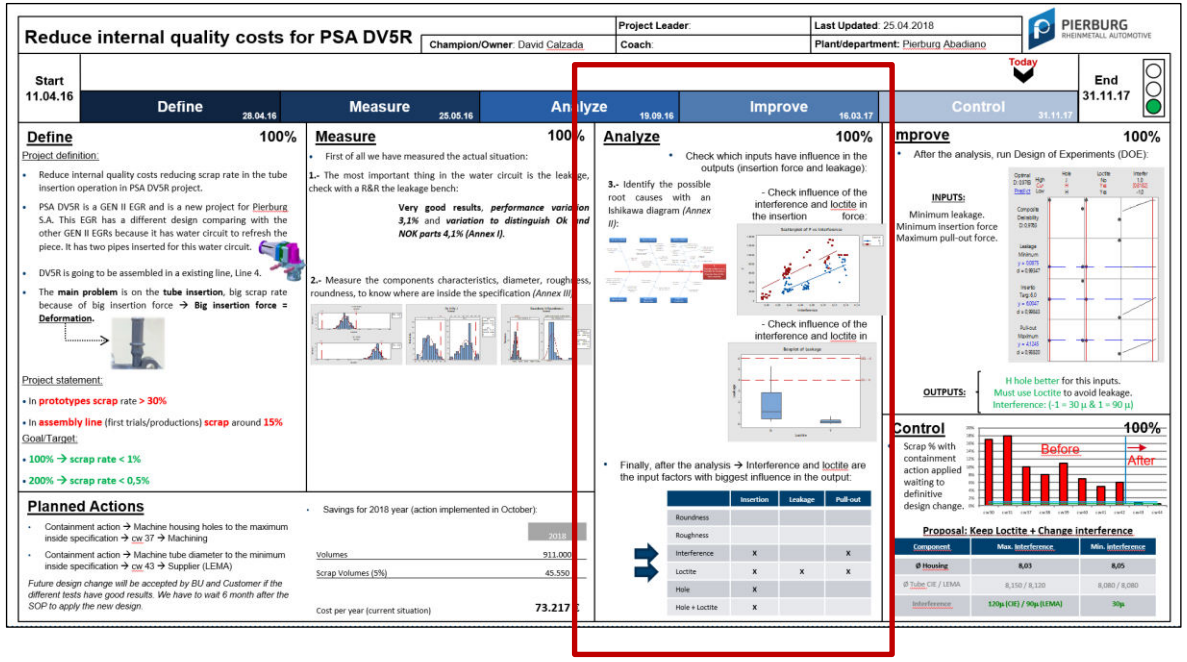
DMAIC Documentation: Difference Elimination Strategy vs Six Sigma

Elimination Strategy



Solution Tree

Six Sigma



Ishikawa Diagram

- The main difference is in the Analyze phase.
- The Elimination Strategy uses the Solution Tree. While Six Sigma uses the Ishikawa diagram.

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	Project Leader:	Last Updated:	
Champion/Owner:	Coach:	Plant/department:	

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Finishing Projects
DMAIC Contents

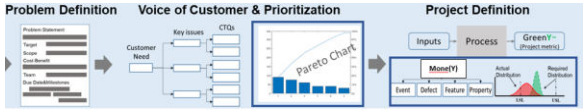
Today

Measure 25.05.16	Analyze 19.09.16	Improve 16.03.17	Control 31.11.17	End 31.11.17

Define (Define improvement project) 100%

Main Tasks
- Define project (project statement)
- Describe initial situation (briefly) + graphic
  → What is the problem (cartooning)?
- Create process overview

Tools (Selection)

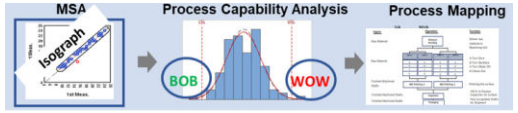


Results (Define)
- Responsibilities are assigned
- Responsibilities are assigned
- Measured variable(s)!
  Only what can be measured can be improved! Rejects are not a suitable metric.
- Possible savings [€]

Measure (Determine actual state) 100%

Main Tasks
- Record process in detail
- Evaluate measurement parameter (suitable?)
- Record actual state by means of existing or measurement of current data (sigma value, Cpk, ...) and evaluate

Tools (Selection)



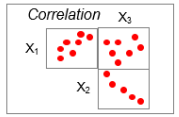
Results (Measure)
- Fact-based understanding of the situation to be improved
- Data basis of the current situation (graphics, key figures, ...)
- Detailed process understanding and representation
- Possible influencing variables are recorded
  → Basis for measurements (data collection) and analysis

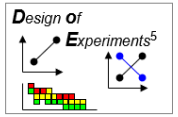
Analyze (Define relevant root cause) 100%

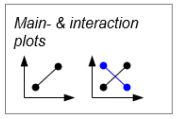
Main Tasks
- Identify main influencing variables
- Identify and present cause-effect relationships
- Identify causes

Tools (Selection)











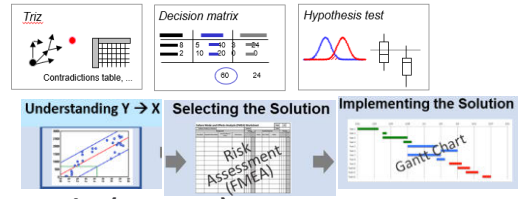
Results (Analyze)
- Confirmed relationships between cause & effect
- Graphical representation of the correlations
- Confirmation Test*

Results (Control)
- Closing (project presentation)
- Completion in JIRA (= report, knowledge management)
- Financially and qualitatively evaluated result

Improve (Develop & Test solutions) 100%

Main Tasks
- Explain the physics Y → X
- Develop, evaluate & select solution variants with date
- Test selected solution and demonstrate effectiveness with date
- Plan & implement implementation (date)

Tools (Selection)

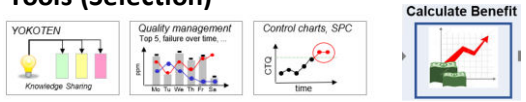


Results (Improve)
- Optimized & proven solution
- Release for implementation

Control (Implement & secure optimized solution) 100%

Main Tasks
- Finalize solution organizationally
- Lessons Learned & Read Across
- Close project

Tools (Selection)



Savings: €

Planned Actions

*Optional

Finishing Projects

DMAIC Documentation (Example): BOB

EVP Pump current > 11,5A

Champion/Owner:	Project Leader:	Last Updated: 17.06.2019
	Coach:	Plant/department: Werk Hartha

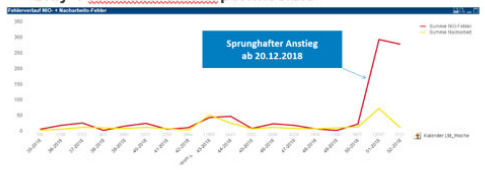
PIERBURG RHEINMETALL AUTOMOTIVE

Start 02.01.19 | Define | Measure | Analyze | Improve | Control 29.05.19 | Today | End 29.05.19

Define 100%

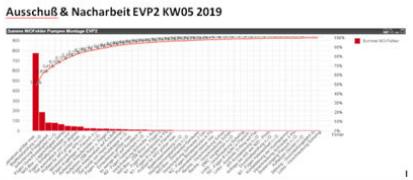
Project definition:

- Increased scrap at the EOL of both EVP lines since 21.12.208
- Current at EOL > 11.5A
- Only 1 remeasurement permissible



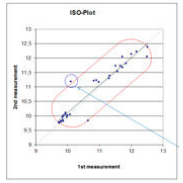
Project statement:

- approx. 20.000 € scrap costs/week



Measure 100%

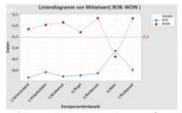
- MSA from EOL available



- Elimination potential material influence through component exchange

With NOK parts
Not OK → parts do not reproducible
→ Repetition with IO parts

Analyze 100%

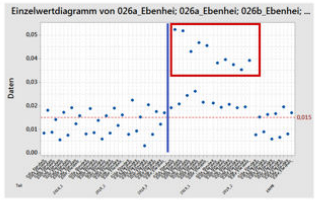


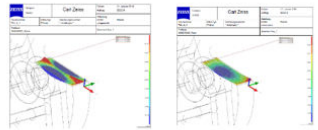
Component Search



Solution Tree

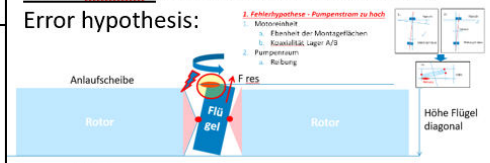
- Analysis of rotor measurement





Main cause: evenness of rotor slots NIO

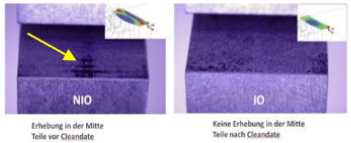
Error hypothesis:



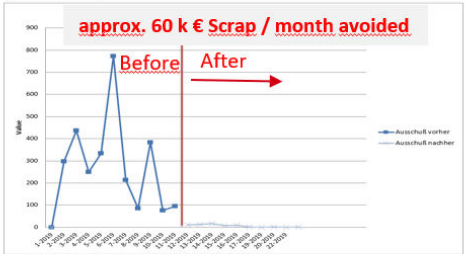
Increased friction of the wings by centrifugal force (video evidence)

Improve 100%

- Evaluation of the pump flow via Qlikview
- Traceability of subcontracted parts improved (SSM)
- Method for root cause analysis clearly defined (component swapping) → Template
- Drawing errors detected → Drawing changed via ECM (evenness rotor slot added)
- Immediate action Supplier: Tool inserts changed + new tool change cycles defined



Control 100%



Savings: 720 k € / year

Planned Actions

- Traceability of subcontracted parts improved continuation SSM is running in a separate project
- Drawing adaptation via ECM is carried out rejection of drawing by SGL Measurement coordination for evenness of wing slots - Deadline: 05/19 / Status expired.

Finishing Projects

DMAIC Documentation (Example): BOB

Stator eOP DB 35 → Risse Kontakt		Project Leader:	Last Updated: 06.08.20
	Champion/Owner:	Coach:	Plant/department: Werk Hartha

PIERBURG RHEINMETALL AUTOMOTIVE

Start 03.07.20 | Define | Measure | Analyze | Improve | Control | Today | End 16.07.20

Define 100 %

Projektdefinition:

Der Kunde Renault hat bei der eOP DB 35 Ausfälle mit dem Fehler: "Not working with nominal" (Mitte Mai 2020). In Livorno wurden nach der Shaker Prüfung Risse im Haken der Statore detektiert.

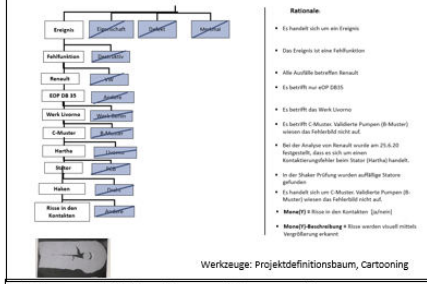


In Hartha wurden von 1.000 Statore visuell geprüft und 20 Statore hatten keine Risse (visuelle Prüfung).

Projektdefinitionsbaum

Projektstatement:

Finde und elminiere die Ursache (BIG X) für den Fehler: Risse im Haken, der in der visullen 100 %-Endprüfung entdeckt wird, um Bandstillstand und Kosten in Höhe von ... Renault zu vermeiden.



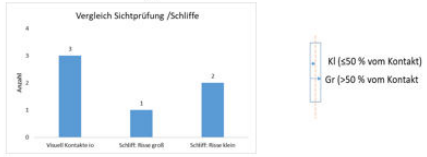
Werkzeuge: Projektdefinitionsbaum, Cartooning

Measure 100 %

Messsystem

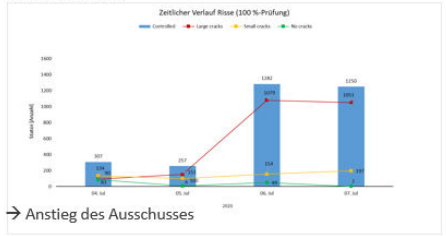
Am 4.7.20: Einführung von Sortierkriterien.

In der Visuelle Prüfung wurde mit Schliffen verifiziert.



Bei visueller io-Prüfung heißt nicht, dass die Kontakte rissfrei sind. Zur weiteren Analyse wird mit visueller Bewertung fortgefahren.

Zeitlicher Verlauf



→ Anstieg des Ausschusses

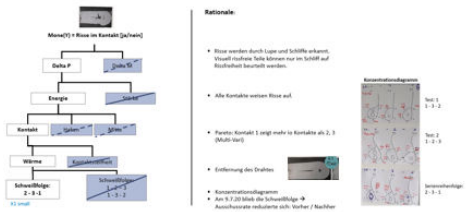
Werkzeuge: Bar Chart, Verlaufsdiagramm

Analyze 100 %

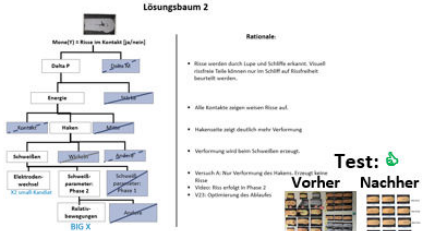
- Kontakt 1 ist robuster als 2 und 3.
- Hakenseite hat weniger Risse als Kontaktseite, aber mehr Verformung



Lösungsbaum 1



Lösungsbaum 2

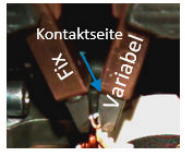


Werkzeuge: Mult-Vari, Konzentrationsdiagramm, Lösungsbaum

Improve 100 %

Mone(Y) → BIG X

Durch die Veränderung des Parameters wird eine erhöhte Stabilität während des Schweißvorganges in Phase 2 erzeugt. Stabilität: Anlage Kontaktseite Teil + keine Relativbewegung in Phase 2



Phase 1 (Biegen)

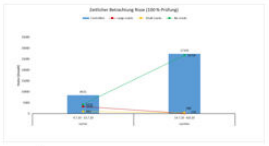


Phase 2 (Schweißen)

Maßnahme(n)

Am 3.7.20 wurde der Versuch V10 (kurzer energiereicher Impuls) eingeführt. Ab 14.7.20 wurde die Suchkraft wurde von 150 N auf 400 N erhöht. Schweißreihenfolge: 2 - 3 – 1. Jedoch muss weiter optimiert werden → S. Bilder Mikrorisse (< 50 % Gabeldicke) → Testergebnis.

Control 100 %



Die Maßnahme ist wirksam.

Lessons Learned

Start: Lessons Learned Workshop 22.7.20

Werkzeuge: Bar Chart

Savings: 541 T€

Planned Actions/Remarks

- Die Ursache für das Kundenproblem "Not working with nominal" wurde verursacht durch eine nicht ausreichende Anbindung des Drahtes im Haken → Nicht korrekt in der Höhe eingestelltes Elektrodenpaar. Die Maßnahme: Elektrodeneinstelllehre wurde am ? in die Fertigung eingeführt.
- In der Validierung bestandene Statore (1.000 h) hatten alle große Risse.
- Einführung eines schweißbareren Materials (Datum: Offen)
- Tausch der „fixen" Elektrodenseite (Datum: Offen)

Finishing Projects

DMAIC Documentation (Example): BOB

DMAIC18001	Project Leader:	Last Updated: 08.11.2018
	Champion/Owner:	Coach:	Plant/department: Pierburg Abadiano

PIERBURG RHEINMETALL AUTOMOTIVE

Start 16.04.18 | Define | Measure | Analyze | Improve | Control | End 08.11.18 (Today)

Define 100%

Project definition:

- Reduce high level of FTQ and scrap in operation 125 because of high closing time of the EGR valve in Line 4. FTQ around 20% in February of 2018.
- The pieces stay open in "fail safe" mode.

Project statement:

- Value in €/100units in March in op 125: **20,55€/100uds.**
- % FTQ value in March: **18,21%.**

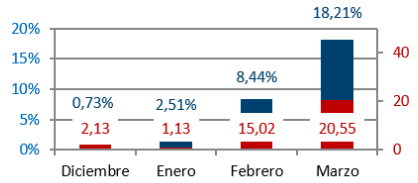
Goal:

- **100% < 2 €/100units & 1% FTQ**
- **200% < 1 €/100units & 0,5% FTQ**

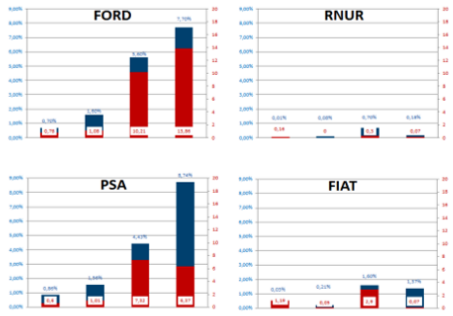


Measure 100%

- Evolution of the FTQ and scrap in op 125 during the year:

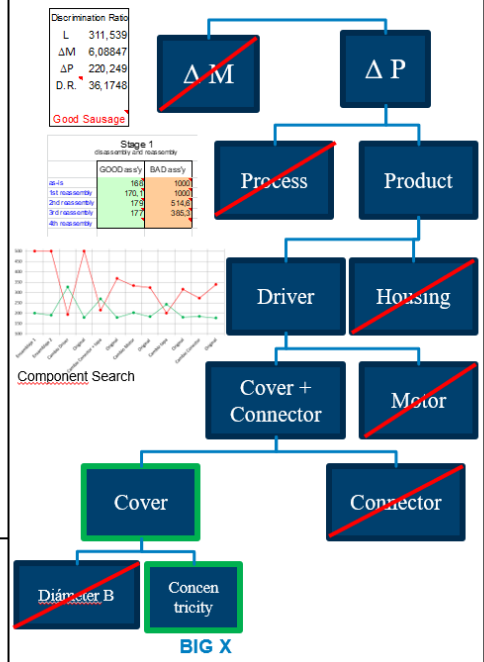


- Evolution of the FTQ and Scrap depending of the reference:



Analyze 100 %

- Analysis of the root cause using Elimination strategy:

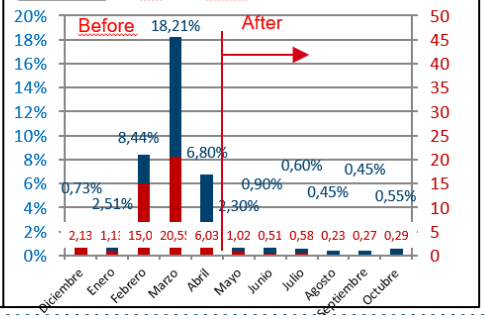


- Root cause the assembly between Cover and Connector.
- Concentricity of the Cover is creating a misalignment in axis.

Improve 100%

- Measurement of 30 pieces in Lab and in assembly line to check the measurement system → Cw15 → Quality.
- Assembly and disassembly four times Wow and Bob pieces in assembly line to check the process → Cw 17 → Production & Quality.
- Swap components between Wow and Bob pieces → Cw 18 → Production & Quality.
- Measure covers from different cavities and select covers from the good cavities (containment action) for the serial production → Cw 19 → Production & Quality.
- Improvements in injection mold → Supplier → Cw 25.
- Check in serial production cover with the improvement in the injection mold → Supplier →
- Medir piezas de las diferentes cavidades de tapa y conector → Cw41 → Metrología.
- Montar tapas y conectores de diferentes cavidades y ver el rechazo de la línea en op 125. → Cw42 → Producción.

Control • Azúl → % scrap & Rojo → €/100uds: 100%



Planned Actions

- Ahorro tras el cierre del proyecto en 6 meses:

	2018
Volúmenes (Volumes)	414.488
Volumen chatarra (Scrap Volumes)	4.145
Ahorro en 6 meses (6 Months)	**27.705 €**

Finishing Projects
DMAIC Documentation (Example): WOW

DMAIC Status Report

[redacted] | Last Updated: 04.02.2021

Champion [redacted] | Plant/department: Hartha

PIERBURG RHEINMETALL AUTOMOTIVE

Start 04.01.2021 — Today — End 06.08.21

Define | Measure | Analyze | Improve | Control

Define 100%

Project definition:

WUP4 - Fügevorgang Spalttopf in Stator

Analyse Prozess:

- Ursprüngliche Auslegung „kraftloses Fügen“

→ Einsetzen/ Vorfügen Spalttopf in Stator via Roboter

- derzeitige Kraftaufbringung von ca. 80N notwendig
- Auslegung Bauteilaufnahme und Roboterhandling nicht für Kraftaufnahme ausgelegt, daher kommt es zum Verdrücken der Aufnahme
- Aktuelle Ausschussquote von 30 – 35%

Mögliche Ursachen:

- Falsche Auslegung Fügeprozess
- Anlieferung Stator montiert nicht nach Spezifikation
- Fehlende HSB vom Lieferant des Stator bez. Toleranzeinhaltung Innendurchmesser

Ziel:

- Prozesssicherer Fügeprozess
- Reduzierung Ausschussquote

Measure 100%

1. Vermessung Bauteile Stator umspritzt und Stator montiert
2. Erstellung Fehlerbaumanalyse Statormontage
   - ☑ → Sind Maße vom Kaufteil und verarbeiteten in Toleranz?
   - ☑ → Werden Maße durch Prozess beeinflußt?
   - ☑ → Fehlerhäufung Formnester/ Fertigungslosen/ Mitarbeiter?
   - ☑ → Gibt es Änderungen im Prozess?
   - ☑ → Zusammenhang zwischen hohen Fügekräften RSR?
   - ☑ → Sind Bohrungen Kontakttaschen kleiner geworden?
   - ☑ → Sind Kontakte größer geworden?
   - ☑ → Kalibrierung bei RSR Fügen möglich?
3. Erstellung Fehlerbaumanalyse Pumpenmontage
   - → Wie weit wird Spalttopf gefügt?
   - → Winkelfehler zw. Gehäuse und Spalttopf
4. Anpassung Fügeprozess an Ist-Stand
   - ☑ → Klärung max. zulässige Fügekraft
   - ☑ → Klärung aufkommender Kosten für Integration
5. Klärung kurzfristige Maßnahmen zur Reduzierung des Ausschuss in Pumpenmontage
   - ☑ → AWE H010/21 „Backen“
6. Werkzeugänderung Stator/ Kontakte
   - → offen, Angefragt bei Lieferant in KW3

Analyze 100%

☑1. Gegenmessung Stator umspritzt und Stator montiert liegt vor - Auswertung

☑2. Fehlerbaumanalyse Statormontage liegt vor - Auswertung

☑3. Fehlerbaumanlayse Pumpenmontage liegt vor - Auswertung

☑4. Technisches Gespräch mit Fa. Akon erfolgt, Angebot liegt vor

☑5. Kurzfristige Maßnahmen definiert

- ☑ → „Backen“ Stator montiert, Muster für Verbauversuche aufgebaut, Verbau positiv, AWE erstellt - **Umgesetzt**
- → „Kalibrieren“ Stator montiert, Muster für Verbauversuche aufgebaut – **verworfen**

☑6. Werkzeugänderung bei Lieferant Domel angefragt

Improve 25%

- ☑ Einsatz **Kurzfristmaßnahme** Stator montiert mit zusätzlicher Temperaturbehandlung bei 150°C backen im Ofen, AWE Freigabe bis 31.01.21 (H010/21)
- ☑ **Mittel- und Langfristmaßnahme** offen

Control 0%

Planned Actions

- Folgebesprechung mit Team in KW06

→ Entscheidung zur mittelfristigen und langfristigen Lösung

→ Entscheidung Einsatzdauer Kurzfristmaßnahmen

Finishing Projects

What information should a DMAIC contain?

Stator eOP DB 35 → Cracks Contact

Champion/Owner: [redacted] | Project Leader: [redacted] | Coach: [redacted] | Last Updated: [redacted] | Plant/department: [redacted]

RHEINMETALL

Start 11.04.16 | Define 28.04.16 | Measure 25.05.16 | Analyze 19.09.16 | Improve 16.03.17 | Control 31.11.17 | End 31.11.17

Today

Define 100%

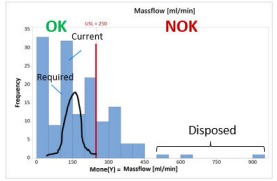
Describe initial situation & Project Statement (briefly)!

What is the problem? → Cartooning

Initial Point
On 20.1.21 a R@R was held by the customer DAF.
For EGR Valve 7.08797.10.0. the test was not passed
→ ~19 % failure rate.

Project Statement
- Identify and control root cause (BIG X) for high leakage rate at EOL (OP 160.3) @line 21.
- Required leakage rate: < 250ml/min
- Target Scrap rate: < 0,5%
- Pot. Savings: ?

Strategy: Malfunction Event

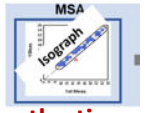


Measure 100%

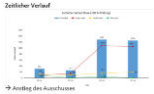
What does the process look like? Where do we discover the problem?



Is the measuring system capable?



What does the time course look like or what does the data show us?



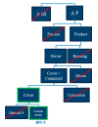
Analyze 100%

How did we approach the root cause(s)?

Six Sigma **Or** Elimination Strategy

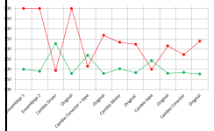


Ishikawa

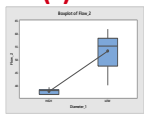


Solution Tree

Which experiment(s) led us to the root cause(s)?



Component Search



BIG X candidate:
BOB WOW Comparison

Confirmation* Test/Risk/Result

Concentricity was confirmed by a B vs W test with a 5% risk.

What is/are the root cause(s)?

Concentricity is the root cause (BIG X).

Improve 100%

Mone(Y) → BIG X Explanation of physics

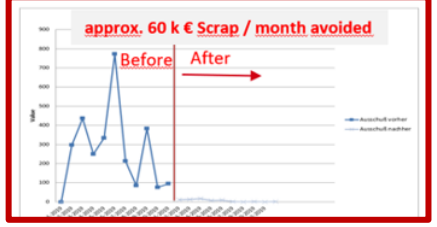
Graphic!



What measure(s) do we implement & when?

Grinding of the bars on 12/1/20

Control 100 %



Savings: 721 T€

Planned Actions

Which measures are planned and when?
Who is responsible?

If no → Slash

*Optional

Module 8 Summary

Tolerancing / Finishing Projects I

- Tolerance targets are set to produce desirable events; limits are set to avoid undesirable events.
- If a Mone(Y) tolerance cannot be met, this is due to either an inability to control the BIG X or too much “all else” variation.
- Tolerance Parallelogram is the means by which we are able to target only the BIG X variation to achieve Mone(Y) tolerances.
- The achieved benefit must be evaluated and communicated at the end of a project.
- Management must actively push the gained results into the organization to maximize value of the performed project work.

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9 Summary

Module 9 Objectives

Summary

Understand:

- Next steps in developing BIG X problem solving skills
- BIG X certification

Be able to:

- Document a project following the DMAIC roadmap

PSQ Green Belt Class

In this class, you have learned how to discover the hidden causes keeping your systems from performing their best.

Concepts:

- BIG X
- Dictionary Game Thinking
- Investigative Cycle

Strategies:

1. Understanding system structure for features, properties, defects and malfunction events.
2. Documenting that structure on a Strategy Diagram.
3. Planning and executing dictionary splits.
4. Converging to the BIG X.

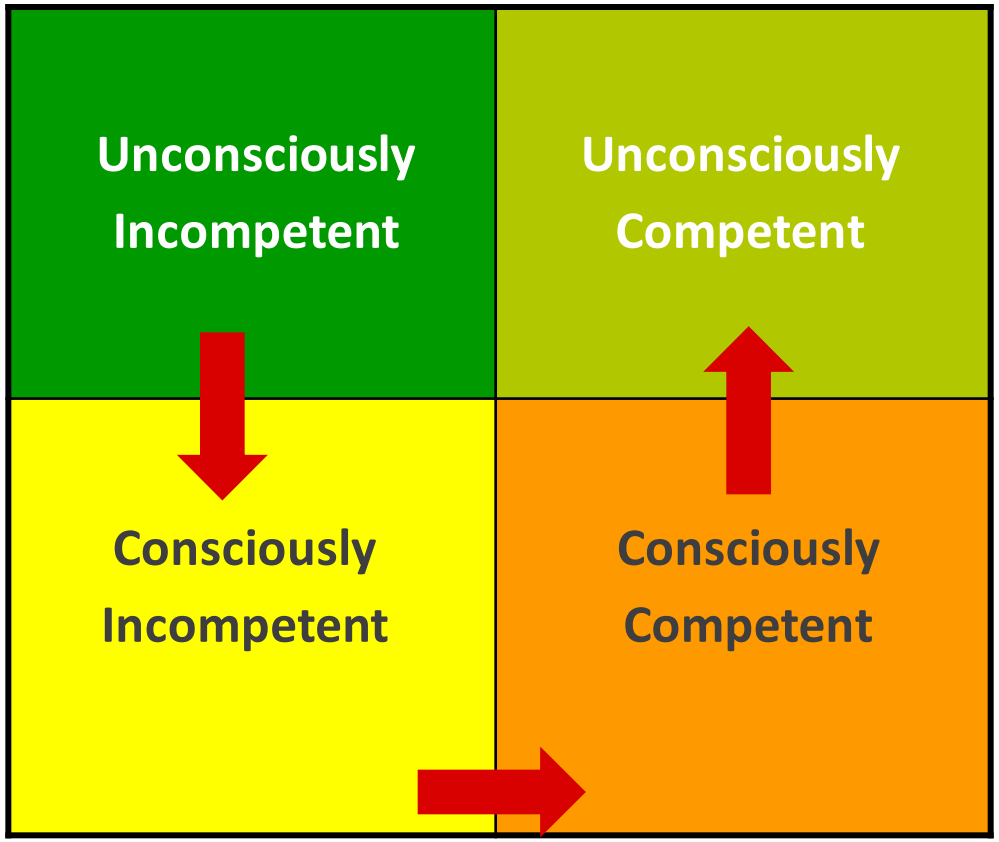
Tools:

- To isolate the BIG X family.
- Search within the BIG X family.
- Confirm the BIG X.
- Establish a tolerance for the BIG X.

You now need to transform that knowledge into a skill.

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Skill Development Steps



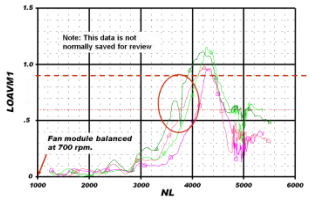
http://en.wikipedia.org/wiki/Four Stages of Competence

Example: Project Nesting

Measured Response or **Undesirable Response**

System Level Problem

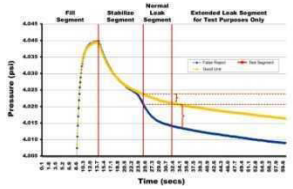
Vehicle
Air Plane
Machine



- Error Code
- Vibration
- Noisy
- Leaking
- No Function
- etc.

Sub-System Problem

Brakes
Generator
Pump

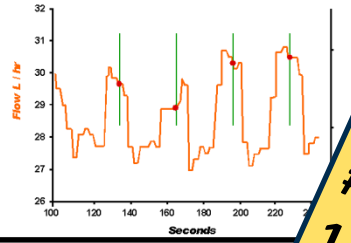


- Error Code
- ration
- oisy
- eaking
- No Function
- etc.

DMAIC # 1

Component Problem

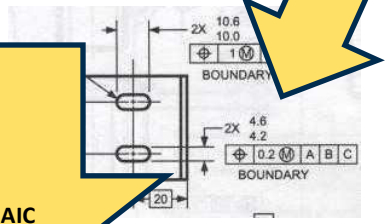
Rotor
Circuit Board
Sensor



- Vibration
- Noisy
- Leaking
- No Function
- etc.

Feature/Property/Defect

Diameter, distance from dat
True Position
Porosity



- Defect
- Damaged or Changed Feature

DMAIC # 2

Process Parameter

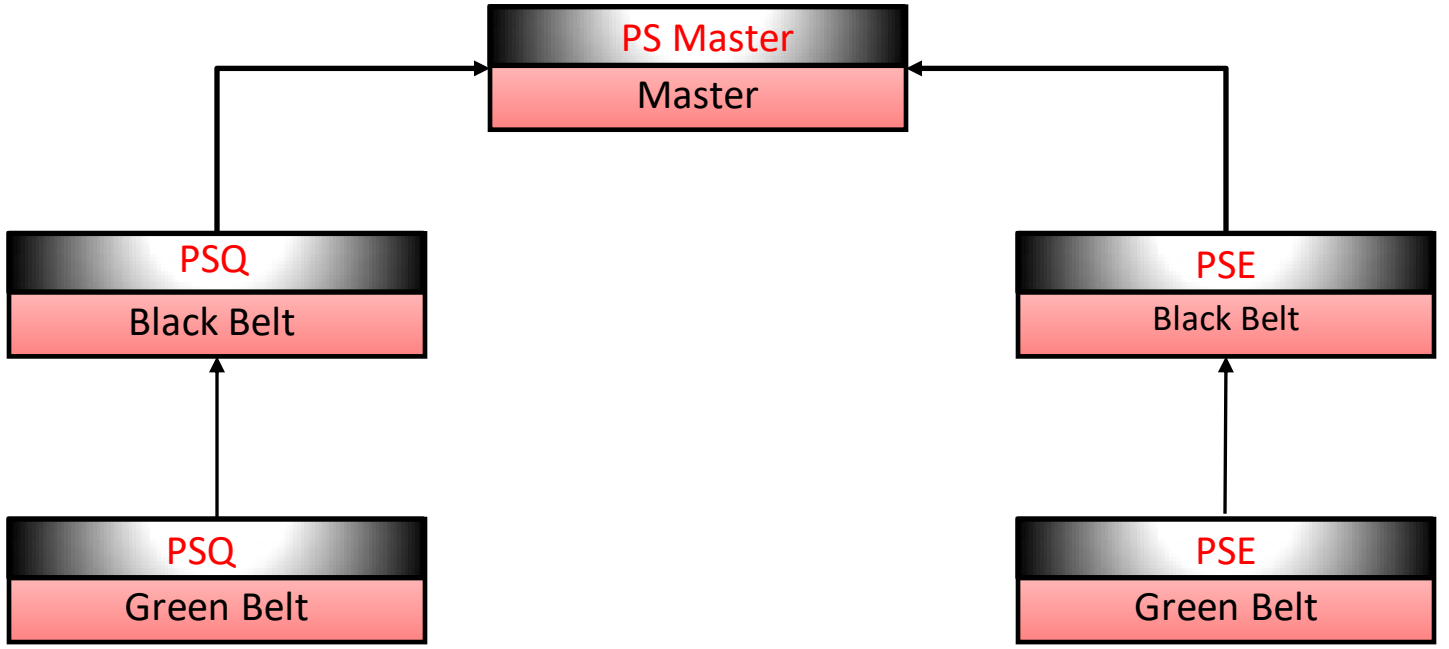
- Process Setting
- Assembly Procedure

Outlook PSQ Black Belt Content

- Destructive Event Strategies
- Malfunction Event Strategies II
- Operation Searchs
- Event to Energy Transform
- Measurement System Analysis:
  - Binary Measurements
  - Destructive Measurements
- Energy Accounting
- Function Models
- Confirmation Testing II
- Multi Factor Experiments

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Problem Solving Certification Requirements



Manufacturing & Process Quality Path

Engineering & Customer Quality Path

PSQ Green Belt Certification

A PSQ Green Belt is able to use basic BIG X tools and to complete a project under the direction of a coach. PSQ Green Belt certification serves to recognize that the candidate has achieved a significant milestone towards PSQ Black Belt certification.

Actions Required:

1. Attend the PSQ Green Belt class (3 Days).
2. Complete one reactive project as the team leader or team member with coaching support from PS Master.

Skill Requirements:

1. Demonstrate proper use of tools deployed in first project.

Time Limit:

Class attendance and project completion (control number issued) must be completed within a one year time window.

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PSQ Black Belt Certification

A PSQ Black Belt is a skilled technical problem solver. A PSQ Black Belt has been recognized for demonstrating the ability to develop and execute problem solving strategy effectively.

Actions Required:

1. Attain PSQ Green Belt certification.
2. Attend the PSQ Black Belt class (4 days)
3. Lead a successful PSQ Black Belt project to completion under the guidance of a of a PS Master.

Skill Requirements:

1. Complete PSQ Black Belt Toolbox.
2. PSQ Black Belt Interview from a PS Master.

Time Limit:

Class attendance and project completion (control number issued), toolbox and interview must be completed within a one year time window.

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PSQ Black Belt Certification

PSQ Black Belt Toolbox

Required tools

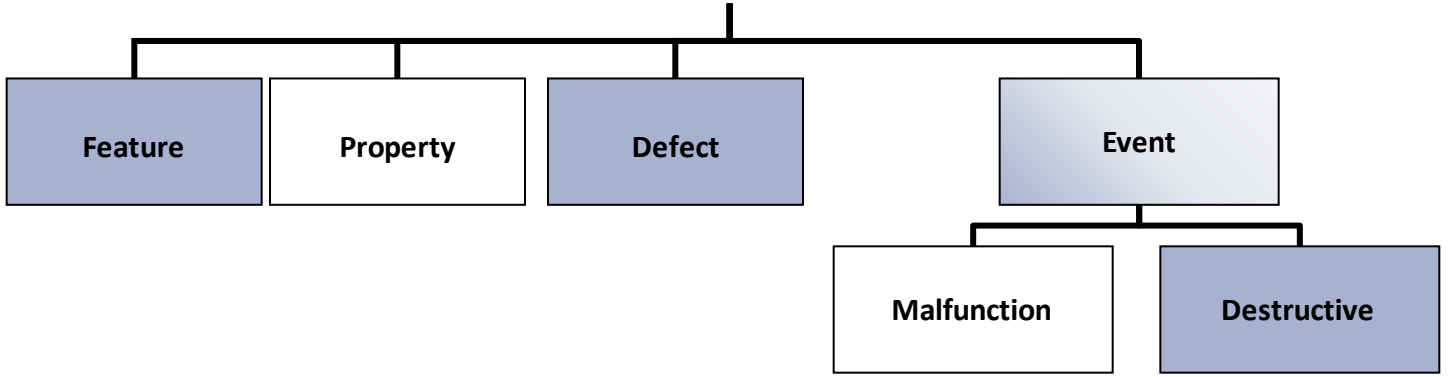
- Strategy Diagram
- Concentration Diagram
- BIG X Confirmation Tests
- Isograph or Component Search – Stage 0
- Tolerance Parallelogram or Concept Diagam

Plus one of the following groups

- Process Flow Diagram and Multi-Vari
- Component Search and Event to Energy Transform

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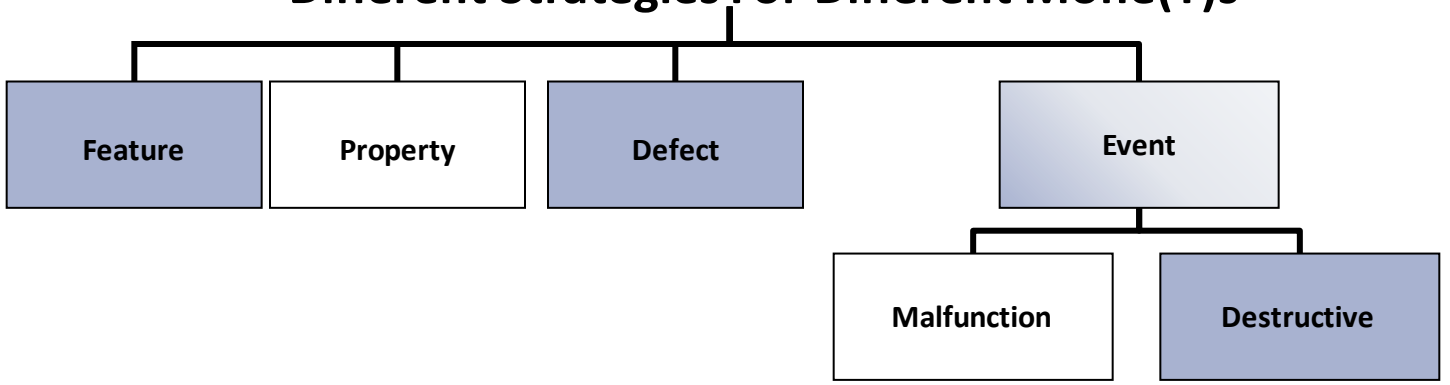
Different Strategies For Different Mone(Y)s



Management assigns a technical project that can be expressed as a type of Mone(Y).

	▪ Radius ▪ Length ▪ Position ▪ Thickness	▪ Density ▪ Hardness ▪ Strength ▪ Concentrations	▪ Dirt in Paint ▪ Ripples ▪ Porosity ▪ Blow Holes	▪ Noise ▪ Leaks ▪ Power Output ▪ Vibration	▪ Cracks ▪ Dents ▪ Welded ▪ Melted	Physical nature of the Mone(Y) (examples)
M	▪ Variable Data ▪ Use Isograph ▪ What' the datum?	▪ Variable Data ▪ No Datum ▪ Measurement Systems are often Events	▪ Location, not magnitude ▪ What does Defect look like? ▪ Can we tell BOB from WOW?	▪ Think energy ▪ Stage 0 ▪ Look at Event over time	▪ Map location of failure ▪ Think Energy ▪ Cut samples into 2 slices	Measurements $\Delta P/\Delta M$
A	▪ Process Flow Diagrams ▪ Strategy Diagrams ▪ Multi-Vari	▪ Process Flow Diagrams ▪ Strategy Diagrams ▪ Multi-Vari	▪ Process Flow Diagrams ▪ Strategy Diagrams ▪ Concentration Diagrams	▪ Energy model and Strategy Diagram ▪ Component Search ▪ Paired or Group Comparisons	▪ Energy model and Strategy Diagram ▪ Concentration Diagram ▪ Event to Energy Transform	Tools for making Solution Tree Splits

Different Strategies For Different Mone(Y)s



Management assigns a technical project that can be expressed as a type of Mone(Y).

A	B vs W 6 Pack Test; Spike B vs W Test; Full Factorial Design Experiment; χ^2 Test; 5 Penny	Tools for Confirmation Testing
I	Tolerance Parallelogram; Concept Diagram; Response Surface Map	Tools for tolerancing the BIG X
I & C	Mone(Y) Run Chart or p-Chart; Barrier B vs. C Test; B vs C 6 Pack Test; SPC; Precontrol; Design Change / Process Change; Corrective Action Tree; Corrective Action Matrix	Tools for Confirming & implementing corrective actions
C	Project Write-up; Leverage Matrix	Tools for reading across the gains

G Glossary

Glossary

1. **4 Ps:** The four elements (Product/Process, Problem, Place, Pain) of the Project Statement.

2. **Assignee:** The individual assigned by the Champion to lead the project. The Projectleader's role includes (a) executing strategy, (b) bringing barriers to sponsor's attention, (c) communicating progress to sponsor, (d) developing corrective action plans.

3. **BIG X:** The cause with the strongest leverage as the Mone(Y) changes from BOB to WOW. It may be a single variable or the interrelationship among several variables, i. e., an interaction. The BIG X becomes a BIG X only after passing the confirmation phase.

4. **BOB:** Best of the Best and Worst of the Worst are the extreme samples that can be readily taken from a distribution of the Mone(Y). The BOB is not necessarily the optimum value, and the WOW is not necessarily the poorest value. The optimum may be at either extreme or at any other point in the distribution.

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Glossary

5. **Champion:** The champion assigned to assure successful completion of a project. Middle- or senior-level executive / manager who sponsors a specific improvement project, ensuring that resources are available and cross-functional issues are resolved. A person most interested in the expected improvement delivered by the project. Usually the process owner or the superior of the process owner → problem or process owner. (Process) Owner = person accountable for the quantity & quality of the process output (e. g. line supervisor). A manager who monitors the day-to-day progress of a project, helps develop and execute strategy, and removes barriers to progress. The champion assigned to assure successful completion of a project. The Sponsor's role includes (a) selection of team; (b) provision of required resources to the team, (c) removal of barriers, (d) review of project definition, strategy selection and execution, (e) review of corrective action implementation, (f) leveraging the resulting assets across the organization.

6. **Continuous Improvement:** The concept that any business must continually get better at what it does in order to maintain its competitive position. It is often misinterpreted to mean improvement can only occur gradually.

7. **Contrast:** Difference observed when comparing the Mone(Y) across different design or process families.

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Glossary

8. **Defect:** Product imperfections that occur during manufacturing. They are localized deviations. Evidence left behind after an undesireable event. One time non-repeatable events. Can be quantified as attributes (good/bad).

9. **Delta M (Measurement error):** Variation in measured values due to characteristics of the measurement process. The measurement process includes instruments, procedures, fixtures, part locating features, and operator influences.

10. **Delta P (Product variation):** Variation in measured values due to a true difference in the feature, defect or product performance.

11. **Deployment Champion:** Senior executive / manager who sponsors the overall Business Excellence / Six Sigma / Continuous improvement initiative. Ensures that business metrics are linked to Top 5 problems and takes responsibility for leveraging, across functional areas, knowledge gained during project work.

12. **Design Family:** Families of variation, including a contrast and a field of vision, that are observed within a single part. Design families describe part shape.

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13. **Destructive Event:** Energy has permanently altered the product and has left behind a trace.

14. **Dictionary Game:** It is a progressive search for the BIG X by leveraging logic. The objective is to ask logical questions that split the dictionary into families of variation to progressively search for the BIG X.

15. **Discrimination:** The ability to see differences.

16. **Discrimination Ratio (ΔP/ΔM):** The ratio of product variation to measurement error; a measure of the ability to see differences. Higher ratios are better.

17. **Event:** Usually a transient and requires activation energy to occur. An Event happens over time (i. e. beginning, middle, end). Desirable events represent the functions the product was designed to perform; undesirable events occur when products do things other than intended. Can be measured with either attribute or variable (number) measurement systems.

Glossary

18. **Feature:** Physical dimensions of a product that describe shape. To be useful for problem solving, features must relate to the physics of the processes that created the shape. Measured with variable measurement systems.

19. **Function:** The activity or purpose that a part or process is expected to perform.

20. **Isograph:** A graphical technique for separating product variation (Delta P) and measurement error (Delta M) to evaluate a measurement system's ability to discriminate.

21. **Leverage:** Strategic advantage. During problem solving and prevention, leverage is often gained from the contrast between BOB and WOW. In the case of Top 5, leverage is gained by identifying problems that, when solved, will have far-reaching effects on your ability to profitably design and manufacture products.

22. **Malfunction Event:** Product does not work at the expected level of performance. Evidence of the events disappears as energy supply is removed. Energy can be applied to attempt to reproduce the malfunction.

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Glossary

23. **Mone(Y):** The response that provides engineering leverage for effective problem solving. To be useful, a Mone(Y) must relate to physics, features, or properties.

24. **Paradigm:** A mental model or lens through which the world is viewed.

25. **Pareto Chart:** Bar chart with categories arranged in rank order by frequency of occurrence.

26. **Pareto Relationship:** Frequency of occurrence of each category is typically at least 50 % greater than that of the next less frequent occurrence for the first 5 or so occurrences and approximately equal thereafter.

27. **Problem:** A failure or risk of failure to meet a goal.

28. **Problem Definition Tree:** Scope tree that starts with a problem as seen by management. Through a series of converging splits it focuses on a project to address the problem.

29. **Process Families:** Families of variation, including a contrast and a field of vision, that involve multiple parts and are defined by the capital structure of the manufacturing operations.

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30. **Project Definition Tree:** The Project Definition Tree defining the problem begins with the problem defined for an engineer and translates it into Statistical Engineering terms. It starts with the Project Statement (4Ps) and ends with a clear Mone(Y) and description.

31. **Property:** A material characteristic associated with a part.

32. **Solution Tree:** The Solution Tree organizes the investigation after the project has been defined. The Solution Tree that represents the planning and execution of dictionary game thinking. It starts with the Mone(Y) event or feature distribution or defect statement and through a series of converging splits, focuses on the BIG X. The solution tree ends with the identification of the BIG X and a confirmation.

33. **Strategy:** The action plan to accomplish a specified objective.

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Glossary

34. **Top 5:** A management methodology characterized by (a) selecting the most important business problem(s) using Pareto analysis and a Scope Tree, (b) converting the problems to a project using a Problem Definition Tree, (c) focusing resources and using a disciplined process of elimination with a Project Definition Tree and a Solution Tree, (d) implementing a permanent solution, (e) leveraging the knowledge gained across the organization.

35. **WOW:** Worst of the Worst and Best of the Best are the extreme samples that can be readily taken from a distribution of the Mone(Y). The BOB is not necessarily the optimum value, and the WOW is not necessarily the poorest value. The optimum may be at either extreme or at any other point in the distribution.

PASSION FOR **TECHNOLOGY.**